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Plenary Session – TRALI Update

PL1-02

HOW DIFFERENT ANIMAL MODELS HELP US UNDERSTAND TRALI

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The 2016 FDA report shows that transfusion-related acute lung injury (TRALI) is still one of the leading causes of transfusion-associated morbidity and mortality. This is despite increased awareness of TRALI and the implementation of various strategies (e.g. use of predominantly male plasma products) to minimize its incidence. TRALI may be either immune/antibody-mediated or non-antibody mediated. But understanding the mechanism behind TRALI has been challenging because TRALI events can be difficult to identify, the incidence is not high, and many of the clinical reports have been retrospective studies. Animal models provide a means of investigating the mechanism of TRALI in a controlled and systematic fashion. Small animals have the advantage of being easier to handle, have knock out strains and are relatively lower in cost. Large animals have the advantage of being more closely related in size to humans and support multiple sample collections.

One of the earliest TRALI animal models demonstrated the immune-mediated mechanism. An *ex vivo* rabbit lung model revealed the role of anti-5b (anti-HNA-3a) and granulocytes (Seeger *et al.* 1990). Mouse *in vivo* models infusing anti-MHC Class I monoclonal antibody (the 34-1-2s clone), and also rat and pig models, have provided further insights into the mechanisms of immune-mediated TRALI. Murine models have shown the importance of neutrophils (Looney *et al.* 2006; Kelher *et al.* 2009), neutrophil Fc receptors (Looney *et al.* 2006), neutrophil extracellular traps (Caudrillier *et al.* 2012), platelets (Looney *et al.* 2009;), monocytes (Strait *et al.* 2011; McKenzie *et al.* 2014), and endothelial cells (Looney *et al.* 2006; Strait *et al.* 2011; Bayat *et al.* 2013;). Elevated levels of C-reactive proteins (Kapur *et al.* 2015) and mechanical ventilation (Vlaar *et al.* 2010) may predispose to TRALI. Furthermore, protective roles have been identified with aspirin administration (Looney *et al.* 2009; Ortiz-Munoz *et al.* 2014), and with lymphocytes (Fung *et al.* 2014), T-regulatory cells and dendritic cells (both Kapur *et al.* Blood 2017). Mouse models have shown that IL-10 infusion either protects from TRALI if administered prophylactically or successfully treats an on-going TRALI reaction if administered therapeutically (Kapur *et al.* 2016).

An *ex vivo* rat lung model by Silliman and co-workers introduced the non-immune mechanism of TRALI (Silliman *et al.* 1998). This non-immune mechanism has further been investigated using *in vivo* rat (Kelher *et al.* 2009; Silliman *et al.* 2011) and sheep models (Tung *et al.* 2011 and 2012). These models have demonstrated the importance of a priming event (e.g. endotoxin administration) and a role for the protein and lipid biological response modifiers that accumulate during routine blood product storage.

This lecture will chronologically review the key animal models of TRALI to show how each has helped us understand the immune and non-immune mechanisms behind TRALI. Better understanding of the underlying mechanisms is essential to guide effective TRALI minimization strategies.

Haemovigilance – Towards Safe Blood Transfusion

2A-S01-01

PATIENT SAFETY: ADVERSE BLOOD TRANSFUSION EVENTS IN CHINA

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Blood, a special and precious healthcare resource, has been in clinical use for over centuries. Today, blood transfusion still remains as an essential medical treatment. The process control of blood safety encompasses the entire blood supply chain from vein to vein. It is potentially high-risk and with health hazard and can face social and technical challenges. Sustained efforts have been made strategically in China. This paper illustrates the national strategies, progress made and challenges faced on blood safety in China.

Blood donation has successfully transitioned from paid donation, then compulsory donation, to voluntary non-remunerated donation in China. The voluntary non-remunerated donation rate has reached 96.3% and its donations given by donors under 45 reached 85%. The implementation of the Blood Donation Law and improvements of the legal system symbolised the maturation of blood services in China under the framework of law and regulations. In addition, academic training, continue education, and disciplinary development have enhanced the technical expertise and capability of the service providers.

Financial support from the government has ensured funding for blood safety. A blood service system that serves the entire nation has been established. With 452 blood centers, the service has reached remote areas such as Tibet. The volume of blood donations has been steadily increasing for the past two decades from fewer than 1,000 tons in 1998 to 4,658 tons in 2016, meeting the basic demand for blood supply.

Thanks to the optimisation of donor screening and blood testing strategies, blood safety has been improving. Positive rates of HIV, HBV, HCV and syphilis screening were merely 0.17%, 2.25%, 0.36% and 0.63% respectively.

Management of clinical transfusion at medical institutions continues to improve. Medical personnel have become better educated on clinical use of blood products. The transition from whole blood transfusion to component therapy has made clinical transfusion more effective. Whole blood use has decreased from 100% to 0.3%. The change from liberal to restrictive transfusion has greatly reduced unnecessary use of blood. Units of red cell components transfused per discharged patient, per operation, and per bed were 0.26U, 0.66U and 9.9U respectfully. Pediatrics, gynecology and obstetrics accounted for 9% of total blood usage.

Adoption of the quality management and informatics systems has decreased the adverse incidence, and blood quality has been improved. National and regional supervisory inspections have encouraged continuous progress on blood safety and technologies. Adverse events have been better controlled since The Blood Donation Law was enacted in 1998.

Amid rapid social and economic development, people now have higher demands on blood safety, creating many new challenges for the blood supply and transfusion services. Development of blood transfusion field calls for unyielding efforts of all parties. It is expected that >=15 per thousand persons will donate blood by 2020. A blood supply and transfusion service in keeping with the socio-economic development and a blood quality and safety support system in line with the "Healthy China" initiative will be put in place, propelling leapfrog change in China's blood services. The Chinese blood supply and transfusion service will face new opportunities in the new era. The Chinese experiences on blood safety may lend lessons to other developing countries, especially countries along the Belt and Road routes, and contribute to global improvement on blood safety.

2A-S01-02

IMPROVEMENT OF THE NEPAL BLOOD TRANSFUSION SYSTEM AN ASSESSMENT OF TRAINING PRIORITIES

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Background: The Nepal blood supply system is centrally coordinated by Central Blood Transfusion Service of the Red Cross Society. In the aftermath of the 2015 earthquake a need for improvement was identified. With support of the Non Resident Nepali Association During a fact finding mission with the actual quality situation and related priorities for improvement were assessed. This assessment of priorities is used as input for the next steps, development of a tailor made training program of the three years project.

Aims: Improvement of Nepal's blood transfusion system.

Methods: After two years of preparation Sanquin in collaboration with CBTS, during a fact finding mission, the actual situation was assessed to identify the next steps for improvement. To facilitate participatory collaboration a one day workshop was held with key stakeholders of the Nepal blood transfusion system. Including representatives of blood banks, the Min of Health, Nepal RCS and hospitals. The outcome of the meeting was a listing of priorities for improvement in five groups.

After the workshop this list was sent to main stakeholders in the country as a questionnaire. Stakeholders were asked to score the main priorities for improvement and add their opinion whether results are feasible for each priority within 2–3 years.

Results: Answers of key stakeholders revealed the priorities for the five groups on a National level for Nepal that are as summarized:

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1. Management & Organization: 1.1 Cost effective work process, 1.2 Quality Management, 1.3 Blood Supply management using software.
2. Infrastructure & Equipment: 2.1 Water, electricity supply and backups., 2.2 Basic lab equipment (directive and maintenance).
3. Donation:3.1 Blood Donor selection, recruitment SOP,3.2 Voluntary Donors, Promotion Material and database, 3.3 Donor care & Counselling.
4. Products: 4.1 Processing, storage and release., 4.2 Components, 4.3 logistics and cold chain.
5. TTI: 5.1 Screening strategies., 5.2 Hepatitis B, C and HIV., 5.3 Blood Group Serology and antibody screening.
Total response 43 of 58 including collective responses ## Priorities selected: high priority and considered doable in 2 to 3 years by respondents. Considered not doable: "centrally coordinated system", "nationwide LIMS", "typing special donors", "clinical transfusion" etc.
The outcome is used for the next phase of the project in a stepwise approach including tailor made training and technical guidance where appropriate. The already existing collaborations of the NRCS with other partners like Global Advisory Panel, are considered in each step. The first training is a Quality Management Training which is recently conducted from 17th to 26th July in Nepal by selecting 15 "champions" working in the different blood banks in Nepal.
Summary/Conclusions: During a workshop priorities for five groups were listed and prioritized by key stakeholders and followed by a questionnaire. Organization, training and basic blood bank processes were feasible priorities. These will be used as basis for training in Nepal and the Netherlands in 2017. Implementation of new technologies showed lower priority or found not doable. More reference: https://www.facebook.com/bloodstudy/?ref=bookmarks.

2A-S01-03

THE EXPERIENCES OF A JOINT AUSTRALIA-CHINA QUALITY PROGRAM IN CHINA

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Background: In China, there are over 430 independent Blood Transfusion Services (BTSs) with broad diversity in operational scale. Since the majority of BTSs conduct blood screening for transfusion transmissible infections within their own laboratories, many of which differ in management and professional experience, EQAS is one of the key mechanisms to measure laboratories' performance and improve the reliability and standardization of the blood screening results. In order to further improve the standard and quality of blood screening in China, a joint Australia-China quality program named China International Transfusion Infection Control (CITIC) was established in June 2011 through collaboration between Shanghai Blood Center (SBC) and NRL, Australia.
Aims: To review the experiences of the operation of CITIC EQAS from 2011 to 2016 and assess the testing results of blood screening laboratories in China based on 5 years data.
Methods: CITIC acts as a marketing and customs support department for National Serology Reference Laboratory, Australia (NRL) EQAS in China, but works beyond these fields, including EQAS orders, importation, transportation, data collection, report translation and troubleshooting, all with a focus on establishing a quality assurance system in China based on the CITIC EQAS platform. A series of training workshops, intercommunication activities and collaborative scientific research works are also integral to the system. EQAS data were analyzed and summarized annually.
Results: Totally 129 registered CITIC participants have participated in the past 5 years. CITIC provides both NRL EQAS for blood screening serology testing and nucleic acid testing (NAT) and local designed EQAS for blood screening serology testing and blood component bacterial testing. EQAS yielded lots of transcription errors (0.1%~0.8%), false reactive results (0.33%~0.66%) and false negative results (0.01%~1.40%) by participants, but unacceptable results decreased yearly. 2 training and intercommunication workshops were carried out annually, which one was international workshop held by NRL, the other was local workshop in China.
Summary/Conclusions: CITIC EQAS was established in China successfully. It is considered valuable to further extend the experiences of the joint Australia-China quality program to developing countries to ensure blood safety.

2A-S01-04

A NEXUS BETWEEN VOLUNTEERISM AND BLOOD DONATION: A CASE STUDY OF PMAS ARID AGRICULTURE UNIVERSITY RAWALPINDI AND QUAID-E-AZAM UNIVERSITY ISLAMABAD PAKISTAN STUDENTS

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Background: Volunteering behavior is socially based and happens at different rates in various geographical areas. Various barriers may continue to exist in community like concerns of family members or friends, religious or ethnic believes that influence the potential of voluntary non-remunerated donors but increase education about health and safety concerns may prove effective.
Aims:
-To understand the motivational factors behind blood donation activities and response of family to blood donation act.
-To find out the links between blood donation behavior and other volunteer activities of the university students.
Methods: This study was conducted in two most famous universities of Pakistan i.e. Quid-e-Azam University, Islamabad and PMAS Arid Agriculture University, Rawalpindi. From both universities total 200 students (also the members of blood donor's societies) were randomly selected and face to face interviews were conducted for the collection of data. All the collected quantitative data were analyzed by SPSS 19.0.
Results: The major age group of our studied population was 23–27 years (n = 152). Respondent belongs to all provinces of Pakistan but the majority was of Punjab province (n = 136). Results generated against questionnaire about the motivational factors and response of their families shows that; 57% of respondent's mothers were illiterate, 77% became voluntary donors due to convincing by their friends, 68% students did not tell their parents about their voluntary activity, When further question was asked by the interviewer that why they did not told their parents about this volunteer act than 52% replied that due to the fear of restriction, 69% participate were also involve in other routine and emergency volunteer activities, in our studied population on 17% participants donated blood as a replacement donation, 34% participants donated blood >4–5 times, 100 percent respondents replied that there is no religious restriction regarding donation of blood voluntarily, 92% replied that the patient can receive blood of donor having different religion, they further stated that for blood transfusion the relationship is only of humanity.
Summary/Conclusions: The study suggests that perceptions toward voluntary blood donation could be influenced to a large extent by socio-demographic variable. Socio-cultural barriers to voluntary blood donation exist in predominantly illiterate rural communities of the country but the young generation of Pakistan is the altruistic in nature. There is requirement of education, stimulation and motivate at national level especially to illiterate people of society to increase the number of voluntary blood donation. There is also need to make the Blood transfusion system more transparent in Pakistan.

2A-S01-05

ESTABLISHMENT AND UTILITY OF A SHARED DEFERRAL SYSTEM FOR HIV POSITIVE DONORS IN YANGTZE RIVER DELTA REGION: A 7-YEAR REVIEW

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Background: The HIV/AIDS epidemic in China has progressed through the entry and localized spread phases, and has reached the rapid spread phase. The overall HIV prevalence among blood donors in Yangtze River Delta Region (Shanghai, Zhejiang and Jiangsu province) was increased progressively from 2003 to 2016. So far, there is no multicenter, cross-provincial donor deferral registry (DDR) to interdict donations made by confirmed-HIV donors in China.
Aims: The aims of this study were to create an effective multicenter, cross-provincial donor deferral system to interdict the return donors previously confirmed positive for HIV in other services in China, and to analysis the role of DDR retrospectively.
Methods: The Yangtze River Delta Region consists of 26 regions located across Shanghai, ZheJiang province and Jiangsu province. All donations were first screened for anti-HIV-1 and HIV-2. Samples reactive for anti-HIV 1 + 2 were confirmed by

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Western blot. Information from three provinces were combined and compooled by uploading the cloud platform. The platform and database from all blood services regions were forwarded to the blood center of ZheJiang province. All confirmed-HIV donors were notified. Such donors were deferred permanently for future blood donations as part of the donor records in Yangtze River Delta Region. All blood services conducted predonation screening with DDR records.
Results: There is an estimated population of 200 million and 2 million blood donors every year in Yangtze River Delta Region. From April 2010 through July 2017, a total of 2455 donors were confirmed positive for the presence of HIV-1 in the study. No HIV-2–positive donors were found. All positive donors were deferred permanently. Three HIV positive donors concealed health and deferral history and attempted to donate at a second blood center. All three donors had been defended by our donor deferral system and motivation of their donation was malicious.
Summary/Conclusions: The shared donor deferral system in Yangtze River Delta Region was valuable at the regional level to prevent deferred blood donors from donating at other blood services. It is urgent to establish a national donor deferral system in China.

Academy Session - Young Donors

2A-S02-01
YOUNG DONORS: GETTING THEIR ATTENTION AND KEEPING THEM ENGAGED

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Last year, Taiwan had an outstanding blood donation rate of 74.9 donations per 1000 people. However, Taiwan is transforming into an aging society with very low birth rates. Recent data showed a dramatic decrease in the number and participation rate of young donors. Therefore, in response, the nationwide "Young Blood Program" launched on June 14th, 2015. The aims of the program were to recruit and encourage young people aged 17 to 20 to donate blood 10 times within 4 years, and to motivate them to be regular donors. We proposed the following strategies:
(1) To conduct promotional activities and educational programs in schools to let students know why, how, when and where to donate blood.
(2) To obtain buy-in and support from school principals, responsible staff and student representatives.
(3) To improve blood donation satisfaction, especially among first time donors.
(4) To recruit a popular singer or celebrity as a donor ambassador.
(5) To share short films of transfusion recipient and donor stories on YouTube, Facebook, LINE, etc.
(6) To set up blood collection sites on campus as much as possible.
(7) To study donor behaviors and attitudes, in order to find factors that influence young donors' willingness to donate blood.
(8) To invite students to visit blood centers and use student volunteers to further enhance youths' understanding of the importance of blood donation.
(9) To develop a new mobile app, which reminds donors to come back.
Results showed that 196,142 young donors joined the program, with approximately 43% returning for further donation during the past 2 years. Interestingly, the blood donation rate and return rate for female youth (47%) were higher than for male youth (40%). The numbers of young donors have slowly decreased since the implementation of the Young Blood Program. However, the young donor participation rate has stopped declining and has even shown a little growth. We will continuously improve our strategies and aim to succeed at meeting clinical demand by increasing the pool of regular young blood donors.

2A-S02-02
WHAT ARE THE SPECIAL ISSUES FOR YOUNG DONORS?

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Young donors contribute significantly to the blood supply in most countries. In Singapore, the teenagers are allowed to donate blood at age 16 years old (with parental consent if <18 years old). In Singapore, donors who are aged between 16–25 years old are considered "young donors". A total of 19,515 young donors donated blood in 2016 (27% of the donor pool). Blood donation is generally safe but not completely free of risks. Local and international studies have showed that adverse donor reactions are more common in young, especially first time donors. These may range from mild (dizziness, pallor, hyperventilation) to severe (Loss of consciousness) and may even sustain injury as a result. There are also some concerns that they may be more prone to iron deficiency due to the physiological need for growth.
Vasovagal reactions: Vasovagal reaction (VVR) is the most common blood donation related adverse reaction. It may occur during haemoglobin testing, during phlebotomy, immediately after blood donation, or after donor has left the premises (Delayed VVR). Most VVs occur within 12 hours of blood donation. It is believed that both physiological and psychological factors play a part in the development of VVRs. Donor Haemovigilance data from local and international blood centres have consistently shown that VVRs are inversely correlated with donor age, and young donors are more susceptible to VVR. First time donor status, donors with smaller Estimated Blood Volume, low systolic and/or diastolic blood pressure are more likely to develop VVR. VVRs may be precipitated by anxiety or fear (finger prick, upon seeing blood flowing out from self or others). Due to hot and humid weather in Singapore, VVR occur more frequently at outdoor collection sites than if the blood collections are held in air-conditioned room. Several international studies showed that adverse events during blood donation affect donor return rates. Since young blood donors are more vulnerable to adverse reactions, it is very important for blood services to develop strategies to mitigate donor adverse reactions. Educating prospective donors on what to expect during blood donation may help to ease the fear of first time young donors. Adequate hydration before and after blood donation, muscle tension exercise can also reduce the incidence of VVRs in young especially first time donors.
Low iron stores: Blood donation has been known to affect iron store in blood donors because a blood donation of 450 ml removes about 200 mg of iron. Teenagers have lower baseline iron store compared to adults but have higher daily requirements for body growth. Young female are even more vulnerable because of the additional menstrual loss as well as lower dietary intake. Therefore it is essential for blood services to monitor and prevent iron deficiency in young donors. The strategies could include lengthening of interdonation interval or limit number of donation per year, provide iron supplements and iron-rich refreshment and educational materials on iron deficiency anaemia.
Conclusion: Regular blood donors provide safest blood. It is important for blood services to provide excellent blood donation experience and ensure that donors remain in good health, particularly the young first time donors. Regular review of haemovigilance programme and focusing on donor safety are the responsibility of all blood services. A positive experience will encourage them to return and donate regularly for many more years.

Best Abstracts

2B-S03-01
MINIMISING UNDOCUMENTED INFORMATION ON THE DONOR REGISTRATION FORM

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Background: The national blood bank of Singapore under the purview of the Health Sciences' Authority, has incorporated the requirement for staff signature and time entry, to various portions of the Donor Registration Form. this is important for documenting the completion of each stage of the donation process and for attaching accountability to the various staff who had attended to the donor. Routine checks through the Donor Registration Form has led the collection team to discover many incidences of omission in signature and time entry. However, to do a 100% daily re-checking and follow -up is not practical and time-consuming. Hence, the team decided to gather the forms of successful donations from all collection sites, for three consecutive months for the year 2016. The intention is to study weakness areas which often led to omission of signature or time-entry.

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Aims: To ascertain the cause or reason for omission in documentation and propose a solution to minimise it.
Methods: Three consecutive months' worth of Donor Registration Forms from all collection sites including mobile drives, adding up to a total of 29,268 copies were examined. From the data that was collected, it was found that the most common areas for omission of signature and time entry were.
-Signature of the staff
-Time the donor was received
-Indicating the blood mixer number
-The last donation date (which is more significance to mobile drives)
A survey was also carried out among 64 staff to test their knowledge on the rationale for documenting certain information on the Donor Registration Form. Without making major changes to the original Donor Registration Form, the collection team redesigned the format and made draft copies for a new form. To promote accuracy and limit the likelihood of omission, provisions were made for more spaces as well as labeled boxes for entry of staff signatures and time. Related or similar information were also identified by specific colors such as user groups from registration staff, medical screening and donation room. Hence, any omission will be easily noticed. The new form was also careful not to cram too much information into one page. Rigorous feedback were solicited from all the staff and managers. A role play simulating the whole donation process was conducted using the newly designed forms. Overall turn-around-time at each stage of donation was also taken into consideration.
Results: The survey showed that overall the staff have good knowledge of the rationale behind documenting certain information. It also showed that a busy donation session or staff's years of experience in the blood bank does not contribute to omission of information required in the form. The reason why it was still omitted could possibly be because the design of the Donor Registration Form did not provide sufficient spaces for signatures and time entries through all the stages of the blood donation process. Staff feedback about the new form however, cited that it was neater, more organised and easier to check for omissions.
Summary/Conclusions: The Impetus for redesigning a new Donor Registration Form Stems from the discovery that many staff omitted their signatures and time-entry as a result of the lack of properly labeled boxes or spaces. This led to difficulty in conducting a thorough look-back if needed. With the newly designed form however, it suited the needs of daily operations as well as allow for display and accessibility of information in an efficient manner.

2B-S03-02

COMPARING THE EFFICACY OF PRE-HYDRATION AND APPLIED MUSCLE TENSION IN PREVENTING ADVERSE EFFECTS RELATED TO BLOOD DONATION

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Background: Adverse effects, especially vasovagal reactions (presyncope and syncope), suffered in relation to blood donation have negative impact on donation behavior of old as well as new donors.
Aims: The study aims to assess the effectiveness of prehydration with different fluids or applied muscle tension in preventing or decreasing the severity of adverse effects related to blood donation.
Methods: Consenting and eligible voluntary blood donors were randomly allotted to Control, prehydration with 500 ml plain water (PW), prehydration with 500 ml oral rehydration solution (ORS), prehydration with 400 ml fruit juice (FJ), or applied muscle tension (AMT) group. Donors of prehydration groups ingested the respective fluids in 5 min and donated blood after 20 min while Control and AMT groups received nothing in the 20 min waiting period. AMT group donors practiced active tensing of lower limb muscles during blood removal time. Heart rate (HR) and blood pressures (systolic - SBP and diastolic - DBP) were recorded at the time of recruitment (baseline), and before, during, 5 min after, 10 min after, 15 min after, and 20 min after blood donation. Adverse effects were determined by SBP and HR measurements (presyncope and syncope) or donors' complaints (subjective). Group characteristics, incidence rates, and hemodynamic response parameters were compared by Chi square and ANOVA, at 95% significance levels.
Results: Total 448 donors were grouped into five experimental groups - Control (n = 115, 25.7%), PW (n = 97, 21.7%), ORS (n = 71, 15.8%), FJ (n = 74, 16.5%), and AMT (n = 91, 20.3%). The groups were comparable by gender distribution, mean age, height, weight, and number of past donations.
Subjective adverse effects occurred in 2.6% in Control, 5.2% in PW, 4.2% in ORS, 2.7% in FJ, and 3.3% in AMT group donors. Presyncope occurred in 1.7%, 8.2%, 7%, 2.7%, and 4.4% donors respectively. Syncope occurred in 1.7%, 3.1%, 1.4%, 0%, and 1.1% donors respectively. Incidence of presyncope in PW was significantly higher vs Control ($P = 0.026$). Other group and adverse effect differences were not significant ($P > 0.05$).
Blood donation was associated with fall in SBP and rise in HR. Compared to mean SBP fall of 13.88 mmHg in Control, significantly more decreases were observed in PW (21.65 mmHg, $P = 0.035$) and ORS (21.65 mmHg, $P = 0.000$) but significantly less falls in FJ (7.67 mmHg, $P = 0.001$) and AMT (9.17 mmHg, $P = 0.006$). Similarly, compared to 2.33 beats per minute (bpm) mean increase in HR in Control, HR increase was significantly more in PW (5.03 bpm, $P = 0.015$) but not significant ($P > 0.05$) in ORS (4.08 bpm), FJ (3.88 bpm), and AMT (3.82 bpm) groups.
Summary/Conclusions: Blood donation challenges the donor's hemodynamic stability which is evidenced by falls in BP and rise in HR. Of the different measures to prevent or decrease severity of adverse effects, consuming 400 ml fruit juice 20 min before donation or practice of applied muscle tension during blood removal are more effective. The pressor response to ingestion of 500 ml plain water is short lasting and is hugely challenged by the stress of blood removal. As a result, presyncope is commonest in prehydration with plain water.

2B-S03-03

MICRORNA130A/PKLR AXIS REGULATES HCV AND HBV REPLICATION

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Background: HBV and HCV are important blood transfusion transmitted viruses. It has been shown to regulate miR-130a expression in infected patients and in infected cultured cells.
Aims: In this study, we identified and characterize miR-130a target genes and investigated the mechanisms by which miR-130a regulates HCV or HBV replication.
Methods: We predicted the possible target genes of miR-130a using four bioinformatics software including miRanda, TargetScan, PITA and RNAhybrid. Dual-luciferase reporter gene assays were performed to valid the target genes. miR-130a and its target genes were overexpressed, or knocked down by siRNA or by CRISPR/Cas9 gRNA in Huh7.51, JFH1, HepAD38 and primary human hepatocytes, respectively. Selected gene mRNAs and their proteins, together with HCV replication, were monitored by qRT-PCR and Western blot, respectively.
Results: The results showed that PKLR is a direct target gene for miR-130a. miR-130a overexpression down-regulated PKLR mRNA and protein levels while miR-130a inhibitor and gRNA increased PKLR expression. miR-130a mimic and PKLR knockdown inhibited HCV and HBV replication, while miR-130a gRNA and PKLR overexpression increased HCV and HBV replication. Supplemental pyruvate increased HCV and HBV replication and rescued HCV and HBV replication inhibition by miR-130a mimic and PKLR knockdown.
Summary/Conclusions: miR-130a regulates HCV and HBV production through its targeting of PKLR and its subsequent pyruvate production. Our data provides novel insights into understanding miR-130a regulated key metabolic enzymatic pathway steps, which are subverted by HCV and HBV replication.

2B-S03-04

RHD EPITOPE EXPRESSION OF FOUR NEW VARIANT RHD ALLELES IDENTIFIED IN THE CHINESE D VARIANT INDIVIDUALS BASED ON ERYTHROBLAST EXPRESSION SYSTEM *IN VITRO*

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Background: Four novel variant RHD alleles defined by the missense and deletion mutations (c.1154G>T, p.Gly385Val; c.79_81delCTC, p.Leu27del; c.689G>A, p.Ser230Asn and c.710C>T, p.Ala237Val) were identified in the D- or D variant Chinese donors in our previous study. The serological epitope typing was conducted in two donors with *RHD*79_81delCTC* and *RHD*710T* alleles respectively using the anti-D reagent (D-Screen) when red blood cells were available. In this study, transfection analysis of the four variant RHD alleles *in vitro* cultured erythroblasts from

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D- donor were conducted to confirm the effect of the variant alleles on the expression of RhD epitopes.

Aims: To confirm the effect of the four variant *RHD* alleles on the expression of RhD epitopes by using the cultured erythroblast expression system *in vitro*.

Methods: The four *RHD* mutations were introduced into a lentiviral vectors carried a wild-type *RHD* coding sequence, which was established by Sanquin Research, by using QuickChange II XL Site-Directed Mutagenesis Kit (Agilent). Then, the lentivirus carried the wild-type or four variant *RHD* alleles was prepared by transfecting HEK 293 cells with helper plasmids using the calcium phosphate method. The supernatant of the cultured HEK 293 cells containing the virus particles was then harvested. Meantime, the erythroblasts from three different Caucasian donors with D- phenotype were separated and cultured from the peripheral blood mononuclear cells and kept in the expansion medium for 2–3 days (van den Akker, 2010). The erythroblasts were then lentivirally transducted with the *RHD* wild-type construct or the four variant constructs. After 48 h transduction, the cells were differentiated and tested for RhD epitope expression by flow cytometry using 12 kinds of in-house monoclonal anti-D reagents and additional six monoclonal anti-D from the ALBAclone RhD typing kit (ALBA bioscience) against 11 D epitopes.

Results: RhD epitope expression data *in vitro* for *RHD*79_81delCTC* and *RHD*710T* tested by flow cytometry were consistent with the RhD serological result obtained by D-Screen to show a partial D phenotype. For the other two variant *RHD* alleles lacking of serological typing data from the red blood cells, the erythroblasts transfected with the *RHD*689A* construct showed a partial D reaction pattern, while the erythroblasts transfected with the *RHD*1154T* construct showed a weak and partial expression of D epitopes tested by flow cytometry using the 18 monoclonal anti-D reagents.

Summary/Conclusions: Expression study *in vitro* confirmed that the four novel variant *RHD* alleles identified in the Chinese Han previously could abolish or suppress the expression of D epitopes to result in a variant D phenotype.

2B-S03-05

AN INHIBITOR OF EQUILIBRATIVE NUCLEOSIDE TRANSPORTER 1 FUNCTION, S-(4-NITROBENZYL)-6-THIOINOSINE (NBMPR), ENHANCES EXPRESSION OF ANTIGENS IN THE AUGUSTINE BLOOD GROUP SYSTEM

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Background: The low-prevalence At^a negative phenotype in the Augustine (AUG) blood group system is found in individuals homozygous for the variant, c.1171G>A, in *SLC29A1*. This missense mutation results in a p.Glu391Lys amino acid substitution in equilibrative nucleoside transporter 1 (ENT1), a multipass transmembrane glycoprotein on the red blood cell (RBC) membrane. A proposed new low-prevalence antigen in the AUG system, identified via a case of hemolytic disease of the newborn (HDFN), is associated with c.1159A>C in *SLC29A1* predicting p.Thr387Pro in ENT1. A nucleoside analogue, S-(4-Nitrobenzyl)-6-thioinosine (NBMPR), has been shown to inhibit ENT1 nucleoside transport *in vitro*. We hypothesized that expression of AUG blood group antigens on human RBCs may be modified at NBMPR levels that inhibit transport activity.

Aims: To examine the effects of NBMPR on the interaction of antibodies against blood group antigens by flow cytometry.

Methods: NBMPR (Sigma-Aldrich) was tested at the minimum level reported to inhibit nucleoside transport [Takano, Drug Metab Pharmacokinet, 2010]. RBC suspensions (0.8%, 20uL) were prepared from two At^a positive random blood donors and incubated with NBMPR (20uL for final concentration of 1.5 mM, 2 min, room temperature (RT)) followed by two examples of polyclonal anti-At^a, or examples of human polyclonal anti-D (for D-positive) or anti-K (for K-positive) (10uL, 30 min, RT). FITC mouse anti-human IgG (BD Biosciences) was added (20uL, 30 min, RT) for detection by flow cytometry. In addition, the maternal alloantibody implicated in the case of HDFN was tested against RBCs from two family members (father and first son) carrying the novel c.1159A>C mutation in *SLC29A1* and serologically positive for the proposed AUG antigen. Median fluorescence intensities (MFI) for samples without NBMPR were designated as the baseline (100%) and differences in the presence of NBMPR were determined.

Results: NBMPR did not modify binding of anti-D or anti-K. In the presence of NBMPR, increased MFI for anti-At^a was evident with an average of 126% for the first example and 138% for the second example. For the two examples of RBC expressing the proposed new low-prevalence antigen, the increase in MFI for the maternal alloantibody was 121% for the first example (father) and 119% for the second (first son).

Summary/Conclusions: At levels reported to block nucleoside transport, the ENT1 inhibitor, NBMPR, enhanced binding of antibodies to the At^a antigen and antibodies to a proposed new low-prevalence antigen in the AUG blood group system. The mechanism of this enhanced antigen expression is unclear, however, the presentation of antigen sites from the ENT1 protein conformational structure in the presence of NBMPR may assist in antibody binding. That both are similarly enhanced may be expected as residues 387 and 391 are both in the same region of the ENT1 molecule, the fifth extracellular loop. Alloantibodies to antigens in the AUG system are associated with transfusion reactions and HDFN; this assay, using a readily available inhibitor, provides a novel addition to the serologists' investigatory arsenal when antibodies to antigens in the AUG blood group system are suspected.

2B-S03-06

DEGLYCOSYLATED ANTI-CD36 ANTIBODY FOR TREATMENT OF FNAIT CAUSED BY MATERNAL CD36 ANTIBODIES IN A MURINE MODEL

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Background: Foetal/neonatal alloimmune thrombocytopenia (FNAIT) is bleeding disorder of foetus/neonates caused by maternal antibodies which recognize paternal-derived antigen on foetal platelets leading to platelet destruction. The majority of FNAIT cases in Caucasians are caused by anti-HPA-1a alloantibodies. Recent data indicate that anti-CD36 antibodies seem to be the most frequent antibodies responsible for FNAIT in Asian populations. Until today, however, little is known about the pathomechanism and treatment of anti-CD36 mediated FNAIT.

Aims: In this study, we established a mouse model of FNAIT induced by maternal anti-CD36 antibodies and to prove the efficacy of de-glycosylated anti-CD36 antibody for the treatment of FNAIT.

Methods: Female homozygous CD36-/- or heterozygous CD36$^{+/-}$ mice were immunized two or three times with $1x10^8$ wild-type (WT) platelets at weekly intervals. After immunization, anti-CD36 antibodies in serum of immunized mice were detected by flow cytometry using WT platelets as targets. Subsequently, the immunized CD36-/- or CD36$^{+/-}$ female mice (as control) were bred with WT-male mice. During pregnancy and following delivery, miscarriage, platelet counts and bleeding disorders were analyzed. Polyclonal or monoclonal anti-CD36 IgG antibodies were purified from a pool of immunized mice or hybridoma, respectively, and deglycosylated with PNGase F enzyme. Deglycosylated anti-CD36 antibodies were subsequently injected intravenously into the immunized female CD36-/- mice at the 7th, 12th and 17th day after breeding with the WT-male mice. The platelet counts and the red blood cells as well as bleeding severity of the pups after delivery were analysed.

Results: Anti-CD36 antibodies were developed in all fourteen CD36-/- female mice, but not in CD36$^{+/-}$ female mice (0/4) after immunization with WT-platelets. When immunized female CD36-/- mice were bred with WT-male mice, and the clinical pictures of the heterozygous pups (n = 63) were examined after delivery, mild thrombocytopenia (platelet count: 361.2 ± 28.08 $X10^9$/l, n = 22) and mild anaemia (red blood cell count: $3.40 \pm 0.16X10^{12}$/l, n = 22) was observed in the survivor cohort. 4/63 (6.35%) of the pups, however, developed severe bleedings and intracranial haemorrhage, and interestingly, significant higher mortality (24/63; 38.10%) and miscarriage was found in the pups delivered. In contrast, no dead pups and miscarriage were observed in the control cohort. After prenatal treatment of CD36-/- mothers with de-glycosylated anti-CD36 IgG, both polyclonal and monoclonal, the mortality of pups was significantly reduced compared to untreated group (38.10% versus 5.26%; $P < 0.005$). However, no significant increase of platelet and red blood cell counts was observed.

Summary/Conclusions: In this study, we have established a mouse model of FNAIT that reproduced the symptoms of human FNAIT induced by anti-CD36 antibodies. Although only mild thrombocytopenia was observed, maternal anti-CD36 antibodies can cause severe bleeding in the foetus and miscarriage. These severe clinical pictures could be markedly reduced by introduction of de-glycosylated anti-CD36 antibodies although this treatment did not significantly increase the platelet and the red blood cell counts. This approach seems to prevent the fatal effect of maternal anti-CD36 on foetus survival during pregnancy, although the precise mechanism is not known yet.

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Academy Session – Transfusion Leadership

2B-S04-01

TRANSFUSION LEADERSHIP IN THE HOSPITAL

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Health care in the hospital environment, including across the various multidisciplinary transfusion processes, is increasingly complex. Transfusion practice involves different organizations [government/jurisdictional funders, the blood supplier, hospitals, professional bodies, regulators, accreditors] and individuals including donors, staff groups such as managers, junior and senior medical and nursing, scientists, transport and clinical quality and also importantly, patients and carers.

In the hospital setting a comprehensive, robust, collaborative, inclusive and effective clinical governance framework for oversight of local transfusion/blood management activities is essential for patient safety and optimal patient outcomes. Haemovigilance activities should include integrated systems, processes, leadership and culture applicable to all aspects of the hospital transfusion chain; the latter is intrinsically high risk, given the possible severe adverse outcomes of error, multiple potential vulnerabilities and often at least some areas not being supported by electronic systems.

Strong and informed clinician leadership is a vital focus for inspiration, motivation and support to staff to work and aspire to excellence in transfusion practice; key leaders should ideally be involved with patient care at some level and facilitate a culture of collaborative teamwork, accountability, openness, engagement and improvement.

The hospital transfusion/blood management committee [TC], meeting regularly with a focus on actions, outcomes, improvement and staff/consumer engagement, is fundamental to good local haemovigilance. It should be integral to the hospital's overall governance structure with clear accountability to executive so that effective change can be supported, resourced, implemented and maintained.

Recommendations are available as to appropriate TC membership, activities and reporting lines; these can be adapted to the individual environment to enhance functionality. Key clinical personnel representing areas of high blood use, the laboratory, clinical quality and crucially senior administration all contribute; it is also invaluable to have relevant external agencies present such as the blood supplier. Finding acceptable meeting times for busy clinicians is challenging; it is important to accommodate them and make effective use of their time when present. The TC leader should have accountability for and advanced content knowledge of transfusion, possess leadership qualities as above and work closely with a Transfusion Nurse/Officer; this might be a haematologist but a clinician from another discipline such as surgery or intensive care might act in this role to provide a different perspective and expand the capacity/experience of transfusion locally.

The agenda should encompass regular items and reports e.g. usage/wastage/compliance/audits and policy review but also consider illustrative cases, improvement and other projects, educational opportunities, new guidelines/research and information-sharing/benchmarking with other groups. Small TC working groups are critical to progress between meetings.

Enablers and barriers to good transfusion practice should be actively managed; these include system issues, evidence barriers and individual behaviours/beliefs. Staff/consumer engagement is facilitated by a multi-pronged approach; on-line learning modules, decision support tools/apps, alerts, case discussion, newsletters, improvement/research projects, unit/grand round presentations, unit/ward audits/feedback, brochures, displays etc. may all be useful.

2B-S04-02

BLOOD SERVICE LEADERSHIP

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Background: This abstract is prepared for the Abstract Book for the 28th Regional Congress of the ISBT, in the academy session Transfusion Leadership. The session is designed to recognize the roles of leadership in blood (transfusion) services (BTS's) and hospitals, and how leadership could be developed and measured.

Aims: to define the roles of leadership and management–to illustrate the differences on the leadership between blood services and institutions of other professions–to explore elements needed in BTS's leadership, and means for leadership development and measurement.

Summary/Conclusions: The leadership is about the strategic thinking for the missions of a institution, and values to its customers and communities, and employees as well. While the management deals with means and efficiencies toward the missions. For institutions in industry sectors and other for-profit professions, market principles are decisive in defining leadership. While blood program is based on voluntary donations and indispensable in life-saving services, the leadership of BTS's is then distinctive in terms of determining missions and values of the services, and strategies in resource mobilizations and talent competing. The concept and competence of leadership could and shall be developed in all possible means whenever possible within the local situations, by combining with and adapting theories into practices, and attending tailored courses. One of such examples is the Educational Course for Blood Service Leaders, initiated by Shanghai Blood Center and WHO Collaborating Center for Blood Transfusion Services in 2008, with CSBT, AABB and ISBT as co-hosts.

Clinical Transfusion

2C-S05-01

HAEMORRHAGE PROTOCOLS

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Much attention is understandably focussed on trauma since this accounts for ~5 million deaths per year worldwide with haemorrhage being an important contributory factor. However it is important to recognize that major haemorrhage can be encountered in many clinical settings and while there are some key common principles, a discipline specific approach is also needed e.g. trauma, obstetric, gastrointestinal, paediatrics, oral anticoagulation and the logistics of the mass casualty setting also need to be considered.

Timely and systematic intervention is essential. This starts with the challenge of prompt recognition of serious life threatening haemorrhage followed swiftly with appropriate resuscitation coupled with any urgent action needed to control bleeding. The importance of haemorrhage protocols in supporting prompt resuscitation is commonly highlighted. Such protocols need to reflect the multidisciplinary approach required with emphasis on clear communication. In particular a breakdown of communication between clinical and laboratory teams is an important source of adverse events in this highly fraught situation. Accordingly, development of such protocols should be undertaken as a joint initiative between teams with a clear consensus on roles responsibilities and actions.

National guidelines based on review of current evidence where available can help underpin local guidelines. Hospital Transfusion Committees should help ratify such local guidelines with overseeing mechanisms for effective implementation including training of all staff involved supported by drills. A designated Team Leader, generally the most senior clinician on the scene, can help co-ordinate effective team working together with identification of a further member to facilitate clear communication with the transfusion laboratory and other support services.

Specific challenges to the development of haemorrhage protocols begin with determining appropriate triggers for activation. While there are several arbitrary definitions of major haemorrhage, e.g. loss of one blood volume within a 24-hour period, a more practical and pragmatic approach is needed for local protocols. Early use of tranexamic acid is supported by high quality evidence in trauma and now also obstetrics. There has been a shift to early use of empirical blood and component ratios in particular in trauma. While local protocols should enable the release of blood and blood components promptly to minimize delay to initial resuscitation, ongoing liaison is essential together with clarity around the role of laboratory testing or near patient testing to guide further issue.

Many centres employ haemorrhage protocols but compliance where studied has been reported to be variable. Some largely retrospective studies indicate some potential benefit in the use of such protocols but on the background of concerns about increased component wastage. In addition to audit of deployment of haemorrhage protocols, important lessons can be learnt with regular review of adverse events and by reporting these to haemovigilance schemes (www.shotuk.org).

There remain important research questions around early control of bleeding e.g. role of early fibrinogen replacement, use of whole blood, use of cold stored platelets being subjected to further scrutiny. Further active research is also needed to help define the optimum approach to management of patients with significant haemorrhage seen in a variety of clinical disciplines.

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2C-S05-02

PRELIMINARY FINDINGS SUGGEST BLOOD PRODUCT TRANSFUSION MAY CONTRIBUTE TO ADVERSE PATIENT OUTCOMES FOLLOWING CORONARY ARTERY BYPASS GRAFTING

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Background: Blood product transfusion has the potential to modulate the recipient inflammatory response resulting in adverse patient outcomes, a phenomenon referred to as transfusion-related immunomodulation (TRIM). Coronary artery bypass grafting (CABG) is a complex cardiac procedure that triggers an overall systemic inflammatory response hypothesized to be a result of exposing patient blood to the bypass circuitry. These processes are thought to contribute to adverse patient outcomes including prolonged intensive care unit (ICU) length of stay (LOS) and ventilation time (VT). Dysfunction of immunoregulatory cells including dendritic cells (DC) and monocytes, may also contribute to adverse outcomes post-surgery, and changes in these cells may serve as predictive biomarkers to guide patient intervention. Understanding the role DC and monocytes play in mediating patient outcomes post-transfusion following cardiac surgery is limited.
Aims: We aimed to characterize the patient DC and monocyte phenotype at 5 timepoints (admission, peri-operative, ICU, day 3, day 5) in CABG patients (n = 25) and further investigate clinical outcomes in a sub-group of patient that received packed red blood cell (PRBC) and/or platelet concentrate (PC) transfusion (n = 3).
Methods: Using an ex-vivo whole blood culture model, the DC and monocyte phenotype was assessed (CD9, CD38, CD40, CD80, CD83, CD86, IL-6, IL-8, IL-10, IL-12, IL-1α, TNF-α, MIP-1α, MIP-1β, MCP-1, IP-10) with lipopolysaccharide (LPS) in parallel as a model of bacterial complication. Differences in inflammatory responses were associated with clinical outcomes for the entire patient cohort (ICU LOS: short ≤24 h, long >24 h; VT: short ≤8 h, long >8 h; Kruskal-Wallis, $P < 0.05$) and outcomes were also assessed in the sub-group of transfused patients. Potentially confounding factors were included in the analyses (age, sex, smoking history, diabetes status).
Results: For all 25 patients, cardiac surgery resulted in suppression of the DC and monocyte phenotype with evidence of immunoparalysis following exposure to LPS. We found that dysfunction in cytokine and chemokine response was associated with adverse patient outcomes. In particular, we found lack of DC and monocyte response to LPS stimulation at admission was associated with prolonged ICU LOS and VT. Unstimulated DC and monocytes also had a reduced capacity to produce inflammatory mediators during surgery and was also associated with prolonged VT and ICU LOS. Age, sex, smoking history, nor diabetes status were associated with ICU LOS or VT. Of the 25 patients studied, 3 received blood product transfusion. Our preliminary results on this small patient sub-group suggest that patients receiving transfusion were more likely to have poor outcomes with 68% of transfusion recipients having long ICU LOS (vs 36% for non-transfused patients) and 100% of transfusion recipients having long VT (vs 59% for non-transfused patients).
Summary/Conclusions: A reduction in the production of DC and monocyte inflammatory mediators following CABG surgery was associated with prolonged ICU LOS and VT. Using LPS to assess immune competency at admission may enable for early prediction of adverse outcomes prior to surgery. Preliminary findings from our prospective study of CABG patients suggest blood product transfusion may be associated with prolonged ICU LOS and VT in this patient cohort. This ongoing study will provide a comprehensive assessment of immune modulation in CABG patients and contribute further evidence in relation to transfusion outcomes as more patients are recruited.

2C-S05-03

IS LOWER RBC UTILIZATION AMONG CHINESE HOSPITALS COMPARED TO OTHER COUNTRIES EXPLAINED BY LOWER TRANSFUSION THRESHOLDS?

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Background: Multiple clinical trials have demonstrated that transfusion thresholds 70–80 g/l are safe in most patients and these thresholds have been widely adopted in many countries. However, the number of red blood cell (RBC) units transfused per capita in China is lower than in western countries and the reason(s) for the difference is unknown. There have not been published systematic analyses on how Chinese physicians make decision about RBC transfusion. We performed a case-control study in five Chinese hospitals to evaluate RBC transfusion use.
Aims: To assess the pre-transfusion hemoglobin thresholds and other clinical factors associated with RBC transfusion in 5 Chinese hospitals.
Methods: We identified all patients who received at least one RBC transfusion and then randomly chose 1000 patients from each of five hospitals and characterized the underlying demographics, co-morbidities, surgical procedures (if performed), RBC characteristics, and pre-transfusion hemoglobin concentrations. The transfused patients (cases) were placed into six diagnostic groups based on underlying disorders. Un-transfused patients were randomly selected as controls for each group matched for the same underlying diagnosis. We assessed the pre-transfusion hemoglobin trigger and other clinical characteristics associated with transfusion. We used logistic regression to assess independent risk factors associated with transfusion. The risk factors evaluated were age, gender, underlying diagnosis, cardiovascular disease, hospital, and pre transfusion hemoglobin level.
Results: From the ~5000 patients who received transfusion, we chose 6 diagnostic groups: Surgical: Orthopedic surgery (case/controls; 312/216), general surgery (367,332); Medical: solid tumor (230,232), gastrointestinal bleeding (212/221), cardiovascular (87,77), and hematological malignancies (100,101). The mean hemoglobin pre-transfusion was 80.3 g/l for postoperative surgical patients and 63.8 g/l for all medical patients. Pre-transfusion hemoglobin yielded the strongest association with transfusion in both surgical and medical patients. For example, compared to 100 g/l, the odds ratio (OR) (95% confidence interval) for pre-transfusion hemoglobin of 70–79 g/l in surgical patients was 27.0 (13.9–52.7), and in medical patients was 71.5 (37.8–135.4). In surgical patients only pre-transfusion hemoglobin was associated with transfusion. In medical patients, besides pre-transfusion hemoglobin threshold, the independent factors associated with transfusion were primary diagnosis, hospital and age 75 + . Compared with patients with solid tumor, the odds of transfusion were greater in those with underlying cardiovascular disease OR=2.5 (1.2–4.9), and less frequent in those with hematological malignancies, OR=0.4 (0.2–0.6) and gastrointestinal bleeding, OR=0.5 (0.3–0.8).
Summary/Conclusions: The pre-transfusion hemoglobin was the most important factor associated with transfusion in both medical and surgical patients. However, other clinical factors including age, cardiovascular disease, and primary diagnosis were only associated with transfusion in medical not surgical patients. Transfusion practice in 5 Chinese hospitals appears comparable to western countries in surgical patients although medical patients have lower pretransfusion hemoglobin levels. A clinical trial comparing transfusion thresholds common in China (60 g/l) versus Western countries (70–80 g/l) in medical patients would advance our knowledge about appropriate RBC transfusion thresholds in developing countries.

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2C-S05-04

RED BLOOD CELL ALLOIMMUNIZATION AMONG THE TRANSFUSION RECIPIENTS IN SANGLAH HOSPITAL DENPASAR BALI

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Background: Red blood cell alloimmunization is blood transfusion complication among the recipient. This is related to the frequency of antigen-positive blood transfusion received by the recipient as well as the genetic heterogeneity in populations. The immune response generated from the genetic differences between donor and recipient will induce the formation of allloantibodies. The impact of red blood cell alloimmunization for patients is the difficulty to obtain compatible red blood cells and higher risk of hemolytic transfusion reactions. Bali populations have little information about red blood cell alloimmunization.
Aims: The purpose of this study was to determine and analyze the characteristics of red blood cells alloantibodies in the recipient getting red blood cells transfusion in Sanglah hospital Denpasar Bali.
Methods: We performed a cross sectional study among adult recipients with a history of red blood cells transfusion at least 3 times and willing to participate in this study during the period of December 2016 to March 2017 in Sanglah hospital.
Results: A total of 40 recipients were subjected in this study. Red blood cell alloantibodies detected in 5% of recipients and all of them were multiple alloantibodies with specificity of anti-K antibodies (5%), anti-Kp^a (5%), anti-E (2.5%), anti-C^w (2.5%), anti-Lu^a (2.5%) and anti-M (2.5%). Overall alloantibodies were detected in women aged 46–65 years.
Summary/Conclusions: This is the first study on red blood cell alloimmunization in Bali, and we found frequently anti-K and anti Kp^a, so it could be a consideration, especially in Bali to perform erythrocyte antigen phenotypes, especially Kell antigen and doing cross-match for Kell system to prevent alloimmunization. More data are needed to examine incidence of red blood cell alloimmunization in Bali.

2C-S05-05

ESTIMATING THE IMMUNOGENICITY OF PLATELET ANTIGENS AND DEDUCING THE PREVALENCE OF ALLOANTIBODIES IN CHINESE

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Background: Platelet antigens and antibodies play important roles in several immune platelet disorders. The production of platelet antibody is determined predominantly by antigenic immunogenicity and mismatching, but the immunogenicity of platelet antigens remains unclear to date. The data of platelet antibody, especially in China, are extremely lacking.
Aims: The aim of this study was to elucidate the immunogenicity of human platelet alloantigens (HPAs) and CD36 antigen, and deduce the prevalence of platelet antibodies in Chinese.
Methods: Literatures concerning prevalence of platelet antibodies in Caucasian were reviewed and analyzed. The allele or antigen frequencies of HPA-1 to 6bw, 15, 21bw, CD36 and other low-frequency HPAs (LFHPAs, regarded as one antigen here) of Caucasian and Chinese were collected to calculate mismatching possibility of each antigen. The relative and absolute immunogenicity of each antigen was estimated and normalized using HPA-1a antigen as a reference antigen. We defined relative and absolute immunogenicity of HPA-1a antigen as 1.0 and 0.1015, respectively (anti-HPA-1a production in 10.15% of HPA-1a-negative pregnancy, referred to Turner, Transfusion, 2005). The antibody prevalence in Chinese was deduced using the immunogenicity derived from Caucasian and antigenic mismatching possibility of Chinese. The work was supported by National Natural Science Foundation of China (81570170) and Zhejiang High-Level Innovative Health Talents.
Results: The ranking of antibody prevalence in Caucasian was: 1a > 5b > 1b > 5a ≈ CD36 > 3a > LFHPAs > 15b > 15a > 2b > 4a > 3b > 4b ≈ 6bw ≈ 21bw > 2a (percent range, 66.5% - 0%). The mismatching possibility of 16 antigens (including HPA-1a/b to 5a/b, 6bw, 15a/b, 21bw, CD36 and LFHPAs) in Caucasian was 0.0247, 0.2071, 0.0085, 0.1454, 0.1270, 0.2348, 6.414E-7, 0.0016, 0.0111, 0.1604, 0.0033, 0.2005, 0.1738, <0.0009, 0.0003, <0.0020, respectively. The corresponding possibility in Chinese was 2.30E-5, 0.0095, 0.0030, 0.0954, 0.1446, 0.2237, 6.965E-6, 0.0052, 0.0002, 0.0269, 0.0291, 0.1757, 0.1987, 0.0184, 0.0030, <0.0020, respectively. The relative immunogenicity of above 16 platelet antigens was 1.0, 9.5E-3, <3.2E-4, 1.1E-3, 4.5E-3, 3.5E-4, 1.7E+2, 1.7E-2, 5.9E-2, 4.2E-2, 8.3E-3, 1.5E-3, 3.0E-3, >3.0E-2, 2.2, >2.7E-1, respectively. The absolute immunogenicity of each antigen equals the product of relative immunogenicity multiplied by 0.1015. The ranking of immunogenicity of 16 antigens was: 4a > CD36 > 1a > LFHPAs > 5a > 5b > 21bw > 4b > 1b > 6bw > 3a > 15b > 15a > 2b > 3b > 2a. The deduced prevalence of antibodies against 16 antigens in Chinese was 2.3E-6, 9.2E-6, <9.8E-7, 1.1E-5, 6.6E-5, 7.9E-6, 1.2E-4, 9.1E-6, 1.2E-6, 1.1E-4, 2.5E-5, 2.7E-5, 6.0E-5, >5.7E-5, 6.6E-4, 5.5E-5, respectively. Total prevalence of platelet antibodies in Chinese was about 33% of that in Caucasian (0.0012 vs 0.0038).
Summary/Conclusions: The immunogenicity of 15 dominant platelet antigens and cumulative effect of LFHPAs was determined. HPA-4a and CD36 antigens showed stronger immunogenicity than HPA-1a, and antigens in HPA-3 and HPA-15 systems showed relatively weak immunogenicity despite their high heterozygosity. The expected prevalence of antibodies against 1a, 1b and 5a was obviously lower in Chinese than in Caucasian, but antibodies against 21bw, 4a, CD36 and 6bw presented an opposite trend owing to higher mismatching possibility in Chinese. The cumulative prevalence of all platelet antibodies was lower in Chinese than in Caucasian on account of difference in antigenic mismatching possibility.

Immunohematology – Novel Technology

2C-S06-01

APPLICATIONS FOR KODECYTES IN IMMUNOHEMATOLOGY

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Red cells used in immunohematology are mostly limited to those that nature provides. Very few techniques are available for adding new functionalities to cells, without affecting their intrinsic functionality or vitality. Kode Technology is a surface modification technology that uses amphipathic function-spacer-lipid constructs to rapidly and harmlessly attach bioactive material to cell surfaces (creating kodecytes) and non-biological surfaces (koded surfaces). Originally designed to attach blood group glycans onto red blood cells for quality control use, the technology has since expanded to modification of any type of cell, enveloped virus, liposome and non-biological surfaces (including plastics, metals and glass). Today Kode Technology and the resultant kodecytes are being used in a range of cell-based diagnostics, as powerful research tools, and most recently as a potential immuno-oncotherapeutic agent; soon to enter human trials. Immunohematology applications and opportunities to use the technology in the form of kodecytes range from quality control kits, competency training panels, diagnostic reagents with synthetic rare blood group antigens or infectious disease markers. The constructs can be used for solid-phase antibody mapping and also have potential as therapeutics, including *in vivo* neutralization of ABO antibodies. Together with a large range of R&D constructs Kode Technology remains the most extensive and easy to use technology for adding bioactive material onto the surface of cells for research and diagnostics.

2C-S06-02

DYE-BASED STRATEGY FOR RAPID BLOOD GROUPING

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The blood grouping techniques have been developed for over 100 years, yet no any approach for simultaneous ABO forward and reverse grouping is commercially available although great attempts had been delivered in improving these techniques. The main obstacle lies in the completely opposite analytes existed either on the surface of red blood cell (RBC) membrane or plasma. As a result, a centrifugation step is inevitable for the ABO reverse grouping to remove any red blood cells. In this report, a brand-new generation of rapid blood grouping system has been promoted by rapidly and conveniently observing the color changes induced by different blood types. If the blood with specific antigens is loaded, the preloaded antibodies would capture corresponding RBCs and only plasma could keep swimming to the terminal to react with the dye, following by a prompt naked-eye visible color change. Whereas, if O type blood (without specific blood type antigens) is present, the whole

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blood will swim to the terminal and change to brown, which can be easily discriminated from the former one using naked eyes. The whole turnaround time is less than 30 s for ABO forward test and 2 min for both forward and reverse tests. Furtherly, machine learning strategy could be utilized to automatically recognize measured spectroscopy data and make accurate decision on the distinctive blood groups. With this in mind, automated device could also be developed to fulfill high throughput and multiplex detection, making this strategy a versatile platform suitable for both blood donation screening in the field and routine blood grouping test in central hospitals.

2C-S06-03

DNA OF ERYTHROID ORIGIN IN HUMAN PLASMA

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Background: There is much interest in the tissue of origin of circulating DNA in plasma. Data generated using DNA methylation markers have suggested that hematopoietic cells of the white cell lineages are important contributors to the circulating DNA pool.
Aim: We would demonstrate that cells of the erythroid lineage would also release DNA into the plasma.
Methods: Using high-resolution methylation profiles of erythroblasts and other tissue types, three genomic loci were found to be hypomethylated in erythroblasts but hypermethylated in other cell types. We developed digital PCR assays for measuring erythroid DNA using the differentially methylated region for each locus.
Results: Based on the methylation marker in the ferrochelatase gene, erythroid DNA represented a median of 30.1% of the plasma DNA of healthy subjects. In subjects with anemia of different etiologies, quantitative analysis of circulating erythroid DNA could reflect the erythropoietic activity in the bone marrow. For patients with reduced erythropoietic activity, as exemplified by aplastic anemia, the percentage of circulating erythroid DNA was decreased. For patients with increased but ineffective erythropoiesis, as exemplified by β-thalassemia major, the percentage was increased. In addition, the plasma concentration of erythroid DNA was found to correlate with treatment response in aplastic anemia and iron deficiency anemia. Plasma DNA analysis using digital PCR assays targeting the other two differentially methylated regions showed similar findings.
Conclusions: Plasma erythroid DNA thus represents a noninvasive biomarker for differential diagnosis and monitoring of anemia.

Plenary Session – Therapeutic Possibilities for Thalassemia

PL2 - 01

OPTIMISING IRON CHELATION THERAPY WITH DEFERASIROX FOR NON-TRANSFUSION-DEPENDANT THALASSAEMIA PATIENTS: THE THESIS STUDY

No abstract available

PL2-02

STEM CELL THERAPIES FOR THALASSEMIA

No abstract available

PL2-03

RED BLOOD CELL ANTIGEN GENOTYPING FOR HEMOGLOBINOPATHIES: ARE THERE DIFFERENCES BETWEEN THALASSEMIA AND SICKLE CELL DISEASE?

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Chronic red blood cell transfusions remain an essential therapy for patients with thalassemia and sickle cell disease. Management and prevention of alloimmunization is a critical aspect of chronic red cell therapy. DNA-based genomic approaches for extended red cell antigen profiling of both patients and blood donors now offer substantial improvements to mitigate alloimmunization complications inherent with transfusion therapy. Since most red cell antigens result from single nucleotide polymorphisms (SNPs), typing by DNA methods is relatively straightforward and has become an important part of the practice of transfusion medicine. The results are highly correlated with serologic phenotyping, and are superior in situations of recent transfusion or the presence of a positive direct antiglobulin test (DAT). The ability to determine the patient or donor antigen profile in one assay increases accuracy, provides information for antigens for which no typing reagents are available and for some institutions, can replace serologic typing at lower cost. Routine ABO and RhD typing is not done by DNA methods as the sole method due to the large number of different mutations that result in a Group O or a RhD negative phenotype. Genomic approaches to transfusion management provides new opportunities to mitigate morbidity and mortality due to alloimmunization. This lecture will discuss the practical integration and the value of a genomic approach to improve outcomes and prevent complications in patients with hemoglobinopathies. Differences between management of patients with thalassemia and sickle cell disease will be discussed.

TTID – Hepatitis

3A-S07-01

HEPATITIS C VIRUS: FROM DISCOVERY TO CURE

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Hepatitis C Virus (HCV) infects 71 million people worldwide. Up to 85% will develop into chronic infections. Since its discovery, numerous achievements have been made in the field of HCV research, especially in sequencing HCV whole genome, in developing HCV replicon and *in vitro* cell culture (HCVcc) system, in understanding the molecular mechanisms of HCV resistance to interferon-based therapy and in developing novel treatment regimens that specifically target HCV protease and polymerase. All these achievements make HCV infection a curable viral disease. A brief history of HCV from its discovery to cure will be presented in line with author's own scientific contributions to HCV research, especially in identification of the host genes that can be used to predict treatment response and the role of ISG15/USP18 signaling pathway in HCV resistance to interferon-based therapy. Experience learned from HCV research that may help us manage other transfusion-transmitted infectious diseases will also be discussed in this talk.

3A-S07-02

PREVALENCE STUDY OF HEPATITIS E VIRUS (HEV) IN SINGAPORE DONOR POPULATION WITH THE USE OF PROCLEIX HEPATITIS E VIRUS ASSAY ON PROCLEIX PANTHER SYSTEM

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Background: HEV is classified into 4 major genotypes. Genotypes 1 and 2 are transmitted primarily through contaminated water and fecal-oral routes, while genotypes 3 and 4, commonly found in domestic and wild pigs, are the cause of human infections associated with consumption of undercooked meats and shellfish. HEV Genotype 3 or 4 infections in humans are largely asymptomatic and are considered as an emerging threat to blood safety. Immunosuppressed recipients who acquired HEV via blood components and transplant are at risk of chronic liver diseases.

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Reports of transfusion-transmitted HEV infections have been documented in several countries in Europe and Asia. In Singapore, there are currently no studies of HEV prevalence among the blood donors.
Aims: To study the prevalence of HEV infection in Singapore's blood donor population.
Methods: Nucleic Acid Test (NAT) were performed on 12,541 donations using the Procleix HEV Assay on the Procleix Panther System from 20th February to 26th March 2017. The Procleix HEV Assay is a qualitative NAT assay that can detect all 4 genotypes of HEV using Transcription-Mediated Amplification technology. In this study, testing was performed by individual sample testing. Samples which were found reactive were repeated on the Procleix HEV Assay and further tested for HEV antibody (IgG and IgM) and an alternative NAT by real time PCR. In addition, genotyping was done for the NAT reactive samples.
Results: 28 donor samples were tested reactive on the Procleix HEV Assay. Out of which, 17 samples were NAT repeatable reactive (RR) and tested positive on the HEV real time PCR test. Only 4 NAT RR samples were tested HEV IgG and IgM positive. 2 weak positive HEV PCR (with low level viral load) results were found in the remaining 11 initial reactive (IR) samples and the other 9 IR samples were all negative for HEV PCR. The 2 weak positive HEV PCR samples were HEV IgG and IgM positive while 1 IR NAT/HEV PCR negative sample was tested positive for IgG only. 8 of the 19 PCR positive samples were successfully genotyped and all are determined to be Genotype 3.
Summary/Conclusions: The HEV prevalence rate for the blood donors in Singapore is established at 0.15%. The present results demonstrated a higher prevalence when compared with the infection rates reported in several countries who has conducted studies using similar assays. The findings in this study will be incorporated in the Risk-Based Decision Making Framework for Blood Safety by BSG to assess the need for HEV NAT screening of asymptomatic donors to ensure blood safety in Singapore.

3A-S07-03
MOLECULAR EPIDEMIOLOGY OF HEPATITIS B VIRUS GENOTYPES IN BLOOD DONORS IN ISLAMABAD, PAKISTAN

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Background: Hepatitis B virus (HBV) is a major causative agent of early, severe and prolonged liver infection that subsequently leads to cirrhosis of liver and hepatocellular carcinoma. Blood safety is a serious issue all over the world because of Transfusion Transmitted Infections and whole blood has to be screened before transfusion against these infections including HBV. HBV sequence is characterized by > 8% nucleotide differences for genotype, and 4%>8% nucleotide differences for sub-genotype, as HBV reverse transcriptase lack proof reading activity.
Aims: The aim of this study was to evaluate the molecular epidemiology of HBV genotypes and comparison of serological assay performance versus PCR in HBV screening.
Methods: Blood samples of 8,517 healthy blood donors were collected during the period of November 2016 to April, 2017 from Blood Bank of Shaheed Zulfiqar Ali Bhutto Medical University, Islamabad. Samples were screened for HBsAg assay by using technique of Chemiluminescence Immunoassay on ARCHITECT *i*2000SR Immunoassay Analyzer and protocols provided by Abbott Diagnostics, USA. PCR of positive samples was carried out using already reported genotype specific primers by Naito and colleagues (2001). The results were confirmed by visualizing genotype bands.
Results: The study confirmed the presence of HBV in 2.5% of blood donors and PCR confirmed the presence of HBV-DNA in 92 samples. The genotyping was done by PCR using type-specific primer sequences. PCR was dogged to check six genotypes, i.e. A, B, C, D, E and F. The results of this study shows high levels of genotype D is this region, i.e. 52.17% with less dominating genotype C, which is 16.30% with decreasing ratio of genotype E (14.13%), genotype A and B (9.78%) and mixed D+E (2.17%). The presence of coinfection is found at lowest rate. Due to high percentage of HBV/D, it is concluded that D genotype is common in our population.
Summary/Conclusions: The most prevalent HBV genotype in Islamabad region was genotype D which is responsible for liver cirrhosis and hepatocellular carcinoma. Efficacy of drugs varies with variation in genotypes of hepatitis B virus and also with geographical distribution. So before treatment, full flagged information of HBV genotype is required to start a beneficial and efficient treatment. And there is also need to design drugs on the basis of genotypes.

3A-S07-04
HBV DNA PREVALENCE BY NAT SCREENING IN SICHUAN UMBILICAL CORD BLOOD DONATIONS

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Background: Umbilical cord blood was widely used in hematological malignancy and some other diseases. Hepatitis B virus (HBV) can be transmitted by umbilical cord blood transplantation. In order to reduce the risk of HBV transmission, HBV NAT screening for HBV DNA is used in addition to testing for hepatitis B surface antigen (HBsAg) in many Chinese blood centers, but NAT screen haven't detected in umbilical cord blood banks in China.
Aims: We retrospectively analyzed the results of HBV assays in umbilical cord blood donors.
Methods: All mothers who donated umbilical cord blood were tested for HBsAg, seronegative samples were run on Roche Cobas(®) TaqScreen MPX test, version 2.0 in pools of six, Reactive donors were followed up for further confirmatory testing to rule out false-positive results. Quantitative analysis of HBV DNA, as well as sequencing and mutation analysis of S and pre-S genes, were performed for all NAT-reactive donors.
Results: A total of 25000 HBsAg seronegative umbilical cord blood donors from January 2016 to March 2017 were tested for HBV by cobas S201 platform with TaqScreen assay on mini pools of six donations (MP6). The HBV NAT was 0.188% (47/25000) in umbilical cord blood donors. Among them, HBV viral load were very low, positive anti-HBc was found in 33/47 (70%). We further genotyped HBV S gene in these 47 samples, 28 sample were successfully genotyped, 3 were genotype A, 19 was genotype B, and 6 was genotype C. We found that some of genotype B and C have mutations in the MHR even in the "a" determined S region.
Summary/Conclusions: The HBV seronegative and HBV DNA positive umbilical cord blood donors were relatively higher than in blood donors in anywhere in China. This study implicated that umbilical cord blood have risk for transmitted transfusion infections, additional NAT test should be evaluated to improve the safety of cord blood.

3A-S07-05

Abstract has been withdrawn

Academy Session – Transfusion Medicine Education

3A-S08-01
ACADEMIC EDUCATION OF TRANSFUSION MEDICINE IN CHINA

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The need for physicians and technical consultants specialized in transfusion is huge in nearly twenty thousand hospitals and five hundred blood stations or centers in China. However, the progress made in academic education of transfusion medicine as a specialty has been relatively slow in China. In this presentation, we are going to review the performance of Chinese transfusion medicine academic education. The information of diploma education in transfusion medicine specialty has been provided by the medical universities and institutes nationwide. During the past decade, approximately 1260 undergraduates with BSc and only 10 postgraduates with a MSc or PhD in transfusion medicine graduated from Chinese medical colleges or universities. Currently a few medical universities and institutes are engaged in providing a diploma in transfusion

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medicine as a stand-alone medical specialty for the Chinese undergraduate or postgraduate. Generally, two types of Transfusion medicine training programs are offered in medical universities. The program chosen by three universities is within the general context of laboratory medicine and awards a BSc. A single university organized its program integrated within clinical medicine and awards bachelor degree in medicine. The postgraduate training program available in few universities or institutes is conducted in the self-designated discipline of transfusion medicine, which is listed below the first grade of clinical medicine. We intend to summarize the achievements on Transfusion medicine academic education and the difficulties faced currently in those universities including the Southern Medical University in Guangzhou where I have worked in the past 10 years. We will further discuss future developments in this specialty still in infancy.

3A-S08-02

TRANSFUSION MEDICINE EDUCATION FOR BLOOD CENTRE STAFF

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More than 2.9 Million units of blood are collected annually in Korea, among which about 92% is collected by the Korean Red Cross blood services (KRCBS). These collections are processed further into blood components and the KRCBS supplies about 5.9 Million units of blood components each year. To accomplish this task, the KRCBS operates 15 blood centers, 138 fixed donation sites and 3 blood laboratory centers nationwide and employs more than 2100 people. Blood service operation in Korea is regulated under the Blood Management Law and KRCBS has standard operating procedures (SOP) in place to facilitate consistent conformance to technical and quality system requirements. Continuous education and acknowledgment of the competence of each staff member are essential to the quality and safety of donor management and also to ensure the consistency in the quality and integrity of blood components supplied. To ensure that each staff member has both theoretical and practical knowledge required for his/her position, the KRCBS operates educational programs to train new members of staff and also to refresh the knowledge of the more experienced ones. Passive forms of education like didactic lectures have their limitations. For continuous improvement, it's important to keep staff members empowered and engaged to think of ways to improve processes in their area of work. For this purpose, the QnS Jump-up project has been launched in 2012 wherein staff members are invited to suggest ideas to improve the services provided to customers and the quality of daily work processes. Because high routine workload does not allow all members of staff to participate in face-to-face educational programs, and because blood centers other than KRCBS do not have the capacity to provide comprehensive educational programs for their personnel, the Korea Centers for Disease Control & Prevention (KCDC) has developed an e-learning educational program in collaboration with the KRCBS. This presentation will give an overview of educational strategies developed and implemented in Korea for staff members of blood services.

3A-S08-03

TRANSFUSION EDUCATION FOR NURSES AND TRANSFUSION PRACTITIONERS IN AUSTRALIA

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Transfusion risks, and the principles and benefits of patient blood management (PBM), are well documented in the literature. PBM aims to achieve improved patient outcomes by promoting appropriate transfusion practice, including effective conservation and management of a patient's own blood. One of the challenges is to ensure that all multidisciplinary team members who work in these areas are aware of the risks, benefits and principles associated with PBM and administration of blood products, and undertake their practice accordingly.

Context: Nurses and transfusion practitioners, working in areas where blood/blood products are transfused, or where PBM is practised, need to have both knowledge and skills to undertake their roles safely & effectively. This is achieved predominantly through education/training across the continuum of the nursing career, and aligned with the practitioner's scope of practice and career path. The focus of education/training commences with acquisition of basic knowledge and development of competencies for safe administration of blood components, through to critical-thinking and leadership skills required to implement transfusion and/or PBM practice improvement initiatives within a department/health service. Patients and carers should be informed of the risks and benefits of blood and blood products, and any available alternatives, when a plan for treatment is developed. Nurses play a key role in providing this information to support patients and carers, so that they can be active partners in transfusion decision-making as part of their care and treatment.

Delivery modes: There is a range of education opportunities offered in Australia. Broadly these include:

-Undergraduate curricula that provide basic transfusion knowledge, skills development, simulation and experiences during work placement. The scope and breadth of these vary between institutions as we do not have a national defined curriculum framework. Continuing professional development is provided by organizations such as BloodSafe eLearning Australia, the Australian Red Cross Blood Service and local hospital educators. In hospitals and healthcare organizations this is driven by hospital accreditation standards and guidelines. On an individual professional level, nurses are required to meet continuing professional development obligations for ongoing registration, and transfusion education can assist them to meet these requirements.

-Postgraduate qualifications offered through Blood Matters/The University of Melbourne. This course is primarily aimed for healthcare professionals to build their skills and knowledge, to include more advanced concepts, along with practical approaches to change management, audit and leadership.

Australian education and training is provided in a range of delivery modes, for example: informal/formal situational learning, workshops and tutorials, didactic lectures, synchronous/asynchronous online learning, case study reviews, research projects and professional conferences.

Conclusion: Education is fundamental to ensure that staff working in the area of transfusion and PBM have the knowledge and understanding to provide high quality, effective and safe patient care. Education is provided in a variety of formats to cater for different stages of the nurses' career, learning styles and individual needs. Education is also an ongoing process and requires regular updates and reinforcement to keep pace with changes in the sector. The outcome of this should be a reduction in errors and risk, and improved patient outcomes.

Immunohematology – Abstract Session

3A-S09-01

A HAND-DROP METHOD FOR DETECTION OF ANTIBODY MEDIATED PLATELET PHAGOCYTOSIS BY MONOCYTES

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Background: Monocyte monolayer assay (MMA) has been established for over 30 years and well accepted for prediction of the outcome of red blood cell (RBC) transfusion. However, due to high tendency of platelet to adhere and aggregate, the same assay is not applicable to test platelet phagocytosis. Flow cytometry assay has been used for *in vitro* assay for platelet phagocytosis by monocytes using PBMCs isolated from fresh human blood or cell lines such as THP-1. But adherence and aggregation are not eliminated which may lead to unreliable results.

Aims: We use cyropreserved human PBMCs and a hand-drop method to establish an assay to detect antibody mediated platelet phagocytosis by monocytes with simplicity, reliability, wide application and high reproducibility.

Methods: Pooled PBMCs isolated from fresh human blood and cyropreserved with (RPMI 1640 plus 10% FBS) complemented with DMSOare thawed quickly in 37°C prior to use. (0.5–2) x 10^7 of pooled platelets from six O-type donors are labeled with Cell Tracker™ CMFDA. After opsonization with different dilutions of anti-human HPA-1a serum (NIBSC code 03/152) (1:1; 1:2; 1:4), these platelets are mixed with (1–2) x 10^6 PBMCs and incubated in a hand-drop manner at 37°C, 5% CO_2. Cells are collected and stained with APC conjugated anti-human CD14 antibody for subsequent flow cytometry analysis. Among gated cells for monocytes, $CMFDA^+/CD14^+$ cells are regarded as monocytes with phagocytosed platelets and the percentage of these cells in total $CD14^+$ monocytes are calculated. At the same time, flow cytometry and MASPAT test are also performed to verify the potency of these dilutions of antibody.

Results: In phagocytosis assay, higher concentration of antibody leads to enhanced phagocytosis of platelets by monocytes (phagocytosis percentage of 95.13% ± 0.86%, 23.90% ± 3.41%, 8.19% ± 1.70% for the 3 dilutions of 1:1, 1:2, and 1:4, respectively, while control AB type serum leads to 10.06% ± 1.64% of phagocytosis), which is consistent with flow cytometry and MASPAT results, that is, antibody of higher concentration has higher potency.

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Summary/Conclusions: We have established a hand-drop method for *in vitro* platelet phagocytosis study, where the adhesion and activation of cells are expected to be inhibited as cells are totally suspended without contacting any surfaces. The percentage of $CMFDA^+/CD14^+$ cells can be calculated to evaluate platelet phagocytosis mediated by antibodies in human serum in comparison to controls. The results are stable among tests using different batches of PBMCs. Furthermore, as is performed with cyropreserved PBMCs and flow cytometry, this assay is a simple and quick method which may find wide application in evaluation of platelet phagocytosis and effects of platelet transfusion.

3A-S09-02

ANALYSIS OF microRNAs IN ERYTHROCYTE DURING ITS AGING PROCESS

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Background: During aging *in vivo*, erythrocytes experience morphological and physiochemical changes. Previous studies have shown that Red blood cells have series of changes in protein levels during aging. But recent reports have shown microRNA in red blood cells contributes most of microRNA in whole blood. However, the role of microRNA in the process of red blood cell senescence is not clear.
Aims: We identified differentially expressed regulatory microRNAs in RBC (Red blood cell) during its aging process, which can also be correlated to the changes in RBC during its senescence. Therefore, four aspects of RBC changes widely known to occur during the aging process were selected for the study, which were PK (pyruvate kinase) activity, RBC indices, 2,3-DPG (Diphosphoglyceric acid) and ATP levels.
Methods: Five RBC layers were received after Percoll density centrifugation. We made comprehensive microRNA expression analysis of RBCs collected from enriched mature RBCs on five density layers. We used pyruvate kinase tests to determine the age of red blood cells. We also analyzed the PK activity, ATP and 2,3-DPG levels of five RBC layers. And the indices of RBCs from five layers were analyzed using an automated hematology analyzer. We used Pearson correlation analysis to confirm the microRNAs which can be related to the PK, MCV, 2,3-DPG and ATP levels of RBC during its aging. Finally, we made biologic network prediction of the potential microRNA targets by using TargetScan Human and miRanda databases.
Results: We found that middle-aged red blood cells had the highest ATP level and 2,3-DPG level, and the PK activity and MCV of red blood cells was getting smaller during their aging process. The analysis identified microRNAs whose changes in the expression levels were correlated with the selected four aging processes that are PK activity, ATP loss, 2,3-DPG, and changes in RBC indices. Pearson correlation analysis revealed that a total of 18, 32, 19, and 18 microRNAs had exhibited expression levels significantly correlated with changing PK, ATP, 2,3-DPG and MCV. Bioinformatics analysis identified potential targets and biologic functions of these microRNAs which are significantly associated with"cell morphology" and "cell differentiation", or involved in apoptosis- and hypoxia-related signaling pathways.
Summary/Conclusions: Overall, this study demonstrates that changes in the levels of microRNAs of senescent RBCs correlate with some of the aging events and thus they have the potential to serve as the markers of RBC aging.

3A-S09-03

THE REACTIVITY OF ANTI-HPA-15 ANTIBODIES WITH ENDOTHELIAL CELLS

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Background: Immunization against the Human Platelet Antigen (HPA)-15 system residing on glycosylphosphatidylinositol-anchored CD109 glycoprotein can lead to the production of anti-HPA-15a and anti-HPA-15b alloantibodies. These antibodies are known to be responsible for platelet destruction in different clinical settings of immune mediated thrombocytopenia including platelet transfusion refractoriness (PTR) and fetal neonatal alloimmune thrombocytopenia (FNAIT). The detection of anti-HPA-15 antibodies, however, is still difficult because of the low and instability of CD109 expression on platelet surface. Besides on platelets, CD109 is also found on the surface of endothelial cells. Currently, little is known about the relationship between CD109 and endothelial cells.
Aims: In this study, we asked the question whether endothelial cells are important target for anti-HPA-15 antibodies. The CD109 expression on HPA-15 phenotyped endothelial cells was analyzed by flow cytometry. Binding of anti-HPA-15 antibodies in patient's sera with endothelial cells was measured by modified antigen capture assay (MAIPA). The effect of anti-HPA-15 antibodies on endothelial function was studied by tube formation assay.
Methods: The HPA-15 expression was determined on the human umbilical vein endothelial cells (HUVECs) by flow cytometry. Analyze anti-HPA-15 alloantibodies with HUVECs by MAIPA assay. Detect the effect of the anti-HPA-15 alloantibodies on endothelial function by tube formation assay.
Results: The average MFI of CD109 expressed on nine different HUVECs is 1730.40 ± 218.26 (negative control: 108.46 ± 32.1). No significant difference was observed between different HPA-15 phenotyped. Analysis of anti-HPA-15b antibodies with phenotyped HPA-15aa and HPA-15bb endothelial cells showed stronger reaction when compared to platelets. Antibodies against CD109 inhibited endothelial cells tube formation. In the control experiment, antibodies against platelet specific αIIbβ3 complex did not alter endothelial tube formation. In contrast, antibodies reacted with endothelial αvβ3 complex inhibited significantly tube formation.
Summary/Conclusions: Our results indicated that endothelial cells may represent a better target for the analysis of anti-HPA-15 antibodies. Anti-CD109 antibodies can impair the angiogenesis of endothelial cells and may play the role by the development of endothelial dysfunction.

3A-S09-04

DEVELOPMENT OF MEMBRANE ANTIGEN ARRAYS FOR VALIDATING RARE BLOOD GROUP ANTIBODY REAGENTS

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Background: Cross-reactivity of antibody reagents for rare blood group may result in spurious serological results and risk during blood transfusion. However, mapping the specificity and cross-reactivity of antibody reagents is restricted without the availability of an assay with high-sensitivity and multiplexed detection ability.
Aims: To address this need, we developed a Red Blood Cell (RBC) membrane antigen array.
Methods: First, we established an efficient approach to extract antigens from RBC membranes and characterize their antigenicity; Second, we fabricated RBC membrane antigen array with three panel antigens, optimized the beads coated with antigens and compared the assay performance of antigen array with routinely used microcolumn gel method. Third, we screened antibody reagents of 15 rare blood groups.
Results: We found the assay sensitivity of RBC antigen arrays can be improved 1736 and 1931 folds comparing to microcolumn gel method, respectively. We also found that 60% (9/15) of antibodies showed expected signals to their correspondent antigens, 37% (5/15) antibodies had cross-reactivity to one of three antigens and 6% (1/15) of antibodies had cross-reactivity to two of three antigens. Those cross-reactive antibody reagents were validated using a agglutination inhibition assay.
Summary/Conclusions: All the results indicate that our RBC membrane antigen arrays hold great promising in the validation of antibody reagents for rare blood groups in future.

3A-S09-05

A NOVEL DI*02(T853 M) ALLELE ABOLISHING THE EXPRESSION OF DI2 ANTIGEN

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Background: The Diego blood group system consists 22 antigens, which is expressed on the Band 3 protein that is encoded by the solute carrier family 4, anion exchanger, member 1 (*SLC4A1*) gene. The antithetical antigens, DI1 (Di^a) and DI2 (Di^b), are encoded by the amino acid p.854Leu(c.2561T) and p.854Pro(c.2561C), respectively. Compared with the rare distribution of DI1 antigen in the Caucasian population, there is a polymorphic distribution in most Mongoloid populations including the Chinese. Alloantibody against the DI1 and DI2 antigens could cause mild to severe hemolytic disease of fetus and newborn. Currently, the commercial antiserum against DI1 and DI2 antigens are not available in China, several genotyping methods including polymerase chain reaction sequence–based typing (PCR-SBT), high-resolution melting (HRM) and polymerase chain reaction with allele-specific primers (PCR-SSP) have been developed for DI1/DI2 genotyping analysis. In our previous study, one novel *DI*02(T853M)* allele accompanied with one wild-type *DI*02*

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allele was identified in one Chinese donor having a normal DI:−1,2 phenotype when the DI1/DI2 genotyping analysis was conducted by using PCR-SBT in the Chinese Southern Han donors.
Aims: To investigate the effect of the novel *DI*02(T853M)* allele on the expression DI2 antigen and Band 3 protein through the transfection and expression study *in vitro.*
Methods: The mutation (c.2558C>T, p.Thr853Met) defined the novel mutant *DI* allele was introduced into a pcDNA3.1 vector carried a wild-type *SLC4A1* coding sequence, which was established by Sanquin Research, by using a site-directed mutagenesis kit (Agilent). Human HEK 293 cells which do not express endogenous *SLC4A1* gene were cultured and transiently transfected with the wild-type *DI*02* construct or mutant *DI*02(2558T)* construct. After 48 h culture, the transfected cells were harvested, the expression of DI2 antigen and Band 3 protein on the surface of the cells was analyzed using a monoclonal anti-DI2 (clone R97-008) and anti-Band 3 (clone BRIC6), respectively by flow cytometry. Besides, a high melting curve (HRM) analysis developed for DI1/DI2 genotyping was also conducted in this donor.
Results: The transfection and expression analysis of the mutant *DI*02(T853M)* construct in the HEK 293 cells showed the complete absence expression of DI2 antigen and weaken expression of Band 3 protein. Besides, the sample with the *DI*02 (T853M)/*02* genotype was shown having a melting curve different with the controls with a *DI*02/02* genotype but almost overlapped with the control sample having a *DI*01/02* genotype in the HRM genotyping analysis.
Summary/Conclusions: Expression study *in vitro* proved that the novel variant *DI* allele could abolish the expression of DI2 antigen and suppress the expression of Band 3 protein. The effect of the weaken expression of Band 3 on the structure and function of the red blood cells still needs further investigation. The mutation (c.2558C>T, p.Thr853Met) which encodes the amino acid is located next to the one (c.2561C>T, p.Pro854Leu) encoding the DI1 and DI2 antigens, resulted in the false positive genotyping result in the DI1/DI2 HRM analysis.

3A-S09-06
ESTABLISHMENT OF HUMAN HYBRIDOMA CELL LINES CAPABLE OF PRODUCING ANTIBODIES AGAINST MILTENBERGER BLOOD GROUP ANTIGENS

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Background: Miltenberger, a family of blood groups, is the result of gene recombination and the encoding hybrid molecule of glycophorin A and B. The occurrence of Miltenberger blood group is rare (<0.01%) in Caucasian and Black, however, they are relatively high (2–7%) at Southeast Asia region. There are currently 11 variants (Mi I- XI) of Miltenberger phenotype and each is shown to express from 1 to 6 antigens such as Mi^a, Mur, MUT and others. In Taiwan, antibodies to Miltenberger blood group have been found to be the most frequent red cell alloantibody with clinical significance. Since their complexity and the lack of commercial antiserum, these lead to the difficulty in detecting Miltenberger blood group. Since their complexity and the lack of commercial antiserum, these lead to the difficulty in detecting Miltenberger blood group.
Aims: The aim of this study is to make monoclonal antibodies to detect Miltenberger blood groups.
Methods: Human sera containing Miltenberger antibodies with undetermined specificity to the individual antigenic site are collectively named anti-"Mi^a". Blood donors detected as having anti-"Mi^a" antibody were identified during routine antibody screening and their whole bloods were obtained. Donors' B lymphocytes were transformed by EB virus and fused with JMS-3 cells. Fused hybridoma cells were selected in HAT medium. GP.Mur red cells were used to select the subclones of hybridoma cell lines. The specificities of anti-"Mi^a" in donors and the produced monoclonal antibodies were identified by use of stored panel cells of the phenotypes GP.Vw (Mi I), GP.Hut (Mi II), GP.Mur (Mi III), GP.Hil (Mi V), GP.Bun (Mi VI), and GP.HF (Mi X).
Results: Three hybridoma cell lines that produce monoclonal antibodies against Miltenberger blood group antigens were developed from blood donors. The specificities of these antibodies were identified as anti-Mi^a, anti-Mur, and anti-MUT, and the agglutination titers to Mi.III were 2,048, 32,768, and 512, respectively, as detected in standard saline phase. All these antibodies were determined to be IgM antibodies.
Summary/Conclusions: We have established three hybridoma cell lines from human B lymphocytes. Serological evidences showed these three hybridoma cell lines produce monoclonal antibodies with the specificities of anti-Mi^a, anti-Mur and anti-MUT. In this study, we reported the first human-sourced hybridoma that produce antibodies against the Mi^a antigenic site of Miltenberger blood group. This supports the presence of Mi^a antigen, which has been once considered not presented as a single antigenic site in human. Further, the established cell lines provide a promising solution for the detection of Miltenberger phenotype in Southeast Asia.

Advances in Cellular Therapies

3B-S10-01
MANUFACTURING RED BLOOD CELLS USING BIOREACTORS

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Background: *In vitro* production of stem cell-derived red blood cells (RBCs) is a promising alternative to donor blood as the blood donor pool decreases. Evolving from the conventional 2D plate culture, large scale manufacturing using bioreactors is an inevitable process for better productivity and efficiency in making RBCs. Bioreactors are the only equipment that can manage the tremendous number of cells necessary and meet the requirements for clinical trials.
Aims: This study aims to develop robust automated production systems and continuous process assessments using bioreactors and supporting culture media for successful culture of the RBC products in clinical settings.
Methods: Cord blood CD34 + cells were isolated and differentiated to erythroid progenitor cells in 2-dimensional plates. Then, the cells were transferred and cultured in differentiation media with various additives using a stirred bioreactor. To develop the best parameters, several culture conditions and media were compared.
Results: We successfully developed a RBC production process using a bioreactor that showed very effective maturation and enucleation up to 94% with high viability. The produced RBCs showed similar functions compared to donor fresh RBCs and could be stored for one month.
Summary/conclusions: The optimized culture process using bioreactors would make possible to produce transfusable RBC products in a large scale.

3B-S10-02
AFFECTION OF FBXL10 ON HEMATOPOIETIC CELLS FROM HUMAN CORD BLOOD

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Background: Histone demethylase F-box and leucine-rich repeat protein 10 (Fbxl10, also known as Jhdm 1b or Kdm2b) forms complexes with polycomb-group proteins, essential regulators of hematopoietic stem cells (HSCs).
Aims: $CD34^+$ hematopoietic cells from human cord blood are the important resources of HSCs in clinic transplantation for hematologic diseases. In this study, we explore how Fbxl10 affects this cell population.
Methods: We constructed the recombinant plasmid of PTT5-TAT-FBXL10. Tat-Fbxl10 fusion protein was acquired by eukaryotic expression system and added to the cell culture system. The impact of Fbxl10 on CD34 + hematopoietic cells was analyzed by western blottings, cell counts, flow cytometry, *in vitro* methylcellulose colony assays and CCK8 assays.
Results: Fbxl10 was imported into CD34 + hematopoietic cells by Tat, a kind of cell penetrating peptide. Fbxl10 encodes a demethylase specific to the histone H3 mono/di-methylated at lysine 36 (H3K36me1/me2). After 7 days, the relative H3K36me2 level in hematopoietic cells from cord blood was obviously decreased, which indicated that Tat-Fbxl10 fusion protein had normal bioactivity. Compared with control groups, Fbxl10 neither expanded the number of CD34 + hematopoietic cells nor maintained the expression of CD34. Moreover, Fbxl10 could not augment the colony-forming capacity of CD34 + hematopoietic cells. However, in 21-day culture, the activity of cells treated with Tat-Fbxl10 was significantly increased ($P < 0.05$) in CCK8 assays.
Summary/Conclusions: Our findings implicate that Fbxl10 increased the vitality of CD34 + hematopoietic cells in long-term culture, thus highlight a role of histone demethylation in the epigenetic regulation of CD34 + hematopoietic cells from human cord blood.

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3B-S10-03

THE UBIQUITIN-MODIFYING ENZYME A20 PLAYS AN IMPORTANT ROLE IN REGULATORY T CELLS THROUGH IL2-SIGNALING PATHWAY

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Background: A20, also known as TNFAIP3 (tumor necrosis factor α-induced protein 3), is a ubiquitin-modifying enzyme, which has a key role in the negative regulation of inflammation and innate immune response. Previous studies reported that A20 is highly expressed in lymphoid organs, such as thymus, spleen and lymph nodes, and as a susceptibility locus for human inflammatory and autoimmune pathology, including rheumatoid arthritis (RA), systemic lupus erythematosus (SLE), inflammatory bowel disease (IBD), psoriasis, type I diabetes, coronary artery disease, rheumatic heart disease, and systemic sclerosis. However, A20's function in T cells, especially in regulatory T cells, is still unclear.
Aims: To assess the role of A20 in T cells and to show that the relationship between expression of A20 and Treg proliferation and apoptosis.
Methods: Crossed A20 floxed mice with CD4-Cre Tg mice, and got $A20^{flox/flox}$ mice as control, CD4-Cre $A20^{flox/flox}$ mice as cKO mice. 6–8 weeks mice were sacrificed for isolating thymus, spleen and lymph nodes. Treg cells' were analyzed by flow cytometry for the frequency and number in thymus and periphery, the signature molecules CTLA-4, CD73, FR4, GITR. Injected mice with 2 mg BrdU by i.p., overnight and injected 2 mg BrdU again 1 H before scarified mice to assess the proliferation. Treated cells with IL-2 (R&D) 100ug/ml at 37 degree centigrade for 20 min, then stained with pStat5, pERK1/2 and pAKT antibody to assess IL-2 signaling pathway. Three kinds of T cells were sorted by FACS Aria from different types of mice, and then injected into different Rag $1^{-/-}$ mouse respectively, weight loss was checked twice per week to assay the suppression function of Treg.
Results: Although the number of Treg in A20 deficient cKO mice is increased, the expressions of Treg surface molecules are stable. Weight loss of mice in different injection groups had no significant difference which show that A20 deficient Treg maintain the suppressive function *in vivo* as control. More $Nrp1^+$ Treg in periphery were found in A20 deficient mice, illustrating that the increased Treg were mostly derived from thymus. A20 has no big effects on the proliferation and apoptosis at the generation of tTreg on the thymus. Enhanced phosphorylation of Stat5 in A20 deficient mice, while there is no significant difference of ERK1/2 and Akt, which illustrate that A20 may influence the expression of several Stat5-target genes by regulating phosphorylation of Stat5. This high sensitivity of IL-2 is known be crucial for Treg survival.
Summary/Conclusions: Our experiments revealed that there were no clear defects appeared in the conventional T cell development in A20 conditional knockout mice. Interestingly, the frequency and number of $Foxp3^+$ regulatory T cells (Treg) were significantly increased in thymus and periphery of A20 deficient mice. This observation suggests that A20 may influence the number of Treg. A20 deficient Treg in periphery were mostly derived from thymus, and exhibited the normal suppressive function as control. This increase may be related with the enhanced both IL-2 signal and TCR signal and its sensitivity. These results illustrate that A20 plays an important role in regulatory T cells and its survival, and more precise mechanisms still needs further study.

3B-S10-04

CLINICAL SIGNIFICANCE OF HLA-E ALLELIC GENOTYPE AND HLA-E SURFACE/SOLUBLE LEVELS BETWEEN HEALTHY INDIVIDUALS AND ACUTE LEUKEMIA PATIENTS

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Background: HLA-E is one of the non-classical HLA molecules with limited polymorphism. HLA-E serves as a ligand of natural killer (NK) cell surface receptor CD94/NKG2 which can inhibit NK cell activity by CD94/NKG2A and play a crucial role in immune surveillance. It is reported that HLA-E expression alteration or allele specificity are related to several clinical manifestation such as virus infection and tumors, but rarely combined genotype and expression data for their correlation study.
Aims: In this study, we detected both of the surface and soluble expression levels as well as their correlations with the genotype of HLA-E in Chinese Leukemia patients. We compared all three indexes with the healthy individuals from the same population, aimed to offer a systematic visual for the correlation of HLA-E with acute leukemia.
Methods: By sequencing exon 3, genotype frequency of HLA-E were firstly observed respectively in 182 Chinese healthy controls and 136 acute leukemia patients. Both of the surface expression and soluble secretion in 64 healthy controls 54 acute leukemia patients correlates their genotype were analyzed. The surface expression of HLA-E on lymphocytes of acute leukemia patients and healthy individuals was detected with flow cytometry and depicted per subject as fold expression using anti-human HLA-E-PE and mouse IgG1-PE isotype control antibodies (The mean fluorescence intensity (MFI) value of HLA-E/MFI isotype control). The soluble expression of HLA-E in plasma of per subject was detected and quantified by a specific sandwich Elisa.
Results: In 182 healthy individuals, we detected 77 HLA-E*01:01, *01:03 heterozygous cases (42.3%), 64 cases were homozygous for HLA-E*01:03 (35.2%), and 41 cases were homozygous for HLA-E*01:01 (22.5%). The frequency of nucleotide change (T/C) in exon 3 which differentiate HLA-E*01:01 (43.6%) and HLA-E*01:03 (56.3%) is consistent with Chinese population in 1000 genome project (http://browser.1000genomes.org/index.html). In acute leukemia group, we detected 55 HLA-E*01:01, *01:03 heterozygous cases (40.4%), 68 cases were homozygous for HLA-E*01:03 (50%), and 13 cases were homozygous for HLA-E*01:01 (9.6%). The frequency of HLA-E*01:03 homozygous individuals was significantly higher in patient group than that of in control group (X^2=7.056, P = 0.008, OR=1.845), while the frequency of HLA-E*01:01 homozygous individuals was much lower in patient group than that of in control group (X^2=9.286, P = 0.002, OR=0.363). The relative surface expression of HLA-E*01:03 homozygous individuals is significantly higher than of in HLA-E*01:01 homozygous individuals in both leukemia and control groups, but there is no significant change between the corresponding genotypes in two groups. On the other hand, the soluble expression level of HLA-E is significantly higher in acute leukemia patient group than of in control group, but there is no allele-specific expression in both of the control and patient groups.
Summary/Conclusions: Our study conclude that HLA-E*01:03 homozygous is a susceptible gene for acute Leukemia and HLA-E*01:01 homozygous is protective. Surface molecule of HLA-E on lymphocytes is allele-specific expression in both acute leukemia patients and healthy individuals. For soluble HLA-E, the expression in acute leukemia patients is higher than that in healthy individuals but not in an allele-specific manner. Our study imply that soluble HLA-E secretion and HLA-E*01:03 homozygous genotype that brings higher surface expression might play important roles in the mechanisms of tumor escape for acute leukemia.

3B-S10-05

ESTABLISHMENT AND APPLICATION OF A QPCR METHOD FOR KIR3 DL1 GENE COPY NUMBER DETERMINATION

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Background: Killer cell immunoglobulin like receptors (KIRs) are glycoproteins expressed on the surface of natural killer (NK) cells and a few subsets of T cells playing significant roles in immunity, pregnancy and transplantation. KIR genes exhibit high diversity in allele copy numbers and haplotypes, which have been shown to correlate with protection from certain viruses such as HCV and HIV. But, routinely used sequencing based technology cannot detect KIR copy numbers.
Aims: To establish a copy number assay method by quantitative polymerase chain reaction (qPCR) to determine copy number for KIR3DL1 locus, and a cohort of individuals was tested to detect and evaluate capability for precise copy number discrimination.
Methods: Genomic DNA of a random sample was diluted to 12.5, 2.5, 0.5, 0.1 and 0.02 ng/ul, respectively. KIR3DL2 which always has two copies in a diploid genome was used as internal control and amplified by qPCR, and Sybr green was used as fluorescent dye. 162 blood samples were collected and amplified by qPCR and diluted to unified concentration according to concentration-Cp standard curve of 3DL2. Cp values of 3DL1 for all diluted samples were obtained by a second qPCR. △Cp (Cp3DL1 - Cp3DL2) were calculated and compared with a reference sample of fixed copy number to obtain the copy number of each sample. All the PCR reactions were performed in 20 μl volume containing about 50 ng genomic DNA in 1 × LA buffer, 0.5 μmol/l of each primer, 200 μmol/l of each dNTP, 2.0 mmol/l $MgCl_2$ and 0.5 unit of LATaq DNA polymerase and 10ul SYBR Green. The PCR reaction condition was 95°C for 3 min, 45 cycles of 94°C for 20 s, 61°C for 40 s. The genotype frequencies and allele frequencies were calculated by direct counting.
Results: Sequencing results showed that, of the 162 individuals, 151 were 3DL1 positive, and of which, 89 were homozygous and 62 were heterozygous. Copy number assays showed that the 62 heterozygous samples all contained two copies of 3DL1 allele, 43 of 89 homozygous samples contained two copies of 3DL1, 46 of 89 homozygous samples contained only one copy of 3DL1. The distributions of 3DL1 genotype and allele frequencies calculated by copy number assays were different from that without copy number assays. The frequencies of 3DL1 containing two copies were 41.06% without copy number assays, but were 69.54% when tested by copy number assays. Twenty-one genotypes were obtained according to copy number assay and sequencing results. But, there are only 19 genotypes if the copy number is not calculated.

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Summary/Conclusions: Copy number assay affects frequencies of KIR gene and allele, which provides accurate copy numbers of KIR genes that cannot be tested directly by sequencing based technology. The established copy number assay method can be used for disease association studies in future.
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Academy Session – Quality Management in the Laboratory

3B-S11-01

QUALITY MONITORING AND RISK ASSESSMENT IN BLOOD TRANSFUSION SERVICES

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Quality monitoring and evaluation in transfusion medicine is particularly important for several reasons: biological origin of blood components which implies variability of both the initial material and the final products, specific risks associated with transfusion treatment, complex algorithms of donor selection and testing, many inter-connected segments combining laboratory medicine, clinical medicine and pharmaceutical production.

The Aims are to have a harmonized and optimal suited quality monitoring and risk management system. Standards and criteria for quality monitoring and risk assessment have been addressed based on (1) a survey platform on quality management by the ISBT working party on quality management, (2) the critical control points (CCPs) and cross references given in the EuBIS manual and guide and (3) the level-based accreditation model developed by the AfSBT. These include inter-alia good practice guidelines (GPG) (EDQM, Directive 2016/1214) and good manufacturing practices (GMP) to cover blood, blood components and medicinal products for human use.

Quality monitoring requires well defined quality indicators and a systematic process release parametric system. Differences between processes have to be addressed including equipment, facilities and training of staff with respect to the complexity. Quality monitoring will also require to set-up a standard operating procedure (SOP) and documentation system site-by side with GMP/GPG related change control processes (CCPs) covering validation and qualification. 78% of the blood establishments surveyed responded the need to improve areas of quality with 65% related to national requirements. From the QM survey results, particular emphasis will also include monitoring premises and infrastructure requirements. Installation (IQ), operational (OQ) and process (PQ) qualification, systematic process identification using risk assessment matrixes and/or the Ishigawa fishbone diagram technique. In order to assist quality monitoring of processes, a core set of 11 out of 36 QI have been defined by the working party of quality management comprising (1) venipuncture failures, (2) Donor adverse reactions (3) Donor complaints (4) Positive findings on blood product bacteriological testing (5) Wrong blood product issue (6) Blood product complaints (7) Product withdrawal from the market (8) Patient sample non-conformities (9) Non-conformities in the requests for pre-transfusion testing (10) AB0/Rh(D) discrepancies (11) Serious adverse events (SAE). Furthermore, standards and criteria for SOPs/QMS as well as inspection of these have been defined via the EuBIS guide to be used as a tool to assist blood transfusion services to promote the safety of blood and blood components or for competent authorities, who are authorizing and inspecting those blood establishments.

In conclusion, quality monitoring and risk management in transfusion medicine is especially important, considering the biological origin of blood components and specific risks associated with their use. These activities should be strategic and proactive and not reactive. Risk management is important for the safety of donors and patients but also for the reduction of wastage.

Acknowledgement: We would like to thank the contribution given by the members of the ISBT working party on quality management. This work has been supported in part by funds of the European Commission initiating the EuBIS Academy (grant 2006202 and contract 2011/S 167-275198) and funding of the ISBT to the working party.

3B-S11-02

PREPARING YOUR BLOOD CENTER FOR ACCREDITATION

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Modern health care system is now operating in a rapidly evolving social, economic and technological and somewhat political environment where stakeholders demand heavily for quality and safety together with accountability and transparency. Significant pressure is seen at the health care providers who need to proactively work in meeting these expectations. Accreditation appears to be one of the approaches to demonstrate to the stakeholders that health care system is driving for quality, safety and continual improvement. With this, there are growing numbers of health care facilities including blood centers seeking accreditation in the last two decades in addition to the licensing requirement of their regulatory authorities.

Blood centers are establishments who operate to collect blood from voluntary non-remunerated donors, process and test blood collected to safe and quality blood and blood components for clinical transfusion. Depending on the blood center and their customers' needs, accreditation can be in the form of ISO quality management system (including ISO 15189 medical laboratory testing), or others standards i.e. the Current Good Manufacturing Practice (cGMP) and professional one e.g. that of the American Association of Blood Banks (AABB). In all scenario, top management's commitment and support to the accreditation is of utmost important as this provides the foundation towards ongoing journey in quality and accreditation where staff can follow through.

To prepare and achieve accreditation, a task force or committee is needed to clearly determine the goals and objectives, the timeframe and the resources involved. Key personnel should then obtain the relevant standards and information in order to prepare an agreeable timetable that incorporates milestones for accreditation. Training should not only be conducted for implementation personnel on how to use the standards to assess current practice and evaluate the level of compliance but also to staff to understand the organizational direction and importance of the accreditation. A gap analysis is usually come out following the review of the current practice against the standards. At this stage, it is not surprising to have a number of follow up actions that require alignment or even modification of the existing practice and documentation to the required accreditation. Besides, further clarification from the accreditation body may be necessary to ensure proper understanding of the requirement. However, if deviation from the accreditation standards is inevitable, the blood center would need to provide solid evidence to justify.

Once these preparatory works are completed, the blood center should now ready to go through a self-assessment through trained internal assessors and then follow by an on-site assessment by the accreditation body. Again, non-conformances may be identified but bear in mind that these could also be treated as opportunities for better understanding and compliance to the accreditation standards. Moreover, demonstration of ongoing quality improvement under the accreditation framework is of equally importance to the accreditation body and stakeholders.

To conclude, accreditation is an internationally accepted framework in demonstrating continual quality improvement of blood centers. Although significant resources are often necessary, the observed benefits are not only evidenced by having a sustainable quality and safe blood supply but also gaining the public confidence and trust on a reliable organization through commitment towards continual improvement.

3B-S11-03

SELF-INSPECTION AND CLASSIFICATION OF NON-COMPLIANCE

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Self-inspection is a key part of blood establishment (BE) quality management system (QMS). By self-inspection BE shall identify problem and deficiencies or non-compliances (NCs) and verify whether or not it is compliance with the internal SOP, guidelines, standards and regulations. Self-inspection is an ideal tool for optimizing

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processes, implementing preventive measurements and improving the safety and quality of blood. Self-inspection is also the base of different types of assessments.
To set up the self-inspection procedure for whole facility, to familiarize staff with internal-audit process, carry out the self- inspection on a regular basis and find what is still needed to do for improving the QMS.
The secret to a successful and efficient self-inspection is adopted standards, inspection plan, result assessment and correction action. Usually identify the contents of standards/legislation, quality manual, SOPs as self-inspection criteria. Before inspection to learn the number and severity of NCs in the previous inspection/assessment can provide detail self-inspection points. Self-inspection planning should include the objectives and scope of the inspection, the inspector, the detail inspection procedure and the timetable. In organization those respected technical experts and senior staff member who has practical experience in operational area, are familiar with processes and has ability to communicate effectively can be trained as inspector to review procedures and identify problems. After the self-inspection completion, the responsible person, quality manager, inspector and other related personnel should discuss on any relevant findings observed, classify NCs and decide necessary immediate measures. Afterwards draw up a detailed inspection report, compile action plan, take measurements and follow up. The structure of the official inspection report include Inspection scope, findings, NCs, observations, time to corrective action, and conclusions with clear statement about the outcome.
According to the severity degree, the NCs found in self-inspection are classified into 4 categories: critical non-compliance, major non-compliance, other significant non-compliance and observation. Identify NCs with clear classification and examples, references, pictures, copie can help to plan action to implement corrective measures. Urgent measures may be needed under the urgent decisions. Deficiencies that have been corrected during the inspection should be included in the inspection report with a statement that it has been corrected. BE's inspections, control measures, corrective and preventive actions should be a dynamic, active and continuous process, aiming quality and safety. Following the implement of new preventive measurements, a new inspection may be needed and scheduled. NCs start high and will reduce over time with handling the NCs.
In conclusion, self-inspection can help the BE to correct NCs as quickly as possible, use self-inspections as part of learning process, recognize efforts of staff, evaluate the facility's quality and operational systems to determine whether the service they provide is appropriate and in control.
Acknowledgement: We would like to thank the contribution given by the members of the ISBT working party on quality management. This work has been supported in part by funds of the European Commission initiating the EuBIS Academy (grant 2006202 and contract 2011/S 167-275198) and funding of the ISBT to the working party.

Immunogenetics – Abstract Session

3B-S12-01

PHENOTYPIC CHARACTERISTICS OF NOVEL RHD VARIANTS IN INDIANS

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Background: The RhD antigen being unarguably the most clinically significant blood group antigen has led to it being studied in several comprehensive studies in Caucasians, Black Africans and East Asians. Several phenotypic studies have been carried out in Indians but an extensive study was lacking. India being a formidable heterogeneous population a study of the phenotypic variability would definitely add to the repertoire of RhD variants and a better understanding of the same.
Aims: To phenotype and study novel RhD variants in Indians.
Methods: Samples (n = 223) presenting with a weak D phenotype by serological analysis using routine commercial anti-D reagents were genotyped for the *RHD* gene by conventional molecular approaches, i.e. by melting curve analysis and sequencing the *RHD* coding regions. Quantitative multiplex PCR of short fluorescent fragments (QMPSF) assay was performed for assessment of *RHD* exon copy number variations. The novel D variant samples identified were further characterized by using the Advanced Partial RhD Typing Kit (Alba Bioscience, Edinburgh, UK). Serological data of D antigen epitope profile and antigenic sites on these D variant samples was also evaluated.
Results: Out of the 223 RhD variants identified in the study seven variants were found to be novel. They are as follows: Exon 3 duplication (n = 130), *RHD*(G63C) (3), *RHD* (A59T) (1), *RHD*(A237V) (1), *RHD-CE(5:E223Q,V238M,V245L)-D (1)*, *RHD(T201R)-CE(5)-D(I342T) (1)*, *RHD(L214F)-CE(7)-D (1)*. Majority of the exon 3 duplication variants were detected in the IgM phase with weaker strength and showed presence of the 'C' antigen. They also showed a heterogeneous pattern of agglutination with varying strengths on testing with panel of anti-D reagents from ALBAclone Advanced Partial RhD Typing Kit (Alba Bioscience, Edinburgh, UK). Absence of epitopes 1.2, 8.1 and 6.3 in different combinations was observed. Additionally, flow cytometry analysis in a subset of samples showed a range of D antigenic sites (range: 1276–6050 D antigens/cell). Another novel weak D mutation RHD(A237V) predicted to be present on the top of transmembrane domain eight which is close to the extracellular domain showed absence of 1.1, 1.2, 6.3, 8.1 and 9.1 epitopes and was found to be responsible for the production of anti-D. For three samples showing weak D mutation RHD(G63C) predicted to be located in the second transmembrane domain the mean antigenic density was found to be in the range of 5088–7406 D antigens/cell.
Summary/Conclusions: All the novel RhD variants gave weaker reaction at IgM phase of testing and did not require indirect antiglobulin test for detection. Overall we describe a major novel weak *D* allele resulting due to exon 3 duplication in Indians. The varying pattern of agglutination, strength and the antigenic density observed for this novel variant can be attributed to the altered transcript biosynthesis. Preliminary data also suggest interindividual variation in the relative amount of transcripts produced. The study contributes to improve Rh blood group diagnostics significantly in more than one billion Indians.

3B-S12-02

IDENTIFICATION OF A LARGE GENE FRAGMENT DUPLICATION INVOLVING THE SMP1 AND PART OF THE RHD AND RHCE GENES IN ONE CHINESE DONOR

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Background: A series of complex molecular events has been reported to resulting in the normal, absence or weaken/partial expression of D antigen, such as point mutation, short or large deletion, insertion, splicing site mutation and hybrid allele, but no duplication has been identified before. The developed Multiplex Ligation-dependent Probe Amplification (MLPA) genotyping assay contains probes specific for *RHD* and *RHCE* consensus sites and some mutations that could not only to determine the presence of consensus and mutant *RHD* or *RHCE* sequence but also its copy number. In previous study, one Chinese donors having three copies for all probes specific for the exon 2–10 of the *RHD* gene but with two normal copies for the probe specific for *RHD* 5′-UTR was identified by MLPA.
Aims: To confirm the existence of the duplication of the *RHD* gene and clarify the exact region of duplication in the Chinese donor using the next generation sequencing (NGS) method.
Methods: Genomic DNA of the donor was extracted and then send to a Chinese biological company (the Beijing Genomics Institute, Wuhan, China) for the whole genome sequencing using Illumina HiSeqTM 2000 platform. The raw data was aligned to the reference sequence (Human genome 19, hg19) using Burrows-Wheeler Aligner (BWA) software. Then, the further bioinformatics analysis of variant calling around the *RHD* gene on the chromosome 1 was performed to define the two ends of the duplication region, by using the CNVs (copy number variations) analysis which correspond to relatively large regions of the genome that have been deleted (fewer than the normal number) or duplicated (more than the normal number).
Results: A large fragment with a length of 93,800 bp was identified to be duplicated having a ratio of 1.49 for the copy number shown by the CNV analysis, which indicates three copies existence compared with the normal control with a ratio of 1.0 having two copies for this region. The large duplication region of 93,800 bp was annotated from IVS1-862 of the *RHD* gene to IVS7-2081 of the *RHCE* gene covering the whole genomic sequence of the *SMP1* gene located between the *RHD* and *RHCE* genes. Besides, a short deletion region within the *RHCE* gene with a length of 6000 bp was also found in this donor with a normal CCDee phenotype. The copy number for the region was shown with a ratio of 0.17 indicating a deletion by the CNV analysis. The deletion region was defined from IVS2-1064 to IVS1-940 of the *RHCE* gene involving the whole coding sequence of the exon 2 of the *RHCE* gene.
Summary/Conclusions: A large duplication fragment involving the *SMP1* and part of *RHD* and *RHCE* gene accompanied with a deletion covering the exon 2 of *RHCE* gene were firstly described. How the large duplication and the short deletion happened and its effect on the expression of D, C and e antigens are needed a further investigation.

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HYBRID GLYCOPHORIN AND RED BLOOD CELL ANTIGEN GENOTYPING IN AMERICAN BLOOD DONORS WITH MIA PHENOTYPE

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Background: The Mi^a antigen is carried on six different hybrid glycophorins in the Miltenberger Subsystem. The GP.Mur (Mi.III) glycophorin with the Mi^a phenotype is relatively common in the Southeast Asian populations. Antibodies against the hybrid glycophorins can cause both hemolytic transfusion reactions and hemolytic disease of the fetus and newborn. Cross-reactivity of the epitopes among different classes in the Miltenberger Subsystem makes the precise identification of the glycophorins using serology challenging.
Aims: To determine the hybrid glycophorin distribution in serologically Mi^a(+) Asian American blood donors using molecular techniques, as well as the red cell minor antigens in the same cohort.
Methods: A total of 4052 random Asian American blood donors primarily from the Gulf Coast Regional Blood Center (Houston, TX) have been screened by using the NOVACLONE™ anti-Mi^a mouse monoclonal IgG typing reagent. The Mi^a(+) donors identified in the initial screening were confirmed serologically by using supernate from anti-Mi^a GAMA210 monoclonal IgG antibody. The Mur(+) and the Mi^a(+) donors were identified by using anti-Mur 64D6 monoclonal IgM antibody, which were confirmed by using two polyclonal human antisera specific for the Mi^a and Mur antigens. Genomic DNAs extracted from the Mi^a(+) donors were amplified in the Polymerase Chain Reaction (PCR) using the *GYP* hybrid gene sequence specific primers (SSP). The hybrid glycophorins were identified by bi-directional DNA sequencing. The frequencies of GP.Mur, GP.Bun, GP.Vw and GP.Hut glycophorins in the Mi^a(+) cohort and among the donors were determined. The hybrid glycophorin phenotyping between serology and molecular testing were compared. In addition, phenotypes for the 35 minor blood group antigens were determined by using the PreciseType™ HEA Molecular BeadChip Test. The genotypes and zygosity of the *GYP.Mur/GYP.Bun* alleles in the Mi^a(+) donors were also determined by using the real time (RT) PCR method and the prototype BeadChip assay.
Results: At the Gulf Coast Regional Blood Center, 236 Mi^a(+) samples (5.82%) were identified by initial screening of 4052 Asian American donors using the anti-Mi^a NOVACLONE IgG typing reagent. At Immucor, only 215 Mi^a(+) samples could be confirmed with both anti-Mi^a GAMA210 IgG and anti-Mur 64D6 IgM due to the condition of some samples that precluded serological testing. One discrepant result between the donor center and Immucor had been investigated by SSP-PCR, which indicated that one Mi^a(-) sample was mistyped as Mi^a(+) at the donor center. Genotyping of the 236 Mi^a(+) samples showed that 91.10% (215/236), 6.36% (15/236), 0.85% (2/236) and 0.42% (1/236) were *GYP.Mur*, *GYP.Bun*, *GYP.Vw* and *GYP.Hut* with frequencies of 5.31% (215/4052), 0.37% (15/4052), 0.05% (2/4052) and 0.02% (1/4052), respectively, in the donor population. There was 99.53% (214/215) concordance between results from serology and *GYP* hybrid gene SSP-PCR coupled with sequencing analysis. One GP.Vw sample was mistyped as Mur(+) by using the anti-Mur 64D6 IgM antibody due to weak antibody reactivity (1 +). The minor red cell antigen profiles in the 236 Mi^a(+) donors were consistent with previously studies in Asian American donors. The hybrid glycophorins identified in the prototype BeadChip assay were found to be 100% concordant with genotyping by DNA sequencing.
Summary/Conclusions: The GP.Mur glycophorin is the most frequent phenotype of the Miltenberger Subsystem in the Mi^a(+) Asian American blood donors. This study provides an efficient process to characterize Mi^a(+) donors using a combination of serological and molecular methods.

3B-S12-04

HYPERMETHYLATION OF CPG ISLAND IN THE ABO GENE LEAD TO AN ABW PHENOTYPE DONOR

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Background: The ABO blood group system is highly important in clinical transfusion and transplantation medicine. Many of ABO subgroup have been discovered, which were caused by any mutation in the coding region or splicing site are known. Expression of ABO gene was dependent upon DNA methylation of the promoter region. However, rare donors with variant phenotype were reported due to the hypermethylation of CpG island.
Aims: This study aims to investigate the molecular basis of ABO gene in variant with serologic ABO blood group discrepancy.
Methods: The phenotype of ABO variants were typed with standard serologic method. The full coding regions of ABO gene for these variants were amplified with polymerase chain reaction and then directly sequenced. The level of methylation in ABO gene promoter CpG island was analyzed by bisulfite sequencing method. The ratio of each CpG site methylation was analyzed by BiQ Analyzer software. Then, the data analysis was performed by GraphPad Prism5.
Results: No mutations in the coding region or splicing site were found in the individuals with ABO subtypes. Compared with methylation level of each variant and the normal samples, hypermethylation of CpG island was existed in one case of AB_W phenotype. 40.3% of the CpG sites were methylation through the region. Both the amount of CpG site and the average ratio of CpG methylation extended from the region −700 to 200 showed significant difference with the other individuals ($P < 0.05$).
Summary/Conclusions: The hypermethylation located in the proximal promoter region was crucial for elucidating the mechanism of variant phenotype donor and offering useful information in blood transfusion.
This work was supported by the Science Research Foundation of Zhejiang Province (LY12H08001, LY17H080003), the Medical Science Research Foundation of Zhejiang Province (2016RCB006, 2017KY315, 2013C33193).

3B-S12-05

INVESTIGATION OF HLA-DPB1 MATCHING STATUS IN UNRELATED HEMATOPOIETIC STEM CELL TRANSPLANTATION BY EPITOPE-SPECIFIC TYPING METHOD

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Background: Allogeneic hematopoietic stem cell transplantation (HSCT) is a well-established treatment for advanced hematologic malignancies. The impact of HLA-DPB1 match after HSCT remains controversial. HLA-DPB1 match between donor and recipient can be analyzed at the allelic level, single amino-acid level or classified as permissive/non permissive HLA-DPB1 disparities. The epitope-specific typing (EST) method for HLA-DPB1 is an innovative protocol for rapid, targeted detection of DPB1 TCE disparities.
Aims: To analyze the matching status of *HLA-DPB1* in unrelated Hematopoietic stem cell transplantation by the epitope-specific typing method.
Methods: Two pairs of specific primers as previously reported (EST-AD-F: CCGCAGAGAATTACGTGC; EST-AD-R: TCCGGCACTGCCCGCTC; EST-CE-F: GGGCGGCCT GATGAGGAC; EST-CE-R: TCGTAGTTGTGTCTGCATAC) were applied to amplify the high variable region sequences of *HLA-DPB1* for 106 samples (53 pairs of unrelated donor-recipient),the HGH gene was used as an internal control in both reactions. The amplification products were detected by 2% agarose gel electrophoresis and the genotype was assigned. All the samples were detected simultaneously by PCR-SBT method in parallel so as to evaluate the accuracy of the EST method.
Results: The two specific target fragments were observed with the PCR amplification, and good results were obtained after directly electropherogram. Among 106 samples (53 pairs of unrelated donor-recipient), 71 individuals were negative for both of the two primers, 6 individuals were positive for both of the two primers, 18 individuals were only positive for EST-AD primers,23 individuals was only positive for EST-CE primers. The distribution between donor and recipient is no evident different according to the results of Chi-square test ($P > 0.05$). Of 53 pairs detected in our study, 36 pairs were matched and the other mismatched, which is identical to the result of PCR-SBT method as previously reported.
Summary/Conclusions: The epitope-specific typing method for *HLA-DPB1* is not only low-cost and timesaving but also reliable for large-scale retrospective clinical studies of HLA-DPB1 in unrelated hematopoietic stem cell transplantation.
This work was sponsored by National Science Foundation of China (81401732), Zhejiang Provincial Program for the Cultivation of High-Level Innovative Health Talents, and medical science and technology foundation of Zhejiang province (2013RCB003 and 2015RCA008).

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ESTABLISHMENT OF SEQUENCE-BASED TYPING ASSAY FOR KIR2DS4 GENE AND POLYMORPHISM ANALYSIS IN CHINESE HAN POPULATION AND IDENTIFICATION OF A NEW ALLELE KIR2DS4*016

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Background: Killer-cell immunoglobulin-like receptors (KIR) are members of the immunoglobulin receptor family, expressed by nature killer cells and part of the T cells with high polymorphism level on both of the gene content and allelic variation. KIR gene consists of 14 functional genes and 2 pseudogenes, their nomenclature are depending on the number of extracellular immunoglobulin-like domains (2D or 3D) and the length of the cytoplasmic tail (L or S), and transducing inhibitory (L) or activating signals (S) for NK cells. In general, KIR gene are divided into two haplotypes, haplotype A and B and KIR2DS4 is the only activate gene in haplotype A. KIR2DS4 gene contains 8 exons and 7 introns, the polymorphisms are mainly located in exons 4 and 5. At present 31 alleles been found according to the latest published data of the International IPD/KIR Database, resulting in 14 different amino acids (KIR2DS4*001-KIR2DS4*016). Depending on the difference of the exon 5, it can be divided into KIR2DS4 normal and deletion type. The deletion type of KIR2DS4 delete 22 bp from the nucleotide position 454 to 475 in the exon5 and has not been found to have an activity so far.
Aims: In this study, we aimed to establish a reliable sequence-based typing method for KIR2DS4 and study its allelic polymorphism in Chinese Han population.
Methods: Two hundred and twenty-two Chinese non-numerated blood donors were recruited in the context of informed consent in this study and checked as Han Minority. Firstly, we use PCR-SSP method to detect the positive/negative of KIR2DS4 gene in 222 random Chinese Han individuals. KIR2DS4 positive individuals were amplified by primers using high fidelity enzyme and long-fragment PCR procedure. The sequences of exons 4 and 5 were obtained by Sanger sequencing from both direction, and then analyzed by Seqman software compared with existed allele sequences on the IPD/KIR Database.
Results: We successfully amplify the fragment of 3.2 kb length contains exons 4 and 5 of KIR2DS4 and detect allele frequency in Chinese Han population. In 222 random Chinese Han individuals, 209 are KIR2DS4 positive (94.1%). By sequence-based typing, we identified 12 genotypes and 7 alleles of KIR2DS4. The highest frequent genotype is KIR2DS4*00101/011 with the frequency of 50%, followed by the genotype KIR2DS4*00101/011,*010 with the frequency of 14.9%. The six known alleles and their detection frequency is as follows: KIR2DS4*00101/011 (180, 81.1%), KIR2DS4*010 (53, 23.9%), KIR2DS4*004 (34, 15.3%), KIR2DS4*003 (15 and 6.8%), KIR2DS4*006 (2, 0.9%) and KIR2DS4*015 (1, 0.5%). In this study, we found a new allele, KIR2DS4*016, with the difference in exon 5 comparing its most similar allele KIR2DS4*010. In the exon 5 of KIR2DS4*010, there is a 22 bp-deletion, while the exon5 of KIR2DS4*016 is normal. This is not a rare allele because it was detected 3 times in studied population and with the frequency of 1.4%. The sequence of the new allele sequence has been submitted to GenBank (accession no.: KC414890) and the IPD-KIR database (submission no.: IWS40001804), and was nominated by WHO Nomenclature Committee for Factors of the HLA System.
Summary/Conclusions: In this study, we successfully established a sequence-based typing method for the exons 4 and 5 of KIR2DS4 gene. Using this method, we identify both of the genotype and allele polymorphism of KIR2DS4 gene in Chinese Han population and detect a novel common allele KIR2DS4*016, which enrich the polymorphism data of KIR2DS4 in Chinese Han population and supply a basic information for KIR2DS4 population genetics and disease association studies.

Immunohaematology – Red Cell Genetics

3C-S13-01

A BROAD UTILIZATION OF NEXT-GENERATION SEQUENCING FOR RED BLOOD CELL GENOTYPING TO SOLVE PHASING, GENE ARRANGEMENTS AND COMPLEX SEROLOGY FINDINGS

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Background: Red Blood Cell genotyping has been developed to predict phenotypes and to reduce alloimmunization in transfusion. It also provides distinguish information for serology complications. ABO subtypes, RH variants and Miltenberger series are most frequent serological complications seen among blood donors in our center, along with para-Bombay and JK nulls. Due to the vast variety of the genetic changes may be presenting, a platform capable of valid genotyping is desired. With the advances in Next-Generation Sequencing (NGS) and bioinformatics analysis, NGS is now the ideal platform for high-throughput, accurate, extensive and rapid genotyping.
Aims: Utilize next-generation sequencing result from customized panel for RBC blood group genes to determine genotypes and to solve complex serology findings.
Methods: Genotyping: A probe-based capture panel targeting blood group genes associated with 15 clinically relevant blood group systems including ABO, RH, MNS, DUFFY and KIDD were tested. The enriched DNA was sequenced on Illumina MiSeq platform to generate 300 bp paired-end reads. RBC genotype is determined after bioinformatics analysis for SNVs, CNVs, SVs, and phasing is introduced to reduce ambiguities between alleles for ABO subtypes.
Phenotyping: RBC antigen serology typing with Bio-Rad ID system for ABO, RH, KEL, MNS, FY, JK, LE, LU, DI and in house anti-sera for "Mi^{a}" with gel test. Also ABO subtyping with ad/el or saliva testing, weak D testing was applied when necessary.
Results: A highly accuracy genotyping result from NGS is obtained (99.3%), including complex cases resolutions. Haplotyping can be achieved with sequenced reads phasing to determine *ABO* heterogeneity, read depths from *RHD*, *RHCE* can be used to determine the zygosity of *RHD* and hybrid *RHD-RHCE* complex cases. D variants are determined and help classified Del from D negatives. Serology inconclusive Miltenberger series are distinguished through NGS genotyping as *GP.Hut*, *GP.Vw*, or *GP.Mur* resulted from gene rearrangement. However, with blood group genes that produce transferase but not antigen itself, such as P1PK or Lewis system, the prediction accuracy yield lower than other blood groups.
Conclusions: We set up a customized NGS panel for blood group system prediction, established an automatic analysis pipeline. Both open source bioinformatics tools and in house developed scripts allow us to detect antigen determine SNPs, potential CNVs, SVs with gene hybridization, low percentage mosaicism and also novel SNVs for possible gene expression alteration. We are confident to validate NGS genotype result having high concordance with phenotype data and utilize it to solve complex cases.

3C-S13-02

Abstract has been withdrawn

3C-S13-03

MOLECULAR GENETIC ANALYSIS OF PARA-BOMBAY PHENOTYPE REVEALING A NOVEL ALLELE IN THE FUT1 GENE

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Background: The para-Bombay phenotype has a very low frequency in world population. The para-Bombay phenotype is characterized by the absence of weak expression of ABH antigens on red blood cells, but ABH substances are present in saliva. The underlying molecular basis is most commonly a mutated *FUT1* gene present with or without an active FUT2 gene.

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Aims: In this study, Samples from 15 Chinese individuals serologically typed as para-Bombay were investigated. Among these, a novel *FUT1* mutation was identified.
Methods: Standard serologic techniques were used. Genetic mutations of *FUT1* and *FUT2* genes were analyzed by DNA sequencing. The haplotypes of *FUT1* were sequenced prior clone the genes into T vector.
Results: Five *h* alleles were observed from 15 para-Bombay phenotype probands, i.e. *h1* (nt547-552Δag), *h2* (nt880-882Δtt), *h3* (nt658c→t), *h9* (nt424c→t) and *hnew*-2(nt661 c→g). Seven *FUT1* genotypes, i.e. *h1/h1*(n = 4), *h1/h2* (n = 3) and *h1/h3* (n = 4), *h1/h9* (n = 1), *h1/h6* (n = 1), *h2/h2* (n = 1), *h2/hnew* (n = 1) and three functional *FUT2* genotypes, i.e. $Se^{357}Se^{357}$ (n = 10) and $Se^{357}Se^{357,716}$ (n = 4) $Se^{357,716}/Se^{357,716}$ (n = 1) were identified in 15 probands.
Summary/Conclusions: A novel non-functional *FUT1* allele (661 C>A, Arg >Gly) was identified. Our data add to the growing database of mutations in the *FUT1* gene and confirm previous reports regarding molecular analysis for para-Bombay phenotype.

3C-S13-04

IDENTIFICATION OF A RHAG*(R191Q) ALLELE ASSOCIATED WITH A WEAK RHD AND NORMAL RHCE PHENOTYPE

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Background: The molecular background of majority of the D variant phenotypes are missense mutations, insertions, deletions in the *RHD* gene, or *RHD-CE-D* hybrid alleles. In rare cases, mutations in genes encoding proteins that are required for proper membrane expression, such as *RHAG* and *ANK1,* can also lead to absence or weakened expression of Rh antigens.
Aims: To unravel the molecular basis of weak RHD expression in a Chinese blood donor.
Methods: A blood sample with D variant phenotype was collected from the Guangzhou Blood Center. RhD serological typing was performed using two monoclonal anti-D reagents(clone Rum-1 and TH-28/MS-26) and a commercial panel of monoclonal anti-D reagents(D-Screen, Diagast) by a tube method. The RhCcEe antigens were typed using monoclonal antibodies (anti-C, MS-24 and P3 × 25513G8; anti-c, MS-33; anti-E, MS-258/MS906; anti-e, MS-16/MS-21/MS-63 and P3GD512/MS63). The *RHD* and *RHCE* genotypes were analyzed by the developed *RH* multiplex ligation-dependent probe amplification (*RH*-MLPA) assay; all 10 coding exons and promoter of the *RHD* gene were amplified and analyzed by direct sequencing. A customized Ampliseq panel was used for next generation sequencing (NGS) of all exons, flanking introns regions and untranslated regions of 10 genes including *RHAG* and *ANK1* in this donor. DNA preparation and sequencing were performed on the Ion S5 sequencer system (ThermoFisher Scientific) according to the manufacturer's instructions. Bioinformatics analysis was performed using Torrent Suite Software in combination with resources such as the Human Gene Mutation Database (HGMD), ClinVar and Pubmed.
Results: The donor was identified with a weak D phenotype but a normal CCee phenotype by serological methods ((4 +)MS-24 and (4 +)P3 × 25513G8 for anti-C, (4 +)MS-16/MS-21/MS-63 and (4 +)P3GD512/MS63 for anti-e). Red blood cells from this donor has a negative agglutination with the IgM anti-D (Clone Rum-1 for epD6.1), while weakly reacted with IgM/IgG anti-D ((2 +), Clone TH-28/MS-26 for epD6.4 and epD9.1). For the reaction with the panel anti-D (D-Screen), weak positive agglutination with all nine kinds of anti-D was detected ((1 +)HM10 for epD6.6; (1 +)HM16 for epD6.4; (2 +)P3*61 for epD6.1; (1 +)P3*290 for epD5.4; (1 +) P3*212 for epD9.1; (1 +)P3*241 for epD5.4; (1 +)P3*290 for epD3.1; (w+)P3*212 for epD8.2; (w+)P3*290 for epD3.1). No variant was identified in this donor analyzed by the *RH*-MLPA technique and DNA direct sequencing of all *RHD* exons and promoter. NGS analysis of the *RHAG* and *ANK1* genes revealed a homozygous mutation (c.572G>A, p.Arg191Gln) located on the exon 5 of the *RHAG* gene in this donor, which resulted in an amino acid substitution in the intracellular loop of RhAG protein.
Summary/Conclusions: The homozygote *RHAG*(R191Q)* allele was identified to associate with a weak D phenotype but a normal CCee phenotype by NGS technology. This finding strongly suggests that this aminoacid change interferes with the correct association of RhAG and RhD proteins, while it is not involved in the association with the RhCe polypeptide. Preliminary *in vitro* expression studies in which the different alleles were co-expressed in 293T cells are in agreement with this hypothesis.

3C-S13-05

Abstract has been withdrawn

Academy Session - Haemovigilance

3C-S14-01

EXPERIENCE OF HEMOVIGILANCE IN JAPAN

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Japanese Red Cross (JRC) is the only facility in Japan that deals with blood business from blood collection to product delivery. It collects around 4.8 million of blood donations annually. 940 thousand patients receive blood transfusion annually in Japan.
A hemovigilance system was established in Japan in 1993 by the JRC headquarters, initial purpose of which was to collect reports of adverse transfusion reactions from medical facilities. Although this is a voluntary reporting system, the annual number of reports increased from 228 in 1993 to 1,939 in 2004 and has since remained between 1,500 and 1,700. Currently JRC, in cooperation with medical facilities, collects almost all of the serious adverse reactions caused by blood transfusion. Reports on non-hemolytic reactions account for >90% of all reports and those of transfusion-transmitted infection (TTI) account for 5–6%.
One of the characteristics of JRC hemovigilance system is that it is supported by a laboratory analysis conducted in the JRC Central Blood Institute. The JRC has collected pre- and post-transfusion blood samples from patients with adverse reactions. These samples together with repository blood samples obtained from corresponding donated blood were assessed for the pathogenesis of the adverse events. Data obtained using these samples were helpful in solving issues in transfusion medicine such as transfusion-transmitted infection (TTI), transfusion-related graft-versus-host disease (TR-GVHD), some type of anaphylactic shock or TRALI. For example, for each of the suspected TTI cases, we have determined the causal relationship between the infection and blood transfused. Filing up and analyzing these data, we evaluated the effectiveness of the current screening system and estimated residual TTI risks. We were also able to establish the infectivity of blood collected at several stages of infection. All these data have greatly contributed to the planning of strategy for new TTI screening.
Through the hemovigilance activity, we understood that TR-GVHD is not a rare event in Japan, which led us to the quick implementation of almost universal X-irradiation of cellular products. We were also able to verify the mechanism of anaphylactic shock caused by anti-haptoglobin antibody present in the blood of transfused patient.
Hemovigilance also includes the function of blood donor vigilance where adverse reactions occurring on blood donors during or after blood donation are collected and analyzed. It led us to implement a muscle tension exercise to decrease the rate of vasovagal reactions or mitigate the reactions.
One of the drawbacks of JRC hemovigilance system, however, is that cases of medical malpractice or near miss case such as incorrect blood transfusion or major mismatch transfusion are hardly reported, which is the limitation of the voluntary reporting system. Although intense collaboration between blood center and medical facilities is essential in this regard, a strong leadership by national transfusion society or new legislative measures are needed to establish comprehensive national hemovigilance.

3C-S14-02

HAEMOVIGILANCE IN CHINA: SHANGHAI EXPERIENCE

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Haemovigilance (HV) is a continuous process of data collection and analysis of transfusion-related adverse reactions and events (AR/AE) in order to investigate their causes and outcomes, and prevent their occurrence or recurrence. HV plays an essential role in ensuring the safety of donors and recipients with regard to blood transfusions. HV systems have been implemented effectively in most of high-income countries throughout the world, however, there has been not a functional HV system

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in China, either national or regional. By 2013, the first guideline on HV in China was released by the blood quality management committee of Chinese Society of Blood Transfusion (CSBT), and revised in 2015. For blood establishments and hospitals looking for help with setting up new plans or further improving blood transfusion safety, the guideline provides ample information on definitions, methodologies, and documentation protocols. Based on the guideline, a pilot program of transfusion reactions report was developed in Shanghai, which is a passive reporting model, but in spite of the delays and limits of the data, the program provides useful information to address a variety of questions in transfusion practices. HV is also considered to be used for other important objectives, such as the training of transfusion practitioner, the implementation of patient blood management (PBM) and the development of clinical guidelines. With the integration of HV into the existing expert symposiums and education courses in the past few years, hundreds and thousands clinician and blood bank practitioner got training and the importance of HV was recognized widely, and the principles of non-punitive principle and learning from errors. In 2017, the working party on haemovigilance of CSBT was set up, which is a milestone for the progress of HV in China. Through the working party, we can study the practices of antecedence countries and learning their experiences, by formulating policies, evaluating modes and mechanisms, compiling standards and guidelines, improving information management and standardization from one region to the other, to establish a national HV systems step by step in China.

3C-S14-03

HAEMOVIGILANCE IN JIANGSU PROVINCE OF CHINA: PRACTICE OF ESTABLISHING A WIDE AREA HAEMOVIGILANCE NETWORK

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In 2016, the WHO Global Database on Blood Safety reported that national haemovigilance (HV) system had been established in 70 countries, among which 38 countries were members of the International Haemovigilance network by the end of August 2017. In China, Haemovigilance is still in the early stage except for 2 HV systems in Hong Kong and Taiwan, the wide area haemovigilance network (HWAN) of whole journey and two-way tracking of transfusion information has not been established in mainland China. So it is difficult to systematically report and intervent transfusion adverse events between blood services and hospitals at Multi-city-level. Therefore establishing a regional HV system that covers the entire transfusion chain is necessary to ensure the safety and supply of blood. To explore, build and operate an exemplary HWAN based on cloud techniques, that covers the entire transfusion chain and to realize transfusion information control, sharing, exchange and statistical analysis from blood donors to recipients in the whole blood transfusion chain, and to monitor, intervent and manage effectively the transfusion adverse events. The HWAN software and a standardized transfusion database were developed under the latest country laws & regulations, the existing international standards, the HWAN information technology standard, data element standard and basic function criteria. The data of transfusion chain and other public platform were collected and transfusion information control, sharing, exchange and statistical analysis were realized by the application of the Middleware Technology combined with manual reporting way. Transfusion adverse events were reported, analyzed and intervented by establishing a reporting and analysis system for transfusion adverse events, an unified procedure for their early warning, reporting and disposal, as well as an assessment method of prevention/corrective measures. The warning information and disposal advice were immediately issued and an efficient emergency response system was built through the SaaS services portal which linked together the relevant blood services, medical institutions and management departments within the scope of HWAN. The pilot HV work was designed and carried out in 6 representative blood center, central blood station, basic-level blood station, secondary hospitals and tertiary hospitals of three cities of Jiangsu province, and an exemplary HWAN that covers the entire transfusion chain have been explored. Jiangsu's HV cloud is China's first regional HV network, which can realize bidirectional vein-to-vein monitoring and intervention, encourage joint activities between the members of transfusion chain at higher level and wider range, provide reference to ensure the blood supply& quality& safety, appropriate use of blood resources, and improve the response capability in blood emergency.

3C-S14-04

DONOR HAEMOVIGILANCE IN SINGAPORE

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Donor Haemovigilance is a surveillance process to monitor the undesirable events during and after blood or blood components donations. In Singapore, we started to collect data on bruising and haematoma rates which was used to measure staff performance since 2003. A more comprehensive donor haemovigilant programme was set up in 2007. All adverse events were captured and collated. For the past 10 years, the haemovigilance rate for whole blood donation was about 2–3% per year. The main adverse event was vasovagal reaction, and mostly affecting the new young donors. The haemovigilance rate for apheresis donation was lower (0.7–1.7%). The lower adverse reaction rate for apheresis donation could be due to apheresis donors were all frequent donors. Most of the vasovagal reaction were mild to moderate (>99%) according to the ISBT Donor Haemovigilance Workgroup definition.

3C-S14-05

INTRODUCTION OF THE KOREAN HEMOVIGILANCE SYSTEM

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Background: The first regulation on hemovigilance in South Korea was first established in 1999 by the Blood Management Act. The head of medical institutions must report specific transfusion reactions (death, disability, requirement of hospitalization, transfusion-transmitted infection) to the Ministry of Health and Welfare within 15 days of confirmation. The purpose of this act was mainly focused on management of transfusion-transmitted infections.
Aims: A more general form of hemovigilance system was needed for surveillance on the transfusion chain.
Methods: A research funded by the Korea Centers for Disease Control and Prevention was initiated in 2007 for development of the hemovigilance system. Review on hemovigilance systems of other countries and development of draft report forms were proceeded on the first year. Pilot operation of the system with 46 participating institutions was conducted during 2008–2009 and a website for reporting was developed during this period.
Results: The system was formally launched in a nationwide scale in 2010. It is supported by the Korean Ministry of Health and Welfare but is independently operated by the Korean Society of Blood Transfusion (KSBT). Although participation and reporting is voluntary, it is included in the accreditation checklist by the Korea Institute for Healthcare Accreditation and Assessment for Certification in Laboratory Excellence by the Laboratory Medicine Foundation. Participation is also required for institutions to be eligible for receiving blood management fee. These requirements have led to a significant increase in the number of participating institutions. Hemovigilance correspondents or blood bank directors report near-misses, incidents, and all types of adverse transfusion reactions. Reports can be sent by email but online reporting via the website is generally recommended. Explanatory meetings are held annually for participating institutions. Categories of adverse events and their severity were revised according to the definition by the ISBT Working Party on Hemovigilance and a simplified algorithm for discriminating acute adverse reactions was developed in 2015–2016. Imputability grading was included in the pilot operation system but is currently excluded. Annual reports in Korean language can be accessed via the website. Among 215 participating hospitals, 169 reported 3,375 adverse events during the fiscal year 2016. The majority was adverse reaction without incident (97.5%), followed by near miss (2.2%), incident (0.2%), and adverse reaction with incident (0.1%). Among 3,293 cases of adverse reactions, febrile non-hemolytic transfusion reaction (54.1%) and allergic reaction (24.9%) accounted for most of them. Most of the adverse reactions were non-severe (85.1%), and red cells (69.9%) were the most frequently related blood component.
Conclusions: A government-supported hemovigilance system was introduced in Korea from 2007. We have successfully operated the system for ten years including a two-year pilot study and gained substantial increase in the number of cases reported. We hope to further enhance the system qualitatively regarding case validation and standardization of case reporting.

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Plenary Session – Platelets

PL3-01

ROLE OF CD36 IN IMMUNE THROMBOCYTOPENIA AMONG ASIAN POPULATION

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CD36 (GPIV, Nak[a]) belongs to the class B scavenger receptor family, which is a highly glycosylated 88-kDa protein and is expressed widely in different cells such as platelets, monocytes, macrophages, endothelial and epithelial cells. There are two types of CD36 deficiency; type I with total absence of CD36 expression on platelets and monocytes and type II lacking of CD36 on monocytes but not on platelets. CD36 deficiency on platelets is more frequent in Asians (about 3–11%) and Africans (about 8%) than in white people (<0.4%). In south of China, the frequency of CD36 deficiency on platelets is about 2% (1.8% in Guangzhou; 2.16% in Shanghai), however, a little bit higher in Guangxi (4.13%). Meanwhile, more than 30 mutations underlying CD36 deficiency have been described in Asian population. 1228-1239delATTGTGCCTATT, 329-330delAC and C1156T seem to mainly responsible for CD36 deficiency in Guangdong population. In contrast, C268T, 949insA, 329-330del AC and 1228-1239delATTGTGCCTATT are the most common mutations in Japan. Recent data indicated that more than 0.5% of individuals with CD36 type I deficiency may be at risk of developing anti-CD36 antibodies after receiving transfusions or during pregnancy in China and Japan. Since the first case of platelet transfusion refractoriness caused by anti-Nak[a] antibodies was reported in a Japanese patient, the impact of anti-CD36 antibodies has been reported in several clinical conditions of immune-mediated thrombocytopenia including fetal/neonatal alloimmune thrombocytopenia (FNAIT), post-transfusion purpura (PTP) and platelet-transfusion refractoriness (PTR) as well as transfusion related acute lung injury (TRALI). Compared to other platelet antibodies against HPA systems, anti-CD36 antibodies seem to be more frequently found in PTR and FNAIT cases among Chinese population. Recently, two cases of FNAIT and five PTR cases induced by anti-Nak[a] antibodies were found in Guangzhou Blood Centre. However, the incidence of thrombocytopenia caused by anti-CD36 antibodies in Guangxi (four PTR and six FNAIT) is higher than other regions of China, which may be associated with the high frequency of CD36 deficiency. For the treatment of thrombocytopenia caused anti-CD36 antibodies, transfusion with CD36*null* platelets is necessary to improve or prevent bleeding. Therefore, a donor registry of CD36*null* donors should be established for the treatment of immune mediated bleeding disorder caused by anti-CD36 antibodies. Furthermore, testing of anti-CD36 antibodies should be considered in immune mediated thrombocytopenia, and probably in suspected TRALI as well. To improve the treatment of these disorders, further studies on the pathomechanism of anti-CD36 mediated immune disorders is necessary.

PL3-02

MULTIFACETED REGENERATIVE LIVES OF EXPIRED PLATELETS

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There is great interest in using expired (5 – 7 days old) platelet concentrates, no longer suitable for transfusion due to bacterial contamination concerns, as source material for producing human platelet lysates (hPL) for cell therapy procedures and direct clinical use in regenerative medicine. The scientific rationale behind this trend lies in the richness of platelets in a myriad of growth factors and cytokines, including PDGF-AB, TGF-b, VEGF, EGF, HGF, BDNF, and many other cell growth-promoting biochemical entities. Expired platelet concentrates collected by blood establishments are typically frozen at −20°C or colder when reaching the expiry date and stored frozen until hPL preparation. The main, now well established, use of hPL is to replace fetal bovine serum (FBS) at different stages of cell therapy protocols, providing xeno-free nutritional supplementation of interest for development or use of *ex vivo*-expanded cells for human transplantation. Blood establishments in some countries already produce pooled hPLs as clinical-grade material for use in cell therapy trials. It is also available, for research use from commercial suppliers. To improve batch-to-batch consistency, these hPLs are manufactured by pooling the lysates from approximately 40 to 50 platelet concentrates. Numerous studies have demonstrated hPL capacity to expand primary mesenchymal stromal cells from different tissue sources, as well as various differentiated primary cells. hPL avoid immunological and zoonotic infectious risks linked to the use of FBS, and in a number of situation, exhibit more potent capacity at expanding cells, shortening the cell population doubling time under conditions where cells maintain their inherent physiological, biochemical, and functional properties. Another application of platelet-derived biomaterials (typically referred to as platelet-rich-plasma or PRP, or platelet gels) is for regenerative medicine. In these applications, the lysate is currently prepared from an autologous donation enriched in platelets, or using a therapeutic-grade platelet concentrate available from hospital blood banks. The scientific rationale is based on the expectation that the multiple growth factors released by platelets can contribute clinically to enhancing the healing and repair of damaged or injured tissues. The main indications encompass healing of recalcitrant diabetic ulcers, burns, repair of bone defects, reinforcement of implants (especially in maxilla-facial surgery), release of osteoarthritis pains, or treatment of dry eye syndrome. Although there is an overall belief that these growth factors-rich platelet materials are clinically beneficial, the variability in the characteristics of these products does not generally allow drawing evidence-based conclusions and recommendations on best clinical practices. Future trends may possibly include the development of pharmaceutical, small- or large-pool virally-inactivated platelet lysate preparations, made from expired platelet concentrates. Those could be manufactured following controlled conditions and meet quality control specifications tailored-made for dedicated clinical applications (healing of soft tissue, bone augmentation, treatment of osteochondral defects, neuroregeneration, etc.). In conclusion there is a strong scientific rationale for making optimal use of expired platelets available in blood establishments to produce growth factor preparations for both cell therapy procedures and dedicated regenerative medicine applications.

PL3-03

A CLINICAL TRIAL OF FROZEN PLATELETS

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Introduction: Platelets have a short shelf life (5 days) and up to 20% of platelet units may be discarded due to expiry. Smaller Australian hospitals often cannot justify keeping many (or even any) platelet units on-site, and cryopreserved platelets may alleviate this problem. A single controlled clinical trial of cryopreserved platelets in cardiac surgery patients has previously shown that transfusion of cryopreserved platelets was associated with less bleeding and fewer red cell transfusions, with no adverse effects. Before cryopreserved platelets can be approved for civilian use, more information regarding their efficacy, cost-effectiveness, and safety is required. Therefore a pilot clinical trial is being conducted to demonstrate the effectiveness and feasibility of using cryopreserved platelets for the management of post-surgical bleeding, to inform a future definitive trial.

Methods: Cryopreserved vs liquid platelets for surgical bleeding (CLIP) is a multi-site, blinded randomised controlled effectiveness trial comparing cryopreserved platelets to liquid-stored platelets. Cardiac surgery patients at high risk of perioperative bleeding are identified using a validated scoring method. Consented patients are randomised to receive either cryopreserved or conventional platelets. Platelets are cryopreserved at the Australian Red Cross Blood Service and shipped to hospital sites. When required, the platelets are thawed and reconstituted in thawed FFP immediately prior to transfusion. Outcomes include acceptability to clinicians, a sufficient trend towards greater efficacy in reducing blood loss and transfusion requirements, the rate of adverse effects, incidence of venous thromboembolism, cost per patient, length of ICU stay and 28-day mortality.

Results: To date 107 patients in four metropolitan tertiary hospitals have been randomised. Of these, 40 patients have been transfused. No adverse events have been reported to date, and no reported transfusion delays. One of the challenges has been the low overall conversion rate from randomisation to transfusion (37%). Access to monitored −80°C freezer storage has also been challenging. As thawing platelets is not a routine blood banking activity, training large numbers of hospital blood bank staff in this process and maintaining their competency over the duration of the trial was also demanding.

Conclusions: There have been many challenges associated with introduction of a novel blood component in a clinical trial, all of which have been overcome. Upon completion data will be analysed, and together with other learnings from the trial, will be used to inform the design and sample size calculations for a larger definitive trial.

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TTID – Screening

4A-S15-01

BLOOD SCREENING STRATEGY IN INDONESIA

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Indonesia is the biggest archipelago country in Asia with more than 400 Blood Centres and has been classified into medium to high endemic of hepatitis B country. The safety of blood is a big challenge due to geographic and infrastructure difficulties and budget limitation for blood services.

To describe the evolution of safety of blood towards transfusion transmitted infections (TTIs) in Indonesia.

Analyzing the data of blood screening on TTIs in Indonesia from 2006 to 2015 to describe the impact of policy and regulations on the improvement of blood safety in Indonesia. The Government Regulation followed by the Ministry of Health Decrees on Blood Services have been developed since 2009 to improve a quality of blood services in Indonesia. Those regulations directed more stringent donor selection and blood screening towards antigen and or antibody of HIV, hepatitis B virus, syphilis and hepatitis C virus on all donated blood either by Rapid Test or ELISA or Chemiluminescence methods. While blood screening for malaria was only run in the endemic areas using a rapid test. Moreover, to increase the safety of blood, Nucleic Acid Test (NAT) was added in 12 Blood Centres.

There was a significant decrease of HBV seropositivity (from 2.13 to 1.4%) and relatively constant of HCV seropositivity (0.4 to 0.59%) which possibly due to better donor selection. However, the seropositivity of HIV was fluctuate increased from 0.02 to 0.27% as well as syphilis from 0.37 to 0.86%. The increase of HIV seropositivity could be due to the usage of four generation of ELISA or Chemiluminescence methods that were known to be more sensitive.

Implementation of more stringent donor selection and sensitive blood screening methods proved that government regulations improved blood safety in Indonesia. However, the variation of blood screening methods used, need to be reduced to standardize blood safety. Consolidation of blood screening may be one strategy to solve this problem, although some part of Indonesia has difficulty in geography and infrastructure.

4A-S15-02

A PILOT SEROSURVEY OF BABESIA MICROTI IN CHINESE BLOOD DONORS

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Background: *Babesia* spp. are tick-borne, intraerythrocytic protozoan parasites that can cause disease (i.e., babesiosis) in humans. Babesiosis is an emerging transfusion-transmissible infection, usually attributable to infection with *Babesia microti*. B. *microti* has the ability to establish asymptomatic, infection that can persist for months to years, escaping notice at time of blood donation. Transfusion transmitted babesiosis (TTB) poses serious risk for certain patient subsets such as neonates, patients aged >50 years, the asplenic and the immunocompromised (e.g. HIV, cancer and those on immunosuppressant therapy), that are over-represented among transfusion recipients; such is the likely basis for the high all-cause mortality (~19%) associated with TTB. Reports of human babesiosis in the People's Republic of China (PRC), Mongolia,' Korea and Japan, suggest that Babesia is regionally endemic yet is underreported and/or misdiagnosed. To date, no blood donor surveillance for Babesia has been undertaken in PRC. If Babesia is present, there could be implications for the Chinese blood supply as TTB has been shown to incur significant morbidity and mortality in transfusion recipients.

Aims: To determine the rates of *B. microti* seroreactivity in a sample of blood donors in PRC.

Methods: Residual (i.e. unused) blood (8 ml each) from donor samples that were routinely collected as part of the donation process at Mudanjiang Blood Center (MBC) in Heilongjiang province, PRC (19th July 2016-15th August 2016) were shipped to Institute of Blood Transfusion (IBT), Chengdu for testing. The study population represents a convenience sample of community blood donors (n = 1100) that were negative for routinely screened infections (i.e. HIV, HBV, HCV and *T. pallidum*). The data were deidentified following collection. MBC was selected given that it was considered to be a potentially endemic area: *B. microti* and B. *venatorum* have previously been reported in Heilongjiang; there has also been local reporting of Lyme disease and other tick-borne infections. Diagnostic assessment was confined to indirect fluorescent antibody (IFA) testing. The IFA slides were prepared at the Holland Laboratory of the American Red Cross (Rockville, Maryland, USA), and shipped to IBT prior to initiation of the study. The slides were prepared using *B. microti*-infected hamster blood; Samples that were initially positive to a titer of 64 were subjected to repeat IFA testing. Final seroreactivity was based on a repeat reactivity to ≥64.

Results: A total of 1100 individual donor samples were collected for the study. At time of writing 503/1100 (45.7%) have been evaluated, comprising 416 whole blood and 87 platelet donors. Seven out of 503 (1.39%) were seroreactive for *B. microti* (1/87 [1.15%] platelet donors and 6/416 [1.44% whole blood donors), all at titers of 64 (n = 4) and 128 (n = 3). The remaining testing is currently underway.

Summary/Conclusions: *Babesia* spp. are transfusion-transmissible pathogens of global importance that to date have largely been understudied. Our preliminary findings suggest a need for further Babesia surveillance in Chinese blood donors to evaluate the risk to the blood supply. Additional testing using newly developed assays that have been developed for blood donor screening, including molecular evaluation, will enable the detection of additional Babesia species with discrimination between past exposure to the parasite with concomitant clearance, from active infection.

4A-S15-03

TRANSFUSION-TRANSMITTED INFECTIONS IN THE ERA OF NUCLEIC ACID TESTING

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Background: Nucleic acid testing (NAT) of all blood donations for human immunodeficiency virus (HIV), hepatitis C virus (HCV) and hepatitis B virus (HBV) has been performed in Taiwan since February, 2013. The implementation of a comprehensive NAT is expected to reduce transfusion-transmitted viral infection (TTI) and improve blood transfusion safety accordingly.

Aims: We evaluated the use of NAT in determining the number of NAT-reactive donors among those seronegative for anti-HIV, anti-HCV and hepatitis B surface antigen (HBsAg). We estimated the residual risks of HIV-, HCV- and HBV-infectious donations and reported the investigation of suspected cases for the evidence of TTI before and after NAT implementation.

Methods: We performed triplex NAT on blood donations in minipools (pools of 8 donations) using Procleix Ultrio Plus Assay. Additional testing including individual and discriminatory NAT were performed for samples that constituted a reactive pool. We also performed serologic tests of anti-HIV, anti-HCV and HBsAg using Murex immunoassay. We identified a new infection defined as a repeat tested donor with NAT and immunoassay non-reactive at baseline donation and being either discriminatory NAT-reactive or confirmed immunoassay-reactive for subsequent donation. The confirmation of immunoassays was performed using western blotting assay for anti-HIV, anti-HCV, and neutralization test for HBsAg. The window periods used for risk estimation were 9.1 d for HIV-1, 7.4 d for HCV and 30 d for HBV. For investigation of TTI, the implicated donors were followed for the development of a reactive test result and/or analyzed the stored samples of transfused components by using individual NAT. We also examined the similarity of viral gene sequence between the implicated donors and the blood recipients. Sequencing of Env C2V5 region for HIV, C/E2 plus NS5B region for HCV, and Pre-S and S gene for HBV was performed for similarity analysis.

Results: Among 7,001,347 donations during 2013–2016, we identified 6, 52 and 1927 seronegative NAT-reactive donors of HIV, HCV and HBV, respectively. The NAT yield rate was 0.086 (0.02–0.15), 0.74 (0.54–0.94) and 27.52 (26.29–28.75) per 10^5 donations correspondingly. The estimated residual risk per million donations after NAT was 0.64 for HIV, 0.88 for HCV and 39.04 for HBV, respectively. Transfusion-transmitted HIV infection was reduced from 0.77 (0.44–1.11) per million of immunoassay-screened donations (21 TTIs of 27,111,970 donations, 1999–2012) to 0 of 7,001,347 donations after NAT implementation. Prior to NAT, transfusion-transmitted HBV infection was estimated to be around 100–200 per million and transfusion-transmitted HCV infection was in the range of 0–1,352 (178–2,882) per million donations. After NAT implementation (2013–2016), a total of 2 and 10 clinical cases have been submitted for HBV and HCV TTI investigations. None of these cases were confirmed as attributed to transfusion transmission based on the evidence that the stored sample of the transfused components as well as the follow-up samples of the implicated donors were negative for the pathogen.

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Summary/Conclusions: Universal NAT implementation unquestionably improve blood safety in Taiwan. NAT has contributed the striking reduction of TTIs, especially for HBV and HCV.

4A-S15-04

RECOMBINANT EXPRESSION OF G5P AND ANALYSIS OF IT IN THE DETECTION OF SCRAPIE PRION PROTEIN

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Background: Human prion diseases are infectious and invariably fatal neurodegenerative diseases, including sporadic Creutzfeldt-Jakob disease (sCJD), and variant CJD (vCJD). vCJD has already been transmitted from human to human by blood transfusion. Development of a biochemical assay for the detection of prions (PrPSc) in the blood of patients affected by prion disease is a top medical priority to increase the safety of the blood supply. The gene 5 protein (g5p) of the bacteriophage Pf1 is a 144 residue single-stranded (ss) DNA binding protein involved in replication and packaging of the viral DNA. As previous studies have demonstrated that in the transforming of normal prions (PrP) to abnormal prions (PrPSc), some small agent like nucleic acids may participate in forming change, g5p may has ability to bind with the prion-nucleic acids complex, which have be reported in some researches. g5p may be used as a novel tool for the study, detection, and treatment of prion diseases, which should be investigated.

Aims: To establish a method to gene synthesis and recombinant expression of g5p and investigate it's ability in the detection of PrPSc.

Methods: Basing the gene information of g5p offered in the GenBank, g5p gene was constructed by Overlap Extension PCR and cloned into pET-28a-c(+) plasmid and transformed into E. coli. Expression of g5p was induced with IPTG and purified by Ni^{+}-NTA affinity chromatography. The purified g5p (100 μg) was conjugated to 7×10^{8} tosyl activated magnetic beads in 1 ml of PBS at 37 centigrade for 20 h. The g5p-conjugated beads were incubated with 0.1% bovine serum albumin (BSA) in PBS to block non-specific binding. The specific capture of scrapie prion protein by g5p was performed by incubating brain homogenate from CJD or non-CJD and g5p-conjugated beads (10 μg g5p/6×10^{7} beads) in 1 ml binding buffer (3% Tween-20, 3% NP-40 in PBS, pH7.5). After incubation with constant rotation overnight at room temperature, the PrP-containing g5p-beads were collected with magnetic force and unbound molecules in the solution were removed. Following three rinses in the wash buffer (2% Tween-20 and 2% Nonidet P-40 in PBS, pH 7.5), the g5p beads were resuspended in SDS sample buffer (3% SDS, 2 m M EDTA, 10% glycerol, 50 m M Tris, pH 6.8) and was detected by Western blot analysis with 3F4 as primary antibody for probing the PrP molecule.

Results: g5p gene was successfully synthesized and highly expressed in E. coli BL21 (DE3) under the optimal induction of IPTG at 0.8 mmol/l for 6 h. g5p precipitated PrP from CJD brains, but not from normal control brains, showing it's specific capture ability of scrapie prion protein. Moreover, the amount of the PrP captured by g5p decreased in a concentration-dependent manner after the addition of salmon DNA, further arguing that the capture of PrP by g5p is mediated by, or can be competed with nucleic acid. To gain insight into the PrP bound to the g5p, we treated the captured complex with PK and found that PrP bound to the g5p were PK-resistant further supporting it's specific capture ability of scrapie prion protein but normal prion protein.

Summary/Conclusions: We successfully established a method to gene synthesis and recombinant expression of g5p, which can capture PrPSc with a high level of specificity and affinity while not recognizing PrPC at all. Thus it may has potential value for the blood safeguarding of prion diseases, it may improve the sensitivity of protein misfolding cycle amplification for blood detection, or develop a novel filter to discard the PrPSc in blood.

4A-S15-05

COMPARING ASSAY PERFORMANCE OF ELISA AND ELECTRO-CHEMILUMINESCENCE IMMUNOASSAY(ECLIA) IN DETECTING TREPONEMA PALLIDUM SPECIFIC ANTIBODIES

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Background: Syphilis is a major concern to transfusion safety with increasing incidence in several countries including China. Therefore, it is very important to search for a screening test with sufficient sensitivity and specificity in syphilis diagnosis. ELISA is a traditional assay used by blood banks in China to screen serum for the presence of Treponema pallidum (TP) specific antibodies. In recent years, electrochemiluminescence immunoassays (ECLIA) have been used for the detection of serum total antibodies to TP; However, research on the consistency of the performance for TP detection by ELISA and ECLIA are limited.

Aims: To evaluate the performance of ECLIA for TP specific antibodies detection and its comparison to ELISA.

Methods: A total of 1337 serum samples from 16 blood bank centers of China were included. 670 anti-TP reactive samples were served as positive control and 667 samples with elevated ALT levels and non-reactive to anti-TP were considered as negative control. All of these samples were tested with one ECLIA test and eight commercially available ELISA tests (1 to 8). Samples showing consistent results among all of these assays would be considered as either positive or negative. Discrepant results were further confirmed by Western blot (WB) or TP particle agglutination assay (TPPA) and any of these two tests with a positive result, would be considered as positive.

Results: Excluding 14 indeterminate results, in the 1323 samples, 272 plasma samples had discrepant results between ECLIA and ELISA assays. The TPPA and WB analysis showed that in the discrepant samples the consistency rate of ECLIA and confirmatory test was significantly higher than that of any ELISA test and confirmatory test (ECLIA:76.84% vs 8 ELISAs: range 19.4%~33.33%, $P < 0.01$). Consistency analysis showed that Kappa coefficient between ECLIA and confirmatory test (0.917) was higher than that between any ELISA test and confirmatory test (range 0.767~0.891). Further analysis showed that the false-negative rate of ECLIA was similar to any ELISA assay ($P > 0.05$), but the false-positive rate of ECLIA (4.72%) was lower than that of most ELISAs (6 ELISAs: range 11.66%~ 6.99%, $P < 0.05$) and similar to the other two ELISAs (ELISA 3: 5.01%, $P > 0.05$; ELISA 5: 4.98%, $P > 0.05$). The area under the receiver operating characteristic curve (AUC) of ECLIA was significantly higher than that of 6 ELISA assays (ECLIA versus ELISA 1: 0.963 vs 0.940, $P < 0.05$; ELISA 2: 0.963 vs 0.954, $P < 0.05$; ELISA 3: 0.963 vs 0.939, $P < 0.05$; ELISA 5: 0.963 vs 0.932, $P < 0.05$; ELISA 6: 0.963 vs 0.949, $P < 0.05$; ELISA 7: 0.963 vs 0.940, $P < 0.05$), indicating a better detection efficiency for ECLIA. Although the sensitivity of ECLIA (97.4%) had no significant difference against most of the ELISA tests (ELISA 1: 98.4%, $P = 0.130$; ELISA 2: 97.8%, $P = 0.06$; ELISA-3: 98.6%, $P = 0.114$; ELISA 5: 98.0%, $P = 0.603$; ELISA 8: 97.7%, $P = 0.752$), the specificity of ECLIA (95.5%) was significantly higher than the ELISA tests (ELISA 1: 89.7%, $P < 0.05$; ELISA 2: 93.3%, $P < 0.05$; ELISA 3: 89.2%, $P < 0.05$; ELISA 5: 88.3%, $P < 0.05$; ELISA 8: 93.1%, $P < 0.05$). ECLIA also showed higher sensitivity than ELISA-6 (95.3%, $P = 0.034$) and ELISA-7 (93.7%, $P = 0.001$), but similar specificity (94.5%, $P = 0.439$ and 93.5%, $P = 0.050$).

Summary/Conclusions: Compared with ELISA, ECLIA is easy to use, automated with high throughput and more specific to detect antibodies to TP. In the future, ECLIA may be an alternative test with high sensitivity, and higher specificity to ELISA as a screening test for syphilis diagnosis in China.

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Academy Session – The Transfusion Team: the role of the Transfusion Practitioner

4A-S16-01

THE ROLE OF THE TRANSFUSION PRACTITIONER IN THE MULTIDISCIPLINARY TEAM

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Transfusion and patient blood management (PBM) processes are complex; while transfusions can be lifesaving, equally there may be associated morbidity and mortality. The safety and PBM culture in Australia is propelled by obligations to comply with mandatory governance frameworks. These frameworks support appropriate use of blood/blood products, and that there are adequate levels of safety at all points on the vein to vein journey. A key to the success of blood management governance is the health service multidisciplinary Blood Management team/Governance Committee (BMC). Their structure depends on the size of the health service. Importantly, representation includes healthcare executive, clinical governance, and consumer/s, with the transfusion practitioner (TP) as a key resource. Other members include staff (medical/laboratory/nursing) from areas that regularly undertake transfusion and PBM activities, such as: perioperative suites, emergency departments, clinical wards, infusion centres, general practices, and laboratories.

Engagement with relevant healthcare stakeholders is important. The TPs skillset focuses on 'staff and patient education, adverse events, transfusion governance and monitoring of transfusion practices within organisations. . .to ensure current clinical practices align with state, national, and international guidelines and standards.' (Miller 2012) Benchmarking through key performance monitoring, and sharing of ideas at both a local and international level allows for system improvements, and ensures efficiency and safety is maintained. The TP, working in a multidisciplinary capacity across the clinical spectrum at all levels and specialties, is often seen as the driving force for change within the healthcare system in areas affecting blood transfusion and PBM.

In Australia, the TPs influence can be within an individual organization, across multiple sites, or health networks, in metropolitan or rural/regional areas. The TP conducts a critical role pulling together resources, promoting exchange of information, encouraging engagement and empowering colleagues to facilitate change. Highly developed communication skills assist the TP to engage the many different stakeholders and clinical environments.

Successful PBM requires a coordinated approach to care across many specialities. Each clinical discipline has a role to play in the assessment and management of: pre-operative anaemia, bleeding and thrombosis risk, and tolerance of anaemia. The TP is often the key link between these clinicians, the patient, and the planning process.

For any multidisciplinary team to function effectively, collaboration is essential. Highly functional teams recognize knowledge and experience, and utilize each member's skills to work together to deliver the best possible outcome for patients. Examples of working together include the development of protocols, education, auditing and review of compliance and patient outcome. TPs often undertake haemovigilance activities, and work together to follow-up reactions, and act on any recommendations with support of the BMC.

There is growing body of literature available supporting multidisciplinary teams, and the TP role, in the implementation of PBM, and reducing unnecessary transfusions, and thus improving patient outcomes.

Conclusion: Effective transfusion and PBM practice requires a systematic cross-specialty approach to ensure success. The TP are the essential link in the multidisciplinary chain. However, they require strong support and leadership to potently effect change and enhance practice.

4A-S16-02

WORKING TOGETHER ON TRANSFUSION AUDITS TO IMPROVE CLINICAL PRACTICE

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Background: For many years, great efforts have been put in place to ensure quality and safety of blood and blood products during collection and processing. In recent years, the focus has shifted to enhancement of clinical transfusion processes. Conducting audits is one effective mechanism of evaluating the ongoing blood administration process against set standards. When conducted jointly with the transfusion nurse, these audits can identify gaps in the transfusion process from the clinical perspective.

Methods: Random transfusion audits were conducted jointly by both Blood Bank technical staff and the transfusion nurse in Paediatric and Obstetric wards using a checklist. This checklist consisted of essential procedures and checks during blood transfusion.

Results: Many good transfusion practices were observed. In the majority of cases, there was positive identification of patients before transfusion by doctor and nurse. However, there were still areas for improvement. The transfusion set up in the Obstetric ward was not consistent with the Paediatric ward. There was non-compliance of donning of personal protective equipment by nurses and ensuring signed consent before transfusion. All findings were reviewed and interventions were undertaken. The checklist was improvised for future audit use and the institution's clinical transfusion policy was revised for better clarity.

Conclusion: The initiatives implemented after reviewing all the findings from the transfusion audits improves practice, thereby enhancing patients' safety.

Donation and Processing – Abstract Session

4A-S17-01

CELL-FREE NUCLEIC ACIDS IN BLOOD PRODUCTS MODULATE GENES INVOLVED IN INFLAMMATION

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Background: Cell-free nucleic acids (CFNA) are present in human plasma. We recently described their presence in manufactured blood products eligible for transfusion but little is known about their biological activity on human cells. Previous experiments performed with peripheral mononuclear cells suggested that CFNA could modulate genes related to innate immunity.

Aims: The aim of this study is to investigate if macrophages, the key initiators of innate immunity and chemoattraction of leukocytes during the inflammatory reaction, are also sensitive to CFNA.

Methods: Macrophages were produced through *in vitro* exposure of monocytes to GM-CSF. They were then exposed to CFNA extracted from RBC, FFP and platelet concentrates (PC), respectively. A gene expression array analysis was then performed. Three distinct macrophages cultures were basically tested individually to include biological replicates.

Results: 976 genes were found to be significantly regulated (up regulation and down regulation) after CFNA exposure in all tested products (FFP and RBC and PC). 131 genes were regulated after exposure to RBC-CFNA and FFP-CFNA. 41 were shared between RBC-CFNA and PC-CFNA. 118 genes were shared between PC-CFNA and FFP-CFNA. Finally, 81, 166 and 120 regulations were seen only with one product. Together, these observations indicate that several genes are significantly regulated after CFNA exposure, with a large common part between the three blood products. Genes involved in innate immunity are notably present. The Three main represented functional families are (i) genes related to immune response and inflammation (ii) apoptosis (iii) angiogenesis. IL-8, a potent inducer of inflammation, is notably up regulated as well as VEGF which is also known to harbor proinflammatory properties.

Summary/Conclusions: CFNA present in blood products have a biological effect *in vitro* on macrophages, with regulations of genes involved in the innate immunity and inflammatory response. Free nucleic acids are then identified as components of blood products possibly interfering with the recipient's immune system. Thus, free nucleic acid can be considered as new potential candidate participating to some transfusion side effects.

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4A-S17-02

THE MECHANISM OF ENZYMOLYSIS OF BLOOD GROUP A ANTIGEN WITH A-N-ACETYLGALACTOSAMINIDASE FROM ELIZABETHKINGIA MENINGOSEPTICUM

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Background: Alpha-N-acetylgalactosaminidase (NAGA) from *Elizabethkingia meningosepticum* is a hydrolase which can convert both group A1 RBCs (containing A1 antigen) and A2 RBCs (containing A antigen) to O RBCs. How does NAGA cleave both A antigen and complex A1 antigen, or NAGA is an exoglycosidase or a novel glycoside hydrolase having both endo and exo-glycosidase activity, which is a key scientific problem to be solved. In addition, the another scientific question to be answered is the safety of clinical application of A-ECO RBCs.
Aims: To reveal the mechanism of enzymolysis blood group A antigen with a-N-acetylgalactosaminidase from Elizabethkingia meningosepticum and evaluate the function and safety of these A-ECO RBC *in vitro*.
Methods: The residual carbohydrate in A-ECO RBCs solution was purified by centrifugation, ethanol precipitation and Sevag method, and the carbohydrate type was determined by TLC, NMR, electrospray ionization mass spectrometry, infrared Spectroscopy and high performance gel permeation chromatography (HPGPC) analysis. The A-ECO RBCs were also typed by traditional typing in test tubes, gel column agglutination technology and fluorescence-activated cell sorting (FACS) analysis. The morphology of native and ECO RBC was observed by scanning electron microscopy. The function of A-ECO RBCs were evaluated by physiological and metabolic parameters, including osmotic fragility, erythrocyte deformation index, levels of 2,3-diphosphoglycerate, ATP, methemoglobin, free Na+, and free K + . Manual cross-matching was applied to ensure blood compatibility.
Results: The RBC agglutination tests and FACS results showed that A RBCs were efficiently converted to O RBCs. The products of the oligosaccharides obtained from the enzymatic hydrolysis of A1 RBCs were confirmed as monosaccharide N-acetylgalactosamine. The FCAS results showed that the surface of A ECO RBCs contains a large number of type 3H antigen. These data demonstrate that NAGA is an exoglycosidase. Functional analysis suggested that the conversion process had little impact on the physiological and metabolic parameters of the RBC. About 50% of group O and 35% of group B sera reacted with the A-ECO RBCs in a sensitive gel column cross-matching test.
Summary/Conclusions: This study revealed the mechanism of enzymolysis blood group A antigen with a-N-acetylgalactosaminidase, that is type 3 chain H antigens were generated. Cross matching test confirmed that the A-ECO RBCs still reacted with some group O and B sera, which may partly reflect the complexity of group A1 antigens. More research on the safety of A-ECO RBCs is necessary before the application of these RBCs in clinical transfusion.

4A-S17-03

APHERESIS EDUCATION EXPERIENCE

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Background: Workers need to be qualified and competent for their job. Scholing to perform specific jobs is needed.
Aims: Evaluation of the results of an international course for apheresis nurses/operators and physicians.
Methods: Learning objectives for anatomy, adverse events, apheresis basics, calculations, cell collections for cellular therapy, donor apheresis, hematology, nursing, pediatrics, physiology, quality and therapeutic apheresis for a course for apheresis nurses were described by a group of experienced apheresis nurses and physicians (Joint Task Force for Apheresis Education and Certification; JTFAEC). Based on these learning objectives, the authors built a course with 17 different lessons. Also tests with at least 100 different and independent questions of the types TRUE/FALSE, multiple choice and open questions were made. Tests are made prior and after the course. In co-operation with colleague authors and editors, ASFAs 5 and 6th edition of "Principles of Apheresis Technology" was written to cover the learning objectives, and can therefore applied as manual for the JTFAEC course.
Results: Since Dec 2013, the course was given 19 times in 8 countries to 278 nurses (from Austria, Belgium, Cambodia, Hong Kong, Indonesia, Italy, Netherlands, Norway, Singapore, South Africa, Sweden, Switzerland, UK and USA) and 102 physicians (from Belgium, Indonesia, Netherlands, Norway, South Africa, Sweden, and UK). Four trainers from the South African National Blood Service and 10 trainers in Hong Kong were trained as train-the-trainer. The newly trained trainers gave, supported by the 2 authors, successful courses. In the post-course tests, 99.3% of the apheresis nurses and 100% of the physicians showed a significant increase of knowledge in apheresis compared to the results in the pre-course tests. 99.3% of participating apheresis nurses and operators could be certified according to the JTFAEC requirements.
Summary/Conclusions: This program provides an approach to educate and certificate apheresis nurses, which can be of help in audits. It is also shown that this course can be of value for physicians increasing their knowledge regarding apheresis. Up to now, all courses were given in English and Dutch. The course is being translated into German. By giving this course in a train-the-trainer situation, courses like this can be given in local languages for better understanding for persons for whom English is not the native language. At present, training of apheresis nurses (and physicians) according to the concept of this course is planned in Switzerland (Oct 2017), Netherlands (Feb 2018) and Sweden (Oct 2018).

4A-S17-04

THE EFFECTS OF STORED RED BLOOD CELLS ON DENDRITIC CELLS' MIGRATION ABILITY, FUNCTION AND UNDERLYING MECHANISM

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Background: The interactions between stored red blood cell (stored RBC) and immature dendritic cells (imDCs) involve that how the immune system identifies and clears the stored RBC and how stored RBC affects the function of immune system.
Aims: To study above-mentioned interactions, the inflammation responses initiated by stored RBC, the maturation, migration ability and function of dendritic cells after co-incubation were monitored as indicators to shed light on the specific correlations.
Methods: We used C57BL/6J mice fresh RBC, stored RBC and PBS separately stimulated imDCs derived from mouse bone marrow at 37°C for 10 h with a low dose of LPS *ex vivo*. Co-culture supernatants were collected for batch analysis of cytokines and chemokine secretions by ELISA. DCs were then collected for phenotype analysis by FACS and the *in vivo* distribution and migration dynamics by bioluminescence imaging after adoptive transfusion, and the proliferation and activation of antigen-specific CD8 + T cells elicited by DCs were also detected.
Results: The results showed that the expressions of costimulatory molecules CD40, CD80 and CD86 and chemokine receptors CCR7 on imDC treated by stored RBCs were significantly higher than those of fresh RBC. ImDCs treated by stored RBCs produced higher levels of pro-inflammatory cytokines IL-6, TNF-α, IL-12p70, IL-1β and chemokine MCP-1 than the fresh RBC pretreated group, as well as lower levels anti-inflammatory cytokines IL-10 expressions. After being injected subcutaneous or intravenous injection, the homing percentage and speed of DCs pretreated by stored RBCs to LNs were higher than the fresh RBC group. On the contrary, the proliferation, activation and IFNγ secretion of antigen-specific CD8 + T cells in the stored RBC group was lower than the fresh RBC group.
Summary/Conclusions: These results demonstrated that although stored RBC promoted DC maturation, migration and homing, they did reduce the imDC's ability to cross-present antigen and activate antigen-specific CTL. Further investigations suggested that the down-regulated expression of CD47 on stored RBC induced phagocytosis through interaction with the inhibitory immunoreceptor SIRPα expressed on imDC compared with fresh RBC. Stored RBC also enhanced the production of ROS in the co-incubation system and proinflammatory cytokines secreted by imDCs, which promoted DCs' maturation. The Rho/ROCK, PI3K/Akt and NF-κB signaling pathway were involved in stored RBC-induced dendritic cell cytoskeleton rearrangement, enhanced DCs motility ability, promoted the ability of DCs' migration and homing to lymphoid tissue.

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4A-S17-05

PLATELETS INHIBIT THE GROWTH OF *STAPHYLOCOCCUS AUREUS* BY DAMAGING BACTERIA STRUCTURE

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Background: Platelets are classically used in the clinic to maintain hemostasis. Recent evidence has supported the important roles of platelets in host inflammatory and immune responses, and platelet-rich plasma has been shown to inhibit the growth of bacteria *in vitro* and *in vivo*. However, very few studies have shown platelets can inhibit bacterial growth directly, and the related mechanisms have not been elucidated.
Aims: Use washed platelets co-culturing with *Staphylococcus aureus* to find out whether platelets could display antibacterial effect directly.
Methods: Platelets from human peripheral blood were washed and purified, and then the washed platelets were co-cultured with *Staphylococcus aureus in vitro*, and the bacterial growth was evaluated. The transmission electron microscopy (TEM) was used to probe *Staphylococcus aureus* ultrastructure after cocultured with or without platelets. *In vivo* study, fresh separated murine platelets were transfused to the *Staphylococcus aureus* infected mice to explore the antibacterial defense effect, while a platelet depletion murine model was administrated with *Staphylococcus aureus* infection directly. Histological section analysis and blood routine examination were used to evaluate the infection severity of mice.
Results: Platelets significantly inhibited the growth of *Staphylococcus aureus* when co-cultured *in vitro*, and the ultrastructure of *Staphylococcus aureus* was seriously damaged after co-cultured with platelets. And these antibacterial features were completely different from plasma. The murine *Staphylococcus aureus* infection model showed that depletion of platelets exacerbated the severity of infection, whereas transfusion of platelets alleviated infection.
Summary/Conclusions: Purified platelets inhibited the growth of *Staphylococcus aureus* directly and that platelets played an important role on antibacterial defense.

Donor Management – Donor

4B-S18-01

MANAGEMENT OF BLOOD DONATION IN AN EMERGENCY SITUATION

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Background: The NRCS BTS was established in 1966 and assists responsibility to provide countrywide blood and blood products for the people of Nepal. The NRCS BTS consists of 105 blood collection centres in 72 districts across Nepal. The NRCS is the major provider of safe and secure blood and blood products for the people of Nepal, and it does so supporting a policy of 100% of voluntary non-remunerated blood donation.
Aims: The demand for blood and blood components has steadily risen each year since establishment. Last year 2016 Nationwide 236,799 units were collected and 337,321 units of blood and blood components were supplied. The Central Blood Transfusion Service, which is located in Kathmandu coordinates the national NRCS blood program, and provides management, recruitment of blood donors and technical oversight to all NRCS blood centres. It also provides a large proportion of the total national blood collection, processing and distribution activity.
Methods: On 25th April 2015 a devastating earthquake (7.8 magnitudes) occurred in Nepal, the epicenter being 77 km North West of the capital Kathmandu. The impact was felt in 39 of 75 districts of the country. A series of aftershocks occurred causing further damage and intensified pre-existing vulnerabilities. As a result, 57 districts out of 75 have been affected finally. The strongest aftershock, measuring 7.3 magnitudes, struck on 12 May 2015. More than 8,400 people lost their lives and nearly 20,000 were injured.
In addition to loss of lives, the two quakes have caused extensive destruction and damage to infrastructure, housing and livelihoods. It led a drastic reduction in living conditions, income and access to basic services. More than 1.7 million families (approximately 8.5 million people) have been affected. Up to 600,000 houses were damaged, about 2.8 million people displaced.
Results: The earthquake and aftershocks have intensified pre-existing vulnerabilities and there were urgency of needs and the challenges for relief and recovery operations, casualties with need of blood across the widespread affected area. From very next day 26th April 2015 many volunteer blood donors and organization donor group came to BTS and donated blood. Many organizations and individuals were willing to support blood service coming to blood center and even with telephonic communication. Continue communication from all NRCS District Chapters and Blood centres to support with blood units. All over the country Blood Service started collecting blood. Many organizations requested for blood donation but as blood stock were managed in the centres and all were listed in cue. All Blood centres collected day to day blood from volunteer blood donors and all stakeholders. There were sufficient blood stock but demand was less from all Hospitals. All Blood Centers provided free supply of blood and blood products to all earth quake victims for almost I month.
Summary/Conclusions: Blood was well managed during the disaster time as many organizations, individual blood donors and motivators were willingly supporting. Many of blood donors and organizers were listed in cue with request to donate blood whenever in need as could not let them to donate as there were excess of blood stock in all blood centres.

4B-S18-02

A LINKAGE ANALYSIS BETWEEN SOCIAL ACTIVITIES AND VOLUNTARY BLOOD DONATION IN SHENZHEN

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Background: Shenzhen is the first city in China to start voluntary blood donation, and a city without planned blood donation, mutual blood donation, and blood management office. Voluntary blood donation was started since 1993 and it becomes the only way to donate blood in Shenzhen. From 1998, We fully met the clinical blood use for 19 consecutive years. Street voluntary blood donation is the main mode whereas collective blood donation is auxiliary. However, donation on street is strongly affected by weather conditions, especially in winter and summer. The insufficiency of blood supply occurs occasionally from time to time.
Aims: Shenzhen is a city with a history of 38 years and currently it becomes a super large city with a population of more than 21 million. Due to uncertainty in clinical blood use, a serious strain of partial blood type is occurred infrequently. In addition, the clinical use of blood in Shenzhen is increasing with an average 10% growth annually. Could we fulfill the growing clinical demand if we only keep on the strategy to rely on street voluntary blood donation?
Methods: Our strategy is by two ways: 1) To build more blood donation sites on streets; 2) to increase the intensity of the collective blood donation. We took the long-term activities of collective blood donation with "the red action" in winter whiles "blood month for angels in white" and "fueling life" in summer.
Results: After several years of practice, we have successfully overcome the blood insufficiency supply. In the past 5 years, top organizations with number of collective blood donors includes social charity organizations, universities, enterprises, civil servants, hospitals and so on. The ratio of collective voluntary blood donation increased from 11.6% in 2012 to 20% in 2016. These organizations believed that: voluntary blood donation is not only the responsibility for local blood bank, but also for whole society. A significant change in the mode of propaganda and recruitment of voluntary blood donation occurs in Shenzhen. The social organizations, such as the Shenzhen lions Association, have been made firm contribution on propaganda and recruitment of voluntary blood donation. The participation of social organizations in different fields fully reflected the public welfare, sociality and universality of voluntary blood donation work.
Summary/Conclusions: In this analysis of our achievements, we can conclude the following reasons: 1) A substantial support by Shenzhen municipal government in the legislative, financial and other aspects. 2) We precede the blood donation insisting in two principles "voluntary and free". In propaganda, we pay more attention to the humanitarian spirit of blood donors, humanity and the traditional morality. 3) Setting good examples by blood center employees. Employees in our blood center from the leadership to the general staff give regular blood donation. There are more than 50 peoples in our blood center who have won the nationwide voluntary blood donation Awards. 4) The harmonious environment in Shenzhen promotes the attention of all walks of life and actively participates in voluntary blood donation.
In summary, we have proved that our long-term adherence to the principle of "voluntary and free" is correct. Blood donation without material stimulation promoted the public image, and improved the sense of happiness and pride. More organizations join the teams for blood donation. It also promoted the progress of city's civilization, and to ensure the sustainable development of voluntary blood donation in the future.

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4B-S18-03

EFFECT OF REGULAR PLATELETPHERESIS ON DONORS' CALCIUM LEVEL AND BONE DENSITY IN NATIONAL BLOOD CENTRE, KUALA LUMPUR

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Background: Plateletpheresis is a process of selectively removing platelet from whole blood and returning the remainder to the donors. Citrate is used as the anti-coagulant to ensure the blood fluidity throughout the procedure. However, citrate chelates calcium and causes hypocalcaemia. Calcium is essential in bone remodelling. Chronic effect of frequent exposure to citrate anti-coagulant and hypocalcaemia state to donors' bone density is not known.

Aims: The aim of this study is to investigate the effect of regular plateletpheresis on donors' calcium level and their bone density status. The pre- and post- donation calcium levels were compared in regular plateletpheresis donors and compared between the groups of regular donors based on the frequency of donation. The effect of regular plateletpheresis on donors' bone density status were determined and compared between the groups of regular donors based on the frequency of donation.

Methods: Fifty plateletpheresis donors in National Blood Centre, Kuala Lumpur were selected to participate in this study from 15th January till 31st March 2016. Those donors were divided into two groups based on their number of plateletpheresis donations; donated less than 20 and more than 50 times. Each donors' bone density status was measured through dual emission X-ray absorptiometry (DEXA) scan, followed by pre-sampling of liver function test (LFT), calcium and magnesium levels prior to plateletpheresis. A post-procedural blood sampling was also taken to measure for calcium and magnesium levels.

Results: This study had involved 50 patients with the median age of 35.0 years and 45.2 years respectively for plateletpheresis less than 20 and plateletpheresis more than 50. Majority of participants for both arms (96.0%) were male. Based on the race, most of the patients were Malays for both groups. There were improvements of the median corrected calcium and magnesium levels for plateletpheresis less than 20 group as compared to median for plateletpheresis more than 50 group. The level of albumin was higher for plateletpheresis less than 20 group with 96.0% of the cases had a normal DEXA scan findings. There was a significant difference in age ($P = 0.006$) between plateletpheresis less than 20 and plateletpheresis more than 50 cases. There were significant differences ($P < 0.05$) of magnesium level for both group at baseline and after plateletpheresis procedure. However, there were no significant differences based on gender ($P = 1.000$), race ($P = 0.413$), the corrected calcium for both groups at baseline and after plateletpheresis procedure. There were no significant differences of albumin level ($P = 0.167$) and DEXA scan ($P = 1.000$) between plateletpheresis less than 20 and plateletpheresis more than 50.

Summary/Conclusions: This study showed that regular plateletpheresis had no effect on the calcium level and bone density among donors in both arms of the study. However, donors' magnesium levels in both groups were affected. Addition of magnesium supplement to regular plateletpheresis donors on the other hand, need further investigations to ascertain that it may be of benefit to our regular plateletpheresis donors.

4B-S18-04

ANALYSIS OF THE IRON STATUS OF MALE BLOOD DONORS AT FIRST-TIME AND ONE- TO FIVE-YEAR MAXIMAL (MAX.) DONATIONS IN TAIWAN

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Background: In Taiwan, up to 70% are repeated donors. Regular blood donors are at risk of iron deficiency but characteristics that predispose to this condition are poorly defined.

For detecting iron status, we used tests including ferritin, serum iron, and total iron-binding capacity (TIBC).

Aims: Present study has determined the iron profile of regular voluntary blood donors to evaluate their iron status. The objective of this study is to investigate the iron status of first-time and regular male blood donors.

Methods: In Taiwan, the majority (59.1%) of blood donors are men; therefore, only men were included in our analysis. The whole blood donors selection criteria for men include age between 17 and 65 years old, body weight ≧50 kg, and hemoglobin ≧13.0 g/dl. Furthermore, the maximum amount of the whole blood donation (max. donation) among male donors is restricted/limited to 1,500 milliliters per annum. Pre-donation hemoglobin assessment was done by copper sulfate density method; serum ferritin, serum iron, and TIBC were tested by indirect ELISA (enzyme-linked immunosorbent assay). Transferrin saturation (%) was calculated as serum iron divided by TIBC. Low ferritin and low transferrin saturation were defined as serum ferritin level less than 23.9 ng/ml and transferrin saturations less than 20%, respectively. In this study, there were 111 first-time blood donors and 1,250 one- to five-year blood donors with max. donation. Two-sample t-test was used to compare group means, including ferritin level and transferrin saturation.

Results: The first-time blood donors had significantly higher ferritin level and transferrin saturation, compared with those of the one- to five-year blood donors with max. donation (ferritin: 217.9 vs 32.6~40.3 ng/ml; transferrin saturation: 36.5 vs 24.8~27.5%; both $P<0.0001$). In first-time blood donors, the prevalence of low ferritin and low transferrin saturation were 0.9% and 4.5%, respectively. One- to five-year blood donors with max. donation had high prevalence of low ferritin (32.9, 42.7, 42.9, 39.3 and 36.5%) and low transferrin saturation (27.2, 34.1, 39.9, 28.6 and 25.9%).

Summary/Conclusions: These findings showed that regular blood donors had lower ferritin and transferrin saturation but remained stationary. So, we think regular blood donors had better understanding of their iron status, and may seek some way to keep their health to keep donation. Early recognition and reversal of excessive iron loss may avoid symptomatic iron store depletion in blood donors and reduce volunteer loss due to iron deficiency (ID) anemia.

4B-S18-05

THE EFFECT OF PSYCHOLOGICAL CARE ON ANXIETY OF BLOOD DONORS AMONG COLLEGE STUDENTS

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Background: Nowadays, psychological care plays an important role in our society. College students are the most major blood donation population. But still many of them are easy to be anxious or have adverse reaction at the time of or after blood donation because of the deficiency of medical knowledge. According to the research before, mental stress is the main cause of the adverse reaction. How to alleviate the anxiety of the blood donors among college students has become an important problem.

Aims: To investigate the effect of psychological care on anxiety of medical students and non-medical students, to explore the most appropriate mode of promotion and education for voluntary blood donation, to enhance the proportion of fixed blood donors from college blood donors, and to ensure blood supply to meet clinical use demand.

Methods: 200 students from a comprehensive university in Wuhan city, aged between 18 and 22, who volunteer to donate blood were randomly divided into observation group and control group, 100 students each group. In observation group, there are 58 medical students and 42 non-medical students. In control group, there are 52 medical major students and 48 non-medical major students. Control group was given routine blood donation nursing and observation group was given psychological care. Before and after the treatments on control group and observation group, the scores of Self-rating Anxiety Scale (SAS) were collected and the cases of adverse reaction were recorded respectively.

Results: In observation group, the SAS scores collected from medical students and non-medical students were significantly decreased after donations ($P < 0.05$), respectively. In control group, the SAS scores had no obvious change before and after donations ($P > 0.05$), whether of medical students and non-medical students. The incidence of adverse reaction of control group is higher than that of observation group ($P < 0.05$). The incidence of adverse reaction of non-medical major volunteers is higher than that of medical major volunteers in observation group and control group, respectively ($P < 0.05$). The incidence of adverse reaction of medical students in control group is higher than that of medical students in observation group ($P > 0.05$).

Summary/Conclusions: Psychological care can effectively alleviate the anxiety emotion of volunteers from universities, especially non-medical major students, so that most of the adverse reactions can be preventable. During donations, different methods of nursing including conventional care and psychological care should be chosen according to the actual situation. Timely, effective and holistic psychological care deserved the promotion to alleviate the anxiety emotion and the incidence of reverse reaction of volunteers. It is necessary to establish a scientific and effective mode of promotion, education and psychological counseling for voluntary blood donation.

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Academy Session: The Role of Molecular Testing in the Transfusion Laboratory

4B-S19-02

BASIC ASPECTS, APPLICATIONS AND PLATFORMS CURRENTLY AVAILABLE FOR MOLECULAR BLOOD GROUPING OF DONOR AND PATIENT POPULATIONS

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The characterization of the molecular basis for the myriad alleles within the currently recognized 36 blood group systems have allowed for the widespread use of molecular based red blood cell (RBC) grouping. The majority of allelic variants among the blood groups are attributable to single nucleotide polymorphisms, which are immensely amenable to routine genotyping methods such as PCR-RFLP, SSP, SSO and fluorescent-based real-time PCR assays. These techniques are feasible in a low to medium throughput setting. Multiplexing and use of solid surfaces or beads for probe capture have further improved the throughput and turnaround time allowing multiple alleles to be interrogated simultaneously. Apart from numerous reports of in-house developed assays, several medium throughput commercial assays based on chip or bead arrays, for the routine transfusion laboratory are currently available. Novel approaches, as in the use of MALDI-TOF mass spectrometry appears as an attractive option for accurate, cost-effective, high throughput SNP genotyping. The drawback of many of the techniques currently in use is that new novel variants may be missed. In addition, direct genotype-phenotype correlation may be misleading in some blood groups. Genotyping is certainly effective in predicting the expressed antigen phenotype where the gene directly contributes to the expression of the antigen. However, in certain cases, antigen expression may be altered by promoter mutations, modifying genes and interacting proteins. The presence of homologous genes and genetic events such as crossover, recombination and inactivating mutations further complicates the matter. If genotyping assays are not developed to take these into account, one may erroneously assign the blood group. Massively parallel targeted- sequencing is likely the next step in the evolution of RBC genotyping as it would be able to circumvent many of the issues raised earlier. Parallel developments in bioinformatics and rapid large-data computing and processing could eventually move this technology from the research bench to routine workflow.

In the hospital setting, genotyping is unlikely to supplant serological typing but would rather be complementary to it. Some scenarios where it would play an important role would be in the transfusion management of transfusion dependant patients such as in sickle cell anaemia or in situations where RBC phenotyping cannot be undertaken as in auto-immune haemolytic anaemia or where anti-sera is not readily available or is of weak potency. Where molecular typing is likely to show its might is in wide-scale donor genotyping to establish pools of fully typed donors as well as identifying donors with rare blood groups. Such an exercise will certainly change the landscape of how we will manage transfusion of patients, and will facilitate prompt delivery of best-matched blood for patients, and in future individualized transfusion based on the patient's genomic data.

4B-S19-03

PRACTICALITIES AND PITFALLS IN THE MOLECULAR LAB

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Molecular immunohematological techniques has been applied in clinical laboratories and blood centers. The application in the transfusion medicine mainly focus on the identification of blood group antigens in the complex cases in China, such as having a strong positive direct antiglobulin test (DAT) or after a recent incompatible blood transfusion for the recipients, ABO subgroups, D variants or having a need to typing the Asian specific and common antigens (Miltenberger series antigens and Di^a) but lacking of the corresponding commercial antibodies. For the identification of the complicated antibody in the transfusion recipients, the comprehensive red cell genotyping for the common antigens could provide a phenotype profile of the recipient to exclude the alloantibodies that could not produced and provide clues for the antibody identification and could also reveal the antigen variants to confirm the antibody is an alloantibody rather than an autoantibody. Besides, red cell genotyping has well implemented in the obstetric medicine, in particular for fetal *RHD* genotyping. In the past decades, cell-free fetal DNA (cffDNA) in the maternal plasma replaced the cell pellet of amniotic fluid to be used for routine fetal *RHD* genotyping in D- pregnant women in Europe for the purpose of administrating the antenatal RhIg only when the fetus is *RHD* positive. Fetal *RHD* genotyping for the Chinese pregnant women with a true D− phenotype is of comparatively less importance. Because the lower frequency distribution of hemizygous *D* genotype (6%) was present in the Chinese population resulting in only a 3% probability for a true D− Chinese pregnant women to have a D− baby with a husband having a D+ phenotype compared with 40% of probability for the D− Caucasian women. In comparison, the routine testing for the "Asian type" DEL, which having a frequency of 17–30% in the Chinese, Korea and Japanese with a preliminary D- phenotype, is more necessary in the East Asian population for the administration of antenatal RhIg. An increasing number of evidences indicate that "Asian type" DEL defined by the synonymous mutation (c.1227G>A, p.Lys409Lys) is of a complete DEL phenotype carrying the intact epitopes of D antigen and could not produce alloanti-D immunized by D+ red cells. The molecular methods to detect the *RHD*1227A* is more accurate and applicable for routine testing of "Asian type" DEL for the traditional adsorption/elution testing is not only labor and time-consuming but also could not distinguish the complete DEL and partial DEL.

Plenary Session – TTID

PL4-01

PATHOGEN REDUCTION TECHNOLOGY

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Until recently, transfusion transmitted infection has been controlled through a combination of donor selection and screening and laboratory testing. As more infections emerge, such testing is becoming increasingly complex and burdensome. An alternative approach is the use of pathogen reduction technology, either to replace, or to supplement testing. Methods for treatment of platelets and plasma are widely available. Additionally there is active development of methods for treatment of red blood cells or whole blood. These methods will be discussed, along with their benefits and potential areas of concern. Selected examples of successful implementation and use of the technology will be presented.

PL4-02

IMPACT OF VECTOR-BORNE INFECTIONS IN TTID

No abstract available

PL4-03

HIGH INCIDENCE OF HBV INFECTION REMAINS IN YOUNG CHINESE REPEAT BLOOD DONORS VACCINATED AT BIRTH

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Background: This study was to determine the incidence of HBV infection in the young generation of blood donors born after mandatory implementation of hepatitis B vaccination in 1992.

Methods: Repeat blood donors born between 1992 and 1997 who gave blood twice or more during the past three years were enrolled. Donors were tested for HBV markers of HBsAg, anti-HBc and anti-HBs by immunoassays (EIAs) and viral DNA by nucleic acid tests (NAT).

Results: 14,937 young repeat donors 18–23 years old were tested for HBsAg by EIAs (9/14937; 0.06%). HBV DNA yield was 1:1,494 (10/14,937) in repeat donors using Ultrio NAT (limit of detection [LOD] 10.4 IU/l), while further HBV DNA was detected in 1:192 (9/1732) anti-HBc+ repeat donors using Ultrio Plus NAT (LOD

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3.4 IU/l) (Figure 1). Most cases were characterized as occult HBV infection (OBI).Of 14,937 repeat donors, 20.9% were anti-HBc+ positive, while approximately 50% of 12,024 donors were anti-HBs negative or with levels <100 IU/l. HBsAg+ or OBI strains were classified as wild-type of genotype B or C. Incident HBV infection in repeat donors was approximately 1:18.5 person-years (1.1%/y).
Conclusions: HBV vaccination appears largely protective of incident HBV infection, but undetectable or low anti-HBs level may allow incident new infections. A boost of hepatitis B vaccine for adolescent prior to age 18 and implementation of more sensitive NAT of blood donation are expected to improve HBV safety in blood transfusion.

Management and Organisation Organisational Issues

P-001

Abstract has been withdrawn

P-002

MAINTENANCE MEASURES OF RH (D) NEGATIVE BLOOD TYPE BY THE EMERGENCY BLOOD DONATION TEAM

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Background: In the past ten years, the blood supply of Rh (D) negative blood type red blood cells in Shijiazhuang region showed a rapid growth trend, and blood output increased by 1.18 times, and the number of corresponding blood donors increased by 1.25 times. By the end of 2013, only 2.5% of the blood donors who had been in contact with the center for more than 5 years had been active. This requires us to set up a contingent of rare blood groups that can keep in touch for a long time, and take reasonable maintenance measures to develop and expand.
Aims: To study the maintenance measures and comparative study of Rh (D) negative blood emergency blood donors in Shijiazhuang area, and to study more effective maintenance methods.
Methods: 1) use advanced information technology to communicate and publicize. The use of the central blood donation service platform SMS function, the first time to send care information. Make regular telephone visits to understand the recent health status and life status of donors. Set up a rare family of blood loving micro signals and public platform for donors to build a WeChat exchange of information, learning knowledge, self-help and mutual assistance platform. 2) warm and thoughtful blood donation service. To provide a full range of services at the center of blood donation to rare blood donors, blood recruitment department by special reception, notify the Department with the whole blood, provide considerate service for blood donors. And encourage blood donors to take part in the next blood donation with full enthusiasm. 3) love association activities, held a quarterly "rare blood type" knowledge of science lectures, regularly organized rare blood type emergency team expansion training, rare blood type essay photography activities and rare blood type fun games. 4) file management, each quarter of the rare blood type emergency team archives combed, the newly added rare blood type files classified arrangement. 5) the development of blood donors. In a timely manner to the donors feedback on the examination and blood test information, put forward the corresponding suggestions and guidance, to meet the donor's physiological and psychological needs; to preach the preferential policies and blood compensation; clear blood for clinical use value, enhance the sense of social identity and social responsibility; good scientific blood, reduce the loss rate of blood donors for the first time they will, efforts into the development of rare blood emergency blood donation team.
Results: Since the establishment of the rare blood type emergency team, the number of blood groups in the emergency ranks has been greatly increased by adopting the above-mentioned maintenance measures. The total number of rare blood type emergency teams grew from 334 in early 2014 to 561 in the first half of 2017, an increase of 227.
Summary/Conclusions: After 2014–2017 years of statistics, the use of information technology for communication and advocacy and rare blood group love friendship activities two ways to develop rare blood type emergency team effect is most obvious. The development of blood donors of 168 people, accounting for the total number of blood donors in 84.8%. and the development of more effective use of maintenance measures, the development of rare blood emergency team can be better, the capital of Rh (D) emergency appointment negative blood for clinical use and provide a solid guarantee.

P-003

Abstract has been withdrawn

P-004

HOW EFFECTIVELY THE DO THE 28 VOLUNTEER TEAMS WORK FOR BLOOD DONATION SERVICE IN GUANGDONG, CHINA?

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Background: Volunteer teams for blood donation often help blood banks to promote blood donation in China. Most of teams are funded by local government. It is reported that those teams with more volunteers may help recruit more blood donor. However, some research shows that the work effectiveness of volunteer teams is more significant to improve the promotion of blood donation. In China, blood donation has developed unevenly among different cities. The work effectiveness of volunteer team varies as well, which may related to their administration and management.
There are 28 registered volunteer teams in Guangdong. They help local blood banks to organize social public to donate blood. They carry out the events to recruit blood donors. The volunteers are in different age, gender, occupation, education level and blood donation experience. However, some volunteer teams worked more effectively than others by recruiting more donors.
Aims: 1)To find out whether the work effectiveness of volunteers is related to the recruitment of blood donor. 2)To learn how effectively the 28 volunteer teams work for blood donation service in Guangdong Province. 3)To search obstacles of volunteer management among 31 blood banks.
Methods: This is a secondary data analysis for blood donation service volunteer teams in Guangdong in 2014. The data included the volunteer names, date of estimation, and working location of each team, the number of registered volunteers and their working time. Volunteers who participated at least 3 times per year are recognized to be active volunteers. The Effectiveness Ratio of volunteer team (ER) is introduced. ER = the number of active volunteer/the total number of volunteer registered. The annual total number of blood donors is also summed up in different blood banks, which recruited by the local volunteer teams. Blood banks are divided into 5 groups by the donor number (n): Group 1 ($n \geq 200{,}000$), Group 2 ($50{,}000 \leq n < 200{,}000$), Group 3 ($20{,}000 \leq n < 50{,}000$), Group 4 ($10{,}000 < n \leq 20{,}000$), Group 5 ($n \leq 10{,}000$). The One Way ANOVA is conducted to access the association between ER and the annual total donor number in different blood banks.
Results: The average ER is 41.93% in Guangdong Province in 2014. There were 13122 volunteers registered and only 5489 volunteers have participate donor recruitment. According to the size of ER, the ranking of 5 blood bank groups is Group 1 > Group 3 > Group 2 > Group 4 > Group 5.
Summary/Conclusions: The volunteer teams for blood donation service in Guangdong are working less effectively than the national average. Generally, more active volunteers help recruit more donors. However, medium size blood banks worked effectively (with higher ER), but recruited less blood donors than some other blood banks. It can be improved in volunteer management, organizational issues, recruiting methods and financial support.

P-005

INVESTIGATION OF THE ESTABLISHMENT OF THE GROUP EMERGENCY BLOOD DONATION TEAM

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Background: Since 1998, our nation's voluntary uncompensated blood donation system came into effect, volunteer blood donation depends mainly on voluntary individual behaviour. By plenty of exploration and practice, group recruitment of volunteer blood donation has become one of the important links of blood collection and supply. Blood donors of most blood transfusion service came from blood donors of sporadic blood donation housing and mobile blood collection vehicle in street.

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We found many social groups in practical word, who attended to blood donation. Especially, the leaders of some enterprises and institutions and social groups wanted organized staff to collectively donate blood for the purpose of enhancing team consciousness and team spirit and improving sense of collective honour, but they could not achieve devotion of love due to busy work or inconvenient traffic. Recent years, we steadied existing group volunteer blood donation team, meanwhile we developed the new blood donation mode of group recruitment and acquired a remarkable effect. Group volunteer blood donation mainly replied emergent events of using blood, satisfied blood reserves and corrected blood type deviation, which resolved the supply on blood for clinical use and ensured blood safety.

Aims: All groups of provincial capital satisfied requirement of blood for clinical use in emergency situation by establishing sound group emergency blood donation team. It should come as soon as called and be competent in emergency situation.

Methods: We put up the platform of voluntary service, established Hebei provinicial youth's emergency volunteer blood donation team and carried out emergency exercises by organizing and carrying out the activities of volunteer blood donation. We also sustained and enhanced emergency volunteer blood donation team of Shijiazhuang by policy intervention, official support and high-quality conversazione of college student volunteers.

Results: The number of the group emergency blood donation team of Shijiazhuang has been growing every year. The number of blood donors in 2016 increased by 30% compared with that of 2015. For now, the established group emergency blood donation team of Shijiazhuang includes 2216 persons.

Summary/Conclusions: Group recruitment was a mode of the recruitment of blood donors and played an important role in ensuring blood for clinical use, especially in slack season of blood donation (including severe cold, sweltering summer and the vacation of college and secondary school). The group emergency blood donation team was an important direction of future development of volunteer blood donation recruitment.

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P-006

THE ANALYSIS OF REFUND POLICY OF BLOOD EXPENSES BASED ON BIG DATA OF BLOOD COLLECTION AND TRANSFUSION IN HEBEI PROVINCE

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Background: Since 2012, Hebei province on the basis of the information networking platform for blood collecting and supplying, realize the provincial blood costs in return and the discharge of blood charges reported, easing the "easy reimbursement of blood donation difficult" criticism [2]. But in the course of the actual operation of the blood charge return, blood donation to the donor body according to local blood donation to meet the regional standards to determine whether blood donors enjoy free blood conditions, the original purpose is to protect the blood donor's location of the need for the use of the territory, this practice has become a common practice in the country. The implementation of the "blood donation law of the People's Republic of China" in Hebei Province article 20th stipulates that the amount of blood donated by a blood donor is three times exempt from the prescribed fee; Blood donation amount to 1000 ml, lifelong enjoyment free charge of blood. The province data show that some blood donors in the different cities in province, the total amount of blood has reached or more than 1000 ml, but in order to return the blood time, did not consider the blood donation in the province of the total amount of blood donation to 1000 ml when the reimbursement, this paper summarizes and analyzes the data of the former blood fee return in the province, and understands the distribution of the beneficiary population, in order to improve the system of reimbursement for blood expenses, and to safeguard the rights and interests of unpaid blood donors [4], and provide reference for providing quality services for unpaid blood donors.

Aims: The paper aims to analyze the big data of blood donors in Hebei province, investigate the blood donors which donate more than one thousand ML blood, improve the refund system of blood expenses, safeguard the rights of blood donors, provide the high quality service to donors, and assist the technical support for the blood expenses refund system across the region.

Methods: Based on the blood collection and transfusion system of Hebei province, the researchers indentify the donor which is accordant with name and ID number by SQL querying, compare and analyze the big data which were created from October 1998 to January 31th 2017.

Results: The blood donors, who donate in various regions, increased in recent years. About 22% of volunteers donate one thousand ML blood totally in various regions. Those just can get triple expense back in every region, and could not have the benefit of free blood for life in Hebei.

Summary/Conclusions: "Big data" is a useful message to "cleanse" the use of massive, high growth rates and diversified information assets. The "Big Data" of the related data platform of blood donation in our province should be dug in depth, and the donation policy is more beneficial to the blood donors. According to large data, the realization of provincial and provincial blood donation time combined with the National Unpaid blood donation award is worth learning. 1. Further improve the relevant policy basis. 2 on the basis of data platform, perfect unified blood expense reimbursement rule. 3. Establish a technical basis for the reimbursement of transregional blood costs.

P-007

APPLICATIONS AND CONSIDERATIONS ABOUT THE PROPORTION CONCEPT OF FIXED BLOOD DONORS

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Background: Fixed blood donors, refers to those who donated blood 3 times in total, and one time within the past one year, usually as an important index to evaluate the degree of stability of blood donors' team building and capability of blood support. "The Method of Commending and Reward of Blood Donors (2014 Revision)" takes the local fixed donors to blood donors more than 50% as one of the important conditions of an advanced province (city). Based on the actual concern of fixed blood donor's population size or the capability of blood supply, domestic blood supply agencies have different understanding and application, some calculate according to the ratio of the number of fixed blood donors. (hereafter referred to as the ratio of number of fixed blood donors), some calculate according to the ratio of person-time of fixed blood donors), and some calculate according to the ratio of volume of fixed blood donors (the ratio of volume). 3 proportions are defined differently, and the results are quite different. What indicators can best reflect the actual effect of fixed blood donors to local blood supply, we analyzed the ratio of the number ratio of person-time and the ratio of volume of the Blood Center of Hebei Province, discuss the best concept of ration of fixed blood donors which can reflect the capacity of blood supply.

Aims: Analysis on fixed number of blood donors and changing volume of blood collection from Blood Center of Hebei Province in recent four years will help to explore the application effect and practical significance concept of blood donors proportion, and offer reference for developing donors team and enhancing the efficiency of blood collecting.

Methods: Extracting data from the formula definition of blood donation volume, number and person-time of donors during 2012 to 2016 to compare with the aggregation of each year to measure and calculate in proportion. In accordance with statistics, $P < 0.05$ shows a statistically significant difference.

Results: The blood collection volume was increasing every year, and the proportion of fixed donors, person-time and blood volume had been general uptrends with different levels, which were inversely associated with the total growth.

Summary/Conclusions: The proportion of donors is inadequate to evaluate the capability of sufficient blood collecting. Instead, regarding the proportion of fixed donors number as a consideration to analyze the reservation utility and satisfaction degree of local demand would have more guiding significances.

P-008

A CASE OF INSPECTION ERROR CAUSED BY LABEL ERRORS OF BLOOD COLLECTING NURSE

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Background: January 2016, there was a case of errors in blood samples collected by blood collecting nurses, which resulted in invalid test results of 2 specimens. Because a number of people at the same time to donate blood on the blood car, in order to catch up with the progress of the work, the nurses violated the operation rules of blood collecting, and the last blood specimen of blood donors was taken before labeling, and didn't timely check. In addition, the day they went back to the blood station later, specimens received by the staff on duty, but the staff just counted the number, didn't check the specimens one by one. The next day, when the laboratory staff handover, found a specimen without label, so he asked blood collecting nurse. The nurse did not find the specimen was duplicated, and she thought she forgot to sign. Inspection personnel cut the pigtail catheter sample from the

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blood bag for test.The result of specimen was initially rechecked is different from the result which was be tested again of the ABO blood type, the working part of the investigations found that covered the label. By reverse tracking each work link, then they found the blood collecting nurse made the label of 1 specimen repeated, and the other label of specimen was forgot. So it caused "one blood with two labels" and "one blood has no label". This paper aims to find the potential safety problems of the label of blood collecting and the work of handover in this blood station, and remind the staff to attach importance to the labeling, checking and handing over in the process of blood collection, and intensify the training of professional training, and take effective measures to prevent such incidents from happening again.

Aims: This paper discusses the causes of the errors of labeling, which caused by the blood collecting nurse, and the consequences of the examination errors, then puts forward some improvement measures to prevent such incidents from happening again.

Methods: According to analysis of the case that the nurse made the label mistake of the blood sample of 1 donors, resulting in detection result of two blood specimens was invalid, this paper further clarifies The importance of corresponding labeling and checking work in blood collection.

Results: By this case of error analysis, finding the human factors and objective factors for false label in the key link of blood collection and supply, putting forward the corresponding improvement measures according to the potential problem of quality and safety.

Summary/Conclusions: In actual work, we should increase the management of blood collecting staff and professional work training for blood collecting nurse, strict labeling management procedures, and strengthen the process control, eliminate the staff doesn't have strong sense of responsibility, check carelessly and other causes of quality error in the process of blood collection, strengthen source control, ensure the safety of blood.

P-009

APPLICATION AND REVISION ON THE CONCEPT OF BLOOD DONATIONS PER THOUSAND

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Background: WHO regards donations per thousands as an important indicator to evaluate the level of blood supply in countries and regions. However, there are different understandings on the concept and measurement methods of donations per thousands interiorly, which will make different effects on the application of this concept in blood source development.

Aims: Based on the analysis of the population structure in Shijiazhuang, and the application results of the concept of blood donation in the collection and delivery of thousands of population is discussed. Combined with the development trend of blood transfusion, the reasonable development direction of blood donor recruitment is set up, and the blood collection efficiency is effectively improved.

Methods: The population of Shijiazhuang area in 2012–2015 was divided into two categories by setting the proper age as the boundary, the blood donation rate (‰), the blood donation volume (‰) and the blood donation rate (‰) of the population in proper age were calculated. The population data were analyzed statistically, $P < 0.05$ was statistically significant.

Results: In recent years, the proportion of the population of the underage group and the elderly group in Shijiazhuang increased year by year, and the proportion of the population of the proper age group was declining, the number in the next few years continued to decrease.

Summary/Conclusions: Data shows significant differences in the ratio of the number of blood donors, person-time and the volume of blood donations per thousand, which is based on the resident population. The ratio among the number of donors, person-time and volume in the blood donation population of proper age could reflect the blood collection capacity of local blood banks, which is also suitable for the blood industry as a comparison between the indicators. The blood transfusion services should take the blood donation rate of the proper age group as the core indicators, and adopt the appropriate measures to improve the blood collection efficiency by increasing the frequency of blood donation and the 400 ml-sampling rate.

Information Technology

P-010

INTRODUCTION OF BLOOD INFORMATION MANAGEMENT SYSTEM USING SMART DEVICE

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Background: Korean Red Cross Blood Service Headquarters has added various confirmation procedures for ensuring the safety of blood donors and blood from blood collection sites. However, the nurses at the blood collection site needed frequent movements to handle various tasks, so they skipped some confirmation steps that can check the quality of blood only by desktop computers. Therefore, It has been emphasized to meet the needs for a new system that it is easy to carry and use while working, and it is possible to check and process the data such as 'whether the blood donor is qualified to donate blood ', 'whether the donor's ID is identical with blood-bags used during blood collection ', and 'the side effect records of the donor while collecting blood'.

Aims: Researchers examined the device that can meet 'portability' which enable staffs to carry and work with one hand, 'functionality' which enable staffs to process various important data in real time, and 'high security' for the importance of blood management works that process sensitive and important data including personal information.

Methods: In order to meet 'portability' and 'functionality' among the above three purposes, we have applied the smartphone to our work. We have selected specific models based on strict criteria and applied Mobile Device Management System (MDM), a security solution for mobile device management, to minimize part to manage. we also allow smartphone to utilize the LTE or 3G only, and limited the communication method by only the service through the VPN (Virtual Private Network) to secure the communication section.

Finally, an application related to blood collection task was developed with in-house method using the native development language, and distributed through MDM.

In this way, security-authenticated infrastructure that only the authorized user and the authorized device can use the service using the encrypted end-to-end communication that has been checked for security has been established.

Results: Smart BIMS (Smart Blood Information Management System) system was applied to the blood collection site in mid-June 2013. Some of the functions of BIMS (Blood Information Management System) that were only available through full-browsing of desktop computer have applied to the first Smart BIMS, and have become possible without exception at all blood collecting sites.

The nurses that adapted to a new working environment suggested new functions such as 'checking donor's past blood donation symptoms' and 'checking equipment information used for blood collection' adding to 'whether the blood donor is qualified to donate blood' and 'whether the donor's ID is identical with blood-bags used during blood collection'.

Furthermore, 'task schedule table' and 'various statistics program' are added to convenience of the person in charge.

The Smart BIMS is currently being developed for the convenience of its powerful and convenient functions, and has been developed for using in street blood collection.

Summary/Conclusions: The Korean Red Cross Blood Service Headquarters had preparatory process for three years, so even if it is difficult for the nurse in charge to use the desktop computer for the blood collection, it is possible to access the BIMS anywhere in the mobile communication network. Smart BIMS, which uses smartphone as terminal and end-to-end encryption of communication section to satisfy both 'portability', 'functionality' and 'security', has high user satisfaction. Thus, various add-ons are under review.

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P-011

AUTOMATION OF TRACEABILITY IN A BLOOD TRANSFUSION CENTER

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Background: The Egyptian National Blood Transfusion Services (ENBTS) has started a pilot implementation for the Blood Management Information Technology System (BMS), between its head quarters National Blood Transfusion Center (NBTC) and Tanta Regional Blood Transfusion Center in September 2016. The project will include another 15 sites. All of them will be connected through a secure Wide Area Network (WAN). One of the vital functions of the software is the facility to track different errors easily.
Aims: To illustrate how the applied software has facilitated the process of tracking several errors, through unit tracking and user tracking.
Methods: Traceability is done using different options:
1. Unit Tracking: It is one of the simplest ways used by the user to track a unit. All the user has to do is only enter the Donation Identification Number (DIN) and a detailed history for the unit will be displayed. For example:
-The Unit Current Status: It displays the components separated from the entered DIN, Is it available for issuing to hospitals and patients or not? Is it still pending any laboratory results? In case it is pending any results the user can access the results screen, to know which lab hasn't entered the results yet. Is it pending the final ISBT label to be printed?
-Unit Current Location: Is the unit still in the BTC, is it reserved to a specific patient, is it issued to a certain patient/hospital or is it available at a certain fridge/freezer?
2. Process Tracking: The users can track when certain process took place for a certain DIN. The results are displayed by user name, date and time of occurrence.
Both of the above tools are used by users according to the permissions granted to them.
3. Database: At some stage we need the vendor company to interfere, in case we have problem in the traceability process. They interfere, by tracking the processes or the unit in the database, by running certain queries.
Results: There are two common problems that kept appearing especially at the beginning of the implementation, for example:
1. Unknown Unit: When the users at the issuing department try to issue a certain unit to a hospital/patient, they receive the message unknown unit. When they track the unit they find out that either the label was not verified or the unit was not released.
2. Donor In Process: The donor gave his donation, and at the next visit when the registrar tries to register a new donation, the donor status is still in process. Using the traceability function the user may find out that certain laboratory results were not entered by the users. Sometimes the DIN was entered wrong so the user couldn't find the donation to which the result should be entered.
Summary/Conclusions: The implementation of the BMS is a positive step towards the automation of the ENBTS. The implementation of the BMS had a great advantage in terms of improved traceability quality and accountability. The traceability function in the software has been of great support to different user categories, especially the quality members, as it helped them to track down different errors raised.

P-012

Abstract has been withdrawn

P-013

CONSTRUCTION OF PUBLIC OPINION MONITORING AND CRISIS PREVENTION AND CONTROL SYSTEM IN BLOOD TRANSFUSION SERVICE

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Background: With the rapid development of we media, such as WeChat、micro-blog、network broadcast, the spread and influence of the public opinion have been greatly improved. Inevitably, the crisis caused by the negative public opinion shows great explosive and destructive force.
Aims: Dependent on the influence of public opinion, especially in the voluntary blood donation work, the Blood Transfusion Service should establish an efficient management of public opinion crisis.
Methods: By analyzing the characteristics of public opinion, tracking several network public opinion crisis management cases, using the methods of survey, literature and experience summary.
Results: Combined with practical work, this article summarizes a public opinion management system, whose main line is" public opinion Collection - public opinion dealing – recovery". This system is based on a clear organizational structure, using modern network monitoring, and its principle focuses on prevention and early intervention. Meanwhile, six mechanisms have been established throughout the management system.
Summary/Conclusions: Through the implementation of this series of measures, the Blood Transfusion Service has improved its coping ability which lays a foundation for the steady development of voluntary blood donation work.

P-014

DISCUSSION ON HOW TO DO THE INFORMATION MANAGEMENT OF BLOOD STATION LABORATORY

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Background: Blood station laboratory is the major department to provide the information of blood collection and supply. Therefore, its information.
management is of particular importance.
Aims: To improve the information management of laboratory of blood station, ensuring the security of information detection, software system,
internal network and data transfer is significant parts of the daily work of blood station laboratory.
Methods: According to the source of the blood station laboratory, the writer divided information into three categories: first, the resulting information blood donors; second, operating parameter information on laboratory equipment and laboratory information management system; third, the native operation software of equipment and computer.
Results: According to the classification, information management of blood station laboratory presents different characteristics. It has management difficulties and weak spot, such as the imperfection on relevant laws, regulations and the systems; as for the requirements in terms of access, the suppliers of computer information management system are not inadequate; the ability of emergency processing and recovering to the normal operation for accident of computer management system needs to be strengthened.
Summary/Conclusions: Based on the problems above, the author puts forward some suggestions to improve the information management system of blood station laboratory. First, to improve the information management regulations, level-to-level administration and responsibility assignment should be conducted. Date backup needs to be well-established. Second, we should formulate information management standards, put forward specific requirements and other measures to strictly control the access of administrations to the information management system. Much emphasis should be put on the information management of blood station laboratory to ensure the information security of blood collection and supply, to protect the staff and to provide better services for the patients.

Training and Education

P-015

Abstract has been withdrawn

P-016

TRAINING AND EDUCATION SITUATION OF TRANSFUSION PRACTITIONERS IN GUANGZHOU

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Background: Blood transfusion medicine is becoming more and more close to clinical medicine. Much work of transfusion practice pays more attention to "Blood" decades ago shift to "Patient" in recent years. Transfusion practitioners not only focus on blood but also responsible for clinical consultation, blood management and even treatment. There are various training courses organized by hospitals and blood centers all year around. However, the training pattern is conservative and backward.
Aims: The characteristics and shortages of current situation are discussed and explore new training strategies for transfusion practitioners.
Methods: This study is based on the empirical research method and questionnaires were allocated after annual training in the form of lecture. Totally 457 valid questionnaires were collected.

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Results: Most transfusion practitioners started to work in blood centers and transfusion department in hospital after graduation, lacking of clinical practice. Training and education is mainly "one-to-many" in instead of "one-to-one" teaching. However, from the questionnaires, majority (99%) transfusion practitioners show satisfactory with the training course and no suggestions on contents and methods of training. The other suggestions include innovation of the content and training pattern, improve the dietary during training and shorten the training time.
Summary/Conclusions: The result shows a passive acceptance of training knowledge in transfusion practitioners. Meanwhile they have no ideas with learning reinforcement between transfusion and clinical practice. The concept of transfusion is hysteretic that there is a general deficiency of clinical knowledge in transfusion practitioners. The core of teaching method is to ask learners to be automatic learning instead of passive learning.

P-017

DISCUSSION ON-THE-JOB TRAINING MODE OF EMPLOYEES IN BLOOD BANKS

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Background: Staff training and assessing must be proceeded according to the measures for the blood banks management, Blood bank staff can go to work officially after specially training and assessing and reaching the requirements of post qualification.
Aims: To understand the on-the-job post training situation and possible factors, and to establish relevant training system.
Methods: To collect and analyze the situation of on-the-job training persons who have ever attended the training in Changsha Blood Center from 2005 to 2016.
Results: After establishing a perfect on-the-job training system, the examination result is obviously better than before.
Summary/Conclusions: To establish on-the-job training system is the foundation to proceed smoothly with working on blood banks, and it is an important method to improve not only for quality system establishment of the blood banks, but also the quality of the collected blood.

P-018

THE IMPORTANCE OF DIVERSIFIED TRAINING IN THE DEVELOPMENT OF BLOOD STATIONS

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Background: As the knowledge structure of blood bank staff is uneven, they always focus on the training of professional knowledge, and pay great attention to the lack of professional ethics, corporate culture, public relations etiquette and so on.
Aims: In order to strengthen the management of blood bank, to provide better service for blood donors and patients, to promote further development of blood station, to improve the comprehensive quality of blood station staff, and to adopt diversified training method.
Methods: By increasing the training frequency and time, from 75 h to 100 h, on the basis of general training, diversified training, innovative training forms. For example: send personnel to visit the higher level units, hire professional training team, guide public relations etiquette, improve professional ethics, Once a month between middle-level leadership of classical culture, historical stories, essays on modern inspirational Reading Festival full of poetry and prose, speech contest, held on the blood of special enterprise culture forum and so on. Assessment evaluation method innovation of training effect, make full use of information technology, the opening of the paperless training assessment system, implementing flexible evaluation.
Results: Through the diversification of training, business knowledge has been strengthened, the worker's occupation moral realm and professional technical level has been greatly improved, flexible and innovative evaluation methods to stimulate the staff's curiosity and enthusiasm, a one-time training examination qualified rate of 100%. Public relations etiquette training, the establishment of corporate culture, shaping the blood civilized image, promote teamwork, core cohesion greatly enhanced work efficiency, rapid promotion, service quality has been a qualitative leap, donor satisfaction reached ninety-nine percent, with 100% satisfaction Blood Institute.
Summary/Conclusions: Diversified training plays an important role in improving the comprehensive quality of the blood bank staff and improving the service quality. The blood bank should innovate the form of training and carry out diversified training.

P-019

DEVELOPING THE BLOOD TRANSFUSION – TRAINING AND EDUCATION ARE TOP PRIORITY

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Background: Now the cultivation of talents is closely related to the development of blood stations, in quality and professional work management, talent is also the most important factor, therefore, it is urgent to establish and perfect the training and education mechanism.
Aims: Put forward reasonable establishment and improvement of talent training and education mechanism in blood station, suggestions and strategies for the development of blood stations.
Methods:
1. Establish the goal of training/A training target.
Stratified and staged targets are established:
Stratified targets: According to the blood station agencies and the department of training needs, combine the business work of blood station and the specialty of personnel to set up curriculum. Training are divided into national level, provincial level, station level and department level. Staged targets: Set up short term and long term training targets, conduct systematic and procedural training, review, evaluate and summarize the achievement of the target in time. The special person is responsible for gathering, finishing all kinds of training information, guidance department a planned training, forming a strong learning atmosphere, make employees be willing to learn.
2. Establish diversified training methods.
Changing the single teaching-learning method, interactive training, hands-on training instead. Now, the work is tense and heavy, to ensure full participation, we can use the Internet platform, using WeChat training, video training and other training methods. These training methods require the level of trainers to be improved, therefore, we must strengthen the training of knowledge and vision for trainers.
3. Perfect training evaluation mechanism.
Training evaluation, refers to the training program, training process and effectiveness evaluation. We must establish a sound training evaluation mechanism, put an end to ineffective training and take formal training, diversification of assessment methods, instead of the old test paper, knowledge competition, practical test, observation, questioning, case study and other effective methods can be adopted.
4. To develop the training incentive mechanism.
Spirit of the award: For individuals and departments with high quality training, give spiritual forms "red flags" for example, or other forms. By using these methods and measures, gradually change from the previous passive state "make me learn" to active state "I want to learn", to ensure the full implementation of the established training target.
Material rewards: Material incentives for completing training are also essential, give achieve personal and department higher learning materials or other meaningful rewards.
Results: 1. Complete the training plan successfully; 2. Improve the quality of training; 3. Improve staff learning efficiency; 4. Improved staff learning enthusiasm.
Summary/Conclusions: In a word, the establishment and implementation of grassroots blood station training and education mechanism, we should pay attention to the unity of the development strategy of blood station and staff growth. Designing training education system with scientific methods, in order to improve the quality of blood station and staff as the main target, designing training and education courses that meet the working unit, using a variety of training methods, to ensure the quality of training.

P-020

STUDY OF STANDARD OPERATING PROCEDURES USING VIDEO

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Background: Standard operating procedures (SOP) are an important part of the quality management system. The SOP training effect of employees is closely related to their performance.
Aims: To study the feasibility of standard operating procedures change to video type.
Methods: Writing standard operating procedures with theory and actual. Using modern media equipment to shoot standard operating procedures video, during the process of filming, timely find and modify the standard operating procedures deficiency, eventually forming the feasible standard operating procedures.

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Results: Finished some standard operating procedures video, and applied to staff operation.
Summary/Conclusions: Standard operating procedures video can verify the SOP operability, standardized staff operation; video data easy to save, just in time to see, and solve the problem at any time. Standard operating procedures video is easy to training new employees.

P-021
THE IMPORTANCE OF ETIQUETTE TRAINING IN BLOOD DONATION SERVICE
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Background: Blood stations should provide open service to the society for the blood donors. The good behavior of the staff represents the Image of blood stations directly and deeply. Standard degree of statements and actions, professional degree of medical techniques and satisfaction of service level are convincing evaluation indexes to the blood stations.
Aims: The staff should master skilled technical expertise and be highly educated and cultivated. To improve the whole character of the staff, we strengthen their professional skills and ethics, reinforce the etiquette training and take steps to guarantee the training results in the continuing education.
Methods: Making training plan, taking versatile forms in the training course, attracting the staff's participation, building the atmosphere of learning, improving the training results. After centralized training, let the staff apply them to practice. In order to guarantee the training results, evaluation mechanism, supervision and administration are needed.
Results: The blood donors' satisfaction is improved year by year with the carrying out training continually.
Summary/Conclusions: Professional etiquette is an important way for medical staff to publicize their professional image and win social recognition. In the face of unpaid blood donors, blood station staff should be grateful and give them respect and admiration and give them the best service. For blood stations, it is an effective way to maintain good impression in the blood donors to make the etiquette throughout all the services. It can lay a solid foundation to develop fixed unpaid blood donors, win the trust and respect of them and reduce the risk of blood usage.

P-022
PRELIMINARY ESTABLISHMENT OF THE INTERNAL QUALITY ASSURANCE SYSTEM OF TRANSFUSION MEDICAL NETWORK COURSE IN BEIJING UNION MEDICAL COLLEGE
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Background: Although blood transfusion medicine education is increasing emergence, the level of teaching materials lags far behind the development of blood transfusion medicine. The refresh speed of the current domestic teaching materials is so slow, and meanwhile, most of the teachers haven't received the systematic training of transfusion medicine, these cases lead to the situation that the teaching theories fall behind the real practice.
Aims: Preliminary inquiry into the method of the internal quality assurance system of transfusion medical network course, establish the Yale remote course dedicated web site, to further improve the Union Medical College in blood transfusion medicine higher education resources.
Methods: According to the concept of ICDE review, establish a complete internal quality assurance system of transfusion medical network course, which based on "development - implementation – checking – feedback - rectification". The purpose of this paper is to explore five aspects, such as objectives, curriculum design and learning resources, teaching staff, teaching process and assessment.
Results: Established the internal quality assurance system of transfusion medical network course, and further improved in the teaching process.
Summary/Conclusions: Blood transfusion medical network course content is an important part of transfusion medical higher education in Union Medical College, establishment and improvement of the internal quality Established the internal quality assurance system of transfusion medical network course, and further improved in the teaching process. assurance system is conductive to guarantee and improve the quality of transfusion medical higher education.

Risk Models, Standards and Regulation

P-023
IMPLEMENTATION OF INFECTION CONTROL STANDARDS IN EGYPTIAN NATIONAL BLOOD TRANSFUSION SERVICE
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Background: Health care associated infections are a worldwide problem. They occur across all points of health care delivery. Thus, the fundamentals of infection control need to be employed regardless of constraints in resources and support. These fundamentals are essential to protect the donor, patient, clients and provider against exposure to infectious microorganisms and against the morbidity and mortality associated with these agents.
Aims: Implementation of infection control standards in Egyptian blood transfusion services.
-to ensure prevention and control of health-care associated infections.
-to minimize occupational hazards associated with the delivery of health care.
-to promote sound infection control practices focusing on injection safety and safe blood transfusion.
Methods: The infection control team developed an action plan for implementation of the infection control program in the Egyptian national blood transfusion services. It included maintaining an appropriate infrastructure, ensuring the availability of supplies, establishing effective infection control practices, improving the occupational health management system, development of surveillance for nosocomial infection and training of health care providers. The plan is reviewed and updated annually. In 2013 the infection control team initially formulated infection control standard operating procedures and distributed them to one central and 24 regional blood transfusion centers. They referred to Egyptian national infection prevention and control guidelines and policies for health care services as well as American association of blood banks standards. The standard included infection control practices concerning hand hygiene,waste management, personal protective equipments, safe injection &prevention of needlestick and sharp injuries,environmental cleaning,-post-exposure prophylaxis, aseptic techniques, reprocessing of instruments, surveillance of nosocomial infection, facing emerging and re-emerging illness, laboratory working instructions and dentistry working instructions. The infection control team provided initial training and guidance to a total of 181 individuals including infection control officers, heads of different departments and health care providers of national and regional blood centers. Periodic assessment, supervision, monitoring and training of health care providers as well as annual evaluation of infection control program were performed.
Results: Improvement of the infrastructure as regards construction of hand washing basins and establishment of central sterile services area for reprocessing of instruments to ensure high standards of decontamination. Introduction of new supplies of infection control as disposable aprons and disposable tourniquets. Vaccination coverage of health care providers for hepatitis B is more than 95%. Training of the majority of health care providers. Statistical evaluation of the implementation of the infection control program has increased from 33.5% in 2013 to 80% in 2016. Increasing the compliance of the health care providers to infection control standards from 30.4% in 2013 to 82% in 2016.
Summary/Conclusions: Infection control is a necessary component of safe, high quality blood supply. The infection control program is not only committed to prevent adverse outcomes such as health care associated infections but also contributes to qualitative improvement of blood transfusion services.

P-024
ANALYSIS OF THE CAUSES OF DEFERRAL OF BLOOD DONATION IN NINGXIA AND TAKE EFFECTIVE MEASURES FROM 2010 TO 2016
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Background: There are 5 blood stations in Ningxia district, one of which is the blood center of the province (grade), and 4 [1] of the blood station of the city (grade) center. Since the 2010 implementation of blood concentration detection in

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Ningxia, Ningxia blood center, give full play to the advantages of centralized testing center laboratory by analyzing blood during the operation of unqualified reason, take preventive measures in time, greatly improving the detection accuracy, effectively avoid the blood resource waste, ensure the safety of blood.
Aims: Analyze the causes of disqualification of blood donation in Ningxia and take effective measures to reduce the scrap rate.
Methods: Statistical analysis of the reasons for the disqualification of blood donation in Ningxia from 2012 to 2016 uses of disqualification of blood donation in Ningxia and take effective measures to reduce the scrap rate.
Results: From 2012 to 2016 in Ningxia to develop the whole blood concentration detection, blood scrap rate reduced year by year, from 3.4% in 2010 to 1.72% by 2016, including inspection items scrap ratio decreased from 2.95% in 2010 to 1.51% in 2016. The scrapping rate of non-test items also declined from 0.45% in 2010 to 0.21% in 2016.
Summary/Conclusions: Strengthen the district personnel operation training, the implementation of integration management, communication meeting at regular blood stations in the business analysis, strict screening blood donors and ALT, HBsAg, TP screening measures such as before, can effectively reduce blood unqualified ratio, reduce resource waste blood.
gauge: ZH-CN;mso-bidi-language:AR-SA'> in Ningxia to develop the whole blood concentration detection, blood scrap rate reduced year by year, from 3.4% in 2010 to 1.72% by 2016, including inspection items scrap ratio decreased from 2.95% in 2010 to 1.51% in 2016. The scrapping rate of non-test items also declined from 0.45% in 2010 to 0.21% in 2016.

P-025

THE POTENTIAL IMPACT OF NAT INTRODUCTION – AS MAJOR PILLAR IN TRANSFUSION STRATEGY IN ROMANIA – REGARDING BLOOD DONATION, BASED ON THE ASSESSMENT OF 2014–2016 ACTIVITY

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Background: The study design was already applied to regional level (Bihor County, Romania) and presented at ISBT Copenhagen 2017; being taken into account the previous study conclusions, the present one is a logical and more complex continuation at national level, about the impact of NAT introduction for donated blood, and measures which should be taken for increasing the transfusion safety and the number of blood donors, both major objectives for national strategy.
Aims: The paper presents sequentially, the panel and ratio of pre-donation rejection criteria of potential blood donors and post-donation non-validation criteria, the impact of chemiluminescence screening test and NAT introduction, management of professional attitude and the future policy which should be developed according to regional/national/European realities.
Methods: The study is a descriptive and analytic one, being evaluated the potential/donors' files and medical records afferent to the period of time 01.01.2014–31.12.2016. Due to the large number of assessed individuals (more than 400,000 donations/year), the study is a population one, being taken into account the number/gender/age and blood donor category, along with number/(non)medical rejection criteria of potential/blood donors. Taking into account the transfusion network, the blood centers were divided according to the annual number of donations, the threshold being 10,000 donations (26 BTCs with less than 10,000, and 15 BTCs with more than 10,000).
Results: Were considered rejection/allowance blood donation criteria, being proven that the panel of diseases which rejected (temporary or permanent) the individual is very large, the predonation (for potential blood donors) level of hemoglobin and "other medical reasons" significantly fingerprinted the rate of temporary deferral in blood centers with less than 10,000 donations/year, and for those with more than 10,000 donations/year the large majority was represented by "other medical reasons". In case of postdonation (for blood donors) rejection criteria, for both BTCs categories, the majority of discarded blood units was due to increased values of ALT, significant from the statistic point of view, followed by the viral markers, mainly by AgHBs positive results.
Summary/Conclusions: The large postdonation rejection rate, due to increased value of ALT -with negative viral markers-, along with the discordance between annual viral reactivities number (done by ELISA)/confirmed viral infections number, obviously showed the negative impact, upon the final validated number of blood donations. When about increased ALT values, the discarded rate reduced the annual total number of blood donations, till 3% at the national level, up to 10,000 units/year. When about the discordance of viral reactivities/confirmed infections, on one hand, showed a high rate of nonspecific reactivities, due to ELISA method limits, on the other hand, the permanent deferral, irrespective infection confirmation – induces a frustration, a confusion about health status of the involved individuals. According to these aspects, the major steps ahead when about national transfusion safety, are chemiluminescence method -as screening test-, and NAT introduction, with both individual and general positive impact, regarding the transfusion infectious residual risk, and number of total validated blood donations, as well.

P-026

STUDY ON THE RISK FACTORS OF GETTING INFECTION AMONG BLOOD COLLECTING NURSES FOR BEING STABBED

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Background: According to the World Health Organization (WHO), the rates of HIV, hepatitis B virus (HBV) and HCV infections upon needlestick injuries are 0.3%, 3.0% to 6.0% and 1.8%, respectively. Needlestick injuries often have a serious negative psychological impact on the infected people, and it is easy for these people to be pessimistic. Those, who are infected by HBV or HIV, will often develop long-lasting psychological disorders. Some will shout and scream while some will talk less or even become suicidal. If medical care staff, who work on the firing line, are mentally not stable, it is not only difficult for them to focus on blood collection, but their condition might also influence their working skills. Meanwhile, it will produce a certain degree of negative emotion on other staff working at the blood bank, which may lead to errors in blood collection, even leading to panic.
Aims: To analyze the risk of infections following needlestick injuries among blood collecting nurses.
Methods: We observed and analysed the complications occurred during blood draw.
Results: After needlestick injury, nurses could be infected by a small amount of HBV virus. Moreover, the HIV prevalence was from 0.2% to 0.4%, as a result, the chances of HIV infection is 19% to 40%, and the infection ratio by HIV virus is from 0.2% to 0.6%.
The main causes of these needlestick injuries are the long working hours, the crowded working environment and the inappropriate operation.
Summary/Conclusions: In order to prevent future needlestick injuries and virus infection, it is necessary for nurses to improve their working environment, urge them to strengthen professional study and personal qualities, and develop their individual theoretical level, operational proficiency and working responsibilities.

P-027

IMPLEMENTATION OF HIGHEST LEVEL CONDITIONS AND STANDARD AGAINST INFECTIOUS AGENTS

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Background: Highest level of safety is required for work with dangerous and exotic agents that pose a high individual risk of aerosol-transmitted laboratory infections, agents which cause severe to fatal disease in humans for which vaccines or other treatments are not available.
Aims: Achieving a good level of safety when working with dangerous and exotic agents.
Methods: When dealing with biological hazards at this level the use of self-contained oxygen supply is mandatory. The entrance and exit of this level biolab will contain multiple showers, a vacuum room, an ultraviolet light room, and other safety precautions designed to destroy all traces of the biohazard. Multiple airlocks are employed and are electronically secured to prevent both doors opening at the same time. All air and water service going to and coming from a highest level of safety lab will undergo similar decontamination procedures to eliminate the possibility of an accidental release. All the following conditions which is mentioned in the results section must be implemented.
Results: The laboratory supervisor must enforce the institutional policies that control access to the laboratory. Eating, drinking, smoking, handling contact lenses, applying cosmetics, and storing food for human consumption must not be permitted in laboratory areas. Mechanical pipetting devices must be used. Policies for the safe handling

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of sharps, such as needles, scalpels, pipettes, and broken glassware must be developed and implemented. Perform all procedures to minimize the creation of splashes and/or aerosols. Decontaminate work surfaces with appropriate disinfectant after completion of work and after any spill or splash of potentially infectious material. Decontaminate all wastes before removal from the laboratory by an effective validated method. A sign incorporating the universal biohazard symbol must be posted at the entrance to the laboratory when infectious agents are present. An effective integrated pest management program is required. The laboratory supervisor must ensure that laboratory personnel receive appropriate training regarding their duties.
Summary/Conclusions: Individual risk of infectious disease by dangerous and exotic agents can be controlled by implementation of highest level conditions and standards in microbiological practices.

P-028
TO EXPLORE THE YANGZHOU AREA SHIELDED UNPAID BLOOD DONORS APPLY FOR REJOINING STRATEGIES AND FOLLOW-UP SURVEY

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Background: In recent years, with the mining, the cause of the further development of blood supply, blood quality and blood transfusion safety has been more and more attention and attention, blood collection institutions to increase the investment in testing equipment, reagent sensitivity has also been greatly improved. Highly sensitive reagents can lead to false reactivity. The result of the counterfeit reactivity of the reagent will be treated permanently and will cause psychological and mental stress to these shielded donors.
In order to protect the rights and interests of donors, the US FDA published in July 2005 the "HIV-1 and HCV nucleic acid detection, blood treatment and blood donors shielding and return to the guidelines", China in September 2013 by the China Blood Transfusion Association blood quality management The Working Committee has issued a Guide to Screening and Rejoining Respondents. However, since the above guidelines are not mandatory, the implementation criteria are not uniform and there are difficulties in implementation, and some of the blood collection agencies are still out of blood safety considerations, With a cautious attitude, leading to the current self-blood donors in the field of rehabilitation is difficult to reunify.
Aims: To explore the Yangzhou area of unpaid blood donors to shield the rejoining strategy for the results of the blood donors responsible for the test.
Methods: Primary screening colloidal gold method to detect HBsAg, TP results of reactive blood donors, the blood ELISA test results to determine the treatment. Previously, due to ELISA (HBsAg, anti-HCV, anti-HIV) single reagent reactive screening of blood donors, specimens sent to Jiangsu Province, the blood center to further confirm.
Results: 33.47% of the colloidal gold detection of reactive blood donors were tested by ELISA double reagent was negative; once tested by ELISA single reagent to respond to blood donors eligible for returnees in 182 cases, accounting for 37.22%.
Summary/Conclusions: It is necessary to take a shield and rejoin the blood donors who have been tested for colloidal gold and who have been tested by ELISA. They can restore some of the blood donors, protect the blood donation rights of the blood donors, reduce the damage and disputes to the blood donors, Is conducive to blood and blood donors who communicate.

Blood Supply Management and Utilization

P-029
POST BASELINE BLOOD SAFETY SITUATION ANALYSIS OF MALAWI: SUCCESSES AND CHALLENGES OF THE RESOURCE LIMITED SUB-SAHARAN AFRICA COUNTRY

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Background: Malawi is a small, land locked country in Southern Africa with a total area of 118, 484 square kilometres of which 94,246 square kilometres is land area. Its neighbours are Tanzania in the north and northeast, Mozambique in the east, south and southwest while Zambia lies to its west and northwest. Mid-year 2015, it had a projected population of 17,101,849. Efforts to improve blood safety in sub-Saharan Africa started in the last century and are still ongoing. The Malawi Blood Transfusion Service (MBTS) was established in 2004 as a nationally coordinated blood service with a mandate to ensure the safety and adequacy of blood and blood products for the country. As one of way of monitoring blood safety, the MBTS periodically conducts situational analysis surveys on blood safety in the Country. The baseline survey was conducted in 2007 and the post-baseline survey was carried out in 2016 to assess blood safety issues for 2015.
Aims: The main goal of the survey was to assess the blood safety situation in the country in order to track progress made over the years. This will assist in planning, monitoring and evaluation and reprogramming of interventions for future improvement of blood safety.
Methods: A cross-sectional and retrospective survey was carried out in all 87 hospitals in the country over a month. Quantitative and qualitative data was collected by trained laboratory technicians in September 2016 based on January to December 2016 data. Data analysis was done by a team of independent statistical consultants using SPSS version 20.
Results: Estimated national blood needs have increased from 80,000 to 120,000 whole blood units per annum; voluntary unpaid blood donation at national level has increased from 44% in 2007 to 70% in 2015. When compared with 2007, there successes were documented in hospitals in these areas: infrastructure, equipment, reagents, personnel, blood storage, consumables, laboratory procedures, compatibility testing and blood issuing, essential stationery, blood bank personnel safety, blood donor safety, screening for infectious markers and ordering of reagents. However, Challenges were noted in the following: hospital transfusion committees, use of the recommended IAT cross match, use of cross match labels and availability of back-up power. Unlike in 2007, the following documents are now available in Malawi: a blood policy; guidelines for safe blood transfusion and guidelines for the clinical use of blood and blood products. However, there is still no legislation to regulate blood transfusion.
Summary/Conclusions: This post-baseline situational analysis has provided information that will be useful for planning and for monitoring and evaluation of the strategies being implemented to achieve blood transfusion safety in Malawi. The current annual blood need for Malawi is 120,000 units per year. Further challenges still exist in the areas of quality systems at hospital level. These include lack of a sustainable maintenance program for hospital equipment; weak systems for ensuring an uninterrupted supply chain for reagents and supplies and record keeping and data management.

P-030
A REVIEW OF D NEGATIVE RED BLOOD CELL UTILISATION IN THE PUBLIC SECTOR HOSPITALS IN HONG KONG

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Background: Among Hong Kong Chinese blood donors, 0.29% are Rh(D) negative (D-). Demand for D- red blood cell (RBC) products increased by 38.9% from 2007 to 2016 (c.f. 26.8% increment in all RBC products). In 2016, 1857 units of D- RBC units were distributed to both public and private hospitals, which accounted for 0.77% of total RBC issues.
Aims: To determine the fate of D- RBC issues and to understand the inventory management and utilisation practices in the public sector.
Methods: Fate data of RBC units distributed in 2016 were collected from Blood Transfusion Network System (a database developed by Hong Kong Hospital Authority for blood transaction operation and storage of records), which included the outcome of RBC, components age at transfusion, patient blood group and transfusion indication.
Results: A total of 1560 D- RBC units (including 30 units of mini-packs) were distributed to public hospitals from January to December 2016 with 25.3% being group A, 21.9% group B, 46.2% group O and 6.7% group AB (c.f. ABO blood group frequency in local donors: 25.8%, 26.8%, 40.8% and 6.7% respectively). 1.41% and 0.51% of D- RBC units were wasted due to expiration and exceeding the "30-min" rule respectively (c.f. 0.25% and 0.39% for all RBC). 51.4% of D- (all ABO groups) RBC units were transfused into D- patients (group A: 63.0%; group B: 36.5%; Group O: 56.2%; group AB: 22.6%). 10.9% of group O D- RBCs were transfused to non-group O recipients. Regarding the product age when transfused, the median was 31 days (mean±SD: 27.9 ± 11.5 days). There was a bimodal distribution with the first peak on Day 12 and the second on Day 40. Product age at the time of transfusion to D- patients was significantly younger than that to D+ patients (median: 19 days vs 39 days; $P < 0.0001$). These 1560 RBC units were transfused into 821 patients with a median of 2 units being transfused per individual (range: 1–47 units).

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Forty-four individuals (5.2%) who were given 5 or more D- RBC units consumed 583 units (37.4%) in total. The main indications for transfusion were end-stage renal failure, haematological and non-haematological malignancies and Cooley's anaemia. These included 2 D+ patients with haematological disorders who developed anti-LW in the course of the disease and were given D- RBC empirically as LW antigens expressed weaker in the absence of D antigen. A total of 125 (8.0%) D- RBC units were transfused to non-Chinese patients (including 8 neonates) and 22 neonates of ethnic Chinese.

Summary/Conclusions: These data reflected the current status of inventory management and utilisation practices of transfusion of D- RBC in Hong Kong. To cater for emergency situations, obstetric and neonatal indications, it is not uncommon to over-stock D- RBC into the inventory of hospital blood banks with the outcomes being either the units transfused into D+ patients near the end of the product shelf-lives or expired. Re-evaluation of hospital blood bank inventory holding policy may lower the demand of RBC products of this infrequent phenotype.

P-031

BLOOD INVENTORY MANAGEMENT AT A BLOOD TRANSFUSION SERVICE OF A SPECIALIZED LIVER CENTRE

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Background: Blood inventory management is core of any Blood Transfusion Services (BTS). The key to a successful BTS is the timely provision of required blood components while avoiding wastage of the same. It is even more challenging for hospital based BTS, which has to balance both the collection and issue of blood components.

Aims: To develop an algorithm to determine the ideal inventory levels for the BTS of a specialized liver centre.

Methods: The study was retrospective, analytical and non-interventional. Data on various parameters of inventory management such as blood collection, requests, cross-match, issue and wastage were collected and analyzed. Thirty six cross-sections of data (2 per month) were randomly analyzed to find the median age of blood units, group wise availability and issue details. Data on zero stock/low stock and fluctuations in demand and supply were collected and analyzed. We derived two formulas to determine ideal inventory levels and the required blood collection. The inventory levels for shortage/ideal/excess inventory were determined and colour coded to improve monitoring of inventory and take appropriate action to maintain it.

Results: During the study period, a mean of 684 donations per month were collected, of which 10.8% were discarded due to TTI positivity. The collection exceeded the issue for packed red blood cells (PRBC), random donor platelet concentrate (RDPC) and single donor platelet concentrate (SDPC) while the issue of cryoprecipitate (CP) and fresh frozen plasma (FFP) exceeded the collection (this skew was due to the fact that our BTS caters predominantly to liver disease patients). A total of 9.67 PRBCs, 62.7 FFPs, 16.01 RDPCs, 2 SDPCs, and 6.4 CPs were discarded per month.

The issuable stock of PRBCs was 16.68, 7, 56.7, 8.5, 32.2, 5, 39.1 and 10 units for A positive, A negative, B positive, B negative, AB positive, AB negative, O positive and O negative blood group units, respectively and the issuable stock index was 4,10.5, 3.8,11.8,14,10, 5 and 11 days for the same. There were on an average 1.015 donations/issue/month. The overall PRBC stock per issue was 5.44 or 5.44*I (issue) and the overall issuable stock index were 3.84 days. The cross-match held units were 1.6*I (CT*HT; cross-match*holding time) and cross-match/transfusion was 1.94. Effective holding time was, therefore, 0.82 days (1.6/1.94).

The ideal stock levels were calculated to be 5–7 IS (issuable stock) and was colour coded green. IS at 2–4 and 0–1 were coloured coded as yellow and red, respectively and higher values of 7 and above were coded blue to reduce wastage. The Ideal stock of PRBC, FFP, CP, RDPC and SDPC was calculated to be 600, 823, 295, 289 and, 33 units per month, respectively. The required blood collection was approximately 34/day. This would still leave us with a shortfall of 225 units of FFP which would have to be procured from outside.

Summary/Conclusions: This study was an example of a practical approach to develop an optimal inventory level in a standalone hospital based BTS. It also highlights the issues faced in inventory management in specialized institutes with skewed requirements.

P-032

Abstract has been moved to 4B-S18-01

P-033

A STUDY OF RED CELL REQUESTS AND USAGE PATTERN IN DISTRICT GENERAL HOSPITAL-HAMBANTOTA SRI LANKA IN 2016

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Background: District General Hospital (DGH) Hambantota Sri Lanka, which has a bed strength of 635 and total admissions of 63000 per year. It has most of the major specialities including Medicine, General Surgery, Paediatrics and Gynaecology and Obstetrics.

Blood and blood product usage has increased significantly during the last decade due to epidemics of dengue haemorrhagic fever and increasing facilities in surgical specialties.

Aims: To minimize the wastage of Red Cell Concentrates (RCC) and to optimize blood usage and to discuss the findings with relevant stakeholders at the annual Hospital Transfusion Committee Meeting.

Methods: Number of cross matches and issues for each specialty were collected retrospectively from the cross match request forms for RCC and RCC issue Register and analysed. The study period was 01st of January to 31st of December 2016.

Results: A total of 7972 request forms were evaluated during the study period and 14638 cross matches performed during the same period. Among the four major specialities, C: T ratio was highest in the Gynaecology and Obstetrics (C: T 9.27:1) and General Surgical units (C: T 3.24:1), whereas it was up to the recommended ratio in Medical (C: T 1.60:1) and Paediatric units (C: T 1.62:1).

Among the other specialities the highest C: T ratio of 37.86:1 is in the Otorhinolaryngology unit, whereas lowest C: T ratio of 2.17:1 is in the Intensive Care Unit. The C: T ratios of Orthopaedic and Oral and Maxillofacial units were 7.42:1 and 17.67:1 respectively.

Out of requests 70.87% were routine, 29.03% were urgent and only 0.1% were emergency requests. Only 2574 requests eventually ended up needing a transfusion and only 4334 of cross matched blood units were actually transfused. The overall C: T ratio for the hospital was 3.38:1.

Monthly average of RCC discards due to past dates were 34 units in the year 2016.

Summary/Conclusions: Inappropriate reservation of RCC for various types of surgeries which hardly needs blood transfusions results in greater wastage of RCC in the DGH-Hambantota. Implementation of Maximum Surgical Blood Ordering Schedule is of utmost importance and group, antibody screen and serum saving method should be implemented without any delay in order to assure appropriate clinical use of blood and blood products and also to minimize the unnecessary workload of the blood bank staff.

P-034

ENHANCING INVENTORY MANAGEMENT OF PHENOTYPED RED BLOOD CELL (RBC) BY USING LEAN SIX SIGMA

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Background: The Blood Services Group (BSG) Singapore secures the nation's blood supply, ensuring that all patients in Singapore have access to adequate and safe blood. Provision of specific phenotyped red blood cell (RBC) for patients with blood group antibodies is essential to prevent blood transfusion reaction and alloimmunization. BSG keeps a small inventory of specific phenotyped RBC to ensure these blood are readily available upon requests. However, shortfall of the phenotyped RBC inventory has been observed in BSG.

Aims: Our aim is to study the possible root cause of the shortfall phenotyped RBC inventory and to enhance the management of these blood.

Methods: Lean Six Sigma tool was used to identify the potential possible causes contributing to the shortfall of phenotyped RBC inventory. Retrospective data was collected from 1st September to 31st October 2016. Statistical analysis was performed using SigmaXL software. Gage Repeatability and Reproducibility (Gage R&R) study was conducted to assess the inventory officers' skill on the decision making when arranging phenotyped RBC.

Results: 3 possible causes namely (i) improper issuance of requested blood, (ii) phenotyped blood storage arrangement and (iii) insufficient phenotyping were identified.

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Results showed that there is no significant relationship between the high shortfall of phenotyped RBC with improper issuance of requested blood ($P > 0.05$). The Gage R&R Study revealed that phenotyped RBC arrangements into a specific blood tray varied among the inventory officers, resulting rarer blood type not being replenished to the specific blood tray, hence contributing to the shortfall of phenotyped RBC inventory. Historical data showed that phenotyping of RBC had increased by 8 folds over the past 5 years. However, the increase did not significantly improve the phenotyped RBC inventory level. Phenotyping workflow analysis showed that most of the newly tested phenotyped RBCs were not placed into the phenotyped blood inventory but distributed as normal blood. This was because these blood were not being relabeled with phenotype printed on the label.

Summary/Conclusions: Varied RBC arrangement method contributes to the shortfall of the phenotyped inventory. A guideline based on the prevalence of RBC phenotype of our population demographics would be provided to guide inventory officer when arranging phenotyped RBC. A new workflow was proposed to hold the RBC units selected for phenotyping and to be relabeled when the blood unit is found to be suitable for topping up the phenotyped RBC inventory.

P-035

PREPARATION OF A MAXIMUM SURGICAL BLOOD ORDERING SCHEDULE FOR THE DISTRICT GENERAL HOSPITAL HAMBANTOTA SRI LANKA

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Background: A study on red cell requests and usage pattern in District General Hospital (DGH)-Hambantota in 2016 was found to have cross match to transfusion (C:T) ratios of 3.24:1 and 9.27:1 among General Surgical and Gynaecology & Obstetrics units respectively. High C:T ratios were noted in other surgical specialities and therefore it highlighted the necessity of the Maximum Surgical Blood Ordering Schedule (MSBOS) for the hospital.

Aims: To find out the C: T ratio, Transfusion Index (TI) Transfusion Probability (% T) for the surgeries/procedures that are performed at DGH- Hambantota and to group surgeries/procedures which have a C: T ratio $\geq$ 2.5:1 and/or TI of $\leq$0.5 and/or % T $\leq$ 30% in to Group, antibody screen and Save (G&S) category.

To group others into the category that needs blood to be cross matched and to prepare the MSBOS for DGH Hambantota.

Methods: Retrospective study was done using the cross match request forms for Red Cell Concentrate (RCC) and RCC issue Register and analysed. The study period was 01st of January to 31st of December, 2016. 7972 request forms were evaluated to find out total surgeries/procedures during the study period. Even though total of 189 surgeries/procedures was categorized during the study period, 60 surgeries/procedures were performed only once in the year 2016 and they were excluded from the study.

Results: All Surgeries has C:T ratio $\geq$ 2.5:1 except for the Above/Below Knee Amputations (C:T 2.34:1, TI=1, %T = 100%)) and Open Prostatectomy (C:T 2.5:1, TI=1.3, %T = 100%).

In Laparoscopic surgeries (C:T 79:1, TI=0.02% T = 1.52) Thyroid surgeries (C:T 77.7:1, TI=0.02% T = 1.13%) and Caesarean Section (C:T 74:1, TI=0.01, %T = 1.34%) all 3 parameters are as mentioned.

The Medical Induction in miscarriages and blood reservation for major/minor degree Placenta Previa patients also had a very high C:T ratio of C:T 77:1 and C:T 82.3:1 respectively with low TI and %T of TI=0.01, T%=1.38% and, TI=0.04 T%=2.64 respectively. Lap and Dye Test, Laparoscopic Cholecystectomy, Hemithyroidectomy/ Thyroid Lobectomy, Mastectomy±Axillary Clearance, Laparoscopic Appendicectomy did not use any transfusion even though 143,119, 86, 34 and 28 RCC units were cross matched respectively.

Summary/Conclusions: When considering all three parameters, Laparoscopy Surgeries with suspected ectopic pregnancy, Oesophagectomy, Wide Local Excision of Buccal Mucosa, Anterior Resection, Wound Toilet and Haemorrhoidectomy are grouped into surgeries that need one blood unit to be cross matched. Above/Below Knee Amputations and Open Prostatectomy need 2 blood units to be cross matched.

Laparotomy, almost all Laparoscopic surgeries excluding suspected ectopic pregnancy, Caesarean section,Thyroid, Breast, Urological and Colorectal surgeries excluding Anterior Resection and Haemorrhoidectomy, Splenectomy, Internal Fixation of Fractures, Hemiarthroplasty, abdominal and vaginal hysterectomy, Medical Induction in miscarriages, Placenta Previa can be grouped into G&S category which does not need blood units to be cross matched if the antibody screening is negative. Even though the study shows that the Placenta Previa and Laparotomy comes under G & S category after a negative antibody screening, request of 3 blood units is the practice in DGH Hambantota.

P-036

A SURVEY OF BLOOD SUPPLY IN CHINA DURING 2012–2014

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Background: China is a middle-income country, which hosts more than 20% of world population. Increasing of the blood supply in China, along with increased health care coverage raises many challenges.

Aims: The aim of this study was to understand the status of blood products supply in China during 2012–2014.

Methods: The survey questionnaire about blood center activities were sent to all blood centers of 32 provinces via internet. The data were collected from responses and analyzed using Microsoft Excel 2013.

Results: The total supply of whole blood and RBCs in 2012 was 18,644,700 units, in 2013 18,985,800 units, and in 2014 19,658,800 units. A similar trend of the total platelets (PLTs) supply was also observed during the same period of time of 2012 to 2014: 1,019,100 units in 2012, 1,168,400 units in 2013 and 1,276,200 units in 2014. Similarly, the plasma supply was 27,529,300 units in 2012 and 27,657,600 units in 2013, which rose to 28,307,500 units in 2014. The total cryoprecipitate supply were 1,653,900, 1,891,300 and 2,366,500 units in 2012, 2013 and 2014, respectively. When the blood supply was analyzed according to geographic regional population, large differences of rates of blood supply between regions were evident.

Summary/Conclusions: The blood products supply in China was steady increasing. Blood centers in China continue to face the challenges of providing sufficient blood supply in the future.

P-037

INCREASING EFFICIENCY AND EFFECTIVENESS BY MODIFYING BLOOD BANK POLICY: A SINGLE CENTER EXPERIENCE

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Background: Reserving cross-matched blood units leads to excess waste, acquisition, and use of personnel resources. Analysis of actual supply of units reserved revealed some departments had as low as 10% effective use. The standard policy of ordering cross matches at will for non-trauma patients, and then reserving the cross matched units for 72 h led to the afore mentioned problems. This study hypothesized that a new policy with a rule that no cross match would be performed for these patients when hemoglobin is greater than or equal to 10 g/dl with a negative antibody screen could increase efficiency.

Aims: This study examined the effect of changing the standard operating policy in the blood bank on cross-matched blood units and waste.

Methods: Cross matches, cross matched units supplied, and waste of disqualified blood units were analyzed for two years previous to, and three years following, the new policy implementation.

Results: 25% decrease in cross matches, 50% decrease in disqualified blood units, and 25% increased use of cross matched reserved units followed implementation of the new procedure.

Summary/Conclusions: The new standard operating policy saved personnel hours of cross matching and unnecessary reservation of blood units, leading to increased monetary savings and more effective blood supply.

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P-038

ANALYSIS OF BLOOD COLLECTION AND SUPPLY IN ZHENGZHOU AREA

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Background: Zhengzhou is the capital of the most populous provinces in China. Recent years, the amount of blood supply is at record highs every year, but the actual demand of medical institutions is quite large, which leads to the phenomenon of insufficient blood supply and partial blood type lasts a long time in a year. This situation has affected patients with rapid blood service. To explore a breakthrough has become an urgent problem to be solved.
Aims: Through the analysis of the blood collection and blood usage of medical institutions, and the development of the local population and medical institutions, to further assess the future trend of the local blood supply, to study effective long-term donation development measures, and to improve the blood supply situation.
Methods: Statistical analysis the volume of blood collection and supply in Zhengzhou area, and the development of population size in this region and the development index of medical institutions, from 2012 to 2016.
Results: From 2012 to 2016, the population of Zhengzhou increased from 9.3 million to 9.7 million, the average annual growth rate of 1.9%. The number of medical institutions increased from 3807 to 3963, the increase in the number of 156, the increase was 4.1%. The number of beds in medical institutions increased from 57894 to 85914, the average annual growth rate of 10.4%. The number of blood donors increased from 190 thousand to 210 thousand, the average annual growth rate of 2.1%. Thousands of blood donation rate remained at between 18.8 to 21.6. Blood donors donate blood volume increased from 64352600 ml to 66198700 ml per year, the average annual growth rate of 0.7%. During 2015, the group booking blood donors donate blood volume accounted 14.6% to total blood volume, individual voluntary blood donors accounted for 85.4%. During 2016, the group booking blood donors donate blood volume accounted 30.7% to total blood volume, individual voluntary blood donors accounted for 69.3%. Medical institutions use blood volume increased from 310 thousand units to 400 thousand units. The amount of cryoprecipitate, irradiation of red blood cell, apheresis platelet and other blood components used increased significantly, followed by an increase of 246%、145%、120%. In contrast, the amount of whole blood and concentrated handmade platelets used decreased significantly, followed by an increase of −68.1%, −99.3%.
Summary/Conclusions: In recent years, the number of people who donate blood, blood donation rate, blood donation volume, the utilization rate of blood components and so on all have been greatly improved. In the past, most of the volume of blood is collected from individuals in the streets, Nowadays, group reservation donors play a more and more important role in voluntary blood donation. The amount of Washed red blood cells、 irradiated red blood cells、 apheresis platelet and cryoagglutinin use is increasing. The amount of whole blood that easily stimulates transfusion reactions use has fallen sharply. However, the blood supply growth rate is lower than the blood demand growth, lead to the blood supply and demand is still serious, This means that the next few years will be more serious. Development of long term mechanism of voluntary blood donation needs Support and help from all walks of life, needs to effectively integrate the province's blood resources, to solve the provincial capital supply pressure.

P-039

RED CELLS EXPIRY: TOWARDS SAVING OUR PRECIOUS RESOURCE

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Background: Donated blood is limited and expensive resources with only 2% of Malaysian population are regular donors. Waste of a single unit of blood will result in a wasted voluntary effort of a blood donor in addition to other incurred cost such as compatibility testing, automated or manual labor and time spent in the procedures.
Aims: To determine the magnitude of red cells expiry rate in Hospital Serdang.
Methods: Data for the audit was obtained from the LIS system and manual count of daily stock. Most of the expired red cells were whole blood, notably due to the ready stock of fresh whole blood to fulfill the demand of transfusion practice in dengue patients. After a change of the transfusion practice in dengue cases from whole blood to packed cells the expiry rate of red cells reduced tremendously. In addition to that, several others strategic steps were undertaken. The "first in, first out policy was tightened. Medical officers were made responsible to follow up the cases in wards and the Transfusion Specialist emphasizing about the judicious use of blood and blood products by doing Continuous Medical Educations to various departments in Hospital Serdang and also implementation of Patient Blood Management approach to optimized red cells usage.
Results: In 2014, red cells expiry rate was noted to show an increasing trend from 3.79% in January to 8.10% in December. However, in 2015 from January to December, the rate was significantly reduced from 3.49% in January to 0.42% in December. The successful of maintaining low expired of red cells continued in 2016 with none of red cells expired in December 2016. In January 2017 to June 2017, every month the expired of red cells were below 0.05%.
Summary/Conclusions: We should aim for zero or near zero percent of wasted blood. There are still shortfalls in transfusion practice that can still be improved. It is hoped that with the strategies implemented, the rate of red cells wastage in Hospital Serdang will be further reduced.

P-040

HOSPITAL BLOOD WASTAGE: AN 11-YEARS OF EVALUATION OF BLOOD WASTAGE CHANGES IN IRAN (2005–2015)

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Background: Although blood components are spacious resources, their wastage has been continuing to be a problem at hospitals all over the world. As there is not comprehensive study about hospital wastage in Iran, the first aim of the study was to identify the WAPI "wastage as percentage of issues" of RBC, FFP and PC during 2005 to 2015 and the second goal focused on reasons of the blood product wastage.
Aims: The first aim of the study was to identify the WAPI "wastage as percentage of issues" of RBC, FFP and PC during 2005 to 2015 and the second goal focused on reasons of the blood product wastage.
Methods: In this project, during mentioned period, wastage as a percentage of issues (WAPI) was calculated for RBC, PLT, and FFP separately. In addition, for each product, the percentage of wastage was calculated as divided to total of the unit wasted.
Results: Rom 2005 through 2015, wastage rate of RBC, FFP and PC were 5.7 ± 0.7, 1.4 ± 0.4 and 3.2 ± 0.5 respectively. The main cause of wastage of RBC, FFP and PC was expiry of date and among in date wastage, the major reason was unused units. Although the outdated units of RBC and FFP decreased, there was an increase in outdated units of PC. In contrast to the outdated units, the percentage of unused RBC and FFP increased, whereas unused PC decreased.
Summary/Conclusions: This study indicated the WAPI of RBC, FFP and PC and focused on reasons of the blood product wastage. By understanding the reasons of discarded units, hospital transfusion committee can develop plans to improve performance and minimize the number of discarded blood to a reasonable rate.

P-041

ANALYSIS OF CHANGSHA BLOOD CENTER BLOOD SUPPLY FROM 2010 TO 2016

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Background: With continuous development of Blood donation work, there has been significant progress in clinical blood component transfusion, but poor clinical blood supply was happen occasionally.
Aims: To investigate the situation of quantity and variety of the blood supply in Changsha blood center from 2010 to 2016.
Methods: By using retrospective study method, we collected the date about Changsha blood center supplying blood to medical institutions in the whole Changsha area from 2010 to 2016. We studied the variation trend of clinical blood application.
Results: The total blood supply of Changsha blood center from 2010 to 2016 maintained rising tendency, increased by 33.05%. And the average annual growth rate is 5.1%. It is declined slightly from 2010 to 2012, and then it steadily grew after 2012. It is hard to transfuse whole blood, but the proportion of Clinical blood component utilization reaches to 99.78%, more to 99.99% since 2014. Red blood cell and plasma are supplied steadily, increased a little, and the annual average growth rate

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is 3.3% and 2.68%, respectively. However, the supply of blood platelet and cryoprecipitation grow rapidly and the annual average growth rate is 22.66% and 50.44%, respectively.
Summary/Conclusions: This paper shows that clinical blood component supply has further developed to be scientific and reasonable, but the supply levels of red blood cell and plasma were growing up slowly. It also provides scientific basis to draw up rational plans for safety, time, effective blood-supply and formulate thorough strategies of recruiting blood donors.

P-042

ANALYSIS IN EMERGENCY SUPPORT FOR BLOOD SERVICE ABOUT JUL 21 GUDUN ROAD EXPLOSION

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Background: In recent years, public emergencies have occurred frequently in the world, the blood service is an important part of medical rescue. It has been a hot issue how to react quickly in unexpected public events to protect the clinical blood effectively in the study of emergency management of domestic blood service institutions.
Aims: Analyzing the characteristics of emergency support of blood service to supply strategy of blood supply in Jul 21 Gudun Road explosion.
Methods: Through blood service and routine drill of the G20 Summit and 7.5 bus arson, the emergency plans for blood services are gradually improved in Zhejiang Province Blood Center. Jul 21 Gudun road explosion has happened ,we start emergency blood supply plan immediately, opening green channel, supplying the blood for hospital that receive the wounded, guaranteeing needs of clinical rescue blood. Starting emergency blood sampling plan, adding blood collection staff and site service personnel, adjusting the layout of blood donation sites Rationally, extending the opening hours of blood donors, reinforcing news propagation, Increasing material delivery and serving people enthusiastically to donate blood.
Results: July 21 to 24, a total of more than 3500 people collected blood donation total of more than 5200 U. July 21 10:00 to July 25 at 8:00, a total of more than 41,000 ml of red blood cell suspension to the hospital, frozen plasma more than 175,000 ml, 335 units of platelets alone, cold precipitate coagulation factor 10 units, effectively protect the treatment of the Explosive of the wounded.
Summary/Conclusions: Emergency plan of blood service of built on the basis of exercise, that can play an active role in blood In public health emergencies, after the practice of emergencies.

P-043

THE ANALYSIS AND PREDICTION OF RED BLOOD CELL USE IN FUZHOU

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Background: Concerns have been raised that the reform of the medical and health system will increase the demand for red blood cell (RBC) products. Modeling can be applied to predict RBC demand and aid future planning for donor recruitment and transfusion services.
Aims: To analyze RBC products used in hospital, Fuzhou, and use mathematical model to predict the demand of RBC products.
Methods: RBC supply in Fuzhou from 2007 to 2016 was retrospectively analyzed, and autoregressive integrated moving average (ARIMA) model was used to model RBC transfusion in 2007–2016 and predict the RBC demand in 2017.
Results: 1) Blood component transfusion rate has been 100% since 2010 (except for one case used 6 unit whole blood in 2014); 2) The average growth rate of RBC transfusion from 2007 to 2016 was 1.8%, A、 B、 O、 AB type RBC accounted for 28.1%(27.5–28.4%), 24.6%(24.4–25.0%)、40.7%(40.4–41.3%)and 6.6%(6.2–6.9%) respectively; 3) The average monthly RBC usage was 10488.7 unit from 2007 to 2016, RBC transfusion in January、 February、 July and August was lower than average; 4) The average RBC usage of every hospital bed was decreased from 7.64 U/bed in 2008 to 4.39 U/bed in 2016, decreased by 42.6%; 5) Use the ARIMA model to predict the RBC demand in 2017, except for AB type, the A、 B and O type RBC predict demand value from January to June was matched the actual usage value perfectly, the average relative error was <10%; The ARIMA model predicts an increase of RBC demand by 8.7% in the second half of 2017 compare with the same period in 2016.
Summary/Conclusions: Predictions show that the demand of RBC has increased rapidly. It highlights the need for taking proactive measures to improve donor recruitment and retention, along with implementation new medical techniques, such as restrictive transfusion practice and patient blood management to reduce RBC used.

P-044

A MULTI-OBJECTIVE MODEL FOR BLOOD INVENTORY DYNAMICS OPTIMIZATION MANAGEMENT IN BLOOD CENTER

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Background: In a relatively long time, the quantity of blood collection and supply, the level of blood inventory in blood center, have often been determined by experience. With respect to the blood inventory management in blood center, practically experience can work well in most cases. Theoretically there is still some optimization space in this regard, even though the space would be little in most people view. Moreover, it should also be concerned that experience cannot work effectively sometimes in complex circumstances. Therefore, basing on advanced optimization theory, studying on the optimization method for blood inventory management in blood center would be valuable.
Aims: In order to realize more effective dynamic optimization management of blood inventory in blood center, a novel multi-objective optimization model and its solution algorithm are proposed.
Methods: Considering the dynamic optimization management of blood inventory in blood center, an optimization principle is designed, which takes account of both the highest extent of blood supply guarantee and the lowest extent of blood inventory cost. Basing on the principle, a multi-objective optimization model for blood inventory dynamic management is constructed, which includes objective function and constraint sets. In the model, the quantity of blood supply is taken as decision variable, and the level of blood inventory as status variable. Further, basing on the Particle Swarm Optimization algorithm, an intelligent algorithm suitable for the solution of the model is formed.
Results: With simulation backgrounds for the test of model and algorithm being designed, which contains a series of weight coefficient combinations of different blood supply guarantee extents and different blood inventory cost control extents, and the algorithm's parameter being selected, also blood demand quantity sequence and blood collection quantity sequence being given, efficiency and effectiveness of the model and its solution algorithm are inspected. It is showed by the inspection that the model can be solved by the algorithm efficiently, and the solution results tend to be reasonable.
Summary/Conclusions: According to the results gained by the proposed model and algorithm, relatively reasonable dynamic optimization management plan for blood inventory can be obtained, which could be the useful reference for decision-making of blood inventory management in blood center.
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P-045

Abstract has been withdrawn

P-046

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P-047

RESEARCH ANALYSIS OF THE DEPARTMENT OF BLOOD SUPPLY IN HEBEI PROVINCE BLOOD BANKS SYSTEM AND A COMPARISON WITH HEBEI PROVINCE BLOOD CENTER

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Background: The author investigates the basic situations such as the amount of blood supplied per year, inventory level per day, the maximum space for blood storage, responsibility of departments, distribution of staffs in positions, size of the place and networking with hospitals in the 12 blood banks in Hebei province. There is difference in the amount of blood supplied per year, space for blood storage, working space size, distribution of staffs or responsibility of departments between blood centers. There is a considerable difference in the workload per person between blood centers. Missing of information scanning during delivery is hidden in the blood centers lack of persons, equipments or work place. The report is as follows.
Aims: To understand the status of the department of blood supply in Hebei province blood centers through investigating the basic situation of the department of blood supply in 12 blood centers of Hebei province in order to provide reference for the construction of the department of blood supply for the future.
Methods: Questionnaires were sent out to the 12 blood banks in Hebei province. The contents of questionnaires included the amount of annual RBC and apheresis platelet and plasma and cryoprecipitate sent out, inventory level per day, the maximum amount of storage capacity for the qualified product inventory, responsibility of the department of blood supply, the number of staffs in the position and the professional qualifications, size of the working place for the department of blood supply, situation of computer management, the number of hospitals to which the blood was directly delivered and situation of networking with hospitals from 2014 to 2016. Each blood product was calculated based on units, daily working intensity per person was used to reflect the workload of staffs, the size intensity was used to reflect the size of the working place, the hoarding days was used to reflect the quantity. The data were analyzed statistically, and P values less than 0.05 was consider to be statistically significant difference.
Results: The inventory level of RBCs decreased year by year, while the inventory level of platelets and plasma increased year by year. There were serious deficiencies in the storage space for plasma in the province. The professional qualifications of the staffs in the blood supply position or in the management position were senior. There were statistically significant differences in the working intensity per person in the blood supply position and the intensity of size of the working place for the department of blood supply between the other blood banks in the province and Hebei Province Blood Center ($P<0.05$), and the working intensity of the staffs in Hebei Province Blood Center topped the list among the blood banks in the province.
Summary/Conclusions: Plasma backlog problems are serious in each blood bank, and these problems appeal to the nation who can provides support in aspect of the policy. The staffs in the blood supply position should strengthen the professional knowledge training for blood transfusion and provide advice and guidance service for scientific rational clinical use of blood in addition to the simple blood delivering job. Installing RFID system in the department of blood supply is an effective defensive measure from the point of system management mistakes and effectively guaranteeing the safety of blood in all the blood banks regardless of the grade.

P-048

ANALYSIS OF STORAGE PERIOD OF LEUKOCYTE-REDUCED RED BLOOD CELLS TRANSFUSED IN PATIENTS IN HOSPITALS IN SHIJIAZHUANG AREA

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Background: The scope of blood supply for Hebei Province Blood Center includes more than 100 hospitals in whole areas of Shijiazhuang.The Center sends out blood products strictly based on the distribution principle "first in, first out".Storage days and scrapped blood units of RBCs transfused in patients in the hospitals requiring blood supplied by the Center are collected and analyzed statistically from 2013 to 2016 to assess the blood supply ability of Hebei Province Blood Center, the degree of the transfused blood and the management level of blood inventory plan in hospitals, and to provide the reference data and prescriptive recommendations for reasonable regulation of RBC inventory turnover volume and standardized blood distribution principle in the Center.
Aims: Storage days and scrapped blood units of RBCs transfused in patients in the hospitals requiring blood supplied by Hebei Province Blood Center, were analyzed statistically from 2013 to 2016, in order to assess the blood supply ability of Center and the degree of the transfused blood and to provide the reference data and prescriptive recommendations for reasonable regulation of RBC inventory and standardized blood distribution principle in the Center.
Methods: The data of leukocyte-reduced RBCs delivered to hospitals of different grades each year and the storage period were collected, the average storage days of RBCs and different blood groups were calculated and analyzed statistically, and P values less than 0.05 was consider to be statistically significant difference.
Results: The total volume of blood delivered and sent out and amount of data observed (U) were gradually increased over the 4 years. The volume of blood required in hospitals in urban areas accounted for more than 2/3 of the total volume of blood sent out, and the volume of blood required in hospitals in suburban areas accounted for less than 1/3 of the total volume of blood sent out. In addition to blood group AB RBCs of which the average storage days were lower in suburban hospitals than in urban hospitals, the average storage days of the other three blood group RBCs were higher in suburban hospitals than in urban hospitals. The average storage days of each blood group RBCs showed a continuous increasing trend year by year. The average storage days of blood group O RBCs transfused were about 18 days, blood group B RBCs nearly 20 days, and blood group A and AB RBCs more than 20 days. The average storage days of blood group A, B, O and AB RBCs were 8, 6, 6 and 4 days higher than that on the day of delivery in the blood center, respectively.
Summary/Conclusions: First, RBC inventory is regulated at an appropriate level by the blood center. Second, the blood with a shorter storage period should be selected when delivered to suburban hospitals in order to reduce the delaying effect of the storage period and scrapped rate due to exceeding the time limit. Third, the hospitals make blood inventory reservation plan and haemovigilance mechanism. Fourth, for some patients with special diseases, blood centers and hospitals should consider the distribution principle "last in, first out" in order to ensure the efficacy of blood transfusion for the these patients. Fifth, the center strengthens the supervision degree with respect to the TMIS system specification use in order to obtain more useful data; the department of blood supply supervises the blood inventory of each hospital using TMIS system, participate in the blood regulating between hospitals and minimize the scrapped rate of blood due to exceeding the time limit at the same time.

P-049

Abstract has been withdrawn

P-050

Abstract has been withdrawn

P-051

Abstract has been withdrawn

P-052

HOW TO DEAL WITH THE NEGATIVE PUBLIC OPINION OF BLOOD BANK

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Background: In recent years, many Chinese blood banks repeatedly faced negative public opinion crisis that caused the blood donation work encountered Waterloo; blood donor reduced sharply leading to the shortage of the blood supply for clinical use.
Aims: To solve the crisis of negative public opinion of blood bank.
Methods: This paper introduces four principles to cope with the negative public opinion of blood bank: the principle of timeliness, the principle of honesty, the principle of openness and the principle of responsibility.
Results: The negative public opinion of blood bank has reduced by four principle.
Summary/Conclusions: There are four effective principles can be done to the the crisis of negative public opinion of blood bank.

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P-053

AN OPTIMIZATION METHOD OF BLOOD SUPPLY WHEN PAROXYSMAL BLOOD DEMAND OCCURS

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Background: Paroxysmal blood demand is often urgent and with large quantity of blood demand. In particular, the paroxysmal accident such as earthquake or war, requires the blood center cooperation so as to guarantee the blood supply. In this case, the blood deployment is controlled by a hub such as National Health and Family Planning Commission (NHFPC) in China. Each blood center is like a node in a social network, and NHFPC is the hub of the network and each blood center is directly connected to and controlled by the hub. However, each blood center also faces blood demand of prior arrangement before the paroxysmal event. What is the best blood supply strategy is not easy to get. On the one hand, the strategy might be the best for a certain blood center, but not the best for the total network consisted by the blood centers. And in fact the blood shortage is always the case, even though there is no paroxysmal events. So how to get optimize the blood supply of each blood center is important.
Aims: Traditionally the blood supply quantity is figured out by simply summing all the blood center that can supply. But it might not be the best strategy, the aim of this work is to find the best strategy of blood supply for each blood center so that it will be the best strategy for the total network when the paroxysmal event occurs.
Methods: Taking the blood transportation time to the place where paroxysmal event occurs (place) into consideration. Then each blood center's blood storage subtract the blood demand before the event, and divide by the transportation time of each blood center. This number gives the blood supply per unit time (BSPUT) to the place. Comparing the BSPUT to some threshold which is given by transfusion medicine doctor. Just summing up the "qualified" BSPUT which are more than the threshold. These blood center are called "qualified blood center". Then figure out the blood center with "non-qualified" BSPUT to the place but "qualified" to the qualified blood center. These blood centers are called "less qualified blood center". The blood supply from the less qualified blood center can substitute the storage in the qualified blood center. And the steps can be repeated several times until we find the "less less. . . qualified" blood center. Summing up all the blood supply per unit time then this is the optimized total blood supply per unit time. All the estimation can be valid because of heavy blood shortage. That is why we can use the blood storage in blood centers farther from the (paroxysmal) place to substitute the blood demand before the paroxysmal event in the blood centers nearer to the place. Numerical simulation is used to check this method. Without loss of generality, we set the "effective" blood storage (blood storage subtract the blood demand before the event) as 1, and the corresponding stochastic storage information are put into the transportation time.
Results: The numerical results shows that then number of our method is larger than that in traditional way. This means that the strategy we proposed is more effective than that in traditional way. And it can avoid the perish of the blood to some extent.
Summary/Conclusions: The optimization of the blood supply when paroxysmal event occurs is more important than simply transferring all the storage in the blood centers. What the hub should do is to calculate the "qualified" BSPUT, rather than simply summing up the storage from each blood center, and to make the "non-qualified" BSPUT in the "non-qualified blood center" "qualified", since blood shortage is heavy. This way can make the BSPUT more than the traditional method.

P-054

USAGE OF CLINICAL APHERESIS PLATELETS THE CENTRAL BLOOD STATION IN HANDAN

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Background: Now more than machine adopt platelets to leukocytes single platelets, the collection process is very strict, stinging arm parts should be strictly sterilized blood donors, blood on the advanced acquisition instrument airtight pipeline circulation and separation, are the use of disposable machine platelet consumables, avoid the cross contamination. The use of platelets is expensive. The retention period of platelet special blood bags for platelets is 5d, so the amount of platelets and the corresponding clinical dosage is particularly important. Scientifically planning machine adopt platelet collection capacity and inventory, in ensuring clinical under the premise of demand for fresh machine platelets, reduce the cost of the management of blood, put an end to waste blood has the positive significance.
Aims: To analyze regular usage of apheresis platelets, to establish a mathematical model on the basis of time series analysis method, to make predictions, and to guide the recruitment, collection and storage of apheresis platelets.
Methods: A mathematical model was established on the clinical supply of aphaeresis platelets in Handan city every month from January 2007 to December 2013. Data was entered in Epidata 3. 0 software. The IBM SPSS statistics 21 were imported. Using the time series analysis method, a mathematical model was applied to predict the supply and demands of clinical apheresis platelets each month from January to December 2014. Errors were detected and analyzed.
Results: Expert model on the determination of usage of clinical apheresis platelets was identified to be the Winters additive model, meaning that residual noises were random noise sequences (P>0.05).Diagnosis was conducted on the model that extracted all the information from the original sequence. Comparison of the predicted results and actual values in 2014, the actual values are falling into a predictive value of 95% confidence interval. Relative errors were small. The model was in its optimal condition..
Summary/Conclusions: Through the establishment of a mathematical model, subsequent data can be help guide blood agencies on the recruitment and collection of apheresis platelets. Science can predict future trends in blood supply and demand.

P-055

STANDARDIZED MANAGEMENT PRACTICE OF BLOOD RADIOMETER IN BLOOD STATION

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Background: Blood transfusions have always been a necessary and important means in clinical medical treatment. Using blood irradiation apparatus to irradiate the blood or blood products which are used for transfusion, is the most effective way to prevent the transfusion associated graft-versus-host disease, TA-GVHD. Due to the potential hazards of the radioactive sources used in the equipment, it is particularly important to standardize the management for the blood irradiation apparatus.
Blood transfusions have always been a necessary and important means in clinical medical treatment. Using blood irradiation apparatus to irradiate the blood or blood products which are used for transfusion, is the most effective way to prevent the transfusion associated graft-versus-host disease, TA-GVHD. Due to the potential hazards of the radioactive sources used in the equipment, it is particularly important to standardize the management for the blood irradiation apparatus.
Aims: To prevent radiation accidents caused by protection or improper use by standardized management of blood irradiators.
Methods: Take strict management for the blood irradiating apparatus, and establish a clear management organization, corresponding rules and regulations, obtain administrative license, standardized daily management, emergency drill, etc.
Take strict management for the blood irradiating apparatus, and establish a clear management organization, corresponding rules and regulations, obtain administrative license, standardized daily management, emergency drill, etc.
Results: Over the past years, the blood center of Zhejiang province took the strict management and established an internal control system for the irradiation apparatus., no accident happens these years.
Over the past years, the blood center of Zhejiang province took the strict management and established an internal control system for the irradiation apparatus, no accident happens these years.
Summary/Conclusions: Taking strict management for the blood irradiation apparatus could avoid the radiation accident.

P-056

INVENTORY MANAGEMENT OF BLOOD SUPPLY

Z Cuicui

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Background: With the continuous improvement of medical technology and level, the demand for blood is also increasing. In China, the clinical blood demand increased from 7,563,400,000 ml in 2012 to 8,642,800,000 ml in 2016, with a growth rate of 14.3%. The annual blood consumption recommended by the WHO is 8 ml. At present, the per capita annual consumption of developed countries is more than 20 ml, and about 4 ml in developing countries. But in China, it was 2.9 ml from 2012 to 2015, and there was a great difference between different regions. Large population base, uneven regional development are the reasons for this phenomenon.
Aims: According to the monthly supply of blood in Tianjin Blood Center during the past 3 years (2014–2016), statistical analysis of the most suitable inventory range, and formulate relevant measures,in order to meet clinical needs.

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Methods: Using computer systems, the amount of blood delivered per month in 2014 to 2016 was measured. Statistical analysis of the data, and draw the change curve, observe the law of change. In 2014, the average monthly supply was 3,523,383.3 ml, the maximum and minimum monthly supplies were 3,775,800 ml and 2,824,000 ml. In 2015, the average monthly supply was 3,544,850.0 ml, the maximum and minimum monthly supplies were 3,813,900 ml and 2,573,900 ml. In 2016, the average monthly supply was 3,558,658.3 ml, the maximum and minimum monthly supplies were 4,005,500 ml and 2,502,500 ml.
Blood supply decreased in January, February, August and September. In the blood shortage months, the inventory is relatively low, the average number of days out of the blood is about 7–10 days. In the other months, the inventory is relatively high, the average number of days out of the blood is about 14–20 days.
Data calculation and arrangement:: Sum the average monthly supply during the past 3 years, then divide 90 days. We can get the result-118,076.5 ml. Take the same approach, we can know the other results: 87,782.2 ml and 128,835.5 ml.
Results: The optimum inventory is 1,180,765 ml (118076.5 ml*10 days); The inventory range is 614,475.4 ml (87,782.2 ml*7 days) – 1,803,697.0 ml (128,835.5 ml*14 days).
Summary/Conclusions: Through the above statistical analysis, we can draw up the appropriate inventory scope, and make reasonable blood supply and deployment. Some methods can be used for reference: Develop fixed blood donors. When the amount of blood collection drops or the demand increases, the blood demand of clinic cannot be satisfied, and the blood donors in the team are promptly notified to donate blood. When the total inventory volume is too high, or the proportion of a blood type inventory is too high, we should take the "limited blood collection" measures in a timely manner to avoid the occurrence of waste of blood. When the blood inventory is low, we should report to the superior leadership and departments in a timely manner, and take corresponding measures, such as notification of fixed blood donors, increase Street recruitment, communicate with the hospital surgery scheduled for the medical staff to make the model to encourage blood donation…… in order to adjust the inventory, increase blood inventory, timely to ensure the clinical blood supply. According to the amount of blood collection, reasonable allocation of blood donors, timely preparation for off-season inventory. Reasonable inventory control, not only to avoid waste of blood, but also to avoid the lack of blood supply, affecting medical treatment.

P-057

CURRENT SITUATION OF BLOOD COLLECTION AND SUPPLY IN CHONGQING

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Background: Chongqing, with a famous name of "mountain city" covered by 76% hills and mountains, is a municipality directly under the central government. In 2016, Chongqing has 3016.55 million resident population and covers an area of 824000 square kilometers, including 38 districts. There are 1 blood center and 17 blood stations respectively responsible for the blood collection and supply tasks in different areas. A blood collecting and supplying service system covering urban and rural areas has been established horizontally to the edge and vertically to bottom. For the past few years, with the rapid growth of the demand of clinical blood, blood supply is facing new pressures and challenges and presenting a normalized tendency of tension supply. In order to better understand the current situation of the blood supply and grasp the overall development trend of the blood supply in Chongqing, it is necessary to analyze the voluntary blood donation in Chongqing.
Aims: The research is based on the current contradiction between blood supply and demand in Chongqing, and the current situation of the contradiction development from local and seasonal to structural and normalized. By studying the blood collection and supply service status and the potential need of blood in these medical resources, it is necessary to find out the suggestions of how to satisfy clinical needs.
Methods: Using cluster stratify for 18 blood services in Chongqing, the data were collected, including information of the blood collection ,blood supply in clinical, the blood stations performance and so on, and analyzed with SPSS20.0 software.
Results: The blood donation rate and blood collection of the whole city increased. The proportion of blood collects amounts of blood center was nearly half.. Blood donation mobilization is based on individual voluntary blood donation, the group donated blood to supplement.. The collection model of 400 ml is Accounted for the largest. In Chongqing, there are 30 million permanent residents, The number of beds, patients, in-patients, surgical performance and the blood using for patient increased. The proportion of blood center is more than one-third to more than half The staffs in the blood station of the whole city are including a half permanent employees and a half unauthorized ones. The proportion of education level with universities and colleges is up to three quarters. The proportion of nurses were the largest. Each blood station with 5000U blood collection exceeded the blood center. The cost of whole blood is increases year by year. All blood collection and supply processes of 18 blood centers are performed by information management. The information system has not yet with the network connection with the hospital of blood usage among the city. The Chongqing blood center has its own network with six primary blood centers.
Summary/Conclusions: 1) Blood voluntary donation is unsuitable for the social and economic development. The imbalance of blood collection and supply area is existed among the blood service system. The level of blood collection and blood supply fail to satisfy with the development of medical and the demand of clinical blood use. 2) The number of staffs is not enough to satisfy the needs. Professional education is limited.. The positions in different centers are showed big difference. 3) With the increasing blood cost year by year, it affects and restricts the development of blood collection and supply of the blood service. 4) The lack of the whole city's network on blood information management system affects the deployment efficiency and the trace ability of blood information.

P-058

TENTATIVE EXPLORATION ON THE LINKAGE MECHANISM OF CROSS REGION BLOOD COLLECTING AND EMERGENCY SUPPLY

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Background: Blood stations break administrative division limit, to carry out all-round cooperation, is to enhance regional blood to disasters or accidents emergency rescue capabilities, the important measures for this view with the industry consensus, but in how to establish coordination mechanism, reasonable allocate resources, provide perfect, perfect the cross-regional logistics information network data exchange and the blood effective application and improve the efficiency of collaboration needs to be further study.
Aims: To Guangxi Liuzhou, Laibin, Hechi establish cross-regional blood collecting and supplying linkage emergency safeguard mechanism to explore, in order to in the corresponding region between the blood concentration detection, blood collecting and supplying drill, personnel and technology deployment, important guarantee for the role of blood emergency relief, etc.
Methods: Three adjacent regional blood services leadership in the practice of the long-term cooperation, constantly sum up experience, to reach consensus, is feasible and effective way to agreement, formation of institutionalized work mode. At the same time, the relevant departments and personnel of the blood supply department should be trained to clarify the responsibilities, working methods and the way of completion in the linkage mechanism.
Results: Through three blood stations emergency safeguard mechanism to establish linkage, Guangxi blood center on the lower central blood stations business further play guiding role, the corresponding region to emergencies blood collecting and supplying emergency disposal ability has been strengthened. For cross-regional maternal emergency blood use, emergency treatment of acute massive bleeding, or some special personnel, such as after chemotherapy in patients with blood, especially in machine, emergency use of rare blood type of blood platelet, play a crucial role. In addition, it is also obvious that the blood supply agencies in the corresponding regions can optimize the blood inventory management and prevent blood waste. This is especially important for a blood station with relatively small size and low blood inventory to adjust.
Summary/Conclusions: For a short period of time when disaster or emergency blood surging demand, only by a single blood station is difficult to meet the emergency blood supply of accident rescue, should break administrative division limit, according to the nearby is urgent to establish the principle of cross-regional blood collecting and supplying linkage mechanism. Modern blood stations management objectively requires collaboration between blood stations is comprehensive, but to really do a good job in this collaboration, to establish a cross-regional (provincial) blood information sharing platform is very necessary and urgent, it is the significant support of modern information management system of blood stations. Health and family planning committee of Guangxi is building blood information sharing platform, it is completed and put into use, will greatly improve the management level blood collecting and supplying in Guangxi, and to the development of business collaboration between blood stations within the region that provide technical support and broad prospects.

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Quality Management

P-059

FOUR YEARS EXPERIENCE OF A PETITE LABORATORY IN AN EXTERNAL QUALITY ASSURANCE PROGRAMME

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Background: A participation in the inter laboratory comparisons is one of the specific criteria for laboratory accreditation scheme of Malaysia (MS ISO 15189). This program provides information on overall performance of laboratory services including competency of laboratory personnel, methodology assessment and safety of patient care. Our petite laboratory has participated in the Royal College of Pathologists of Australasia (RCPA) Australia since 2013.
Aims: The aim of the study was to evaluate our four years performance in an external quality assurance programme.
Methods: We performed a retrospective overview on our RCPA test results for four years from 2013 to 2016. We analysed and documented the test results which include ABO and D Grouping, Direct Coombs' Test (DCT), antibody screening, compatibility testing and transfusion decision. Antibody identification is beyond our scope of accreditation.
Results: The accuracy of our current method to determine ABO and D grouping was challenged significantly as twice we failed to detect mixed field (MF) reaction on the issued samples in 2013 and 2016 respectively. The former was a sample from post-stem cell transplant patient and the later was from subgroup of A patient. The discordances could be attributed to the insensitivity of tube technique and the inability of laboratory personnel in detecting MF reaction. Nevertheless, we managed to secure concordant results for all tests using gel method which include DCT, antibody screening and compatibility testings. In deciding a transfusion, we were able to give concordant transfusion decision in 95.8% (68/71) of cases but failed in three different occasions.
Summary/Conclusions: These failures have helped us to detect the lack of competency among our young and inexperienced laboratory personnel as we are a new operating laboratory of a new medical centre and the disadvantages of our current tube method. Following this overview, we strongly support our decision in changing our current tube method to gel method as it provides better sensitivity and objective reading as well as a stable result that permits us to read and verify at a later time. For transfusion decision, a good decisive practice has to take in place for a better patient blood management and inventory management. For continuous monitoring of laboratory performance, we strongly recommend our laboratory counterpart to join at least one external quality programme.

P-060

VALIDATION OF BECKMAN PK7300 AUTOMATIC BLOOD GROUP ANALYSIS SYSTEM

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Background: To validate the applicability of the capacity on Beckman PK7300 automatic blood group analysis system.
Aims: To validate the applicability of the capacity on Beckman PK7300 automatic blood group analysis system.
Methods: ① According to the recommends of manufacturers and other laboratories, we determined 6 parameters including incubation temperature,reagent red blood cell concentration,sample red blood cell concentration, sample diluents ratio, diluted sample volume, determination parameters.The other 3 main parameters (including incubation time,antibodies of standard serum, dilution of sample plasma) were evaluated with matching experiments. ② The sensitivity and specificity were validated by using of blood types known and special samples which were consisted of unconformity of forward and reverse typing and RhD(-)samples. ③ The consistency and suspicious rate were validated by compared with microplate technique.
Results: (1) The 3 main parameters (incubation time,antibodies of standard serum, dilution of sample plasma) were determined as 60 min,1:40 and 1:2.5, respectively. (2) Using the evaluated parameters, the 3930 samples which known ABO blood group were done forward typing and reverse typing detection.The rates of correct interpretation were >95%.Simultaneously,the 70 cases and 4 cases special samples were detected correctly. (3) The rate of consistency was >95% and suspicious rate was <2%.
Summary/Conclusions: It indicated that the sensitivity, specificity,consistency and suspicious rate of the analysis system can meet the requirements.

P-061

THE EXTERNAL QUALITY ASSESSMENT OF BLOOD BANKS IN JIANGSU PROVINCE FROM 2015 TO 2016

Y Ji and X Chen

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Background: In 2012, the Ministry of Health and the Standardization Administration of China had issued the new "Blood stations technology operation procedures ". In order to adapt to the new rules, Jiangsu Province Blood Center readjusted the "blood services quality control laboratory system of quality assessment (EQA)". A total of 12 blood banks in Jiangsu Province were on board with the project. After 4 years, makes sense to study the results of the project.
Aims: To investigate whether the quality and detection level of the new system to the whole province blood stations improved, and to analyze the increased comparability test results.
Methods: Quality control samples were allocated to each of the quality control labs for coagulation test (FVIII:C, Fbg) and residual quantity of methylene blue, which were added to the EQA from 2014, and the results were collected for statistical analysis. Reports were sent to each lab in time.
Results: The PT percent of pass of EQA results for coagulation test (FVIII:C, Fbg) and residual quantity of methylene blue in 2015 were 77.77% and 80.00%, respectively. After the evaluation report suggestions were given, accuracy and precision of the quality control labs had obvious progress, according to the results in 2016. Results of all the three items had an obviously up-trend with a 83.33% and 87.50% pass rate.
Summary/Conclusions: The results showed that quality tests of blood are the same important as security detection. It must be run through all aspects of the laboratory quality management. In order to ensure the accuracy of test results, laboratory should regular calibration instruments, pay attention to personnel's training and continuously improve the quality of laboratory management.

P-062

ESTABLISHMENT AND EVALUATION OF THE INTERNAL QUALITY CONTROL METHODS IN BLOOD NUCLEIC ACID SCREENING SYSTEM

Y Jiang and G Zhang

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Background: At present, nucleic acid detection technology (NAT) has been widely popularized, and the internal quality control is of great significance to ensure the reliability of nucleic acid detection results. However, a more unified nucleic acid quality control system has not yet been established between the different blood laboratories. The experimental results are quite different and comparatively poor between different laboratories. Therefore, it is urgent to establish a standardized quality control method for NAT in the laboratory.
Aims: To establish and evaluate the method of internal quality control (IQC) in blood screening by KHB nucleic acid test (NAT) systems.
Methods: The QC standard HBV DNA (200 IU/ml), HCV RNA (2000 IU/ml) and HIV RNA (2000 IU/ml) were diluted with negative plasma for blood donors to the detection limit. The suitable concentration of IQC was determined. The same batch IQC samples were tested 20 times, and the mean (x), standard deviation (SD) and coefficient of variation (CV) were calculated. With $x \pm 2SD$ as warning, $x \pm 3SD$ as rejected, the NAT screening IQC frame was established and analyzed. The Levey-Jennings QC chart was drawn, the IQC results were analyzed 180 times and the feasibility and effectiveness of internal quality control model were evaluated using the Westgard multi-rule QC method.
Results: Take the HBV as an example, the correlation coefficient between IQC and Ct value was $r = -0.920$, which was highly correlated at $P = 0.05$ level. Selecting a concentration of 100 IU/ml of IQC, the same batch IQC samples were tested 20 times, x was 31.76, SD was 1.10, and CV was 3.46%. And the same batch IQC samples were tested 180 times continuously, x was 32.02, SD was 1.13, and CV was 3.53%.
Summary/Conclusions: It is feasible that the Ct value was used in NAT IQC. Weak positive IQC selected by our laboratory can enhance the reliability of the experimental results, and it also can eliminate the error caused by the staffs and other experimental conditions changes, so as to monitor the effectiveness of the nucleic acid extraction and amplification detection.

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P-063

Abstract has been withdrawn

P-064

THE DISCUSSION ON THE MANAGEMENT MODE OF UNPAID BLOOD DONORS WHO WERE TEMPORARILY DEFERRED IN HAINAN PROVINCE

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The Blood Centre of Hainan Province, Haikou, China

Background: As the contradiction between the increased blood volume for clinical use and the difficulties of unpaid blood donation recruitment increased year by year, under the condition of the blood tests ability to constantly improve, it is necessary to effectively manage for the temporarily blocked unpaid blood donors, follow up detection, and provide personalized follow-up services, to retain, consolidate and reduce the loss of blood donors.
Aims: To establish a computer system of effective management for unpaid blood donors who are temporarily deferred, provide a basis for the temporarily deferred donors to reenter and unify the process, and also provide quality follow-up services for the reentered blood donors.
Methods: The temporarily deferred blood donors states of blood donation were identified in their profiles in our province, and make them in the state of unable to collect whole blood but only can collect he blood specimens, the specimens can be collected in any blood collection points of our province to follow-up detection, and every time the test results were recorded in the blood donation profile, if the serological and nucleic acid test are qualified after two times tests with each method, their rights of blood donation can be restored.
Results: There were 2935 qualified specimens after the test of 3446 reentered blood specimens (85.2%). Among them, 840 of the 990 reentered specimens due to the reactivity of HBsAg were qualified (84.8%), 589 of the 659 reentered specimens due to the reactivity of Anti-HCV were qualified (89.4%),. 603 of 715 reentered specimens due to the reactivity of Anti-HIV were qualified (84.3%), 583 of 717 reentered specimens due to the reactivity of Anti-TP were qualified (81.3%), 320 of 365 reentered specimens due to the reactivity of NAT were qualified (87.7%).
Summary/Conclusions: Through the information system of modern blood station management (SHINOW9.0) for the temporarily deferred blood donors, the whole process of the identification, collection and testing results of the reentered donors was effectively managed, the loss of blood donors was reduced, and it was beneficial to the result consultations of blood donors and the follow-up care services.

P-065

ANALYSIS OF THE PROBLEMS WHICH WERE FOUND IN THE SITE INSPECTIONS IN 2016

Y Cui and F Zhao

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Background: Because blood collection and supply work is a repetitive task and it starts from scratch every day, the error is unavoidable. Blood stations have different management mode for error management. Our Blood center takes standardized operational inspection as the core of patrol as an important measure of site management and process control. The business department set up a special position and personnel, in charge of monthly site patrol of 24 blood collecting and supply departments and sites, focusing on implementation of operating procedure and evaluation of training effectiveness. The problems found in sites should be communicate with the staffs and keep records, the problems which needed to be coordinate should be reported to the relevant management department and urge to solve as soon as possible, all the problems which found in the patrol should be collected and feedback to the responsible departments, and linked to performance appraisal.
Aims: to analyse of the problems which found in the site inspections in 2016 from the center, to seek appropriate countermeasures to promote the continuous improvement of quality management system of blood collection and supply.
Methods: By summarize site inspection records and found problems according to quality management elements and analysis method, to propose the countermeasure, and then according to the five elements of man, machine, material, method and environment to classify and analyse from the aspects of quantity and proportion in 2016.
Results: Personnel factors account for the largest proportion, the second one is process.
Summary/Conclusions: To analyse the site inspections in 2016, to pay close attention and take corresponding measures to solve relevant problems, in order to further standardize and improve business processes, so as to provide qualified blood for the clinic and provide a good environment and services for blood donors.

P-066

STATISTICAL PROCESS CONTROL ANALYSIS TO VOLUME OF RED CELLS IN ADDITIVE SOLUTION

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Background: Statistical process control (SPC) is used to monitor the performance of blood component collection and production processes in the UK and elsewhere. <Quality requirements for whole blood and blood components>(GB18469-2012) In china require the volume control ranges for whole blood are 200 ml±20 ml, 300 ml±30 ml, 400 ml±40 ml, and the volume control ranges for red cells in additive solution are labeled ml (±) 10%. There are differences in the separation and preparation process of blood components in elsewhere, and part of the quality characteristics of blood components products depend on the biological characteristics of local donors. There is no definite quantitative standard for the qualified range of suspended red blood cell, so that the capacity control of quality sampling is lack of definite basis.
Aims: The volume index of red cells in additive solution collected from Urumqi blood center was analyzed by statistical process method. Assess the extent to which capacity control occurs during the operation of blood collection and processing in Urumqi blood centre. Identify possible problems in the monitoring process and generate signals to analyze the continuous quality performance of the monitoring process.
Methods: Red cells in additive solution was prepared by the whole blood that was centrifuged by hand splint. The volume was calculated by random sample weighing, dividing by product density, after deducting plastic blood bag skin weight. The product volume data of 1U, 1.5 U and 2U were statistically analyzed with Minitab software.
Results: The volume values of 3 sizes of red cells in additive solution were 151 ± 9.93 ml, 227 ± 11.95 ml, 304 ± 13.51 ml, and the volume values all accorded with the normal distribution. The Cpk of 1U, 1.5U and 2U red cells suspension control in the volume range of ±3SD (0.9, 1, 1) were higher than in range of ±10% (0.50, 0.64, 0.74). The detection sensitivity of the CUSUM control chart is higher than that of the Shewhart control chart.
Summary/Conclusions: The capacity of volume control for 1U red cells in additive solution was relatively poor, which may be related to the sex distribution of the sampling donors. The ability to control the volume of the suspended red blood cells should be improved. The data of CUSUM control graph still needs to accumulate, to fix the frequency and number of sampling and optimize the related parameter setting.

P-067

STUDY ON FUNCTIONS OF PROVINCIAL BLOOD CENTER ON REGIONAL BLOOD QUALITY MANAGEMENT

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Background: In China, there are over 430 independent Blood Transfusion Services (BTSs) under the jurisdiction of local governments at different levels. A BTS belongs to provincial government named Provincial Blood Center, while a BTS belongs to local government named Central Blood Bank or Blood Bank. The testing proficiency and the blood quality of local blood banks were limited by the relatively poorer management and less professional experience. Although most of blood banks have already established quality assurance system, there are still a lot of problems in quality internal audit in local blood banks. Law has been regulated that provincial blood center has the duties on business guide for local blood banks on regional blood quality. But faced on numerous local blood banks with broad diversity, how could a Provincial Blood Center effectively fulfill the duties, especially for those blood banks without any administrative relationship?
Aims: To share the experiences from Jiangsu Provincial Blood Center on how to develop the functions on regional blood quality management.
Methods: A blood quality management system covered 18 blood banks all over the province has been established. The system is including 3 parts, a provincial EQAS

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for quality control laboratories, a joint blood quality internal audit mechanism, and quality control business and technique training for the technicians from blood banks. Blood screening EIA testing (HBsAg, Anti-HIV1/2, Anti-HCV and syphilis serologic testing), biochemical quantity testing (ALT, Total plasma protein, Sodium ion and Potassium ion, Methylene blue), blood cell quantity testing (Hb, Hct, Count of WBC, RBC and PLT) and blood coagulation quantity testing (Eighth coagulation factor and Fbg) were included into EQA programs.
Results: Most of laboratories achieved good results in EQA, except for some laboratories had some problems on biochemical quantity testing and blood coagulation quantity testing. EQA can improve the testing proficiency of blood quality laboratories of blood banks, and blood quality can be improved based on the strict and accurate quality testing by blood quality control laboratories. The joint blood quality internal audit mechanism can help to enhance the function of internal audit for finding the potential problems of the blood banks. The training, of course, was good for improving ability of the staffs on blood quality control and management.
Summary/Conclusions: Although the blood quality management system still need further improvement, Jiangsu Provincial Blood Center has find a good way to enhance the functions on blood quality guidance for regional blood banks.

P-068

TREND ANALYSIS OF LEUKOREDUCED ERYTHROCYTES BY RATIO ANALYSIS

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Background: The 2012 edition of "technical regulations for blood stations" puts forward the trend analysis of the sampling results of blood quality control, but the specific methods of trend analysis are not clearly defined.
Aims: The ratio trend analysis method is used to unify the test results of different items by ratio conversion, and the variables of different units are drawn in one graph. Through horizontal comparison with blood between different projects, and longitudinal comparison of varieties of blood,the complex and tedious variable relationship will be reflected simply. Through analysis the change of each variable trends in chart, the risk and the problem among them will be found, it may offer some guidance of blood collection, preparation, testing, provision, so as to help improving the blood quality.
Methods: Standard deviation rate = (measured value-standard value)/standard value, in%; with standard rate of zero baseline value. Take the blood bar code as the abscissa, deviation standard rate as the ordinate, draw broken line trend chart. According to the "Blood and blood components quality requirements ", the exchange rate of leukoreduced red blood cells as the standard of quality control: capacity (±10%), hemoglobin (Hb) content is more than 0, hematocrit (HCT) (±14%), residual leukocyte is less than 0, the final storage hemolysis rate <0.
Results: According to the 2016 ratio trend chart of leukoreduced erythrocytes quality control data, all the items met the requirements except that the capacity of data appeared to exceed interval for 19 times. Among them, the capacity has negative deviation with 40% data beyond interval, combined with the blood volume trend positive deviation and steady trend line analysis that the filter manufacturers, maintenance liquid categories and changes of preparation process which lead to the high capacity internal control standard of current set. At the same time, during the blood preparation process, the residue in the filter and the multi bag catheter,the two centrifugal loss in the plasma preparation are also factors of negative deviation. Hb content present positive deviation which indicates that the blood donors' health consultation is reasonable. At the same time, the quality control standard of blood Hb content is low because it is far away from baseline. The HCT trend was stable and distributed on both sides of the baseline, indicating that the blood donors' health consultation and component preparation is reasonable.
Negative deviation and away from the baseline of leukocyte residual indicate that leukocyte filtering link is reasonable, while because of the detection method of blood storage time, the changes of leukocyte morphological, personal experience and different skills of operators, which all result in high fluctuations of the trend. Comparing with the whole blood, the trend fluctuations of the final hemolysis rate in leukocyte suspending erythrocytes is bigger. Considering the preservation of blood, the filtration of ingredients and the plasma fractionation,which all cause damages in red blood cells.
Summary/Conclusions: Using ratio analysis method to analyze the trend of component blood, which can not only be classified according to blood type, but also classified by human, machine, material, method, surroundings, a variety of complex relationships.It can make presentation easily and concisely in one trend map. It not only fully improves application efficiency of the quality control results, but also provides help for improving the quality of blood preparation.

P-069

THE REGULAR MAINTENANCE AND CALIBRATION OF BLOOD STATIONS SEMI-AUTOMATIC BIOCHEMICAL ANALYZER ROLE IN LOWERING BLOOD ALT SCRAP

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Background: Current semi-automatic biochemical analyzer has been done in various blood stations before donating blood ALT screening applications, no matter dry or wet all achieved good effect, greatly reduces the for ALT unqualified cause blood collection and scrap. The standard of the 2015 version of blood station technical operation procedure for the determination of alanine aminotransferase is "ALT no more than 40 U/l to be ALT no more than 50 U/l", and this adjustment will also greatly reduce the failure rate of ALT. Currently, there are many reports on the comparison of biochemical analyzer, and the results are very good. However, there is no uniform method for the problems of the subsequent use of semi-automatic biochemical analyzer. We adopted the methods recommended in the CLSI ep15-a guide. gave us a good effect for nearly five years of experiments on 16 semi-automatic biochemical analyzers.
Aims: In order to reduce the blood for ALT unqualified scrap rate through regular maintenance and calibration of a semi-automatic biochemical analyzer.
Methods: Every 3 months CLSI EP15 - A guidelines recommend method for validation: 5 day, an analysis of a day, daily 2 levels, each level the same specimen repeat 3 times. If because of the quality control program, or operation to a batch is out of control, eliminating data, and perform additional batch. The operator shall be carried out in accordance with the manufacturer specification operation.
Results: After nearly five years of experiment semi-automatic biochemical analyzer was found after every 3 months of maintenance and maintenance, ALT of scrap rate continues to decrease 5.15%(3 828/74 320) 2.14%(1 633/76 313) 2.22%(1 759/79 232) 1.57%(1 228/78 238) 0.68%(574/84 406) ($P < 0.01$).
Summary/Conclusions: The CLSI EP15 - A guide to methods recommended by regular maintenance and calibration for semi-automatic biochemical analyzer could have a effectively control blood ALT scrap rate, which is worthy of reference for colleagues.

P-070

COMPARISON OF 3 PRETREATMENTS FOR ELIMINATION HYPERLIPIDEMIA ON THE DETECTION OF ALT

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Background: To assess the function of 3 different pretreatments for reducing the interference on the determination of ALT in the hyperlipidemia blood samples .
Aims: To assess the function of 3 different pretreatments for reducing the interference on the determination of ALT in the hyperlipidemia blood samples .
Methods: (1)To detect ALT and triglycerides (TG) concentrations of the different simulation lipid blood specimens. (2) The ALT of high concentration of lipid sample (lipid concentrations of more than 5%) were detected that were pretreated by 3 different pretreatments. (3) 46 limpid samples were selected randomly and ALT concentrations were detected by 3 different pretreatments.
Results: (1) Low concentration of lipid sample has no significant effect on ALT, and lipid concentrations of more than 5% has serious interference on the ALT detection. (2) The effects of the 3 different pretreatments on the high concentration of lipid samples are different. By contrast, the 10% PEG and ultracentrifugation are better than ether extraction. (3) ALT concentrations of 46 limpid specimen randomly collected were post-treated by the 3 different pretreatments,the results showed no significant different between PEG and ultracentrifugation and original value,but the ether extraction was lower than the other 2 methods and the original value.
Summary/Conclusions: (1) When the lipid concentration reaches to 5% or more,the ALT test will be interfered.(2) The study showed that PEG and ultracentrifugation can effectively eliminate the influence of lipid on the detection of ALT,but compared to the PEG method was more economy and convenient.

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P-071

RETROSPECTIVE ANALYSIS OF EQAS RESULTS ON BACTERIAL TESTING IN JIANGSU BLOOD BANKS FROM 2015 TO 2017

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Background: Bacterial testing is one of the sampling observation programs of blood products in blood banks in China. Due to the small number of sampling per month and the enforcing strict disinfection procedures before blood donation, lots of blood banks have not detected bacteria over the years. CITIC EQAS on Bacterial Testing can answer the wonder about the reliability of results. From July 2015, 12 blood banks in Jiangsu Province had participated in CITIC Bacterial Testing EQAS.
Aims: To summarize and analyze the results obtained from CITIC EQAS on Bacterial Testing for Blood Screening from 2015 and 2017, and to improve bacterial detection performance in the quality control laboratory of 12 blood banks in Jiangsu Province.
Methods: The EQA samples were tested and results were reported from July 2015. The bacterial strains from the positive culture flasks were identified. The results of 12 blood banks' laboratory were summarized and compared with the expected results. The timing of the positive results was also analyzed.
Results: Five batches of bacterial strains samples were detected by 2017. Two false positive results and three false negatives results were identified. The coincidence rate were gradually increased from 96.6% to 100%, while the variable coefficient range of the detection time of the positive samples were 8.4% to 38.1%.
Summary/Conclusions: Participating in the CITIC EQAS on Bacterial Testing for Blood Screening is helpful to determine the factors leading to false negative and false positive results and to improve the accuracy and reliability of bacterial detection. The retrospective analysis of EQAs results facilitates the standardization of specifications in the testing process.

P-072

RISK ANALYSIS AND CONTROL OF REGIONALIZED CENTRALIZED NUCLEIC ACID DETECTION

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Background: In 2015 December Yunnan Kunming Blood Center began to nucleic acid centralized detection of Yunnan Province, this research explore that how to safely and effectively carry out the centralized testing.
Aims: Establish and improve the quality system of centralized detection by the risk analysis and risk control, To ensure the smooth implementation of centralized detection, and to explore the application of risk management in the centralized detection process.
Methods: Formulate a index matrix of risk evaluation, Identify the risk points that may occur during centralized detection, put forward risk control measures, reduce the risk of centralized detection.
Results: Divided into three levels of risk points to acceptable risk, increased risk and unacceptable risk by the index matrix of risk evaluation, develop appropriate control measures for different levels risk. Retrospectively analyze the effect of control measures we found that every point residual risk is reduced.
Summary/Conclusions: Establish and improve the quality management system what conform features of centralized detection and key process, can reduce the detection risk caused by various reasons. To ensure implemented centralized detection with high quality, efficient and orderly manner.

P-073

THE INTERVAL ANALYSIS OF THE IDENTIFIED/RESOLVED RESULTS OF THE TESTS IN TWO NUCLEIC ACID DETECTION SYSTEMS

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Background: Most blood services are currently carrying out the nucleic acid detection technology, nucleic acid detection technology has improved the detection sensitivity, at the same time, also inevitably produce false positives, cause the waste of blood to resources. The identification/resolution rares of the e-n + are generally not high. This paper aims to analyze whether the identification/resolution rares of the e-n + are different in different intervals.
Aims: To analyze the signal ratio (SR) and CT value distributions of the test results of Grifols Procleix TIGRIS system and Shanghai haoyuan CHitas Bss1200 system, and provide the basis of increasing the rate of E-N+ identification/resolution and perfecting the detection strategy.
Methods: A nucleic acid test was conducted for the 102176 serology negative samples of the blood donors from January 2016 to December 2016, using the detection systems of Procleix TIGRIS and CHitas Bss1200, the reactive samples in the first test were identified/resolved, and analyze the results according to the intervals.
Results: The identification rate of the group of $15 \leq SR<20$ was significantly higher than that of the group of $10 \leq SR<15$ ($P<0.05$) in the Procleix TIGRIS system; the resolution rate of the group of $39 \leq CT<41$ was significantly lower than that of the group of $37 \leq CT<39$ ($P<0.05$), there was no statistically significant difference between the group of $35 \leq CT<37$ and $37 \leq CT<39$ ($P > 0.05$).
Summary/Conclusions: The identification/resolution rares of two nucleic acid detection systems have significant differences in different intervals, it is necessary to follow up these blood donors and verify, perfect the detection strategy and increase the identification/resolution rate.

P-074

APPLICATION OF TREND ANALYSIS IN THE STATISTIC ANALYSIS OF THE BLOOD QUALITY CONTROL

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Background: It has been stipulated in Blood Station Technical Operation Procedure since 2012, that the trend analysis should be carried out for the sampling results of blood quality control, however, the analysis method is not specified. For apparent volume results, the mean value of monthly weighing data can be calculated to draw a line chart with an error bar for trend analysis. For sampling that needs to open blood bags, such as Hb and Hct in red blood cells in additive solution leukocytes reduced, etc., eligibility is currently determined by equal to or greater than 3 bags of qualified products in 4 bags of detected products. Due to small sample size, this determination method lacks persuasiveness. In this paper, results were determined by the t-distribution table, P25 values of monthly data were calculated to draw a line chart, and trend analysis was conducted after combining with the scatter diagram drawn using monthly data.
Aims: Taking the statistical analysis of the quality control data of red blood cells in additive solution leukocytes reduced as an example, the application of trend analysis in quality control was illustrated.
Methods: The quality control items for red blood cells in additive solution leukocytes reduced including Hemoglobin (Hb) content, Hematocrit (Hct) and apparent volume data were detected and extracted, a scatter diagram and a line chart were drawn to describe the trend of numerical changes, and monitoring was performed based on the trend chart.
Results: Results of each quality control item for blood cells in additive solution leukocytes reduced were qualified. P25 value of monthly Hb content was greater than 18 g/U, showing a slight increasing trend. Hct showed deviations twice, which were determined as qualified by the t-distribution table, with a trend fluctuating within the qualified range. Monthly data of apparent volume fluctuated up and down around the labeled amount.
Summary/Conclusions: Trend analysis can be effectively applied in quality control.

P-075

DWYER PRESSURE MEASURING INSTRUMENT USED IN BLOOD STATION NUCLEIC ACID LABORATORY PRESSURE MEASUREMENT

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Background: Blood station nucleic acid testing laboratory pressure system is an important guarantee to prevent contamination of nucleic acid laboratory. Most blood stations now rely on laboratory pressure gauges to display laboratory pressure.
Aims: Effective real-time monitoring of nucleic acid screening laboratory pressure gradient, Verify the effectiveness of the pressure display system installed in the laboratory, Prevention of laboratory pollution caused by stress systems.

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Methods: Use the handheld Dwyer - 477 to measure the pressure between the partitions and the corresponding buffers, Compared with the screen display pressure device, the error rate is calculated.
Results: The Dwyer - 477 measuring instrument has a margin of error of 10.0%>22.7% compared with the original pressure detection device, which can be used within the agreed range. After the biological safety cabinet is opened, the pressure in the preparation room (2 room) of the specimen is greatly affected, so as to avoid pollution, try not to open the door and transfer window in this state, reducing the flow of personnel.
Summary/Conclusions: Dwyer - 477 pressure measuring instrument with high precision, easy to carry, the advantage of LCD digital display and can be calibrated, one is for our regular inspection and calibration screen system, better monitoring pressure distribution of the entire lab, 2 it is also can be used as a backup device screen system, better prevent DNA laboratory pollution, ensure quality of blood test.

P-076

ANALYSIS ON THE EFFECT GENERATED IN QUALITY MANAGEMENT OF BLOOD STATION BY QUALITY ANALYSIS MEETING SYSTEM

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Background: After establishment of quality analysis system, to identify problems and potential risks through dynamic monitoring of the whole process of the blood collection and blood supply, and take timely measures to achieve continuous improvement and explore the role of quality analysis meeting plays in improving quality management levels of blood station.
Aims: After establishment of quality analysis system, to identify problems and potential risks through dynamic monitoring of the whole process of the blood collection and blood supply, and take timely measures to achieve continuous improvement and explore the role of quality analysis meeting plays in improving quality management levels of blood station.
Methods: Problems and potential risks were detected through weekly, monthly and yearly three-level quality analysis during the whole process of blood collection and supply, rectifications were made or preventive measures were taken.
Results: Problems and influencing factors were detected through quality analysis and improvement plan was raised to ensure the quality of blood and promote the continuous improvement of quality management system. The scrapping rate of each index was significantly reduced ($P < 0.01$), saving valuable blood resources and ensuring blood quality.
Summary/Conclusions: The application of quality analysis meeting in quality management of blood station makes the data analysis of each work an important means to improve work quality and management skills. It fully reflects the principle of prevention first, plays the driving role of quality analysis meeting in blood quality, achieving continuous improvement in work quality of the blood station, ensuring full implementation of the quality management and playing an important role in ensuring the blood quality.

P-077

NUCLEIC ACID TESTING IN THE INNER MONGOLIA BLOOD CENTER

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Background: With the continuous development of blood transfusion in China, the prevention and control of transfusion-related infectious diseases and ensuring the safety of blood products are the most important tasks in the work of the national blood supply agencies. Therefore, the detection of blood nucleic acid with high sensitivity and "window period" is the inevitable trend of the development of blood screen in the future.
Aims: In order to raise the level of blood safety, our center has been preparing nucleic acid test since the end of 2010, and the nucleic acid test is officially started in August 2011.
Methods: During the construction and management process of nucleic acid testing laboratory in recent years, we have explored and summarized the study carefully. It not only prevents the laboratory pollution, but also guarantees the quality of nucleic acid test, and also accumulates a lot of theory and practical experience on nucleic acid detection.
Results: This article from the laboratory Settings, file system building, personnel qualifications and training, specific experimental operation, between the chamber and qualitative evaluation of indoor quality control and pollution prevention in the laboratory and a few aspects, such as the center of the nucleic acid detection are introduced in detail, quality control and management experience.
Summary/Conclusions: Nucleic acid laboratory should identify when conducting nucleic acid testing laboratory quality control and management of the key links, in addition to establishing various rules and regulations, standard operating procedures and strict training operators, for different detection system to be targeted to take effective measures, to ensure that the experiment of nucleic acid detection quality.

P-078

BLOOD STATIONS ENZYME FROM LAB QUALITY CONTROL IN LABORATORY ANALYSIS

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Background: To improve laboratory testing ability under the condition of fine management in laboratory.
Aims: The laboratory quality control of enzyme free laboratory was analyzed.
Methods: The quality control status of HBsAg, HCV and HIV in the enzyme free laboratory from 2014 to 2016 was analyzed retrospectively.
Results: The average CV value in 2016 was obviously higher than that in 2014 and 2015, the difference was statistically significant ($P < 0.05$), and the CV value decreased gradually from 2014 to 2016.
Summary/Conclusions: In recent years, I stand enzyme free laboratory, indoor quality control basically qualified, there is still room for improvement.

P-079

THE ANALYSIS OF 69 CASES OF POOR QUALITY BLOOD SPECIMEN

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Background: Unpaid donors to donate blood, the quality of blood specimen is very important, such as blood specimens does not comply with the requirements, directly affect the testing results of light, to cause blood scrap was unable to complete testing. Weight loss leads to the failure of clinical blood to cause major medical accidents. It is well known that the test link is very important in the whole blood supply. The results directly affect the quality of the finished product.
Aims: This paper discusses the rate of non-conformity of the quality of non-remuneratory blood donors and their causes.
Methods: I stand in the 69 cases of substandard blood samples collected from May 2011 to May 2016, and all the relevant professionals are examined and analyzed for the reasons of deferral.
Results: The causes of poor blood samples were analyzed, 40 of them were specimens of hemolysis, accounting for 58.0%. 12 cases of fat blood, 17.4%; Seven cases were inadequate, accounting for 10.1%, and six for test tube tags, accounting for 8.7 percent. Four cases of untimely centrifugation (test tube) were 5.8 percent.
Summary/Conclusions: In blood transmission of CaiGongXie system disease, blood specimen collection and test process is a very important basis. In the process of blood specimen collection, need each of standard operation procedures, to ensure accurate test results.

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Blood Donation
Blood Donor Recruitment

P-081

THE STATUS AND NEW TENDS IN ADVERTISING AND RECRUITMENT FOR VOLUNTARY NON-REMUNERATED BLOOD DONATION IN CHINA

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Background: With the development of medical technology and the implementation of the policy of universal health care in China, the demands on blood for clinical use have presented a gradual increase as well. Although the blood supplied by medical institutions of different regions keeps rising, the rate of blood donation in China up to now is only 10‰, and the growth rate of blood supply is still slightly lower than that of blood demands. Meanwhile, the regional and seasonal blood shortage happens from time to time, which indicates that the blood shortage has become the main challenge that all blood donation and supply institutions need to face.
Aims: To provide a safe, adequate and effective blood supply for clinical use by strengthening the advertising and recruitment for voluntary non-remunerated blood donation, and attaching the importance of the innovation of new theories and practices for the blood donors' recruitment and service.
Methods: Apart from laying emphasis on different methods of advertising and recruitment of blood donors, such as news report, blood donation publicity on streets, telephone and messages as well as volunteers' advertising, all blood donation and supply institutions are supposed to create new methods by joining hands with Wechat, Microblog, brand projects, such as brand Items "Red Summer" and "Warm Winter held by Beijing Red Cross Blood Center since 2009", "Red Action" at Shenzhen Blood Center since 2011 as well as the brand licensing activities at Shanghai Blood Center since 2014.
Results: Among the voluntary non-remunerated donors, the number of whole blood donors rises from 11.890 million in 2012 to 12.915 million in 2016, the growth rate of which is 1.08%,1.13%,1.31%,4.89% from 2013 to 2016 each year, and the units of whole blood rise from 20.017 million U in 2012 to 21.513 million U in 2016, the growth rate of which is 0.52%,1.77%,1.65%,3.35% from 2013 to 2016 each year. Meanwhile, the number of blood component donors rises from 0.675 million in 2012 to 0.949 million in 2016, the growth rate of which is 12.30%, 8.44%, 7.79%, 7.11% from 2013 to 2016 each year, and the units of blood component rise from 0.965 million U to 1.477 million U, the growth rate of which is 18.13%, 9.65%, 9.84%, 7.57% from 2013 to 2016 each year. The amount of blood donation in China has been increasing steadily, among which the donation of whole blood speeds up while the blood component slows down a little.
Summary/Conclusions: Centered on blood donors, scientific and effective advertising and recruitment, which plays a pivotal role in solving the problem of blood shortage, are greatly advised to make full use of traditional and new media, brand project and licensing activities. Additionally, it has become a trend that the future recruitment, which aims to attract growing social participation and support, will be distinctive owing to the unique features of local conditions. Besides, making appointment to donate blood may become a reality in the near future.

P-082

IRON DEFICIENCY AND ANEMIA AMONG BLOOD DONORS IN CHINA

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Background: Iron deficiency is a global health issue estimated to concern as many as 2.7 billion people and explain half of the cases of anemia worldwide. Iron deficiency in blood donors is taken more attention in developed countries due to the interval and frequency of whole blood donation. In China, the interval of whole blood donation (WBD) is six months and the minimum hemoglobin (Hb) level is 115 g/l for women and 120 g/l for men. No report on iron deficiency as a result of a whole blood donation in China is available.
Aims: This study aimed to estimate the prevalence of iron deficiency in voluntary blood donors and provide basic data to determine whether it is reasonable to shorten the interval of WBD in Jiangsu, eastern China.
Methods: The study was conducted at the mobile vehicles in Jiangsu Province Blood Center and involved 149 men and 80 women who passed the Health Check Questionnaire including of Hb level by copper sulfate assay. Serum ferritin was determined by ELISA kit (Elabscience, China) on Hamilton ELISA working platform (SRAR line, Switzerland) and red blood cells (RBC) parameters of complete blood count was determined anemia by a hematology analyzer (SYSMEX KX-21, Japan). We have analyzed the correlation between the RBC parameters (RBC count, Hb, HCT, MCV and RDW). Socio demographic information and donation frequencies were collected.
Results: After passing the Cu2SO4 assay, Hb level of the male donors were all up to grade by hematology analyzer and the average was 149.36 g/l. All of them had normal RBC count, HCT and RDW, only one of them had low MCV (70 fl). The average Hb level of woman donors was 127.93 g/l and three were lower than criterion (1 was 99 g/l and 2 were 110 g/l). Among them, only 1 with low ferritin level (8.67 g/l) and MCV (78.6 fl) was anemia. All of them had normal RBC count and RDW. Seven of them had low HCT (<36%). The prevalence of iron deficiency (ferritin < 12 μg/l) was 8.00% (18/225); 13 were female and 5 were male. Iron deficiency prevalence between them was significantly difference ($\chi 2 = 14.98$, $P = 0.0001$). No difference in iron status was between the first donation and repeat donation. ($\chi 2 = 0.15$, $P = 0.6985$).
Summary/Conclusions: Anemia, iron deficiency and iron-deficiency anemia prevalence are not higher than expected and that of other countries in current donation interval in China. It should be pay more attention on iron deficiency of female donors. Further investigation need to observe whether the prevalence of iron deficiency and anemia increase after the interval is changed to three months, especially among regular blood donors and female blood donors.

P-083

A NOVEL APPROACH BY SERUM FERRITIN AND HEMATOLOGIC DISORDER SCREENING FOR IMPROVING BLOOD DONOR HEALTH AND DONATION SAFETY OF HEMOVIGILANCE IN THAILAND

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Background: Making safety of blood donation is major role. A significant disadvantage of blood donation is the potential risk of iron deficiency.Blood donation leads to iron loss, as about 0.5 mg iron is lost per each milliliter of blood donate.In Thailand iron deficiency is a common public health problem,and the various thalassemia and hemoglobinopathies are prevalent.
Aims: This paper aims to study the integrate the serum ferritin determination, and the thalassemia trait with hemoglobinopathy screening program to improve a donor health and safety donation in routine hemovigilance program in Thailand.
Methods: The volunteer Thai blood donors 49 male and 80 female donors were preliminary studied. The donors were screening for Hb determination by Hemocue. The blood samples were investigated for various hematological parameter on automatic hematological analyzer (Sysmex Xt 1800 i). Serum ferritin was estimated using chemiluminescence principle (Abbot Architect i 600).Before transfusion, all blood were test for HIV, hepatitis B and C by NAT (Roche cobas TaqScreen MPX test) and syphilis (Cobas).
Results: In 129 individuals it was found 20.15% of non acceptable blood donation that depleted iron stored, DIS (SF < or = 15 ng/ml) was found 69%. Interestingly,the acceptable blood donation was found DIS 35.22% which Hb E trait 13%,beta thalassemia trait 2.91% and hereditary persistence of fetal hemoglobin (HPFH) 1 case.
It was revealed that DIS was not significantly different of number donation (17.84% in first vs 18.18% in repeated donation), and was found increased in female (27.27%) than male (7.95%). Nevertheless, microcytic anemia was found 3.40% in first blood donation and 6.81% in repeated blood donation, respectively.
Summary/Conclusions: This paper could suggest that integration of determination iron store status and thalassemia trait and hemoglobinopathy examination are necessary to be an additional tests of routine blood donor screening in Thailand.

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P-084

INTERVENTIONS TO PROMOTE BLOOD DONATION IN NON-REGULAR DONORS-A SYSTEMATIC REVIEW

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Background: The recruitment and retention of blood donors are the key to blood safety and sufficiency worldwide. Evaluating interventions promoting blood donation is a prerequisite for motivating blood donors. Given the different determinants of donation behavior between high baseline motivation and low baseline motivation donors and that two-third of first-time donors would stop after their first donation, it is much more meaningful to motivate non-regular donors to become regular donors.
Aims: To evaluate the effectiveness of interventions promoting blood donation targeting non-regular donors with low baseline donation rate.
Methods: Data sources: Pubmed, CINAHL, EMBASE, Cochrane, PsycINFO and the reference list of previous reviews. The search strategy consisted of five key words with their variations: randomized trial, blood, donor, intervention and promotion. In all databases, keywords are combined by Boolean operators "AND", "OR" and "NOT"(Transplantation related studies were excluded using "NOT"). Study eligibility criteria: only RCT or quasi-experiments are included, population are non-regular donors including potential donors from the healthy population, novice donors (1 to 4 donations), temporarily deferred donors and lapsed donors. All interventions targeting blood donation promotion are included, outcomes should include donation behavior or incidence of negative reaction, language is English. Study appraisal is done using the Jadad score. The original 2 × 2 table was used to calculate OR. Heterogeneity was analyzed using a $\chi 2$ test (Cochran's Q) and I2, pooled effect sizes (OR) were calculated in subgroups with at least 3 interventions using R.
Results: There were totally 7192 studies identified, 24 studies with 35 different interventions (S = 35) were included in the review ultimately. "Cognitions based" group (S = 11, pooled OR=1.74, 95%CI [1.17,2.57]) and "Reminder" group (S = 3, pooled OR=1.55, 95%CI [1.30,1.86]) showed significant effect with small effect sizes (Cohen's d in each group were less than 0.20). "Altruism" group (S = 8, pooled OR=1.30, 95%CI [0.99,1.71]), "Measurement of cognitions" group (S = 4, pooled OR=1.08, 95%CI [0.98,1.20]) and "Incentive" intervention group (S = 4, pooled OR=1.22, 95%CI [0.72,2.06]) had insignificant effect. More studies with high validity were needed to draw conclusions in the remaining three groups with five interventions.
Summary/Conclusions: This review includes four types of donors sharing the common characteristic of low baseline donation rate or having not formed the habit of blood donation in other words, so that they all should be a target of promotion. But there are three limitations including small number of studies in three categories, relatively high heterogeneity in three groups and low Jadad scores in some studies, which implies that validity needs to be improved in future studies. This review finds that "Reminder" and "Cognition based" interventions are two effective interventions. As for interventions of "Modeling", "Foot-in-the-door/door-in-the-face", each subgroup in behavior intervention and "Implementation intention", more studies with high validity are needed to draw conclusions about their effectiveness in promoting blood donation among non-regular donors due to the limit number of study.

P-085

Abstract has been withdrawn

P-086

ANALYSIS OF DONORS GIVING BLOOD AT FIXED SITES AND THROUGH MOBILE CAMPAIGNS, WITH THE AIM TO DEVELOP A STRATEGY FOR EFFECTIVE DONOR RECRUITMENT AND RETENTION

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Background: Increased aging population could cause imbalance of need and supply of blood in the near future; therefore, making efforts to increase donation rate of young adults and increase regular donors in the donor pool are mandatory. Previous studies indicated that there were differences in donation patterns between donors donated in fixed site and mobile site. To suggest effective strategies for recruiting and retaining donors, we need to know the characteristics of donors donating at different donation sites.
Aims: 1. To characterize blood donors donating at different donation sites in Taiwan.
2. To develop a collecting strategy for blood centers, aimed at long-term development, in keeping with a proper contribution of blood between first time and regular donors.
Methods: Donors who donated whole blood (WB) at least once in 2014 were included in the analysis. The type of donation sites were classified into three groups: fixed sites (FS), donation campaigns such as social groups, companies or other organizations held in mobiles sites (MO), and donation campaigns at the military camps or the schools held in mobile sites (MMS). The distribution of donor types (including the first-time donors in 2014, the regular donors who donated blood at least twice in recent two years, and the returning donors who did not donate blood in 2013 but returned in 2014), averaged number of WB unit per donor, and the first and second return rate in two-years among the first-time donors were calculated by the type of donation sites. Multiple logistic regression analysis was used to estimate odds ratios (OR) and their 95% confidence intervals (95%CI). All data analyses were conducted using SAS 9.4.
Results: In 2014, there were 456232, 471996 and 161696 WB donors contributed 739949, 662663 and 194351 WB collections in FS, MO and MMS, respectively, and 47.9% of the first time donors were recruited from MMS. The average units of WB donation per donor was significantly higher in FS (2.38, 95% CI: 2.37 to 2.38) than that in MO (2.03, 95% CI: 2.03 to 2.03) and in MMS (1.51, 95% CI: 1.51 to 1.52), and this phenomenon is the same in all age groups in both genders. Highest percentage of the regular donors was observed in FS (72.5%) than in MO (64.8%) and MMS (43.8%), and regular donors are safer blood source than others with the lowest confirmed HBV, HCV or HIV positive rate (0.09 per 1,000 vs 1.72 per 1,000 in returning donors and 16.91 per 1,000 in first time donors, $P < 0.0001$). In addition, the first return rate of the first time donors recruited at FS (50.0%) was significantly higher than that at MO (44.8%) and at MMS (46.1%), and the adjusted OR was 1.28 (95% CI: 1.25–1.32) at FS and 1.10 (95%CI: 1.06–1.13) at MO compared with that at MMS. Furthermore, among the first time donors who had their first return donation, the second return rate was significantly higher in donors whose first return donation was done at FS (OR=1.4, 95% CI:1.36–1.45) compared with those at MO or MMS.
Summary/Conclusions: Blood centers should have different strategies for collecting blood suitable for local conditions. Mobile is still the main donation site to recruit new donors, especially for young donors from MMS. Blood collections from FS were observed with a main contribution to the demands of blood, the safety of blood, and the retention of blood donors; therefore, FS has advantage of long-term donor management.

P-087

THE INTENTION OF RETURN DONATION AMONG THE LAPSED YOUNG DONORS AFTER THEIR FIRST DONATION – SOUTHERN TAIWAN'S EXPERIENCE

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Background: In Taiwan, young donors play a major part among our first-time donors. In 2012, among all the 164368 first-time donors, the number of aged 17–24 young donors was 119653,which accounted for 72.8% of all first-time donors. However, only 30.7% of these young donors would keep donating blood next year, which means that nearly 70% of them didn't become regular donors after their first-time donation. There is a need to identify factors explaining why some first-time young donors stop donating blood.
To address this phenomenon, in 2015, Taiwan Blood Services Foundation launched a campaign called "Young Blood", aiming to encourage the continuous blood-donation of first-time young donors.
Aims: Discuss the connection among lapsed return intention of young first-time donors, attitude of blood-donation (ABD), experience of blood-donation (EBD), and knowledge about blood-donation (KBD); analyze if demographic attributes will impact the donors' return intention; investigate the current situation of young donors' continuous donation in Taiwan after "Young Blood" campaign.
Methods: A total of 13339 first-time donors, who donated in 2013 and were disappeared in 2014, were invited to participate the study. We mailed them a questionnaire assessing their intention of blood donation, ABD, EBD and KBD. Decision tree analysis and one way ANOVA test analyses were adopted to assess the effects of ABD, EBD and KBD on intention of blood-donation. The relationship between intention of blood-donation and donor demographics was explored.
Since the promotion of "Young Blood" campaign in 2015 by Taiwan Blood Services Foundation, we have designed brand new visual identities, adopted new activity themes, assigned endorsers, produced a series of brand-marketing-based micro films, and sent after-donation gifts suitable for young donors. We hope to find out

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solutions to bridge the gap between benign intention and behavior, so that first-time young donors can be encouraged to donate blood regularly.
Results: We received 937 answered questionnaires (7%).Age between 17–30, college graduates, and females are 3 groups of donors that account for the majority of study participants and age between 17–22,college graduates have higher return intention ($P < 0.01$). The ABD, EBD and KBD are positively associated with their return intention ($P < 0.05$).Among all ABD, "Altruistic behavior" is the most associated with return intention than "Selfish behavior"($r = .364$, $P < 0.01$). Study participants who worked in service industry, younger than 22 years old, and donated on mobiles, these groups of donors have shown higher return intention ($P < 0.05$).
After the campaign "Young Blood" was launched by Kaohsiung Blood Center, in 2016, the number of young blood donors increased by 1.2%.(26364 persons in 2015, 26675 persons in 2016);blood donations increased by 0.3%.(35401 donations in 2015,35498 donations in 2016).
Summary/Conclusions: This study proved the relationship between first-time young donors' intention, attitude, experience, and knowledge of blood donation. The finding helps us in understanding the first-time donors' database and to adapt our donor recruitment and retention practices accordingly. As the promotion of "Young Blood" campaign in 2015 by Taiwan Blood Services Foundation, the situation of lapsed young donors has been gradually improved.

P-088

ANALYSIS OF PLATELETPHERESIS DONOR PERMANENT DEFERRAL CAUSES IN CHINA CHONGQING AREA: THE SEROLOGICAL STUDIES OF TRANSFUSION-TRANSMISSIBLE INFECTIONS CHARACTERISTICS

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Background: Apheresis component collection is a rapidly growing area in the blood collection field. Single donor plateletpheresis has numerous advantages over random donor platelets which include decreased risk of the transfusion-transmitted infections, bacterial contamination and all immunization. Because of these advantages, plateletpheresis supply is increasing year by year, which need for more plateletpheresis donors. In developing states like China, shortage also exists and the major plateletpheresis donors are voluntary donations and a few of family replacement blood donations. To ensure a safe blood supply, plateletpheresis donors screening is particularly important. In this study, we conducted a retrospective analysis for plateletpheresis donor permanent deferral in Chongqing blood center. The prevalence of transfusion-transmissible infections hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency (HIV) and treponema palladium (TP) among family replacement donors and voluntary were analyzed.
Aims: To ensure an adequate and safe blood supply, necessary changes in donor recruitment and retention policies could be achieved and it is essential to analyze the reasons for donor permanent deferral.
Methods: The study was undertaken in Chongqing blood center during the period from January 2015 to December 2016. ELISA and NAT were applied to assess the hepatitis B virus (HBV) surface antigen, antibodies against hepatitis C virus (HCV), human immunodeficiency virus (HIV) and treponema palladium (TP) of plateletpheresis donors.
Results: A total of 17,342 plateletpheresis donors in Chongqing blood center were enrolled in this study during the period from January 2015 to December 2016. Of those, 13,151 (75.83%) were males and 4,191 (24.17%) were females, male-female ratio being observed as 3.14:1. Among all the plateletpheresis donors, 3,642 (21.00%) were under the age of 25, followed by 26–35 years age group (7,143/17,342, 41.19%), 36–45 years age group (3,895/17,342, 22.46%), 46–55 years age group (2,423/17,342, 13.97%) and 56–60 years age group (239/17,342, 1.38%). Replacement and voluntary donors contributed 5,305 (30.59%) and 12,037 (69.41%), respectively. Among all the plateletpheresis donors, 194 (1.12%) were deferred because of seropositive serology. Replacement and voluntary deferred donors comprised 109 (2.05%) and 85 (0.68%), respectively ($P < 0.05$). Among the deferred donors, 194 (1.12%) were seropositive for HBV (77/17,342, 0.44%), followed by HCV (49/17,342, 0.28%), TP (42/17,342, 0.24%) and HIV (26/17,342, 0.15%). Prevalence deferred females contributed 67 (1.60%), while males contributed 127 (0.97%) of the deferred cases. Deferral rate was highest among 46–55 years age group (40/2423, 1.65%) followed by 36–45 years age group (63/3895, 1.63%), In the other groups are <1%.
Summary/Conclusions: The results demonstrated that the low-risk plateletpheresis donor population was voluntary. To ensure an adequate and safe blood supply, we should strengthen to publicize blood donation knowledge before plateletpheresis donation. Increasing the number of voluntary donors is the best way to achieve the safest blood, cause of they are younger and the low-risks of transmitting different viral infections.

P-089

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P-090

EVALUATION OF REPLACEMENT OF COPPER SULFATE GRAVIMETRIC METHOD BY A NON-INVASIVE METHOD FOR BLOOD DONOR HEMOGLOBIN SCREENING

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Background: Measurement of hemoglobin (Hb) prior to whole blood donation ensures the donors' health and the quality of blood components. Except invasive methods, there are currently several point-of-care (POC) non-invasive methods available on the market to determine Hb level. In Taiwan, the fingerstick copper sulfate (CuSO4) gravimetric method has been the primary Hb screening test for whole blood donation for several decades. Although the CuSO4 method is fast, cost-effective, and easy to perform, it has disadvantages, such as being a non-quantitative measurement of Hb, having the difficulty of performing quality control, and some other drawbacks, like environmental contamination problem due to bio-hazardous material disposal, risk of needle sticks of health care personnel and potential infection to donors, etc. In addition, the painful experience resulting from sampling capillary blood may reduce the willingness of recruitment and retention of blood donors.
Aims: The purpose of this study was to evaluate the accuracy of a non-invasive method for hemoglobin screening in blood donors, and to assess whether it has the potential to replace the currently used CuSO4 method in Taiwan.
Methods: Pre-donation Hb levels of blood donors were simultaneously measured by Pronto-7 (Masimo, USA) and CuSO4 methods. The performance of these two methods was evaluated by comparison with the reference method, the KX-21N automated hematology analyzer (Sysmex, Japan). The cutoff Hb values of 12 g/dl for females, and 13 g/dl for males were used for minimum requirement of donor acceptance in Taiwan. If passed the CuSO4 test, the participant was eligible to donate blood, and an EDTA tube was filled with 2 ml blood sample from the diversion pouch of the blood bag, and subjected to Hb determination by the KX-21N analyzer. If failed the CuSO4 test, the participant was not eligible for blood donation, and blood was drawn by additional venipuncture for Hb determination.
Results: A total of 603 blood donors were enrolled in the study. Coefficients of variation in duplicate measurements were 2.34 for Pronto-7 and 1.01 for KX-21N. The mean Hb values were 13.87 g/dl (95% CI, 13.76–13.98 g/dl) by Pronto-7, and 14.61 g/dl (95% CI, 14.49–14.73 g/dl) by KX-21N respectively. The Pronto-7 test underestimated Hb values without statistical significance (mean bias −0.75 g/dl; 95% limits of agreement, −3.08, 1.58) compared to KX-21N test. Regarding the false acceptance rate, no statistical significant difference ($P > 0.05$) was found between Pronto-7 and CuSO4 methods, in which 1.3% donors by Pronto-7 method and 1.5% donors by CuSO4 method were inappropriately accepted to donate blood, compared to the reference method. However, the false deferral rate of Pronto-7 method with 11.8% was much higher than that of CuSO4 method, with only 0.3%. A significantly higher deferral rate of Pronto-7 compared with CuSO4 was observed ($P < 0.05$).
Summary/Conclusions: The replacement of CuSO4 method by non-invasive methods is promising and should be encouraged, since they have advantages of eliminating pain and reducing the fear and stress compared to invasive methods. In this study, the Pronto-7 showed a low false acceptance rate, but a relatively high deferral rate. Therefore, to raise the blood donor acceptance, a second method is necessary to retest those with Hb values below the cutoffs.

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ANALYSIS OF KNOWLEDGE REGARDING TRANSFUSION TRANSMITTED DISEASE AMONG SCREENING POSITIVE BLOOD DONORS WHO DONATED BLOOD INTO NATIONAL BLOOD CENTRE, SRI LANKA

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Background: Blood donors are the key element of transfusion service. Sri Lanka totally depends on voluntary Non remunerated blood donation. To minimize the risk of TTI, all blood donations are tested for HIV 1/2, HBV, HCV, Syphilis and Malaria. In the National Blood Centre, all donations are screened by serological as well as molecular testing (NAT).
Aims: To determine the knowledge of TTI among the donors who were screening positive for HIV 1/2, HBV, HCV and Syphilis.
Methods: This is a descriptive cross sectional study during the period from 01st August 2016 to 31st January 2017. Total number of 171 donors who donated blood in to National Blood Centre and screening positive for HIV 1/2, HBV, HCV and Syphilis were participated.
Results: A total of 171 screening positive donations were recorded during the study period. 118 (69.9%) were males and 53 (30.1%) were females. 82 (47.9%) donors were belongs to 26–35 years age category.
Among the study population, 130 (76%) were first time donors and 41 (24%) were repeat donors.
168 (92.8%) donors answered that diseases could be transmitted through blood transfusion. But 110 (64.3%) donors mentioned only one example for blood born diseases and 51 (29.8%) mentioned none of diseases. 166 (97%) donors gave positive answers regarding sexually transmitted diseases and 122 (71.3%) donors mentioned only one example.
As risk behaviors 40 (23.3%) donors had tattoos and 11 (6.4%) donors had experience with sex workers. 170 (99.4%) donors had knowledge that transfusion service investigated donated blood for infectious diseases. Out of this number, 140 (81.8%) donors mentioned only one example and 9 (5.2%) donors mentioned none of disease. All donors donated blood voluntarily and none of donors donated for checking their health status.
Summary/Conclusions: Majority of donors had general knowledge regarding blood born diseases and sexually transmitted diseases. But most of donors mentioned about HIV. They couldn't mention other diseases which were tested in transfusion service. None of donors donated blood for checking their health status.
The knowledge of blood born diseases and sexually transmitted diseases should be improved. This could be achieved by organizing the lectures, small discussions prior to or during blood donation campaigns. For general awareness, leaflet distribution and display of banners could be arranged. Conducting monthly blood collection campaign organizer meeting these massages could be easily convey to the blood donors. Publishing small articles, massages, and reminders in to the social media may be a successful alternative during awareness.

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MEASURES TO PROMOTE THE CONSTRUCTION OF REGULAR APHERESIS DONORS IN ZHEJIANG BLOOD CENTER

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Background: The WHO reports in developed countries, blood from regular donors accounted for 83.9% of the total donated blood. By the end of 2012, the number of regular apheresis donors and the times of regular apheresis donation in the Zhejiang blood center accounted for 41.7% and 66.5% of the total blood respectively. Blood donation of regular apheresis donors accounted for 69.0% of the total blood volume, while the times of regular apheresis donation accounting for 17.6% of the newly increased apheresis donors. The qualification rate of primary screening and donated blood tests were 71.1% and 99.1% respectively. And the proportion of first-time blood donors as high as 72.6%, the proportion of times were 39.4%. In the face of the arduous task of the construction of the regular apheresis donors, we took various measures in 2013 to carry out the construction of the regular apheresis donors team.
Aims: To sustainably develop the regular and voluntary non-remunerated apheresis donor team to make regular apheresis donation account for more than 90.0% of total blood.
Methods: The regular apheresis donor team construction initiatives included: ① To focus on publicity and education, all staff to participate in recruitment ② To optimize the process to rationalize and humanize the staff allocation, group collaboration, before-and-after collection security, equipment and material configuration arrangements. ③ To locate the position of medical staffs, reducing the shift and transactional interference, and to implement the whole-process management and the continuous overall care. ④ To pay attention to extended after-donation services, including routinely visit and greeting to the first-time apheresis donors and the establishment of a "platelet family" online group, by which we can timely communicate with blood donors. ⑤ To implement precise recruitment and retention. To provide personalized education, encouraging reserved donation; to promote health education and behavior intervention to failed blood donors so as to retain them; to carry out "1 + 1 Love Relay Activity ", recruiting new members. ⑥ To establish a blood donor incentive-care retention mechanism. Set up a dedicated website and the Hall of Fame, to provide social networking activities, donation gifts in birthday month, accident insurance and medical card for those donate more than 10 times per year; usually send messages of thanks, birthday and festival greetings. ⑦ To standardize the management of reservation, the effectiveness of recruitment and retention implement performance appraisal. To carry out innovation and research for recruitment and retention. To research health education path management of the first-time apheresis donors. To carry out the application research of the health promotion technology of the unqualified blood donors.
Results: By the end of 2016, the number of regular apheresis donor and the times of regular apheresis donation accounted for 67.5% and 89.4% of the total blood respectively. The blood donation of the regular apheresis donors accounted for 91.6% of the total blood volume, and the times of regular apheresis donation increased to 45.0% of the newly increased apheresis donors. The qualification rate of primary screening and donated blood test increased to 85.5% and 99.6% respectively. The proportion of first-time blood donors dropped to 52.6%, the proportion of times dropped to 15.9%.
Summary/Conclusions: The implementation of the strategy of the construction of regular apheresis donor is to ensure the steady growth of the apheresis donation and provide the basis of safe and adequate blood.

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P-098

A PRELIMINARY STUDY ON THE EFFICIENCY OF APHERESIS PLATELET DONORS RECRUITING BY TELEPHONE

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Background: With the increasing demand for apheresis platelet, it is the priority work to establish a sufficient and effective group of fixed component donors. To find out the effective strategies to improve the proportion of fixed donors, we contacted with the apheresis platelet blood donors who no longer come to donate in last 3 years by telephone.

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Aims: To investigate the causes of the loss of apheresis platelet donors and evaluate the effect of telephone recruitment.
Methods: A total of 1,467 members of the blood donors were given telephone communication to summarize the reasons for the loss of their donations and to mobilize them to donate blood again. The results were analyzed statistically.
Results: The total loss rate of blood donors in Jiangsu Blood Center was 46.1% from 2014 to 2016, with an annual average loss rate of 29.7%. The reasons for the loss included telephone number changed, been not local, had no time and without intention. A total of 209 (14.2%) of the donors agreed to donate again, but the total number of successful donors was 87 (5.9%).The longer the loss, the lower rate of successful donation.
Summary/Conclusions: The recruitment strategies should be targeted for component blood donor, improve the satisfaction of blood donation, and to maintain the blood donors with moderate contact and communication. Blood donation service work should be improved to establish a stable and sustainable development of component blood donors team.

P-099

THE CHARACTERISTICS OF PLATELETPHERESIS DONORS IN GUANGZHOU, CHINA

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Background: Plateletpheresis (P-) donations have gradually become indispensable resources in modern blood transfusion practices worldwide. The P-donation system and the apheresis technology has been developed during the past two decades in China. However, the increasing supply has not kept pace with the rapidly increasing clinical demand for platelet. The P-supply relies on the sufficient P-donors recruitment.
Aims: The better understand of the Characteristics of P-donors may help to find out the motivation of P-donation. Base on different motivation, blood banks can tailor-make different kinds of strategies to recruit P-donors in different characteristics.
Methods: This is a descriptive study on the demographic profile of P-donors in Guangzhou, China. The factors include demographic factors of donors and the characteristics of donation experience. The demographic factors include age, gender, education level, registered residence (whether resident population in Guangzhou or not) and occupation. The characteristics of donation experience refer to the numbers of past donation, recruitment methods and donor's career stages (i.e. first time donor, repeated donor and lapsed donor). Logistic regression is used to identify whether the above independent factors (rates) are associated with different factors.
This study uses the sample taken from the donation record at Guangzhou Blood Center from 2010 to 2015. These factors can be classified into different segments across the donation types and their different donor's career stages.
Results: In this sample, the proportion of male blood donors was 66.9%,which was much higher than that of women. Blood donors in 26–35 years accounted for 42.1%. Workers and students were the main part of donors. 57.8% were first-time donors, and 42.2% were repeated donors. 54.5% reported an education level of primary or secondary school. About 1.0% of whole blood donors were P-donors.
Summary/Conclusions: Young and middle-aged male are the main blood donors in Guangzhou. We should try to preserve young and middle-aged blood donors. Young college students, repeated whole blood donors and female also show greater potential for P-donation. To improve the recruitment of P-donors, we need to develop effective advocacy planning and recruitment strategies, and to establish a repeated and flexible group that promise to donate P.

P-100

ASSOCIATED FACTORS FOR BLOOD DONOR AMONG THE UNDERGRADUATE STUDENT IN A PUBLIC UNIVERSITY

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Background: Statistics data provided by the Malaysia National Blood Centre have indicated that only 3% of the population donated blood in the past 5 years when the targeted value is 10%. Amongst of the reasons that people do not become blood donors are due to lack of knowledge and false belief about donation program. Blood donor pool should be sustained with young and loyal donors. Getting and recruit younger people to donate blood and to maintain donor pool is a challenging task. Study on the factors that influence the practice of blood donation among young people is needed to increase more participation from this group of youngsters.
Aims: The aim of this study is to assess the associated factors for potential young blood donors among undergraduate students in a public university.
Methods: This cross sectional study had involved 304 respondents. Participants were selected by convenient sampling technique. The respondents were given a set of questionnaires consists of knowledge and perception towards blood donation. Data were entered, cleaned and analyzed using Statistical Package for Social Science (SPSS) software version 22.0. Chi-square test was used to determine any significant association between blood donor group and the factors studied. A P-value of <0.05 was considered as statistically significant.
Results: A total of 304 respondents with a mean age of 19.28 ± 0.71 (18–25 years old) with ratio 1:5 male to female. As many of 18.4% (n = 56) of respondents was a regular/previously blood donor. There were five factors significantly associated with blood donors. The factors were sex (male 38.5% vs female 14.3%, P = 0.00), age group ($\leq$19 years old 12.9% vs $\geq$20 years old 26.2%, P = 0.03), perception do they know where is the location of blood bank (yes 30.8% vs no 15.1% vs I don't know 6.4%, $P < 0.001$), perception that blood donation is a religious duty (yes 19% vs no 23.6% vs I don't know 8.2%, P = 0.046) and also perception that blood donation is harmful (yes 6.7% vs no 22.8% vs I don't know 4.3%, P = 0.003). The other factors such as level of knowledge, family monthly income, perception towards altruism, national duty and health effect were not statistically significant.
Summary/Conclusions: In conclusion, age, sex, perception towards harmful effect, a location of blood bank and religion play associated factors among undergraduate students in a public university.

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STUDY OF THE CHARACTERISTICS OF VOLUNTEERS REPORTED FOR THE BLOOD DONATION AND PREVALENCE OF VARIOUS REASONS FOR DEFERRAL AT A TERTIARY CARE CENTER IN RAJASTHAN, INDIA

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Background: The donor should be in good health in order to avoid any untoward effect to the donor or to the recipient. A scientific approach towards blood donor deferral is very important to ensure blood safety and to maintain safe donor pool for future blood availability. The most important steps, to ensure that blood and its products for transfusion do not have pathogenic virus/bacteria, are the proper selection of blood donors and laboratory investigations of blood bags for transfusion transmissible infections. However, donor deferral leads to loss of precious blood components which could have been made available for transfusion. To prevent this, we should have the knowledge about the reasons of temporary and permanent deferral and this should be practically implemented for safe donor selection. Donor deferral can be categorized as temporary short term (1–56 days), temporary long term (57–365 days) and multiple years/permanent (more than 365 days).
Aims: The present study has been aimed at screening subclinical communicable/non-communicable illnesses prevalent in the apparently healthy adult population; strengthening the efficient blood safety program and planning for public health programs depending upon the prevalence of various reasons of blood donor deferral. We focused our objectives to investigate frequency and prevalence of various temporary and permanent reasons of blood donor deferral in the western part of Rajasthan (North India) among the apparently healthy volunteers, who reported for the blood donation.
Methods: This prospective study has been conducted at the Department of Immuno-Haematology and Transfusion Medicine, Sardar Patel Medical College and Associated Hospitals, Bikaner. The apparently healthy volunteers, who reported to donate blood as voluntary or replacement donors at our blood bank and in outdoor blood donation camps organized in the western part of Rajasthan arranged by various organizations over a period of 2 years (i.e., from July 1st 2015 to June 30th 2017), were included in this study. Data were analyzed for sex, age, donation frequency, donor type and donor deferral prevalence. Focus was kept on analyzing frequency of various reasons among deferred volunteers and further classifying them into temporary/permanent reasons of deferral. Microsoft excel and SPSS software version 16 were used for statistical analysis and suitable tests of significance were applied.
Results: Out of total 69,069 volunteers; 55,601 (80.5%) were found to be fit for blood donation as per donor selection criteria while 13,468 (19.5%) were deferred.

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93.8% were deferred due temporary reasons and 6.2% were deferred permanently. Among total deferrals, 31.4% were due to low-hemoglobin-levels; 15.4% due to low-body-weight; 14.9% were deferred due to medications; 9% due to chronic/major illnesses. 8.3% due to H/o blood donation <3 months; 7.8% due to H/o illnesses related to Hepatitis B or C, syphilis or malaria; 5.6% due to high-BP; 3.4% due to hyperglycemia; 2–3% were deferred due to high risk behavior and 1–2% due to other reasons.

Summary/Conclusions: Early diagnosis and treatment along with follow-up facilities must be strengthened at Blood Bank level to maintain Safe-Donor-Pool for future blood availability by decreasing drop outs of the potential donors. Anemia and low body weight being the leading causes of donor deferral, public health measures must be carried out to correct them in general population.

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BLOOD DONOR MOTIVATIONS, SATISFACTION AND LOYALTY: A STUDY IN KUNMING, YUNNAN PROVINCE OF CHINA

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Background: In modern healthcare, blood supply has become an important social problem. The blood donor system is only depends on voluntary donors in the Kunming, Yunnan Province of China .Voluntary blood donors may have different motivations for donating blood, and there are differences in satisfaction with blood donation. Donor recruitment and retention are crucial for sustaining blood safety and sufficiency.

Aims: Aims are to explore the motivations, satisfaction and loyalty of Kunming blood donors, update of blood donor recruitment and retention .

Methods: The study was conducted at the blood collection sites of Yunnan Kunming Blood Center. Active donors randomly (N = 1031)between May and August 2016 filled in a validated questionnaire during donation. SPSS software was used to statistically analyze the responses. Univariate analysis was done using Fisher's exact test. A multivariate model was constructed controlling for gender, age, occupation and education background, including the variables that were significant in univariate analysis.

Results: Factor analysis of 7 genre blood donation motivations revealed three factor named "social responsibility and devotion of love", "reserve blood and self-benefit","-convenient and easy" reached statistical significance. The results with regression analyses showed that only "social responsibility and devotion of love" had an significant effect independent of gender, age, occupation, and education on blood donation. The overall satisfaction of blood donation was 99.89% among both women and men. The satisfaction to waiting time before donation was the lowest (90.11%), and to service attitude of stuff at blood collection sites was the highest (93.54%).

Summary/Conclusions: It showed that blood donation motivation not only linked to a high degree of altruistic reasons, but also to a combination of some self-regarding motives. By analyzing the motivation and satisfaction of blood donation, we can improve the service process and promote the recruitment and retention of blood donation.

P-103

ANALYSIS ON INFORMING ON BELOW STANDARD BLOOD TESTS RESULTS AND IMPROVEMENT MEASURES

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Background: Based on *People's Republic of China Blood Donation Law*、 *Blood Center Administration Measures*, *Blood Center Quality Management Rules*, blood donors are required to do the compulsory free health checks before donating blood, and should agree to follow the principles of voluntary donation, and informed and consent. All the services are provided by the blood center. Meanwhile, we are required to give feedback to the donors based on the results of the blood tests [1-3]. According to the test results, we divided those into "qualified blood" and "unqualified blood ". But if our staff does not properly inform the unqualified result, it is easy to cause medical disputes, and even legal proceedings. Therefore, all the blood stations pay much attention to the great importance of this work, and all staff are suggested to contact donors by telephone, SMS, mailing or letters, interviews in time. All the unqualified results should be informed personally and confidentiality. Recently, we have put lots of efforts on combining various specific practice and performed several analysis in order to improve the skills and methods of informing, enhance the satisfaction of donors, provide better services for blood donors, and to promote the healthy and sustainable development of voluntary blood donation in this region.

Aims: To improve the method of informing blood donors of disqualified blood detection results, as well as to promote healthy and sustainable development of regional blood donation.

Methods: Modern management information systems, such as SHINOW9.0, of ALT, HBsAg, anti -HCV, anti -HIV, anti –TP, were used to give responses to the donors regarding their 'below standard' blood test results within 1 month of the blood extraction. The following conditions were regarded as failure to inform: when people gave an empty or wrong number, a number that does not correspond with the donor, and when people did not leave contact information. In addition, another condition that if the donor could not be contacted over three different times during a day.

Results: Between the 1st of February, 2015 and the 31st of January, 2016, the number of 'below standard' blood test results amounted to 3,695 people, of which 3,302 people were informed successfully and the inform success rate was 89.36%. On the other hand, 393 people had failed results, and the inform failure rate was 10.64%. All the failures were due to cancelled, switched off or non-answered phones, wrong numbers and not leaving contact information.

Summary/Conclusions: In order to ensure we have the accurate and complete information about our blood donors, we need to control the information collection from the starting point. Secondly, the construction of laboratories should be enhanced and external interference should be eliminated. Thirdly, we need to improve our communication methods and skills by giving information. We should also enhance satisfaction in blood donors, and reduce the occurrence rate of medical disputes.

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A PRECISE RECRUITMENT STRATEGY FOR REPEATED BLOOD DONATION ANALYSIS IN ZHEJIANG BASED ON REPEATED BLOOD DONORS

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Background: Unpaid blood donation is a common problem, in the past few years, many developed and developing countries scholars have carried out the attitude and motivation of blood donors to study. Some scholars believe that the possibility of repeated blood donation and the first and second blood donation interval. For the analysis of the timing of recruiting blood donors, there are few reports in the literature at home and abroad. To this end, we in 2006 to 2015 in Zhejiang Province to participate in unpaid blood donation blood donation blood donation frequency, the relationship between the time of donation and demographic characteristics of the statistical analysis.

Aims: To understand the frequency of blood donation and the behavior of donation in Zhejiang province from 2006 to 2015, and to find out the best intervention time and recruiting people to develop accurate recruitment strategies.

Methods: The relationship between blood donation behavior, blood donation frequency, recurrence time and demographic characteristics was statistically analyzed by descriptive statistics and Cox regression.

Results: Among the total blood donors, 69.23% were the first donors, and 2 were donated twice. Blood donation for the first time and the second interval, the third and second interval or the fourth and third interval, again blood donation time in half a year to 1 year high, accounting for 40.79%, 47.55% and 53.07%. Sex is unstable in the probability of repeated donation of blood donors again. In the second and the first repeated blood donation, for example, in the blood donation age, relative to the age of 18–25 years of age, 26–35 years old, 36–45 years old, 46–55 years old, 56 years of age repeated blood donation the probability of blood donation is 1.198, 1.515, 1.468 and 1.003 times respectively; in the educational level, junior high school, high school, university, graduate students and others are primary school education degree of repeat blood donors once again the probability of blood donation 1.074, 1.121, 1.201, 0.937, 0.963 times; in the occupational type, staff, workers, farmers, medical workers, national staff, soldiers and other students were repeated blood donors again the probability of blood donation 1.502, 1.574, 1.852, 2.152, 2.072, 1.359, 1.384 times.

Summary/Conclusions: It is necessary to carry out repeated blood donation research to improve the frequency of donation again. It should be classified for different blood donors. It is necessary to pay attention to the demographic effect of repeated blood donors and take intervention measures to improve the repeated blood donation rate.

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APPLICATION OF SOCIAL MARKETING/PLANNED BEHAVIOR TO CONVERT AND RETAIN REPLACEMENT DONORS INTO VOLUNTARY DONOR

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Background: In Pakistan the blood donation and transfusion services are still exceptionally promising comparing with worldwide practices. Voluntary Blood Donors are the indispensable source of safe blood. Presently the attitude of voluntary blood donation in Pakistan is extremely disappointing. It is reported the only 10% of people donate blood voluntarily while 90% are family or replacement donors. According to WHO all countries should aim their transfusion services based on voluntary non remunerated blood donation so that by 2020. In order to achieve this goal, social marketing concept may positively promote voluntary blood donation rate. Social marketing is a new idea and considered as a successful tool to promote public health programs through planned behavior theory. In present study, we have adopted various factors for changing concept of replacement donation into voluntary blood donation.
Aims: The aim of our study was to promote voluntary donation rate in Pakistan by implementing of social marketing concept.
Methods: This single center descriptive study was conducted at Blood Bank of Rawalpindi Institute of Cardiology, Pakistan. Theory of planned behavior and different social marketing strategies (create awareness and interest, information about need of blood, positive attitude of blood bank staff, motivation, extra attention to donors and decreasing risk of fear) applied on the target population of replacement donors. The data and response from 250 participants were collected through designed questionnaire and analyzed by SPSS 16.0.
Results: In our studied population the minimum age limit was 18 years whereas maximum age was 49 years with the mean value of 29 $\pm$ 10.7. By implementing theory of planned behavior; out of 250 responded participants, 93.0 individuals (37.2%) showed a changing tendency of voluntary donation (They were registered as voluntary donors).
Summary/Conclusions: The present study provides some useful information regarding boosting of voluntary blood donation rate in Pakistan. Voluntary donation can be improved by changing the behavior of blood bank staff, giving extra care to donors and by appointing a social marketer who are trained to sensitize, educate, and bring awareness about voluntary blood donation in family/replacement blood donors.

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PREVALENCE OF ABO AND RHESUS BLOOD GROUPS AMONG BLOOD DONORS IN NATIONAL BLOOD BANK SERVICE, ADDIS ABABA, ETHIOPIA

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Background: ABO and Rh blood groups are the most studied blood group systems among human populations due to their clinical, genetic and anthropological importance. Together these two systems have proved to be the most important, for blood transfusion purposes. The knowledge of distribution of ABO and Rhesus (Rh) blood groups at local and regional levels is helpful in the effective management of blood banks and safe blood transfusion services for donor recruitment and also stock management.
Aims: To assess the ABO and RH blood group distribution among blood donors who donated blood in national blood bank service, Addis Ababa.
Methods: Facility based cross-sectional retrospective record review was conducted at National Blood Bank Service. A structured questionnaire used to collect data from July 1/2015 to June 30/2016 and a total of 48,212 voluntary non-remunerated blood donors data were reviewed. Data was entered and analyzed using SPSS version 20. Descriptive statistics, including frequencies and percentage were used to present data in text and tables.
Results: The predominant blood donors belonged to age groups between 18–24 years (52.5%). Male donors were more than female donors, ratio approximately 2:1. Majority of the blood donors were private workers (65.1%) followed by students (30.7%). The others are civil servants (4%) and unemployed (0.1%). The most common blood type is group O (41.7%) and least common being AB (6.4%). The other groups account A (29.2%) and B (22.7%) The prevalence of Rhesus positive an negative distribution in the studied blood donors was 92.5% and 7.5% respectively. Blood groups frequency with respect to ABO and Rhesus status was found to be shown by formula O>A>B>AB.
Summary/Conclusions: Knowledge of frequency of the different blood groups is very important for blood banks and blood transfusion services that would contribute significantly to the national health system. The finding from this study will also serve as an input for National Blood Bank Service to recruit and mobilize blood donors based on age and also occupation if there is a blood shortage in general and also if there is a specific blood group shortage in particular.

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APPLICATION OF ANISODAMINE IN THE TREATMENT OF SEVERE BLOOD DONATION

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Background: The severe blood donation response seriously affects the health of blood donors. How to use drugs to quickly restore blood donors to new research topics
Aims: To investigate the effect of anisodamine in the treatment of severe adverse reaction in effect, provide the basis for treatment.
Methods: A total of 30 cases of severe blood donation from 2012 to 2016 were collected and grouped, control group and experimental group with 15 cases in each group, the former using conventional processing methods, which in addition to conventional treatment, intramuscular injection of anisodamine to 10 mg. Two groups were observed and recorded from blood donation reaction to restore their own after the departure time, and on the second day follow-up results were compared.
Results: The control group in 13 cases of recovery time ranging from 0.5–4 h, 2 cases still do not recover for more than 4 h and need further treatment in the hospital. 11 cases in the experimental group within 0.5 h of recovery, 4 cases recovered within 0.5–1 h; visit almost all in control group, fatigue in second days, dizziness and other symptoms, 2 cases still need to stay in bed, while the experimental group only individual fatigue, dizziness, and No other discomfort.
Summary/Conclusions: Anisodamine can effectively relieve the symptoms of severe adverse reaction and shorten the reaction time, it is worthy of popularization and application.

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ANALYSIS OF ATTITUDES OF REGULAR BLOOD DONORS REGARDING BLOOD DONATION AND DONOR RETENTION WHO DONATED BLOOD INTO THE NATIONAL BLOOD CENTER, SRI LANKA

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Background: Sri Lanka has centrally coordinated blood transfusion service and donations totally depend on voluntary non remunerated blood donors. Regular blood donors are the backbone of transfusion service. While maintaining continuous blood supply, the service meets challenges in recruiting and retention of blood donors. These challenges make the exploration of factors affecting blood donation in order to identify effective strategies for increasing the donor base and fulfilling future blood needs.
Aims: To analysis of attitudes regarding blood donation and donor retention strategies among regular blood donors in Sri Lanka.
Methods: A prospective community based study was carried out. Total numbers of 3629 donors were participated. All donors who donated blood two times or more (n = 3981) in to the National blood centre within 1st January 2015 to 31st March 2015 were eligible for this study. Out of 3981 regular donors, 151 were refused to participate and 201 were not responded.
Results: Among the total study group 2579 (71.1%) were males and 1050 (28.9%) were females. 1332 (36.7%) donors believed that others will identify himself as a important person when he/she be a blood donor. But 1351 (37.2%) donors were against it and 946 (26.1%) had no answer. 2919 (80.4%) donors thought that blood donation is a responsibility and 328 (9%) donors challenged it.
As a habit, 2086 (57.5%) donors would like to donate blood. 912 (25.1%) donors were disagree the concept and 631 (17.4%) donors were not answer. 3059 (84.3%) donors would like to donate for helping other people but 236 (6.5%) were against it.

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1195 (46.3%) donors thought that blood donation may explore their own strength and 1000 (27.6%) donors were not believe it. 1092 (30.1%) donors would like to donate blood because their family members or friends thought blood donation is an important activity but 1908 (52.6%) donors were against.
For benefit of their own health, 2676 (73.7%) donors were donate blood but 292 (8%) donors were against. As a purpose of free health check, 703 (19.4%) donors would like to donate blood. But 2443 (67.3%) donors were not expecting free health check.
Summary/Conclusions: Nearly same percentage of donors was for and against regarding blood donor is an important person. Vast majority of blood donors donated blood for self satisfaction and as a responsibility. Most of regular donors would like to donate as a habit. Almost all donors believed that they were helping other people by donating blood. Majority of donors mentioned that blood donation explore their own strengths. Vast majority of blood donors thought that blood donation may helps to good health. As a purpose of free health check, very few number of blood donors were donate blood. Influence of family members or friends regarding blood donation was not identify as a key factor for blood donation and donor retention.

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FIRST TIME DONORS: WHAT MOTIVATE AND HINDER DONORS FROM BLOOD DONATION

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Background: Blood donation in Malaysia is practised as voluntary non-remunerated donation. However recruiting and retaining blood donors remains a challenge. Over the years deferral rate has also increased. If we are able to understand the motivating and hindering factors, then modifiable deferral rate can be improved especially among those first time donors.
Aims: In view of the increase in number of temporary deferred donors, insight about factors which motivate and hinder whole blood donor will help us in blood bank to strengthen the strategy to retain donors. This study is conducted to identify aspects that contribute to improve the status of regular donors. As a result of this study, NBC should be able to improve and strengthen their recruitment and collection state.
Methods: Four hundred and eighty first time temporary deferred whole blood donors from National Blood Centre (NBC), Kuala Lumpur, participated in this study from January 2014 till December 2014. Donors were randomly extracted from Blood Bank Information System by computer software. Based on the inclusion and exclusion criteria, the donors were categorized into 2 main groups of which one group of donor who return for blood donation and the other who did not return for donation after first time deferral. Standard questionnaires were develop and donors were contacted for question and answer session with options to choose. Questionnaires were developed based on NBC donor criteria guideline and pilot study conducted for validation.
Results: The total number of temporarily deferral at NBC between year 2010 – 2014 is 147, 087 donors out of 803,431 which represents about 18.3% of total temporary deferral. A total of 480 donors were included in this study. 240 donors who returned and 240 donors who did not return for donation following first time temporary deferral. Donors who return were significantly younger than donor who did not return (29.5 years versus 35 years, $P < 0.001$). Socio- demography also significantly differs between return and did not return. Females had higher proportion (61.3%) for returned compared to males (38.8%). Singles (68.3%) were more likely to return for donation compared to married donors (31.7%). Relating to occupation; comparing return versus did not return donors professionals (38.8% versus 33.3%), student (26.4% versus 20.4%), general worker (21.9% versus 21.7%), uniform body (6.4% versus 9.6%) and housewife/unemployed (6.3% versus 15%). Donors who lived in urban areas were more likely to return for donation compared to donors who lived in rural areas (34.6%).
Summary/Conclusions: This study, clearly showed that younger generation, single and professionals from urban areas contributed to higher rate of return donor. Thus focusing on this category of donors to become regular donors will enable us to enhance and improve transfusion service in Malaysia.

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A LONGITUDINAL ANALYSIS OF ANTECEDENCES OF BLOOD DONATION BEHAVIORS IN EAST CHINA SAMPLE 2009–2015

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Background: To improve the efficiency of blood recruitment is urgent for blood collection agencies. How to make good use of the big data of blood donors is a good chance more than a challenge for blood collection workers and researchers for its implication in blood recruitment.
Aims: This longitudinal analysis sought to identify the antecedents and tendency of blood donation behavior with a big data.
Methods: First, Using an East China sample 2009–2015, it investigated the demographics of donors, including of age, gender, profession, education as well as the environmental factors, which influence blood donation, such as the time of a day, the days of a week, the days in a month, the months of a year, the change of years. Furthermore, it built a simulation model to predict the blood donation of a sample 2016.
Results: The results evidenced some donor demographics. Male tends to be more willing to be donors and be more generous than females. Educated individuals tend to be the latent donors rather than those uneducated. Self-employed citizens were found to be the most charitable donors and civil servants including teachers, public medical staff to be the least charitable ones among all personnels. Some environmental factors influencing donation were also found. 1:00 and 2:00 PM of a day, weekends of a week, early of a month, were found the efficient time to recruit latent donors. The simulation model well predicted the donation of year 2016.
Summary/Conclusions: Blood donation is influenced by demographics as gender, age, education, profession and also by environmental factors as time of a day, weekends of a week, month of a year and yearly fluctuation. The simulation model is proved to well predict the blood donation.

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FACTORS INFLUENCING THE WILLINGNESS TO DONATE: HAN NATIONALITY AND THE ETHNIC GROUPS COMPARISONS

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Background: Blood donation rates in many minority areas are under the average level of 9‰ in China. To increase blood donation by the minority, understanding factors that influence the willingness to blood donation plays an vital role in the development of targeted campaigns for collection centers. The influence of these factors on the minority and the Han people have seldomly been investigated before.
Aims: To investigate effects of factors influencing the willingness to blood donation on Han People and the ethnic groups.
Methods: A kind of voluntary, anonymous survey on willingness to donate blood was administered to the public who have never donated before in Jinghong City, Xishaungbanna Dai nationality autonomous prefecture. The sites we choose are restaurants, shops, super markets, streets and squares.
Results: 70% participants have a willingness to donate blood. The most commonly reported reason willing to donate is to help people in need of blood (66% ethnic groups, 64% Han nationality).As for the reasons they finally do not donate are included: "because my physical condition do not meet the requirements" (37% ethnic groups, 27% Han nationality, $P < 0.05$) and "because I am afraid to get infected with diseases"(12% ethnic groups, 23% Han nationality, $P < 0.05$) and "I am worried about the safety of the apparatus "(11% ethnic groups, 20% Han nationality, $P < 0.05$).
Summary/Conclusions: Both ethnic groups and the Han nationality have a strong willingness and a similar motivation to donate blood, While barriers to blood donation are different. These factors might be useful for designing a campaign to recruit and retain the minority donors.

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EFFECT OF AIDET COMMUNICATION MODE ON THE BLOOD DONORS

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Background: AIDET is an acronym for 'Acknowledge, Introduce, Duration, Explanation, Thank you' and AIDET communication mode is widely and effectively used in field of patient care. Building the standardization of communication procedure and language through AIDET communication mode is to improve the effect of blood donors history questionnaire.
Aims: To investigate the effect of AIDET (Acknowledge, Introduce, Duration, Explanation, Thank you) communication mode on the blood donors history questionnaire.
Methods: The AIDET communication procedures and template were established since 2016 in our blood center, the donor satisfaction and the retention of donors were valued after the application of the AIDET communication mode.
Results: Blood donors' scores about communication values improved significantly and more donors were willing to donate again through the implementation of the communication mode.
Summary/Conclusions: AIDET communication mode can effectively improve the effect of blood donors history questionnaire.

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ANALYSIS REPORT OF APHERESIS PLATELET AND WHOLE BLOOD DONORS IN 2014–2016, DALIAN

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Background: In recent years, with the rapid increase of blood volume in clinical application, the difficulties and pressure of blood donation became stronger that the recruitment of blood donation was particularly important.
Aims: Therefore, we made a detailed investigation and analysis the donors' on two kinds of blood donation form, in order to finding the different forms of recruitment methods, and striving to expand whole blood and platelet donation volunteers.
Methods: We try to analysis on the occupation, educational background, age and gender of voluntary donors in 2014–2016 by statistical methods.
Results: 1. people engaged in other work was mainly donors that the team of the platelet, and the workers, farmers, students, employees of enterprises and institutions were assistants. The whole blood donor team was mainly people engaged in other work and students, workers, farmers, employees of enterprises and institutions were auxiliary. As the proportion of group blood donors increased, the proportion of workers, farmers and students in whole blood donors showed an upward trend, and caused the loss of some apheresis platelet donors. 2. College degree or above accounting for about 30% of the apheresis platelet donors, while for about 50% of the whole blood donors; The special secondary school education and below were nearly 70% of the apheresis platelet donors, while for about 50% of the whole blood donors; 3. In terms of gender, male donors accounted for nearly 70% of the apheresis platelet donors, while for about 60% of the whole blood donors, in recent years, male donors in both whole blood and apheresis platelet had shown a downward trend, while on the contrary, female had shown an upward trend. Between the ages of 25–30 and 30–35, both male and female donors had shown a downward trend in whole blood and apheresis platelet. The age group of 35–55 and above, both male and female donors showed an upward trend, in the apheresis platelet donors, the upward trend was more obvious.
Summary/Conclusions: There were differences in the composition of whole blood and apheresis platelet donors, and we should take different recruitment according to the characteristics of their respective groups.

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INVESTIGATION OF FEASIBILITY OF POLICY ON 300 ML UNPAID BLOOD DONATION IN HAINAN PROVINCE

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Background: All the time, there are two types of the amount of blood donation in Hainan Province: 200 ml and 400 ml. Generally, the proportion of the 400 ml blood donation is greater than that of the 200 ml blood donation in the total amount of blood donation and in the donors from street blood donation, but the percentage of the 200 ml blood donation is greater than that of the 400 ml blood donation in the donors for the first time blood donation and the ones from group blood donation. In order to encourage more 200 ml donors to donate more blood, the policy on 300 ml unpaid blood donation was promoted in Hainan Province from August 2016.
Aims: To investigate the feasibility of policy on 300 ml unpaid blood donation and provide the information for the further promotion.
Methods: The data, from August to December 2016, about the amount of blood donation, the blood donation person-time in various groups of donors, and the insufficient amount of blood donation after the policy on 300 ml blood donation was carried out, were collected, while the same kind of the data from the same period of last year were collected. The information, about the proportion of the blood donation person-time in each group of donors with different amount of blood donation as well as the development tendency of the amount of blood donation and the insufficient amount of blood donation, was statistically analyzed.
Results: The total proportion of the 300 ml blood donation was up to 17.9 per cent from August to December 2016, after the policy was carried out. The increase of donation volume is larger than that of blood donation person-time. The growth of the average amount of blood donation was 1.6%, which in the first time donors, the repeat donors, the donors of the street blood donation, and the donors of the group blood donation was 2.7%, −0.1%, 2.7%, and 1.3%, respectively. The proportion of the insufficient amount of blood donation fell from 2.0% to 1.7%, and the result shows the statistical differences.
Summary/Conclusions: The total amount of blood donation rises up and the insufficient amount of blood donation drops after the policy about the 300 ml blood donation, which demonstrates that the policy is reasonable and safe.

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STUDY ON THE EFFECT OF COLD STIMULATION INTERVENTION ON REDUCING BLOOD DONATION

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Background: At present, 100% of the blood used in our country comes from the voluntary blood donors. It is necessary to expand the blood donors, protect the physical and mental health of blood donors, avoid blood discarding, and fully meet the clinical use of blood, which is the focus of blood bank work. Blood donation reaction is one of the reasons of blood donors fear, is one of the main reasons for inadequate blood collection, blood will not only cause the scrap, will let the fear of blood donors, blood loss caused by the source, and the cause of blood donation to the healthy development of the negative impact. It is an urgent problem for the blood bank to take reasonable measures to reduce the incidence of blood donation and to avoid the loss of blood donors.
Aims: To explore the role of cold stimulation intervention in reducing blood donation reaction.
Methods: October 2003 - October 2013, Handan blood center were found in 2352 cases of blood donors, blood donation reaction delayed or first blood donation reaction occurs in the blood before blood donation reaction, including selection of blood donation reaction history, delayed blood reaction history or for the first time in the blood before blood donation but 200 cases of blood donation reaction of blood donors on the basis of the blood donation reaction type, according to the odd and even number randomly, 3 patients in each group had a history of blood donation reaction including blood donors in 90 cases, delayed reaction to blood donation history of blood donors, but the first blood donation in the blood before the appearance of 7 cases of blood donation reaction. The intervention group was divided into the intervention group and the control group according to the cold stimulation intervention, 100 cases in each group. The control group received voluntary blood donors with routine treatment (normal temperature + psychological counseling), and the intervention group (cold stimulation intervention + psychological counseling). Comparing the intervention group and the control group, the blood donation reaction of the voluntary blood donors decreased or disappeared.
Results: The decrease or disappearance rate of blood donation of the intervention group was significantly higher than that of the control group ($P < 0.05$). The intervention group in blood donors was decreased or disappeared was higher than that in spring and autumn ($P < 0.05$), was significantly higher than that in winter ($P < 0.01$), which may be related to the summer hot weather, many people of untoward reactions occurred in winter, people less about blood donation reaction occurred, and season, temperature was positively related to blood donation reaction, also demonstrated the occurrence of cold stimulation can effectively reduce blood donation reaction.

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Summary/Conclusions: Cold stimulation intervention can significantly reduce the incidence of blood donation reaction and reduce the severity of blood donation reaction. This method is simple, safe, easy to implement and easy to be accepted by blood donors.

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RELATIONSHIP BETWEEN DONATION FREQUENCY AND IRON STATUS IN DONORS AT PMI YOGYAKARTA

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Background: In 2013, only about 50% of blood supply fulfilled the total need of blood. The high demands for blood enhance the need for donors to repeat donation. Blood donation has a profound impact on body iron stores and is an important factor for iron deficiency in blood donors. In blood donation, 200–250 mg iron are lost from each collection l procedure of 425–475 ml blood. Higher donation frequency can result in iron deficiency and anemia. Assessment of hemoglobin level can prevent donation by individuals with anemia, but it is not a good predictor for iron status of donors.
Aims: To determine the relationship between donation frequency and iron status in donors at PMI Yogyakarta.
Methods: This was a cross sectional study conducted for 3 months, from September to November 2014. The study involved 120 donors from blood donation center at PMI (Indonesian Red Cross) Yogyakarta. Sample was selected based on the selection criteria and collected consecutively.
Results: A total of 98 subjects that met the criteria were studied, consisted of 74 male donors and 24 female donors. From 98 subjects, 49 subjects (50%) had 1 donation in the previous year, 21 subjects (21.4%) had 2 donation, 21 subjects (21.4%) had 3 donation and 7 subjects (7.1%) had 4 donation. The blood donation frequency per year significantly related to serum ferritin level ($P = 0.005$; $r = -0.280$) and transferrin saturation ($P = 0.034$; $r = -0.214$).
Summary/Conclusions: There is a relationship between donation frequency and iron status (serum ferritin and transferrin saturation) in donors at PMI Yogyakarta.

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RESEARCH ON BLOOD DONORS DISTRIBUTION AND ITS CHANGING TREND IN DALIAN

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Background: In Dalian recruitment and retention of sufficient numbers of blood donors continues to be a challenge. Understanding who donates blood will help blood centres to target more effective recruitment and retention strategies.
Aims: This study aims to analyze the gender, age and source (organizations or individuals) of distribution about blood donors in Dalian and varied trend from 2011 to 2015, evolve and improve reference basis for donor recruitment measures.
Methods: Changes in gender, age and source (organizations or individuals) distribution among donors were analyzed through comparison of the volunteer donor population between 2011 and 2015, before and after adjustment for demographic changes of the general population in Dalian. Data analysis was carried out using SPSS 18.0 statistical software. Descriptive data was summarized and analyzed.
Results: A proportion of total Male donors increased from 61.6 percent in 2011 to 63.2 percent in 2015, inversely female donors decreased from 38.9 percent to 36.8 percent., and the difference was statistically significant ($P < 0.01$). The mean age of donors has increased from 28.7 years old to 31.2 years old ($P < 0.01$). From 2011 to 2015,the donation proportion of age 18–25 years decreased gradually (49.2 percent VS 38.7 percent, $P < 0.01$);no significant differences were found in those of age 26 to 35 (26.1 percent VS 25.9 percent, $P > 0.05$); donors of age 36 to 45 gradually increased from 17.2 percent to 21.6 percent, $P < 0.01$; and 46 years older increased by >13.8 percent, ($P < 0.01$). During 2011 to 2015, the ratio of individual blood donation decreased from 82.4 percent to 72.8 percent ($P < 0.01$), the ratio of organization blood donation increased from 17.6 percent to 27.2 percent ($P < 0.01$).
Summary/Conclusions: The distribution of voluntary blood donors in Dalian, such as gender, age and source (organizations or individuals), has its own characteristics. The donation proportion of voluntary blood donors from different population shows great volatility. A severe shortage of blood and blood components may be foreseeable future unless we strengthen propaganda education to blood donors from different age and education levels, to establish reasonable blood donor recruitment measures and ensure enough blood supply.

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Abstract has been withdrawn

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MEDICAL DEFERRALS DUE TO ANAEMIA AMONG BLOOD DONORS

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Background: Screening for anaemia is important for blood donation to protect and maintain the health of donors and improve the aim of red blood cell transfusion in the recipient. Haemoglobin levels should be above 110 g/l and 120 g/l in women and men respectively. One of the main reasons for the medical clinical deferrals of prospective blood donors is anaemia.
Aims: The aim of this study is to identify the prevalence of anaemia among prospective blood donors.
Methods: Prospective records of all the reported donors were collected from January 2015 to June 2017 at the Scientific-Production Center of Transfusiology, Astana. Haemoglobin estimation was performed by the rapid diagnostic method on haemoglobinometer Hemo Cue HV201+ (Sweden), using capillary blood samples.
Results: Overall 130877 potential blood donors reported to the blood centre. Prevalence of anaemia in prospective blood donors was 3.5%. It was significantly higher in female donors compared with male donors (86% vs 14%). The level of anaemia was mild in 46% of the female donors while 35% had a moderate level and 4.3% had severe anaemia. Mild anaemia was seen in 10% of the male donors, followed by severe and moderate anaemias 2.1% and 1.9% respectively.
Summary/Conclusions: In conclusion, deferred anaemic donors should be informed about the prophylactic intake of iron supplements and referred for further follow-up so that they can be appropriately treated. In addition, it is important to determine the serum ferritin in regular blood donors.

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DISTRIBUTION CHARACTERISTICS AND COUNTERMEASURES FOR UNQUALIFIED FEMALE VOLUNTARY BLOOD DONORS WITH HEMOGLOBIN LUOYANG IN 2015–2016

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Background: The main reason of female donors whose Hb were unqualified were affected by age, occupation and education, the inappropriate cognition and behavior in the way of life. More blood physiological knowledge should broadcast in recruiting, through appropriate intervention and quickly corrected, more female donors could return to blood donation team.
Aims: Analyzing the female voluntary blood donors basic characteristics whose hemoglobin (Hb) were unqualified in Luoyang, to carry out the targeted health

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education and broadcast the knowledge of voluntary blood donation, to prevent the loss of voluntary blood donors and improve the proportion of female voluntary blood donors.
Methods: Statistical analysis was used in female voluntary blood donors' age, occupation, education level, Hb and the times of voluntary blood donation in the past whose Hb were unqualified before donation from 2015 to 2016 in Luoyang, using SPSS17.0 software by χ^2 test and linear related analysis $P < 0.05$ were statistical significant.
Results: In 2652 donors whose Hb were unqualified before donation from 2015 to 2016 in Luoyang, including 2599 female donors, accounted for 98%. Among those female donors, 41–50 years old accounted for 42.42%, farmers accounted for 48.84%, middle school and below accounted for 57.22%, Hb more than 100 accounted for 81.41%, and there were not related with the times of voluntary blood donation in the past.
Unqualified before donation from 2015 to 2016 in Luoyang, using SPSS17.0 software by χ^2 test and linear related analysis $P < 0.05$ were statistical significant.
Summary/Conclusions: The main reason of female donors whose Hb were unqualified were affected by age, occupation and education, the inappropriate cognition and behavior in the way of life, physiological cycle and diet would cause them. More blood physiological knowledge should broadcast in recruiting, through appropriate intervention and quickly corrected, more female donors could return to blood donation team.

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IMPACT AND ANALYSIS OF A NOVEL "FALL IN LOVE" RECRUIT METHOD ON SUCCESS RATE OF VOLUNTARY BLOOD DONATION VERSUS CONVENTIONAL RECRUIT METHOD

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Background: Recruitment of first-time donors among the pedestrians is essential to maintain and enlarge a stable blood supply in China. Our center has set up 11 sites in Changsha city to recruit the pedestrians to join the voluntary, non-remunerated blood donation in recent years.
Aims: This study aims to assess the impact of a novel "fall in love" street recruit method on voluntary blood donation success rate versus the conventional recruit method and analyze the possible reasons.
Methods: The recruit team was divided into two groups. One group used a novel "fall in love" street recruit method, while the other group used the conventional method to recruit the donors. After 6 months, we compared the success rates between these two groups and analyzed the possible reasons.
Results: In 6 months, the novel method group successfully recruited 20,336 blood donors among 37,931 pedestrians with the success rate of 53.6%, while the conventional method group successfully recruited 18,283 blood donors among 39,497 pedestrians with the success rate of 46.3%. The success rates between these two groups showed statistical significance.
Summary/Conclusions: The novel "fall in love" street recruit method can greatly increase the success recruit rate of pedestrians to join the voluntary, non-remunerated blood donation. The reason for the higher success recruit rate may be the integrated use of psychology, language and behavior to motivate the pedestrians.

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COMPARISON OF TENDENCY OF BECOMING A REGULAR DONOR BASED ON ABO AND RH D BLOOD GROUP AND PLACE OF DONATION AMONG DONOR POPULATION AT DISTRICT GENERAL HOSPITAL HAMBANTOTA SRI LANKA FOR YEAR 2016

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Background: Hambantota district is a coastline district in the south-eastern part of Sri Lanka, with a total population of 662,419. District General Hospital (DGH) Hambantota is the main medical institute of this district, serving the highest number of patients in the area.
Rh D Negative blood groups are rare. Only around 6–7% of world population has Rh Negative blood. This again differs among countries and ethnicities. While among Europeans these numbers are as high as 20%, the Asian countries have counts <3%. In far eastern countries like China and Japan, these numbers fall as low as below 1%. With this reduced numbers, the task of finding Rh D negative blood in Asian countries is even more challenging.
Aims: To find out the percentages of regular Rh D negative donors and find out whether there is any significant number of Rh D negative donors becoming regular donors compared to Rh D positive donors handled by the DGH Hambantota blood bank from 1st of January 2016 to 31st of December 2016 (both days included). And to find out the composition of the donor population for the above period, based on different blood groups. To find out what percentage of in house donors become regular donors compared to mobile donors.
Methods: Retrospective descriptive study on donor population received by DGH Hambantota over 1 year period to find out whether there is a higher tendency for Rh Negative Donors to become regular donors, from the records available at the blood bank at DGH Hambantota. In house and mobile donations were analysed separately.
Results: The total number of donors during the study period were 2155, out of which 1950 (90%) were mobile donors and 205 (10%) were in house donors. 1468 (75%) of mobile donors were regular donors whereas 482 (25%) were first time donors. There were 1394 (95%) Rh D positive donors among the regular mobile donors and 74 (5%) were Rh D negatives. 456 (94%) first time mobile donors were Rh D positive while 26 (6%) were Rh D negatives. There was no statistical significance between the two groups.
Among in house donors, 183 (89%) were regular donors and 22 (11%) were first timers. 164 (89%) of regular in house donors were Rh D positive while 19 (11%) were Rh D negatives. 21 (95%) of first time in house donors were Rh D positives and 1 (5%) was Rh D negative. Again there was no statistically significant difference.
There was more chance of an in house donor to be a regular donor compared to a mobile donor which was extremely statistically significant ($P < 0.0001$). This was true for both Rh D positive group ($P < 0.0001$) and Rh D negative group ($P = 0.04$).
There were 969 (45%) O positives, 48 (2%) O negatives, 526 (24%) B positives, 44 (2%) B negatives, 453 (21%) A positives, 19 (0.8%) A negatives, 87 (4%) AB positives, and 9 (0.4%) AB negatives. The number of B negative donors compared to B positive donors were statistically significant ($P = 0.0106$).
Summary/Conclusions: The in house donors are regular donors compared to mobile donors. However there is no increased chance of much rare Rh D negative donors to become regular donors either in house or at mobiles. This may be due to a lack of proper feedback and reminding mechanism to retain negative blood group donors within the donor pool. The increase occurrence of Group B Rh D negative donors in the donor population may be a local variation and will require more in depth study.

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BLOOD SECURITY STRATEGIES FOR INTERNATIONAL CONFERENCE BASED ON THE INVESTIGATION INTO THE TREND OF UNPAID BLOOD DONATORS IN HANGZHOU, CHINA FROM 2011 TO 2015

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Background: The infusion of blood and corresponding products saves millions of lives every year, including conflicts, natural disasters and other emergencies. They can help the patients inflicts with diseases that threaten their lives live longer, improve the quality of life, and support some complex medical and surgical operations.
Aims: To conduct a statistical analysis of the structure of unpaid blood donators in Hangzhou, China from 2011 to 2015, and provide a solid scientific basis for the blood security on international conference and healthy development of unpaid blood donation.
Methods: Collect the data of unpaid blood donators in Hangzhou from 2011 to 2015, analyze the percentage of unpaid blood donators of different ages, genders, careers and academic degree to the total number of donators at the same period by means of GraphPad Prism, and carry out a short-term investigation into the situation of blood donation among the people nearby unpaid blood donation vehicle.
Results: The number of male donators is obviously larger than that of female ones. The young people of 18–34 years old, people of junior college degree and higher and those with uncertain careers constitute the main part of unpaid blood donators. From the investigation into the people surrounding the blood donation vehicle, it is found that 74.2% interviewees support unpaid blood donation. Based on the analysis of the trend of unpaid blood donators as well as the survey, the blood security strategies are thus formulated and the task of blood collection and security during the summit is completed.

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Summary/Conclusions: G20 Hangzhou Summit's blood security task is satisfactorily completed. The practical act is of great significance for improving the ability of handling severe emergency public health events and guaranteeing blood security in domestic and international large-scale activities.

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CORRELATION BETWEEN RED CELL INDICES AND IRON STATUS (FERRITIN SERUM) IN DONORS AT PMI YOGYAKARTA

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Background: About 350 mL blood was drawn in each donation. Each milliliter of blood contains approximately 0.5 mg iron. Hemoglobin measurement with cut off 12.5 mg/dl could not exclude donor with iron deficiency. Ferritin serum is a sensitive parameter to detect iron deficiency in early stage. Unfortunately in Indonesia, the test device to evaluate ferritin serum is not always available and costly. On the other hand, evaluation of red cell indices (MCV, MCH, MCHC) has been used for long time and can be done with simple equipments.
Aims: The aim of this study is to investigate the correlation between red cell indices and ferritin serum in blood donors at PMI (Indonesian Red Cross) Yogyakarta.
Methods: The study used a cross sectional design. At correlation test, subject were divided based on gender, age, and number of donation in past 1 year. Correlation between red cell indices (MCV, MCH, MCHC) and ferritin serum was tested with Pearson correlation test.
Results: The Pearson correlation test showed significant correlation between red cell index (MCV) and ferritin serum in male donor who donate three times in past 1 year ($r = 0.581$; $P = 0.015$).
Summary/Conclusions: There was significant positive correlation between red cell index (MCV) and ferritin serum in a group of male donor who donate three times in the past 1 year.

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A PROBE INTO THE RE-RECRUITMENT STRATEGY OF LOST BLOOD DONORS

F Ou

Voluntary Blood Donation Office, Qinghai Province Blood Center, Xining, China

Background: To analyze the reasons for loss of unpaid blood donors, and through re-recruitment of lost blood donors, to reduce the loss of blood donors and expand the team of fixed blood donors so as to promote long-term stable development of unpaid blood donation
Aims: To analyze the reasons for loss of unpaid blood donors, and through re-recruitment of lost blood donors, to reduce the loss of blood donors and expand the team of fixed blood donors so as to promote long-term stable development of unpaid blood donation
Methods: Blood donors that had blood donations over 3 times (including 3 times) during 2012–2014 but didn't have a donation in past 1 year were selected by using Tangshan modern blood information management software, and were randomly followed up by phone. Reasons for loss of blood donors were analyzed and targeted propaganda and re-recruitment were conducted.
Results: Statistics of factors for loss of unpaid blood donors showed that busy work and forgetting the time for donations has significantly higher constitution (69.8%) than other factors. The designated personnel conducted re-recruitment of lost blood donors, looked for the reasons for their failure to more donations and worked to solve them during the recruitment, and the blood donors are often willing to have more blood donations.
Summary/Conclusions: Fixed blood donors guarantee adequate and safe blood supply of the blood stations, and through re-recruitment the lost blood donors are often willing to have more donations, which reduces the loss of blood donors and expands the team of fixed blood donors.

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ANALYSIS THE INFLUENCE FACTORS OF VOLUNTARY BLOOD DONORS' RE-DONATION IN LUOYANG

K Xiao

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Background: Using the questionnaire of "Related behavioral factors of voluntary blood donors' to analyze the influence factors of voluntary blood donors' re-donation in Luoyang.
Aims: Investigation and analysis the influence factors of voluntary blood donors' re-donation in Luoyang, it would provide the broad thinking in recruiting and attracting more people to participate involuntary blood donation and strengthen the blood donation teams.
Methods: The questionnaire of "Related behavioral factors of voluntary blood donors' were conducted and collected in Luoyang, using two Logistic regression to analyze the influence factors of voluntary blood donors' re-donation.
Results: 1) In 1472 questionnaires, including 867 males, accounted for 58.91%;there were 477 donors in 25–34 years old, accounted for 32.40%, 408 donors in 35–44 years old, accounted for 27.72%; 629 donors' were college/undergraduate, accounted for 42.73%; 432 donors were middle school and below, accounted for 29.35%; In the occupation, there were 448 farmers, accounted for 30.43% 378 workers, accounted for 25.68%; 1003 people would repeat donation, accounted for 68.13%. 2)The influence factors of voluntary donors' re-donation in Luoyang were the age, educational background, knowledge of voluntary blood donation, satisfaction of the voluntary blood donation environment and the service attitude.
Summary/Conclusions: Retaining 25–34 and 35–44 age groups and keeping 55–60 age groups blood donors, mobilizing all positive factors of donors with college/undergraduate or middle school and below, deepening and expanding publicity, providing good environment for blood donation and improving the quality of service, all of those would have the positive significance in retaining more donors become the fixed blood donors.
Helvetica; color: black; mso-font-kerning:1.0 pt;mso-ansi-language:EN-US;mso-fareast-language:ZH-CN; mso-bidi-language:AR-SA'>influence factors of voluntary blood donors' re-donation in Luoyang

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THE RESEARCH AND APPLICATION OF DATA ANALYSIS OF VOLUNTARY BLOOD DONATION RECRUITMENT IN BIG DATA ERA

F Jing

Quality Control, Tianjin Blood Centre, Tianjin, China

Background: With the popularization day and day that the computer employed, the management information system (MIS) of increasing constantly has been set up in the departments of enterprises and institutions progressively, the MIS has already become the indispensable important component in modern management activity as the important sign of the modernization of management of information. In order to meet the requirement of National Health and Family Planning Commission of the People's Republic of China, must change recruiting the mode and work pattern of already existing non-remunerated blood donation. According to the recruitment of blood donation blood center, the blood management system further development, combined with the analysis of the age of big data, the use of micro media platform, try to promote the recruitment of blood donation.
Aims: To analysis the possible application of data in the recruitment of voluntary blood donation in big data era.
Methods: Back review the significance, model and application of the data analysis in the era of big data. To design and develop the blood donation APP, combined with the application of micro media platform, try to carry out the trials.
Results: Research and development of voluntary blood donation APP, through the wechat public number to send links propaganda, comprehensive consideration of the

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application of APP. Design statistical cluster variable analysis scale was used to evaluate knowledge-attitude-practice (KAP)questionnaire and lifestyle variables of voluntary blood donation, Determine the amount of blood collection, blood sampling, blood distribution and other factors in the next 6–12 months. To statistics differences data between 2 years before and after the application of APP, and constantly improve and perfect the application of APP.
Summary/Conclusions: Through the application of micro media platform of the free blood donation APP, is conducive to supply the side structural reform, improve the supply of blood quality.

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TURN AROUND TIME OF BLOOD DONATION IN BLOOD TRANSFUSION CENTRE, FACULTY OF MEDICINE, KHON KAEN UNIVERSITY, THAILAND

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Blood Transfusion Centre, Blood Transfusion Centre, Faculty of Medicine, Khon Kaen, Thailand

Background: Non- remunerated Blood Donors need to be given the best care. Because there are many steps for blood donation. Therefore, the process to secure and satisfy must be concern
Aims: The aim of this study was to evaluate turn around time for non-remunerated blood donation in Blood Transfusion Centre, Faculty of Medicine, Khon Kaen University, Thailand.
Methods: Retrospective study of turn around time for non-remunerated blood donation (N = 100) in Blood Transfusion Centre, Faculty of Medicine, Khon Kaen University, Thailand. The data was collected by walk in non-remunerated blood donor application. Statistical methods for Mean ± SD time period of (I)Waiting time for the screening room (II) Waiting time for blood donation (III) Time for blood donation (IV) Waiting time for rest in bed and (V) Waiting time for blood pressure check.
Results: Turn around time for blood donation is 44 ± 12 min. It can divide into (I) Waiting time for the screening room (II) Waiting time for blood donation (III) Time for blood donation (IV) Waiting time for rest in bed and (V) Waiting time for blood pressure check were 6 ± 4, 5 ± 3, 7 ± 2, 6 ± 4 and 19 ± 10 min, respectively.
Summary/Conclusions: Forty-four minute for once blood donation is effective. The time for rest after blood donation is 19 ± 10 min, the non-remunerated blood donation will take only 25 min for finish the procedure. The satisfaction rate were more than 90%.

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A BRIEF DISCUSSION ON THE EFFECT OF "GREEN" BLOOD DONATION IN SUMMER ON THE ALLEVIATION OF BLOOD SOURCE TENSION

X Su

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Background: A lot of people are reluctant to go out in the hot weather, which can lead to blood pressure and decrease in blood supply.
Aims: To analyze how to make blood donors under the hot weather factors actively want to donate blood, improve blood work in summer's response, so as to improve health, reduce blood tense situation.
Methods: By visiting, issue the questionnaire to investigate the proposal of 200 blood donors on the improvement measures of "green" blood donation in summer.
Results: Provide green information, 80% of the most frequently used by investigators mentioned WeChat communication software with mobile maps inform blood donation car parked about time, location, improve blood donation member found for accuracy of blood donation, avoid blood donation member summer around looking for blood donation nervousness; Provide the green channel, 75% of the blood donation hope more than ten people donate blood groups have blood donation vehicles to the scene and can online booking directly to the site, avoid high temperature the blood donors go to donation points have uncomfortable journey and saving time also; Green facilities, 95 percent of the respondents responded to the green hardware condition, the blood donation vehicle should be cool and ventilated to ensure that the blood donation is best suited to the temperature; Green decoration, 68 percent of the respondents mentioned improving the appearance of blood donation vehicles, the summer anxiety saw the green car accessories and red propaganda more pleasing to the eye; Green action, 67% of the investigators noted that blood stations can organize some fixed donors green exchange activities, some of the outdoor living collective activities actively, let fixed donors increase publicity of blood donation.
Summary/Conclusions: In response to the hot weather in summer, we should know the needs of the blood donors, promote the green blood donation, keep the blood source, raise the blood volume, and relieve the blood supply stress in summer.

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ANALYSIS OF THE STATUS OF UNPAID BLOOD DONORS IN DALIAN WANDA BLOOD DONATION HOUSE FROM 2015 TO 2016

P Xin and G Yahui

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Background: There is one of the important therapeutic methods to Safe lives in clinical is safe blood transfusion. With the rapid development of the national economy, the overall improvement of the medical level, blood transfusions has increased sharply. This phenomenon brings great pressure to blood transfusion services. The serious problems faced by blood transfusion services are to explore the potential of blood donors, expand the blood donation teams and develop the permanent blood donors. It is very important to analyze the regional blood donors to make specifically propaganda and recruitment for blood donation.
Aims: Dalian Wanda blood donation house was completed in late August 2014. This article mainly understands the status of unpaid blood donors in Dalian Wanda blood donation house from 2015 to 2016, which explores the effective way and benign mechanism of Wanda blood donation house unpaid blood donation work, and provides scientific basis for ensuring the future healthy development of blood donation in this area.
Methods: The data of unpaid blood donors in Dalian Wanda blood donation house were collected by SHINOW9.0 system. The data were analyzed by SPSS19.0 according to sex, age, educational level and occupation, which were compared by X^2 test, which was statistically significant by $P<0.05$.
Results: Male blood donors were significantly higher than female donors. 18–24 years old, undergraduate degree, students were the main blood donors without repayment in this region.
Summary/Conclusions: Wanda blood donation house is located in high-tech park. The main blood donors without repayment in this region had significant characteristics in terms of gender, age, educational level and occupation, to the surrounding college students as the main unpaid blood donors. Therefore, in the formulation of the recruitment strategy for blood donation and the work of blood donation, the appropriate recruitment strategies should be established according to the students. However, it is not to be neglected to high-tech park staffers who will be the potential recruit, which to promote the comprehensive development of blood donation without compensation in Wanda blood donation house.

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THE PRACTICE AND REFLECTION ON THE CREATIVE PROPAGANDA WORK OF VOLUNTARY BLOOD DONATION

M Wu

Blood Station of Jiaxing City, Jiaxing, Zhejiang Province, China

Background: To innovate and expand the content, ways of voluntary blood donation propaganda, will promote the sustained and healthy development of voluntary blood donation.
Aims: To innovate and expand the content, ways of voluntary blood donation propaganda, will promote the sustained and healthy development of voluntary blood donation.
Methods: Cooperating with the mass media, innovating the content of propaganda. Then, we will choose the topic of activities carefully, and optimize different kinds of promotion ways. Finally, we are excepting the whole community could engage in these actives and improve the propaganda ways together. In order to evaluate the

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effect of different kinds of propaganda ways, the annual volume and number of blood donation were collected and analyzed.
Results: From 2012 to 2016 in Jiaxing City, the total number of blood donation increased from 41934 to 47806, the average annual growth rate was 2.73%, there were statistically significant differences between growth rate ($P < 0.01$); whole blood volume increased from 13168.43 to 14859.3 L, apheresis blood volume increased from 2618.5 U to 4691.4 U; the number of blood donors who from Colleges and universities in this city, increased from 2556 to 4137, volume increased from 753.53 to 1217.28 L, the difference of blood donation and blood donation was significant ($P < 0.05$).
Summary/Conclusions: Applying omni-directional, multi-channel and multi-dimensional propaganda and education of voluntary blood donation, the number of voluntary blood donors and volume of blood donation can be increased continuously.

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INVESTIGATION ON BLOOD DONATION OF INSUFFICIENT BLOOD DONATION IN CHANGSHA IN 2016

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Background: Blood resources are precious, Blood physical waste (including hemolysis, blood fat, clots, breakage, precipitation, deficiency, etc.) has many conditions,
The blood deficiency occupies a certain proportion, Although it cannot be eliminated, it can be prevented and controlled. Through statistics, in January 2016–2016 December, different blood donors were not sufficiently distributed, and the basic situation of insufficient blood donation was mastered, To analyze the causes of insufficient blood donation, and to explore the prevention of insufficient blood donation, Reduce the rate of blood loss and the negative psychological impact on unpaid blood donors.
Aims: To understand the situation of insufficient blood donation in Changsha area, analyze the distribution of insufficient blood donation and discuss the reasons.
Methods: The statistical data of blood collection in 2016 was analyzed, including the insufficient blood donation, to assess the proportion and distribution of insufficient blood donation in different donors.
Results: The proportion of insufficient blood donation in Changsha area was 1.32% in 2016. The distributions of insufficient blood donation, among donors in different genders, ages, occupations, blood quantities, and characters, had statistically significant differences.
Summary/Conclusions: The reasons for insufficient blood donation were analyzed and corresponding preventive measures should be taken. Improving the blood donation service, collection technology and the blood collection environment are helpful to reduce the occurrence of insufficient blood donation and save blood resources.

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Abstract has been withdrawn

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DEVELOP MEDICAL SOUVENIRS AND RECRUIT MORE BLOOD DONORS

X Jun

Human Resources, Changsha Blood Center, Changsha, China

Background: In this age of scarcity, there has been no form of propaganda that can remain invincible for a long time. In recent years, the Imperial Palace Museum in Taipei has developed a lot of tourists through the development of cultural and creative products, which has greatly spread the traditional history and culture of the Imperial Palace.
Aims: Blood centers should develop more souvenirs to recruit more blood donors.
Methods: Development of creative blood donation souvenirs to grasp the three elements (1)Culture: Culture is the core. We should cultivate free blood donation and love culture, and make souvenirs a good quality carrier of new blood donation culture. Through the blood donors' cultural impression on the blood donation souvenirs, we can realize the dissemination and reappearance of the love culture.
(2)Innovation: the so-called blood donation is the donation of souvenirs, elements and outstanding creative combination of products, and it is quite common for people to rebuild, has become the new combination product blood culture, in blood donors and blood donation to establish a new relationship between cultures.
(3)Practical: souvenirs are carriers. The souvenir of blood donation should be widely applied to people's daily life, and realize the extension and cultural appreciation of the voluntary blood donation culture in the circulation.
Results: Combining the needs of blood donors with the functions of blood donation souvenirs, satisfying the mental needs and emotional appeals of donors, and satisfying their mental and spiritual well-being.
Summary/Conclusions: The souvenir of blood donation carries the expression of social respect and gratitude for the blood donors. It has the unique significance and value in use. From a certain point of view, the design, the color of the blood donation issue full of souvenirs, the implementation of the new consumer love culture in blood donors, can play the effect of interpersonal communication, to recruit newer and more potential donors.

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SOLUTIONS ON HOW TO DEAL WITH THE BOTTLENECK PROBLEM OF NON-REMUNERATED BLOOD DONATION IN TIANSHUI CITY

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Background: 1 Citizen remunerated blood donation system was abolished on 1 October 1998. Our country is taking full voluntary non-remunerated blood donation presently. Government finished helping blood centers to do the voluntary blood donation from paid blood donation, and no longer restrained it with administration.
2 For a long time, recruiters have been over depending on government administrative orders and documents to urge the administrative enterprises and institutions to participate in unpaid blood donation, which is rigid, and with no new ideas.
3 Recruitment of non-remunerated blood donation is simplified. General public awareness of blood donation is not strong. What's more, negative news reports and blood service staff attitude directly affect the non-remunerated blood donation recruitment work. Therefore, non-remunerated blood donation ran into a bottleneck stage.
Aims: The Recruitment of Voluntary Blood Donors aims to meet the needs of blood and blood components for clinical use. In order to ensure the safety of blood use for patients in hospital and completely make blood and blood components for clinical use free of charge, it is necessary to eliminate paid blood donors.
Methods: 1. Promote blood donation organizations in parties and governments, enterprises and institutions, colleges. 2. Let the blood donors do propagandists obligations. 3. Let the public know the meaning of voluntary blood donation illustrated by the media of radio, television, newspapers. 4. Thanks by telephone or text messages to voluntary blood donation crowd and encourage and mobilize the donation again.
Results: Through the above methods and nearly 10 years' efforts, voluntary blood donation in our city has reached 100%, completely eliminated the paid blood, prevented blood borne infectious diseases, and solved the difficult problem of unpaid blood donation.
Summary/Conclusions: Implementation of voluntary blood donation of 100% can be achieved by mutual discussion, idea collections, careful planning, and enhancement of propaganda. However, blood donation is a long-term and arduous task. With the technical progress of medical science, increase of the amount of blood components, plus the state banned blood donation policy issued; many units have no longer cooperated with the blood donation work. At the same time, the rise of rural cooperative medical care and the clinical use of blood increased significantly, blood donation has the bottleneck problem. How to find effective measures in order to meet the needs blood for clinical use and solve the problem of recruitment of unpaid blood donors and make it go smoothly? It is a question to be discussed.

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FEATURES OF REGULAR BLOOD DONORS' IRON METABOLISM

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Background: Since 2010, a state program to develop voluntary blood donation has been implemented in the Republic of Tajikistan, but a shortage of donors still persists. One particularly important problem is donation refusal for medical reasons. One of the most frequent causes of withdrawal from blood donations in the Republic of Tajikistan is low hemoglobin (Hb) rates. The frequency of withdrawals for this reason has ranged in different years from 3.9 to 7.4% and represents 29 to 36% of the total number of withdrawals. These high rates of withdrawal from donations are associated with the prevalence of latent iron deficiency, especially in women. Determining serum ferritin concentration, erythropoietin and soluble transferrin receptors amongst regular and withdrawn donors allows monitoring iron reserves.
Aims: To determine the metabolism of iron deficiency in regular blood donors.
Methods: The level of serum ferritin was defined in 800 regular donors (400 men, 400 women). 400 of them received additional iron for the prevention of anemia (group I), 400 – did not receive prophylaxis (group II). Hb concentrations in male donors were >130 g/l, female >120 g/l. In those donors who had withdrawn from blood donations due to low Hb (n = 100; 50 men and 50 women), serum ferritin, erythropoietin and soluble transferrin receptors were investigated. Hb levels in males were <130 g/l; <120 g/l in women. Donor's ages ranged from 18 to 60 years old. In the group of donors who were rejected from donating blood due to low Hb levels, some were selected for the study by random sampling from the number of potential primary and regular donors. The concentration of serum ferritin was determined using radioimmunoassay, while erythropoietin and soluble transferrin receptors were determined by fluorescent immunological analysis.
Results: Serum ferritin levels in regular male donors were 65–130 ng/l; female – 45–100 ng/l. In group I donors, who were treated with 250–320 mg of preventive iron after blood and plasma donations, 250 donors observed an increase of serum ferritin levels, 138 donors observed unchanged concentrations, and there was a reduction observed in 12 female donors. In group II donors who did not receive iron supplementation, an increase in serum ferritin levels was observed in only 15 donors, while a decline was observed in 200, and levels were unchanged in 185 donors (mostly men). Hb levels in male donors who had been removed from donation were 114–126 g/l; in female donors – 92–116 g/l. A decrease in serum ferritin was diagnosed in 91 out of 100 donors; the mean value for males was 1.8 and for females 4.1 times lower than in the control group. In 74 donors serum ferritin did not exceed 14 ng/ml. Elevated levels of soluble transferrin receptors were detected in 36 out of 100 donors, which was also combined with a low content of serum ferritin. Soluble transferrin receptors values ranged from 27.62 to 54.22 nmol/l (standard 7.89–29.51 nmol/l), and the median exceeded the control level by 2.6 times and amounted to 40.57 nmol/l vs. 15.72 nmol/l in the comparison group. The concentration of erythropoietin was higher than the threshold in 45 out of 100 donors with low ferritin values. In 20 cases this was combined with a high rate of soluble transferrin receptors. The rest of the erythropoietin levels were within the normal values, however, the median was 2.4 times above the median of the control group.
Summary/Conclusions: Prophylactic iron use in regular blood donors provided the opportunity to minimize blood donation rejections associated with reduced hemoglobin levels.

Blood Collection Incl. Apheresis

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ANALYSIS ON THE TRENDS AND CHARACTERISTICS CHANGE OF PLASMA COLLECTION IN CHINA

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Background: The shortage of resource plasma is still the main factor that restricts blood products in China. The restriction and influence of government policy has an important impact on the plasma collection. It would have a huge effect on plasma collection that a series of policies of some provinces in recent years, including shutting down parts of plasma collection centers in Guizhou and providing a initiate promotion measure in Guangdong. This study analyzed the trends and the characteristics change of plasma collection under the effect of local policies of China in recent years.
Aims: To find out the characteristics change of the plasma donors in China under the influence of local collection environmental changing. We analysed these results and trends to predict the development of plasma collection in future.
Methods: Comparing and analysis were carried out for the national plasma collection data from 2011 to 2015 by statistical methods. The national data included the number of plasma donors, the quantity of plasma and the occupation distribution of the donors from 2011 to 2015.
Results: The total amount of the national plasma collection in 2015 was nearly 1.5 times that of 2011. The average annual growth rate of national plasma collection was 10.65%, and the average annual growth rate of the number of donors was 15.69% during the 5 years. The top five provinces (except Guizhou) with the volumes of plasma collection remained unchanged, but their proportion decreased from 66.9% to 53.5% during the period of 2011 to 2015. The annual growth rate of traditional plasma collection region has decreased year by year, while the growth rate of other provinces has increased year by year like Guangdong, Xinjiang and Chongqing. In particular, the growth rate of Guangdong province reached 127.91% in 2015. The annual national frequency of plasma donation per person was 5.6 during the period of 2011 to 2015. And the annual frequency has decreased from 6.1 to 5.0 since 2012. This phenomenon appeared in most provinces, and the average fall of these provinces was 23.3% in 2015 compared with 2011.
Farmers is still selected as the major objects for plasma donation in China accounting for more than 80% of the total amount of donors, but the occupation proportion has begun to change significantly due to the proportion of farmers which has decreased from 89.11% in 2011 to 82.96% in 2015. The proportion of workers and other occupations rose from 1.84% to 4.32% and from 6.84% to 10.71% respectively. The proportion of farmers in most provinces has been decreased in different degrees except the three provinces in the northwest China and a few agricultural provinces.
Summary/Conclusions: The new pattern of plasma collection in China is taking shape process. The traditional plasma collection provinces in the southwest of China may be replaced by the emerging regions in the future. The occupation proportion of plasma donors is also undergoing positive changes, although it is mainly farmers in recent years. But it is would be replaced by workers and other professionals if possible with the expansion of coastal provinces like Guangzhou. At present, the bottleneck of plasma collection in China is still how to retain the fixed donors. The results showed that national policy has significant effect on plasma collection in China. How to formulate the policies and supervise policy implementation will lead and decide the development of plasma collection in the future.

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INVESTIGATION AND ANALYSIS THE BASIC SITUATION OF APHERESIS PLATELET DONATION IN WENZHOU CITY

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Background: With the rapid development of the blood transfusion medicine and technology, the traditional transfusion fundamental change, the concept and mode components become the development direction of scientific and safe blood transfusion, and machine by platelets in the treatment and prevention of significant curative effect of the blood coagulation dysfunction, is widely applied in clinical treatment. In recent years, Wenzhou region machine adopt platelet clinical rising demand, the obvious contradiction between supply and demand, in order to fully understand the present situation of the machine adopt the platelet donors team, improve the work of recruiting retention, guarantee the adequate and safe for clinical use Wenzhou region of the author in 2014–2016 machine adopt platelet donors analysis of the basic situation of the investigation.
Aims: To know the basic situation of apheresis platelet donation in local area, to provide reference for the establishment of a stable team of blood donors.
Methods: The study of 21, 113 blood platelet donors who successfully donated platelets during the 2014–2016 period, using the software of Zhejiang province blood management and control information system BIS2.0, to investigate and analyze the basic situation of apheresis platelet donation.
Results: The gender, age, cultural degree and occupational all influence the blood donors in Wenzhou area, between 2014 and 2016, the proportion of men and women in the Wenzhou area was maintained at 15.7:1–13.6:1, among them the age of 26–45 was 71.3 percent of the blood donors, the cultural degree of blood donor

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was high school and secondary school (31.79%) in the first, the percentage of low-educated blood donors is decreasing year by year (see table 2), the highest proportion In terms of occupational distribution is employees of the enterprises, freelance residence In the second (29.90%), in the last 3 years, the rate of double platelet donation was increased from 4.11% to 20.01%, however, there was no statistical difference in the rate of blood donation reaction.

Summary/Conclusions: Among blood platelet donors in the Wenzhou area women, higher education, students and institutions have a lower proportion of employees, further mobilization is needed in the future, double platelet donation is lower, still have tremendous room for improvement, there are some unreasonable places in the structure of blood platelet donors, should appropriate adjustment and improvement of current recruitment priorities, develop a local advertising recruitment strategy, to ensure adequate and safe blood supply.

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CONTRAST ANALYSIS BETWEEN RELATIVE BLOOD DONORS AND NON-RELATIVE BLOOD DONORS

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Background: With the continuous increase in clinical blood, all around free of charge. Blood donation has been unable to meet the needs of clinical blood, in this case, mutual blood donation as a first aid and Clinical health needs of a special model, but there are still some differences with the group of free blood donors, so to expand the blood donation team with the continuous increase in clinical blood, all around free of charge Blood donation has been unable to meet the needs of clinical blood, in this case, mutual blood donation as a first aid and Clinical health needs of a special model, but there are still some differences with the group of free blood donors, so to expand the blood donation team.

Aims: To analyze the differences between relative blood donors and non-relative blood donors.

Methods: To observe and compare the characteristics of relative blood donors and non-relative blood donors, and to analyze the related factors of relative blood donors and non-relative blood donors.

Results: Compared with the staff, relative blood donors were more man than woman, the composition of the staff has nothing to do with the age and education level, and the enthusiasm of non-relative blood donors was positive correlation to their educational levels. 2. Comparison of blood from the motivation, the first blood donation more than repeated blood donors ($P < 0.01$). 3. From the psychological attitude comparison, family members donation blood donation motive to "family friendship" mostly, significantly more than other reasons ($P < 0.01$), voluntary unpaid blood donors are "dedicated love" majority, significantly more than other reasons ($P < 0.01$). 4. The rate of blood donation was significantly higher than that of voluntary blood donation.

Summary/Conclusions: Aim of relatives to give their blood may be to help their relatives to cure diseases. It would be only supplementary while the blood supply is shortage. So we should volunteer more and more people to join the blood donation group include the relatives of patients.

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BLOOD DONOR DEFERRAL PATTERN IN ISLAMABAD, PAKISTAN

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Background: The increasing demand of blood and blood products in developing counties like Pakistan reflect the need for mobilization of more voluntary blood donors. The donor selection process requires continuous monitoring to ensure that it achieves its objectives of ensuring donor and recipient safety and providing a sufficient supply of blood and blood components. The blood donors are deferred from donating blood for numerous reasons, either permanently or temporarily. Knowledge of the reasons for donor deferral can help in planning more effective recruitment and retention campaigns aiming at the availability of safe donors. Blood donor deferral leads to loss of available blood units for transfusion.

Aims: To assess the blood donor deferral pattern in the capital city of Islamabad.

Methods: This was a retrospective study conducted through the data reported to IBTA from the 20 licensed blood banks of Islamabad during the period Jan – Dec 2016. Data were entered into SPSS (version 2.0) and frequencies of each causes of deferral were calculated

Results: Data of 71,795 blood donors who came to donate at the 20 licensed blood banks or the camps organized by these blood establishments of Islamabad were analyzed. Among these donors, 68,274 (95.1%) were selected for blood donation after physical and behavioural screening. 3,521 (4.9%) blood donors were deferred (temporary n = 2779/permanent n = 742) from blood donation at this stage. In addition, 2,538 blood donors were deferred permanently after post donation serological screening on the basis of Transfusion Transmissible Infection reactivity. The pre-donation deferral rate was considerably higher among females (59.5%) compared to (40.5%) males.

The causes of pre-donation deferral were Anaemia 39.0% (n = 1,113), Under Weight 16.6% (n = 474), Inappropriate Height Weight Ratio 0.8% (n = 23), Menstruation 5.2% (n = 148), Low Blood Pressure 1.1% (n = 30), Taking Medicines 1.5% (n = 43), H/O Fainting/Vertigo 0.5% (n = 15), Recent Vaccination 1.2% (n = 35), Recent Blood Donation 4.5% (n = 127), Underage/Overage 5.7% (n = 163), Flu/RTI 1.4% (n = 39), Gastroenteritis 0.6% (n = 18), Typhoid 0.5% (n = 13), Hypertension 1.6% (n = 45), Arthritis 0.4% (n = 12), Chest Pain 0.01% (n = 1), Difficult Vein/Obesity 0.8% (n = 22), Blood Phobia 0.9% (n = 25), Fever 1.1% (n = 31), Fasting 0.4% (n = 11), Fear of Needle 0.2% (n = 6), Without Breakfast 3.1% (n = 90), Known H/O Hepatitis B and C 0.4% (n = 11), Asthmatics 0.8% (n = 24), and others (e.g. heart diseases) 15.2% (n = 435). With regards to the causes of post-donation deferral, 110 (0.16%) were reactive for HIV, 894 (1.31%) for Hepatitis B and 1,187 (1.74%) were reactive for Hepatitis C virus. 0.14% were reactive for Malaria and 0.72% for Syphilis.

Summary/Conclusions: Credible scientific data of an important urban city, Islamabad, generated by a regulatory authority indicates that about 5.1% of the blood donors who presented for blood donations were unfit for donations temporarily or permanently. This study therefore underlines the need to focus on retention of suitable voluntary as well as replacement donors in addition to further strengthening donor management systems.

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STUDY ON THE INTERVENTION EFFECTS OF CHYLEMIA IN APHERESIS PLATELET DONORS

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Background: According the China national standard GB18469-2012 *Quality Requirements for Whole Blood and Blood Components* that blood components shall be no severe Chylemia which has evidence to impact the effect of blood transfusion. In Beijing Red Cross Blood Center, chylemia cause 25.5% of apheresis platelet (AP) donor deferral, which waste the blood resources and impact the passion of AP donors.

Aims: To study the effects of different intervention measures on AP donors who have chylemia at first screening, and specify priority intervention measures to increase the platelets collection.

Methods: To intervene of AP donors who have chylemia with the diet education, drinking more water, and do exercise at our blood center from April 2015 to November 2016, and to analyze the intervention effects.

Diet education: Provide diet education to AP donors who have chylemia. To do the test again at next day and evaluate the intervention effects.

Drinking-water method: donors drink water (3–5 cups) within 2 h and do the test again.

Exercise and Drink water method: donors do exercises (brisk walk for 20 min.) and drink water (3–5 cups). Do the test again 2 h later.

Results: For AP donors who have mild chylemia, the donation rate after diet education and exercise method were 90% and 93.8%, respectively. There was no significant difference between two methods. The success donation rate of the drinking-water method was 77.4%.

For AP donors who have moderate chylemia, the donation rate after intervene of three methods were 87%, 83.3% and 82.9%, respectively. There were no significant differences among the three methods.

For AP donors who have severe chylemia, the intervention effects of the drinking-water method and the exercise method were similar, with the success donation rate of 51.4% and 54.2%, respectively. While 90% of AP donors who intervention with the die method donate AP successfully when they come back to donate at next day; obviously, the intervention effect was much better than those of the other two methods.

Summary/Conclusions: Deferral of apheresis platelet donors caused by chylemia can be intervened. There are different chylemia intervention methods. Consider the operation conventional, For AP donors who have mild chylemia, the drinking-water

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method can be used. For those who have moderate chylemia, the drinking-water method or exercise method may be used. The dietary intervention method should be considered as the first choice for those who have severe chylemia. Intervening in chylemia could reduce the occurrence of AP donation deferral, save cost, and increase AP collection volume. At our blood center, the AP donors who have chylemia, but intervention successfully increased 2% AP collection since 2015.

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Abstract has been withdrawn

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FACTORS AFFECTING PLATELET YIELD DURING PLATELETPHERESIS DONATIONS

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Background: High-yield plateletpheresis enables the collection of multiple units from the same donor in a single donation and may have the potential to reduce donor exposure, risks inherent with allogenic transfusion, effective management of platelet inventory and cost effectiveness of the procedure
Aims: This study was conducted to assess the factors affecting platelet yield and quality of product during Plateletpheresis donations.
Methods: A total of 500 plateletpheresis donors were included in this prospective study. All procedures were performed using a fully automated platform of plateletpheresis, Trima Accel version 5.1 (Gambro BCT), Amicus Crescendo version 3.2 (Baxter Healthcare Corp.).The parameters assessed were donor physical characteristics, donor hematological profile (pre donation) and the procedural variables with the platelet yield of the product harvested and divided into two groups Group-I ($\leq 4 \times 10^{11}$/unit, n = 318) and Group-II ($>4 \times 10^{11}$/unit, n = 182).
Results: A total of 500 prospective plateletpheresis donors (age range 18–56 years) were enrolled. In our study actual platelet yield was positively correlating with the weight ($r = 0.155$) of the donors [Group-I (72.47 ± 9.72) and Group-II (75.37 ± 9.98)], and height ($r = 0.129$) of the donors [Group-I (170.31 ± 4.29) and Group-II (171.46 ± 5.14)] and is statistically significant ($P < 0.01$). Actual platelet yield harvested is negatively correlating with the Hb ($r = -0.15$) and Hct ($r = -0.59$) but was not statistically significant ($P > 0.05$). [Group-I (14.53 ± 0.98 g/dl) and (43.60 ± 2.74%) and in Group-II (14.49 ± 0.98 g/dl) and (43.22 ± 2.64%)]. Actual platelet yield was positively correlating with the pre donation platelet count ($r = 0.624$) of the donors and was statistically significant ($P < 0.01$). [Group-I ($2.19 \pm 0.436 \times 10^5$/μl) Group-II ($2.86 \pm 0.505 \times 10^5$/μl)]. Procedural parameter, blood volume processed ($r = 0.706$) [Group-I (2486.73 ± 389.698 ml), Group-II (3320.10 ± 502.87 ml)], was highly correlating with the actual platelet yield harvested and was statistically significant (P value < 0.01).
Summary/Conclusions: Pre donation platelet count and blood volume processed are the main predictables of platelet yield. Hence donors with pre procedure platelet counts ($\geq 2 \times 10^5$/μl) with higher body surface area can be considered for higher yield platelet products.

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THE INFLUENCE OF DONOR LEUKOCYTE CLASSIFICATION ON COLLECTION OF PERIPHERAL BLOOD MONONUCLEAR CELLS

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Background: The malignant tumor seriously endangers the human body health, and receives the widespread attention. At present, there are four main methods for the treatment of malignant tumors, including surgery, chemotherapy, radiotherapy and immunotherapy. Dendritic cell based immunotherapy has become an important and effective way to fight cancer. At present, the main way to obtain dendritic cells is through the differentiation and culture of peripheral blood mononuclear cells. The application of blood cell separator for peripheral blood mononuclear cells has a very important application prospect. Peripheral blood mononuclear cells, including monocytes and lymphocytes, are the two main cells in the classification of leukocytes. In this study, peripheral blood mononuclear cells were collected by five blood cell separators, and the effect of donor leukocyte classification on peripheral blood mononuclear cells was investigated.
Aims: To investigate the effect of donor leukocyte count and classification on peripheral blood mononuclear cells collected from five kinds of blood cell separator.
Methods: With MCS+, COBE Spectra, Amicus, COM.TEC and Spectra Optia five kinds of blood cell separator MNC acquisition program to collect peripheral blood mononuclear cells, compare the total number of white blood cells and classification in products and volunteers collected before, analysis the effect of white cell count and classification on the number of MNC in product acquisition using five kinds of blood cell separator.
Results: The total number of white blood cells had no significant effect on the MNC number of five kinds of blood cell separator. COM.TEC, Amicus, MCS+ three kinds of blood cell separator MNC number of products and pre harvest lymphocyte count was positively correlated, correlation coefficient were 0.829, 0.782, 0.672, $P < 0.05$, the difference was statistically significant.
Summary/Conclusions: Using five kinds of blood cell separator to collecting peripheral blood mononuclear cells, volunteers before harvest white blood cell classification and product MNC number are closely related.

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COMPARISON OF THE AMOUNT OF GRANULOCYTE INTERFUSION BETWEEN FIVE BLOOD CELL SEPARATORS FOR COLLECTING MONONUCLEAR CELL PRODUCTS

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Background: Peripheral blood mononuclear cells are used for biological treatment of tumor patients or for culturing dendritic cells for the study of cellular immunity vaccines, etc. However, the researchers found that the incorporation of granulocytes and platelets in the initial cultures of mononuclear cell (MNC) greatly affected the differentiation and development of DCS. The characteristics of platelets are different from those of mononuclear cells, which are easier to remove, but the effects of granulocytes are difficult to remove. Therefore, in this study, we compared the amount of granulocyte incorporation in peripheral blood mononuclear cells from five blood cell separators in order to provide the basis for optimizing the collection of peripheral blood mononuclear cells.
Aims: To compare the amounts of granulocytes mixed in peripheral blood mononuclear products collected by MCS +, COBE Spectra, Amicus, Spectra Optia and COM.TEC five kinds of blood cell separators.
Methods: Peripheral blood mononuclear cells of volunteers were collected by MNC program of five kinds of blood cell separators, MCS+, COBE, Spectra Amicus, Spectra Optia and COM.TEC. Measured the amount of granulocyte interfusion in the mononuclear cell products prepared by five kinds of blood cell separators respectively, and compared the difference of the amount of granulocyte mixing in the products collected by the blood cell separators.
Results: In the mononuclear cell products collected by five kinds of blood cell separators, the amount of granulocyte in the products prepared by MCS+ was more than that of the other four kinds of blood cell separators. The difference was statistically significant ($P < 0.05$). Moreover, the percentage of mononuclear cells collected by MCS+ was lower than that of the other four blood cell separators ($P < 0.05$). In the acquisition time, Amicus acquisition time was the longest, MCS+ acquisition time was second, COBE, Spectra, COM.TEC and Spectra Optia acquisition time were shorter, the difference was statistically significant ($P < 0.05$).
Summary/Conclusions: Because of the performance and type of the blood cell separator were different, the quality and efficiency of collecting the mononuclear cell products were also different. In the premise of ensuring the quality and efficiency, we should choose the type of blood cell separator with a small amount of granulocytes to collecting peripheral blood mononuclear cells. At the same time, adjust the machine corresponding acquisition parameters, strengthen the quality control, could make the amount of granulocytes lowest mixing in the products.

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FREQUENT PLATELETPHERESIS AND ITS IMPACT ON THE HEMATOLOGICAL PARAMETERS: AN OBSERVATIONAL STUDY

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Background: Frequent plateletpheresis donors are assets for the blood banks. The well-being of these donors has been a matter of concern.
Aims: In our study we intend to analyze the effect of plateletpheresis on the hematological parameters of these donors assessed prior to each subsequent procedure. We also try to compare the effect 3 cell separators used for plateletpheresis on the post donation hematological parameters.
Methods: The study was conducted during February 2016 to March 2017 on all the repeat plateletpheresis donors coming to the Department of Transfusion Medicine for the 2nd time within a month of the first plateletpheresis. The values of the hematological parameters including red cell and platelet indices tested prior to each plateletpheresis were entered into the excel sheet and gap between each donations were calculated. The plateletpheresis were done either on Hemonetics MCS+ separator (Hemonetics Corporation, Braintree, Massachusetts, USA), Fresenius separator (COM.TEC), DN (Fresenius HemoCare GmbH, Bad Homburg v.d.H, Germany) and Gambro Trima Accel, software version 5.0 after taking consent from the donors. The target collection of each procedure was a dose of 3×10^{11} platelets in 200–250 ml of plasma. To compare the effect of the cell separators on the hematological parameters due to the plateletpheresis, parameters at 2 consecutive donations within 7 days were considered. Data was analyzed by Stata 14. Within change in the continuous variables were assessed by paired t- test and between two groups comparison was done by independent t-test or Wilcoxon Rank Sum test. The comparison among the cell separators was done by Kruskal-Wallis test or one way ANOVA. P-values < 0.05 were considered significant.
Results: Of the 98 donors, 35 repeated the plateletpheresis within a week (group I) and 63 underwent 2nd plateletpheresis within 8–30 days (group II). No significant alteration was found in the red cell or the platelet indices within either group but a significant difference in the variation of platelet counts of the 2 groups ($P = 0.025$). Though above the eligibility cut-off of 1.5 lakhs/μl, platelet counts were lower than baseline in group I donors whereas it was higher at 2nd plateletpheresis in group II donors. There were 49 donors who presented to us for the 3rd time for plateletpheresis with a mean gap between 1st and 3rd plateletpheresis being 46 days. No significant difference in the parameters assessed prior to any of the plateletpheresis was found except the platelet distribution width ($P = 0.000$). Plateletpheresis through all the 3 cell separators had similar effects on the hematological parameters.
Summary/Conclusions: There was no significant change in the hematological parameters in the plateletpheresis donors who underwent frequent plateletpheresis. Post donation follow-up hematological parameters were not affected by the cell separators used for plateletpheresis

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THE METHOD AND EFFECT OF IRON RESUPPLY OF LONG-TERM PLATELET DONORS

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Background: Most plateletpheresis donors are the relatively fixed blood donors. More care should be given to the blood donors, so that to establish a stable platelet donors.
Aims: The change of blood parameters of long-term platelet donors were detected after iron resupply, to analyze the necessity of iron resupply. More care should be given to the blood donors, so that to establish a stable platelet donors.
Methods: 42 blood donors who donated platelet more than 5 times in Henan red cross blood center were chosen, include 22 males and 20 females. The blood donors took 1 capsule iron daily (included 50 mg iron), The blood parameters were measured before and after taking the medicine 10 days.
Results: The HGB and HCT were increased significantly ($P < 0.05$) in male and female blood donors after they taking the medicine, but PLT levels in donors showed no significant difference after they taking the medicine.
Summary/Conclusions: Taking iron can improve the hematological indices of long-term platelet donors, reduced the occurrence of iron deficiency, improved the blood quality.

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ANALYSIS AND PREVENTION OF NON - STANDARD VOLUME BLOOD

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Background: Every year a large number of non-standard volume of blood is collected, these non-standard volume of blood collection will result in a huge waste of blood resources
Aims: To investigate the causes of non-standard volume of blood, to provide a basis for targeted measures.
Methods: The statistical analysis was made on the causes of 272 cases of non-standard volume blood in 55374 donors from January 2015 to January 2017.
Results: Blood donation-related side effects accounted for 61.8%, blood collection technique problems accounted for 30.1%.
Summary/Conclusions: Effective relief of blood donors 'tensions and improved nurses' blood collection techniques are important to reduce the collection of non-standard volume blood.

Donor Adverse Events

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DONOR HEMOVIGILANCE: HAVE WE GOT THE FIGURES RIGHT?

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Background: Donor hemovigilance is an important aspect for any blood transfusion facility. Based on the time of occurrence it can be classified as acute or delayed. Delayed adverse reactions occur 24 h after the donor has left the donation site. The transfusion service is aware only in case a donor informs back to the facility leading to an underreporting of the donor adverse events.
Aims: This prospective study aims to know the incidence and types of delayed donor reactions and the risk factors associated with it, amelioration of which would improve the donor return rate.
Methods: The study was conducted prospectively for a period of May 2016 to December 2016 at the Department of Transfusion Medicine, Indraprastha Apollo Hospitals, New Delhi. One thousand blood donors who donated blood at our center during the said period were randomly selected and a telephonic call was made to enquire about the feedback and the central tendencies were analyzed. Root cause analysis (RCA) was done to identify the risk factors associated with the adverse events. Their previous donations and any previous adverse events, if any, were also enquired about.
Results: Thirteen hundred and fifty four donors were called of which 354 (26%) did not respond to the telephonic calls. Forty-eight (13.5%) telephone numbers were incorrect and the rest did not answer the call. Of the 1000 donors who responded, 984 (98.4%) were males and 16 (1.6%) females. The mean age group was 30 (range: 18–61 years). Four (0.4%) donors reported hematoma formation. Other causes of hematoma formation like lifting heavy weights was ruled out. Of them, 2 had moderate pain for which they took medication. There was no other sensory or motor deficit. The hematoma faded within 4–7 days of time in all of them. On RCA it was found that the hematomas were a result of the new nursing staff appointed. She was counseled and trained further to prevent any such adverse event.
Summary/Conclusions: Call back policy was effective in bridging the gap between the actual donor reactions encountered and the one reported. It led to the RCA which strengthened our lacuna and improve our services to retain the blood donors in the pool and prevent their drop outs because of adverse events.

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Abstract has been withdrawn

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EVALUATION OF ADVERSE EVENTS ASSOCIATED WITH BLOOD DONATION AMONG THE BLOOD DONORS

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Background: Although the blood donation process is usually safe and uncomplicated, occasionally adverse events of variable severity may occur during or after the collection. Adverse events have negative impact on donor recruitment and retention.
Aims: To assess the frequency and type of adverse events at our center, we analyzed all the adverse events associated with the blood donation among the whole blood and apheresis donors.
Methods: Blood bank based cross sectional analysis of adverse events was done in blood donations during the period of July, 2015 to June, 2017. These included 55,601 blood donors who were analysed and subsequently compared for common adverse events like hematoma, vaso-vagal reaction or syncope, nausea, vomiting, tetany, etc. The data were statistically analyzed using SPSS version 16 and suitable tests of significance applied.
Results: Total 1018 adverse events were observed among 55,601 whole blood donors (1.83% Incidence) and 2 adverse events were observed among 253 apheresis donors (0.79%). VVR (syncope) with mild intensity were most commonly observed in whole blood donors (1.31%) specially in first time donors (0.97%) followed by- hematoma (0.40%), nausea (0.09%), vomiting (0.02%) and others (0.01%). Among the apheresis donors 1 mild and one moderate VVR was observed. The incidence of adverse events in all the categories of whole blood donors, compared with male versus female donors, was not found statistically significant; ($P > 0.05$). The incidence of adverse events was significantly higher among first-time-donors (4.6%) compared with those having H/o previous donation (2%); ($P < 0.05$). Among the apheresis donors who had adverse reactions, 1 was first time donor and 1 had a H/o donation in the past.
Summary/Conclusions: Vasovagal reactions (Syncope) of mild intensity was found to be the most common adverse event associated with blood donation which was significantly higher ($P < 0.05$) among first-time-donors compared with those having H/o previous donation. Obtaining such data on incidence of adverse events enables to train and prepare personnel in the phlebotomy area to respond quickly to those reactions to improve safety and comfort of the donor. Also it helps in minimizing negative impact on donor recruitment and retention, thereby improving blood donor return rate.

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ANALYSIS OF VASOVAGAL REACTION OCCURRENCE RATE IN DIFFERENT AGE OF BLOOD DONORS IN YINCHUAN REGION FROM 2014 TO 2016

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Background: The safety of donor is an essential prerequisite to increase voluntary blood donation.
Aims: To investigate the occurrence rate of vasovagal reaction among the blood donors during the last 3 years in Yinchuan region, and to analyze how the donor characteristics (age, gender, first-time/repeat donation) associated with such adverse event, in order to provide a scientific basis for exploring preventive measures and reducing the occurrence of adverse reactions.
Methods: 208 cases of vasovagal reaction recorded from January 2014 to December 2016 of Yinchuan region were collected for statistical analysis of blood donor adverse reaction rates in different years, frequency, gender and age groups.
Results: From 2014 to 2016, the adverse reaction of blood donation in Yinchuan decreased year by year. The adverse reaction occurrence rate of first-time donor was higher than repeat donor. In general, females had a higher rate than males, especially for those aging from 18 to 24. There was an obvious difference among different range of ages.
Summary/Conclusions: Adverse event analysis helps in identifying the blood donors at risk of donor reactions and adopting appropriate donor motivational strategies, pre-donation counseling, and care during and after donation. For donor retention purpose, prevention and coping techniques should take gender and age differences into account.

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VASOVAGAL REACTIONS IN WHOLE BLOOD DONORS OF BLOOD TRANSFUSION CENTER, FACULTY OF MEDICINE, KHON KAEN UNIVERSITY

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Background: The blood donation protocol was based on the standard for blood banks and transfusion service guidelines of The National Blood Centre, Thai Red Cross Society. Although whole blood donation is generally considered to be a safe procedure, vasovagal reactions (VVRs) may sometimes occur during and after blood donation. Most VVRs are non-severe reactions, while VVRs with loss of consciousness are an increased risk to injury.
Aims: To evaluate VVRs data, risk factors and the severity of VVRs occurring in whole blood donors of Blood transfusion center, Faculty of Medicine, Khon Kaen University.
Methods: The study included the data of whole blood donation between September 2015 to April 2017. The VVRs reported by the staff at the indoor blood collection site and the mobile sites were analyzed for the demographics data. Data analysis: frequency and percentage were calculated by using excel function and formula.
Results: The total of whole blood collected was 52,333 units. 33,052 donations collected from the indoor site and 19,281 donations from mobile sites. 7,704 were first-time donations and 44,629 repeated donations. The prevalence of vasovagal reactions was 1.3% (674/52,333), from 666 donors. The rate of VVRs in whole blood donations was 0.8% (266/33,052) occurred at the indoor site and 2.1% (408/19,281) at the mobile sites, 3.8% (290/7,704) in first-time donation and 0.9% (384/44,629) in repeated donation. The characteristics of donors with VVRs, 82.3% (548/666) were <30 years old and most of them were female 71.5% (476/666). During the period of study, 42.9% (286/666) occurred only in the first-time of blood donation, 56.5% (376/666) in repeated donation and 0.6% (4/666) in both episodes. In all repeated donors, 20.0% (76/380) had experienced VVRs in previous donation. VVRs occurred 17.1% (115/674) during donation, 54.0% (364/674) after donation and resting, 25.5% (172/674) in the refreshment area and 3.4% (23/674) outside donation site. The severities of VVRs were 95.8% (646/674) non-severe and 4.2% (28/674) severe reactions. In severe cases, 4 donors were with fall and injury and only 5 donors required medical care.
Summary/Conclusions: From the results, VVRs was high in first-time, female and young donors. Most VVRs were non-severe reactions and occurred after donation and resting. High prevalence occurred at the mobile sites may be the effect from uncomfortable, high temperature and crowded environment. Therefore, to improve blood donor safety, donors in the high risk group and factors contributing to VVRs should be given closely attention to decrease the reaction, especially in the severe case.

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THE EFFECT OF RIGOROUS SCREENING IN PREVENTION OF APHERESIS ADVERSE EVENTS BEFORE BLOOD COLLECTION

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Background: Adverse events often occur in apheresis, we should be familiar with the causes, types and clinical manifestations of adverse events caused by apheresis platelets, and targeted screening, in order to reduce the adverse effects of blood donation by apheresis. It is significant for preventing and reducing the adverse events of apheresis platelets donation.
Aims: In this paper we would like to explore how to prevent and reduce the adverse events of apheresis donation, through investigating and analyzing the causes and clinical manifestations of adverse events of apheresis donors.
Methods: According to "Whole blood and component donor selection requirements" GB 18467-2011, and "Quality requirements for whole blood and blood components" GB 18469-2012, we would like to select donors who meet the conditions of apheresis platelets, as well as collect single (double) therapeutic doses platelets, analyzing the reasons by selecting blood donors who took the blood donation adverse events.
Results: Adverse effects of citrate are the main types of apheresis platelets donation, January 2016~December 2016, apheresis donors were 5669 people, only 1.2% of donors have mild side effects, when screening was performed before blood collection.

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Summary/Conclusions: Strict screening of blood donors is the key to prevention. Following the working principle of blood cell separator, we can see the higher the platelet counts of apheresis donors, the less the amount of blood circulation needed to collect the same amount of platelets, and the cycle times, citrate dosage, apheresis time goes down as well, the incidence of blood donation adverse events is also decreasing, at the same time the safer for blood donors.

Therefore, blood donors should be screened strictly before blood collection. We must ensure the weight, height, age, metabolism and psychological state of the donor in the best condition. If the donors are underweight, aging, history of syncope, fatigue, fasting condition, the incidence of adverse events was higher among these blood donors, apheresis are not suitable for donation. When double bag platelets are collected, it's very important for screening of blood donors' weight and platelet counts, blood hct (weight > 55 kg, plt $\geq 220 \times 10^9$/l, hct ≥ 0.36). In the process of collection, we should not only observe the indexes of the blood cell separator closely but also often ask the blood donor's feelings. Early detection the various signs of adverse events to blood donation in order to timely treatment. Only in this way can the health and safety of donors be guaranteed and the adverse events of blood donation will be reduced to the utmost.

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THE ROLE AND SIGNIFICANCE OF COMPREHENSIVE NURSING WITH INCREASE THE AMOUNT OF BLOOD DONATION AND SATISFACTION DEGREE AND REDUCE THE INCIDENCE OF BLOOD DONATION REACTION

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Background: China implements unpaid blood donation system since the promulgation of "blood donation". The crowd of voluntarily joining the unpaid blood donation team is increasing. But with the sharp increase in clinical blood volume, difficulties and pressures are growing foe the unpaid blood donation work. It is the primary task of a blood station to ensure the safety of clinical use of medical institutions. In order to make good use of this valuable resource foe free blood donation, implement comprehensive nursing measures have an important role and significance. Comprehensive nursing measures are based on modern medical model and providing comprehensive care services in the physical, psychological, social, environmental and other aspects for care objects. Comprehensive nursing measures can fully coordinate the physiological, psychological and behavior with the surrounding environment, social culture, contemporary culture to adapt and make people to achieve the best health status.

The object of this paper come from fixed blood donation point and blood donation point of blood donors of Hebei Province Blood Center. The blood donation and blood donors' satisfaction were improved through the implementation of comprehensive care measures for blood donors, providing comprehensive services in the physical, psychological, environmental and other aspects and issuing questionnaires. While the incidence of blood donation has decreased. The implementation of comprehensive care measures can help to keep more blood donors and meet the growing demand for clinical blood. While The implementation of comprehensive care measures have an important role and significance for the future of blood supply work.

Aims: Comprehensive nursing measures were used for analysis and research against increase the amount of blood donation and reduce the incidence of blood donation reaction.

Methods: Select the experimental group from October 2016 to November 2016, 115 blood donors who donated blood and donated blood at the blood center of Hebei province, the experimental group of unpaid blood donors to implement comprehensive care measures and issue questionnaires. Select the control group from December 2016 to January 2017, 103 blood donors who donated blood and donated blood at the blood center of Hebei Province, the control group of blood donors to provide routine blood donation care services and issue questionnaires. Observe the differences between the experimental group and the control group in blood donation of blood donors, blood donation reaction and satisfaction degree.

Results: After comparison between groups, The blood donation rate and blood donation reaction rate of experimental group were 88.70% and 3.50%, the blood donation rate and blood donation reaction rate in the control group were 68.93% and 8.74%, $P < 0.05$, the experimental group was significantly better than the control group, the difference was statistically significant. After comprehensive nursing measures, blood donors have improved their satisfaction with a total of seven satisfaction degrees, such as advice consultation, physical examination service, blood collection services, blood collection technology, environmental sanitation, post-blood donation services and comprehensive services .

Summary/Conclusions: Comprehensive nursing measures can effectively improve the blood donation of blood donors and satisfaction degree, reduce the incidence of blood donation reaction, help to expand and stabilize the ranks of unpaid blood donors.

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THE COMPARISON AND PREVENTION OF THE ADVERSE REACTION OF PLATELETS FROM BLOOD CELL CENTRIFUGES

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Background: Accidental adverse reactions during apheresis platelet collection affect not only the quality of platelet, but also the retention and development of apheresis blood (AP) donors.

Aims: Through the retrospective analysis the adverse reaction number and types of three kinds of blood cell separators during blood donation, to explore the causes, formulate the relative prevention measures, reduce the incidence of adverse reactions, retention of blood donors.

Methods: According to the requirements of the health check blood donors in Changsha blood center, from January 2015 to June and from January 2017 to June, to retrospectively analyze the qualifications for blood donors, each 6000 people, according to the three blood cell separator randomly assigned to each group of 2000 people, between January 2015 and June, those who do not take special measures to set as the control group, from January 2017 to June, those who take special measures to set as observation group, the proportion of statistical comparison of adverse reactions occur.

Results: The incidence of adverse reactions in special measures in 2017 was significantly lower than that in 2015 ($P < 0.05$).

Summary/Conclusions: The correct preventive measures can prevent adverse reactions effectively, improve the comfort level of blood donors, and play an important role in the retention of blood donors.

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STUDY OF THE TREATMENT FOR CARELESS PUNCTURING BLOOD DONORS' ARTERY IN HAEMOSPASIA

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Background: If the vein puncture artery puncture error, take immediate needle, stop blood donation, to adhere to the voluntary blood donors, select the other side of the arm, again venipuncture complete blood donation. However, observed and studied by the author in the practice of collecting blood by artery puncture must immediately stop blood put forward different views: after a long period of work experience and exploration, in the face of collecting blood mistakenly puncturing artery, not into the needle, immediately stop blood. But a close observation of blood donors was asked, without discomfort until the successful completion of blood donation.

Aims: Research purpose discuss the treatment for careless puncturing donor's artery in haemospasia, in order to give better service to blood donors.

Methods: careless puncturing donor's artery when collecting blood from donor, do not stop the blood collection, but complete the work through following treatments instead, such as close observation, ask the blood donor for any discomfort and so on. We divided blood donors into artery group and vein group, through follow-up phone calls to keep track of blood donors' physical condition after donation, observe and compare the adverse reaction of artery group and vein group.

Results: Artery group has 49 donors, four donors (8.16%) among appear bruising and two (4.08%) have slightly discomforts. Vein group has 50 donors, five (10.00%) donors among appear bruising and two (4.00%) have slightly discomforts. After blood donation, both the two group has a small number of donors appear bruising or slightly discomforts, difference between artery group and vein group has no statistical significance ($\chi^2 = 0.099$, $P > 0.005$, $\chi^2 = 0.02$, $P > 0.005$).

Summary/Conclusions: careless puncturing donor's artery when collecting, there is no need to stop blood collection or doing venipuncture again. If the donor do not appear any discomforts, we can complete blood donation by direct artery puncture. This treatment method alleviates the pain caused by repeated puncture, as well as reduces the consumption of blood collection.

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CAUSE ANALYSIS, PREVENTION AND TREATMENT OF ADVERSE REACTION OF BLOOD DONATION

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Background: Observed from blood donation work, adverse reactions occurred during the blood donation caused by various factors damp donors' enthusiasm and are not conducive to retain the donors.
Aims: To analyze the causes of adverse reactions of blood donation, to optimize the quality of blood donation service and improve the ability of treating blood donation adverse reactions.
Methods: Based on a retrospective analysis of a total 1087 cases of adverse reactions occurred during the blood donation from January 2015 to December 2016.
Results: Among the 1087 cases of adverse reactions of blood donation, 726 (66.8%) were caused by mental factors, 184 (16.9%) were caused by physiological factors and 76 (7.0%) were caused by environmental factors. 58 (5.3%) were caused by blood collection operation factors, and the rest 43 (4.0%) were caused by other reasons.
Summary/Conclusions: The most common cause of adverse reactions occurred during the blood donation in our center is mental factors, followed by physiological factors, environmental factors, blood collection technology factors. We should give blood donors a positive psychological treatment, consult about donor's physical status carefully, improve the blood donation environment and our blood sampling staff's operational level. We can bring the donors a safer and more comfortable experience by providing quality services, preventing adverse reactions and conducting correct treatment of adverse reactions.

Blood Products Blood Processing, Storage and Release

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APPLICATION OF MODERN DAY TECHNOLOGIES – EVERY STEP COUNTS

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Background: Standardized processing protocols and automatization is a link to the efficacy of blood components. A positive effect on a recipient is the most important feature of them. Initial step in processing is drawing blood on automatic blood mixers. Automatic strippers, sealers and automatic separation provide high quality blood components, together with GMP in routine work. Top & bottom (T/B) quadruple blood bags are suitable for completely automatized production of all components.
Aims: There was continuously improving quality of blood component production from 2011 to 2016 as we turn to processing of RBC "buffy coat"(BC) removed in additive solution (AS) and RBC leucocyte depleted (LD) in AS, with either docabile filters or produced from quintuple top & top (T/T) blood bags with "in line" filters. We implemented PC (platelet concentrates) processing from pooled BC followed by filtration. There was constant increasing number of produced components with same staff number but on automatic separators.
Methods: Units were collected by standardized mixing procedure on automatic blood mixers -Bagmatic® (LMB). After centrifugation we used sealers with adjustable sealing process time depending on different blood bag tube diameters and the possibility to a segment application-Sealmatic®(LMB). All separations were done on NOVOmatic® and LUXOMatic®, machines with automatic valve breakers (LMB). The filtration process was monitored with automatic measurements of filtration time and weight on Filtramatic® (LMB). During this period we processed totally 47 626 blood units with constant increase from 7969 to 8668 units yearly. Legislation for quality control (QC) requirements for RBC, RBC BCR-AS, RBC LD-AS, FFP, cryoprecipitate and PC, are per "Guide to the preparation, use and quality assurance of blood components", 18th Ed.
Results: In the past 6 years we produced: 8068 RBC, 35 260 RBC BCR-AS, 3 184 RBC LD-AS (filtrated in line or with docable filters), 12 767 PC recovered single unit from BC, 2 987 platelets recovered pooled leucocyte depleted, 9 282 cryoprecipitate units, 45 578 FFP units. There was constant decrease in processing of RBC. Since the implementation, in last 2 years with quadruple and quintuple T/B and T/T blood bags, we have been processing only RBC BCR-AS and RBC LD -AS. From 2013 we started processing pooled leucodepleted PC. The required QC of RBC, FFP, and cryoprecipitate met demanded standards. The required demand for QC in the platelet content met 82.75% PC units, all units met necessary QC in volume and residual leucocytes. Detection of bacterial contamination was negative in all tested PC units and the pH at the end of stocking period was in acceptable range in all tested units.
Summary/Conclusions: The condition of centrifugation, choice of initial separation step as well as using initial filtration influences the choice of methods for further processing. Separation of blood components from whole blood can be automated by different devices. Components are standardized, with increased work efficiency. Automatic separators allow simultaneous separation of two blood units. With automatic valve breakers there is less haemolysis. Touch screen is easy accepting modern technology on all devices which also have possibility of integrating data to the existing quality management information system. With automatization we shortened separation period and achieved complete standardization. Our goal is to increase production of pooled PC-LD and implement pathogen inactivation, as an answer to constantly increasing demands for PC.

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REVEOS SYSTEM IN BLOOD TRANSFUSION CENTRE, FACULTY OF MEDICINE, KHON KAEN UNIVERSITY, THAILAND

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Background: The Reveos automated blood processing system has been developed for whole blood units automated, the machine was full whole blood automation. This system has been installed in Blood Transfusion Centre, Faculty of Medicine, Khon Kaen University, Thailand.
Aims: The aim of this study was evaluate the product specifications and quality of components produced by the Reveos system. Since September 2016 to April 2017.
Methods: Whole blood was processed using the Reveos system and compared according to validate the use of the first installation. Reveos red cells were leukoreduced and stored in SAGM at 4°C. Reveos plasma was frozen at −30°C and factor activity was assessed after thawing. Leukoreduced platelet concentrates Reveos were prepared by pooling 4 iso-group interim platelet units or 1–2 random platelet concentrate from top-bottom method and 3–2 interim platelet units pooled by Reveos pooling set, Comparison the value of volume, hematocrit, platelet contents and white blood cell contamination of LDPRC and LDPC at first installation to date with SPSS statistics.
Results: 408 whole blood was processing with Reveos system. Average of fresh frozen plasma, interim platelet and leuko-packed is 217.52, 62.40 and 10.2 mL., respectively. The platelet index more than 60 cells/u is 75.8%. The quality of LDPRC; hematocrit equal 55.8%, volume equal 313 mL. and white cell contamination equal 0.0 × 106 cells/u. Comparison the value with SPSS statistics were found that the hematocrit, volume and white blood cell408 whole blood was processing with Reveos system. Average of fresh frozen plasma, interim platelet and leuko-packed is 217.52, 62.40 and 10.2 mL., respectively. The platelet index more than 60 cells/u is 75.8%. The quality of LDPRC; hematocrit and volume were 55.8% and 313 mL. and white blood cells contamination was 0.0 × 106 cells/u. Comparison the value with SPSS statistics were found that the hematocrit, volume and white blood cell
Summary/Conclusions: The production of Reveos system; LDPRC and LDPC at first installation provides reached the recommended quality of Council of Europe (EU), American Association of Blood Banks (AABB) and National Blood Centre, Thai Red Cross Society (TRC).

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EVALUATION OF THE QUALITY OF BLOOD COMPONENTS PREPARED BY REVEOS AUTOMATED BLOOD SEPARATION SYSTEM

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Background: The Reveos system (TerumoBCT, Lakewood, Colorado, USA) is a fully automated blood separation system which can process simultaneously four whole blood (WB) units per cycle into red blood cell (RBC), plasma and interim platelet units (IPU). This automated system has the advantages of shortening handling time, saving laboratory space and producing products of more consistent quality.

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Aims: To evaluate the quality of blood components (RBC, plasma and platelets) prepared by Reveos system.
Methods: Fifty-one WB donations (450 mL ± 10%) were collected into Reveos collection bag (Reveos 3C Kit, TerumoBCT) and processed by Reveos system using "3C" protocol on Day 1 after overnight-held at 18–24°C. RBC units prepared were leucodepleted with integrated leucofilters, suspended in SAGM and stored in DEHP-PVC containers. These products were tested for quality control (QC) parameters including total haemoglobin (Hb), residual white blood cell (WBC) count on Day 1 and haemolysis at expiry on Day 42. The plasma prepared were tested for cellular contamination (platelet, RBC and WBC), volume and coagulation factor VIII (FVIII) activity. Four units of IPU were selected for pooling based on platelet yield index (PYI) and leucodepleted using TerumoBCT platelet pooling set with inline leucofilter (ATREUS PLT Pooling set, LRFXL) and suspended in 200 mL of platelet additive solution (T-PAS+, TerumoBCT) in butyryl trihexyl citrate-PVC container. The pooled and filtered platelets (PFPLT) were tested for absolute platelet count and pH at expiry (Day 5) and residual WBC count on Day 1. Four units of randomly selected leucodepleted RBCs and 4 units of PFPLTs were cultured for bacterial screening (BacT/ALERT, bioMerieux) on Day 2.
Results: Regarding leucodepleted RBCs, 92% of the 51 units showed haemolysis <0.8% (mean ± 1 SD = 0.4% ± 0.3%) after storage at 2–6°C for 42 days. The total Hb and residual WBC count were 51.54 ± 6.06 g/unit (mean ± 1 SD, n = 51) and 0.23 ± 0.19 × 10^6/unit (mean ± 1 SD, n = 51) respectively. The plasma units showed low numbers of contaminating platelet (mean ± 1 SD = 18.5 ± 8.5 × 10^9/l, n = 51), RBC (mean ± 1 SD = 0.03 ± 0.19 × 10^9/l, n = 51) and WBC (mean ± 1 SD = 0.0 ± 0.0 × 10^9/l, n = 51). The plasma volume and FVIII activity were 260.0 ± 25.9 mL (mean ± 1 SD, n = 51) and 79.7 ± 26.1 IU/100 mL (mean ± 1 SD, n = 6) respectively. The plasma volume so produced was significantly ($P < 0.001$) higher than that prepared by the in-use semi-automated method with Compomat G5 blood extractor (Fresenius-Kabi, Bad Homburg, Germany) (mean ± 1 SD = 206 ± 43 mL, n = 266, data extracted from our blood components QC database, July-December 2016). The PFPLTs showed an absolute platelet count of 275.3 ± 41.4 × 10^9/unit (mean ± 1 SD, n = 12) and pH 7.18 ± 0.05 (mean ± 1 SD, n = 12) after 5 days of storage at 20–24°C with continuous agitation. Residual WBC count in the PFPLTs was 0.01 ± 0.00 × 10^6/l (mean ± 1 SD, n = 12). No bacterial growth was detected in the leucodepleted RBCs and PFPLTs.
Summary/Conclusions: The Reveos system provides full automation in blood components separation and the manufactured blood components meet all the quality criteria stipulated by American Association of Blood Banks Standards and Council of Europe Guidelines. More plasma (26.2%) was gained and amenable to use in plasma fractionation when processed by Reveos system.

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ASSESSMENT FOR THE EFFECT OF PREPARING CRYOPRECIPITATE USING AUTOMATED BLOOD COMPONENT SEPARATOR

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Background: Preparing cryoprecipitate using manual method was long time consumption and inefficiency.
Aims: Two preparation methods were compared: preparing cryoprecipitate using manual method (i.e. manual method) and preparing cryoprecipitate using automated blood component separator (i.e. instrumental method). Effect of preparing cryoprecipitate with instrumental method was assessed in step of preparation process, time consumption of preparation and quality of cryoprecipitate.
Methods: 48 bags of cryoprecipitate were prepared by manual method in 2013. 48 bags of cryoprecipitate were prepared by instrumental method in 2016. The quality of cryoprecipitate was compared between two groups by statistic methodology. Effect of preparing cryoprecipitate with instrumental method was assessed in step of preparation process, time consumption of preparation and quality of cryoprecipitate.
Results: The levels of factor VIII prepared with manual method and instrumental method respectively were: (101 ± 10.03) IU and (123.5 ± 11.56) IU. The difference were of statistical significance ($t = 10.19$, $P < 0.05$). The levels of fibrinogen prepared with manual method and instrumental method respectively were: (167 ± 15.87) mg and (205 ± 17.52) mg. The difference were of statistical significance ($t = 11.14$, $P < 0.05$).
Summary/Conclusions: Cryoprecipitate was of better quality prepared by instrumental method than manual method. Compared with manual method, the time consumption was reduced, the more efficiency was improved. Furthermore, the whole preparation process could be computerized.

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THE APPLICATION OF AUTOMATIC LEUKOCYTE FILTRATION MONITORS IN BLOOD PREPARATION DEPARTMENT

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Background: Leukocyte filtration can effectively reduce the clinical transfusion reaction. The classic mode of leukocyte filtration is to hang the blood bags in a low-temperature cabinet making the blood flow through the filter by the action of gravity. It is, however, difficult to standardize the manual operations, especially when filtering blood bags in large quantities. Quality of the blood products may also face certain problems, such as the low erythrocyte recovery rate, blood going through the bypass path instead the filter. In 2015, 4 automatic leukocyte filtration monitors were set up in our department, which have been utilized to monitor the whole process of the leukocyte filtration, in order to achieve the standardization and informatization of the filtering operation.
Aims: To discuss the capability of real-time monitoring, data validity and traceability of the equipment, in order to improve the quality of the blood products.
Methods: Blood samples donated between Aug 1, 2016 and Jun 12, 2017 were selected, which included 15502 samples in volume of 300 mL and 17570 samples in volume of 400 mL. Certain parameters of the equipment, including the whole blood weight, blood filtration time and leukocyte residual rate, were collected and statistically analyzed. The actual whole-blood collecting weight range and time of filtration process using different-sized blood bags and filters were calculated. Meanwhile, leukocyte count of 20 specimens from blood braids, 10 cm below the filters for both 300 mL and 400 mL-volumed bags were measured by ADAM-rWBC Cell Counting System. Further comparisons were carried out to evaluate the performance of different filters.
Results: For 300 ml-volumed collections, actual whole-blood weight range was 378 ± 21 g, and filtration time (filters A) was 3'58" ± 52". For 400 ml-volumed collections, actual whole-blood weight range was 495 ± 16 g, and filtration time (filters B) was 9'02" ± 58". The pass rate of leukocyte count for filters A and B was 75%(15/20) and 80%(16/20), respectively.
Summary/Conclusions: Automatic leukocyte filtration monitors can in real time record all the data throughout the process, which also standardized the operation and improve the quality of the blood products.

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PLATELET ADHESION AND BLOOD COMPATIBILITY OF FUNCTIONAL GROUPS GRAFTED PBT FILM AS LEUKOCYTE FILTER MATERIALS

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Background: Poly (butylene terephthalate) (PBT) is widely used as disposable clinical instruments such as leukocyte filters, especially for depletion leukocyte from platelet concentrations (PCs). Most study focus on modifying PBT to improve removal leukocyte and decrease platelet adhesion, however, there are few reports on the effects of modified PBT on platelet function and blood compatibility.
Aims: To evaluate the effects of PBT film grafting with different functional groups (-COOH,-OH and zwitterionic sulfobetaine group) on platelet adhesion, platelet function, red blood cell (RBC) properties and coagulant system, and select suitable filter material for depletion leukocyte from PCs.
Methods: PBT-AA (-COOH), PBT-HEMA (-OH), PBT-SMDB (zwitterionic sulfobetaine groups) films obtained by UV radiation. The physical and chemical properties of modified PBT was characterized by Attenuated total reflectance-Fourier transform infrared spectra (FITR), X-ray photoelectron spectroscopy (XPS), water contact angle, zeta potential and surface roughness. Platelet adhesion was observed by SEM, and platelet aggregation, platelet hypotonic shock rate (HSR), platelet activation (expression of CD62P), RBC deformability, osmotic fragility and aggregation and coagulant system were investigated.
Results: The new absorption at 1047/cm was ascribed to symmetric and antisymmetric C-O-C stretching vibration of HEA, and 1043/m was belonged to asymmetric O = S = O stretching vibration of SMDB, and O/C ratio of PBT-PAA and PBT-PHEA surface was higher than that of original PBT from XPS results, which was confirmed the successful grafting of the functional polymers. In the case of modified PBT film with the same wettability, there was no significant difference of PBT films grafted functional groups on RBC deformability and aggregation, and the surface bearing OH had greater osmotic resilience compared with other films. Platelet adhesion to films decreased in the order: the surface bearing COOH >the surface bearing

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sulfobetaine group >the surface bearing OH, while the effect of functional group on platelet function such as platelet aggregation, platelet hypotonic shock response (HSR) and platelet activation (expression of CD62P) showed that the surface bearing OH had good platelet compatibility. Moreover, the surface bearing OH had no significant effect on blood coagulation, and the surface bearing sulfobetaine group showed anticoagulant property, and exhibited an effect in inhibiting blood coagulation.

Summary/Conclusions: Our results emphasized that surface functional groups had significant effects on the adhesion of platelet, platelet function and blood compatibility, and PBT-PHEA had good blood compatibility and no effect on platelet function. It can help to achieve the design of more effective antiplatelet adherent materials for blood filtration applications.

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CONTINUOUS CONCENTRATION OF R-WBC USING A VISCOELASTIC FLUID

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Background: In blood transfusion, the number of residual white blood cells (r-WBCs) should be limited below approximately ~3.0 cells per μl blood to prevent transfusion complications due to r-WBCs. Therefore, quantitation of r-WBCs is critical for quality control of blood transfusion. Although there have been various techniques used to count r-WBCs including hemocytometer and flow cytometry, those have limitations of low detection sensitivity, high cost and labor intensiveness.

Aims: The purpose of this study is to develop a sensitive residual WBC measurement method using viscoelastic fluid in microchannel chip.

Methods: To concentrate low abundance r-WBCs at the center of the microchannel, 0.1 (w/v)% of hyaluronic acid (HA) sodium salt solution was prepared as a viscoelastic non-Newtonian fluid. WBCs were separated from whole blood by centrifugation using a density gradient solution (Ficoll-Paque PLUS). Briefly, 5 ml of whole blood was diluted with the same volume of PBS. 15 ml centrifuge tube was filled with 3 ml of Ficoll-Paque PLUS and 10 ml of the diluted blood was carefully layered on Ficoll-Paque PLUS to prevent mixing. After centrifugation at 400 g for 20 min, 4 different layers of plasma, WBCs, Ficoll-Paque PLUS and red blood cells (RBCs) were achieved. Collected WBCs were stained with a fluorescent dye (SYBR Green) for visualization and suspended in 0.1 (w/v)% HA solution at the final concentration of 7.5×10^3 cells/ml. A straight microfluidic channel with high aspect ratio (AR = height/width) was used to focus the suspended cells in a single line at the center. The width and the height of the microchannel were 50 and 120 μm, respectively. The microchannel has a single inlet and the trifurcated outlet with three outlet holes. The ratio of trifurcated outlet channels was designed as 10:1:10. Sample injection and suction were controlled by a syringe pump. At the inlet of the microchannel, r-WBC suspension was injected at the flow rate (Q) of 100 μl/min. At the center outlet of the microchannel, focused r-WBCs were collected by suction at 5 μl/min. The collected r-WBCs were counted using a hemocytometer.

Results: Before concentration process in the microfluidic device, r-WBCs in prepared sample were quantified as 7.5×10^3 cells/ml. The numbers of r-WBCs indicate the average counted results from 5 different experiments (n = 5). At Q = 100 μl/min, 1 ml of r-WBC suspension sample could be processed only in 10 min. After concentration, the number of collected r-WBCs at the center outlet was 1.3×10^5 cells/ml. To evaluate the concentration device performance, r-WBC suspension was diluted by 2×, 5× and 10×, respectively. For 2×-diluted sample, the number of r-WBCs was increased from 2.5×10^3 cells/ml to 4.2×10^4 cells/ml. Using the continuous concentration device, the number of r-WBCs was increased by approximately 18-fold. On the other hand, for 5× and 10×-diluted sample, r-WBCs could not be quantified using a hemocytometer due to extremely low abundance. However, thanks to the concentration device, r-WBCs could be quantified using a hemocytometer after concentration process and the concentrations were increased to 2.2×10^4 cells/ml and 1.2×10^4 cells/ml for 5× and 10×-diluted sample, respectively.

Summary/Conclusions: Overall, continuous concentration technique using a viscoelastic non-Newtonian fluid was successfully developed and applied to r-WBC concentration. With further optimization of the channel dimension, flow rate condition, the concentration of a viscoelastic fluid, the device performance can be enhanced to achieve higher throughput and concentration ratio.

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EVALUATION OF THE NEW IN-LINE FILTER BLOOD BAGS WITH AUTOMATED CANNULA BREAKER SYSTEM FOR LEUCODEPLETION PROCESS, COMPONENT QUALITY AND HEMOLYSIS

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Background: The Compoflow® select in-line system has the Compoflow® closure device instead of the traditional bag breaker which is expected to minimize hemolysis, reduce repetitive strain injuries in component laboratory staff as well as lessen the component preparation time.

Aims: To evaluate the performance of Compoflow® select in-line systems used for the preparation of Leukodepleted RCC units from 450 ml whole blood donations processed by the Buffy coat procedure.

Methods: N = 118 Compoflow blood bags were used for component separation by buffy coat method as per the manufacturers instructions. All essential parameters related to the donor, the component separation process, leucofiltration process of PRBC units and quality control parameters of PRBC, platelet and fresh frozen plasma units prepared were analyzed. The plasma free Hemoglobin was checked on Day 1 and Day 7 for all PRBC units. Residual WBC count in PRBC units was analyzed by flow cytometry. 5% of the platelet and PRBC units were submitted for sterility testing.

Results: The mean component separation time on Compomat G5 was 4.7 min. All components quality parameters passed the current regulatory and accreditation criteria.

The component separation process could recover 81.42% of red cells and 63.04% of platelets from the 450 ml of whole blood collected. The mean hemoglobin content in PRBC units was 49.19 g. The residual WBC count in PRBC units was 1.75 × 105 and 8.5% of the product was lost in the leucofiltration process. % hemolysis was 0.22 on day-1 and 0.37 at day-7. None of the PRBC and platelet units showed evidence of bacterial growth on culture. A very favorable feedback was obtained from the donors, phlebotomists and component laboratory technologists about the compoflow system.

Summary/Conclusions: The new compoflow blood bags system enable easy and quick separation of blood components with acceptable quality parameters including efficient leucodepletion of PRBC units and minimal hemolysis.

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Abstract has been withdrawn

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CORRECT TUBE SEALS DONE WITH QSEAL FREE WITHSTAND BREAKAGE AT MAXIMUM SPEED CENTRIFUGATION OF BLOOD BAGS

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Background: To our knowledge, no study about tube seals and the impact of the mechanical stress of centrifugation has been done. A study of this would be of great interest for blood component production, since breaking tube seals when centrifuging packs of whole blood (WB) is a common problem causing unnecessary loss of donated blood. We decided to evaluate whether seals made on PVC-tubes connected to blood bags represent a critical factor when the bags are exposed to high centrifugation speed.

Aims: The aim of the study was to evaluate how much centrifugation stress seals on PVC-tube ends can withstand, and if the strength of the seal is dependent on the content of the plasticizer DEHP in the tube.

Methods: Transfer bags containing either 35% DEHP (BB*T060CM, TerumoBCT, Lakewood, CO, USA) or 32% DEHP (MSV6000LU, Macopharma, Mouvaux, France) (N = 36 for each bag type) were filled with 0.9% NaCl solution (AFAGE1324D, Baxter Medical AB, Kista, Sweden) ranging from 497 to 508 (median = 501) mL, mimicking a standard WB pack of 450 mL ± 10% WB + 63 mL CPD. Tubes (containing

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the same amount of DEHP as respective bag type) were sealed with the cordless sealer QSeal Free (CS646, Conroy Medical AB, Upplands Väsby, Sweden) 12 cm from the bag while containing liquid. Seals were inspected and bags with flawless seals (seals were determined to be flawless when they were *(i)* placed at 90° angle in the centre of the tube, *(ii)* had melted plastic flaps protruding equally on both sides) were packed in centrifuge liners in a standardized way, with the bag label facing the middle of the centrifuge liner and the tube ending facing downwards for maximum pressure on the sealed end. Padding was used to fill up the additional space in the centrifuge liner, to support and protect the transfer bag and secure the position of the tube. Centrifugation was done in centrifuges (MacoSpin, Macopharma) at 4800 × *g* for 11 min with maximum acceleration and semi-hard brake, mimicking an ordinary WB separation program with hardest possible g-force. The seals were inspected after every centrifugation cycle (CC). Any weaknesses in seals, broken seals or broken bags were registered. Unaffected units were re-centrifuged until broken or a maximum of 10 CCs.

Results: No leaks were noted for any of the seals, regardless of DEHP content of the tube. However, leaks were observed in the filled bags, starting at the 8th CC. 27 of the 35% DEHP units (75%) and 34 of the 32% DEHP units (94%) remained unaffected throughout 10 CCs.

Summary/Conclusions: 100% of the seals made with QSeal Free withstood all centrifuge cycles at maximum speed without showing any signs of weakness or leakage. The strain the seals were put through is well above what blood bags are normally exposed to, even though not all seals were exposed to 10 CCs. Therefore, sealing PVC-tubes in a correct manner with an adequate bag tube sealer does not represent a weak point in blood bag centrifugation independent of the DEHP content of the tube. Thus, seals breaking at centrifugation are avoidable. Whether problems with seals breaking in the centrifuge are a result of less quality of the sealing equipment or poorly executed sealings i.e. a need for better staff training, remains to be investigated.

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METABOLIC CHARACTERISTICS OF RED BLOOD CELLS DURING ROUTINE STORAGE IN MAP

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Background: It was debated for a long time that the safety and effectivity of stored red blood cells (RBCs) transfusion. Laboratory results and retrospective clinical evidence indicated that a potential correlation between increased morbidity and mortality with transfusion of RBC units which stored longer than 14 days. The alterations of RBCs stored in vitro, including a wide series of morphologic, structural, and functional alterations, collectively referred to as the "storage lesion". Metabolomic investigations of RBCs stored under refrigerated conditions in saline adenine glucose mannitol (SAGM) and AS-3 additives have revealed the presence of three distinct metabolic phases, occurring on days 0–10, 10–18, and after day 18 of storage. MAP additive is widespread application in china for conserving donor RBCs. The metabolic characteristics of RBC stored in MAP have not been reported.

Aims: We hypothesize that the metabolites in RBC extracts reflect the metabolic characteristics of RBC during routine storage in MAP.

Methods: RBC units produced by 10 U whole blood were stored in MAP. Samples were taken on days 1, 14 and 35. Metabolites in RBCs on days 1, 14 and 35 were extracted in ice-cold lysis and extraction buffer, and mass spectrometry-based metabolomics was performed by LC/TOF-MS system. The raw data of metabolomics analysis from detected by PeakView and MarkerView software. The metabolomic analysis of m/z features, which VIP >1, $Q^2 > 0.5$, $R^2X > 0.5$, and $R^2Y > 0.5$ though OPLS-DA approach statistical analysis and $P < 0.05$ in univariate test statistical analysis, were significantly changed between days 1, 14 and 35. The interaction, construction, and pathway analysis of distinct *m/z* features were conducted with the Human Metabolome Database (http://www.hmdb.ca/) for confirming the metabolic.

Results: Totally 362 metabolites were achieved in RBC extracts, 73 distinct metabolites out of 362 variables were obtained in RBC extracts between days 1 and 14, and 94 distinct metabolites were obtained in RBC extracts between days 14 and 35. Metabolites of glycolytic pathway and pentose phosphate pathway (PPP) were changed significantly in RBCs during routine storage. Glucose in RBC decreased from day 1 to 35. Pyruvic acid increased significantly in RBCs from day 1 to 14, and increased insignificantly from day 14 to 35. Adenine in RBC decreased significantly from day 1 to 14, and decreased insignificantly from day 14 to 35, contrary. Reduced glutathione (GSH) in RBC was decreased significantly on day 35, compared with day 14. Oxidized glutathione (GSSG) and niacinamide were increased significantly at the same storage phase, contrary.

Summary/Conclusions: Comparing the metabolites in RBC extracts between the preliminary, metaphase and terminal stage in MAP, glucose catabolites are channeled through the PPP to fuel the generation of NADPH, in response to the high oxidation environment and oxidative stress during storage.

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PRELIMINARY STUDY OF ANAEROBIC PRESERVATION ON SUSPENDED LEUKOCYTE-REDUCED RED BLOOD CELLS

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Background: The damage of RBCs were increased with the prolongation of preservation, including morphological damage, osmotic fragility increasing, hemolysis. The standard of the United States and the European Union stipulates that hemolysis rate should be <0.8% in the late erythrocyte preservation. The hemolysis rate of red blood cells is 0.2–0.4% for 5–6 weeks. In recent years, oxidative stress caused by red blood cell damage is increasingly concerned about, and anaerobic preservation is expected to reduce this damage.

Aims: To study the effect of anaerobic preservation on suspended leukocyte-reduced red blood cells (RBCs).

Methods: Ten bags of suspended leukocyte-reduced red blood cells were collected and prepared using 200 ml one-time filter blood cell bag. Then, each bag of suspended leukocyte-reduced RBCs was divided into two bags using sterile feeder, and the anaerobic group completed the aseptic exchange of nitrogen 3 times through the 0.22 μm filter. Both groups of suspended RBCs were storaged in 4°C, while the anaerobic group was placed in a stacking manner. The samples were taken at 0, 7, 14, 21, 28 and 35 days, and the hemolysis rate was measured by erythrocyte osmotic fragility test (H50%).

Results: The H50% of the anaerobic group and conventional preservation group showed increasing trends simultaneously, but the anaerobic group was lower in the same preservation time. There were statistically significant differences between the two groups for 28 and 35 days ($P<0.05$).

Summary/Conclusions: The anaerobic preservation can effectively improve the hemolysis of RBCs caused by oxidative stress injury, and extend the preservation time.

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AN IN VITRO STUDY OF COAGULATION PROPERTIES IN REFRIGERATED WHOLE BLOOD AND RECONSTITUTED WHOLE BLOOD

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Background: To rescue the patients with large volumes of lost blood (massive transfusion) requires administration of red blood cells (RBCs), plasma, and platelets (PLTs) or, alternatively, whole blood (WB). However, civilian hospitals now often provide "transfusion packages" containing these components in a 1:1:1 ratio. The aim of these packages is to mimic whole blood, and the transfusion packages may consist of bags in which all the components are mixed or the components may be kept separately. By the way, "fresh" WB has been observed in military application and in pediatric cardiac surgery.The maximum time of WB use in military is 24 h of room temperature storage, and in pediatric cardiac surgery is 48 h of refrigerated storage. Stimulated by the reports from the military and pediatric cardiac surgery, we wanted to elucidate the in vitro consequences of the differences in storage age of the coagulation function and to compare with reconstituted WB to those of what could be regarded as "the golden standard".

Aims: Fresh whole blood (WB) has been observed in military application and cardiac surgery and is associated with reduced blood loss. We undertook an in vitro study of coagulation properties of refrigerated WB stored for 21 days, and compared with reconstituted whole blood which was an intended red cell:platelet:plasma ratio of 1:1:1.

Methods: Ten WB units were obtained from healthy volunteer donors and stored at 4 ± 2°C. Samples were obtained on the day after donation and again on Days 2, 4, 6, 8, 10, 14, and 21. Ten compositions of reconstituted whole blood were prepared with the same ratio of red cells, platelets and plasma as used in local transfusion packages, and the storage time of red cells and platelets was varied in a systematic

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manner. Tests included complete blood count, electrolytes, routine coagulation test, blood coagulation factors, and thromboelastography (TEG).
Results: There was progressive decline in Hb, WBC, PLT, sodium, and coagulation factors, but progressive increase in APTT, PT, and potassium in WB stored at 4 ± 2°C through 21 days. Compared with reconstituted WB, the concentrations of factor(F)V and FVIII, FII, and FX of WB were higher before Day 4, 8, and 14, respectively. FVII, FIX, FXI, and FXII of WB were found to higher than reconstituted WB throug 21 days. TEG variables in all WB units were normal through 10 days; abnormal values in some variables in some units began on Day 14. However, mean R of reconstituted WB was out of normal range for adult values, while mean K, angle, and MA were within normal values. Mean PT, and APTT of WB were shorter than those of reconstituted WB before Day 14, and 10, respectively. Mean Fg of WB was higher than reconstituted WB through 21 days.
Summary/Conclusions: This in vitro study of coagulation properties demonstrates that refrigerated WB remain normal integrated coagulation function to a minimum of 10 days under standard storage condition. The coagulation properties in refrigerated WB were superior than those in reconstituted WB before Day 10.

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STORAGE-RELATED MORPHOLOGICAL CHANGES IN RED BLOOD CELLS

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Background: It is widely accepted that red blood cells (RBC) in storage undergo a progressive morphological transformation from biconcave discocytes to "spiky" echinocytes. The final stage of this transformation, to smaller and denser spheroechinocytes, is irreversible and is linked to a high clearance rate by the liver after transfusion and a decrease in transfusion efficiency. To preserve healthy cells' membrane properties and minimize hemolysis through storage, cells are stored in SAGM (Saline Adenine Glucose Mannitol), a hypertonic solution. As a consequence of storage in SAGM, swollen cup-shaped RBC called stomatocytes are observed.
Aims: In this study we investigated RBC morphology during routine storage in SAGM and the shape change when stored RBC were returned to plasma.
Methods: The morphology of packed RBC, manufactured using standard procedures, was evaluated by microscopic examination weekly during storage (up to 50 days) for six RBC units. RBC of differing morphology were counted following 2 h equilibration, in either fresh SAGM or ABO compatible donor fresh frozen plasma, at 4°C or room temperature (RT). Diameter and morphology were evaluated for more than 200 RBC for each condition.
Results: In SAGM, a majority of RBC maintained a discocytic or stomatocytic morphology though storage. The number of echinocytes in SAGM remained around 2% on average and below 5% at any time. In plasma, an increasing number of the cells adopted an echinocytic morphology during the first 23 days of storage (from 53 ± 28% at day 2 to 73 ± 16% at days 23) and then stabilized at 86 ± 11% between 30 and 50 days. The effect of temperature on the cell shape was seen in plasma before day 23: the percentage of echinocytes was higher at RT than at 4°C (from 70 ± 24% to 81 ± 10% vs 37 ± 23% to 64 ± 17% from day 2 to day 23 respectively, P = 0.0127 paired t-test ANOVA). There was little change in diameter for the sampled population during storage in SAGM and plasma. The apparent diameter of echinocytes reduced due to their increased sphericity. These results were consistent for both 4°C and RT.
Summary/Conclusions: RBC did not develop an echinocytic morphology when stored in SAGM, but maintained a discocytic or stomatocytic morphology. High reversibility of the echinocytic shape was observed between plasma and SAGM until day 50. The percentage of spheroechinocytes unable to restore their biconcave shape in SAGM was below 5%, supposing a low clearance rate after transfusion for the units tested here, even 50 days after donation. This figure is well under the 25% at 24 h requirement from the FDA for the transfusion of red cell concentrate. Donor variation was evident in this study and one of the most important factors explaining the variation in obtained results, confirming the disparity in RBC properties between donors. Examination of RBC returned to a physiological environment provides a measure of RBC quality different from examination in the storage solution.

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PROTECTIVE EFFECTS OF ANTIOXIDANT COMPONENTS FROM GINKGO BILOBA EXTRACT ON STORED ERYTHROCYTES

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Background: Red cell storage injury is common, and it is a task for blood workers to find safe, efficient and simple techniques to prolong red cell storage time and reduce red cell storage damage. The purpose of this study was to investigate the effect of antioxidant components extracted from Ginkgo biloba leaves on red blood cell storage.
Aims: To study the effect of antioxidant components of Ginkgo biloba extract on storage of erythrocyte function.
Methods: Twenty donators with 20 red blood cells were randomly divided into two groups: the experimental group and the control group, 20 rats in each group. All the experimental groups were stored in 4–6°C refrigerator, after 0, 1, 3, 7, 15, 20, 25, 30, 35, 42 d after sampling were detected ATP content, the average volume of red blood cells (MCV), Na+, K+, pH values, and observed red blood cell morphology.
Results: The ATP levels of both groups decreased with the prolongation of storage time, but the degree of ATP reduction and the rate of ATP in the antioxidant intervention group were significantly lower than those in the control group ($P < 0.01$). The normal rate of MCV in the antioxidant intervention group was significantly higher than that in the control group ($P < 0.05$). The levels of Na+ and pH in the antioxidant intervention group were lower than those in the control group ($P < 0.05$), while the K+ was gradually increased with the prolongation of storage time. 0.05). With the prolongation of the control group, erythrocytes gradually lost normal bivalve disc shape, antioxidant intervention group can better maintain the normal form of red blood cells.
Summary/Conclusions: The antioxidant components of Ginkgo biloba extract can protect red blood cells, reduce the damage factors, keep the red blood cell morphology and function, and reduce the red blood cell destruction.

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EFFECTS OF UBIQUITIN ON BIOLOGICAL CHARACTER OF MELANOMA B16 CELLS

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Background: Ubiquitin is a small heat-stable and highly conserved 76 amino acid protein in all eukaryotic cells, our previous studies found that the content of ubiquitin increases significantly during red blood cell storage. While perioperative blood transfusion especially storage blood was associated with higher rates of tumor recurrence and metastasis, and the mechanism behind remains controversial.
Aims: To investigate the effects of extracellular ubiquitin on the proliferation, invasion and early apoptosis of mouse melanoma B16 cells.
Methods: The effects of different concentrations of extracellular ubiquitin (200 ng/ml, 400 ng/ml and 800 ng/ml) on the proliferation of melanoma in mice were detected by CCK8 assay. The invasive ability of mouse melanoma cells treated by different concentrations of extracellular ubiquitin (200 ng/ml, 400 ng/ml and 800 ng/ml) was detected by Transwell. The effect of extracellular ubiquitin on early apoptosis of melanoma cells was verified by flow cytometry.
Results: 1)The results of cell proliferation experiment showed that the absorbance values (OD450) at 96 h of extracellular ubiquitin intervention groups (200 ng/ml, 400 ng/ml and 800 ng/ml), compared with the control group (0.29 ± 0.009), were (0.33 ± 0.007), (0.43 ± 0.008, $P < 0.001$) and (0.40 ± 0.007, $P < 0.001$) respectively. 2) The results of invasion test illustrated that the relative numbers of cells invading through the basement membrane in extracellular ubiquitin intervention groups (200 ng/ml, 400 ng/ml and 800 ng/ml) were (182 ± 12), (332 ± 19, $P < 0.01$) and (271 ± 30, $P < 0.01$) respectively, compared with the control group (165 ± 19). 3) Flow cytometry analysis found that the early apoptosis rate of melanoma cells in the 400 ng/ml ubiquitin intervention group (33.97 ± 1.65)% was higher than that of control group (22.83 ± 2.26)%, $P < 0.01$.
Summary/Conclusions: Extracellular ubiquitin can promote the proliferation and invasion of melanoma cells in mice, and inhibit the early apoptosis. Furthermore, these effects are obvious in 400 ng/ml concentration of ubiquitin.

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THE STUDY OF THE IMPACT OF ECTOINE TO THE FROZEN RED CELLS OF HUMAN BEINGS

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Background: Clinically, generally hospitals need lots of blood when patients suffer a large amount of blood loss, severe anemia or serious infection. It is very important to storage red blood cells (RBC) effectively in vitro to ensure transfusion. The cryopreservation of RBC is one method of storage RBC. Currently high concentration of glycerol are used as the main cryoprotectants for frozen RBC. High concentration of glycerol solution has high osmotic pressure and would be influence the stability of cell. For the glycerol cells, cells need to regulate intracellular and extracellular osmotic pressure and maintain normal form. Ectoine is one of compatible solutes who could regulate osmotic pressure in the halophilic bacteria and halotolerant bacteria. According to other studies, we speculate ectoine could be one of superb cryoprotectants in RBC storage.
Aims: To investigate the effect of osmotic adjustment of RBC with ectoine and to investigate the quality of RBC after thawing.
Methods: RBC kinetics in the isotonic ectoine solution, in the isotonic sodium lactate solution, in the normal saline solution, in ultrapure water and in the isotonic glycerol solution were determined. We ascertained the procedure of glycerolization before freezing RBC and the procedure of deglycerolization after RBC thawing. 1.5%, 3.0% and 4.5% (w/v) ectoine with 57% (w/v) glycerol were set as the experimental groups, while the commercial compound glycerol solution was set as the control group. The recovery rate, the deformability, the osmotic fragility, the content of adenosine triphosphate (ATP), the activity of fructose - 6 phosphate kinase (PFK), the activity of pyruvate kinase (PK), and the activity of lactic dehydrogenase (LDH) were determined for evaluating the quality of frozen RBC.
Results: The RBC kinetics in the isotonic ectoine solution, in the normal saline solution, and in the isotonic sodium lactate solution are similar; the RBC kinetics in ultrapure water and in the isotonic glycerol solution are similar. 3% ectoine with glycerol cryopreservated RBC was better than the commercial compound glycerol solution in the quality of frozen RBC.
Summary/Conclusions: The isotonic ectoine solution has the osmotic adjustment function to RBC. 3% ectoine with glycerol cryopreservated red cells had improved the quality of frozen RBC. This study could be expand the use range of ectoine.

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EVALUATION OF THE ADVANTAGES OF PLATELET CONCENTRATES STORED AT 4°C VERSUS 22°C

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Background: Few studies have investigated the low-temperature storage of platelets.
Aims: We aimed to compare the platelet counts, morphological changes, platelet functions, and platelet metabolisms in platelets stored still at 4°C with those in platelets stored under shaking at 22°C, attempting to provide experimental evidences for the development of a technology suitable for low-temperature storage of platelets
Methods: Manually prepared platelets were stored at 4 and 22°C. The platelet counts, observations with a scanning electronic microscope, blood gas indices, biochemical indices, platelet aggregative function, and expression of surface CD62P were monitored and compared between the two groups.
Results: There was no obvious change ($P > 0.05$) in platelet counts between day 21 at 4°C and day 5 at 22°C. Platelets stored at 4°C for 10–14 days were dramatically activated, had rough surfaces, and showed a significant degree of long pseudopodia formation. At 22°C after 7 and 10 days, platelets were low in quantity and aggregated to a certain degree; moreover, halos appeared around some of them. The pH of the platelets on day 5 at 22°C was lower than that at 4°C, while the lactate dehydrogenase and lactic acid levels in the former group were significantly higher ($P < 0.005$) than in the latter. The maximum aggregation rates induced by collagen and arachidonic acid in the platelets stored at 4°C for 5 days remained >80%, while the rates induced by four inducers in the platelets stored at 22°C were <5%. A higher expression of PAC-1 (activated IIb/IIIa) and CD62P (P-selectin) was noted on the surfaces of platelets stored at 4°C for 10–14 days.
Summary/Conclusions: Platelet counts, cellular morphologies, platelet membranes, cytoplasmic structures, aggregation rates, and hemostatic functions of platelets stored at 4°C for 10–14 days were better than those of platelets stored at 22°C for 5 days.

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DIFFERENTIALLY EXPRESSED MICRORNAS IN PLATELET DURING STORAGE

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Background: MicroRNAs (miRNAs) are small, non-coding, endogenous RNA molecules which possess a multitude of biological functions due to their ability to regulate gene expression via post-transcriptional gene silencing. In recent years, increasing evidence has emerged demonstrating that miRNAs are involved in the regulation of platelet storage lesion.
Aims: The aim of the current study was to investigate the expression level, possible roles and underlying molecular mechanisms of miR-7 and miR-16 in platelet storage.
Methods: The expression levels of miR-7 and miR-16 of room temperature and cryopreserved apheresis platelets after different storage periods were detected using quantitative real-time polymerase chain reaction (qRT-PCR). Biological information software and databases were utilized to predict target genes and molecular functions.
Results: The present study revealed that the expression of miR-7 is significantly downregulated in platelet after room temperature storage for 7 days when compared with fresh platelets. The expression of miR-16 is significantly upregulated in platelet after room temperature storage for both 5 and 7 days when compared with fresh platelets. However, both these two miRNAs showed no expression profile changes in platelet after cryopreserved for 5 and 7 days, as compared with fresh platelets.
Summary/Conclusions: MiRNAs seem to be involved in platelet storage lesion and may serve as predictive biomarkers for stored platelet quality and safety.

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THE EXPRESSION OF BCL-2 FAMILY PROTEINS IN THE INTRINSIC APOPTOTIC PATHWAY OF STORED PLATELETS

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Background: Bcl-2 family proteins play key roles in the intrinsic apoptosis pathway during platelets storage. Researchers have shown that different platelet miRNA profiles are expressed during storage and some of these perturbations are associated with apoptosis regulation. One of these miRNA, let-7b, has specifically been shown to increase during platelet storage and is thought to regulate apoptosis related genes and accelerate the death process.
Aims: The aims of this study are to present the expression of Bcl-2 family proteins and the post-transcriptional regulation of miRNA.
Methods: Three apheresis platelets were collected and stored at 22°C under agitation, and the samples were withdrawn on days 1, 3, 5, 7 and 9 from the platelet bags. The expression of Bcl-2 family proteins was determined by Western blot. Real-Time PCR was performed to determine the expression of miRNA and mRNA. The interaction between Bcl-xL and Bak was assessed by coimmunoprecipitation. The potential target of miRNA let-7b was identified by a dual-luciferase reporter assay.
Results: A significant decrease in anti-apoptotic Bcl-x_L and increase in pro-apoptotic Bak expression were found on the third day of storage. The ratio of Bak:Bcl-x_L was showed a decreased trend during platelet storage. And the expression of miRNA let-7b was upregulated. An interaction between Bcl-xL and Bak was also identified during platelets storage. Luciferase reporter assay demonstrated a potential binding site of the miRNA let-7b in the 3'-UTR of the Bcl-x_L gene. Expressions of Bcl-x_L and Bak protein, but not mRNA, were influenced by let-7b transfection.
Summary/Conclusions: MiRNA let-7b targets Bcl-x_L mRNA, and regulates the interaction of Bcl-x_L/Bak in the intrinsic apoptotic pathway of stored platelets

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MYELOPEROXIDASE LEVEL INCREASES DURING STORAGE EVEN IN LEUKODEPLETED PLATELET CONCENTRATES

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Background: The contaminant leukocytes in PC is an important factor implicated in platelet storage lesion during storage. Leukodepletion is a method to reduce contaminant leukocytes. Myeloperoxidase (MPO) is an enzyme produced by polymorphonuclear cells that have the potential to change structure and function of platelets during storage.
Aims: To analyze the myeloperoxidase level in leukodepleted PC during storage.
Methods: This study was conducted at the Blood Transfusion Services Unit, Dr. Sardjito Hospital, Yogyakarta, Indonesia. Myeloperoxidase level was measured using ELISA method from stored leukodepleted PCs (≤72 h and >72 h), and compared to non-leukodepleted PCs as control. The difference in each group was analyzed using ANOVA and *post hoc* test with significance level of $P < 0.05$.
Results: There were 31 PCs with ≤72 h and 33 PCs with >72 h of storage. The median of MPO level in >72 h was higher than in ≤72 h stored leukodepleted PCs (11.11 ± 3.97 ng/mL vs. 10.69 ± 3.16 ng/mL). In >72 h stored PC group, the median of MPO level was significantly lower in leukodepleted than in non-leukodepleted PCs (11.11 ± 3.97 ng/mL vs. 15.58 ± 7.82 ng/mL; $P = 0.001$)
Summary/Conclusions: The myeloperoxidase level was revealed higher in >72 h than in ≤72 h stored leukodepleted PCs and significantly lower in leukodepleted than in non-leukodepleted PCs

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STUDY ON PLATELET STORAGE DAMAGE AND PARTICLE RELEASE IN APHERESIS PLATELETS

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Background: With the increasing application of platelet, platelet storage lesion is caused more and more attention. Researches had shown that platelet storage lesion was significantly, platelet activation increased and platelet microparticles releasing also increased with prolonged storage time. Platelet microparticles can transfer information between cells and can improve the effect of tumor growth, inflammation and promoting tissue regeneration process. It is an urgent problem to study the platelet storage, the properties and release of platelets microparticles, and how to reduce platelet storage damage and the release or removal of particulates.
Aims: To study platelet storage lesion and the release of platelet microparticles with the prolongation of platelet preservation time, and to avoid the bad effects of clinical transfusion.
Methods: The platelets were made by plateletapheresis by Amicus donated by healthy volunteers from 6 young men, then suspended for 1, 3, 5 and 7 day. The platelet count was tested by Blood Cell Counter; CD62P expressed on platelet surface was detected by flow cytometry; Platelet aggregation was test by CA501 platelet aggregation analyzer. Platelet microparticle concentration in single platelet supernatant was measured with human platelet microparticle ELISA assay reagent.
Results: 1. There was no significant difference in different preservation time of platelet count ($P > 0.05$); but with the preservation of the lactic acid concentration increased gradually; the expression of CD62P on platelet surface increased significantly, suggesting that platelet activation increased; Platelet aggregation decreased significantly, suggesting that platelet storage lesion significantly with prolongation of storage period.
2. The results showed that the concentration of platelet microparticles gradually increased with the prolongation of the storage time, and the platelet particles preserved for 7 days were significantly higher than those preserved for 1 days ($P < 0.05$).
Summary/Conclusions: With the prolongation of storage time, the platelet activation increased, the damage became obvious, and the release of platelet microparticles increased.

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MICROPARTICLES IN STORED APHERESIS PLATELET

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Background: Microparticle (MP) could appear during the process of preparation and storage of blood products. Moreover, the formation of microparticle has been well illustrated in whole blood, but not in stored apheresis platelet.
Aims: The aim of this study is to compare the levels of platelet (PLT)-derived (PDMP), red blood cell–derived (RDMP), leukocyte-derived (LDMP), Monocyte-derived (MDMP,) and endothelial cell–derived (EDMP) in circulating blood and stored apheresis platelet through flow cytometry (FCM).
Methods: The aphaeresis platelets (8 with and 8 without leucofiltration) were obtained from 16 healthy blood donors, and stored in 22 ± 2°C for 5 days. Platelet-poor plasma (PPP)was generated by centrifugation of the circulating blood (from the donor) and the platelet products respectively at 2500 g for 15 min. The PPP was then labeled with CD41a (PDMP), CD235 (RDMP), CD45 (LDMP), CD14 (MDMP) and CD144 (EDMP), which were detected on day 0, day 1, day 3 and day 5 of storage by FCM.
Results: There were significantly higher percentages of TMP (total microparticles) and PDMP (13.45% ± 1.73% vs. 7.39% ± 1.41% $P < 0.05$, and 59.29% ± 6.46% vs. 20.50% ± 6.81% $P < 0.001$, respectively), higher numbers of TMP, PDMP and EDMP (4964 ± 917.7/μl vs. 2794 ± 421.5/μl $P < 0.05$, 4517 ± 936.6/μl vs.879.6 ± 552.0/μl $P < 0.001$, and 39.57 ± 6.2/μl vs. 22.93 ± 2.78/μl $P < 0.05$, respectively) in fresh PLTs (day 0) than those in circulating blood. However, lower percentages of RDMP, LDMP and MDMP (9.42% ± 0.83% vs. 22.38% ± 2.60% $P < 0.001$, 2.79 ± 0.51 vs.16.16% ± 2.55% $P < 0.0001$, and 1.69% ± 0.35% vs.15.94% ± 2.66% $P < 0.0001$, respectively) and lower counts of LDMP and MDMP (146.5 ± 16.13/μl vs. 383.5 ± 46/μl $P < 0.001$, 79.01 ± 9.65/μl vs. 587.3 ± 99.01/μl $P < 0.001$, respectively) were detected in fresh PLTs than those in circulating blood. The numbers of TMP, PDMP and RDMP were respectively increased by 2-fold, 3-fold and 1.9-fold, but the numbers of MDMP, LDMP and EDMP were not obviously increased after 5 days of storage (compared with the levels in day 1, respectively). Additionally, leucofiltration had no significant effects on the MP formation in apheresis platelets.
Summary/Conclusions: The MP formation was different between the apheresis platelet and circulating blood, much higher levels of PDMP and lower levels of RDMP, LDMP and MDMP were detected in PLTs. The TMP, PMP and RDMP accumulated, but the other types of MP were not accumulated in the platelet concentrates during the 5 days of storage.

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STUDY ON LONG TERM CRYOPRESERVATION ON PLATELETS

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Background: In this project, the quantity and quality of platelets after the low temperature condition were monitored continuously.
Aims: To discuss the most reasonable conditions of low temperature preservation and preservation of platelets.
Methods: 1. Preparation of fresh platelet rich plasma (FPRP) from whole blood. 2. Each FPRP will be divided into 19 parts, of which 9 cases preservated in the −80 low temperature refrigerator for 2–18 months. 9 pieces were kept in liquid nitrogen for 2–18 months. 1 samples were detected the quantity and activity of the platelet used as control. 3. During the storage period, each sample was removed from the −80 low temperature refrigerator and liquid nitrogen every 2 months. The test data is compared with the control group and the same conditions the other storage time point sample test data were analyzed, extend the observation of different temperature conditions with the preservation time of platelet. The reasonable period of time clearly different low temperature preservation of platelets.
Results: 1. At low temperature for 2 months, there were some changes in platelet count in FPRP compared with fresh FPRP. 2. Two cryopreservation methods were used to preserve the platelets in 2–18 months. There was no significant difference between the two groups in the same period.
Summary/Conclusions: Liquid nitrogen and low temperature refrigerator can be used for long-term preservation of platelets. During the period of 2–18 months, there was no obvious difference between the two methods.

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THE EXPERIMENTAL RESEARCHES OF THE LYOPHILIZING PLATELETS

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Background: The platelets was used extensively in clinical practice, to improve platelets preservation time and quality has become one of the hot research of transfusion medicine.

Aims: To study the basic protectant and rehydrating solution on the suitable glycerol content of the lyophilizing platelets, to investigate the effects of various kinds of saccharides and its different concentrations on the viability of the lyophilizing platelets. To understand the changes in the function and activity of the platelets, the maximum aggregating rates and the expressions of CD61, 62p and PAC-1 of the platelets before and after lyophilizing was detected.

Methods: According to the various kinds of saccharides added into the platelets lyophilizing basic protectant, the platelets lyophilizing basic protectant was divided into 5 groups. The expressions of the CD61, 62p and PAC-1 of the platelets before lyophilized and rehydrated were analyzed by flow cytometry (FCM), the maximum aggregation rates of the no-induced platelets and induced platelets were detected by the platelet aggregation instrument.

Results: Particularly 7% glycerol plasma protectant for preservation of the platelets lyophilizing agent and water for the rehydrating solution of the lyophilizing platelets, the recovery of the platelets after rehydrating was 85.5 ± 13.7%. The recovery rate of 0.15 g/mL trehalose and 0.10 g/mL sucrose were much higher than the others which were (94.04 ± 6.03)% and (95.35 ± 5.15)%. The maximum aggregation rates of the platelets before lyophilizing and rehydrating were (34.30 ± 33.14)% and (22.10 ± 23.25)%. The results of CD61 expression of the platelets before lyophilizing and rehydrating were (43.17 ± 13.55)% and (48.64 ± 13.24)%. The results of CD62p expression of the platelets before lyophilizing and rehydrating were (17.34 ± 6.47)% and (9.79 ± 4.48)%. The results of PAC-1 expression of the platelets before lyophilizing and rehydrating were (4.92 ± 6.32)% and (4.22 ± 4.64)%.

Summary/Conclusions: The 7% glycerol plasma protectant for preservation of the platelets lyophilizing agent and water for the rehydrating solution of the lyophilizing platelets was the most appropriate. Comparing the recovery rate of the lyophilizing platelets, the results show that 0.15 g/mL trehalose, 0.10 g/mL sucrose and 0.16 g/mL mixed saccharides for the additive agent of the lyophilizing platelets are the more appropriate single saccharide and mixed saccharides. The maximum aggregation rate of the platelets before lyophilizing has showed no significant difference as compared with it of the rehydrated platelets. The lyophilizing platelets still has certain aggregation activity, that showed its has a certain ability of survival. Although the expression of CD61, CD62p and PAC-1 were differences, but the differences of its were no statistical significance. The researches provided a theoretical basis for the establishment of the platelets lyophilizing method.

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THE REMOVAL OF DMSO FROM FROZEN PLATELET CONCENTRATES BY DIALYSIS WASHING

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Background: In addition to freezing injury, it has been reported that some damage to platelets may result from the cell packing that occurs during removal of the cryoprotectant.

Aims: In this research, we try to study the effect of the removal of DMSO from frozen platelet concentrates by dialysis washing on platelet function.

Methods: This study examined DMSO removal by fluid exchange across hollow-fiber (HF) filters as an alternative to centrifugation. 60 copies of fresh platelet concentrates were randomly divided into 2 groups (Group A: dialysis group, Group B: centrifugation group). The frozen platelet concentrates were stored for 1 week, 3 months and 6 months respectively. Then the following parameters of platelets were detected: Platelet counting, MPV, pH, CD62P expression rate, apoptosis, hypotonic shock response test (HSR), platelet aggregation function, platelet recovery (PPR), and DMSO residual.

Results: Platelet counting in Group A is significantly higher than Group B, and the DMSO residual in Group A is significantly lower than Group Bin the storage of 1 week, 3 months and 6 months respectively, the difference is statistically significant ($P < 0.05$). There are no significant differences between other indexes of frozen platelet concentrates between the two groups during the storage at 1 week, 3 months and 6 months.

Summary/Conclusions: Platelet quality was comparable after washing by either technique, but dialysis washing does remove cryoprotectant more rapidly than does centrifugation.

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CORRECTIVE EFFECTS OF 4°C STORAGE ON IN VITRO HEMORRHAGIC MODE OF PLATELETS

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Background: Few studies have investigated the low-temperature storage of platelets.

Aims: The corrective effects of storage at 4°C and agitation at 22°C on the in vitro hemorrhagic mode of platelets were compared to provide an experimental basis for the preservation of platelets for clinical applications.

Methods: Manual collection platelet 200 ml (10 doses) is divided into 2 × 100 ml, storage at 4°C and 22°C respectively. A total of 5 × 200 ml. such a bag (200 ml) of PLT was divided into two sub-bags each with a volume of approximately 100 mL—respectively stored at 4°C and at 22°C. A model of In vitro massive blood loss was established with the in vitro hemodilution method—diluting blood with normal saline (NS) at a ratio of 9:1 (NS: whole blood).The corrective effects of suspended erythrocytes, fresh frozen plasma, and platelets which were prepared on the ratio of 1:1:1 respectively stored at 4°C and at 22°C were evaluated for in vitro hemorrhagic by using Thromboelastography (TEG) test index and routine blood marker levels

Results: Platelets stored at 4°C and those agitated at 22°C for 1, 3, and 5 days were corrected using the in vitro hemorrhagic model prepared using the same blood samples. Platelet counts (×109/l) were corrected from 20–27 to 127–161vs 128–160, respectively ($P > 0.05$), and the TEG-MA (mm) were corrected from 12.7–14.4 to 45–51 vs 47–50, and TEG-R values were corrected from 27.7–9.9 to 4.4–4.3 vs 4.5–4.7, respectively ($P > 0.05$). For platelets stored at 4°C for 7–14 days, their platelet counts (×109/l) were corrected from 18–27 to 162–161, TEG-MA (mm) were corrected from 8.8–14.5 to 46–43, TEG-R (min) were corrected from 24–13 to 5.5–5.2 ($P < 0.05$)

Summary/Conclusions: Correction was performed using the hemorrhagic mode on the suspended erythrocytes, fresh frozen plasma, and platelets stored at 4°C or those agitated at 22°C. The platelet counts, TEG-R, and TEG-MA values were corrected, thus supporting the claim that platelets stored at 4°C for 10–14 days can achieve a good coagulation effect.

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CORRECTIVE EFFECTS OF 4°C STORAGE ON IN VITRO THROMBOCYTOPENIC MODE OF PLATELETS

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Background: Few studies have investigated the low-temperature storage of platelets.

Aims: The corrective effects of storage at 4°C and agitation at 22°C on the *in vitro* thrombocytopenic mode of platelets were compared in order to provide an experimental basis for the preservation of platelets for clinical applications

Methods: Using thromboelastography and platelet count, the *in vitro* corrective effects of thrombocytopenic platelets were evaluated for those stored at 4°C and those agitated at 22°C for various durations

Results: Platelets stored at 4°C and those agitated at 22°C for 1, 3, and 5 days were corrected using thrombocytopenic platelets prepared using the same blood samples. The platelet counts were corrected from 10–30 × 10^9/l to >100 × 10^9/l. Platelets that had been stored at 4°C for 7–14 days achieved similar correction values. Platelets that had been stored at 4°C for 10–14 days and those agitated at 22°C for 5 days were corrected using thrombocytopenic platelets. Their corrected thromboelastography (TEG)-MA values were within the normal range, and thus the corrective effects were achieved

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Summary/Conclusions: Platelets stored at 4°C for 10–14 days and those agitated at 22°C for 5 days were corrected using the *in vitro* thrombocytopenia mode. Based on the platelet counts, the TEG-MA values showed that corrective effects were achieved, thus supporting the claim that platelets stored at 4°C for 10–14 days could achieve good coagulation effects.

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EXPERIMENTAL STUDY OF THAWED CRYOPRECIPITATE AT DIFFERENT TEMPERATURES AND TIME ON FACTOR VIII AND FIBRINOGEN

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Background: Cryoprecipitate (Cryo) stored at 20–24°C for up to 6 h after thawing is a common practice in many US hospitals. However, in China, Cryo were thawed and stored for 24 h at 2° to 6°C.
Aims: To examine the level of activity of coagulant factor (F) VIII and fibrinogen (Fib) in thawed Cyro at different temperatures.
Methods: Cryo were randomly selected, thawed and stored in separate refrigerators at (4 ± 2) or (22 ± 2) . The activity level of FVIII and Fib were monitored and measured in each Cryo sample at baseline, 6 and 24 h.
Results: The activity level of factors showed decline after 6 and 24 h in storage. The activity level of FVIII stored at (4 ± 2) is significantly lowered than that stored at (22 ± 2) . The activity level of Fib stored at at (4 ± 2) is a little lower than that stored at (22 ± 2) , which is relatively more stable than FVIII.
Summary/Conclusions: The changes in activity level for FVIII and Fib in Cryo are comparable at different temperatures. However, decline was most significant when storage time increased. Thus, the thawed Cryo should be stored at a temperature of (22 ± 2) and be transfused as early as possible.

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EVALUATION OF ADAMTS13 ACTIVITY IN THAWED PLASMA STORED AT DIFFERENT CONDITIONS

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Background: The von Willebrand factor (VWF)-cleaving protease, ADAMTS13, is often deficient in cases of thrombotic thrombocytopenic purpura (TTP). Plasma exchange is the primary treatment of TTP to help restore ADAMTS13 activity. However, the stability of ADAMTS13 in thawed plasma stored at different temperature has not been determined.
Aims: The study evaluated ADAMTS13 activity in thawed plasma stored at different temperature for a period of time, aiming at providing information for transfusion services.
Methods: The study included 20 fresh-frozen plasma (FFP) units obtained from healthy donors. After thawed at 37°C water bath, each of the plasma was divided to 6 parts, which were separately placed at 4°C (for 3 days and 5 days), room temperature (for 1 day and 2 days) and 37°C (for 6 and 12 h). The retained activity of ADAMTS13 for each sample was evaluated with the fluorescence resonance energy transfer (FRET) assay kit from America Diagnostica inc.
Results: The initial value of ADAMTS13 was 104.1% ± 18.5%, and the mean retained activity (%) for storing at 4°C\ 3 days, 4°C\5 days, room temperature\1 day, room temperature\2 day, 37°C \6 h and 37°C \12 h was 98.1% ± 18.6%, 101.5% ± 17.7%, 99.9% ± 20.5%, 105.1% ± 23.5%, 106.6% ± 22.4%, 106.7% ± 20.0%. No significant difference was observed between the thawed plasma stored at different temperature for a certain period of time ($P > 0.05$).
Summary/Conclusions: ADAMTS13 is a very stable enzyme, the activity does not diminish in thawed plasma products stored at different temperature. It means that stored thawed plasma for a certain period should be as effective as freshly thawed plasma products.
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Abstract has been withdrawn

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THE CHANGES OF NEUTROPHIL FUNCTION DURING THE PRESERVATION OF NEUTROPHILIC GRANULOCYTES

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Background: The statistical data in recent 10 years showed that clinical albuginea infusion has reached more than 120,000 units and up to 6,000 patients have been treated or cured. It has become an important means of our city to cure ineffective bone marrow severely damaged tumor radiotherapy and chemotherapy, and joint anti-infection .
Aims: To investigate the effect of the preservation status of platelet-rich tunica albuginea on neutrophil (PMN) function over time in 24 h.
Methods: Randomly selected 10 bags of 400 ml healthy adult blood collected within 6 h to separate the platelet-rich tunica albuginea. Each bag of tunica albuginea (40–60 ml) was divided into three groups on average respectively according to the collection time 6 h, 12 h, 18 h group, and placed in 22 °C incubator.PMN was sorted by flow cytometry and the functions of phagocytosis, secretion of reactive oxygen and apoptosis were measured.
Results: (1)The phagocytic function: There was no significant difference when 6 h group and 12 h group were compared. 6 h, 12 h and 18 h group had significant differences. (2)Reactive oxygen: there was no significant difference of 6 h group, 12 h group, 18 h group when each two groups were compared.(3)Apoptosis (apoptosis and necrosis), there was no significant difference when 12 h group and 6 h group were compared. There was no significant difference when 18 h group and 12 h group were compared as well. There was significant difference when 6 h group and 18 h group were compared ($P < 0.05$).
Summary/Conclusions: The phagocytosis and apoptosis of neutrophils begin to change at the beginning of 18 h, and leukocyte collected after 12 h should be transfused as soon as possible.

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Abstract has been withdrawn

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THE METABOLISM EVALUATION OF BLOOD QUALITY AFTER MARINE STORAGE AND TRANSPORTATION

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Background: Glycolysis is the only energetic metabolic pathway for mature erythrocytes to obtain adenosine triphosphate (ATP). Generally, the current methods used to measure RBCs glycolysis are not in living state in real-time, or are destructive to cells or require radioactivity. XF technology can be applied to different types of cells, in which the red blood cells are suspended and the cell shape and size are different from other cells, An extracellular flux (XF) methodology has been particularly advantageous. so application of the XF technology in erythrocytes and exploration of the assay conditions are necessary.
Aims: To evaluate the quality of blood after marine storage and transportation, a quantitative energy metabolism platform supported by extracellular flux (XF) technique is established to determine the glycolysis level of red blood cells. The "functional dose", as a valuable supplement, is therefore involved into the quality evaluation of red blood cells after marine storage and transportation.
Methods: The glycolysis of erythrocytes was determined by XF methodology. The concentrations of glucose and 2-deoxyglucose (2-DG) were optimized to when the extracellular acidification rate (ECAR) of erythrocyte glycolysis was detected. Red blood cells suspensions were sailing for 20 days in the Gulf of Bohai or the Pacific Ocean. The level of glycolysis, ATP contents and free hemoglobin of red blood cells

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were detected before and after marine navigation. Red blood cells suspensions stored at 4°C in blood bank on land were served as control.
Results: The results showed that 40 mM glucose and 400 mM 2-DG were the most suitable concentrations for erythrocyte ECAR assay by XF technique. And the evaluating indictors of blood quality and energy metabolism of red blood cells were detected in twice marine storage and transportation. The blood samples sailing in the Gulf of Bohai have no change significantly, indicating that the sample is well preserved. The energy indicators of the blood samples in Pacific Ocean navigation decreased significantly, and it is recommended that the samples be discarded.
Summary/Conclusions: For the first time, XF technique is applied for the measurement of erythrocyte glycolysis and enables high-throughput and multi-indictor dynamic detection of viable cells becomes possible. The energy metabolism of erythrocyte should be recommended as an important indicator in the evaluating system of blood after marine storage and transportation.

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EFFECT OF TRANSPORTATION ON THE STORAGE OF RED BLOOD CELLS
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Background: Red blood cells are inevitably subjected to vibration during transportation, especially the vibration and acceleration will further aggravate biochemical and functional changes caused by processing and storage. Numerous studies have focused on the storage injury of red blood cells, however, there is less research and attention on vibration factors in transportation. And the present study of transportation vibration has no systematic study on the changes of erythrocyte mass with the storage time extending after vibration, namely the study on storage damage of red blood cells by vibration.
Aims: The discussion on effects of vibration on free hemoglobin (FHb), hemolysis ratio, LDH, K^+ concentration of red blood cells in different storage time.
Methods: Choosing 20 bags (2 U/bag) red blood cells (RBC) stored for 7 days (Control group and Experimental group, 10 bags per group) as the research object, the Control group were stored with conventional methods at 4°C, the combination of wheeled vehicle (three direction vibration of horizontal, longitudinal, vertical) was selected to simulated RBC transportation using electromagnetic vibration test system. The RBC samples were prepared on vibration 0 h, 3 h, storage of 14 days, 21 days, 28 days, 35 days to detect and analyze FHb, hemolysis ratio, LDH, K^+ concentration of samples.
Results: The FHb, hemolysis ratio, LDH, K^+ concentration was significantly increased after vibration ($P < 0.05$), and FHb, hemolysis ratio, LDH presented a greater increasing trend than Control group with the extending of storage time ($P < 0.05$), but there was no significant variation of K^+ concentration in different storage time. Value added of FHb, hemolysis ratio of experimental group in 14 day were greater than control group ($P < 0.05$), there were no significant difference of LDH, K^+ in 14, 21, 28, 35 day between control group and experimental group.
Summary/Conclusions: Vibration resulted in FHb, hemolysis ratio, LDH, K^+ concentration increment in different storage time. The value added of FHb and hemolysis ratio was significantly increased in 14 day, but no difference in 21, 28, 35 day, and there was no difference of value added of LDH, K^+ in different storage time.

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EFFECTS OF EXPOSURE DURATION OF SIMULATED TRANSPORT VIBRATION ON ERYTHROCYTE QUALITY
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Background: The transport vibration may cause damage to red blood cells in the emergency blood transport, due to complex transport conditions. The damage seems to depend on the exposure duration, however, the number of publications involving the effects on RBC quality by vibration exposure duration is still limited.
Aims: To investigate the effect of vibration exposure duration on the quality of red blood cells during transportation.
Methods: The combination of wheeled vehicle (three direction vibration of horizontal, longitudinal, vertical) was selected to simulated blood transportation using electromagnetic vibration test system based upon GJB150.16A-2009. 14 aliquots of suspended erythrocytes stored for 4 days were vibrated for 6 h, and samples were collected before the vibration, 3 h, 4.5 h, 6 h after vibration, to test Free Haemoglobin (FHb), K^+, LDH and haemolysis ratio.
Results: The FHb, K+, LDH and haemolysis ratio of red blood cells after vibration was significantly higher than before vibration ($P < 0.05$). The FHb increased with the exposure time extending of vibration (0.28 ± 0.06 g/l vs.1.30 ± 0.42 g/l vs. 1.66 ± 0.46 g/l vs.2.14 ± 0.54 g/l, $P < 0.05$), and there was a similar tendency in LDH level (69.40 ± 27.90 U/Lvs.507.20 ± 134.70 U/l vs. 679.40 ± 151.10 U/l vs. 838.50 ± 169.90 U/l, $P < 0.05$). the concentrations of K+ 4.5 h, 6 h after vibration were higher than 3 h (9.60 ± 0.72 mmol/Lvs.10.20 ± 0.8 mmol/l, 9.60 ± 0.72 mmol/l vs.10.60 ± 0.84 mmol/l, $P < 0.05$), but there was no statistically significant of K^+ concentrations between 4.5 h, 6 h after vibration (10.20 ± 0.8 mmol/l vs.10.60 ± 0.84 mmol/l, $P > 0.05$).
Summary/Conclusions: The FHb, K^+, LDH and haemolysis ratio of suspended erythrocytes, which were vibrated for 6 h at combined wheeled vehicle vibration level, were significantly changed, the increase of FHb, haemolysis ratio and LDH is related with the exposure duration of vibration.

Blood Components

P-201

Abstract has been withdrawn

P-202
THE RELATIONSHIP OF POST-TRANSLATIONAL MODIFICATION OF HEMOGLOBIN AND HYPOXIC ADAPTATION IN TIBETAN POPULATION
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Background: It was reported that a series of physiological changes will occur in the Han population migrated to Tibetan Plateau. For instance, there are significant enhancement of hemoglobin (Hb) and hematocrit (Hct) in Han migrants ascends to Tibetan Plateau, hypoxia played an important role in this process. However, this symptoms seldom happens in native Tibetan population. Does red blood cells (RBCs) of Tibetan have another way to adapt hypoxic circumstance? To clarify the oxygen delivering capacity differences between Han and Tibetan population, we evaluate the RBCs parameters changes after Han immigrated into Tibetan Plateau (>1 years), and the oxygen-carrying ability difference between Han and Tibetan population.
Aims: To illustrate the mechanism of hematology changes of Tibetan population in adapting hypoxic environment. And our data can provide basic information to support blood component processing and blood storage in Tibetan Plateau.
Methods: Blood samples was collected from Tibet Autonomous Region blood center and Chengdu blood center according to standard procedures. Samples were divided to three group according to donors information: (1) native Tibetan group (8 male, 1 female), (2) Han migrants group (7 male, 1 female), (3) lowland Han group (8 male, 3 female), and their ages ranged from 19–50. The information of weight and height of the donors was not supplied by the blood centers. The 2,3-DPG concentration, RBCs affinity to oxygen (PO_2 at 50% saturation of hemoglobin, P_{50}) and MALDI-TOF MS were detected.
Results: Compared to Han migrants, RBCs of native Tibetan population have higher affinity to oxygen (30.63 ± 1.60 *vs* 24.81 ± 0.74 mmHg, $P < 0.05$), even the 2,3-DPG concentration have no significant difference in those two group (3.15 ± 0.43 *vs* 2.93 ± 0.95 mmol/l). And RBCs affinity to oxygen do not have significant difference between native Tibetan and lowland Han population (24.81 ± 0.74 *vs* 23.95 ± 1.79 mmHg). Combining with MALDI-TOF MS data, we found that the molecular weight of partial hemoglobin β unit was increased (from 15848 to 16169) in native Tibetan. This phenomenon was common in native Tibetan population, but seldom found in Han migrants.
Summary/Conclusions: Combining with MALDI-TOF MS data, we found that there are some changes in hemoglobin of native Tibetan. Considering this phenomenon was common in native Tibetan population, it suggest that it is a kind of post translational modification in hemoglobin. And this modification seems to be associated with hemoglobin affinity to oxygen.

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RED CELL HEMOLYSIS IN A QUALITY CONTROL ANALYSIS. A COMPARATIVE STUDY

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Background: Red cell storage lesion is a known phenomenon that represents a challenge to blood banks striving to provide maximum safety and therapeutic effect of blood components. In vitro hemolysis is a key parameter measured to analyze the quality of the chain of production, and the effect the storage lesion has over red cell concentrates. Visual evaluation of hemolysis can be defective, and it has been reported that using blood bag segments could lead to incorrect readings, thus putting a stress to Quality Control and Quality Assurance analysis. The AABB and the Council of Europe have defined their standards of percent hemolysis for red cell concentrates as 1% (AABB) and 0.8% (CoE).
Aims: The study aimed to analyze the behavior of red cell hemolysis during storage, comparing two blood bag manufacturers, and also to further test the fidelity of blood bag segments with respect of the bag, after careful mixing. The evolution of hemolysis during storage and up to expiration date was also observed.
Methods: 138 red cell concentrates were tested, 70 produced with Terumo-Optisol, 68 with JMS-SAGM. They were distributed into 5 groups, by day of storage: 1–10, 11–20, 21–30, 31–40, and an expiration date group of 41–42 days with 27 bags each group (29 for 42 days). Each bag's segment was uniformally stripped and mixed with the bag, and was immediately sealed with heat, with the middle third segment transferred to an EDTA tube, and centrifuged for 10 min at 3500 rpm. A second sample from the same segment was used to measure hemoglobin and hematocrit with a CellDyn 1800 counter. A paired sample was obtained from the center of bag, and was treated the same way. Free hemoglobin was measured with the HemoCue PlasmaLow Hb photometer.
Statistical analysis included the Wilcoxon rank-test to compare samples between segments and bags, and paired t-Student test to compare manufacturers. Also, statistical description included means and standard deviations analysis..
Results: Hemolysis measured from the segments did not reflect the hemolysis measured from the bags ($W = 118$, $Z = -9.8617$, P-value is 0, significance at $P \leq 0.01$), and actually overestimated it. Means from first subgroups were consistently double measured from the segments than from the bags. For the subgroup at expiration date, means were 0.79 (segments) and 0.54 (bags); segments mean was almost at the Council of Europe standard limit for hemolysis in Quality Control. Segments did faithfully reflect other components parameters, like hematocrit and hemoglobin content.
Progression of hemolysis was observed, as expected from the nature of the storage lesion. However, the peaks for hemolysis in both segments and bags was the group of 21–30 days, which saw a 75% increase from hemolysis of the previous group, and stabilized from day 30 to expiration (day 42).
Summary/Conclusions: Red cell concentrates bag segments are not reliable a source of hemolysis measurement, even if they are thoroughly and uniformally stripped and mixed. Quality control of hemolysis to these concentrates should avoid using them, since they overestimate it. Hemoglobin and hematocrit measurement did not present this phenomena. The question could be open to determine if there could be other quality control parameters that could experience the same situation. There was no difference between the two manufacturers analyzed.

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EXPLORING THE FEASIBILITY FOR ADVANCED PREPARATION OF PLATELETS

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Background: How to make the more effective use of platelet in treating the hemorrhagic disease.
Aims: Explore the feasibility for advanced preparation and gathering of frozen platelets.
Methods: Add DMSO 100% and 75% solution to ready-to-freeze platelets respectively until DMSO solution percentage stands at 5%; use different methods to resuscitate platelets and compare the results.
Results: DMSO concentration, speed of DMSO injection and processing conditions all contribute to platelets quality. When temperature remains 38°C, 75% DMSO solution yields highest resuscitating rate. The results yielded under all conditions in the process show differences that demonstrate statistical significance. Floccules formed in DMSO 100% and 75% solution during the thawing process also show difference that demonstrates statistical significance.
Summary/Conclusions: DMSO 75% solution is more effective for effective preparation and gathering of frozen platelets.

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TOXIC EFFECTS OF DIMETHYL SULFOXIDE ON RED BLOOD CELLS, PLATELETS, AND VASCULAR ENDOTHELIAL CELLS IN VITRO

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Background: Dimethyl sulfoxide (DMSO) is widely used in biological studies as a cryoprotective agent for cells and tissues, and also for cryopreserved platelets (PLTs). However, few data on the toxic effects of DMSO following intravenous infusion of cryopreserved PLTs are available.
Aims: The aim of this study was to explore dose-related effects of DMSO on red blood cells (RBCs), PLTs and vascular endothelial cells in vitro.
Methods: We investigated the effects of DMSO on the membrane stability and function of blood cells and the proliferation of vascular endothelial cells involved in the infusion response, using in vitro blood cells and EAhy926 cells stimulated with DMSO or without DMSO (control).
Results: The results showed that DMSO treatments had significant effects on RBCs, affecting osmotic fragility and increasing hemolysis. Free hemoglobin (FHb) level of RBCs was 0.64 ± 0.19 g/l after incubation for 6 h with 0.6% DMSO, and these levels were elevated compared with controls (0.09 ± 0.05 g/l). Aggregation of PLTs induced by adenosine diphosphate, thrombin (THR), and thrombin receptor activator peptide (TRAP) were inhibited by DMSO treatment because the THR generation capacity was reduced. The intensity of the cytosolic esterase-induced fluorescence response from carboxy dimethyl fluorescein diacetate (CMFDA) in PLTs was decreased about $29\% \pm 0.04\%$ after treatment with DMSO. DMSO also inhibited the proliferation of the vascular endothelial cell line EAhy926 cells by blocking the G1 phase. Apoptosis of EAhy926 cells with 0.6% DMSO stimulation was increased three-fold compared to controls.
Summary/Conclusions: On the basis of these findings, it was concluded that DMSO was toxic to the hematologic system. This should be taken into account when assessing the infusion effects of cryopreserved PLTs or other blood products requiring DMSO as a vehicle, such as cryopreserved stem cells, in order to avoid adverse therapeutic effects.

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EFFECTS OF DIFFERENT FRESH FROZEN PLASMA PREPARATION METHODS ON COAGULATION SYSTEM

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Background: Fresh frozen plasma (FFP) is a kind of precious blood resources, mainly used for the treatment of congenital coagulation disorders and acquired coagulation disorders in patients, its' clinical indication is limited and have the risk of blood transfusion related adverse reactions.
Aims: In order to reduce the risk of blood transfusion related adverse reactions in the patient who is treated with FFP, two improvement methods consisting of Leukocyte reduction filter (LR) and gamma on with 30GY irradiation (GI) were proposed, and the effect was determined by detecting parameters.
Methods: Collected 40 FFP samples from 40 blood donors by The Red Cross Blood Center of Henan Province, the study was carried out with Single blind randomized blocks design. Each sample was divided averagely into four groups: Group A as control group, Group B as LR group, Group C as GI group and Group D as LR plus GI group, then test the four groups by the automated blood coagulation analyzer.
Results: After tested each element of four groups meet national standards for the safe use, but two of them statistically significant: 1. FVIII:C: Group A: 105.8 ± 49.6, Group B: 98.3 ± 44.8, Group C: 105.1 ± 40.31, Group D: 87.4 ± 37.5, FbgC variations between groups: Group C have no significant difference with Group A, Group

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B decreased 7 percent compared with Group A, Group D decreased 11 percent compared with Group A. the difference between Group A an Group D was statistically significant ($P < 0.05$). 2. Fibrinogen contents (FbgC): Group A: 2.287 ± 0.358, Group B: 1.968 ± 0.296, Group C: 2.122 ± 0.331, Group D: 2.122 ± 0.331, FbgC variations between groups: Group C have no significant difference with Group A, Group B decreased 14 percent compared with Group A, Group D increased 8 percent compared with Group B. the difference between Group A an Group B was statistically significant ($P < 0.05$).
Summary/Conclusions: Through the study, we can find out that the FFP were still stratified the general clinical needs though partial clotting factor activity had lost after LR and GI, so for the patients who have blood transfusion related adverse reactions we may have better choices: The treatment of GI may have positive effects in the replacement therapy of FbgC, while After GI FFP will be less effective, FFP will be less effective in the replacement therapy of intrinsic coagulation factors, after LR+GI especially after LR, FFP will be more effective in the replacement therapy of extrinsic coagulation factors. However, there is still a need for a larger sample and multicenter randomized controlled trial to confirm whether clinical popularization is available.

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Abstract has been withdrawn

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DESCRIPTION OF EXPIRED PRODUCT PLATELET CONCENTRATE AT BLOOD TRANSFUSION UNIT BEKASI COUNTY IN 2015 AND 2016

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Background: Since 2005 blood transfusion unit Bekasi county has produced platelet concentrate which is provided from whole blood donation. One obstacle that we had to deal with is lots of platelet concentrate expelled because of expired. It is caused by the expired period of platelet concentrate is shorter than the other blood products (5 days from the donation date). Because of that, we need more accurate forecasting to avoid more expire products.
Aims: To know the amount of expired platelet concentrate, so we are able to do the evaluation to forecasting if it's had been done precisely.
Methods: We collected secondary data in 2015 and 2016 from blood transfusion information system.
Results: Platelet Concentrate Production in 2015,
A Blood Group: 3042 unit
B Blood Group: 3299 unit
O Blood Group: 3869 unit
AB Blood Group: 1149 unit
Total Production: 11.359 unit
Platelet Concentrate Expired in 2015
A Blood Group: 231 unit (7.6%)
B Blood Group: 225 unit (6.8%)
O Blood Group: 171 unit (4.4%)
AB Blood Group: 335 unit (29%)
Total expired: 962 unit (8.5%)
Platelet Concentrate Production in 2016
A Blood Group: 3584 unit
B Blood Group: 4100 unit
O Blood Group: 4878 unit
AB Blood Group: 1304 unit
Total Production: 13.866 unit
Platelet Concentrate Expired in 2016
A Blood Group: 329 unit (9.2%)
B Blood Group: 202 unit (4.9%)
O Blood Group: 112 unit (2.3%)
AB Blood Group: 270 unit (20.7%)
Total expired: 913 unit (6.6%)
Summary/Conclusions: Base on the data above, in 2015 we found 962 units of expired platelet concentrate (8.5% of production), The largest number is in AB blood group (29%) and in 2016 we found 913 units expired platelet (6.6% of production), the largest number is still in AB blood group (20.7%). It means in 2016 the expired platelet has reduced 1.9% compare with 2015. Although the expired number has been reduced, but it still relatively high. In 2015 & 2016 we did forecasting by counting the distribution of platelet concentrate in month for each blood group. In 2017 to meet a demand and reduce the expire platelet concentrate, we change the mechanism of forecasting by counting distribution of platelet concentrate in week, and write every demand of platelet concentrate which is can't be fulfill because of the emptiness of stock.

Plasma Products

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CIRCULATING CONCENTRATIONS OF EOTAXIN AND GDF11 IN CHINESE BLOOD DONORS

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Background: Eotaxin (also known as C-C motif ligand 11, CCL11) is regarded as a potential pro-ageing factor which negatively regulates neurogenesis and cognitive function, and has been found age-related increase in the plasma of mice. Growth differentiation factor 11 (GDF11), a promising rejuvenating factor, can reverse age-related diseases, such as age-related cardiac hypertrophy, impairments in cognitive function, and so on. Furthermore, it was demonstrated that GDF11 in mice plasma decreased with age. These studies suggested a new strategy in the treatment of aged-related diseases, where blood plasma from young donors could be used as an eotaxin-poor and GDF11-rich source to reverse aged-related diseases. However, the concentration of eotaxin and GDF11 in blood donors has not yet been analysed.
Aims: To evaluate the concentrations of eotaxin and GDF11 in the plasma of Chinese blood donors.
Methods: In all, 476 healthy volunteers were enrolled from qualified blood donors aged 18 y–56 y at Jintang Plasmapheresis Center and Meishan Blood Center. Each of the plasma samples was divided into aliquots, and then they were stored at −70°C until analysis. The concentrations of eotaxin and GDF11 were determined by using R&D Human CCL11/Eotaxin DuoSet ELISA and Human GDF-11/BMP-11 DuoSet ELISA, respectively. ELISAs were carried out according to the vendor's instructions. Briefly, all the plasma samples were tested in duplicates. After 2 h incubation in eotaxin or GDF11 capture antibody-coated plates, streptavidin-conjugated horseradish peroxidase was added. After further washing, tetramethylbenzidine was introduced. Finally, the optical density was measured at 450 nm using a spectrophotometer.
Results: The average age of the volunteers was 44.45 ± 9 y (male, 44.16 ± 10.06 y, n = 181; female, 44.67 ± 8.25 y, n = 295; $P > 0.05$), and the samples were arbitrarily divided into the four following age categories: <30 y (18 y–29 y, n = 60), 30 y–39 y (n = 67), 40–49 y (n = 189) and ≥50 (50 y–56 y, n = 160). Mean eotaxin and GDF11 levels were 118.5 ± 57.61 pg/mL and 21.54 ± 17.62 pg/mL, respectively. The concentrations of eotaxin showed significant increases with age, with an overall change of 1.41-fold by >50 y (92.46 ± 44.56 pg/mL, 99.74 ± 56.86 pg/mL, 115.52 ± 58.06 pg/mL and 130.21 ± 53.0 pg/mL, resp.; $P < 0.05$). Consequently, a slight but significant correlation ($r = 0.27$, $P < 0.001$) between eotaxin levels and age was found. However, GDF11 showed a small, but not significant, trend of decrease with age. Interestingly, there was a clear distinction between genders, where eotaxin were observably higher in male than in female (126.51 ± 53.09 pg/mL *vs.* 112.12 ± 51.10 pg/mL; $P < 0.01$). Moreover, no ABO blood type-related variations were observed for eotaxin and GDF11.
Summary/Conclusions: In general, our findings revealed the plasma concentrations of eotaxin and GDF11 in Chinese blood donors. Except ABO blood type, both age and gender showed different effects on plasma levels of eotaxin and GDF11. And these new data are quite helpful for the further study of young blood plasma.

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INFLUENCES OF ABO BLOOD GROUP, GENDER AND AGE ON FACTOR VIII ACTIVITY AND FIBRINOGEN CONTENT IN PLASMA OF BLOOD DONOR

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Background: ABO blood group is a hereditary factor of plasma level of factor VIII. As the growth of the age, the incidence of thrombotic disorders is higher and higher. Fibrinogen and coagulation factor VIII play an important role in the blood agglutination.
Aims: To investigate influences of ABO blood group, gender and age on factor VIII activity and fibrinogen content in plasma of blood donor.
Methods: A total of 193 healthy volunteers, 111 male and 82 female, were enrolled for this study from normal blood donors aged between 18–55 y (mean 31.24 ± 10.66 y) at Fujian Blood Center. Factor VIII activity and fibrinogen content was measured by the clotting assays using commercially available kits (Diagnostica Stago S.A.S., France) on a STA Compact® automated coagulation analyzer (Diagnostica Stago S.A.S., France). The phenotype of ABO blood group was determined by serological tests with monoclonal anti-A and anti-B antibody blood grouping reagent kit (Changchun bode biological technology Co., Ltd., Changchun, China) and red blood cells reagent kit for use in ABO reverse grouping (Shanghai blood biological pharmaceutical Co., Ltd., Shanghai, China) on a Hemotype automated blood group analyzer (HealTek biological technology Co., Ltd., Beijing, China). Values were expressed as means and standard deviation (SD). Multi-group comparisons between factor VIII activities and fibrinogen levels in different ABO groups were conducted by one-way ANOVA followed by LSD *post hoc* test. The comparisons of factor VIII activities and fibrinogen levels between genders were accomplished using two-tailed unpaired Student's t -tests. Bivariate correlation analysis was used to calculate the associations of factor VIII activity and fibrinogen level, factor VIII activity and age, and fibrinogen level and age. A 95% CI [2.5%97.5%] was used and a P-value < 0.05 was considered statistically significant.
Results: The activity of factor VIII was significantly less in group O subjects (722.51 IU/l ± 190.22 IU/l, n = 92) than in group A (888.63 IU/l ± 275.47 IU/l, n = 43, P < 0.001), group AB (837.41 IU/l ± 115.74 IU/l, n = 13, P < 0.05), and group B subjects (933.86 IU/l ± 243.34 IU/l, n = 45, P < 0.001), whereas the fibrinogen level in group O subjects (2.37 g/l ± 0.46 g/l, n = 92) was not significantly different from those in non-group O subjects (2.54 g/l ± 0.65 g/l, n = 43, in group A; 2.37 g/l ± 0.49 g/l, n = 13, in group AB; 2.41 g/l ± 0.54 g/l, n = 45, in group B subjects) (P > 0.05 for all comparison). Comparing with the factor VIII activity in group A, B, and AB subjects each other, there were no significant differences among them. Same as the factor VIII activity, there were no significant differences among the fibrinogen levels in group A, B, and AB subjects. There were no significant differences between the factor VIII activities in the male group and the female group and between the fibrinogen levels in the two groups. Factor VIII activity and fibrinogen level showed significant and positive relationships with age (r = 0.191, P < 0.005 and r = 0.152, P < 0.05, respectively). Fibrinogen level was significantly correlated with factor VIII activity (r = 0.307, P < 0.0005).
Summary/Conclusions: ABO blood group and aging have significant effects on the factor VIII activity in plasma. Fibrinogen level observably increased with age. Fibrinogen content was strongly and positively correlated with factor VIII activity. It is suggested that cryoprecipitate should be prepared from raw plasma from older blood donors in group AB, A and B to get more quantity of fibrinogen and factor VIII.

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ESTABLISHMENT OF THE ACCURATE METHOD FOR DETERMINATION OF ANTI-A**B**42 ANTIBODIES IN PLASMA AND PRELIMINARY ANALYSIS OF A**B**42 ANTIBODIES LEVEL IN PLASMA DONORS

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Background: Alzheimer's disease (AD) has become a global problem. Up to now, few drugs have impeded the neuropathological progression of AD. Intravenous immunoglobulin (IVIg) is another approach that has been examined for treating AD. Although the multiple antibodies including anti-Aβ antibodies in IVIg should give it an advantage over monoclonal antibody for treatment of AD, its encouraging results in pilot studies have not been replicated in larger trials. The development of AD-specific IVIg might be the best chance for IVIg treatment of AD to succeed.
Aims: Establish an accurate enzyme linked immunosorbent assay (ELISA) method for determination of anti-$A\beta_{42}$ oligomers antibodies in plasma and use it to examine anti-$A\beta_{42}$ antibodies level in plasma donors, so as to lay the foundation for preparation of specific IVIg with rich anti-$A\beta_{42}$ antibodies.
Methods: 1) $A\beta_{42}$ oligomers were prepared from synthetic $A\beta_{42}$ peptide at different initial aggregation concentrations and through phosphate buffer saline (PBS) solution, and then $A\beta_{42}$ oligomers were analyzed by SDS-PAGE and western blot. The best aggregation condition was chosen to aggregate $A\beta_{42}$ oligomers. 2) The orthogonal experiment was designed to establish an accurate determination system for measurement of the anti-$A\beta_{42}$ oligomers antibodies. The performance of the determination system was evaluated. 3) The levels of anti-$A\beta_{42}$ oligomers antibodies were tested in plasma donors from different districts, and the characteristics of the anti-$A\beta_{42}$ oligomers antibodies in plasma donors were analyzed.
Results: 1) The amount of $A\beta_{42}$ oligomers rose with the increase of the initial aggregation concentration. Trimers and dispersion zones of medium- and high-molecular-weight oligomers were generated from $A\beta_{42}$ at 250, 125 and 62.5 μmol/l when there was no SDS in PBS. Trimers and medium-molecular-weight oligomers were generated at the same concentrations when SDS was mixed with PBS. 2) The orthogonal analysis showed that the measurement result of anti-$A\beta_{42}$ oligomers antibodies in plasma was heavily influenced by the detection conditions including $A\beta_{42}$ oligomers coating concentration, coating pH, blocking buffer, primary antibody and secondary antibody conditions, and then the best detection system for anti-$A\beta_{42}$ oligomers antibodies was established by orthogonal experiment. Its performance was evaluated. The correlation coefficient was 0.994, the relative deviation of stability was 3.5%, the relative deviation of precision was 4.77%, and the accuracy error was 10.16%. 3) There were great differences in the anti-$A\beta_{42}$ oligomers antibodies levels among plasma donors, ranging from 3.16% to 581.7%. The antibodies in female donors were 10.1% higher than those in male donors and the antibodies increased with age. The level of antibodies in plasma donors from Guizhou was 25.0% higher than that in those from Sichuan. There was no obvious difference in the antibodies between plasma donors with different ABO blood types. The proportion of plasma donors with high antibodies levels from Guizhou is higher than that from Sichuan.
Summary/Conclusions: 1) Appropriate initial aggregation concentrations contribute to the generation of $A\beta_{42}$ oligomers. Addition of 0.05% SDS to PBS could contribute to form stable $A\beta_{42}$ oligomers. 2) The anti-$A\beta_{42}$ oligomers antibodies determination system obtained from orthogonal experiment is stable and can be used to test the plasma samples. 3) The levels of anti-$A\beta_{42}$ oligomers antibodies among plasma donors show great difference and vary with gender, age and region.

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PREPARATION AND EVALUATION OF FREEZE-DRIED CRYOPRECIPITATE

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Background: Concentration of factor VIII from fresh plasma by cryoprecipitation remains the basis for preparation of products used to treat haemophilia A. However, the storage of cryoprecipitate requires −20°C, it's hard to ensure the storage temperature during military and emergency situation.
Aims: To research the method of freeze-dried cryoprecipitate and evaluate its effect.
Methods: Forty units of fresh whole blood (200 ml) were collected from volunteer donors. Two bags of cryoprecipitate with same capacity were prepared from each whole blood by conventional method, one bag was stored at −20°C as frozen cryoprecipitate, and the other was prepared into freeze-dried cryoprecipitate after adding Compound Amino Acid Injection by the lyophilizer and stored at 4°C. Two sets of cryoprecipitates from 20 donors were taken out and analyzed for factor VIII, fibrinogen, and pH after 24 h, and the other cryoprecipitates were evaluated after 12 months.
Results: There are no significant differences of factor VIII, fibrinogen, and pH between frozen and freeze-dried cryoprecipitate after 24 h, and 12 months storage, respectively (P > 0.05).
Summary/Conclusions: The freeze-dried cryoprecipitate stored at 4°C after 12 months remain normal factor VIII and fibrinogen contents.

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INFLUENCE OF PLASMA-DERIVED α-2-MACROGLOBULIN ON RED BLOOD CELLS STORAGE

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Background: Red blood cells (RBCs) transfusion is a potentially lifesaving intervention for millions of chronically or massively transfused recipients worldwide every year. Normally RBCs can be stored up to 42 days on the condition of current administration based on MAP (Mannitol adenine sodium dihydrogen phosphate) storage solution. However, the decrease of Adenosine triphosphate (ATP), accumulation of reactive oxygen species (ROS) and cytokines may bring adverse effects on the RBCs, and additionally influence RBCs' therapeutic efficacy. α2-Macroglobulin (α2-M) is known as a broad spectrum inhibitor of protease, which participates in physiological and pathological processes widely. α2-M also has the potential ability of scavenging free radical and regulating cytokines.
Aims: The aim of this study is to investigate the potential protection influence of α2-M on RBCs during storage period.
Methods: Different concentrations of α2-M were added to RBCs separately, and the final concentrations were 3 g/l (physiological concentration), 9 g/l (3 times physiological concentration), 15 g/l (5 times physiological concentration) respectively. Treated samples were collected at day 7, 14, 30 of storage. Routine blood test, blood gas analysis, crystal osmotic pressure, and free hemoglobin were investigated.
Results: The hemoglobin concentration of samples before treatment is 17.6 g/dl. The hemoglobin concentrations of different α2-M groups ranged from 13.0 to 16.8 g/dl at different time points, and no significant differences were observed. The hematocrit of samples before treatment is 0.54. The hematocrit of different α2-M groups ranged from 0.5 to 0.52 at different time points, and no significant differences were observed. Both hemoglobin and hematocrit met the standard of Chinese blood bank technical operation procedure. However, the hemolytic ratio of medium α2-M concentration group (9 g/l) was >0.8 at day 14, 30 of storage. The hemolytic ratio of high α2-M concentration group (15 g/l) was >0.8 at day 7, 14, 30 of storage. The hemolytic ratio of normal α2-M concentration group (3 g/l) was lower than 0.8 at different time points, which met relative quality standard.
Summary/Conclusions: Therefore, α2-M did not bring better protection effect on RBCs storage on the current experimental condition. The concentration of ATP and the influence of crystal osmotic pressure need to be investigated for further improvement of experimental conditions.

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Abstract has been withdrawn

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COMPARISON OF THE ANTIBODIES AGAINST B-AMYLOID 42 OLIGOMERS CONCENTRATION AMONG IMMUNOGLOBULIN PREPARATIONS FROM THE SAME MANUFACTURER

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Background: Intravenous immunoglobulin (IVIg) has been under study as a potential agent for treatment of Alzheimer's disease (AD).It is reported that some IVIg can significantly decrease the Aβ level in cerebrospinal fluid and can reduced cognitive decline. Antibodies against β-amyloid (Anti-Aβ antibodies) in IVIg were thought to be the important component contributing to the beneficial effect of IVIg for treatment of AD. Now all studies found that IVIg from different manufactures differed in the levels of anti-Aβ antibodies, which may affect their efficacy in treatment of AD. However there are no studies on levels of anti-Aβ antibodies in IVIg from the same manufacturer. We would like to know whether IVIg in different batches from the same manufacturer differ in the levels of anti-Aβ antibodies. In addition, the levels of anti-Aβ antibodies in the specific immunoglobulin have never been taken in account. It is necessary to find out whether there are differences in the levels of anti-Aβ antibodies between IVIg and specific immunoglobulin.
Aims: We detected and compared the levels of anti-Aβ antibodies in IVIg and specific immunoglobulin from the same manufacture to illustrate the possible different efficacy made by immunoglobulin from the same manufacture in treatment of AD.
Methods: We measured the levels of anti-Aβ antibodies in IVIg and specific immunoglobulin by enzyme-linked immunosorbent assay (ELISA), using Aβ42 soluble oligomers as the antigens. 30 batches of 5% IVIg preparations were collected from one Chinese manufacturer, 18 batches of these preparations were prepared by ethanol fractionation based on the Cohn methods, while 12 batches of them were prepared by chromatography process. 8 batches of specific immunoglobulin were collected from the same manufacturer, and 5 batches are hepatitis B immunoglobulin, 2 batches are rabies Immunoglobulin, 1 batch is tetanus Immunoglobulin.
Results: The results showed that the different batches of IVIg which were prepared by the same process showed no significant variations in levels of anti-Aβ42 oligomer antibodies, while IVIgs which were prepared by different processes showed significant variations. The levels of anti-Aβ42 oligomer antibodies in IVIg prepared by chromatography process were remarkably higher than that in IVIg prepared by ethanol fractionation process. The average value of anti-Aβ42 oligomer antibodies in these IVIgs are 33.63 ± 4.13 μg/mL and 6.57 ± 1.78 μg/mL respectively. We need to compare the levels of anti-Aβ antibodies between IVIg prepared by ethanol fractionation process and specific immunoglobulin at the same concentration. When the specific immunoglobulin (8 g/l) is at concentration of IVIg (50 g/ml), the average value of anti-Aβ42 oligomer antibodies in hepatitis B immunoglobulin, Rabies immunoglobulin and tetanus immunoglobulin were 9.27 ± 2.21 μg/mL, 38.23 ± 1.44 μg/mL, 32.48 μg/mL respectively. The levels of anti-Aβ42 oligomer antibodies in Rabies immunoglobulin and tetanus immunoglobulin were remarkably higher than that in IVIg prepared by ethanol fractionation process and the levels of anti-Aβ42 oligomer antibodies in hepatitis B immunoglobulin was slightly higher than that in these products.
Summary/Conclusions: IVIg in different production processes from the same manufacture differed in the levels of anti-Aβ antibodies, and there were significant differences between specific immunoglobulins and IVIg which were similarly prepared by ethanol fractionation process. It is concluded that IVIg even from the same manufacture may lead differentiated efficacy in treatment of AD.

Pathogen Inactivation

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QUALITY OF RED BLOOD CELLS AFTER PHOTOCHEMICAL TREATMENT OF RIBOFLAVIN WITH VISIBLE LIGHT

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Background: Pathogen inactivation (PI) of blood cells (RBCs) could reduce transfusion risks associated with RBCs, such as transfusion-associated graft-versus-host disease (TA-GVHD). Our previous study demonstrated riboflavin with visible light (RB+L) treatment can effectively inhibit lymphocyte proliferation and cytokine secretion. However, the function of red blood cells after treatment remains to be evaluated.
Aims: The aim of the present study was to investigate the quality of RBCs after treatment of RB+L and the feasibility of its utilization on RBC pathogen inactivation.
Methods: Whole blood was collected from randomly selected healthy donors and divided into two aliquots. One was stored at 4°C without treatment, while the other was added by riboflavin with a final concentration of 100 μM and exposed to fluorescent light (centered at 420 nm) from both sides by a total energy of 40 J/ml. After treatment, RBC index such as MCH, MCV and RDW, and metabolic levels of glucose, lactate, and extracellular potassium of treated and untreated samples were assayed at once or per 7 days during storage till Day 35.
Results: The results showed that there were no significant differences in MCH between RB+L treated samples and untreated ones except day 7 and 14. MCV and RDW of both treated and untreated RBCs (n = 9) increased during storage, however, after 14 days, treated RBCs had obvious lower levels of the two indexes compared to untreated ones, which is theoretically better for RBC preservation (MCV: untreated 96.0 ± 4.1 fL vs RB+L 94.7 ± 3.8 fL at day 35, $P < 0.01$; RDW: untreated 15.5 ± 1.2% vs RB+L 15.1 ± 0.8% at day 35, $P < 0.05$). Since one of RBC storage lesions is the RBC shape turning from disc to sphere, resulting in the increase of cell volume and RDW. RB+L treatment might ease up this damage process indicated by our results, and the mechanism of which needs further study. No statistical difference was observed in glucose and lactate contents for treated and untreated RBCs,

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while extracellular potassium of RB+L RBCs was higher than untreated ones (20.64 ± 2.39 vs 15.72 ± 0.36 mM), suggesting changes in Na-K ATPase function.
Summary/Conclusions: These results demonstrated that RB+L treatment did not cause severe impact on the structure and metabolic functions of red blood cells. It might be a promising method for prevention of adverse immune responses caused by WBCs.

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Abstract has been withdrawn

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A MITOCHONDRIAL DNA-BASED QUALITY CONTROL ASSAY FOR VALIDATION OF THE PATHOGEN REDUCTION EFFICACY BY METHYLENE BLUE TREATMENT

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Background: Blood transfusion can result in transfusion transmitted infectious diseases (TTIDs). Pathogen reduction technology is an important tool to further reduce the risk of TTIDs. Among all kinds of methods, methylene blue treatment has been used in plasma for a long time. Although its efficacy has been well validated, with the development of regulation, it is necessary to introduce a quality control assay into the process of pathogen reduction. We intend to use the mitochondrial DNA breakage as a quality control marker to investigate its potential as a novel validation assay for the efficacy of MB treatment.
Aims: To develop a new quality control method for monitoring the efficacy of pathogen reduction by methylene blue treatment.
Methods: Mitochondrial DNA (mtDNA) was extracted from plasma samples before and after methylene blue treatment. Inhibition of PCR amplification of mitochondrial DNA in short and long amplicon target regions, ranging from under 200 base pairs (bp) to over 1800 bp, was measured in treated relative to untreated components.
Results: Inhibition of PCR amplification of long-amplicon mtDNA targets was observed after methylene blue treatment. With the increase of treatment duration, more inhibition was observed. The difference of long-amplicon between untreated and standard-procedure-treated samples was about 1 log. However, the short-amplicon was not affected following MB treatment. Our data from 40 donor samples suggest the difference in crossing point between the long and short fragment might predict the efficacy of MB treatment.
Summary/Conclusions: Mitochondrial DNA-based real-time PCR assay is a promising quality control method that can be used in validating the efficacy of pathogen reduction by MB treatment.

Novel Blood Products

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PROSTAGLANDIN E2 PROMOTES THE DIFFERENTIATION OF EARLY MESODERM AND HEMATOPOIETIC PROGENITOR FROM HUMAN EMBRYONIC STEM CELLS

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Background: Differentiation of human embryonic stem cells (hESCs) into the hematopoietic progenitor cells offers a novel opportunity to obtain pure and therapeutic cell populations. However, the molecular mechanism of mesoderm and hematopoietic cell differentiation remains largely unclear.
Aims: To study the biological functional of endogenous Prostaglandin E2 (PGE2) in early mesoderm and hematopoietic differentiation of hESCs.
Methods: The induced cells from hESCs in the presence of PGE2 were identified by RT-PCR, flow cytometry and colony-forming assays.
Results: Our results demonstrated the PGE2 exposure could alter the gene expression pattern and morphology of hESCs and result in a robust hematopoietic differentiation of them with higher frequencies of CD34+ and CD45+ cells. Using a stepwise induction method, we find that PGE2 signaling plays a critical role in the initiation of the mesoderm differentiation of hESCs. Inhibition of PGE2 synthesis by indomethacin remarkably blocked the BMP4-induced mesoderm differentiation of hESCs. Subsequently, Hemogenic endothelium and hematopoietic progenitor cell differentiation were blocked by indomethacin addition. COX-1 knockdown or selective inhibition of COX-1 activity confirmed that COX-1, one of the key enzymes responsible for PGE2 synthesis, is required for mesoderm specification of hESCs, which is the direct target gene of BMP4 signaling in hESCs. Subsequently, PGE2 stimulated the mesoderm differentiation of hESCs via the EP2-PKA-GSK3β/β-catenin signaling pathway.
Summary/Conclusions: Cyclooxygenase-1-derived PGE2 licenses human embryonic stem cell differentiation into mesoderm and hematopoietic progenitor cells. Our data provide novel insights into the function of endogenous PGE2 signaling in the early lineage specification of hESCs.

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THE RESEARCH ON PLURONIC F68 FOR MEGAKARYOCYTE PROGENITOR CELL DIFFERENTIATION

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Background: Hematopoietic stem/progenitor cells show the potential of self-renewal and differentiation ability, and they can differentiate into megakaryocytes and platelet for clinical cell therapy. However, the differentiation efficiency is low. We attempted to establish an efficient differentiation scheme.
Aims: To observe the effect of Pluronic F68 on megakaryocyte differentiation from hematopoietic stem/progenitor cells in order to obtain more megakaryocyte progenitor cells.
Methods: The cord blood mononuclear cells were isolated and inoculated in cell culture bag or cell culture flask respectively. The WIGGENS shaker and cell culture bags were used to mimic WAVE Bioreactor for three-dimensional (3D) cell culture. At the same time, the cells were cultured in cell Flask by conventional two-dimensional (2D) cell culture as control. The induced cells were detected for morphology, surface marker expression, survival, and cell number on day 14.
Results: In the three-dimensional cell culture, Pluronic F68 was added, and the stained cells were observed by microscope, the nuclear-cytoplasmic ratio was increased, the cell volume became larger and the nuclear shape was irregular, the cytoplasm appeared magenta granules, and the megakaryocyte progenitor cells became more mature. By three-dimensional culture, the expression of CD41/CD61 was (36.30 ± 1.27)% Vs (23.95 ± 1.34)%, hence the differentiation for megakaryocytic progenitor cells was significantly higher than that in the two-dimensional control group ($P < 0.01$). Furthermore, adding Pluronic F68 in three-dimensional culture resulted in highest differentiation efficiency for megakaryocytic progenitor cells ($P < 0.01$). The cell viability and cell number were no significantly difference ($P > 0.05$) between three-dimensional culture group containing Pluronic F68 and control group.
Summary/Conclusions: Three-dimensional culture is beneficial for the differentiation of megakaryocyte progenitor cells, but the cell viability was lower than Two-dimensional control group, but adding of Pluronic F68, the satisfied cell growth and better induction efficiency were obtained. The research result would provide a new method in megakaryocytes production for clinical application.

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PREPARATION, CHARACTERIZATION AND FUNCTIONAL INVESTIGATION OF DEXTRAN-BOVINE HEMOGLOBIN CONJUGATE AS A BLOOD SUBSTITUTE

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Background: Development of blood substitute is a persistent and urgent need in biomedicine. Dextran can be easily derivatized with various chemical moieties, thus can conjugate with hemoglobin (Hb) to increase the size of Hb and overcome the disadvantages of Hb, such as short half-life, renal toxicity and vasoactivity, to produce a promising blood substitute.
Aims: Herein, the dextran-bovine Hb (bHb) conjugate (dex-bHb) was prepared by periodate-oxidized dextran conjugation with the Cys-93(β) protection, and the characterization and function of dex-bHb were investigated.
Methods: The correct ratio of Hb to dextran40 to $NaCNBH_3$ was determined via the SDS-PAGE experiment. The infrared spectroscopy and Raman spectral analysis were carried out to demonstrate the structure. Then the blood compatibility of dex-bHb was tested in vitro and the bio-availability was evaluated in a hamster model of hemorrhagic shock.
Results: The purity and productivity of the product were highest when the ratio of Hb to dextran40 to $NaCNBH_3$ was 1:1:0.5. The dex-bHb preserves the structure features of Hb according to infrared spectroscopy and Raman spectral analysis. In vitro, the dex-bHb showed no obvious effects on red blood cell aggregation and hemolysis rate. After intravenous administration of dex-bHb to hamsters with hemorrhagic shock, the mean arterial pressure and microcirculatory blood flow showed significant recovery. And the heart, liver, spleen, lung and intestine damages of the shock hamsters were alleviated via histopathologic study.
Summary/Conclusions: Thus, dex-bHb was expected to act as a potent blood substitute with microcirculation expansion and organ protection ability.

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THE INTERNALIZATION MECHANISM OF A SERIES OF NOVEL CELL PENETRATION PEPTIDES WITH ENHANCED PROTEOLYTIC STABILITY IN SERUM

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Background: Human Blood is a complex system containing various enzymes and immune active substances, which have been proved to be great barriers for drug delivery. We have developed a series of novel cell penetrating peptides (CPPs) consisting of various numbers of d- and l-arginine residues. These peptides have been proved to be ideal good drug carriers due to high uptake efficiency and enhanced proteolytic stability in serum. Unfortunately, the internalization mechanism of these CPPs is still unclear.
Aims: In this study, we applied FRAP (Fluorescence recovery after photobleaching) to analyze the internalization mechanism of these CPPs.
Methods: All the CPPs were labeled with FITC. The recovering speed of FITC after photobleaching presented the motility of CPPs within the cell membranes.
Results: We demonstrated that the uptake efficiency and proteolytic stability of these CPPs were all significantly affected by the number of d-arginine residues in the peptide sequence. CPPs with more numbers of d-arginine residues showed with enhanced uptake efficiency and proteolytic stability. And the FRAP experiments showed that CPPs with more numbers of d-arginine moved faster within the cell membrane.
Summary/Conclusions: Based on the FRAP experiments, it can be explained that the higher uptake efficiency of CPPs with more numbers of d-arginine maybe caused by their faster movement speed within cell membranes. The results here might provide useful guidelines for the design and application of CPPs in drug delivery.

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INHIBITION OF INFLAMMATORY CYTOKINES BY IVIG IS DEPENDENT OF IGG-FC SIALYLATION IN HUMAN MONOCYTIC THP-1 CELLS

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Background: Intravenous immunoglobulin (IVIg) is used to treat patients with autoimmune and systemic inflammatory diseases, but the exact working mechanism is not known. The prevailing studies, which states that Fc-sialylated IgG plays an important anti-inflammatory role, is currently debated.
Aims: The study aimed to prepare the Fc-sialylated IgG (SA+Fc) fraction from Chinese IVIG preparations, and determine the efficacy of SA+Fc fraction to inflammatory cytokines in Human Monocytic THP-1 Cells.
Methods: IgG molecules were split into F(ab,)2 and Fc fragments by pepsin digestion. Then the Fc fragments were purified using Protein G affinity chromatography column and molecular sieve column. After that, the sialylated IgG (SA +IgG) and Fc-sialylated IgG (SA +Fc) were enriched by Sambucus nigra agglutinin (SNA) lectin fractionation. The efficacy of IVIg to inflammatory cytokines was measured by ELISA after adding SA +IgG or SA +Fc in PMA-treated and LPS-stimulated monocytic THP-1 cells.
Results: The fractions of SA +IgG and SA +Fc had high purity antibodies and intact molecular structure. Compared with the normal control group, TNF-α and IL-6 levels in THP-1 cells were significantly increased by LPS stimulating at 24 h ($P < 0.05$). Compared with the untreated IVIg group and Fc group, the levels of TNF-α and IL-6 were significantly decreased after adding the fractions of SA +Fc at 0.5 mg/ml ($P < 0.05$).
Summary/Conclusions: IgG-Fc sialylation has strong inhibition of inflammatory cytokines in Human Monocytic THP-1 Cells. Our study could provide a new idea for the study of the new natural anti-inflammatory drug.

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PHARMACOKINETIC STUDY OF A RECOMBINANT FACTOR FVIII- FC FUSION PROTEIN (GEN-9208) IN HAEMOPHILIA A MICE

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Background: Prophylaxis for haemophilia A with conventional factor VIII products requires frequent intravenous dosing. Recombinant factor VIII-Fc fusion protein, rFVIIIFc, has a prolonged half-life compared with conventional rFVIII, which would also allow for less frequent injections and thus has better clinical applicability.
Aims: To evaluate the pharmacokinetics properties of a new recombinant factor VIII-Fc fusion protein (GEN-9208), and compare it with a conventional factor VIII products (Control), in haemophilia A mice.
Methods: The pharmacokinetics were evaluated after single i.v. administration of 280 IU/kg of GEN-9208 and Control in haemophilia A mice model respectively. The plasma FVIII:C activity were determined by a validated chromogenic assay method. The pharmacokinetic parameters were calculated using the Phoenix WinNonlin 6.0 software with non-compartmental models. And the statistics of GEN-9208 and Control were compared by *t*-test.
Results: The half-life ($t_{1/2}$) were 18.95 ± 3.90 and 6.82 ± 1.82 h ($P < 0.0001$), the area under the concentration-time curve (AUC_{last}) were 25.87 ± 9.93 and 8.54 ± 3.67 h/IU/ml ($P < 0.01$), the clearance (Cl) were 13.35 ± 12.40 and 37.23 ± 12.40 ml/h/kg ($P < 0.01$), the apparent volume of distribution (Vd) were 402.25 ± 372.41 and 350.79 ± 119.88 ml/kg ($P > 0.05$), and the mean residence time (MRT_{last}) were 21.05 ± 4.22 and 8.41 ± 2.32 h ($P < 0.0001$) for GEN-9208 and Control respectively, after 280 IU/kg i.v. administration in haemophilia A mice respectively.
Summary/Conclusions: The study indicated that compared with conventional rFVIII product, GEN-9208 had obvious a prolonged half-life in haemophilia A mice after 280 IU/kg i.v. administration, which confirmed its long-acting characteristics and deserved to be further investigation.

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Transfusion Transmitted Infections Screening Strategies for TTI

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Abstract has been withdrawn

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DATA ANALYSIS OF CENTRALIZED BLOOD SCREENING ACROSS THE REGIONS OF HUBEI PROVINCE IN CENTRAL CHINA DURING 12 YEARS (2005–2016)

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Background: Centralized blood screening, a transfusion service model with high quality, high efficiency and low cost, which can optimize the resources distribution of blood services and is a necessary measure to guarantee the safety of blood transfusion, has been extensively utilized in developed countries but this practice starts relatively late in China. The government of China has issued a series of policies and measures to prompt centralize blood screening since 2006. Blood banks of Hubei province in central China began to implement this model in five blood services gradually since 2005, including a blood center and four blood banks (Wuhan blood center, Ezhou blood bank, Xiantao blood bank, Tianmen blood bank and Qianjiang blood bank).

Aims: To investigate the current situations and the data of centralized blood screening across the regions of Hubei province. Discuss the advantages of centralized blood screening and its appropriateness and feasibility for blood banks in China.

Methods: A retrospective study was performed using data of centralized blood screening of five blood services from 2005 to 2016 in Hubei province. Descriptive statistics, such as percentages and rates were computed. The sample size, total qualified rate of blood and deferral rate of related indexes was counted and the rates were compared between regions.

Results: Compared to 2005, the number of sample size involved in centralized blood screening in 2016 increased by 86.3%, from 124,685 in 2005 to 232,118 in 2016 and the sample size in these blood services increased with different levels. Differences of blood quality were found between them and the total qualified rate of blood of Wuhan blood center is the highest one ($P < 0.01$). Deferral rates of indexes, including HBsAg (hepatitis B surface antigen), anti-HCV (anti-hepatitis C virus antibody), anti-TP (anti-treponema pallidum antibody) and ALT (Alanine aminotransferase) in different areas of Hubei province were found to be different ($P < 0.01$). Deferral rates of ALT in all blood services have a tendency to decline year by year and deferral rates of anti-TP in all blood services have a tendency to increase after 2011.

Summary/Conclusions: The centralized transfusion model implemented in Hubei province in central China has withstood the test of time and continues to evolve to meet new situations and ensure blood safety. The blood quality of services involved in the centralized blood screening has variance, the health consultation and general examination before blood donation should be strengthened in areas with high prevalence of related indexes. The declination of deferral rates of ALT was attributable to the primary screening of ALT before blood donation. The growth of deferral rates of anti-TP was attributable to the usage of ELISA (enzyme linked immunosorbent assay) after 2011. The laboratory implemented centralized blood screening has the abilities to continue to introduce advanced technology, such as nucleic acid test (NAT) and the screening results can promote the blood services to improve their work processes. It was recommended to extend centralized blood screening in appropriate areas in China to guide the direction and focus of the future work and improve the ability to guarantee blood safety.

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PERFORMANCE OF HEPATITIS B VIRUS (HBV) NUCLEIC ACID ASSAYS IN DALIAN BLOOD DONORS WITH DISCREPANT SEROLOGIC AND MOLECULAR TESTING RESULTS HBV DNA CONFIRMATORY ALGORITHM IS NEEDED

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Background: HBV blood safety remains a major issue in China due to the high prevalence of HBV infection. To reduce the HBV transfusion-transmission risk, a program for HBV, hepatitis C virus and human immunodeficiency virus simultaneous nucleic acid testing (NAT) was initiated in 2010 in several blood banks across China. The rapid technical evolution of NAT assays is challenging for blood banks that have to (re)-evaluate their blood screening strategies according to the benefits but also the limitations associated with the implementation of new assays.

Aims: To compare the performance of multiplex NAT assays to detect HBV DNA in HBsAg positive and negative blood donors from Dalian, China.

Methods: Blood donors were screened pre-donation for HBsAg using a rapid test. Eligible donors were further tested with two HBsAg ELISA assays and for HBV DNA by using either CobasTaqscreen MPX 2.0 (Roche) in 6-samples minipools, ProCleix Ultrio ID (Grifols), or ProCleix Ultrio Plus ID assays. Samples initially reactive (IR) were discriminated according to manufacturer's standard procedure. NAT IR/HBsAg+ and NAT IR/HBsAg- samples were retested twice and discriminated once using both ProCleixUltrio ID and ProCleixUltrio Plus ID systems.

Results: First, 239/121,315 (0.2%) donations were tested IR with Ultrio; 93 (39%) tested HBsAg+ and were not investigated further. Discriminatory assay performed on 146 NAT IR/HBsAg- samples identified 30 (20.5%) HBV DNA+, 1 (0.7%) HCV RNA+ and 1 (0.7%) HIV RNA+. Second, 118/51,247 (0.23%) samples were initially reactive with Ultrio Plus: 23 (19.5%) HBsAg+ and 95 HBsAg-. Discriminatory testing of HBsAg- samples identified 43/95 (45%) HBV DNA+ and 1/95 (1%) HIV RNA+. Based on direct discriminatory testing without repeat multiplex testing, non-reproducible reactivity rates of 78% (114/146) and 54% (51/95) were observed in HBsAg- samples using Ultrio and Ultrio Plus, respectively. In addition, 26 (22%) and 1 (4%) HBsAg+ samples tested NAT non-reactive with Ultrio and Ultrio Plus, respectively. In parallel, 98,539 samples were tested with MPX. Positive pools resolution identified 95 (0.09%) HBV DNA+ (51 HBsAg+ and 44 HBsAg-), 2 (0.002%) HCV RNA+ and 2 (0.002%) HIV RNA+ samples; 19 (27%) HBsAg+ samples were included in NAT non-reactive pools; and no viral nucleic acid was detected in individual HBsAg- samples present in 46% of the pools initially reactive. When retested twice and discriminated once, 17/85 (20%) and 64/85 (75%) HBsAg+/NAT initially non-reactive samples were found HBV DNA positive with Ultrio and Ultrio Plus, respectively. Ultrio Plus also detected HBV DNA in 14/99 (14%) of HBsAg- samples non-repeated reactive with Ultrio but failed to detect viral DNA in 3/42 (7%) of Ultrio repeated reactive samples. The rate of initially reactive/non repeated reactive HBsAg-samples was 70% (99/141) and 34% (11/32) with Ultrio and Ultrio Plus, respectively. Confirmatory assay based on virus concentration was developed to definitively characterize the HBV status of HBsAg-/NAT non-repeated reactive donors.

Summary/Conclusions: Ultrio Plus improved HBV DNA detection in HBsAg- blood donations compared to Ultrio and MPX assays. The combination of highly sensitive NAT and presence of low levels of HBV DNA may result in a high rate of non-repeatable reactive samples. Discrepant serologic and molecular markers are also observed. It is critical to develop more sensitive and specific molecular confirmatory algorithms to limit blood shortage by allowing safe re-entry of donors with discrepant testing results.

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APPLICATION OF ELECTROCHEMILUMINESCENCE IMMUNOASSAY AS A COMPLEMENTATION TEST TO NAT FOR HBV SEROLOGICAL MARKERS IN CHINESE BLOOD STATION

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Background: Nucleic acid test (NAT) is commonly utilized in detecting transfusion-transmitted infections (TTI) in donated blood samples. We applied minipool NAT (MP-NAT) to blood donations for HBV/HCV/HIV detection collected from Nanjing, China since June 2010. However, we found it inconsistent that samples of some pools of MP-NAT positive were resolution NAT negative and some samples of resolution NAT positive were quantity NAT negative.

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Aims: To evaluate electrochemiluminescence immunoassay (ECLIA) as a complementation test to NAT for HBV serological markers for better TTI detection accuracy.
Methods: All blood donation samples went through ELISA test for HBsAg, anti-HCV, anti-HIV, and anti-treponema pallidum, of which ELISA negative samples were further screened by MP-6-NAT. Samples of being ELISA negative but NAT positive (NAT yield) were confirmed by HBV DNA/HCV RNA/HIV RNA quantity tests, and ECLIA of HBsAg, anti-HBs, HBeAg, anti-HBe, and anti-HBc. Samples of being MP-6-NAT positive but resolution NAT negative also went through ECLIA of HBV serological markers.
Results: A total of 275 samples were confirmed as HBV DNA(+)/ELISA(−) from NAT yield, with a median of HBV DNA <100 IU/ml. Based on five ECLIA HBV serological markers, only 19 (7%) samples were all negative, while the rest were anti-HBs and/or anti-HBc positive. Out of these 275 samples, 119 (43%) were HBV DNA negative according to HBV DNA quantity test. The two groups, HBV DNA negative and HBV DNA positive, however, had no statistical differences in each of seven HBV serological marker combinations (□□2 test). By comparing with 693 samples of being resolution NAT negative, both groups showed statistically significant difference ($P < 0.01$) in anti-HBe/anti-HBc, anti-HBc, anti-HBs, HBsAg/anti-HBe/anti-HBc and all HBV serological markers negative, but not in anti-HBs/anti-HBc or anti-HBs/anti-HBe/anti-HBc. We followed five of 15 donors up to 256 days who showed anti-HBs positive only, and observed that two donors experienced seroconversion and two donors without seroconversion still carried detectable HBV DNA.
Summary/Conclusions: ECLIA of HBV serological markers, when combined with HBV DNA quantity test, may serve as a complementation test to NAT by reducing false negative results and informing donors of infection stages.

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THE QUALITY EVALUATION OF FOUR ELISA BLOOD SCREENING REAGENTS IN OUR LABORATORY

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Background: The nucleic acid test has been carried out in all blood centers in China since 2015. However, the blood samples still are detected by one time nucleic acid test and two times ELISA test in all blood centers, which is such a huge waste. So shall we only detect blood samples with one time ELISA and one time nucleic acid test? It's up to the quality of ELISA blood screening reagents. We must evaluate all ELISA reagents objectively. Our lab joined a research and undertook the evaluation of four reagents in our laboratory in 2015.
Aims: This research is to evaluate ELISA blood screening reagents, including home-made reagents and imported reagents. On the one hand we can get the quality evaluation of the reagents. On the other hand, we can evaluate the level of our laboratory and our reagents which were used in the research.
Methods: The trial samples which were collected by 19 blood centers in China were single reagent positive or double reagents positive. There were 1,473 HBV samples, 1,229 HCV samples, 1,036 HIV samples and 1,372 TP samples in total. Every sample was detected with the same reagent by three laboratories. When three laboratories gave the same result, we thought that it's right. If they had different results, we detected them with the confirmatory tests.
Results: 31 reagents were involved in this study, including 24 homemade reagents and 7 imported reagents. The highest sensibility of HBsAg reagent was DiaSorin reagent, which was 96.2%. the highest specificity of HBsAg reagent was Livzon reagent, which was 99.2%. The highest sensibility of Anti-HCV reagent was DiaSorin reagent, which was 98.9%. The highest specificity of Anti-HCV reagent was WaiTai reagent, which was 98.4%. The highest sensibility of Anti-HIV reagent was Bio-Rad reagent, which was 100%. the highest specificity of Anti-HIV reagent was XinChuang reagent, which was 98.7%. The highest sensibility of Anti-TP reagent was WaiTai reagent, which was 99.2%. The highest specificity of Anti-TP reagent was GBI reagent, which was 98.7%. According to the sensibility, specificity and the amount of the reagents, DiaSorin HBsAg reagent, Wan Tai Anti-HCV reagent, Xin Chuang Anti-HIV reagent, and Xin Chuang Anti-TP reagent made a good performance in this research. There were four reagents in our laboratory which were evaluated in the research. Such as: Livzon HBsAg reagent with 84.59% sensibility and 99.21% specificity, Kehua Anti-HCV reagent with 96.53% sensibility and 98.12% specificity, Waitai Anti- HCV reagent with 97.72% sensibility and 98.44% Waitai Anti-HIV reagent with 98.53% sensibility and 96.43% specificity. Obviously, the sensibility of Lizon HBsAg reagent was lower than others. However, the other three reagents were much better than others in this study.
Summary/Conclusions: The results showed the sensibility and specificity of reagents clearly, which can supply suggestions to us. Especially, when we detect blood samples with only one time ELISA and one time NAT, we must make sure to get the accurate results. What's more, our lab will make corresponding adjustments on the basis of the evaluation results. And then, we should evaluate our reagents regularly and make adjustments about our reagents in time. At last, we must underline that there were many uncontrollable factors in this research, such as: the quality of blood samples, the different levels of operators, the different equipments in the laboratories. All of them affected our results directly. Therefore, more objective experimental data is needed urgently in the future.

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ANALYSIS OF INVALIDATED RESULTS BY THE NUCLEIC ACID TEST ON BLOOD SCREENING

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Background: In October 2015, the National Health and Family Planning Commission of China issued a notice that nucleic acid test (NAT) technology would be nationwidely applied in blood screening at all the blood donation and supply institutions by the end of 2015. According to the Blood Donation and Supply Institutions Operation Procedures (2015 edition), it was explicitly stipulated that on the basis of ELISA method, all blood products must be rescreened by NAT. Moreover, based on the monitoring of the testing performance termed 4.8, the real-time monitoring was required to detect any possible changes in handlings and operations, so as to prevent the failure of the batched experiment or the missing detection of weak positive samples.
Aims: To analyze the cause of the invalid results for Cobas's 201 system and formulate corrective actions to reduce the possibility for testing failure and missing detection and ensure the safety of blood transfusion.
Methods: After simultaneously tested by two different ELISA reagents for HBsAg, anti-HCV and anti-HIV1/2, blood samples (except double ELISA positive samples) were also analyzed with NAT. The invalid results were identified and calculated in line with Roche Cobas's 201 kits from January 2013 to December 2016. According to detection time, detection systems, regents Lots, the amplification graphs and test alarm information, the cause of invalid results were further analyzed to formulate a general standard for future work.
Results: Among 52 415 pools (306 208 samples) tested by Roche Cobas's201, 1 372 NAT invalid were found (the total invalid rate: 2.62%). From 2013 to 2016, the NAT invalid rates were 2.01%, 2.03%, 3.92% and 2.74%, respectively ($c2 = 93.047$, $P < 0.05$). Among the cause for invalidation, the invalid rate for reagent quality control non-reaction was highest (38.78%). The NAT invalid rate of MPX2.0 reagent (3.35%) was higher than MPX reagent (1.85%, $c2 = 115.730$, $P < 0.05$). The NAT invalid rates of different MPX2.0 reagent Lots had statistical differences ($c2 = 698.217$, $P < 0.05$), and the reagent Lot 1 was the highest invalid rate (10.44%). The machine operation failure rate was the second type (34.55%), the CAP reagent needle detection error and liquid path error was higher (38.82%). There was statistical difference between the invalidation rates of A and B detection systems ($c2 = 6.279$, $P < 0.05$), and the detection system B was much higher (2.84%).
Summary/Conclusions: Through monitoring the invalid rates for NAT results, it can be timely to identify potential problems in the testing processes of laboratory, formulate procedures for monitoring the running test and verifying the testing reagents and external quality control products, ensure the effectiveness of the blood-donors' results and comparability of the testing reagents Lots, reduce the cost of NAT reagents and evaluate the quality management system operation in our NAT laboratory.

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COMPARISON OF ENZYME-LINKED IMMUNOSORBENT ASSAY AND CHEMILUMINESCENT IMMUNOASSAY FOR HUMAN IMMUNODEFICIENCY VIRUS SCREENINGS IN BLOOD DONORS

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Background: In Chinese Mainland, each screening lab of blood bank was using double enzyme-linked immunosorbent assay (ELISA) screening followed by nucleic acid test (NAT) to detect indicators of human immunodeficiency virus (HIV). In 2016, a new version of national regulations on blood screening tests in blood donors was released in China, in which, beside of classic ELISA, chemiluminescent immunoassay (CLIA) was also introduced as an optional method in blood-donor serological screening.
Aims: To compare the performance of a CLIA system with classic ELISA methods in HIV serological screenings among blood donors.
Methods: From Apr 1 to Jun 30 2017, 23,105 blood-donor samples were detected by two local ELISAs (WANTAI Biopharm and InTec) for antigen/antibody (Ag/Ab) of HIV and then NAT for HIV RNA (Roche cobas201). Nineteen ELISA positive samples and 70 ELISA negative samples were analyzed by a CLIA system (Roche e411) and the former were also sent to CDC for further confirmations. One donor of ELISA negative/NAT reactive was traced and samples were collected every week until fourth week after his donation.
Results: Nineteen ELISA positives and one ELISA negative/NAT positive were detected. All the seventy ELISA negative samples showed negative CLIA results. Among 19 ELISA positives, three samples indicated NAT reactive/CLIA positive results were confirmed as true HIV-positive by CDC. Seventeen of 19 ELISA positives showed non-reactive NAT results were confirmed as false positives, four of which showed CLIA positive results. For the only ELISA negative/NAT positive sample, CLIA test suggested a positive HIV Ag/Ab result (COI: 1.01). And by the fourth week after his donation, HIV Abs were detected from his blood sample by ELISA and then confirmed true positive by CDC.
Summary/Conclusions: We observed that chemiluminescent immunoassay showed good sensitivity and specificity in HIV Ag/Ab screenings among blood donors, especially for donors in window phase of HIV infections. It could be a practicable strategy to replace classic ELISA method with CLIA as serological assay in HIV screening tests.

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PREVALENCE OF SEROLOGIC MARKERS AMONG BLOOD DONORS WHO USE CONFIDENTIAL UNIT EXCLUSION (CUE) IN KURDISTAN PROVINCE, IRAN

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Background: The Confidential unit exclusion (CUE) option has been used to increase blood safety at blood transfusion.
Aims: This study aimed to compare the results of serologic markers between blood donors CUE-positive ("should not use") and negative ("can be used").
Methods: This cross-sectional study was done at the Kurdistan Blood Transfusion Center, Iran, between 2011 and 2014. Serologic tests were performed using commercial products to detect surface antigens of the hepatitis B virus (HBsAg), antibodies against the hepatitis C virus (anti-HCV) and antibodies against the human immunodeficiency virus (anti-HIV). The seropositive results were confirmed using the confirmatory assays.
Results: Of the qualified donors, 98847 donors (91180 male and 7667 female) during 2011 and 2014 gave blood; 30088 (30.4%) donations were from first-time and 68759 (69.6%) donations were from repeat donors. The CUE option was chosen by 918 (26 female and 892 male) donors. Out of this number, 535 (58.2%) were first time donors and 383 (41. 8%) repeat donors. The prevalence of confirmed HBs Ag was 0.5% (5/918) and 0.2% (207/97929) among CUE-positive and negative donations, respectively ($P = 0.03$). The prevalence of confirmed anti-HCV was 0.76% (7/918) and 0.04% (47/97929) among CUE-positive and negative, respectively ($P < 0.001$). The prevalence of confirmed anti-HIV was 0.1% (1/918) and 0.008% (8/97929) among CUE-positive and negative donations, respectively ($P < 0.01$).
Summary/Conclusions: Because of the higher prevalence of serologic marker positivity in donors who chose the CUE option, offering CUE to blood donors could be a potentially useful method for improving blood safety.

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APPLICATE RISK MANAGEMENT TO ESTABLISH THE QUALITY MANAGEMENT SYSTEM OF DETECTION PROCESS IN BLOOD BANK SCREENING LABORATORY

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Background: The purposes of blood bank screening laboratory is ensure blood safety, and prevent transfusion infection by blood transfusion virus. To eliminate the hidden dangers, and prevent quality accidents is the most basic requirement of the quality management. Therefore, risk management should be applied to establish and manage the quality management system of the screening laboratory.
Aims: Applicate the risk management, to identify and evaluate each critical control point of detection process. According to the results of identification and evaluation, establish management measures, and ensure that the detection process management system can be fully effective and feasible operation. Explore the application of risk management in blood bank screening Laboratory quality management.
Methods: According to the characteristics of blood screening laboratory quality management, establish the risk rating table, and use the table to carry out risk evaluation and risk response of the detection process. According to the results of risk evaluation and response, establish the process quality management system, and develop the Standard operating procedures of detection process.
Results: According to the impact of the blood safety, the risk levels is divided into four levels at the risk rating table: mild, moderate, severe and critical. Each level develop the corresponding response principle. Among the 21 critical control points of the detection process, 2 were moderate, 8 were severe, and 10 were critical. According to the different risk levels of critical control point, develop targeted response measures, and draw up the 7 relevant standard operating procedures.
Summary/Conclusions: Explained the process uncertainty clearly by risk management, combined with the characteristics of blood screening laboratory, the dynamic and continuous management of the testing process can ensure the effective and feasible of quality management measures. Risk evaluation and risk response of the critical control point can ensure the systemic and adequacy of the quality management system, and ensure that the system is dynamically and continuously improved.

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A CHEMILUMINESCENCE BASED APPROACH TO NUCLEIC ACID TESTING TO DETECT HBV, HCV AND HIV-1 IN BLOOD SCREENING

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Background: Hepatitis B virus (HBV), Hepatitis C virus (HCV), and human immunodeficiency virus (HIV) are life threatening blood borne infections that can be unintentionally transmitted in transfused blood products.
Aims: To establish a highly sensitive, specific and fast method which is different from the existing methods to detect HBV, HCV and HIV-1 in parallel utilizing a NAT technique that incorporates PCR-chemiluminescence.
Methods: Viral genomes were extracted from donated serum samples using magnetic nanoparticles. The isolated viral DNA and RNA were amplified in a one-step

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parallel RT-PCR reaction. HBV, HCV and HIV-1 were detected using complementary nucleic acid probes and quantified using a chemiluminescent substrate.
Results: The length of the amplified PCR products for HBV, HCV and HIV-1 were 119 bp, 220 bp, and 174 bp, respectively, indicating that the extraction methods had successfully isolated high quality viral genomes from primary serum samples. The probes and reaction conditions used for chemiluminescent detection of HBV, HCV and HIV-1 genomes in unknown primary samples were empirically optimized. Of the 10,422 blood samples screened, 12 were HBV positive and 2 were HCV positive. These results were consistent with those of parallel-controlled study using Roche Cobas TaqScreen MPX Test kits.
Summary/Conclusions: The assay described is fast, accurate and sensitive for detecting HBV, HCV and HIV-1 in parallel that has broad implications for blood screening and epidemiological studies.

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CONFIRMATION OF THE OPTIMAL SIGNAL-TO-CUTOFF RATIOS OF ELISA KIT FOR HIV AND TREPONEMA PALLIDUM ANTIBODY

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Background: Most Chinese blood banks adopt double ELISA for HBsAg, anti-HCV, anti-HCV, anti- Treponema pallidum and borderline to avoid false negative. Any sample of signal-to-cutoff ratio above borderline of either ELISA test is defined as reactive. Although it assures blood safety to some extent, as a result of false positive, a lot of blood was wasted.
Aims: To discuss the necessity of setting borderline of ELISA kit for HIV and Treponema pallidum antibody.
Methods: China national center for clinical laboratories performed a program to assess the sensitivity and specificity of ELISA kit in domestic market. During Jun 2015 to Dec 2015, 17 blood banks and their reactive samples including double-reactive and single-reactive ones were included. A total of 1036 anti-HIV reactive samples were confirmed by nucleic acid test and western blot (WB), while 1378 anti-Treponema pallidum (TP) reactive ones were confirmed by Treponema pallidum particle agglutination assay (TPPA) and western blot (WB). Each sample being reactive with either method was confirmed as positive. Receiver operator characteristics (ROC) curves were performed to determine the optimal S/Co ratios.
Results: Among seven manufacturers' ELISA kits for anti-HIV, the Youden index reached maximum when the S/Co ratios of reagent coded 23003 (Youden Index = 0.98) got to 1,and reagent 23081 (Youden Index = 0.97), 23043 (Youden Index = 0.96), 23001 (Youden Index = 0.98), 23107 (Youden Index = 0.96), 23089 (Youden Index = 0.98) and 23085 (Youden Index = 0.97) to 5. While among five manufacturers' ELISA kits for anti-TP, the Youden index reached maximum when the S/Co ratios of reagent 23003 (Youden Index = 0.87) and 23016 (Youden Index = 0.90) got to 1, reagent 23081 (Youden Index = 0.88) and 23043 (Youden Index = 0.90) to 3, and reagent 23001 (Youden Index = 0.90) to 5.
Summary/Conclusions: Since the optimal S/Co ratios of 7 ELISA kit for anti-HIV and 5 for anti-TP tested are all above 1, there is no need to set borderline.

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INVESTIGATION ON HEPATITIS G INFECTION IN VOLUNTARY BLOOD DONORS IN BEIJING AREA

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Background: Recently, in our hospital some patients after hematopoietic stem cell transplantation were found infected with hepatitis G virus (Hepatitis G, virus, HGV) and caused liver damage. we are not sure the route of infection with HGV. Therefore, we investigated the infection status in different group blood donors in the Beijing area and Discuss the dangers of HGV carriers of blood safety.
Aims: To analyze the infection status of hepatitis G virus (HGV) in different blood donors in Beijing and its influence on blood safety, and to provide the basis for blood screening strategy adjustment.
Methods: HGV RNA were detected by reverse transcription polymerase chain reaction (RT-PCR) in 693 patients with different blood donors. The HGV infection incidence in different blood donors was compared by x^2 test.
Results: Testing of 694 blood donors, 15 were HGV RNA positive. The positive rate was 2%. Among them, HGV RNA were not detected in 93 military blood donors; 3 of HGV RNA positive were detected in 164 of street blood donors, the detection rate was 1.83%, 12 of HGV RNA positive were detected in 437 of platelet blood donors, the detection rate was 2.75%; Compared to the 3 different blood donors, the HGV RNA detection rate was $x^2 = 2.846$, $P = 0.2410$.
Summary/Conclusions: There were no statistically significant difference in HGV infection incidence between the military blood donors, the street blood donors and the platelet blood donors. HGV RNA screening should be performed for some high-risk recipients, such as pediatric recipients and patients after hematopoietic stem cell transplantation.

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EVALUATION OF A MULTIMARKER QUALITY CONTROL SAMPLE FOR MONITORING THE PERFORMANCE OF MULTIPLEX BLOOD SCREENING NUCELIC ACID TESTS

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Background: NRL, Australia in collaboration with Exact Diagnostics (Fort Worth, Texas), developed a new quality control (QC) sample called QConnect TriScreen for use in NRL's QConnect service. QConnect TriScreen is a multimarker QC sample containing 50 IU/mL HCV RNA and HBV DNA, and 250 IU/mL of HIV RNA. The viral load of the QC was determined by digital PCR against the respective WHO International Standards. An international study was conducted to evaluate the suitability of this QC for monitoring multiplex blood screening nucleic acid tests (NATs).
Aims: To assess the performance of the QConnect TriScreen QC sample for monitoring the performance of multiplex blood screening NATs.
Methods: Laboratories using either the Roche cobas® MPX (cMPX) for use on the cobas® 6800 (c6800) or the cobas® 8800 (c8800) Systems; or the Roche cobas® TaqScreen MPX Test, version 2.0 for use on the cobas s 201 system (MPX2) were invited to participate in the trial. Laboratories were provided with single-use vials of QConnect TriScreen Lot 16081703. Participants tested the QC samples each day a test run was performed and entered the cycle threshold (Ct) values for each of HBV DNA, HIV RNA and HCV RNA results into EDCNet for on-line analysis. The coefficient of variation (CV) for the Ct values was calculated for various dataset, including within-laboratory variation.
Results: A total of 16 laboratories participated in the QConnect trial. Of these, five were from India, five from Europe (Belgium, Italy, Greece, Switzerland and the Netherlands), four from Australia and two from Thailand. Nine of the participants used the cMPX (eight using c6800 and one using c8800) and seven used MPX2. Seven laboratories had two instruments and one had three instruments. In addition to the trial, results from six Polish laboratories, each using the MPX2 assay and having tested and entered results for the same lot of QConnect TriScreen into EDCNet were included in the analysis. There was a total of 380, 378 and 385 QConnect TriScreen test results reported for HBV DNA, HIV RNA and HCV RNA testing respectively, using either the c6800 or c8800 Systems. The CV of Ct values reported by each laboratory ranged from 0.95 to 3.40%, 0.62 to 2.99% and 2.36 to 4.40% for HBV DNA, HIV RNA and HCV RNA testing on the cMPX assay. In comparison, there were 445, 438 and 438 QConnect TriScreen results reported for HBV DNA, HIV RNA and HCV RNA testing, respectively, on the MPX2 multiplex assay, with a CV of results ranging from 0.87 to 9.33%, 0.69 to 1.99%, and 2.17 to 4.33% for HBV DNA, HIV RNA and HCV RNA testing on the MPX2 assay. The highest CV of 9.33% reported by a single laboratory testing QConnect TriScreen for HBV DNA on the MPX2 assay was due to a single result of Ct 47.20. Of all results reported, only one non-reactive result was obtained.
Summary/Conclusions: QConnect TriScreen was tested by 22 laboratories in nine countries using either the MPX2 or cMPX assays. The reproducibility of the results reported by these laboratories was excellent, irrespective of the assay or the analyte tested. Marginally greater CVs were reported when testing for HCV RNA compared with the other two analytes. QConnect TriScreen is an appropriate QC sample for the monitoring of the cMPX and MPX2 multiplex blood screening NATs.

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STUDY ON REENTRY EVALUATION MODE FOR BLOOD DONORS USED TO BE HBV REACTIVE

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Background: In China, blood banks implement a screening strategy of dual serology test and then nucleic acid test (NAT). Donors of suspected transfusion transmitted infection are quarantined permanently. To balance blood supply and blood safety, we have developed strict principles for donors reentry in Jiangsu Province, China. Considering high prevalence of HBV in China, we strengthen the reentry principles for donors used to be HBV reactive to identify occult HBV ones.
Aims: To investigate the feasibility of HBV reentry strategy.
Methods: Donors having been quarantined for more than 6 months could propose for reentry application and samples were drawn. All donor samples were routinely screened by dual-ELISA for HBsAg, anti-HCV, anti-HIV and anti- Treponema pallidum and those non-reactive ones were tested by minipool of NAT for three times. To identify occult HBV donors, samples of NAT non-reactive were further tested by ECLIA for HBV seromakers. Donors of only 4 ECLIA patterns were accepted to reentry: all 5 HBV seromarkers negative, anti-HBs only but having history of hepatitis B vaccine injection, anti-HBc only, anti-HBs/anti-HBc with anti-HBs more than 200 IU/l. Additionally, ECLIA detection rate of HBV infection was compared with ELISA and NAT.
Results: During Oct 2016 and June 2017, a total of 166 HBV-reactive donors have applied for reentry, among which 156 persons were quarantined for HBsAg reactive and another 10 were quarantined for HBV DNA reactive. Among the 3 screening methods, the highest HBV yield (40.3%, 67/166) was observed on ECLIA, while only 3.6%(6/166) on ELISA and 5.4%(9/166) on NAT. Increased detection times of NAT from one to three times can help improve detective ratio of OBI donors from 3%(5/166) to 5.4%(9/166). Among 4 ECLIA patterns, anti-HBs reactive only but having history of hepatitis B vaccine injection, amounted the most (41.6%), with a median anti-HBs concentration of 338.5 IU/l.
Summary/Conclusions: Routine screening tests merely based on ELISA and NAT could miss occult HBV donors and may decrease blood safety. Anti-HBs (quantitative)/anti-HBc and history of hepatitis B vaccine injection should be added to evaluate reentry donors used to be HBV reactive.

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THE EFFECTS OF CARRYING OUT NUCLEIC ACID TEST AND SEROLOGICAL TEST SYNCHRONOUSLY

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Background: Since March 1, 2016, the 2015 edition of the technical operation regulations of blood stations has been formally promulgated and implemented, and nucleic acid detection has been carried out in an all-round way.
Aims: By analyzing the effect of nucleic acid detection and serological detection in parallel detection mode, we comprehensively evaluated the role of nucleic acid detection and serological detection in reducing the risk of transfusion-related infection.
Methods: We collected a total of 73559 samples of blood donors from March 1, 2016 to May 31, 2017 and then carried out serological tests (HBsAg, anti- HCV, HIV antigen/antibody, anti-TP, ALT and blood type) and about HBV, HCV, HIV nucleic acid combined qualitative tests. We counted the detection rate of each test, and then compared and analyzed the results of the serological test and NAT.
Results: Of the 73559 samples tested, we found that the number of the result that both serological tests and nucleic acid test were positive was 407 (0.55%).The number of the result that just the nucleic acid test was positive was 489, and its detection rate was 0.67%. After discriminating experiment, 131 samples' result of HBV DNA was positive, not found that HCV RNA and HIV RNA was positive. There were 61 samples of negative nucleic acid test and the test of the two serological reagent tested positive, HBsAg(+)44, anti-HCV(+)17, no HIV antibody/antigen positive sample.
Summary/Conclusions: NAT is effective for detection of HBsAg (-) HBV infection in blood donors. The results of NAT and serological detection are inconsistent and have some complementary effects. After the full implementation of NAT, its role will have to be rigorously validated for the impact of serological reagents and strategy choices.

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TRENDS IN TRANSFUSION TRANSMISSIBLE INFECTIONS (TTIS) SEROPREVALENCE AMONG NEPALESE BLOOD DONORS

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Background: Blood transfusion an important therapeutic option in the life-threatening disease conditions and in sustaining life after severe blood loss. But this life saving procedure may also put people at risk due to infusion of infectious agents from the infected blood of the donors. The most common transfusion-transmissible infections (TTIs) are human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV), *Treponema pallidum* and malaria parasites. The world health organization (WHO) recommends that all donated blood and blood components should be screened for the presence above mentioned TTIs agents before being used for clinical or for manufacturing purposes. TTIs can exist as asymptomatic in their hosts, so donors must be screened for high-risk behavior. Although, in current situation the risk of common TTIs is lower than ever, the supply of safe blood products remains subject to contamination with yet to be identified human pathogens. Only continuous improvement and implementation of donor selection, sensitive screening tests, and effective inactivation procedures can ensure the elimination, or at least reduction, of the risk of acquiring TTIs.
Aims: In Nepal, mandatory screening tests are performed for HIV 1 and 2, HBV, HCV and syphilis, as recommended by the Nepal Red Cross Society (NRCS) and Central Blood Bank Transfusion service (CBTS). However, there is limited information on trends of TTIs in Nepalese blood donors. In an extensive study conducted from 2001 to 2007 in over 500,000 donors, the overall decreasing trend was observed for HBV and HCV seroprevalence, both nationwide and in CBTS. In a different study it was reported that the prevalence of HIV sero positivity among blood donors ranged from 0.019% to 1.26%. In terms of syphilis, the prevalence of sero positivity among Nepalese males was 0.60%. However, these studies lacked data on increasing or decreasing trend for HIV and syphilis in blood donors. Hence, the aim of this study was to examine the trend of sero positivity for TTIs in voluntary blood donors in Nepal that reflects safety of blood products supplied by the transfusion centers.
Methods: A retrospective study for the prevalence of TTIs- HIV, HBV, HCV and Syphilis on 7,744 voluntary blood donors visiting Blood bank of Tribhuvan University Teaching Hospital (TUTH), Kathmandu from 2006 to 2015 was done. Antibodies for TTIs were tested using ELISA and Spot test. The analysis was performed using SPSS.
Results: As reported previously, we also observed a decreasing trend for HBV and HCV sero positivity in Nepalese blood donors. Importantly, we observed similar decreasing trend for HIV (0.26% in 2007 to 0% in 2015) and syphilis (0.3% in 2006 to 0.1% in 2015) sero positivity. The analysis further revealed that in certain years between 2006 to 2015 there were instances when HIV sero positivity among blood donors were null.
Summary/Conclusions: Putting all data together, we conclude that there is a decreasing trend for sero positivity for TTIs among Nepalese blood donors. To our knowledge, this study is the first of its kind that include all four common TTIs as recommended. This observation reflects that people are more aware of such infections that ultimately results in safer blood supply.

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TRENDS IN THE PREVALENCE OF TRANSFUSION TRANSMITTED INFECTIONS AMONG BLOOD DONORS IN A TERTIARY CARE HOSPITAL IN NORTH INDIA: EIGHT YEARS' EXPERIENCE

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Background: The prevalence of transfusion transmitted infections (TTIs) among the blood donors not only represents the prevalence of the disease in the general population but also indicates the quality of donor screening through questionnaire and pre donation counseling.

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Aims: To find the prevalence of TTIs among the blood donors in the last 8 years and to see the prevalence among the 2 sexes.
Methods: The donor samples were screened for HIV (anti-HIV), HBV (HBsAg) and HCV (anti-HCV) tests by the ELISA technique till May 2013 after which it was replaced by the chemiluminescence method. Donors are screened for syphilis by the rapid plasma reagin technique (RPR) whereas for malaria, rapid card tests were used. All the data was entered into excel sheet and analyzed.
Results: The total number of donors screened during the period June 2009 to July 2017 in the Department of Transfusion Medicine of our institute is 40403 of which 97.6% were male and 2.4% were female (39717 replacement; 686 voluntary). The distribution of total reactive samples from 2009 to 2017 is as follows: 4.63% (4 male; 1 female) in 2009, 1.96% (36 male; 1 female) in 2010, 2.68% (76 male; 2 female) in 2011, 2.21% (86 male; 3 female) in 2012, 3.36% (180 male; 5 female) in 2013, 3.13% (190 male; 5 female) in 2014, 3.05% (226 male; 7 female) in 2015, 2.37% (184 male; 7 female) in 2016 and 2.05% (82 male; 2 female) in 2017.
During the 8 year period, HIV accounted for 12.8% (95.7% male; 4.3% female), HCV accounted for 23.4% (96.8% male; 3.2% female), HBV: with the highest prevalence, accounted for 60.9% (97% male; 3% female), syphilis accounted for 3.8% (100% male), malaria: with the lowest prevalence accounted for 0.63% (85.7% male; 14.3% female) of the reactive cases. The reactive cases also included 10 samples of HIV and HBV co-infections, 4 samples of HBV and HCV co-infections, 3 samples of HIV and syphilis co-infections, 1 sample of HBV and syphilis; and 1 sample of HCV and syphilis co-infections. So the overall seroprevalence of HIV is 0.3%, HBV is 1.65%, HCV is 0.64%, syphilis is 0.1% and malaria is 0.02%.
Summary/Conclusions: The analysis of the data reveals that the prevalence of TTIs has decreased from around 4% in 2009 to around 2% in the donors in 2017. However, the seroprevalence of each infection at our center is less as compared to that seen by Memon et al in their blood bank located in the southern part of the country and higher than that of Saini et al. whose blood bank in central India.

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HTLV INFECTION AMONG VOLUNTARY BLOOD DONORS IN ZHEJIANG PROVINCE CHINA

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Background: To prevent the blood transmission of human T-cell lymphotropic virus (HTLV), Zhejiang Province China have introduced anti-HTLV blood screening.
Aims: To assess the prevalence of HTLV infection in Zhejiang Province, China.
Methods: 691,096 blood samples were obtained from voluntary blood donors from 12 blood collection agencies in Zhejiang Province (from March 1, 2016 to March 31, 2017). Initial screening for HTLV-1/2 antibody was performed using the anti-HTLV (1+2) antibody ELISA kit. 238 positive samples identified by ELISA were confirmed by Western Blotting (WB).
Results: 23 (1/30 047) cases were confirmed as positive for HTLV-1. Infected blood donors are located in 5 cities, Jinhua (3 cases, 3/53 302, 0.06‰), Wenzhou (9 cases, 9/102 419, 0.09‰), Taizhou (3 cases, 3/57 569, 0.05‰), Hangzhou (5 cases, 5/156 412, 0.05‰) and Lishui (3 cases, 3/36 623, 0.08‰), the other 7 cities have not yet found positive blood donors. Confirmatory tests were carried out to analyze the bands of 23 positive specimens and found that they belonged to type HTLV-1.
Summary/Conclusions: HTLV infection blood donors in Zhejiang province are in uneven distribution around different city, and residual risk of HTLV infection via blood transfusion should be assessed, which is the basis for different areas' comprehensive screening strategies to further improve the safety of blood transfusion.

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COMPARISON OF TWO NUCLEIC ACID TESTING SYSTEMS AT BLOOD SCREENING LABORATORY

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Background: Nucleic acid technology is a new technology for detecting the nucleic acid of virus. NAT increases the sensitivity of screening, so the risk of infection through transfusion could be reduced. Many countries have already implement NTA in blood donor screening for more than 10 years. After the research and evaluation in some blood centers, National Health and Family Planning Commission said that NAT must be carried out in all blood banks in our country before January 1, 2016. Two kinds of nucleic acid testing systems were used for blood screening after the ELISA screening in our laboratory from January 1, 2016.
Aims: To discuss the role and significance of NAT in blood centers by comparing the application of two different nucleic acid screening systems.
Methods: From January 1 2016 to September 30th 2017, ELISA-negative blood samples were tested by Shanghai Kehua and Novartis TIGRIS nucleic acid screening systems to detect HBV DNA 、HCV RNA and HIV RNA .The reactive samples were determined with the Novartis Ultrio Plus discriminatory tests and Kehua split tests.
Results: A total of 63753 specimens were detected by Shanghai Kehua system, and 46 mixed specimens were determined to be positive (the positive rate of mixed detection is 0.072%), among which 16 pools were determined reactive by split testing. The number of reactive samples was 16, among which 15 was HBV DNA (the positive rate is 0.235‰), 1 was HIV RNA (the positive rate is 0.0157‰). The effective split ratio was 34.8%. For the Novartis Ultrio Plus test, 81 samples out of 69522 were reactive (the positive rate is 0.117%), while 35 specimens of the initial reactive samples were confirmed to be HBV DNA reactive (the positive rate is 0.503‰), and 2 specimens were confirmed to be HIV RNA reactive (the positive rate is 0.0288‰). The positive rate in the discrimination assays was 45.7%.
Summary/Conclusions: Both Novartis TIGRIS and Shanghai Kehua screening system can detect positive samples, but the Novartis TIGRIS system shows an apparently higher sensitivity, specificity and positive rate than the latter. NAT can ensure the safety of blood transfusion, but at the same time, it produces some false positive results. So, further tests must be designed to verify the initial reactive samples.

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THE COMPARATIVE STUDY OF CYTOMEGALOVIRUS IGG SEROPREVALENCE AMONG BLOOD DONORS IN WUHAN IN 2016 AND 1990

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Background: Cytomegalovirus (CMV) can be transmitted by blood transfusion. As early as 1990, the positive rate of CMV-IgG among the blood donors in Wuhan was researched, now, more than 20 years late, in order to explore the change of positive rate of CMV-IgG in blood donors in Wuhan, the same research was again performed. By comparing the data between 1990 and 2016, the change of positive rate of CMV-IgG among the blood donors in Wuhan and relating factors were analyzed and were reported as followed.
Aims: To study the change of cytomegalovirus (CMV) IgG seroprevalence among blood donors in Wuhan in between 2016 and 1990.
Methods: CMV-IgG in 440 plasma samples from blood donors in Wuhan in 2016 were tested and the seroprevalence were compared with the one in 1990. The results were further analyzed by sex, age and occupation.
Results: In 2016, the total CMV-IgG seroprevalence among blood donors in Wuhan was 88.86 %, which were lower than the one in 1990 (95.20%, $P < 0.01$). In 2016, the CMV-IgG seroprevalence among male blood donors in Wuhan was 88.24%, which were lower than the one in 1990 (95.43 %, $P < 0.01$). In 2016, the CMV-IgG seroprevalence among female blood donors in Wuhan was 89.60 %, which were lower than the one in 1990 (94.95 %, $P < 0.05$). Both in 1990 and 2016, whether male or female, the positive rate of CMV-IgG among blood donors in Wuhan increased with the age.
Summary/Conclusions: The positive rate of CMV-IgG among blood donors in Wuhan has fallen, which is related to the improvement of the social and economic status of our country and the only-child policy. With the advent of the two-child policy and Chinese aging, the controlling of CMV infection should be strengthened.

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INVESTIGATION OF HTLV PREVALENCE AMONG BLOOD DONORS IN A REGION OF CENTRAL CHINA

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Background: Human T-cell lymphotropic viruses (HTLV) may be transmitted through blood transfusion and screening all blood donors for HTLV is mandatory in many countries. However, this screening strategy is now being adopted only by several provinces in southern and eastern China. HTLV prevalence data of the blood donor population in other regions or areas of China are lacking currently.
Aims: The aim of this study is to investigate the prevalence of HTLV among voluntary non-remunerated blood donors in a region of central China where centralized blood screening is implemented since 2005.
Methods: Anti-HTLV-1/2 was screened by enzyme-linked immunosorbent assay (ELISA) among blood donors from Wuhan city, Ezhou city, Xiantao city, Qianjiang City and Tianmen city between June and August in 2016. Results were interpreted according to the manufacturer's instructions. Primarily reactive samples were retested by double hole review, and repeated reactive samples were further confirmed by Western Blot (WB) and fluorescent PCR.
Results: During the study period, 43566 samples of blood donors were screened, 11 primarily reactive samples were screened, 4 repeated reactive samples were detected induplicate and none of them were confirmed for anti-HTLV-1/2 by WB and HTLV RNA by fluorescent PCR.
Summary/Conclusions: The results suggested that the Wuhan area is a low-prevalence region for voluntary non-remunerated blood donors in terms of HTLV-1/2 infections. Whether to implement universal HTLV-1/2 screening for blood donors in this area need further inquires on two aspects: cost-effectiveness of blood screening and safety of transfusion. Targeted screening for donors coming from endemic region, such as Fujian province, southeast China in combination with universal leucodepletion treatment of blood components can be considered an effective strategy to prevent HTLV-1/2 virus infections by blood transfusion with cost savings and without affecting blood safety.

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Abstract has been withdrawn

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COMPARISON OF PERFORMANCE BETWEEN TRIFA AND ELISA IN BLOOD DONOR SCREENING

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Background: Screening for hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV) and treponema pallidum (TP) in blood donors is mandatory in China. Currently, enzyme linked immunosorbent assay (ELISA) is practically the serologic test primarily used for the four indexes mentioned above in China. As a convenient, highly sensitive and highly specific method, time-resolved fluorescence immunoassay (TRIFA) is widely used in a lot of fields but its performance and feasibilities on blood screening are rarely discussed.
Aims: To compare the performance of TRIFA and ELISA in analyzing HBsAg (hepatitis B surface antigen), anti-HCV (anti-hepatitis C virus antibody), HIV Ag/Ab (human Immunodeficiency Virus antibody and antigen) and anti-TP (anti-treponema pallidum antibody) in blood donors.
Methods: 2,082 voluntary non-remunerated blood donors were randomly selected in Wuhan Blood Center and the plasma samples were collected and screened for HBsAg, anti-HCV, HIV Ag/Ab and anti-TP with TRIFA on Easycuta Mini platform (SYM-BIO) and ELISA on FAME platform (Hamilton) in parallel. All repeatedly reactive (RR) samples discordant between the two systems were investigated by confirmatory assays. The positive coincidence rates, negative coincidence rates and total coincidence rate were calculated and compared according to the different targets, respectively.
Results: The relatively positive coincidence rates, negative coincidence rates and total coincidence rate of HBsAg, anti-HCV, HIV Ag/Ab and anti-TP between two systems are 90.9%/99.61%/99.57%, 100.00%/100.00%/100.00%, 50.00%/99.66%/99.62% and 87.50%/99.81%/99.76%, respectively. There are 7 HBsAg reactive cases screened by TRIFA were subjected to chemiluminescence microparticle immunoassay (CMIA, Abbott) and 4 cases remained to be reactive. One anti-HCV reactive case screened by TRIFA and double-antigen sandwich ELISA was subjected to recombinant immunoblot assay (RIBA, MPI) and the result is not changed. The result of the same sample screened by indirect ELISA is non-reactive for anti-HCV. The results of HIV Ag/Ab screened by TRIFA presented 7 false positive cases but no false negative existed which were confirmed by CMIA (Abbott). As to anti-TP, there are 4 reactive cases identified by TRIFA were subjected to treponema pallidum particle agglutination assay (TPPA, Fuji) and the results fall into grey zone.
Summary/Conclusions: There are some discrepancies in the results of two systems on screening of HBsAg, anti-HCV, HIV Ag/Ab and anti-TP. We found that compared with ELISA, TRIFA is more sensitive and effectively avoid false negative and the results of TRIFA better match up with those of the confirmatory tests on HBsAg, anti-HCV and anti-TP. This screening method can well meet the demands of blood stations on the performance of the detection of transfusion transmitted infections in terms of sensitivity and specificity.

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THE PRIMARY FEASIBILITY STUDY ON THE RE-ENTRY OF NAT REACTIVE BLOOD DONORS

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Background: Blood safety in China has been greatly improved as the further development of non-remunerated blood donation and technological progress including blood borne disease screening. While false-reactive results inevitably remain, largely due to technology restriction such as methodology, quality of reagents and specimens, and human errors. Therefore blood services shall have the obligation to establish effective and reliable re-entry procedures for donors with false-reactive results, so as to eliminate the consequent adverse effects left on donors' reputation, rights and interests.
Aims: To analyzed the feasibility of re-entry of NAT reactive blood donors.
Methods: ELISA negative and NAT Reactive Blood Donors in 2012–2014 were selected and made an appointment for the first time. Blood samples collected were tested: ELISA (HBsAg, Anti-HCV, Anti-HIV and Anti-TP) and NAT (HBV-DNA, HCV-RNA and HIV-RNA). ELISA-/NAT- samples were tested by ECLA for HBsAg, HBeAg, Anti-HBs, Anti-HBc, Anti-HBe, Anti-HCV, Anti-HIV.
Results: 30 donors were called back, and 11 of them were ELISA negative and NAT reactive, which were deferred permanently. 19 of them were ELISA negative and NAT negative, and the ECLA results were 89.5% ELISA-/NAT- donors were Anti-HBc+ or Anti-HBe+ (12 cases Anti-HBc+/Anti-HBe+, 5 cases Anti-HBe+). Anti-HBc+ or Anti-HBe+ donors had risk of infection, which were deferred permanently.
Summary/Conclusions: In the view of blood safety, it is important that establish a set of program to re-entry of ELISA-/NAT+ blood donors.

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HTLV SCREENING IN BLOOD DONORS IN SHAOXING

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Background: Human T-lymphtropic virus (HTLV) is a retrovirus which can be transmitted through blood transfusion, The effective method to prevent HTLV transmission by blood transfusion is HTLV screening in blood donors. HTLV already screened in many developed countries, while Whether HTLV should be included in the screening of blood donors and how our country should carry out HTLV blood screening were discussed. It was reported that there were HTLV prevalent in some southeast coastal areas in China, But we have no data on the epidemiological characteristics of Shaoxing, one of the southeast coastal city in China.
Aims: To understand the epidemiological characteristics of human T-lymphotropic virus (HTLV) in blood donors in Shaoxing so as to provide the reference for establishing blood screening strategy formulation.
Methods: 47656 specimens from blood donors in Shaoxing were screened by ELISA method, the reactive specimens were confirmed by immunoblotting test.

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Results: 8 reactive specimens were found ,The reactive rate of screening test on HTLV was 0.17‰, while none was confirmed to positive, the positive rate was 0.00‰.
Summary/Conclusions: The prevalence rate of HTLV in blood donors of Shaoxing was very low, there was high false positive rate while HTLV screening by ELISA method.

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ANALYSIS OF SYPHILIS SCREENING POSITIVE POPULATION AMONG GANSU PROVINCE BLOOD DONORS

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Background: According to statistics of legal communicable diseases in Gansu province, the incidence of syphilis in Gansu province was 0.018% in 2013 and 0.020% in 2015, and the province reported a number of the top five class b infectious diseases. So, we should master the distribution and prevalence of syphilis in the blood donation population in Gansu province, we should make the basic data analysis for the screening and returning strategy of syphilis positive blood donors.
Aims: By analyzing the structure of the blood donors syphilis antibody screening positive population in Gansu province, provide the basis for recruiting safe blood source strategy, provide basic data for the development of Gansu province syphilis reactive donors reentry strategy.
Methods: By Collecting syphilis detection results of Lanzhou city in 2011 - August 2016 blood donors, statistical analysis was conducted. By Collecting syphilis detection results of Lanzhou city and the three blood stations in the province in 2015 - August 2016 blood donors, statistical analysis was conducted.
Results: The syphilis antibody screening positive rate of Lanzhou 299969 blood donors were 0.59%, rejects in blood donors of different degrees of education, ages, genders, careers, marriage and nationalities had the statistical significance; rejects in blood donors of different degrees of education, ages, careers and nationalities had no statistical significance between Lanzhou city and the three blood stations in the province.
Summary/Conclusions: Blood donors in Gansu province, the degree high, 18–30 years old, students, staff, Han population is the key of the province to recruit security fixes blood donors; we will carry out syphilis antibody joint screening strategy and discuss Gansu province syphilis reactive donors reentry strategy in our province.

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STUDY ON THE APPLICATION OF ELISA INTERNAL QUALITY CONTROL IN RANDOM WELL

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Background: Enzyme-linked immunosorbent assay (ELISA) is affected by many factors, such as application of sample, incubation, washing and so on. The results would be different if a sample was tested in different well of a microplate.
Aims: To explore the value of testing ELISA internal quality control (IQC) in random well by comparing the results of IQC in random wells with in fixed wells.
Methods: HBsAg, anti-HCV, anti-HIV/HIV P24 and anti-TP were tested by ELISA in Nanjing Red Cross Blood Center laboratory. IQC in fixed and random wells were tested parallel in blood donor screening. The fixed well of positive IQC was in the position of well F1 in HBsAg, anti-HIV and anti-TP tests, and well G1 in anti-HCV and HIV P24 tests. The fixed well of negative IQC was in the position of well H1 in the tests above. The position of random well was derived by function f(x) = RAND() *88 from Excel after rounding to the nearest integer.
Results: There were no significant difference ($P > 0.05$) by comparing the results of negative IQC in random wells and fixed wells in all the tests expect anti-TP ($P < 0.05$). Although the difference were statistically in anti-TP negative IQC test, all the results in random wells and fixed wells were negative. The difference were statistically significant ($P < 0.05$) between the results of positive IQC in random wells and fixed wells in anti-HIV, HIV P24 and anti-TP tests.
Summary/Conclusions: The value of testing ELISA negative IQC in random well was negligible. However, it should be adopted that testing positive IQC by ELISA in random well.

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INCIDENCE OF HEPATITIS B AND C VIRUSES AMONG BLOOD DONORS OF A TEACHING HOSPITAL IN MIRPUR, AZAD JAMMU KASHMIR, PAKISTAN

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Background: Access to safe blood and blood products still remains a major challenge throughout the world. Unsafe blood transfusion is very costly from economic point of view. The situation in Pakistan is not different where Transfusion-Transmitted Infections (TTIs) continue to threaten the blood safety. Blood-borne transmission remains a key vector of the Hepatitis B and C in Pakistan, affecting about 7.4 percent of the population. The Blood Transfusion sector in Pakistan is passing through a system reform from a fragmented network of blood banks to a centralized network of Regional Blood Centres. However, the pre-reform Blood Transfusion Services in Pakistan are mostly hospital based with no functional separation of the process into independent establishments for 'production' and 'utilization'.
Aims: The objective of the current study was to figure out the incidence of HBV and HCV in Mirpur, Azad Jammu & Kashmir (AJ&K), Pakistan.
Methods: This was a retrospective cross-sectional study conducted at the Divisional Headquarters Teaching Hospital, Mirpur, AJ&K, which is providing quality health services since 1960s. All the blood donors [10,460] who donated blood from January 2015 to December 2016 were included in the study. All the donations were screened for Hepatitis B surface antigen, anti-HCV antibodies, and anti-HIV antibodies with ELISA.
Results: Out of 10,460 blood donors, 98% were replacement while 2% were voluntary non-remunerated blood donors (no paid/professional donor). A total of 848 donations (8.1%) were found positive for at least one of the infections. The prevalence of HCV and HBV was found to be 499 (4.77%) and 349 (3.33%) respectively. No case of HIV was reported during the study period. The prevalence of HBV is comparable to previous studies in other parts of the country but HCV prevalence has shown a dramatic increase when compared with figures already reported and needs special attention. This data also suggests that HIV prevalence is infrequent.
Summary/Conclusions: The frequency of HBV is comparatively low. This might be due to vaccination programs and increased public health awareness. The best approach to minimize these TTIs is to have a careful donor selection and mandatory screening.

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CONFIRMATION ANALYSIS OF HBSAG SCREENING REACTIVE RESULTS FOR VOLUNTARY BLOOD DONORS IN CHANGCHUN

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Background: Blood services in China to reduce blood transfusion infection after the residual risk of hepatitis b virus (HBV), screening reagents with high sensitivity, have a certain degree of false positive test results, which might cause the loss of voluntary blood donors, so it should be evaluated through the experimental study to establish a mechanism for the blood donors back.
Aims: To analyze the test results of ELISA method for the detection of HBsAg reactivity specimens in ChangChun area, and evaluate the rejoined mechanism of blood donor and blood safety.
Methods: From December 2013 to August 2014, 307 HBsAg ELISA reactivity samples were routinely screened in the blood center of JiLin province, screening reagents are BIO-RAD hepatitis B virus surface antigen diagnostic kit (ELISA) and Shanghai KeHua hepatitis b virus surface antigen diagnosis kit (ELISA), confirm test for specimens were done by ZhuHai LiZhu hepatitis b virus surface antigen diagnosis kit (ELISA) and ZhuHai LiZhu hepatitis b virus surface antigen diagnosis kit (neutralization test).
Results: Seventy-one positive cases are confirmed after 307 HBsAg ELISA reactive specimen neutralization tests, the positive rate was 23.13%, among them, 98 cases of ELISA two types of reagents were confirmed positive, and the positive rate was 68.37%. Among 209 single-reagent ELISA specimens, 4 positive cases are confirmed, positive rate of 1.91%; difference of confirmatory test between two types of reagent and single reagent is statistically significant ($P < 0.05$). In the ELISA two type reagent reagents samples, $S/CO \geq 5.0$, 82 cases in total; positive cases are 58, positive rate is 70.73%; S/CO between 1 and 5 is 16, positive cases are 9, positive rate is 56.25%, the difference is not statistically significant ($P > 0.05$), Among the 4

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positive cases of the reagent of ELISA single reagent, one of them was the ELISA single reagent gray zone.
Summary/Conclusions: HBsAg ELISA screening test results from voluntary donors have a false reactivity problem, in order to reduce the lose of voluntary donors and ensure the supply of blood, It can be considered for the confirmation test of HBsAg ELISA single reagent, and the establishment of reagent sample for blood donors who have been screened for HBsAg ELISA single reagent. At the same time, ELISA method to detect HBsAg is necessary to set reasonable grey area range.

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A TIME-BASED METHOD FOR RANDOM SAMPLING OF QUALITY CONTROL SAMPLE IN ELISA

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Background: Enzyme-linked immunosorbent assay (ELISA) has been used to screen transfusion transmitted infections (TTIs) in most pretransfusion laboratories. Quality controls are often used to monitor the performance of ELISA. Currently, in most laboratories it is a common practice to run quality controls in fixed positions, which is convenient to the technologist. However, since the experimental conditions such as temperature or chemical reaction may vary from different wells in a plate, failure to place control samples in a random manner may result in failure of detecting testing problems by quality controls.
Aims: To design a method to perform quality control in a random position that more closely simulates an actual donor sample's testing conditions in ELISA sampling.
Methods: Based on a mathematical processing model, the last digit of current minutes time (m) and the last digit of current seconds time (s) were used to generate a random number, which was used to make changes on positions with time during the quality control sampling process. Then we designed a correspondent computer program and performed a test run in Hamilton MicroLab Star liquid handler. We analyzed the distribution feature of random positions being selected over 160 independent run in 73 days.
Results: In the mathematical model, 'm' was first converted to ASCII code, and then multiplied by 10. The obtained number plus the value of ASCII code of 's' to generate a time-based random number, denoted as 'a'. The above process was successfully completed under MicroLab Star programming. In the MicroLab Star internal programming language, we called a function named as MthR01Draw(), which is a random number (between 0 to 1) generator, for 'a' times. The obtained number multiplying the number of donor samples in the given experimental run, after a calling of Mthround function (taking an integer) finally generated the position for quality control sample. When the sampling process to the specific position, the donor sample will be substituted by a quality control sample along with its identifier automatically on the device. The statistical results showed that the random positions were largely uniformly distributed, which means the quality control sample has an equal chance to be placed in any position of a plate as a donor sample.
Summary/Conclusions: Random placement of quality control sample in a plate more closely simulates an actual donor sample's conditions. The method used in this experiment could help to build a complete position-randomized quality control strategy in enzyme immunoassay.

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A NATIONWIDE SURVEY OF BLOOD DONATION IN CHINESE BLOOD SERVICES

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Background: Evaluation of trends in the rate of donation numbers and transfusion-transmitted infections (TTIs) in blood donors is essential for monitoring blood supply safety and donor screening effectiveness.
Aims: The aim of this study was to investigate the rate of donation numbers and the unqualified rate of TTIs sero-markers, including hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), and TP among blood donors in China from 2009 to 2014.
Methods: Data were collected in response to a questionnaire referred to blood centers/banks in China mainland between 2009 and 2014. The numbers of donors and the deferral numbers of HBsAg, anti-HCV, anti-HIV, and anti-TP among blood donors were expressed by donation year and area.
Results: The data of donation numbers was collected referred to 319 blood services from 2009 to 2014. From the data across the country, the number of blood donors kept increased from 10,654,899 in 2009 to 12,470,215 in 2014. The blood donation rate is only 9.12‰ of the whole population, only 9 cities' donation rate was above 2%. The double-positive rate of HBsAg and anti-HIV in donors remained relatively steady, was 0.41% and 0.044% in 2009 and 0.39% and 0.037% in 2014. But double-positive rate of HBsAg showed a decrease from 0.43% to 0.31% in provincial blood centers. Moreover, the double-positive rate of anti-HCV showed a significant decline from 0.22% in 2009 to 0.13% in 2014. The double-positive rate of anti-TP increased from 0.42% in 2009 to 0.48% in 2012 and decreased to 0.40% in 2014.
Summary/Conclusions: The number of blood donors increased dramatically which were employed in recent years in Chinese Blood Transfusion Organization have been effective. Besides the problem of blood availability, there are great challenges in ensuring blood safety. Furthermore, measures for establishing TTIs confirmation laboratory are urgently needed.

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ANALYSIS OF THE DIFFERENT PRESERVATION TIME ON THE TEST RESULTS OF NUCLEIC ACID WEAK-POSITIVE SPECIMENS

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Background: NAT technique is employed to test on specimens tested blood donors is an important measure to provide blood safety. NAT technology of high sensitivity, but if the specimen preservation processing is undeserved, the virus nucleic acid may be degraded, which affect the reliability of the test results, increase the risk of disease transmission by blood transfusion, dangerous for blood safety. Therefore, the influence of nucleic acid specimen on nucleic acid test results is analyzed
Aims: To test and verify the specimen of nucleic acid testing to preserve time before centrifugal 4 h, 12 h and 24 h of nucleic acid test results, for NAT process control, and test strategy to provide data support.
Methods: Three groups weak-positive specimens of HBV, HIV-1 and HCV were prepared in the preparation test. Each item was 10, with total of 90, each of which was 5 ml. 60 specimens weak-positive test can be divided into three groups, each group of each item is 10 teams, respectively at 4°C for 4 h, 12 h and 24 h after save, in 1600 g under the condition of centrifugal 15 min at 4°C. Using Procleix Tigris system, three joint tests and single - item discrimination tests were performed on these 90 test specimens.
Results: The test results of nucleic acid test were not statistically significant ($P > 0$ 0.05), and the samples results of 3 groups were all reactivity.
Summary/Conclusions: Conclusion: prove that the nucleic acid samples can be stored in 4°C, centrifuge within 24 h after blood collected, routine testing, it don't affect the results.

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Abstract has been withdrawn

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SCREENING DONATED BLOOD FOR TRANSFUSION TRANSMITTED INFECTIONS BY SEROLOGY ALONG WITH NAT: A NAKASERO BLOOD BANK EXPERIENCE

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Background: Blood transfusion safety has tremendously improved through testing of donated blood with increasingly sensitive assays compared to the historical serologic testing for transfusion transmitted diseases. Although newer strategies like nucleic acid testing (NAT) have helped further shorten the "window period" no technology exists to completely detect all window period donations and therefore none completely closes the exposure-to- seroconversion window period.
Aims: To establish the relative safety of blood at which screened using NAT and serology. To establish the rate of false positive on serology.

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Methods: 642 donations were screened individually for some TTIs, namely; HIV, HBV and HCV by serology and nucleic acid testing (NAT). All reactive samples were retested (wherever possible). The reactive results of either serology or NAT were followed up, blood units were discarded and donors were notified and counseled
Results: We evaluated 32 (4.98%) cases which were reactive on both NAT and Serology. 28 (4.36%) cases were detected on NAT (1.1% HIV, 0.16% HBV, 0.16% HCV). 4 (0.62%) cases were however reactive on serology and Non detected on NAT (0.16% HIV, 0.31% HBV and 0.16% HCV).
Summary/Conclusions: Our study indicated that it is important to use both NAT and serology to improve on the blood safety, a practice which is not yet embraced at Uganda Blood Transfusion Service. Screening of blood for TTIs using Serology alone is associated with high rate of false positive and increased donor deferral.

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BLOOD MANAGEMENT AND BLOOD SAFETY FOR MULTI-TRANSFUSED THALASSEMIA PATIENTS IN A STAND ALONE BLOOD BANK

A Verma

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Background: Transfusion Transmitted Infection (TTI) and alloimmunization are two major problems in multi-transfused patients .
Aims: This study is aimed to estimate the prevalence of blood TTI and presence of red cell antibodies in multiple blood transfused patients of beta thalassemia major who are taking blood from Rotary blood bank, New Delhi.
Methods: There are 106 Thalassemia Major patients registered with Rotary blood bank, New Delhi, India. All the patients are provided blood without taking any replacement donor and without taking processing charges. This is in accordance with our National guidelines.
Rotary blood bank collects blood from 100% voluntary donors.
The study was also conducted to know the extent of problem of alloimmunization and to find important red cell antibodies in thalassemia patients.
All 106 thalassemia patients were screened. The samples were subjected to red cell alloantibody and autoantibody detection by column gel agglutination technique and by SPRCA. Three cell panel and then 11 cell panel reagent cells were used in screening and identification of alloantibodies respectively.
Results: Out of 106 multiple blood transfused patients –33% patients were infected with TTI on serological test.
Total 93 male patients and 13 female patients were infected with TTI. The sero-reactivity for HIV was 2.83% (03/106); 02/106 were males and (01/106) were females. The seroreactivity for HBsAg was 16% (12/106) were males and females 5/106, HBcAb 2.83% (3/106) The sero -reactivity for HCV was 20.75% (22/106); (17/106) were males and (5/106) were females.
Out of 106 patients 15 (14%) subjects were alloimmunized. All alloimmunized subjects were recipient of more than 20 units of transfusion. Total nine clinically significant alloantibodies were identified. Anti E and anti c were commonest antibodies in 60% patients .
Summary/Conclusions: HIV, HBV, HCV infections are most prevalent TTI among multiple blood transfused patients with thalassemia major.
Red cell alloimmunization is another risk in these patients so to prevent it extended phenotype matched blood transfusion for Rh-c, and Rh-E antigens should be given.
Thalassemia major patients require lifelong transfusion support so extra precautions should be taken.

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HTLV INFECTION SURVEY IN BLOOD DONORS OF HUZHOU CITY, ZHE JIANG PROVINCE

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Background: HTLV is a blood-borne virus. It may cause serious consequences for receiving blood.
Aims: To investigate the infection status of HTLV in blood donors in Huzhou area, and to provide the basic data for taking the appropriate intervention strategy.
Methods: Screening was performed by enzyme-linked immunosorbent assay (ELISA) with double antibody sandwich, and the positive samples were screened again.
Results: 30835 samples were tested, screening positive rate of 0.04% (11/30835). Of which five blood donors have many blood donation experience, suspected asymptomatic infection.
Summary/Conclusions: Huzhou area is not an endemic area of HTLV infection, but the risk of infection has increased the number of suspected blood donors, so it is necessary to carry out HTLV screening in blood donors.

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Abstract has been withdrawn

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VERIFICATION PROCEDURE OF HBSAG ELISA IN BLOOD SCREENING

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Background: To ensure the safety of donated blood, the performance of ELISA is pivotal for laboratory results and should be under effective control. Therefore, the verification of ELISA reagents is an indispensable part for screening laboratory in blood centers.
Aims: To verify the blood screening reagent and guarantee the blood safety, we comprehensively evaluated the performance of candidate HBsAg ELISA in this study.
Methods: Standard materials and real samples were simultaneously tested by current-used reagents as well as candidate reagent, and the results were compared to illuminate the qualification of the candidate, both in lab compatibility and concordance with the background testing.
Results: The repeatability and accuracy of the candidate HBsAg ELISA met the lab requirements. As compared with other two current-used ELISA, the concordance of the candidate was 100%.
Summary/Conclusions: As the final user, our department implemented the verification of test to ensure the integrity and validity of blood screening. In this study, the candidate passed the verification with good specificity and sensitivity.

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DISCUSSION ON THE METHOD OF PERFORMANCE COMPARISON OF NUCLEIC ACID DETECTION SYSTEM

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Background: "*Technical operation of procedures in Blood bank*" provides that the detection of the same project between the different systems need to compare performance, the current our center has three sets of nucleic acid detection system, in order to clearly understand the detection performance of the system and the differences between systems, we explore the methods of Detection system performance comparison.
Aims: To explore the nucleic acid detection system performance comparison methods and indicators, by comparing the performance of nucleic acid detection system, found the differences what between the various systems to ensure the test results accurate and reliable.
Methods: In the same test environment detect the same samples, the same reference substance and the same serum plate by different systems to statistical analysis the sensitivity, specificity, repeatability and coincidence of each system for comparing the performance and suitability of each system.
Results: Two systems which HuaYiMei detection system and GRIFOLS detection system serum plate evaluate what specificity and sensitivity was 3 projects of 100%. KeHua system the specificity of the project was 100%, the sensitivity of HBV-DNA project was 100%, the sensitivity of HCV-RNA project was 92.9%, the sensitivity of HIV-RNA project was 90.9%. The coincidence rate what parallel detection of HuaYiMei and GRIFOLS system was 99.93%, KeHua and GRIFOLS system was 99.94%, KeHua and HuaYiMei system was 99.91%. There was no statistical differences in the pool detection positive rate and the single detection positive rate of two kinds of pool detection system that KeHua and HuaYiMei by χ^2 test.
Summary/Conclusions: Explore establishing the system what performance comparison of nucleic acid detection system for comparison the nucleic acid detection system performance in future. Through comparatives analysis to prove that there was

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no significant difference between the three nucleic acid detection systems, to facilitate the daily detection arrangements, and the detection system can be a backup between each other, to ensure the detection conduct smooth and blood safety.

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SEROPREVALENCES AND TITER OF HUMAN NEUTRALIZING ANTIBODIES AGAINST DIFFERENT TYPES OF ADENOVIRUS DETECTED IN PLASMA

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Background: Human adenovirus (HAdV) had caused several outbreaks of acute respiratory infections all over the world in recent years. In Southern China, the main genotype of HAdVs include HAdV-3,-4,-7,-14,-55. By detecting the positive rates and titers of human neutralizing antibodies (NAbs) against different types of adenovirus, the history of infection and the protective immunity against specific adenovirus types could be understood.
Aims: To investigate the positive rates and titers of human neutralizing antibodies against HAdV-3,-4,-7,-14, and -55 in Guangzhou area and provide data for adenovirus vaccine development as well as guidance for the selection of appropriate immune population and optimal immune time.
Methods: Plasma samples from 270 healthy blood donors in Guangzhou Blood Center were collected and tested by in vitro micro-neutralization assay, the seroprevalences and titers against HAdV-3, -4, -7, -14 and -55 were analyzed.
Results: Of the 270 plasma samples from blood donors, the seroprevalences of NAbs against HAdV-3,-4,-7,-14 and -55 were 47.78%, 41.48%, 47.41%, 40.74% and 38.15% respectively; the proportions with medium-high titer NAbs (128–1152) were 11.48%, 3.33%, 9.26%, 12.22% and 4.07% respectively; the proportions with high titer ≧1152 were 7.78%, 27.41%, 8.52%, 4.81% and 14.44% respectively. The positive rates of NAbs against HAd-4, -7 and -55 increased with age, and there was no significant difference in the proportions of NAbs against all HAdVs between male and female.
Summary/Conclusions: There are similarly seroprevalences of NAbs against HAdV-3,-4,-7,-14 and -55 in blood donors representing general adults in Guangzhou area. More medium-high titers of NAbs against HAd-3, -7 and -14 were found than HAdV type 4 and 55, While there are a lack of high titer protective antibodies against HAd-3, -7 and -14.

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APPLICATION OF NUCLEIC ACID TESTING (NAT) TECHNOLOGY FOR BLOOD SCREENING IN INNER MONGOLIA BLOOD CENTER

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Background: Transfusion of blood products can help save millions of lives. However, as the demand for blood and blood products increases year by year, many countries face severe challenges of blood, quality of blood products and safety. The world health organization recommends mandatory screening of HIV, hepatitis b, hepatitis c and syphilis in all the blood collected. At present, the traditional methods of serology of blood screening are commonly used for blood screening in all major blood supply agencies in China, namely, the enzyme-linked immunoassay (ELISA) technique that we usually use.
Aims: In order to effectively improve the accuracy of the blood Test results, and ensure the safety of the blood, the national health and family planning commission notice requirements within the scope of the national overall Nucleic Acid blood screening Test (Nucleic Acid Test, NAT).Nucleic acid detection as the latest generation of detection technology, direct detection of pathogens of DNA or RNA, therefore has higher sensitivity and specificity, hepatitis b, hepatitis c, HIV/AIDS can be significantly shorten the "window period". This experiment through a year of unpaid donors specimens simultaneously twice enzyme to avoid detection and virus nucleic acid detection, again to the test results of two methods were analyzed, and then discusses the nucleic acid testing (NAT) technology is used to the meaning of blood screening.
Methods: Using enzyme-linked immune detection technology (ELISA) and transcription mediated amplification (TMA) method for the center for a year and all unpaid donors specimens collected simultaneously two enzymes to avoid detection and virus nucleic acid detection, again. The results from the nucleic acid testing were analyzed compared with those from ELISA. The ELISA experiment was based on the fame 2430/2420 automatic enzyme-linked reaction system and the Swiss automatic sample system STAR. NAT uses the American Novartis automatic blood virus nucleic acid detector (Procleix TIGRIS System).
Results: Among all the test results, the positive rate of ELISA detection than NAT. At the same time there is the certain percentage of ELISA(−)/NAT(+). Especially since I center for nucleic acid detection has been successfully captured an AIDS window period infected blood donors, to ensure the quality, to ensure the safety of the blood recipients have played a very important role, but also strongly confirmed to carry out the necessity of nucleic acid detection.
Summary/Conclusions: It can be seen that currently, the two methods of blood screening ELISA and NAT complement each other, which must be adopted simultaneously, which is of great significance to prevent the infection of the recipient "window period". Novartis TIGRIS system of NAT is appropriate for screening the virus of the donors' blood.

Hepatitis B (HBV)

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CONTENT DETERMINATION OF GRANULOCYTIC MYELOID-DERIVED SUPPRESSOR CELLS AND ITS CORRELATION WITH HBV DNA LEVEL

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Background: Myeloid-derived suppressor cells (MDSCs) is a group of novel immunosuppressive cells which can be divided into two types as granulocytic-MDSC (G-MDSC) and monocytic-MDSC (M-MDSC). MDSC can suppress both adaptive and innate immune. In combination with the immunosuppressive status of chronic hepatitis B virus (HBV) infection, as well as the strong immune suppression function of MDSC in acute and chronic inflammation, tumor and other pathological conditions, we hypothesized that MDSC may also be involved in immunosuppression during HBV infection.
Aims: We collected 77 blood donors with different HBV infected status, and studied the content of G-MDSC, markers related to liver function, hepatitis B surface antigen (HBsAg) and HBV DNA, in order to investigate the numbers of G-MDSC and weather it mediated HBV immune suppression in varied HBV infection status thus to provide a theoretical basis and clues for the novel treatment for hepatitis B.
Methods: ALL 77 HBV infected donors were divided into 3 groups according to the varied HBV infection status, including 31 cases of occult HBV infection (OBI) with HBsAg-/HBV DNA+, 36 cases of inactive HBV Infection (IHI) with HBsAg+/HBeAg-/HBV DNA+ and 10 cases of active HBV infection (AHI) with HBsAg+/HBeAg+/HBV DNA+. Lymphocytes were isolated from fresh peripheral blood and G-MDSC (CD33+CD11b+HLA-DR-CD15+) were sorted and counted by Flow cytometry. The possible correlation of G-MDSC with the age, liver function markers, HBsAg and HBV DNA level were analyzed.
Results: The content (Median ± SD) of G-MDSC in OBI, IHI and AHI groups were 0.70% ± 1.39%, 0.65% ± 1.30%, 0.41% ± 0.79% respectively. There was no significant statistical difference among 3 groups ($P > 0.05$). G-MDSC was negatively correlated with HBV DNA level, the r value was −0.492 ($P = 0.003$) in IHI group and −0.637 ($P = 0.025$) in AHI group, however, the correlation had no statistical difference between these two groups. Neither statistically significant correlation was observed between G-MDSC and age, AST, ALT and HBsAg.
Summary/Conclusions: The content of G-MDSC was negatively correlated with HBV DNA level, which suggested G-MDSC involved in HBV immune suppression but played a different role with M-MDSC (data not shown) in the immune process of HBV infection.

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PRE-S/S MUTATIONS OF OCCULT HEPATITIS B VIRUS FROM BLOOD DONORS WITH OR WITHOUT ANTI-HBS

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Background: Occult hepatitis B virus infection (OBI) with genotype B or C strains (OBI_B or OBI_C) has not been characterized for pre-S/S mutation with anti-HBs status of OBI carriers.
Aims: To better understand pre-S/S mutations of OBIs that associated with antibody pressure elicited by vaccination or host immune response to HBV.
Methods: The molecular characteristics of envelope protein mutations was extensively analyzed in corresponding to anti-HBs seromarkers of OBI carriers.
Results: 99 HBsAg-/DNA+ blood donors were identified as OBIs from 310,167 blood donors in Shenzhen blood center, yielding the rate 1:3133. OBI_B was predominant in comparison with OBI_C. The mutations ps_1L84I, ps_2Q2K/R/I and sM103I, sP105R, sG130R/A, sT131N/A, sM133S/I/T and sF134I/L and sV177A occurred in 12%–16% of 42 pre-S1 (ps_1) and pre-S2 (ps_2) and 57 S (s) proteins of OBI_B strains, while higher frequency of mutations ps_1D27G, ps_2T6N/S and sS53L was detected in 33%–50% of 12 OBI_c strains. By stratifying OBI carriers with anti-HBs seromarker, the frequency of pre-S/S mutations was even higher in anti-HBs- than that in anti-HBs+ OBI carriers. The mutations ps_1T68I, sK122R and sQ129R were found uniquely in 5%–8% of anti-HBs+ OBI_B carriers, while the mutations ps_1L45F/N, ps_2D14N, sE2G/D, sA5S/V, sY100S/L, sQ101R/H, sP105R/L, sS113N and sY161F/S were found uniquely or preferentially in 7%–12% of anti-HBs- OBI_B carriers. Nine OBI_B and three OBI_c pre-S/S mutations were commonly detected from 9% to 50% of either anti-HBs+ or anti-HBs- OBI carriers. However, the well-known mutation sG145R/A/V relating to vaccine escape was detected in low frequency (5.3%) from OBI_B carriers with no preferential anti-HBs selection.
Summary/Conclusions: Most pre-S/S mutations of OBIs occurred randomly and only few of mutations might associate with antibody pressure elicited by vaccination or host immune response to HBV, which would be helpful for understanding of OBI occurrence.

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SEROLOGICAL AND MOLECULAR CHARACTERIZATION OF HBSAG-NEGATIVE HEPATITIS B VIRUS INFECTION IN BLOOD DONORS

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Background: Hepatitis B Virus (HBV) infection is the most important factor causing hepatitis, liver cirrhosis and hepatocellular. HBV infects over 350 million people worldwide and the prevalence of hepatitis B surface antigen (HBsAg) is as high as 7.18% in Chinese population. Pre-seroconversion window period (WP) infection and occult HBV infection (OBI) are characterized by the presence of HBV DNA in plasma with undetectable HBsAg using the most sensitive commercial assays, which is a potential risk of HBV transmission through blood transfusion after HBsAg screening. Nucleic acid testing (NAT) for HBV, HCV and HIV has been used in blood donors to decrease the risk since 2010 in China. OBI prevalence varies between different geographic areas as the population infected with different HBV genotype. The generating of OBI strains may be related to mutations in HBV S protein and genomic regulatory elements, which affect the detection of antigen in vitro and viral replication.
Aims: To investigate the serological and molecular characterization of HBsAg-negative HBV infections from blood donors in Shanghai, China.
Methods: Blood donor samples confirmed HBsAg negative and HBV DNA positive were collected by Shanghai blood center. Serum anti-HBc and anti-HBs were tested by enzyme-linked immunosorbent assay (ELISA). HBV DNA loads were determined by Roche COBASAmpliPrep/COBAS TaqManHBVTest v.1.0. HBV S and BCP/PC regions were amplified using nest-PCRs and products were cloned in the TA cloning kit for sequencing. S nucleotide sequences were translated into amino acid sequences according to the open-reading frames by DNAMAN software and HBsAg serologic subtypes were predicted from the amino acids sequences at position 122 and 160. HBV genotypes were determined by Blast in GenBank. S protein and BCP/PC gene mutations were determined by comparing with wild-type HBV sequences in Genbank database.
Results: A total of 78 samples confirmed HBsAg negative and HBV DNA positive were collected, of which 70 samples with anti-HBc and/or anti-HBs positive were classified as OBI and 8 seronegative samples were classified as window period (WP).There were 23 samples positive for anti-HBs and 18 were weakly reactive for anti-HBs (<100 IU/l).HBV DNA loads of all 70 OBI samples were <200 IU/mL and lower than those of WP samples significantly ($P < 0.01$). S regions were amplified in 27/78 samples, including 22 OBIs samples (12 genotype B and 10 genotype C) and 5 WP samples (3 genotype B and 2 genotype C).Of the 22 OBIs, 12 were adw subtype, 7 were adr subtype, 1 was varied to ayw subtype and 2 had unknown changes. S protein mutations were identified in 18 of the 22 OBI samples with relatively high incidences of Q101K/R, M103T/I, T126A/P, F134L and D144A in 10 genotype B, and S114T/A, T118K/R, K141T, S143T in 6 genotype C. The mutation rate and average number of amino acid substitutions of OBI samples were higher than those of HBsAg positive blood donors ($P < 0.05$). No mutations were observed in 5 WP samples. BCP/PC regions were amplified in 32/78 samples (27 OBIs and 5 WPs) and 19 OBIs had nucleotide mutations with relatively high incidences of G1896A, A1762T and A1764A, A1752G.
Summary/Conclusions: We found that all the HBsAg-negative HBV infection blood donors were genotype B and C in Shanghai. Most of them were OBIs with extremely low HBV DNA loads and a small number were WP infections. Amino acid substitution in S protein and nucleotide replacement in BCP/PC region may be important mechanisms for the occurrence of OBI in these carriers infected with genotype B and genotype C HBV.

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COMPARISON OF TWO COMMERCIAL HBV NAT REAGENTS ON THE DETECTION OF SERONEGATIVE BLOOD DONATIONS IN CHINA

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Background: Hepatitis B virus (HBV) nucleic acid test (NAT) screening plays a critical role in preventing HBV transfusion transmission and ensuring blood transfusion safety. As a highly variant DNA virus, there may be a risk of HBV DNA false-negative results using commercial NAT reagents. Cobas® TaqScreen MPX Test, version 2.0 on the cobas s 201 system by Roche Molecular Systems (MPX2.0) and Haoyuan NAT reagents on ChiTaS BSS1200 System are two of mainly used NAT reagents targeting different regions of HBV DNA in blood banks in China.
Aims: By comparing two commercial HBV NAT reagents targeting different regions of HBV genome: reagent from Haoyuan targets S region of HBV DNA and that from MPX2.0 targets C region of HBV DNA, this study discussed the missing detection of two commercial reagents on HBV NAT screening.
Methods: 687 HBV DNA positive/seronegative donations, from 9 different blood centers or blood banks all over China from 2015 to 2016 were included in this study: 354 HBV DNA positive/seronegative donations detected twice by Haoyuan in two different labs and 333 detected twice by MPX2.0 in two different labs. All these donations were detected again by the other reagents, and serological markers (HBV surface antigen (HBsAg), HBV surface antibody (HBsAb), HBV e antigen (HBeAg), HBV e antibody (HBeAb) and HBV core antibody (HBcAb)) on Abbott Architect i2000 immunoassay analyzer in NCCL from a same tube. 21 samples from which were selected and quantitively detected by COBAS AmpliPrep/COBAS TaqMan HBV test v2.0 (HBV2.0) and RealTime HBV Assay on Abbott m2000.
Results: 146 from 354 HBV DNA reactive seronegative donations detected by Haoyuan were nonreactive by MPX2.0 (H+R-), among which 58.9% (86/146) were both HBeAb and HBcAb positive. 61 from 333 HBV DNA reactive donations detected by MPX2.0 were nonreactive by Haoyuan (R+H-), and 26.2% (16/61) were both HBeAb and HBcAb positive. Ct value among different groups were quite different: average Ct value of ID test by Haoyuan from H+R+ was 36.36 (95%CI 35.99–36.74), while that from H+R- was 38.86 (95%CI 38.52–39.21). Similarly, Ct value of ID test by MPX2.0 from R+H+ was 35.43 (95%CI 35.12–35.75) and that from R+H- was 37.13 (95%CI 36.47–37.78), indicating that low viral load may be an important reason for the false-negative results.
Quantitative analysis results from 21 donations (8 from H+R+, 8 from H+R- and 4 from R+H-) showed that HBV viral load of some donations were several times the LOD of the reagents but they still were undetected. The viral load of one R+H- sample was approximate 100 IU/mL (139 IU/mL from Abbott i2000 and 97 IU/mL from HBV2.0), which is nearly 16 times the LOD of Haoyuan; the viral load of another

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sample from H+R- was 14 IU/mL by Abbott i2000 (undetected by HBV2.0), and 6 times the LOD of MPX2.0, these results above indicated that there was something special for these samples: matrix problem or sequence mutations may appear resulting in missing detection. However, we didn't know the exact region that these reagents target, we could not explore it further.
Summary/Conclusions: Although MPX2.0 and Haoyuan are two mainly used NAT screening reagents with good performance on daily work in blood banks in China, we found in this study that the two reagents still suffered missing detection of HBV DNA. Low viral load may be a main and important reason, besides reasons from samples themselves, such as matrix or HBV DNA sequence may also lead to false-negatives.

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VARIATION OF MAJOR HYDROPHILIC REGION AND "A" DETERMINANT OF HBV IN HCV/OBI CO-INFECTED PATIENTS

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Background: Occult hepatitis B virus infection is defined by the presence of HBV DNA in the absence of detectable HBsAg in serum. The mechanisms leading to OBI infection has not yet been researched clearly. Mutations in major hydrophilic region (MHR) and "a" antigenic determinant of the surface antigen was one of the earliest recognized mechanisms leading to occult HBV infection. In HCV/HBV coinfection patients, HCV's inhibition to HBV is another cause of OBI. The HCV core protein can inhibit HBV replication and gene expression and suppresses activity of HBV enhancers. Coinfection with OBI among patients with chronic HCV infection was found in 30% of patients in Italy and up to 80% in Japan.
Aims: To investigate the prevalence of OBI in chronic HCV infected patients in blood donors and drug addict. Elucidate the mechanism of OBI in HCV/OBI co-infection patients.
Methods: TIGRIS high sensitivity nucleic acid test system was used to screen HBV DNA among patients with chronic hepatitis C (anti -HCV+, HCV, RNA+, HBsAg-).86 of HCV infected patients from Guangzhou blood center and 193 of them from Yangjiang prison drug addicts. HBV-DNA detection by PCR was performed using primers specific for the S region of the HBV genome. The surface gene sequence of HCV/OBI co-infected patients was analyzed and compared with 9 wild type HBV and 12 OBI.
Results: Occult HBV DNA were found in the serum of 12 of 279 (4.3%) HCV infected patients, (absence of HBsAg in followed-up specimens, excluding HBV window infection). The positive rate of OBI among HCV blood donors and drug addicts is 5.8% (5/86) and 3.6% (7/193). No significant differences in the positive rate of OBI were observed in these two groups ($P > 0.05$). No significant difference were found in age and gender between the HCV/OBI co-infected patients and HCV infected patients ($P > 0.05$). Two HBV S region (genotypes B) were amplified in 12 HCV/OBI co-infector. The MHR amino acid substitution rate (amino acid variance/total amino acid) of HCV/OBI、 HBV and OBI was 4.2% (5/118) 、0.8% (4/531) and 2.8% (20/708), the "a" determinant amino acid substitution rate was 1.39% (3/216) 、3.82% (11/288) and 4.17% (2/48). The rate of MHR amino acid replacement of HCV/OBI and OBI was higher than that wild-type HBV ($P < 0.05$). There was no statistically difference in the rate of MHR amino acid substitution between HCV/OBI and OBI group ($P > 0.05$). There was no statistical difference in the rate of amino acid replacement in "a" determinant.
Summary/Conclusions: In our experimental conditions, the prevalence of occult HBV co-infection among patients with chronic hepatitis C is higher than blood donors (106/475538, 0.02%) (Wang, R, Z, BMC, Infect, Dis, 2016). Occult HBV co-infection with HCV was associated with mutations in the MHR of HBV.

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Abstract has been withdrawn

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A RECOMBINANT TRUNCATED HEPATITIS B CORE ANTIGEN (HBCAG) AS VACCINE COULD EXERT THE PROPHYLACTIC AND THERAPEUTIC FUNCTION IN A MOUSE MODEL MIMICKING HEPATITIS B VIRUS INFECTION

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Background: Chronic Hepatitis B virus (HBV) infection remains a world-wide health problem, as no therapy available such as IFN-α, nucleos(t)ide analogs, could eliminate the cccDNA (covalent closed circular DNA) of the virus from infected hepatocytes.
Aims: Vaccination of therapeutic vaccine, stimulating the host immune response to control virus infection, is the most convenient method to solved the problem.
Methods: As the immunogenicity of HBcAg elucidated in previous reports, our study devised a recombinant fusion protein containing a truncated HBcAg and a moiety with biotin binding function, core streptavidin (cSA), termed tHBcAg-cSA. This fusion protein was expressed in large amount in *Escherichia coli*, after purification and renaturation, the moiety of the fusion protein could bind to biotin and form large complex. The fusion protein mixed with adjuvant polyIC as vaccine formulation, vaccinated subcutaneously into C57BL/6 mice before or after hydrodynamic tail vein injected with pAAV-HBV-GB1.3 replicon to mimic virus infection.
Results: The HBeAg and HBsAg level of the vaccinated group significantly reduced compared with control group.
Summary/Conclusions: In conclusion, this vaccine formulation containing recombinant truncated HBcAg exert both prophylactic and therapeutic function in the mouse model mimicking HBV infection.

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Abstract has been withdrawn

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PRECISION TRANSFUSION MEDICINE BY COMBINING GENOTYPING AND NAT SCREENING WITH AGENA MASSARRAY 4 PLATFORM

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Background: Recent advances in nucleic acid (genetic) tests have let to the possibility of using a single nucleic acid test platform with high PCR multiplicity and throughput for the blood genotyping as well as the screening for transfusion transmitted diseases in the donors. Typically, a blood center in China operates two sets of labs: one based on immunology and serology techniques for blood typing and chemiluminescence immunoassays for infectious diseases; and the other based on the NAT (nucleic acid amplification tests) techniques for infectious diseases. A NAT test is mandatory because it can detect traces amount of the DNA or RNA of the infectious agents before typical seroconversions can be detected in a donor's blood and thus represents a smaller time window of accidental inoculating infectious agents into transfusion recipients.
Aims: The Aims of this study are to explore the possibility of developing a new multiplexed PCR technique based on the MassArray® NAT platform and to combine it with the blood genotyping panels such that a single instrument might allow safe and precision transfusion medicine for transfusion and cell therapies.
Methods: We tested over 200 blood donor samples for HBV using the MassArray® NAT platform, which is based on MALDI-TOF analysis of the PCR products. We compared the NAT test results with those of the serologic and biochemical features of the samples that were found to contain only HBV DNA and performed threshold measurements.
Results: We identified those donors who were weakly positive for HBV in the serologic and biochemical tests were all confirmed with positive HBV DNA test using the MassArray technique.
Summary/Conclusions: Our tests indicated that the Agena MassArray 4 NAT test platform is a highly sensitive and specific NAT method that can perform both blood genotyping panels and infectious disease screening of the donor. The method is simple, high throughput and fluorescence contamination free.

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PREVALENCE OF HBV NAT REACTIVITY AMONG 6.5 MILLION SERONEGATIVE BLOOD DONATIONS BY COBAS® TAQSCREEN MPX TEST, V2.0 IN CHINA FROM 2015 TO 2016

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Background: Since 2010, blood banks in China have gradually begun to perform nucleic acid test (NAT) screening for Hepatitis B Virus (HBV) DNA, and Hepatitis C Virus (HCV) RNA, Human Immunodeficiency Virus (HIV) RNA, according to the recommendations set by the National Health and Family Planning Commission of China. Several different manufacturers' assays are used for NAT testing in China. In 2015 and 2016, more than 30% of the blood donations were tested by cobas® TaqScreen MPX Test, version 2.0 for use on the cobas s 201 system by Roche Molecular Systems (MPX2.0), which is a qualitative multiplex test that simultaneously detects HIV, HBV, and HCV.
Aims: As China is a highly endemic area for HBV infection, the aim of this study was to analyze the prevalence, trend and distribution of HBV NAT positive/seronegative blood donations by MPX2.0 in China for 2015 and 2016.
Methods: All the blood donations were first detected by two different ELISAs for HBV surface antigen (HBsAg), anti-HCV and anti-HIV and all seronegative donations were then screened by NAT. NAT screening with MPX2.0 was initially tested in minipools (MP) of 6 donation and reactive minipools were further tested by individual donation testing (IDT) to determine which donation(s) was reactive for a viral target. We collected NAT data from all blood banks in China from 2015 to 2016 and analyzed the prevalence of HBV DNA among seronegative blood donations that were screened by MPX2.0.
Results: 33.8% (6,522,333/19,277,036) of seronegative blood donations, from 21 provinces in China, were screened by MPX2.0 from 2015 to 2016; the average frequency of HBV DNA among seronegative blood donations was 1/868, which was slightly higher than the average level of 1/1083 in China. From the results, 10,078 MP were NAT reactive for HBV. 97.6% (9836/10078) of MP were HBV DNA reactive, among which 6,612 (67.2%) donors were HBV DNA reactive by individual donor test (IDT). In all the provinces that had used MPX 2.0 for NAT, Jiangxi province had the highest reactive rate of HBV DNA among seronegative donations (0.118%), while Hebei province was the lowest (0.048%).
Summary/Conclusions: With the high sensitivity of NAT screening in China, the risk of transfusion transmitted infections is decreased. MPX2.0 detected a higher average frequency of HBV DNA among seronegative blood donations (1/868) than the average level of 1/1083 in China. Our study also evaluated the prevalence of seronegative and HBV NAT reactive blood donations using cobas® TaqScreen MPX Test, v2.0 in China.

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SEX DIFFERENCE IN HBSAG SPECIFIC ANTIBODY RESPONSE AMONG HEPATITIS B VACCINE IMMUNIZED POPULATIONS

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Background: Hepatitis B virus infection has been a worldwide health problem. Hepatitis B virus vaccination is still regarded as the most economical and effective method for prevention and control of HBV infection. It seems to be that there is sex-difference in immune response to Hepatitis B vaccine in humans.
Aims: Due to this limited data, it is mandatory to assess sex-differences which might have clinical implications. This study aimed to conduct a systematic review and meta-analysis to assess a more precise estimation of sexual dimorphism of immune development and persistence to hepatitis B vaccine in humans.
Methods: HBV vaccination or anti-HBs or anti-HBs antibody and sexual dimorphism or sex-difference and antibody response combinations were used as medical subject headings for the searches. Total antibody concentrations were analyzed using mean difference with fixed-effects model for a low heterogeneity and random model for a high heterogeneity. And rates of antibody concentration of more than 100, response rates and total positive rates were evaluated using odd ratios (OR) with fixed-effects model or random model. Both fixed-effect model and random-effect model were used for the Mantel-Haenszel analysis. Our included studies examined seroprotection rates, anti-HBs mean geometric titers (GMTs), and anamnestic response with hepatitis B vaccine in male and female populations.
Results: This meta-analysis including 22,863 females and 19,852 males in 42 articles showed that a significantly decreased response to hepatitis B vaccine appeared in males seroprotection rates (OR = 0.75; 95% CI, 0.61 to 0.93; P = 0.007), anti-HBs mean geometric titers (GMTs) (MD = 14.25; 95% CI, 15.04 to 13.47; P < 0.00001) with hepatitis B vaccine.
Summary/Conclusions: Our study indicated that more vaccine dose, supplemental strengthening immunity might be emphasized in male populations.

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ANALYSIS OF HBV INFECTION AND RISK FACTORS AMONG LI MINORITY IN HAINAN

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Background: The infection rate of HBV in China is different in different regions, the lowest in the eastern region, the second in the central region, and the highest in the western region. In 1992, the national HBV epidemiological survey showed that the HBV infection rate in Hainan province was 84.71%, higher than the national standardized infection rate (78.27%), indicating that Hainan province is the high incidence area of HBV infection. In addition, the epidemiological survey also found that the HBsAg positive rate of Han population in Hainan province was significantly lower than that of ethnic minority. Li minority is the most important ethnic minority in Hainan Province, whose population is 88.4% of the total population of the minority nationalities. We speculate that the Li minority population in Hainan may have a higher rate of HBV infection, but there is no literature report. Therefore, the study of HBV infection status and related risk factors of Li minority in Hainan is of great significance to the local HBV prevention and control work.
Aims: To investigate the HBV infection and analyze the related risk factors among Li minority in Baisha country, Hainan Province.
Methods: A total of 1595 individuals of Li minority in Baisha county, Hainan Province were recruited by random sampling method from July 2014 to October 2015. Epidemiological data including baseline characteristics and risk factors were obtained. HBcAb was detected by chemiluminescence method. The difference on age between HBcAb positive and negative group was analyzed by *t* test. The effects of age, gender and related risk factors on HBcAb were analyzed by univariate chi-square test and multivariate logistic regression; P < 0.05 was considered statistically significant.
Results: The positive rate of HBcAb was 71.8% (1145/1595) and no significant difference between male and female was observed ($\chi^2 = 0.134$, P = 0.715). The difference of HBV infection among age groups was statistically significant ($F = 540.769$, P < 0.001). The HBV infection rate was 11.9% in the 12–17 year group, which was significantly lower than that in all the other age groups. The HBV infection rate in the 18–23 year group (28.0%) was significantly higher than that in the 12–17 year group, but significantly lower than that in the other age groups (>85%). Multivariate logistic regression analysis showed that alcohol consumption and tattoo were independent risk factors associated with HBV infection ($\chi^2 = 169.833$, P < 0.001; $\chi^2 = 11.354$, P = 0.001).
Summary/Conclusions: The prevalence of HBV infection of Li minority in Baisha county, Hainan Province is 71.8%. Subjects of 12–17 years old showed the lowest infection rate, followed by that of 18–23 years old, and subjects over 23 years old presented high HBV infection rate. Alcohol consumption and tattoo were independent risk factors associated with HBV infection.

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ROLE OF HLA-DP GENES AND MRNA EXPRESSIONS ON SUSCEPTIBLE OF HEPATITIS B INFECTION

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Background: Hepatitis B virus (HBV) affects approximately 68 million people in China. 10–15% of patients with chronic HBV will develop liver cirrhosis (LC), liver failure and hepatocellular carcinoma (HCC). Chronic HBV infection is influenced by both viral and host factors. In genome-wide association studies (GWAS), the human leukocyte antigen DP (both HLA-DPA1 and HLA-DPB1) gene and their related

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polymorphisms rs3077, rs9277534 and rs9277535 were identified to be associated with susceptibility to hepatitis B infection. HLA genes have been linked to immune response to infectious agents. Genetic variants in HLA genes influence HLA mRNA expression which might also affect antigen presentation and immune response.

Aims: We evaluated the association between HLA gene polymorphisms and the risk for susceptive HBV infection.

Methods: In this study, HLA-DPA1 and HLA-DPB1, rs3077, rs9277534 and rs9277535 polymorphisms were investigated in 168 chronic HBV patients and 217 healthy controls (HC) from Sichuan by using a sequence based typing. The HBV viral load and mRNA of HLA-DPA1 and HLA-DPB1 were measured by real-time polymerase chain reaction (RT-PCR).

Results: The results shown that HLA-DPA1, HLA-DPB1 gene, rs3077, rs9277534 and rs9277535 were all associated with HBV infection in the Sichuan Han populations. DPA1*01:03, HLA-DPB1*04:01 and DPB1*04:02 play a protected role in HBV infection, DPA1*02:02, DPB1*05:01 prone to susceptible to HBV infection. The HBV viral load was significantly lower in rs3077TT compared than rs3077CC, however, the HBV viral load have not association with rs9277534 and rs9277535. Additionally, the 3077CT, 3077TT, 534AA/535AA have significantly lower HLA-DPB1 mRNA expression in HBV group compared that in HC group. DPA1*01:03 and DPA1*02:01 have a significantly lower HLA-DPA1 mRNA expression in HBV infection group compared than HC group. However, DPB1*04:01, DPB1*02:02 and DPB1*14:01 have a significantly higher HLA-DPB1 mRNA expression in HBV infection group compared than HC group.

Summary/Conclusions: It's the first time in our results demonstrate that HLA-DPA1 and DPB1 genes and rs3077C, rs9277534G and rs9277535G allele has a major effect on the risk of HBV infection. rs3077CC with higher HBV viral load, lower HLA-DPA1 and higher HLA-DPB1 expression have a risk of susceptive HBV infection. We suggest that further independent studies should be necessary to strengthen the association of these polymorphisms with susceptible of HBV infection in Chinese populations.

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ANALYSIS OF MONOCYTIC AND GRANULOCYTIC MYELOID-DERIVED SUPPRESSOR CELLS SUBSETS IN DONORS WITH DIFFERENT OUTCOMES AFTER HBV INFECTION

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Background: Myeloid-derived suppressor cells (MDSCs) are a group of heterogeneous cells that have not fully differentiated and could suppress bone marrow-derived immune response. MDSCs have been proved to inhibit T-cell responses in many diseases including hepatitis.

Aims: We aimed to study the correlation of monocytic- and granulocytic-MDSCs (M-MDSCs and G-MDSCs) in peripheral blood mononuclear cells (PBMC) with HBV infected patients and explored the clinical significance of each subset in these patients.

Methods: The study population composed of 31 chronic hepatitis B patients (CHB), 44 spontaneous resolvers (SR), 34 vaccinated donors (VD) and 22 seronegative donors. PBMCs were freshly isolated from whole blood by Ficoll centrifugation and MDSCs were immediately analyzed by flow cytometry (BD canto II) with the following anti-human antibodies: CD11b-FITC, CD33-PE, HLA-DR-PE/Cy5, CD14-PE/cy7, CD15-BV421. The levels of MDSCs between groups were compared by a non-parametric Mann-Whitney U test. Correlations between MDSC and other parameters were analyzed using Spearman's rank test. Statistical analysis were performed using SPSS 16.0 and a P-value of less than 0.05 was considered to be statistical significant.

Results: We found that the frequencies of M-MDSCs and G-MDSCs from CHB patients were significantly increased when compared to seronegative donors (M-MDSC, 0.7% ± 0.6% vs 0.4% ± 0.2%, P = 0.015; G-MDSC, 0.9% ± 0.4% vs 0.7% ± 0.4%, P = 0.025) and VD or SR (data not shown). Besides, the level of MDSCs in PBMCs of CHB patients did not correlate with age, aspartate aminotransferase (AST), alanine aminotransferase (ALT), HBV viral load, serum HBsAg titer or HBeAg status.

Summary/Conclusions: Both G-MDSCs and MDSCs elevated in CHB patients, indicated a role of MDSC in the host immune responses against HBV infection. Since the level of MDSCs did not correlated with HBV DNA load, it appeared that the elevation of MDSCs associate with the immune response caused by the infection rather than the level of active viral replication.

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THE ESTABLISHMENT OF METHOD FOR PREPARATION OF ANTI-HBSAG IGG-BINDING BACTERIAL MAGNETIC NANOPARTICLES

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Background: Recently, new bacterial magnetic nanoparticles (BMPs) synthesized by magnetotactic bacteria have recently drawn great interest due to their unique features. BMPs are interesting as potential carriers for antibodies, enzymes, ligands, nucleic acids, and chemotherapeutic drugs.

Aims: In this study, we used BMPs as carriers of the monoclonal antibodies against HBsAg and prepared BMPs-antibody complexes for the detection of HBsAg.

Methods: BMPs-antibody complexes were prepared at weight ratio of 2:1 (BMPs to antibody) and then added to 0.01× phosphate buffer saline (PBS). Glutaraldehyde was added to a final concentration of 0.25% (V/V). The resulting mixtures were incubated at 25°C for 12 h, and then the desired product was collected as pellet by magnetic separation. Washing buffer (0.01 × PBS) was added, and the washing treatment is repeated three times to remove unbound antibody and glutaraldehyde. The conjugated antibodies were identified by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). The immunological characteristics of the BMPs-antibody complexes were verified by western blot, and the biological activity of antibodies on BMPs was analyzed by enzyme linked immunosorbent assay (ELISA).

Results: The results from SDS-PAGE showed that protein from BMPs and anti-HBsAg IgG antibody complexes had the same band as the standard anti-HBsAg IgG antibody, however, magnetosome membrane protein of BMPs without binding antibodies lacked this band. Western blot and ELISA suggested that BMPs did not affect the biological activity of conjugated antibodies.

Summary/Conclusions: Anti-HBsAg IgG-binding BMPs obtained by this approach could be further applied for the detection of HBsAg.

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HBV INFECTION CHARACTERISTICS OF HBSAG-/ HBV DNA+BLOOD DONORS IN SOUTHWEST OF CHINA

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Background: The real infection status and characteristics of hepatitis B virus of HBsAg negative/HBV DNA positive blood donors is very important for improving blood screening strategy to guarantee blood safety but is difficult to clarify for which supplemental testing and follow-up study are needed.

Aims: To study the real infection status and characteristics of hepatitis B virus of HBsAg negative/HBV DNA positive blood donors in Southwest of China.

Methods: A total of 191HBsAg−/ HBV DNA+ blood donors from three blood centers in the southwest of China were included in this study. Epidemiological questionnaire survey and supplemental testing including high sensitive HBV DNA viral load testing and chemiluminescent immunoassay (CLIA) for HBsAg, anti-HBs, anti-HBc, HBeAg and anti-HBe were conducted. Donors' infection status is defined by the follow-up study, of which some without follow-up was conducted according to the supplemental testing of the original samples. Some of the samples were tested for HBV DNA viral load for six or three times repeatedly to study the possibility of samples with low viral load missed by HBV DNA testing.

Results: Of the 191 donors included in this study, 64.40% (123/191) were male, 81.15% (155/191) were Han, 58.64% (112/191) were above 41 years old, 29.32% (56/191) were below high school education level, 64.92 (124/191) were married and 67.54% (129/191) were repeat donors. 78 donations (40.84%, 78/191) were negative while 113 (59.16%, 113/191) were positive for HBV DNA by high sensitive HBV DNA viral loading testing in which 78(69.03%, 78/113) with a viral load lower than 20 IU/ml. Detailed data was listed in Table 1. Follow-up study was conducted on 36 blood donors, of which, the HBV DNA viral loading of 11 (30.56%, 11/36) changed from positive to negative, and 3 (8.33%, 3/36) changed from negative to reactive. According to follow-up study and supplemental CLIA results, 73 donors were infected with HBV with 16 window period infection (WP), 46 occult HBV infection (OBI) and 11 transient infection. Only 4 (14.81%, 4/27) of the 27 samples were positive for HBV DNA by all repeat testing.

Summary/Conclusions: A longer time follow-up study is needed to clarify the infection status of HBsAg-/HBV DNA+ blood donors. Most of the HBsAg-/HBV DNA+ blood donors are of occult HBV infection and the viral load of HBV DNA is

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very low which may result in the false negative results of HBV DNA testing. Thus, it's very important to improve the HBV DNA assays or adopt repeat HBV DNA testing to increase the rate of detection of OBI donations so as to decrease transfusion transmitted HBV infection by OBI donations.

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SCREENING AND MOLECULAR BIOLOGY RESEARCH ON BLOOD DONORS OF OCCULT HBV INFECTION

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Background: Occult Hepatitis B Virus infection (OBI), a special Hepatitis B Virus infection, which could not easily be detected by conventional ELISA and chemiluminescence method but can be identified by nucleic acid detection characterized by low level of HBV DNA expression, can cause residual risk of HBV.
Aims: To understand the OBI prevalence among blood donors as well as to analysis its molecular cause of occult infection.
Methods: All blood donation samples went through dual ELISA for HBsAg, anti-HCV, anti-HIV and anti-Treponema pallidum and transaminase test, those ELISA negative and ALT less than 50 U/l were further screened by nucleic acid test(NAT) for HBV/HCV/HIV. ELISA negative and HBV DNA positive ones were quantity determined and samples of HBV DNA between 20 and 1000 copies/ml were tested by ECLIA for anti-HBc to choose anti-HBc positive ones for nested PCR. Pre-S and S region of HBV DNA were amplified, sequenced and compared with standard virus sequence, the translation products of which were also compared with standard S protein for genotyping and "a" determinant mutation analysis.
Results: The incidence of occult HBV among blood donors was 0.025‰ (39/157985). The positive yield of nested PCR for Pre-S was 35.9% (14/39). The uniform sequencing results indicated no mixed infection. Several mutation sites, S45D, N68D, Q129R, F134N, V1805, V1901, were found to be related with HBsAg negative. "a" determinant mutation associated with HBsAg not detected or low secretion was not observed.
Summary/Conclusions: The main cause of OBI may be due to low replication of HBV, deficiency of low sensitivity of reagent and virus mutation.

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Abstract has been withdrawn

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CORRELATION BETWEEN HBV-DNA AND HBSAG DETECTION RESULT OF VOLUNTARY BLOOD DONORS IN TIANJIN AREA

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Background: In China the prevalence of hepatitis B is much higher than world level. And the situation of HBV infection in blood donors is complicated. How to prevent the infection of HBV via transfusion is crucial. Although NAT has been widely adopted by blood banks, the traditional ELISA still plays an indispensable role in the screening of HBV.
Aims: To investigate the correlation between the detection results of HBV-DNA and HBsAg of voluntary blood donors.
Methods: Using TMA-chemiluminescence method to detect HBV-DNA of 323 HBsAg positive samples which tested by ELISA.
Results: In 323 cases with HBsAg positive, 7 cases (2.17%) were positive in HBV-DNA; HBV-DNA positive rate of HBsAg S /CO ratio 1.00–1.99 group and 2.00–4.99 group were respectively 2.09% and 2.86%. None HBV-DNA positive samples were found in HBsAg S / CO ratio $\geq$ 5.00 group. The statistical difference between the groups is significant ($P < 0.05$).
Summary/Conclusions: The positive rate of HBV-DNA has some correlation with HBsAg S/CO ratio in blood donors; In China traditional ELISA method still plays an indispensable role in the screening of hepatitis B virus.

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THE ANALYSIS OF HEPATITIS B VIRUS SCREENING BY NUCLEIC ACID MIX-TESTING (NAT)

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Background: Ningxia, an autonomous area of Hui Muslim minority, and it has its own demographic characteristics。To ensure the blood safety, centralized nucleic acid testing has been conducted in the whole area since 2015, Two kinds of reagents were used for serological examination, NATs were then carried out for the serological negative samples.
Aims: To measure the detection of HBV by NAT in Ningxia Hui minority areas. To provide the basis for a rational scientific screening and a strategy for recruiting blood donors.
Methods: A total samples of 134,495 were collected from volunteer blood donors between 2015.1-2016.12.31. The HBsAg was tested twice by WanTai and BIO-Rad Genscreen™ ULTRA HIV Ag/Ab reagents. Negative samples were selected for measurement by NATs using Roche Cobas s201 nucleic acid testing kits. The testing results were collected, grouped, statistically analyzed using SPSS 21.0 software.
Results: Out of a total samples of 134,495 from volunteer blood donors, the positive rate of HBsAg was 0.23% when measured with imported reagents of Bio-Rad; the positive rate of HBsAg was 0.08% with domestic reagents of WanTai; the positive rate of HBsAg was 0.11% with the dual reagents. An overall deferral rate is 0.42% (566/134495) by serological examinations. The detection rate of HBsAg by the imported reagent was significantly higher than that of domestic reagent. The ratio of dual positives accounted for 26.3% (149/566) of all HBsAg's deferral. A sample of a combined mix of 6 serological negative samples were carried out. A total of 80 HBV-DNA positive specimens were detected. The detection rate is 0.06% (80/133744). The 80 specimens were also detected independently by omission of serological examination (omission factor is 0.06%, 80/134495).
Summary/Conclusions: To some extent, NAT can detect serological false-negative HBV samples, in particular, during HBV occult and window period infections. Secondly, the rate of false-positive was higher in serological tests, which should be confirmed, and recruitment of false-positive blood donors is necessary to reduce the waste of blood.

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ANALYSIS OF CENTRALIZED DETECTION RESULTS OF VIRUS NUCLEIC ACID "11+1" PATTERN IN XINJIANG REGION

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Background: Nucleic acid amplification (NAT) is a direct assessment of pathogen nucleic acid on the basis of molecular biology, as a supplement to the enzyme-linked immunosorbent assay (ELISA) blood test, greatly shortening the detection of "window period", reducing blood transfusion Spread the residual risk of disease. Blood centralized detection As a high-quality, high-efficiency, low-cost operation and management model, has been recognized around the world, and developed in the country for many years. China National Health Commission in 2006 promulgated the "one law two rules", issued in 2013, "blood bank set planning guidelines", "to promote blood bank detection of 3 years Action plan "and other clear that the blood center laboratory should undertake regional centralized detection tasks, the provincial health and family planning departments should be based on the number of service population, the number of blood supply and drainage, geographical characteristics, traffic conditions, blood distribution density and detection technology level, The overall plan to assume the task of blood centralized detection of the central blood station. Urumqi Blood Center to assume "11 + 1" centralized detection model, that is, a blood center, 11 central blood stations, Xinjiang vast territory, 11

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blood to the blood center distance of a total of 4200 km, through the detection data analysis Urumqi Blood Center Commissioned centralized testing to improve the quality of laboratory and blood safety.

Aims: Through the analysis of the "11 + 1" centralized test results, the paper discusses the application of centralized testing in Xinjiang, China and the problems to be solved.

Methods: (1) All samples have been excluded from the use of two different manufacturers of ELISA reagents for unpaid blood donors HBsAg, anti-HCV, anti-HIV detection of reactive samples; (2) using Roche nucleic acid blood sieve detection system or Kewei nucleic acid blood sieve HBV DNA, HCV RNA, and HIV RNA were detected by the control system (mixed with 6 mixed or 8). (3) The mixed samples were tested for dissolution.

Results: From December 2015 to July 2017 "11 + 1" centralized detection model were detected 130024 samples (blood center sample 84508, 11 blood pool centralized sample 45516), 25790 pool, split 199 pool, split out 123 (3 copies of anti-HCV, 2 HBsAg), of which 119 were HBV DNA positive, 4 were HCV RNA positive, and 4 were positive for HBV samples. The positive rate of blood samples in Urumqi was 0.07% (55/84508). The positive rate was 0.15% (68%), and the positive rate was 0.15% (68/130024) /45516).

Summary/Conclusions: In Xinjiang, China, the number of blood samples in Xinjiang is very small, only 1–2 times a week to carry out experiments, "11 + 1" centralized testing can reduce the risk of blood transfusion, improve blood transfusion safety, and in reducing costs, reporting in a timely manner There are advantages, but the sample transport cold chain control also need to strictly control.

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BLOOD SCREENING FOR HBV DNA: THE VALUE OF EQA PROGRAM IN CONTINUOUS PERFORMANCE IMPROVEMENT FOR PARTICIPANTS, COMMERCIAL KIT AND THE EQA ORGANIZER

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Background: Hepatitis B virus (HBV) DNA nucleic acid test (NAT) was implemented in blood banks in China since 2010. Since different HBV NAT assays are used by different blood banks, we do not know the performance of each laboratory, let alone their performance.

Aims: This study analyzed the results of an external quality assessment (EQA) from 2015 to 2017 for quality improvement in participating laboratories.

Methods: Each panel, consisting of 19 positive samples ranging from 28.3 IU/ml to 648 IU/ml and 31 negative samples made by National Center for Clinical Laboratory (NCCL), was deployed in April and October each year from 2015 to 2017. A total of 5 panels were tested during this period. The participants were required to test these EQA samples in the same way as donor samples. Concordance rate and cycle threshold (Ct) value in EQA programs were analyzed. For discrepant results, a thorough investigation and corresponding corrective actions were conducted to achieve quality improvement in these laboratories.

Results: A total of 1024 EQA reports were received from 349 laboratories during the 3 years. Firstly, we analyzed the concordance rate for the positive and negative samples. The overall concordant rate for positive samples was 97.3% and negative samples was 97.9%, therefore, around 2% of the participants had inaccurate results. After analyzing the results based on the commercial kit used, the overall concordance rate of the commercial kits for negative samples was higher than 97% while commercial kit 55001, 55081 and 55178 had the lowest concordance rate for positive samples, at 91.6%, 91.3% and 93.0% respectively, suggesting their poor intrinsic performance. EQA panel 161 had the worst concordance rate for negative samples due to the contamination of sample 1611 during sample aliquoting by operators in NCCL. EQA panel 162 had the worst concordance rate for positive samples due to the low concentration of HBV in sample 1624 (28.3 IU/ml). Secondly, Ct values by individual donor (ID)-NAT on positive samples were also analyzed. Ct values of participants that used the same commercial kit were analyzed and Ct values that were out of 2 standard deviation (SD) were defined as outliers. Only data of the 3 widely used commercial kits were analyzed. 2 EQA participants with outlier Ct values were selected in each of these 3 commercial kits. After thorough investigation, 2 laboratories in 55001 group and 1 in 55139 group also had higher Ct values in their routine work compared with their control laboratories, suggesting their consistent poor performance in routine work. This could be due to the delay in sample transportation or processing and should be investigated. 1 laboratory in 55139 group and 2 in 55182 group had approximate Ct values in routine work compared with their control laboratories, whose outlier Ct value in EQA program may be a one-off result. This could be due to the operating personnel (the laboratory in 55139 group), lack of ID-NAT (one laboratory in 55182 group) and occasional cases (the other laboratory in 55182) which was confirmed by telephone interview. For the commercial kits, 55182 has the lowest SD of Ct values on the same sample. For EQA organizer, stringent process should be practiced to prevent contamination during EQA sample preparation. The first and last tube after sample aliquoting, should be verified before deploying to participants.

Summary/Conclusions: As a whole, EQA program could provide a valuable quality improvement tool for participants, commercial kit and the EQA organizer.

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SEROLOGICAL AND MOLECULAR CHARACTERIZATION OF HEPATITIS B VIRUS INFECTION IN BLOOD DONORS BORN BEFORE AND AFTER IMPLEMENTATION OF UNIVERSAL HBV VACCINATION IN CHINA

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Background: Neonatal hepatitis B vaccination program at birth has been implemented nationwide since 1992 in China, which may impact HBV safety in blood donations. With more sensitive triplex nucleic acid tests (NATs) implemented for HBV screening after EIA in Chinese blood center, Testing discrepancies between NAT and EIA are needed to confirm.

Aims: To investigate hepatitis B virus infections between blood donors born before and after the universal HBV vaccination program in Shenzhen, and analyze the serological and molecular Characterization.

Methods: A total of 242 reactive by NAT and/or HBsAg EIA samples from 26320 candidate blood donors were detected, the discrepant results were tested with commercial ELCI: nested PCR;sequencing and a quantitative real-time polymerase chain reaction (PCR) assay.

Results: 132(85.7%) of 154 EIA HBsAg+ samples were confirmed HBsAg positive, The HBsAg confirmed positive rate (0.73%) in first time presuming vaccinated donors is lower than in non-vaccinated donors($P < 0.001$),195 of 242 EIA HBsAg+ or NAT initial reactive samples were confirmed HBV DNA positive, the total positive rate of HBV DNA was 0.741% (195/26320). The positive rates of HBV DNA in the non-vaccinated blood donors (0.78%) is higher than in the vaccinated blood donors (0.56%, $P < 0.05$). The positive rates of Occult hepatitis B virus infection (OBI) in the presuming vaccinated donors was 0.093% (6/6422), and significantly lower than 0.256% (51/19898) in the non-vaccinated donors ($P < 0.05$).95/120 samples are genotype B, 25/120 are genotype C. There is no significant difference in the distribution of genotype between presuming vaccinated donors and non- vaccinated donors ($P > 0.05$).

Summary/Conclusions: HBsAg EIA and Highly sensitive NATs require further Confirmatory testing. The universal HBV vaccination program markedly reduces the risk of HBV infection in blood donors, and provides an effective guarantee for the safety of blood transfusion.

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SEROLOGICAL STATUS ANALYSIS ABOUT 31 CASES OF SPECIMENS WHICH WERE NUCLEIC ACID REACTIVE

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Background: Our country is hepatitis B superpower. Chinese people who carry the hepatitis B surface antigen was 7.18%. Due to the effects of ELISA method, the longer window period and virus subtype variation was the bottleneck of further improve blood safety. Nucleic acid detection of virus nucleic acid amplification directly had better sensitivity and specificity. Nucleic acid detection increasingly popular in the blood detection, played an important role in our country who had a high rate of HBV, especially for avoiding window or occult HBV infection spread of hepatitis blood transfusion. In OBI, anti-HBc usually said after infection acquired immunity together with anti-HBs. In some countries such as Germany and Japan blood was safe, which anti-HBs was detected more than 100 IU/l. Anti-HBs in 20 to 160 IU/l could reduce five times the blood transfusion infection. In HBV infection high prevalence area, taking anti-HBc and anti-HBs detection may be play a important role in clinical blood use.

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Aims: To ensure the safety of clinical blood transfusion, analysis the serological status about 31 cases of specimens which was nucleic acid reactive .After take the nucleic acid detection mode, assessment to reduce blood transfusion residual risk of HBV transmission.
Methods: 59 762 cases of blood donors (excluding ones of the same positive in ELISA, ALT) were detected by NAT and CLIA.
Results: Nucleic acid was detected in 31 cases of DNA HBV reactive specimens, 0 cases of RNA HIV/RNA HCV reactive specimens. 1 case in ELISA method was Higher "negative value" specimens. In CLIA method detection, 10 cases who no response was suspected the window period specimens .The rate of anti-HBc reactivity was 67.7% (21/31) which suspected the OBI specimens .OBI main serotype mode (HBsAg-/anti-HBs + / anti-HBc +) and (HBsAg-/anti-HBs+ / HBe + /anti-HBc +) were 25.8% (8/31) and 25.8% (7/31). According to the relationship between anti-hbs droplet and blood safety of Germany and Japan, the safe blood of > 100 IU/l was found to be 16.1% (5/31). Anti-HBs titer (20-100) IU/l which could reduce blood transfusion infectivity was found to be 16.1% (5/31). The residual total number of 67.7% (20/31) was completely cut off HBV blood transmission by nucleic acid detection.
Summary/Conclusions: First, blood donors detect methods in NAT and ELISA method was complementary, NAT detection can make up the "window period" of ELISA method. Second, NAT and CLIA test results were consistent; Third, OBI may be the main types of HBV DNA NAT detected blood donors in handan area. Fourth, the "negative value" high specimen in ELISA, which affected the quality of blood, reduced risk of blood transfusion infection risk. Fifth, Healthy blood donation population should be encouraged to regular physical examination and inject vaccination of hepatitis B, increased Anti-HBs drops. Sixth, It is suggested that the Anti-HBs detection in the high epidemic area of hepatitis B may be of great significance to further guarantee clinical blood safety.

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THE ANALYSIS OF ABNORMALITY SCREENED IN FREE BLOOD DONATION

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Background: According to the health inspection requirements of blood donors in China, pre-screening test is conducted for blood donors. The main items used for blood donation are HBsAg, ALT, blood type, hemoglobin
Aims: To review the unqualified causes screening of blood donors in Nanchang, and put forward countermeasures to avoid the waste of blood resources
Methods: For the period from May 1, 2015 to May 31, 2016, the total unqualified rate of total and component screening among the whole blood donors and the platelet donors, student and non-student groups, and primary and repeat blood donors
Results: The positive rates of primary screening, HBsAg and ALT were significantly different between the students and the non - students. There was no significant difference in the rate of blood specific gravity. There were significant differences in the total failure rate and HBsAg positive rate between the first and the repeat blood donors, and the ALT failure rate had no significant difference. There were significant differences in total failure rate and HBsAg positive rate between the whole blood donors and the platelet donors, and no significant difference in ALT failure rates
Summary/Conclusions: To formulate a suitable pattern is an effective way to improve the safety of blood transfusion for different types of blood donors. We may consider abolishing the HBsAg gold standard method test screened in team donation of the repeat blood donors, and the abolition of ALT screening test of student groups in team donation.

P-300

ENHANCED DETECTION OF HEPATITIS B VIRUS (HBV) DNA BY A MORE SENSITIVE COMMERCIAL POLYMERASE CHAIN REACTION ASSAY IN HONG KONG BLOOD DONORS

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Background: With the implementation of nucleic acid testing (NAT) for HBV DNA on individual donor samples, a subset of donations with HBsAg positive HBV DNA negative results were identified. They accounted for 8.5% and 4.0% of all confirmed HBV infections in donations screened by NAT using Procleix ULTRIO and ULTRIO Plus assays on the TIGRIS platform (Grifols, Emeryville, CA) during 2007–2009 and 2009–2011, respectively.
Aims: (1) To evaluate the performance of the cobas® MPX test on the cobas® 6800 System (Roche, Pleasanton, CA); and (2) To characterize the serological profile of these HBsAg+ HBV DNA negative samples in the Hong Kong Chinese donor population.
Methods: Fifty-five archive samples (frozen at −18°C) collected from 2011 to 2014 with discordant serological and NAT results (i.e. PRISM HBsAg (Abbott, Abbott Park, IL) positive, confirmed by neutralization; and NAT non-reactive (either non-reactive by IDT screening or non-discriminatory by dHXV, ULTRO Plus)) were tested with cobas® MPX on the cobas® 6800 System. Repeat testing was performed in initial non-reactive samples if sample volume permitted. The archive samples were also tested for extended HBV serological markers, i.e. anti-HBs, IgM and total anti-HBc, HBeAg and anti-HBe.
Results: With regard to demographic data, 27 (49.1%) were from men and 28 (50.9%) from women; 50 (90.9%) from first-time donors, 3 (5.5%) repeat donors and 2 (3.6%) lapsed donors; mean age and SD were 32.8 and 10.0 years respectively. 25/55 samples (45.5%) were reactive for HBV on cobas® MPX on the first run. Of 30 initial non-reactive samples on first run, 25 had sufficient volume for retesting; of which, 6 were tested reactive on the second run. For the other serological markers: (1) anti-HBs: < 10 mIU/ml (51/55, 92.7%), (2) IgM anti-HBc: negative and total anti-HBc: positive (55/55, 100%), (3) HBeAg: negative (54/54, 100%), (6) anti-HBe: positive (47/53, 88.7%). The ratios of sample readings to cut-off value (S/CO) generated from PRISM HBsAg ChLIA for the 2 groups of sample (HBsAg+ MPX initial reactive group and HBsAg+ MPX initial non-reactive group) were significantly different (P = 0.0036). The former group (n = 25) had a median S/CO of 156.11 (Range: 1.23–537.93) and the latter (n = 30) median S/CO of 47.72 (Range: 1.17–505.27).
Summary/Conclusions: In Hong Kong, when donations were screened by a highly sensitive ID-NAT, 4.0% of all confirmed HBsAg+ donations had HBV DNA levels undetectable by ULTRIO Plus. The serological pattern of these donations was overwhelmingly homogeneous: HBsAg+, anti-HBs <10 mIU/ml, IgM anti-HBc-, total anti-HBc+, HBeAg-, anti-HBe+, consistent with chronic infections, where variably large amounts of S-protein are produced and circulated as free protein or aggregated in pseudo-particles, which are detected as HBsAg. With better NAT reagent performance, more of these samples would test DNA positive suggesting that the discordant results are mostly related to analytical sensitivity of the test system. In this respect, cobas® MPX detects more HBV samples with low viral loads. PRISM HBsAg S/CO values were higher in the MPX+ group when compared with the MPX- group, but a bright-line S/CO value could not be defined to predict MPX results.

P-301

EPIDEMIC AND MOLECULAR CHARACTERIZATION OF OCCULT HEPATITIS B VIRUS INFECTION IN BLOOD DONORS OF NORTHERN CHINA

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Background: Occult hepatitis B virus infection (OBI) in blood donor is focused on since nucleic acid testing (NAT) was implemented in blood screening to reduce transfusion transmitted hepatitis B virus infection. Though quite a few studies had involved in OBI blood donor around the world, the prevalence and characterization of OBI in blood donors of Northern China had not been well revealed due to the diversities among different types of HBV.
Aims: To explore the prevalence and analyze the characteristics of hepatitis B virus S gene of occult hepatitis B virus infection in blood donors of Northern China.
Methods: All donors were screened by rapid assay of HBsAg before donation and eligible donors were tested with two HBsAg ELISA kits and Cobas Taqscreen MPX in 6-samples mini-pool or Procleix Ultrio ID for HBV DNA. NAT-yield donations were discriminated the type of virus. Follow-up and archive samples of HBV DNA+/HBsAg- and NAT non-repeat reactive donors were further tested serologic markers of HBV by electrochemiluminescence assay and individual NAT. HBV of OBIs confirmed by follow-up or/and trace back were tested virus load and sequenced S gene as well. HBV S genes of OBIs were compared with that of HBV DNA+/HBsAg+ donors paralleled in Dalian Blood Center.
Results: 158 232 donations were screened between 2rd Dec 2015 and 31th May 2015. Follow-up and trace back of 80 donors among 169 NAT-yields were completed by the end of December 2015. 106 HBV infected donors were confirmed: 79 OBIs (1 in 2003 donations), 5 serological window period infections, 11 acute infections and HBV infective status uncertain in 11 cases. Sero-positivity was observed in all OBIs. 74/79 (93.7%) were anti-HBc+ while 5/79(6.3%) were anti-HBs+ only.

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Fifty-eight S gene sequences of OBIs were obtained. HBV type C and type B were dominant (79.3% and 13.8%, respectively). OBI_C strains showed a significant higher genetic diversity comparing with strains from HBV DNA+/HBsAg+ donors matched ($P < 0.05$), which differed from OBI_B ($p > 0.05$). Several obvious variations in single amino acid sites were observed: T47K/V/A, Q101R/K/H, T118K/R/A/M, K122R/N, P127T/L/H/S, E164D/G, S174N in type C and S31N, Y161F/S, V168A in type B.
Summary/Conclusions: The prevalence of OBIs was observed 1:2003 in blood donors of Northern China. Occult HBV genotype C strains which were dominant showed a higher genetic diversity in gene S. Some variations in single amino acid sites were suspected to be related with HBsAg secretion.

Hepatitis C (HCV)

P-302

THE ASSOCIATION OF CLASS I HLA ALLELES WITH THE NATURAL OUTCOME OF HCV INFECTION AMONG BLOOD DONORS FROM GUANGZHOU, CHINA

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Background: China is one of the countries with high prevalence of hepatitis c virus (HCV) infection. It is reported that about 30 million Chinese people were HCV antibody positive. Infection of HCV have two divergent natural outcomes: spontaneous clearance of the infection occurred in 10–40% of the persons, while the majority may develop persistent infection. Although the mechanisms are not thoroughly elucidated, it has been well-established that the natural outcome of HCV infection is closely associated with host genetic factors.
Aims: The purpose of our study is to understand the correlation between the polymorphisms of class I human leukocyte antigen (HLA) molecules and the natural outcome of HCV infection in Guangzhou blood donors.
Methods: 600 blood donors infected with hepatitis C were randomly selected in Guangzhou blood center. They were divided into two groups: spontaneous clearance group (n = 200) and chronic infection group (n = 400). High-resolution genotyping of classIHLA alleles were performed using HLA-SBT technology. The difference of the frequencies on classIHLA alleles between spontaneous clearance group and chronic infection group was analyzed to identify possible genes that affect natural outcome of HCV infection.
Results: Our results showed that the frequencies of seven HLA alleles were significantly different between the two groups. Five genes (A*02:01[10.3% vs 6.1%, P = 0.011], A*30:01[5.3% vs 2.9%, P = 0.039], B*13:02[5.8% vs 2.9%, P = 0.014], B*15:27[2.3% vs 0.5%, P = 0.013] and C*04:01[5.5% vs 3.1%, P = 0.046]) appeared to be more frequent in spontaneous clearance group. Frequencies of two alleles (A*24:02[11.3% vs 15.6%, P = 0.04] and B*40:01 [11.3% vs 15.6%, P = 0.04]) was higher in the chronic infection group. Logistic regression analysis showed that age ($P = 6.32 \times 10^{-8}$, OR = 1.039), gender ($P = 2.61 \times 10^{-10}$, OR = 0.397) and A*02:01 allele (P = 0.008, OR = 0.54) were independent factors of natural outcome of HCV infection.
Summary/Conclusions: Younger people, women and individuals with A * 02:01 allele were more likely to clear the virus after infection with HCV.

P-303

IDENTIFICATION OF SIX HEPATITIS C VIRUS RECOMBINANTS IN BLOOD DONOR

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Background: Recombination is an important evolutionary process for many RNA viruses, such as influenza and human immunodeficiency virus. Similarly to most RNA viruses, Hepatitis C virus (HCV) displays very high levels of genetic diversity which have been used to differentiate six major genotypes and more than 80 subtypes. However, HCV recombination is considered to be rare. In the previous study, we found sixteen HCV samples from blood donors with different genotype or subtype in Core and NS5B area by phylogenetic analysis. Its suggests that there may be recombination event in those HCV genome.
Aims: Identification Hepatitis C Virus Recombinant events in blood donor.
Methods: HCV RNA was extracted from serum samples of sixteen blood donors with anti-HCV positive. Core and NS5B areas of viral nucleotide sequences were amplified using RT-PCR. Purified productions of PCR were sequenced and sequences were aligned using MEGA4 with reference HCV strains, and all sites including gaps were removed from the alignment. Recombination event analysis was performed with RDP4.85 software. In this paper, the recombination analysis was based on Core and NS5B regions of HCV genome sequence. Recombination in complete genome sequences will be further detected.
Results: Six reconstruction events were detected in sixteen samples. Four recombination breakpoints located in Core region and two in NS5B. Recombination forms included inter-genotypes and intra-genotypes, such as 1b/1b and 3a/1b.
Summary/Conclusions: Although HCV recombination event may not appear to be common in blood donors, a more intense surveillance of recombination event in HCV is necessary. We can better understand this important blood pathogen's evolution, epidemiology, natural history, replication and functional biology by the recombination analysis.

P-304

THE RESEARCH ON THE ORIGIN AND TRANSMISSION PATTERN FOR HCV 6A IN CHINA

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Background: China is one of the countries where hepatitis C virus (HCV) is increasingly epidemic which has imposed a major health problem. Studies have shown that HCV 6a is a relative new subtype in China.
Aims: To reveal the distribution, origin, and transmission pattern of HCV 6a subtype in China.
Methods: 50 HCV RNA positive samples for each province were collected from 29 provinces all over China. E1 and NS5B fragments of HCV were amplified by RT-PCR, followed by DNA sequencing and phylogenetic analysis to determine the accurate genotypes. Bayesian coalescent analysis integrated in BEASTv1.8.0 was utilized to reconstruct the phylogeographic tree for subtype 6a to detect the origin, transmission pattern and trend of HCV 6a subtype. The Bayesian factor (BF) was calculated by Tracerv1.5 program to compare the models in order to choose the statistically more robust one. We also used Tracer program to reconstruct the BSP (Bayesian Skyline Plot) and calculate the tMRCA (Time of most recent common ancestor).
Results: HCV 6a was found to concentrate in the South region (including Guangdong, Guangxi and Hainan) of China (43.68%), followed by East-inland (22.99%), East-coast (12.64%), Central (11.49%), Southwest (6.90%) and rarely found in north part of China (2.30% in North, Northeast and Northwest). GTR+G+I model and Exponential model were found to be the best models for 6a subtype. The tMRCA for 6a was found to be 1990 (95% Confident interval: 1976–1998). BSP curve was plotted using Tracer program to view the epidemic history and the rapid phase (2001–2005) of viral population growth.
Summary/Conclusions: HCV 6a subtype was introduced into South region of China around 1990s, which was originated from Vietnam via Hong Kong, and afterwards spread to the other districts of China. Rapid increase for 6a in China was found during 2001–2005 and then grew gently.

P-305

THE COMPARISON OF HEPATITIS C VIRUS SUBTYPE DISTRIBUTION BETWEEN BLOOD DONORS AND INJECTION DRUG USERS IN GUANGDONG, CHINA

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Background: Hepatitis C virus (HCV) genotypes can serve as a predictor for sustained response to interferon therapy and aid to determine the optimal duration of treatment. Subtype distribution varies among blood donors (BDs) and intravenous drug users (IDUs) in different areas.
Aims: In this study we learn the characteristics of IDUs with HCV infection in Guangdong, and compare HCV genotype distributions between the two different people groups in one place.
Methods: A total of 286 anti-HCV positive IDUs were recruited from a men's prison and a women's prison in Guangdong province of China, from 2009 to 2011. 5 ml of whole blood were draw from the study subjects and sera were isolated immediately.

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The sera were subsequently applied to test for the presence of HCV RNA by nucleic acid tests (NAT). The viral genotypes of HCV RNA positive sera were determined by phylogenetic analysis using E1 and/or NS5B gene sequences. The distribution of HCV genotypes of IDUs was compared with that of the BDs recruited in our previous study. The difference of HCV genotype distribution between the BDs and IDUs was analyzed using the Chi-square test or Fisher's exact test when the chi-square test condition was not satisfied. The continuous variables were compared among different cohorts using one-way ANOVA. Difference in age between the two cohorts was examined with student's *t*-test.

Results: Our results showed that the rate of spontaneous clearance was significantly higher (30.0%, 9/30) in female IDUs than (9.4%, 24/256) in male IDUs. The proportion of HCV 1b (P = 9.8E-23) and 2a (P = 0.030) in BDs were higher than in IDUs. In contrast, HCV 1a (P = 0.037), 3b (P = 3.4E-9) and 6a (P = 1.2E-5) were more prevalent in IDUs than in BDs. Moreover, multivariate logistic regression analysis incorporating factors such as age, gender and intravenous drug use status was performed. Infection with HCV 1b was found to be more prevalent in blood donors (odds ratio [OR] = 0.097, P = 4.9E-14), while subtype 3b and 6a were found to be independently more prevalent in intravenous drug users (OR = 8.477, P = 8.0E-08 and OR = 1.676, P = 0.006, respectively), independently of age and gender.

Summary/Conclusions: In summary, HCV genotype distribution was significantly different between IDUs and BDs, which supported the correlation of different transmission routes with different subtype distribution. Thus, different measures for the prevention of HCV transmission should be implemented according to the transmission pattern of HCV subtype, and the control of drug use and trade in Guangdong province should be enhanced to effectively reduce the prevalence HCV 3b and 6a.

P-306

HLA-DRB1*11:01 AND DQB1*03:01 ASSOCIATED WITH SPONTANEOUS HCV CLEARANCE IN BOTH HAN ETHNICITY AND LI MINORITY OF CHINA

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Background: Human leukocyte antigen (HLA) alleles have been extensively studied to associate with spontaneous clearance of hepatitis c virus (HCV) infection. However, such association seemed to be ethnic specific. Our previous study showed the association of HLA alleles, including HLA-A*02:01, DRB1*11:01 and DQB1*03:01, with HCV clearance. Little is known on this aspect regarding the population of other minorities of China.

Aims: To investigate the association of HLA alleles and HCV spontaneous clearance in Li minority of China.

Methods: We recruited 148 HCV-infected subjects of Li minority from Baisha county of Hainan province where the prevalence of HCV infection was 7.0%. These subjects included 59 with spontaneous clearance and 89 with persistent infection. HLA-A, B, C, DPB1, DQB1 and DRB1 alleles were determined by PCR-SBT as previously described. Individual association of HLA alleles with HCV clearance was examined by the Chi-square test or Fisher's exact test when Chi-square test cannot be applied.

Results: The distribution of HLA alleles were different in Li HCV-infected subjects compared to Han healthy and HCV-infected population (data from our previous studies). HLA-DQB1*03:01 and DRB1*11:01 were found to associate with HCV clearance (odds ratio [OR] = 3.245, 95% confidence interval [CI] = 1.183–8.905, P = 0.017 and OR = 3.592, 95% CI = 1.080–11.947, P = 0.027, respectively). Interestingly, our previous study also found these two alleles also associate with HCV clearance in Han population (OR = 1.464, 95% CI = 1.120–1.915, P = 0.005 for DQB1*03:01; OR = 1.896, 95% CI = 1.230-2.924, P = 0.003 for DRB1*11:01).

Summary/Conclusions: Our results revealed the role of HLA-DQB1*03:01 and DRB1*11:01 in the spontaneous clearance in both Li and Han ethnicities, even the HLA alleles distributed differently in these two population.

P-307

HLA-DRB1 ALLELES WITH HEPATITIS C INFECTION IN DALIAN HAN POPULATION OF NORTH CHINA

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Background: Hepatitis C virus (HCV) infection is a major health problem worldwide. It has infected 71 million individuals globally and according to the World Health Organization, approximately 399,000 people die each year from the complications. Human leukocyte antigen (HLA) plays an important role in the diagnosis and prognosis of many disease conditions including autoimmune diseases, inflammatory and infectious diseases, cancer and adverse drug effects.

Aims: The aim of this study is to assess the relationship between HLA-DRB1 alleles and HCV infection in Dalian Han population, northern China.

Methods: HLA-DRB1 genotyping of 250 patients with chronic HCV infection were performed by sequence-specific oligonucleotide method. As controls, data of 14529 unrelated healthy bone marrow donors who are living in the same region and with the same ethnicity (Shao, Int J Immunogenet, 2016) was used. The comparison analysis was using Fisher's exact test and the relative risk estimate evaluated by odds ratio (OR) at the 95% confidence interval (CI) was performed by SPSS version 21.0.

Results: We found that in these patients, HLA-DRB1*15 followed by HLA-DRB1*07 were the two most common HLA-DRB1 alleles and the presence of HLA-DRB1*12 was significantly lower in HCV patients when compared with the control group (9.0% vs 12.2%, P = 0.035, OR = 0.714, 95% CI = 0.525-0.972).

Summary/Conclusions: This study indicated that HLA-DRB1*12 was a potentially valuable protective allele of HCV susceptibility in the Dalian Han population that may represent the general population in northern China.

P-308

THE RELEVANT RESEARCH OF DETECTION HCV ABOUT ELISA METHOD AND NAT METHOD COLLOIDAL GOLD METHOD IN VOLUNTEER BLOOD DONORS

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Background: Hepatitis c virus infection is a global public health concern. According to the world health organization, more than 170 million people have been infected with HCV, and there is no effective preventive vaccine to prevent it. According to the 2006 national serum anti-hcv antibody epidemic blood survey, the prevalence of anti-hcv antibody was 0.43% in China between 1 and 59 years of age, and the current HCV infection in China was estimated at 10 million. Domestic anti - HCV testing of blood donors by ELISA method or adopt the method of ELISA and NAT anti - HCV and HCV RNA was detected, but there are always some specimens of ELISA detection has reactivity and NAT negative, this study against HCV reactivity and HCV RNA negative blood donors do the additional test specimens with colloidal gold method, according to colloidal gold method and ELISA method for NAT method has higher self-agreement, in S/CO value high, three methods have higher consistency.

Aims: To explore and study the HCV nucleic acid test and colloidal gold method for the study of anti-HCV reactivity serum, and to observe the practical effect of three detection methods.

Methods: The samples of blood donors were tested positive by two ELISA methods, and the results were analyzed by means of nucleic acid testing and colloidal gold.

Results: The positive rate of nucleic acid detection was 37% in 45 cases of anti-HCV reactivity serum, and 69% of the colloid gold test. Reagent 1 and reagent 2 were not detected when the S/CO value was less than 10.00. Reagent 1 and reagent 2 were 70.8% and 62.9% respectively when the S/CO value was greater than 10.00. The detection rate of reagent 1 and reagent 2 colloidal gold was 46.1% and 40% in S/CO value of 1.00–5.00; The detection rate of S/CO value of 5.01–10.00 is 75%; When S/CO value was greater than 10.00, the detection rate was 95.8% and 77.8%.

Summary/Conclusions: The detection of HCV-RNA by ELISA method for detection of HCV-RNA detection could improve the detection rate of hepatitis c. The colloidal gold method is simple but easy to leak detection.

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THE NECESSITY OF GREY AREA IN ANTI-HCV BY ELISA

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Background: The same two positive specimens were found by both RIBA confirmatory test and HCV RNA nucleic acid testing in 167 grey area samples. We must pay more attention to the HCV grey area specimens by ELISA, due to the limitation of ELISA experiment with ELISA kits from different manufacturer.
Aims: To apply two kinds of methods, including RIBA confirmatory test and HCV RNA nucleic acid testing (NAT), to detect grey area specimen (OD: $0.6 \leq s/co < 1$, detected by ELISA),and analyze their results .
Methods: Apply ELISA to detect the blood specimens with two kinds of reagents from Zhuhai Lizhu and Shanghai Kehua, and the specimens whose OD $0.6 \leq s/co < 1$ would be tested again by RIBA confirmatory test and HCV RNA nucleic acid testing respectively .
Results: The same two positive specimens were found by both RIBA confirmatory test and HCV RNA nucleic acid testing in 167 grey area samples.
Summary/Conclusions: We must pay more attention to the HCV grey area specimens by ELISA, due to the limitation of ELISA experiment with ELISA kits from different manufacturer. The samples with grey area could be detected positively by both RIBA and HCV-NAT, therefore, adding NAT to detect blood samples, is of great practical significance to ensure the safety of clinical blood transfusion.

P-311

PERFORMANCE OF NEW GEENIUS HCV SUPPLEMENTAL ASSAY

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Background: The new Bio-Rad Geenius™ HCV Supplemental Assay is a qualitative single-use immunochromatographic test for the confirmation of individual antibodies associated with infection caused by the Hepatitis C Virus (HCV) in human venous whole blood, serum or plasma samples.
Geenius™ HCV Supplemental Assay is intended for use as an additional screening and/or diagnostic test to confirm the presence of antibodies to HCV for specimens found to be reactive by screening procedures.
This assay allows the individual detection of antibodies to Capsid (Core), NS5, NS4 and NS3 proteins. It is associated with Geenius™ reader and software for results automated interpretation.
Aims: The aim of this study, performed by Bio-Rad R&D department was to evaluate the performance of this new assay in terms of sensitivity, specificity and reproducibility on blood bank routine samples, hospitalized patients, chronic samples and seroconversion panels compared to serology and/or supplemental reference assays.
Methods: A validation of this new rapid test was performed. Specificity was evaluated on fresh negative serum, venous whole blood and plasma samples from blood bank (n = 250) as well as on serum samples from hospitalized patients (n = 100) and patients with unrelated medical conditions (n = 75). Sensitivity was tested on samples from chronic HCV infected patients (n = 100) and on clinical anti-HCV positive samples (n = 76). Ten commercial seroconversion panels were tested. All results were obtained using automated reading with Geenius™ system (reader & software).
Results: Regarding specificity, no false positive result was found among blood bank, hospitalized and unrelated medical condition patient samples. Less than 3% indeterminate rate was observed.
Sensitivity of 100% was obtained on clinical HCV seropositive and chronic HCV infected patient samples with similar results to INNO-LIA® HCV Score Assay.
Performance obtained on seroconversion panels was higher than Deciscan™ HCV Plus Assay and equivalent to INNO-LIA® HCV Score Assay.
Assay reproducibility study (intra and inter-reproducibility) showed no status changes.
Summary/Conclusions: Performance obtained during evaluations met the required specifications in terms of specificity and sensitivity. The new Bio-Rad Geenius™ HCV Supplemental Assay is the first unitary assay for confirmation of anti-HCV antibodies presence in serum, plasma and venous whole blood sample with automated reading and interpretation. In addition, the expert software allows full traceability of the results obtained in less than 30 min.

P-312

CLINICAL EVALUATION OF HCV CORE ANTIGEN ASSAY IN BLOOD DONORS

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Background: Chronic hepatitis C virus (HCV) infection is estimated to affect million individuals worldwide. Screening assays for HCV detection are routinely performed with specific anti-HCV antibodies (anti-HCV) assay in order to guarantee safety of blood donation. HCV/RNA detection by nucleic acid testing (NAT) is also performed on blood donors in compliance with national law. In many industrial countries, NAT introduction for blood donor screening has decreased the risk of viral transmission via blood transfusion. Unfortunately, HCV transmission by hemocomponents is still a problem in developing countries in which NAT cannot be implemented for high cost and technical skill requirement. Chronic hepatitis C virus (HCV) infection is estimated to affect million individuals worldwide. Screening assays for HCV detection are routinely performed with specific anti-HCV antibodies (anti-HCV) assay in order to guarantee safety of blood donation. HCV/RNA detection by nucleic acid testing (NAT) is also performed on blood donors in compliance with national law. In many industrial countries, NAT introduction for blood donor screening has decreased the risk of viral transmission via blood transfusion. Unfortunately, HCV transmission by hemocomponents is still a problem in developing countries in which NAT cannot be implemented for high cost and technical skill requirement.
Aims: The aim of this study was to evaluate the performance of HCV core antigen (HCV-Ag) assay as potential alternative to NAT and determine its potential clinical utility in the management of blood donors.
Methods: From 2013 to 2017, n = 63 randomized blood donors, apparently healthy subjects of both sexes, aged 25–60 years (average 37.1 ± 10.2) and resulted anti-HCV reactive, were studied. All serum blood samples were evaluated for HCV detection using Architect anti-HCV assay by Chemiluminescent Microparticle Immunoassay (CMIA) platform (Abbott Diagnostics, Wiesbaden, Germany). Besides, all these samples were tested also for Architect HCV-Ag assay. Samples yielding sample/cutoff (S/CO) ratios between 0.80 and 0.99 were scored as gray zone (GZ) while those yielding S/CO ratios ≥1.00 as reactive samples. All reactive and GZ samples for anti-HCV assay were further checked by confirmatory testing such as HCV specific immunoblot assay (Fujirebio, Italia S.r.l.). The relationship between S/CO and confirmed seropositive samples were analyzed. All blood donations were screened also by NAT for HCV-RNA by the TaqScreen method on the Cobas s201 system (Roche Molecular Systems, Branchburg, NJ, USA): the assay was performed on mini pools of six samples each and with a nominal sensitivity of <20 international units per milliliter (IU/ml).
Results: Of 63 serum samples, n = 50 were anti-HCV reactive with S/CO between 1.00 and 18.65 while n = 13 resulted anti-HCV GZ. Only 10 anti-HCV reactive samples with S/CO between 9.47 and 18.65 were positive also with HCV-Ag assay. All HCV-Ag positive (n = 10) resulted NAT positive. The correlation between HCV-Ag assay and NAT was 100%.
Summary/Conclusions: Our data highlight that HCV-Ag assay correlate highly with NAT especially with those samples yielding an anti-HCV S/CO >10. Thus, a reliable and easy serological marker such as HCV-Ag assay could represent a valid alternative to NAT especially in those countries where NAT is not currently performed for high cost. Then, the implementation of this serological assay could provide a new scenario for blood safety.

HIV

P-313

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HIV INFECTION IN BLOOD DONORS: NATIONAL SURVEYS IN CHINA FROM 2009 TO 2014

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Background: National surveys on blood collection and supply were launched by Blood Donation Promotion Committee, a branch of Chinese Society of Blood Transfusion, every 3 years since 2006. Considering the threat of HIV to blood safety, questions associated with HIV infection in blood donors were decided to add into the questionnaire in the second survey (2009–2011) and then information of gender, age and donating frequency of HIV infected donors were asked in the third survey (2012–2014) in order to search ways to reduce the risk of transfusion-transmitted HIV infection.
Aims: To investigate and analyze HIV infections of blood donors around China in 6 years from 2009 to 2014.
Methods: As requirement by Ministry of Health of China, all samples reactive to HIV markers in blood screening must be sent to local Centers for Disease Control and Prevention to confirm HIV-infected using western blot. Feedback information of HIV infection in blood donors was collected by questionnaires distributed to blood centers or blood banks in China. The data of gender, age and donating frequency of HIV infected donors were analyzed and HIV detection rates were compared according to year and province as well.
Results: The upward trend of HIV infection among blood donors in majority provinces of China were observed. Average HIV detection rate of nationwide increased from 1.45 (0.41–7.16) per 10,000 donations in 2009 to 2.48 (0.94–5.92) per 10000 donations in 2014 paralleled with the number of provinces of which HIV detection rate above 2 per 10,000 donations rising to 20 from 8. Nevertheless, the data were flat in a few provinces and that of Yunnan Province was significant decreased from 7.16 to 3.87 per 10,000 donations simultaneously. Eighty-five to eighty-nine percentage of HIV infected blood donors were male and most of them were confirmed or suspected to be MSM or related to MSM. HIV-infected donors between the ages of 26 and 45 account for 54% in 2014 decreasing from 60% in 2012 while percentage of that between the ages of 18 and 25 raised by 5%. Furthermore, about 66% of HIV infections was first donor but it was worth to notice that repeat donors who had donated no less than three times account for around 12%.
Summary/Conclusions: HIV infection among blood donors showed an upward trend in China. HIV-infected MSM was the major threat to blood safety and accompanied by a younger age. Some measures need to be adapted to improve donor's self-exclusion before donation and initiative notification after donation.

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PEPTIDE ASSEMBLY NANOPARTICLES LOADED WITH HDAC INHIBITOR ACTIVATE LATENT HIV

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Background: At present, antiretroviral therapy is only able to control HIV replication, rather than completely eradicating virus from patients. This is due in part to the establishment of a latent virus reservoir in resting CD4+ T cells, even at the presence of highly active anti-retroviral therapy (HAART). It is believed that forced activation of latently infected cells can induce viral production, allowing cells to be targeted by an immune response. Histone deacetylase (HDAC) inhibitors have been the most widely investigated latency-reversing agents (LRAs), but most of them resulted in only a modest and transient reactivation of HIV, making it an impractical therapy for HIV reactivation. Therefore, a novel approach to improving the potency of latency activation is required. Nanoparticles can provide a number of advantages over more traditional drug delivery methods, including improved drug solubility, stability, drug targeting, and release.
Aims: Our strategy was confirmed effective to improve the hydrophobic HDAC inhibitor delivery for the treatment of latent HIV. Here, we describe a novel cell-penetrating peptide assembly nanoparticle loaded with the histone deacetylase inhibitor panobinostat (PNP-P).
Methods: In order to improve the water solubility and enhance the penetration of cells, a novel cell-penetrating peptide based amphiphilic peptide monomer was designed and synthesized. In vitro analysis was used to measure the viral reactivity of PNP-P in two latent HIV pseudotype virus cell model (J-Lat Tat-GFP Clone A72 and A82). Cells were treated with PNP-P, and post-treatment expression of GFP, which were used as markers of reactivated HIV, was measured.
Results: Nanoparticles loaded with panobinostat were characterized by size, surface morphology, zeta potential, entrapment efficiency, drug release and cytotoxicity. These particles can target and activate latent HIV in vitro, and showed more efficiency than panobinostat used only. Activation efficiency increased by 2.34-fold in A72 cells and 1.36-fold in A82 cells.
Summary/Conclusions: Taken together our study showed the ability of peptide assembly nanotechnology to provide improved drug delivery for activating latent HIV and provide key proof-of-principle experiments showing how novel delivery systems may enhance future HIV therapy.

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EVALUATING THE PERFORMANCE OF THREE HIV BLOOD DONOR SCREENING STRATEGIES: TWO-DIFFERENT-EIAS, A SINGLE-EIA, OR A SINGLE-CMIA?

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Background: Prevention of transfusion-transmitted HIV infection requires a sensitive and specific donor screening strategy. The two-different-EIAs strategy (using two highly sensitive EIAs in parallel) has been the mandated strategy for routine donor screening in China for many years and is the current strategy used by most Chinese blood centers. The recent change in government regulation now allows donor screening using one EIA (single-EIA strategy) or one CLIA (Chemiluminescent immunoassays) (single-CLIA strategy) for HIV antibody/antigen. There have been no published data comparing the performance of the three strategies in Chinese donors.
Aims: To evaluate the performance of current two-different EIA strategy with that of the single EIA and single CLIA strategies for HIV.
Methods: We performed HIV Ag/Ab chemiluminescent microparticle immunoassay (CMIA, Abbott Diagnostics. a version of CLIA) testing on 2,138 samples from four blood centers which have all undergone routine testing using the two-different-EIAs strategy. Results from routine donor HIV EIA screening testing were obtained from the blood centers. 1931 samples were nonreactive by both EIAs, 63 were reactive by EIAs, 102 were reactive by one EIA and 42 were defined as gray zone by eitherone or two EIAs. Western blot (WB) was performed on all samples with one or both EIA reactive or grey zone result as well as on all EIA negative samples with reactive CMIA results for confirmation. Performance of CMIA, different EIAs and the three strategies were evaluated by comparing the testing results to the confirmed HIV infection status defined by WB positivity.
Results: The positive coincidence rate of HIV Ag/Ab tests with HIV infection status was found to be 95.87% (95% CI: 90.62–98.64%) and the negative coincidence rate of HIV Ag/Ab tests with HIV infection status was 99.31% (95% CI: 98.89–99.61%), both of which seemed higher than or even equal to that of EIAs. Good consistency (Kappa value higher than 0.75) with HIV infection status were found by CMIA (0.88), Kehua ELISA (0.78), BioMerieux ELISA (0.77) and Bio-Rad ELISA (0.76); among which CMIA has the best consistency with HIV infection status compared to all EIAs studied. Of the 224 samples with discrepant results by EIAs and CMIA, 4 samples confirmed positive by WB were reactive by single-CMIA but missed by two-different-EIAs strategy while only one sample missed by single-CMIA compared to two-different-EIAs strategy. One infected sample out of the 224 tested as gray zone by two-different-EIAs strategy and reactive by single-CMIA. All 93 single-EIA reactive samples tested nonreactive by CMIA and negative by WB.
Summary/Conclusions: This is the first study to compare the performance of recently allowed single-EIA, single-CLIA and the currently used two-different-EIAs strategy for HIV blood screening in China. Our results suggest that CMIA performed better than EIAs according to the positive coincidence rate, negative coincidence rate and the consistency of these assays with the real infection status. The advantage of the one-CMIA strategy includes lower false negative and false positive rates. Thus, the implementation of one-CMIA strategy may help to further reduce the risk of transfusion transmitted HIV infection, and need to perform multiple assays, while decrease the unnecessary waste of blood caused by false positive results.

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A COMPARATIVE SEROLOGICAL STUDY OF HIV SCREENING IN BLOOD DONATIONS BY ELISA, CLIA AND ECLIA IN CHINA

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Background: The serological test of human immunodeficiency virus (HIV) is crucial to prevent the transmission of HIV through blood transfusion. In China, enzyme linked immunosorbent assay (ELISA) is the only assay approved for HIV screening in blood donations. Completely automated chemiluminescence immunoassay (CLIA) or electrochemiluminescence immunoassay (ECLIA) are emerging technologies for the detection of HIV, therefore it is beneficial to evaluate the performance of CLIA, ECLIA and ELISA.

Aims: There was no prior comparative evaluation between CLIA, ECLIA and ELISA in China, therefore the purpose of this study is to compare these assays and determine which one performs better for detection of HIV in blood banks.

Methods: A total of 1036 plasma samples were collected from 16 locations, including National Center for Clinical Laboratories (NCCL), blood centers and blood banks. 583 plasma samples that were nonreactive for anti-HIV, and elevated levels of alanine aminotransferase (ALT), served as negative control and 483 anti-HIV reactive samples served as positive samples. These 1036 samples were tested by eight ELISAs (B, D, I, L, L (4th), K, W, W (4th)), CLIA (A) and ECLIA (R). True positives were defined by dual positive of CLIA and ECLIA. Samples with discrepant results between CLIA and ECLIA were further tested by nucleic acid testing using the Roche Molecular Systems assay, cobas® TaqScreen MPX Test, Version 2.0 by individual donation testing (ID-NAT). HIV-positive samples that were missed by ELISAs were also confirmed by ID-NAT. Lastly, we compared CLIA and ECLIA against ELISA methods.

Results: From the 483 anti-HIV reactive samples, 137 samples were considered as true positives (CLIA+ & ECLIA+) and subjected to further testing with ELISAs. 7 of the ELISAs did not detect all the 137 HIV positive samples. ELISA (K, I, L) did not detect 4 HIV positive samples, ELISA (W) did not detect 3 HIV positive samples and ELISA (W (4th), L (4th), D) did not detect 2 HIV positive samples. Most ELISAs missed 3 HIV-positive plasma samples (No: HIV000150051, HIV008150021, HIV019150004), but these 3 samples were detected by CLIA, ECLIA and ID-NAT. 2 samples (No: HIV000150051, HIV008150021) were missed by all ELISAs except B and 1 sample (No: HIV019150004) was missed by four ELISA (I, K, L, and W). After removing the uncertain results of ELISAs, several false positive results were found in all methods (A has 7, R has 8, I has 12, L has 17, K has 19, L (4th) has 24, W (4th) has 30, W has 32, D has 51 and B has 79). The sensitivity of CLIA (100%) or ECLIA (100%) had no significant difference, compared with the eight ELISAs ($P > 0.05$) (B, D, K, L, L (4th), I, W, W (4th); 100%, 98.5%, 97.1%, 97.1%, 98.5%, 97.1%, 97.8%, 98.5%). The CLIA (A) showed higher specificity (99.2%) than the most of ELISA assays (A vs L, K, L (4th), W, W (4th), D, B; 99.2%, 98.1%, 97.9%, 97.0%, 96.6%, 96.4%, 94.3%, 91.1%; $P < 0.05$, 0.05, 0.01, 0.01, 0.01, 0.01, 0.01). The ECLIA (R) showed similar specificity (99.1%) with A (R vs L (4th), W, W (4th), D, B; 99.1%, 97.9%, 97.0%, 96.6%, 96.4%, 94.3%, 91.1%; $P < 0.05$, 0.01, 0.01, 0.01, 0.01, 0.01).

Summary/Conclusions: CLIA and ECLIA have successfully detected the HIV-positive samples that were missed by ELISA during the early HIV infection. Therefore, compared with ELISA, CLIA and ECLIA are more specific and sensitive to detect HIV antibodies/antigens in blood donations. In the future, CLIA and ECLIA may be an alternative test to ELISA for HIV screening in blood donations in China, subject to approval by local authorities.

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NRL BLOOD SCREENING EQAS, 2015–2016. HOW IS IT TRENDING?

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Background: NRL provides Quality Assurance programs to laboratories that test for blood-borne infectious diseases. One of the components of NRL Quality Assurance is External Quality Assessment Schemes (EQAS). Participation in EQAS gives participants a means of assessing independently their laboratory performance and comparing their results with those of others in their peer group. It also offers a means of examining the performances of different assays.

Aims: 1. To examine the ability of participants to detect nucleic acid in samples containing different amounts of HIV-1 RNA, HCV RNA and HBV DNA with respect to Blood Screening EQAS (NATA4315).

2. To examine the ability of participants to detect and discriminate correctly between HIV-1 RNA, HCV RNA and HBV DNA positive samples in the Blood Screening EQAS (NATA4315).

Methods: Participants' data from six NATA4315 EQAS test events over 2 years were analysed. In 2015 15,782 results were submitted from an average of 28 assays. In 2016 18,616 results were submitted from an average of 27 assays. The Multimarker Blood Screening NAT EQAS panels consisted of fifteen samples. These panels were composed of well-characterized plasma samples of known concentrations. The samples were calibrated against the WHO international standards for HIV-1 (97/650), HBV (97/750) and HCV (06/102). The results of testing were submitted to NRL using the internet application OASYS (Oneworld Accuracy, Canada). Aberrant results were those that were reactive for samples that did not contain HCV RNA, HIV RNA or HBV DNA (a result of "Detected" for an analyte-negative sample) and were non-reactive for a sample with a viral load (a result of "Not Detected" for an analyte-positive sample).

Results: The highest number of laboratories participating in the NATA4315 EQAS program over the 2 years was 131 with a mean of 120.

In 2015 there were 2590 results submitted for replicate negative samples over the three test events. All negative samples consisted of NHP and were identical in composition, being dispensed from the sample pool of NHP and were negative for HIV RNA, HCV RNA and HBV DNA. Only six of 2590 (0.23%) results were reported as falsely reactive. In 2016 eight of 3132 (0.26%) results were reported as falsely reactive.

Reported false negative results were also analysed for all three analytes. For HIV samples at a concentration of 500 IU/ml, there was a reduction in reported false negative results from 0.79% in 2015 to 0.36% in 2016. Similar trends were seen in HIV samples at a concentration of 250 IU/ml. In 2015 2.16% of reported false negative results were reported compared with 1.23% in 2016. These trends were reflected in HCV samples as well.

Summary/Conclusions: When comparing overall NATA4315 participant results from 2015 with overall NATA4315 participant results from 2016, there was a reduction in falsely reactive results and false negative results over the 2 years. Participation in EQAS allows laboratories to assess their performance over several years and to observe and facilitate improvements.

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APPLICATION OF NAT IN HIV SCREENING OF BLOOD DONORS IN HOHHOT AREA

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Background: In recent years, Hengyang and Fuzhou had blood transfusion infected with HIV reports. To provide the safety blood for clinic have become the biggest challenge for every blood worker.

Aims: To analyze the situation of HIV blood screening in Hohhot, and discuss the application value of NAT in HIV screening, and provide the basis for improving blood safety.

Methods: Routine detection for HIV was conducted on 56684 blood donor screening samples from March 2014 to 2015 in Inner Mongolia Blood Center by FAME2420 system and TIGRIS system, of which, screening positive samples sent to the Inner Mongolia CDC to confirm the test.

Results: A total of 11 cases were confirmed positive, of which,2 cases were ELISA (−), NAT(+), Western Blot(−). After the 2 and 4 weeks, the blood donors were tracked by the Inner Mongolia Blood Center. After 2 weeks, ELISA (+) and WB were uncertain, and WB was confirmed as positive after 4 weeks.

Summary/Conclusions: In order to ensure the quality of blood and the safety of blood transfusion, the laboratory quality management and the test sensitivity must be strengthened. In addition,AIDS-related knowledge should be propagated among young man and applied to the prevention effort on HIV infection.

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ASSESSMENT OF RESIDUAL RISK OF HIV TRANSMISSION VIA TRANSFUSION AFTER BLOOD SCREENING IN TAIYUAN

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Background: Because of the window period of enzyme-linked immunosorbent assay (ELISA) and nucleic acid detection (NAT), limited detection reagent sensitivity, virus strain mutation, silent infection, immune detection system error and other factors, screening for blood related viruses will have a certain risk. Residual risk assessment for transfusion-transmitted diseases was a statistical analysis of this risk by retrospective and direct measures. In 2000, WHO recommended the mathematical model of the Australian National Reference Laboratory (NRL) and the American Red Cross to the global blood transfusion service. Because of differences in blood donors and testing levels, developing countries have reported relatively little about it. There were some reports about the residual risk of blood transfusion in China, but there were some differences in the results due to the differences in economic development, prevalence rate and infection rate of diseases. The study used mathematical models to assess the residual risk of HIV transmission via transfusion in our region, with the aim of increasing vigilance against residual risk.
Aims: To review the infection status of AIDS virus (HIV) among voluntary blood donors in Taiyuan, and assess the residual risk of HIV transmission via transfusion after blood screening.
Methods: The HIV test results of voluntary blood donors from 2013 to 2016 were analyzed retrospectively. To compare the incidence rate of reactivity of initial blood donors and repeated blood donors between male and female, with the prevalence rate or window period mathematical model to evaluate the residual risk.
Results: The blood samples of voluntary blood donors amounted to 346,565 in 4 years. The positive rate of HIV was 0.08% among male initial blood donors, 0.01% among female initial blood donors, 0.04% among male repeated blood donors, and 0.002% among female repeated blood donors. The total residual risk of HIV for voluntary blood donors after anti-HIV screening was 2.19×10^{-6}, of which 2.08×10^{-5} for male and 1.04×10^{-6} for female, 1.40×10^{-5} for initial blood donors and 7.88×10^{-6} for repeated blood donors.
Summary/Conclusions: The prevalence rate of initial blood donors' HIV infection was higher than that of repeated blood donors', male donors' was higher than female donors' in Taiyuan. The residual risk of HIV transmission of male donors was greater than that of female donors', and initial blood donors' was greater than that of repeated blood donors' in current condition.

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PREVALENCE OF HIV AMONG BLOOD DONORS AND THE ANALYSIS OF SCREENING AND CONFIRMATORY TEST RESULTS IN CHONGQING

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Background: Enzyme linked immunosorbent assay (ELISA) has simple operation and high sensitivity. It is widely used in blood screening in the field of blood transfusion. Anti -HIV screening ELISA kit is a variety of HIV antigen reaction to microtiter plates, as long as the detected and antibody antigen binding was judged as positive, the sensitivity is high, there may be a false positive, so we must use confirmatory test specificity for further detection. The Western blot test (WB) is a method of HIV infection identification prescribed by the national technical standard for AIDS detection (2015 Edition), and its specificity is higher than that of ELISA. Therefore, the ratio of false positive to the positive reaction of screening reaction and positive proof can be analyzed, and the relation between the result of preliminary screening and the result of confirmatory test is compared.
Aims: To investigate the prevalence of HIV among blood donors in Chongqing and analyze the relationship between HIV screening test results and confirmatory test results.
Methods: Blood donors' samples were screened with homebred or imported reagents by enzyme-linked immunosorbent assay from January 2014 to December 2015 in Chongqing Blood Center. The samples were sent to Centers for Disease Control (CDC) to conduct confirmatory test as long as positive or gray area by any of the two reagents in screening test. HIV positive incidence rate was calculated and the correlation between screening and confirming test results was analyzed.
Results: During the study period there were 550 HIV positive cases in screening test among 244454 donors, including 169 positive cases in the double reagents test, among which, 154 cases (91.1%) were positive, 2 cases(1.1%) were uncertain and 4 cases (2.3%) were negative in the confirmatory test. HIV infection rate in blood donors of Chongqing was 0.06%.
Summary/Conclusions: The positive result in the double reagents test is remarkable positively associated with the confirmatory test.

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SURVEILLANCE OF HIV INFECTION IN BLOOD DONORS IN PAKISTAN: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Background: Human Immunodeficiency Virus (HIV) epidemic is an increasing health concern in Pakistan. The mode of HIV transmission largely remains heterosexual followed by blood transfusion. In Pakistan, HIV screening is mandatory under the blood safety legislations. The national strategies to minimize the risk of HIV transmission are the use of behavioral screening questionnaires to defer donors at higher risk of HIV infection and screening the blood with highly sensitive and specific laboratory tests. Assessing the pattern of HIV in a blood donor is necessary to ensure blood safety and also to inform the policy makers on the magnitude of the disease in the apparently healthy population.
Aims: To assess the prevalence of HIV among the healthy blood donor population of Islamabad Capital Territory and its catchment area. A literature search was done to compare the results with the previously reported statistics from across the country.
Methods: This study was conducted at the Shaheed Zulfiqar Ali Bhutto Medical University Hospital Blood Bank, Islamabad, from Jan 2015 – Dec 2016. Blood donors were screened for the prevalence of HIV from Jan 2015 – Dec 2016, by chemiluminescence immunoassay on the fully automated Architect i2000 system (Abbott Laboratories, Abbott Park, IL, USA). The results of the study were compared with the previous 10 years data of the same centre to see the trend over this period. A comprehensive literature survey was done to shortlist national/local studies with similar experimental settings, for subsequent comparison of HIV prevalence results among blood donors. Meta-analyses were gathered from the reported HIV incidence in blood donors from across Pakistan during 1988–2016 by searching through Google, PubMed, and PakMediNet (for Pakistani non-indexed journals).
Results: A total of 54,877 blood donors were tested for the presence of antibodies to the HIV during these 24 months. Of these donors, 54,454 (99.23%) were males and 423 (0.77%) donors were females. Out of them, 77 were found initially reactive for HIV. The repeat testing resulted in 75 (0.13%) positive donors, 95% CI 0.0014 (0.0011 – 0.0018). All positive donors were males. The prevalence of HIV positivity during the last 11 years period (2006–2016) showed a significant change over a period of time through the chi square goodness of fit test. The mean prevalence was 0.06%. The mean prevalence of HIV from 2006–2016 was 0.06% while 0.13% in 47 studies conducted during the period 1988–2016.
Summary/Conclusions: The prevalence of HIV in apparently healthy blood donors is steadily increasing in Pakistan. Specific recommendations include the availability of a sufficient volunteer blood donor pool and adequate blood donor screening, information, counseling, and confidentiality. Implementation of standardized screening protocol and use of highly sensitive tests will reduce the risk of transmission.

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ANALYSIS OF HIV INFECTION IN VOLUNTARY BLOOD DONORS IN WUHAN

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Background: HIV (Acquired Immune Deficiency Syndrome) is a kind of infectious disease which seriously endangers human health. Its incidence rate is rising in our country and the world. HIV is one of the infectious diseases that our country focuses on. The Wuhan has reported HIV infection rates increased year by year. Screening of HIV antibodies for voluntary blood donors is one of the mandatory testing items for blood quality assurance.
Aims: Analysis of HIV infection in voluntary blood donors in Wuhan.
Methods: We analyzed 118867 samples which were from voluntary blood donation in Wuhan at first half of 2017. If a sample was positive or single reagent diplopora

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retest was positive, we will send it to Wuhan City CDC confirmed. We had sent 176 samples (118 males and 58 females) to CDC.
Results: There were 151 unilateral positive and 25 bilateral positive. Of the 25 bilateral positive samples, 24 were from male, and 1 was from female. The results as follows: 6 samples' HIV-1 antibody were uncertain (±), 23 samples' HIV-1 antibody were positive (+) (22 male, 1 female), 21 samples' HIV-1 antibody were negative and 125 samples' HIV antibody were negative.
Summary/Conclusions: The above data showed that the number of men suffering from HIV is much greater than women.

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COMPARATIVE EVALUATION OF THE BIO-RAD GEENIUS(TM) HIV-1/2 CONFIRMATORY ASSAY AND THE NEW LAV BLOT 1 AND 2 IN THE JAPANESE POPULATION

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Background: Currently, diagnosis of HIV infection in Japan is carried out in two-step algorithm in which a sample tested reactive on HIV-1/2 antigen/antibody assay is then retested with Western blot and nucleic acid test. However, this algorithm demonstrated some limitations as Western blot sometimes shows false negative, indeterminate results, and cross-reactivity between HIV-1 and HIV-2. In 2014, CDC in the US published revised guideline for diagnosis of HIV infection in which the use of an HIV-1/HIV-2 antibody differentiation assay (Multispot) is recommended after a repeatedly reactive HIV-1/2 antigen/antibody test. In October 2014, Bio-Rad Laboratories received FDA clearance on the Geenius™ HIV-1/2 Supplemental Assay and replaced Multispot in the revised algorithm. Since then there has been a growing interest in Japan in the new HIV diagnostic algorithm using Geenius™.
Aims: We compared performance of Bio-Rad Geenius™ HIV-1/2 Confirmatory Assay and Bio-Rad NEW LAV BLOT 1 and 2 (NLB 1 and 2), Western blot kits for HIV-1 and HIV-2, respectively, to examine if Western blots can be replaced with Geenius™ in future HIV testing algorithm in Japan.
Methods: As HIV-1 positive specimens, 193 plasma specimens were collected from HIV-1 positive individuals, of which 58 specimens collected from untreated patients, 73 from treated patients, 16 from acutely infected patients, 22 from 5 patients sequentially during the early phase, 10 false-positive specimens on HIV-1/2 antigen/antibody assays, and 14 specimens from HIV-1 patients that showed reactivity on HIV-2 Western blot. In addition, 130 HIV negative plasma specimens were collected from healthy individuals. HIV-1 positivity and negativity of the specimens were determined using HIV-1/2 antigen/antibody assays or the HIV-1 nucleic acid amplification test. HIV-2 plasma specimens were obtained commercially. All of the above specimens were tested with Geenius™ and NLB1 and 2.
Results: Of 131 HIV-1 positive specimens from established infections, 130 were HIV-1 positive and 1 indeterminate by Geenius™; 129 positive and 2 indeterminate by NLB 1; 14 positive, 113 indeterminate, and 4 negative by NLB 2. Of 130 negative specimens, 128 were negative and 2 indeterminate by Geenius™; 106 negative and 24 indeterminate by NLB 1; 117 negative and 13 indeterminate by NLB 2. Overall, Geenius™ sensitivity and specificity were 99.2% and 98.5%, respectively; those of NLB 1 were 98.5% and 81.5%, respectively. The specificity of NLB 2 was 90.0%. Disagreement between Geenius™ and NLB 1 was remarkable for indeterminate results (kappa = 0.82). Of 16 samples from acutely infected patients, 7 were HIV-1 positive, 5 indeterminate, and 4 negative by Geenius™; 14 indeterminate and 2 negative by NLB 1. When sequential samples collected during the early phase were examined, Geenius™ demonstrated positive results 0 to 27 days earlier than NLB 1. For 10 HIV-1 false-positive specimens, Geenius™ provided 2 indeterminate and 8 negative results; NLB 1 gave 4 indeterminate and 6 negative; NLB 2 conferred 5 indeterminate and 5 negative. All of the 20 HIV-2 specimens were HIV-2 positive by Geenius™, indeterminate by NLB 1, positive by NLB 2.
Summary/Conclusions: Geenius™ assay is a reliable alternative to Western blots in sensitivity, specificity, and HIV-1/2 differentiation. It will decrease the number of indeterminate results and thus reduce the number of nucleic acid tests needed for additional confirmation. To introduce this assay into the HIV diagnostic algorithm in Japan, medico-economic evaluation is needed.

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HBV, HCV, AND SYPHILIS CO-INFECTIONS AMONG HIV POSITIVE BLOOD DONORS IN BLOOD CENTER OF ZHEJIANG PROVINCE, CHINA: A RETROSPECTIVE ANALYSIS

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Background: Hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), and syphilis still cause high burdens of disease in China.
Aims: The objective of the study was to assess the risk of co-infections with HBV, HCV, and syphilis among HIV positive blood donors in Blood Center of Zhejiang Province to find out characteristic of these donors.
Methods: In this study, we reviewed data from 2,221,131 blood donations in Zhejiang Blood Centre between January 1999 and July 2017. All donations were tested for anti-HIV, HBsAg, anti-HCV and syphilis by serological screening. NAT assay was added since August 2010, and then 956,317 donations were tested for HBV, HCV and HIV. HIV reactive samples were sent to Center for Disease Control and Prevention (CDC) for further identification.
Results: Among the 2,221,131 blood donations, we found that 275 donors were confirmed with HIV infection by CDC using Western blot. Of these 275 HIV-positive donors, 12% (33 cases, confirmed between 2007 and 2017) was HIV-syphilis co-infected, 2.2% (5 cases, confirmed between 2001 and 2003, 1 case, confirmed in 2014) was HIV-HCV co-infected and 0.4% (1 case, confirmed in 2009) was HIV-HBV-syphilis co-infected. CDC's investigation found that all these HIV-syphilis co-infected cases related with sexual transmission. Eighteen of them (54.5%) associated with men who have sex with men (MSM), and the other fifteen (45.5%) associated with commercial heterosexual sex.
Summary/Conclusions: There was a high prevalence of HIV-syphilis co-infection in HIV positive donors, suggested that sexual transmission was becoming the leading route of HIV infection in China, replacing intravenous drug abuse or history of blood (plasma) donation.

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Abstract has been withdrawn

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FOLLOW-UP AND THOUGHTS ON THE PROCESS OF HIV ANTIBODY SEROCONVERSION ON A VOLUNTARY BLOOD DONOR

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Background: Hainan area blood donors of anti HIV positive rate increased, and the young men is the main feature, and in January 19, 2017, the general office of the State Council issued the "China AIDS prevention and containment" 13th Five-Year "action plan" to elucidate the epidemic of AIDS in China is consistent with the trend, increasing the number of people HIV latent infection free blood donation, a great threat blood safety, from low risk blood donors in recruiting blood donors, blood donation of blood donors were detected by HIV, is an important strategy to ensure the safety of blood
Aims: To analyze the status of the process of blood donors' HIV infection and seroconversion, to discuss the influence from uncertain factors in Centralized detection mode on early-stage transfusion-transmitted diseases, and to provide evidence and support for a more scientific blood management method.
Methods: Conduct Routine test, Verification test and validation test on a primary blood sample and a Follow-up blood sample of a blood donor with ELISA, NAT, and WB.
Results: Regarding the primary blood sample, the results from the Routine test and Verification test, the 3rd generation ELISA Test kit on HIV were all negative; the result from the 4th generation Antigen antibody combined detection reagent was positive; and the result for HIV antibody from WB was uncertain. Regarding the follow-up blood sample after 4 weeks, the result from 3rd and 4rd ELISA Test kit were strongly positive; the result from HIV RNA was relatively strong; and the result from WB was positive.

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Summary/Conclusions: The application of 4rd ELISA Antigen antibody combined detection reagent and NAT has successfully shortened the HIV detection window period. The management and supervision of blood sampling and blood supply should be improved. The transmission of HIV through transfusion can be contained and the safety risk of blood can be minimized by recruiting blood donors from low risk population.

Bacteria

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BACTERIAL CONTAMINATIONS OF PLATELET PRODUCTS IN DONGGUAN BLOOD CENTER

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Background: Platelet products pose a special risk of bacterial contamination due to their preservation and collection methods.
Aims: After five prevention measures were taken, the incidence of bacterial contamination of platelet products needs to be assessed to accurately manage the risk of transfusion-transmitted bacterial infection.
Methods: 28711 platelet product's samples from Dongguan Blood Center were examined by bacterial culture tests including initial culture and subculture, if initial culture was positive, subculture was performed, and bacterial identification was needed if subculture was positive. 3073 platelet samples were examined in 2006, From 2007 to 2016, four prevention measures were taken, including sterilizing donation environment, screening donors, depleting white blood cells from pooled platelet, and remove the first 15 ml of collected platelet, a total of 25638 platelet samples were examined. Bacterial culture of recipients and transfused bag for checking the cause of platelet transfusion adverse reactions was performed in hospitals of Dongguan.
Results: 63 (0.22%) positive cases in 28711 platelet samples were found to be contaminated with bacteria from 2006 to 2016. In 2006, the positive rates of pooled platelets and apheresis platelets were 0.72% and 0.47%, respectively. After 2007, four prevention measures were taken, the positive rates of pooled platelets and apheresis platelets brought down to 0.33% and 0.14%, respectively. From 2006 to 2016, a total of 44 positive reaction platelet samples were identified. The isolated bacteria include coccus, bacillus, aspergillus and anaerobic stains. 13 cases of platelet products in 2006 and 37 cases from 2007 to 2016, which were happened platelet transfused adverse reactions, were examined for the cause. Samples included 50 recipients and transfused bag were taken bacterial cultures, none had bacterial contamination.
Summary/Conclusions: Our study underlines the need for taking prevention measures can further reduce the risk of platelet transfusion-associated bacterial infections, bacterial culture of platelet products is limited.

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THE PRELIMINARY ESTABLISHMENT AND APPLICATION OF QUALITY CONTROL EVALUATION METHOD OF BLOOD COMPONENT BACTERIA TESTING

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Background: According to the preparation of 1% per month or 4 bags of various blood components for bacteria detection, because of fewer, and sustained an annual does not appear more has the phenomenon of bacterial growth, the results, to strengthen the quality of blood component samples bacteria detection process control is necessary.
Aims: To ensure the accuracy, reliability and validity of the results of bacterial detection of blood components.
Methods: By joining the evaluation activities of blood component bacteria testing of CITIC, analysis of microorganism culture monitoring system instrument on a regular performance of the instrument and the same strain of different concentrations and different strains of bacteria detection ability, realize self assessment, and to continuous monitoring of laboratory detection ability, to improve its ability of detection. Establish a concentration of 30 cfu/ml staphylococcus aureus, or concentration of 50 cfu/ml E. coli, such as standard strain single bacteria as low concentrations of bacteria detection of internal control, set conditions as follows: the qualitative results for the "+", check out time < 24 h. At the same time, bacterial detection was carried out parallel to conventional blood component samples.
Results: The coincidence rate of the results of the three bacterial test rooms in 2016 is 100%. The conformity rate of the first time in 2017 is 90%; The reason for leak detection is extremely low concentration of "educational" sample E (aspergillus Niger, 10 cfu/ml), which was not detected in the PBA bottle culture of the microbial culture monitoring system for 7 days; The SI value of the positive sample is s less than 2, the optimal rate of time distribution is 100%, and the detection value is controlled in the "mean plus or minus 2SD" range. In routine work, the qualitative results of BPA in low-concentration bacterial detection are "+", and the detection time (h) is 12.92 + 1.94; The qualitative results of BPN are "+" and the detection time (h) is 13.48+ or 2.49, which met the expected requirements of its setting. Between January 2016 and July 2017, we tested the samples of 398 bags of blood ingredients, and the test results of the oxygen culture bottle (BPA) and anaerobic culture bottle (BPN) are all qualified.
Summary/Conclusions: Through the coincidence rate of the results of the interventricular quality assessment report and the time of detection, objective to evaluate the laboratory's ability to detect the bacterial contamination of the blood components, and to carry out continuous monitoring and improvement of the laboratory. The method of detecting internal control with low concentration bacteria is established, and the quality control of each experiment is effectively carried out to ensure the correctness, reliability and validity of the experimental results.

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RESEARCH ON THE POTENTIAL RISK OF TRANSFUSION-TRANSMITTED BRUCELLOSIS IN BEIJING

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Background: Brucellosis is a worldwide zoonosis,which has great economic importance. The major source of infection for brucellosis is the infected domestic animals. There are some reports regarding brucellosis can survive well in stored blood. Brucellosis is now endemic in 25 provinces of China, and also increasing in some areas of Beijing. The positive rate of brucellosis was 6.91% in Huairou, 0.35% in Changping, and increased from 0 in 2008 to 1.72% in 2013 in Pinggu. Blood screening of blood donors in Beijing and surrounding areas is of great significance to the safety of blood. It is necessary to investigate the prevalence of brucellosis among healthy blood donors in Beijing.
Aims: The aim of this study was to investigate the prevalence of brucellosis among healthy blood donors in Beijing, and explore the potential risk of transmission of brucellosis.
Methods: Randomly selected 2657 healthy donors, collect by Tongzhou Center Blood Station from July 2016 to January 2017. Rose Bengal Plate Agglutination Test(RBPT) and Standard Tube Agglutination Test(STAT) were used to detect brucella antibody, and positive samples of the two methods were detected by ELISA.
Results: In 2657 samples, RBPT was positive in 4 cases(0.15%), degree of aggregation are "++". SAT was positive in 4 cases (0.15%), 3 cases with SAT titer ≥1:100 (+++), 1 case SAT titer ≥1:200 (++++), including 2 cases of SAT/RBPT positive. The other 22 cases detected by SAT were suspicious samples, SAT titer ≥1:50 (++). 1 cases with ELISA positive (0.04%), OD=120.0, the concentration is 670.875 ng/l. Among SAT positive and suspicious samples: female> male ($P < 0.005$), pastoral areas > other areas ($P < 0.005$), the difference was statistically significant.
Summary/Conclusions: It is important for the safety of blood with a strict health consultation for blood donors who are infected with Brucellosis, and Brucella blood screening would carried out in Brucellosis endemic areas.

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Parasites

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PREVALENCE OF HAEMOPARASITES AMONG BLOOD DONORS IN NAKASERO BLOOD BANK

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Background: Haemoparasites can be a potential threat once found in a donors blood. However, the prevalence of parasitic infections especially haemoparasites is a serious case which needs to be addressed, some diseases resulting from parasites during the transfusion race due to fatality among which filarial worms, malaria parasites, trypanosomes has been reported to cause.
It is also stated that trypanosomes are found in Africa though the prevalence is very low among the donors as compared to other haemoparasites.
Aims: The objective of the study was to determine the proportion of haemoparasites in donated in Nakasero blood bank, Uganda.
Methods: A cross sectional study was carried out at Nakasero Blood bank between the months of April and July 2016. About 2 ml of venous blood samples were collected into an Ethylene Diamine tetra acetic acid (EDTA) containing bottles for the study.
Aseptic technique was used during the blood collection to prevent any blood contamination.
Results: Out of 384 samples, Positive for Haemoparasite species were 22 (5.7%), the rest being negative 362 (94.3%). Of the proportion of Haemoparasites, Plasmodium species were 16 (72.7%) as per thin film, Microfilaria species 6 (27.3%) were positive. Blood group O had the highest prevalence 14 (63.6%), blood group A 5 (22.7%), blood B 3 (13.7%) and blood group AB 0 (0.0%). No trypanosome species were seen.
The number of male positive for haemoparasites was 15 (68.2%) and females were 7 (31.8%).
Summary/Conclusions: Malaria parasite is the most prominent haemoparasite in Nakasero blood bank with a higher prevalence in male than female. There was insignificant prevalence of microfilaria and much less insignificant in Trypanosomes.

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INVESTIGATION OF BABESIA MICROTI SEROPOSITIVITY BY IN-HOUSE ELISA KITS IN BLOOD DONORS

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Background: The tick-borne pathogen Babesia spp. has become recognized as the leading infectious risk associated with blood transfusion in the United States. Transfusion transmitted Babesia spp. cases were also reported in some countries. In China, there were some investigations of Babesia spp. infection on normal populations in several regions and the current status of the pathogen is insufficiently evaluated. No commercial reagents to screen Babesia spp. are available and there is limited data regarding Babesia spp. infections in blood donors in China.
Aims: The aim of this study was to evaluate the seropositivity of *Babesia microti* (*B. microti*) in blood donors in Jiangsu, eastern China.
Methods: An investigational in-house enzyme immunoassay (EIA) kits for *B. microti* as a screening test were used in donors' blood samples from March to May in 2017 in Jiangsu Province Blood Center. The applied specific human immunoglobulins targeting peptides derived from *B. microti* secreted antigen 1(BmSA1) were coated as antigen in this house optimized ELISA. The EIA positive samples were confirmed by western blot (WB) and nested-PCR targeted 18S ribosomal RNA gene.
Results: Total 950 blood donor's samples including 649 male and 301 female donors were screened and 5 (3 male and 2 female) were positive for BmSA1. Four of them were local residents and one was from other province. The five samples were positive by WB, but they were all negative by nested-PCR.
Summary/Conclusions: This investigation showed that there is *B. microti* infection in blood donors in Jiangsu province. However, larger scale, multicenter studies should be conducted to further evaluates the epidemiology of *B. microti* infections in Chinese blood donors.

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Abstract has been withdrawn

Newly Emerging Pathogens and Other Transfusion Related Pathogens

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NO EVIDENCE OF NOVEL HUMAN PEGIVIRUS 2 ACTIVE INFECTION IN HCV-INFECTED BLOOD DONORS FROM FRANCE AND CHINA

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Background: A novel bloodborne human pegivirus, tentatively named human hepegivirus 1 (HHPgV-1) or human pegivirus 2 (HPgV-2) was recently identified in multi-transfused and hepatitis C virus (HCV)-infected patients and intravenous drug users from the US and UK. Identification of this virus in post-transfusion samples of blood product recipients suggested a potential risk of blood transmission, although the pathogenicity of this virus remains unknown. HPgV-2 active infection prevalence is largely unknown due to the limited number of studies conducted on selected populations and the lack of standardized serologic and molecular screening assays.
Aims: To investigate active HPgV-2 infection in blood donors originating from Western Europe and Asia.
Methods: HPgV-2 RNA detection was performed with a one-step TaqMan-based real-time RT-qPCR assay using primers/probe targeting the NS2-3 region as previously described. Partial NS2-3 sequence derived from the HPgV-2 prototype sequence (GenBank KT439329) was *in vitro* synthesized and used as control in the RT-qPCR assay (limit of detection: 100 RNA copies/reaction).
Results: HPgV-2 RNA was investigated by RT-qPCR in 1,123 plasma samples from HCV-infected (656 HCV RNA+/anti-HCV Ab+ and 467 HCV RNA-/anti-HCV Ab+) and 94 non-infected blood donors collected in France and China between years 2007–2016. Intravenous drug (IVD) injection was reported for 114 (18%) of 636/860 (74%) HCV-infected French donors with documented risk factors. No HPgV-2 RNA was detected in the 1,217 samples investigated, irrespective of the geographical origin, HCV infection status, or IVD user status. Sample quality and/or presence of inhibitors were eliminated by successfully quantifying HCV RNA load ($\log_{10}$ 4–7 IU/ml). The HPgV-2 RNA loads reported so far (range: $\log_{10}$ 2.5–6.6 copies/ml) were above the detection limit of the assay used, minimizing the risk of false negative related to a lack of sensitivity of the method. Genetic diversity of HPgV-2 strains of different geographical origins may negatively affect RNA detection since a ~5% nucleotide diversity was observed between the North American HHpgV-1/HPgV-2 sequences available. However, nucleotide substitutions in the reverse primer and/or probe annealing sites did not impact on detection.
Summary/Conclusions: Absence of detectable HPgV-2 RNA in HCV-infected blood donors from France and China may be related to lower prevalence compared to those reported in donors (~1%) and IVD users (10.9%) from US, possibly related to difference in exposure. However, HPgV-2 exposure in blood donors might be underestimated due to the inability to serologically detect potential recovered infections as a validated immunoassay is currently unavailable. Further investigations using larger cohorts and standardized sensitive assays are required to determine the global epidemiology and the natural history of this new virus to estimate its potential pathogenicity but nowadays impact on blood safety seems limited.

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METAGENOMICS ANALYSIS OF HEALTHY BLOOD DONORS IN CHONGQING OF WESTERN CHINA, WITH IDENTIFICATION OF EMERGING INFECTIOUS DISEASES

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Background: Many emerging infectious pathogens are known to be existed in healthy blood donors, and could be transmitted via blood transfusion or plasma derivatives with potential hazardous consequences against recipients. With more convenient application of high through put sequencing, it becomes much easier to investigate uncultured microbiome in qualified blood donations to identify the pathogens.
Aims: The goal of this study is to investigate the microbiome in healthy donations in Chongqing using metagenomics methods and to reveal some emerging and re-emerging infectious diseases which might potentially threat the blood safety.
Methods: This study used deep sequencing strategy to analyze the metagenomics from pooled sample collected from 5,000 voluntary blood donors from Chongqing. Genomic DNA and RNA was extracted and amplified with random primers PCR respectively in order to construct a 250PE library which would send to do deep sequencing using Illumina Miseq. All reads were trimmed to remove low quality bases and adapter sequences. The fully overlapping paired-end reads passing the quality filter were concatenated using PEAR (v0.9.6, Zhang et al., 2014). A single N was used to join the paired-end reads without overlaps together by concatenating the mates. We classified the final reads using Kraken (v0.10.5-beta, Wood et al., 2014) and a Kraken database made from complete RefSeq bacterial, archaeal and viral genomes, along with the GRCh37 human genome. The unclassified reads by Kraken were aligned to NCBI nt database using BLASTn (v2.2.30+, Altschul et al. 1997) with E-value cut-off $1E^{-5}$. The best alignment hits were used to classify the reads. Krona (v2.5, Ondov et al., 2011) was used to generate all taxonomic distribution plots. Finally, the potential emerging and re-emerging infectious pathogens were identified out of the classified microbiome by experience.
Results: A total of 1.23 GB raw data with 2,450,046 assembled reads were generated in the DNA library. And, the RNA library were 1.98GB raw data with 3,967,242 clean reads. After cleaning the human background, 211 reads from bacteria, 98 reads from viruses, and 341 reads from parasites were identified. The major pathogen were shown as *Toxoplasma gondii* and *Spirometra erinaceieuropaei*. There were 3 kinds of viruses from Anelloviridae were found including Torque teno virus, Torque teno midi virus, and Torque teno mini virus. In addition to the Anelloviridae, there were only one virus named Avian leukosis has been detected. Besides, some pathogens which were already considered as threats to blood safety in some developed countries were also discovered such as *Plasmodium spp*. and *Leishmania infantum*.
Summary/Conclusions: The investigation has revealed the metagenomics structure of the qualified blood donations in Chongqing. The results showed us a thought-provoking discovery of DNA fragments of some pathogens which might threat the blood safety. The displayed serious results let us have to think about regulating some reasonable screening methods as well as donor recruitment strategy in certain epidemic areas or seasons to ensure the blood safety. However, on the contrary, the results should be considered more cautiously because the existing of DNA fragments could not represent the existing of infectious pathogens. The validity of the metagenomics hints were suggested to go through epidemiological investigations and specifically tested under laboratory ways such as bacteria or virus culturing to ensure the vitality of those pathogens.

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Abstract has been withdrawn

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SEROPREVALENCE OF HEPATITIS E VIRUS INFECTION AMONG BLOOD DONORS OF GUANGZHOU, CHINA

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Background: Hepatitis E virus (HEV) can be transmitted through blood transfusion and can cause chronic hepatitis in immunocompromised patients. Little is known on the prevalence of HEV among blood donors in Guangzhou city of China.
Aims: To investigate the seroprevalence of HEV infection in voluntary blood donors who donated in Guangzhou from April to June 2017.
Methods: We screened anti-HEV IgG, IgM and HEV antigen in 3044 plasma samples using enzyme linked immunosorbent assay (ELISA). Samples with the presence of anti-HEV IgM or HEV antigen were further tested for HEV RNA by quantitative real-time PCR.
Results: The seroprevalence of anti-HEV IgG, IgM and antigen was 20.04% (610/3044), 0.89% (27/3044) and 0.03% (1/3044), respectively. However, the HEV RNA was undetectable in all anti-HEV IgM or HEV antigen positive samples. Besides, we found that the prevalence of anti-HEV IgG increased significantly with age and was higher in men (22.95%, 431/1878) than in women (15.35%, 179/1166).
Summary/Conclusions: The seroprevalence of anti-HEV IgG was high in Guangzhou blood donors, but no HEV RNA positive case was found. The risk of transfusion transmission of HEV remains to be investigated.

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USP18 IS A KEY MEDIATOR OF DENGUE VIRUS RESISTANCE TO TYPE I INTERFERON

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Background: Although Dengue Virus (DENV) is mainly transmitted by the infected mosquitoes of the genus *Aedes*, it may also be transmitted by the contaminated blood products. Type I Interferons (IFNs), as the main component of the host innate immunity, play an important role in defending virus infection. Previous studies demonstrated that type I IFNs inhibited DENV replication if the cells were pre-treated with IFNs prior to infection, while cells infected with DENV first developed IFN resistance. The underlying molecular mechanism remains unclear. Ubiquitin-specific protease 18 (USP18) is a negative regulator of the type-I IFN signaling and increased USP18 expression contributed to IFN resistance of HCV. In the present study, we investigate the role of USP18 in IFN resistance of DENV.
Aims: To explore the role of USP18 in IFN resistance of DENV.
Methods: Silencing of USP18 was achieved by siRNA transfection into Hela cells and the silencing efficiency was confirmed by western blot. The expression level of USP18, type I Interferons (IFNα/β) and DENV RNA were quantified by Real-Time PCR and the Jak/STAT signaling pathway was examined by p-STAT1 (western blot), Interferon-sensitive response element (ISRE) activity (dual-luciferase reporter assay), and ISG expression (real-time PCR).
Results: DENV infection led to the increased expression of USP18. siRNA specifically targeted against USP18 in Hela cells infected with DENV make the DENV more sensitive to IFN treatment as shown by the fact that 70% lower level of DENV RNA replication was observed in USP18 silenced cells compared to the irrelevant SiRNA control group. Silencing of USP18 activated the IFNa-mediated Jak/STAT signaling pathway as shown by the increased expression of p-STAT1, enhanced ISRE activity, and elevated expression of ISGs.
Summary/Conclusions: USP18 induced by DENV infection is a critical host factor that is utilized by DENV to confer resistance to type I interferon and USP18 can serve as a potential target for developing novel therapeutic agents against DENV, and potentially other viruses' infections.

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Abstract has been withdrawn

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SEROPREVALENCE OF HTLV- I/II AMONG VOLUNTARY BLOOD DONORS IN VIETNAM

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Background: Infection with HTLV-I, though asymptomatic in most cases, can lead to potentially grave consequences, such as adult T-cell leukemia-lymphoma and HTLV-1-associated myelopathy/tropical spastic paraparesis. Association of HTLV-II with leukemia pathogenesis is not established; however, there is some evidence of an association with a neuro-degenerative disease similar to HAM/TSP and occasionally also with lymphoproliferative disease. Both viruses may be transmitted by whole blood transfusion, from mother to child predominantly through breastfeeding, and by sexual contact. Very little is known about HTLV prevalence among blood donors in Vietnam and hence routine pretransfusion screening for HTLV has not mandatory in the country until now.
Aims: The study was designed to investigate the seroprevalence of HTLV-I/II among Vietnamese blood donors.
Methods: Total of 14,819 samples from healthy blood donors from the Northern, Central and Southern parts of Vietnam and another 1,003 samples from blood donors who were found reactive with one or more markers including HBsAg/anti-HCV/HIV Ag/Ab were included in the study with informed consent. The blood samples were screened for anti-HTLV-I/II antibodies by Chemi-Luminescence Immuno-Assay (CLIA) using Abbott Architect HTLV-I/II assay. The anti-HTLV-I/II repeat reactive samples were further tested by immunoblot method using MP Biomedicals HTLV Blot 2.4 for confirmation and differentiation of HTLV-I/II infection. Pro-viral HTLV genome testing was performed on the available whole blood repeat reactive samples (N = 11). The data was analyzed with MS Excel and SPSS 22.0.
Results: Among 14,819 blood donors, 32 samples (0.22%) were reported repeat reactive with anti-HTLV-I/II Screening assay, but only 1 case was confirmed positive (0.0067%), and 5 cases were classified as indeterminate (0.034%) on immunoblot. The rate of repeat reactive anti-HTLV-I/II samples was 0.399% among samples from blood donors who were reactive with HBsAg/Anti-HCV/HIV markers, but none of them were confirmed by MP Biomedicals HTLV Blot 2.4. Further confirmatory testing of N = 11 repeat reactive whole blood samples was performed by pro-viral HTLV *gag* and *tax* region molecular characterization but none of them were confirmed. The whole blood samples from the donors that exhibited positive reactivity on MP Biomedical HTLV 2.4 blot as well as one donor with indeterminate blot results were not available for further molecular characterization.
Summary/Conclusions: HTLV-I/II prevalence was found to be low among blood donors in Vietnam in this study. However, continuing efforts to monitor the infection trends and paying attention to key regions to control high-risk transmission for a safe blood transfusion is recommended.

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Abstract has been withdrawn

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INVESTIGATION OF HTLV-I/II INFECTION AMONG 30 308 CASES VOLUNTARY BLOOD DONORS IN NANCHANG

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Background: Human T- Lymphotropic virus(HTLV) is a kind of complex retrovirus and widely distributed throughout world. The corresponding information was reported in local area of China. It is necessary to investigate the infection status of human T-Lymphotropic virus among volunteer blood donors in Nanchang, and provide appropriate strategies for blood safely in this area.
Aims: To investigate the infection rate of human T-Lymphotropic virus(HTLV) in Nanchang voluntary blood donors,providing the reference for establishing blood screening strategy formulation.
Methods: 30,308 volunteer blood donors (16,811 males, 13,497 females, range 18–54 years, from April to December in 2016) were enrolled in the study. The blood samples were collected and anti-HTLV-I/II of the serum were detected by double-antigen sandwich enzyme-linked immunosorbent assay(ELISA).All samples of ELISA screening reactive were retested twice, if retested results were still reactive,the samples was confirmed by Western Blot(WB) and PCR.
Results: 15 donors samples were screening reactive (15/30,308, 0.049 5%).And among them 2 cases (1 male and 1 female) were indentified positive by PCR. The positive rate of anti-HTLV I/II in blood donors in Nanchang was 0.006 6% (2/30,308).
Summary/Conclusions: The anti-HTLV I /II blood donors in Nanchang is still at a relatively low lever. It is necessary to expand the amount of monitoring and follow-up, and a critical need to take comprehensive preventive measures.

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THE INVESTIGATION AND STRATEGY OF HTLV INFECTION IN CHANGSHA BLOOD DONORS

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Background: Changsha belongs to HTLV low prevalence area, as a new economy active area, the migrating population increased greatly, and many universities load here. Students are main force of blood donors, and many blood donor are from the coastal area of HTLV high prevalence. Therefore, HTLV blood donation screening can improve the safety of blood transfusion.
Aims: To investigate the infection rate of human T-lymphotropic virus (HTLV) among voluntary blood donors in Changsha, and to provide the reference for developing HTLV antibody testing and taking other relevant measures in Hunan blood center.
Methods: While conducting routine testing, HTLV I/II antibody was detected by ELISA in voluntary blood donors from Hunan blood center. And the positive samples were further confirmed by National center for clinical laboratories. The distribution of voluntary blood donors from 2000 to 2016 in different provinces was analyzed using Sigmaplot drawing software.
Results: 16767 blood donors were collected and tested, of these, 24 likely positive by screening (0.14%), these samples were further confirmed by Western blot, 1 case was identified positive (0.00596%), from Fujian Province, and 1 case was determined indeterminate, from Hunan Province. As can be seen from the distribution of blood donation population, the number of Fujian, Guangdong Province and other coastal areas was more than Hunan.
Summary/Conclusions: The anti-HTLV I/II blood donors in Hunan blood center are at a relatively low level, but the number from high level HTLV donors increases quickly. It is necessary to develop HTLV antibody testing and take other relevant measures in Hunan blood center in order to ensure blood safety and the health of the receptors.

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MANIPULATION OF VIRAL MICRORNAS IS A POTENTIAL ANTIVIRAL STRATEGY FOR THE TREATMENT OF CYTOMEGALOVIRUS INFECTION

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Background: Cytomegalovirus (CMV) infection leads to notable morbidity and mortality in immunosuppressed patients. Current antiviral drugs are effective but are seriously limited in their long-term use due to relatively high toxicity.
Aims: According to in vitro studies, CMV miRNAs play important roles in the regulation of viral replication, immune modulation, and immune evasion. We investigate the antiviral effect of CMV miRNAs on viral production.
Methods: In the present study, we characterized the expression of murine CMV (MCMV) miRNAs both in vitro and vivo.
Results: Although 29 miRNAs were detectable during in vitro infection, only 11 miRNAs (classified as Group 1) were detectable during in vivo infection, and as many as 18 viral miRNAs (classified as Group 2) were less detectable (<50% of animals) in both the liver and lungs. In addition, viral miRNA profiles obtained for the blood revealed unstable, reduced expression. We next explored the in vitro effects of viral miRNAs on MCMV replication. The inhibition of Group 1 viral miRNAs exerted less influence on virus production, but transfected cells overexpressing four miRNAs in Group 2 showed a statistically lower viral loads than those transfected with control miRNA (29%, 29%, 39%, and 43%, respectively, vs control). Finally, we performed hydrodynamic injection of viral miRNA agomirs and observed lower levels

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of MCMV recurrence in the livers of animals overexpressing two miRNAs agomirs in Group 2 than in the control livers, confirming the antiviral effects of viral miRNA manipulation in vivo.
Summary/Conclusions: Thus, the manipulation of viral miRNA expression is an attractive potential therapy and represents a novel antiviral strategy for the miRNA-based treatment of cytomegalovirus infection.

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INVESTIGATION OF THE SIGNIFICANCE OF ALT SCREENING IN VOLUNTEER BLOOD DONORS UNDER THE CONDITION OF PERFORMING HBV/HCV NUCLEIC ACID TESTING

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Background: With the development of nucleic acid detection technology, more and more discussions have been made about whether to cancel ALT detection or not, now re-evaluating the importance of ALT testing for blood safety improvement after nuclear acid testing.
Aims: By analyzing the different range of Alanine aminotransferase (ALT) levels in donor samples to re-evaluate the importance of ALT testing for blood safety improvement after nuclear acid testing.
Methods: 148118 blood donors were collected in Changsha blood center during 2016 and 2017 by rate method for detecting ALT, all samples were analyzed by ELISA, NAT and ALT testing, among which ALT unqualified ones were investigated retrospectively.
Results: Among 148,118 samples, 859 samples were found ALT unqualified of which 4 samples were HBsAg and HBV DNA positive, 3 samples disqualified, Comparing the ALT deferral rate with the ALT deferral rate of HBsAg and anti-HCV, the difference was statistically significant ($P < 0.05$). There were no samples of ALT disqualified and positive for HBV DNA and one sample positive for HCV RNA. There were 8 samples of normal ALT and negative ELISA results but positive for HBV DNA. 847 samples which were only ALT unqualified experienced a retrospective investigation, among which 60 samples were donors who had been previously ALT unqualified for many times.
Summary/Conclusions: ALT testing plays a role to reduce residual risk of transfusion-transmitted diseases. NAT testing can obviously reduce HBV leakage of ELISA screening.

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EVALUATION OF THE SENSITIVITY AND SPECIFICITY OF COBAS® CHIKV/DENV

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Background: Arboviruses such as chikungunya virus (CHIKV) and dengue virus (DENV) are a public health concern. Transfusion-transmitted (TT) DENV has been reported in Asia, South America and the Caribbean; no TT-CHIKV has yet been reported. Currently, no duplex test for CHIKV and DENV is available for blood screening. cobas® CHIKV/DENV for use on the cobas® 6800/8800 Systems is a qualitative in vitro nucleic acid screening test (NAT) for the direct detection of CHIKV RNA and DENV serotypes 1–4 RNA in human plasma. This multi-dye test simultaneously detects and discriminates the two viruses.
Aims: To perform an external evaluation of the analytical sensitivity and clinical specificity of cobas® CHIKV/DENV.
Methods: Testing was performed on a cobas® 8800 System at the American Red Cross. Analytical sensitivity: Evaluated using randomized panels prepared from serial dilutions of CHIKV and DENV heat-inactivated culture isolates (CHIKV concentrations traceable to 2014 CBER panel; DENV to 2013 DENV-1 serotype CBER panel). Multiple dilution series of co-formulated CHIKV and DENV at 5 concentrations (from approximately ~0.125× to 2× the 95% limit of detection (LOD)), and negative panel members were tested. Each concentration (24 replicates of each) was tested with 3 reagent lots. Results were used to calculate the LOD for each target by Probit Analysis. For each viral target, the estimated LOD from the combined lots was compared to the LOD claim for cobas® CHIKV/DENV (difference of $\log_{10}$ LODs (estimated – claimed) to be ≤ 0.3).
Clinical specificity: Samples from ≥10,000 US volunteer donations were de-identified and tested by individual donor testing with cobas® CHIKV/DENV. Reactive samples were to be tested by alternative NAT and evaluation of amplicon by hemi-nested PCR or probe fragment hydrolysis. A reactive result on cobas® CHIKV/DENV would be considered a true positive if the additional testing was positive. Specificity of cobas® CHIKV/DENV was calculated as the percentage of CHIKV/DENV RNA negative samples that were non-reactive.
Results: The percentages of reactive results with the LOD panels across 3 lots were 12.5% (~0.125x LOD), 44.4% (~0.25× LOD), 59.7% (~0.5× LOD), 90.3% (~1x LOD) and 98.6% (~2x LOD) for CHIKV positive samples. The percentages of reactive results were 43.1% (~0.125x LOD), 65.3% (~0.25x LOD), 88.9% (~0.5x LOD), 100% (~1x LOD) and 100% (~2x LOD) for DENV positive samples. Negative panel members were nonreactive. The estimated 95% LOD for CHIKV was 10.5 CBER NAT Detectable Units/ml (95% CI: 8.2 to 14.6 CBER NDU/ml). The estimated 95% LOD for DENV was 52.8 PCR Detectable Units/mL (95% CI: 41.6–75.4 PDU/ml). Difference of estimated minus claimed results was +0.2 logs and -0.1 logs for CHIKV and DENV, respectfully.
There were no donations with cobas® CHIKV/DENV-reactive results. The prevalence of CHIKV/DENV reactive results in US donations was 0% (0/10,528; 95% Exact CI: 0% to 0.035%). The specificity of the cobas® CHIKV/DENV test was 100% (95% Exact CI: 99.965% to 100.000%) for both the CHIKV and DENV targets.
Summary/Conclusions: The analytical sensitivity for both the CHIKV and DENV targets was comparable to the manufacturer's claim.
None of the US donations in this study were reactive with the cobas® CHIKV/DENV test.
The 100% clinical specificity demonstrated in this study supports the use of cobas® CHIKV/DENV as a blood donation screening test for CHIKV and DENV RNA.
cobas® CHIKV/DENV for use on the cobas® 6800/8800 Systems is not commercially available.

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PREVALENCE OF HUMAN PARVOVIRUS B19 AND HUMAN PARVOVIRUS 4 IN BLOOD DONORS OF XI'AN, CHINA

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Background: Human parvovirus B19 (B19V) and human parvovirus 4 (PARV4) are parvoviruses known to infect humans and transmit through the contaminated blood and plasma derived medicinal products (PDMPs). B19V infection is associated with a spectrum of clinical manifestations and could provoke serious complications in some high-risk groups. Although the clinical significance of PARV4 infection remains uncertain, a variety of clinical associations have been reported. PARV4 infection commonly associated with intravenous drug use and seropositivity of blood-borne viruses has also been documented in Northern Europe, the United States and Asia. However, studies concerning the existence and loading of B19V and PARV4 in Chinese blood donors, especially in Northwestern China, were limited.
Aims: To evaluate the prevalence of B19V and PARV4 in blood donors of Xi'an.
Methods: A total of 2000 plasma donations were collected between 2015 and 2016 from Xi'an. Each sample was tested for aminotransferase (ALT), anti-syphilis, anti-HIV 1/2, HBsAg and anti-HCV and has been confirmed to be qualified according to the requirements of Pharmacopeia of the People's Republic of China. Aliquots of plasma from 10 human blood donations were pooled and subjected to nucleic acid extraction using the High Pure Viral Nucleic Acid Kit (Roche Diagnostics, Mannheim, Germany), according to manufacturer's instructions. Then, a multiplex quantitative real-time PCR (qPCR) assay was performed to detect and quantify all three genotypes of B19V and PARV4 DNA in these samples.
Results: Of 2000 blood donors, 9 (0.45%) specimens were found positive for B19 DNA and all specimens were negative for PARV4. The quantitative DNA levels of B19V ranged from 6.80×10^2 to 1.69×10^5 copies/ml.
Summary/Conclusions: PARV4 exhibited significantly lower infection rates than B19V among blood donors of Xi'an. This result was similar to those observed in other studies obtained from the donor population of the Chinese mainland. This study, to some extent, provided a basis for improving blood safety and preventing B19V and PARV4 infection in China. Larger studies are needed to investigate the prevalence of B19V and PARV4 in Chinese blood donors.

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Immunohaematology Red Cell Immunology: Serology

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INVESTIGATION ON IRREGULAR ANTIBODIES OF RED BLOOD CELL OF PATIENTS WITH β-THALASSEMIA MAJOR IN LI NATIONALITIES IN HAINAN, CHINA

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Background: Thalassemia was once a major health problem in Hainan, especially in Li nationalities. Although regular blood transfusion is an important measure to maintain the life quality of patients with thalassemia, Side effects such as red blood cell(RBC) alloimmunization may be caused. RBC alloimmunization antibody detection rate was 9. 64%, 9.4% and 23.0% among the Chinese thalassemia patients requiring long-term transfusion in Shenzhen, Hong Kong and Taiwan. There have been no reports of RBC alloimmunization about the patients with β-Thalassemia Major in Li nationalities in Hainan. The purpose of this study was to investigate its prevalence among Li β-Thalassemia major patients.

Aims: To investigate the irregular antibodies of red blood cell for β-Thalassemia Major patients with long-term transfusion in Thalassemia high incidence population-Li nationalities in Hainan, so as find Corresponding measures to reduce the incidence of irregular antibodies and make transfusion safe and effective.

Methods: The peripheral blood from 113β-Thalassemia Major patients with long-term transfusion was collected. The irregular antibodies were screened and identified by saline test,polybrene test and indirect antiglobulin test .

Results: The positive rate of irregular antibody was 9.73% (11/113), focused mainly on Rh system antibody (9/11), including anti—E(5/9), anti—E and anti—c(4/9), the other two cases are anti-JK^b and anti-Le^a.

Summary/Conclusions: Such patients should be given multiple antigen comparable blood, especially Rh E, c to reduce the high occurrence of Rh system antibody on the base of ABO, RhD matched transfusion.

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RED BLOOD CELL ALLO-IMMUNIZATION IN IRAN, A COMPREHENSIVE REVIEW OF THE LITERATURE

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Background: In most blood transfusion services, phenotyping and pre-transfusion compatibilization of the most immunogenic antigens are normally applied to patients that require regular transfusion. However, the risk of alloantibody development still remains a big challenge, particularly in developing countries like Iran, where high costs of pre transfusion phenotyping do not allow it to be a part of routine transfusion procedure.

Aims: In the current study we tried to have a comprehensive review on the status of red blood cell alloimmunization in Iran. For this purpose we searched for papers investigating alloimmunization in transfusion dependent patients and also in patients with no regular transfusions who were candidate for surgery or who needed blood.

Methods: We searched PubMed, Google Scholar, SID, and MAGIRAN data bases using the following keywords: 'blood transfusion', 'alloimmunization', 'alloantibodies', 'irregular antibodies', 'red cell antibodies', and 'Iran'. No limitation for the date of publication was defined and studies in either Persian or English languages were included. All the identified records were then screened for the relevance and duplication.

Results: A total of 22 papers were included in this study. 10 papers were in Persian and 12 in English. All of the studies were conducted from 1999 to 2016 and providing alloimmunization data from different cities all over of Iran.

Most of the studies included subjects with β-thalassemia. According to these studies, most of the detected alloantibodies are anti-kell (anti-K antigen) and anti-Rh system antibodies mainly anti-E, anti-D, anti-C, and anti-c.

Three out of 22 studies were conducted on patients without regular transfusions who were candidate for elective surgery. These studies which included a higher number of patients may provide more general data of Iranian population. The incidence of alloimmunization in these patients was 0.8–0.9% and all three papers introduced anti-K, anti-E, anti-C/anti-c as the most common detected antibodies.

Summary/Conclusions: The development of anti-D antibody as one of the most common alloantibodies, shows that we have still problems in blood group typing, because D antigen is one the main antigens which should be matched between donor and recipient especially for young female recipients. This problem may reflect clerical and technical errors made by the staff and also inappropriate or incomplete quality control of the reagents and equipments. Inappropriate methods of antibody detection, lack of standard protocols, and lack of an efficient surveillance system are other causes of allo-immunization in developing countries.

Cross match is another critical step in pre-transfusion testing which may be affected by a wide range of clerical and technical errors; such as mislabeled or partially labeled tubes, misinterpretation of hemolysis in serum grouping as negative and blood group discrepancy due to inadequate washing of cells.

Alloimmunization may also occur because many alloantibodies may not be detected, as no further transfusions are required or because the titer of antibodies decrease over time and reaches a non-detectable level prior to testing. Anyway it is an inevitable outcome of transfusion, since the compatibility of all the transfused antigens is impossible.

Our study with gathering the majority of studies on Iranian population from all over the country may provide an overview on the status of alloimmunization in Iran including the total incidence and the prevalence of responsible antibodies.

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Abstract has been withdrawn

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PREVALENCE OF UNEXPECTED ANTIBODIES IN TRANSFUSED PATIENTS AND PREGNANT WOMEN AT TAKSIN HOSPITAL

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Background: It's very important to screening antibody and identification among pregnant women and transfused patients to provide a better care and safety of blood transfusion services. Antibody screening and identification test considered as a standard routine in immuno-hematology laboratory testing in both antenatal care for pregnant women and compatibility test for transfused patients. However, the differentiation of antibody pattern may different due to population and laboratory method applied in the testing routine.

Aims: This study aimed to determine and identify the prevalence of unexpected antibody in transfused patients and pregnant women who attended antenatal clinic at Taksin Hospital.

Methods: A cross-sectional study of the retrospective (2012–2014) data were analysed for the prevalence of allo-immunization in each group.

Results: A combination of microcolumn agglutination and tube test was applied to the antibody identification on 15,297 samples of transfused patients and 12,445 pregnant women.

Summary/Conclusions: Among the total transfused patients in this study, they were 5,233 males (34.21%), 10,064 females (65.79%) and 12,445 cases of pregnant women. The antibody screening results appeared that 142 transfused male patients (2.71%) were positive and identified as 22 Rh, 53 MNS, 16 P1PK, 14 Lewis, 1 Kidd, 2 Duffy, 26 multiple antibodies and 8 unidentified antibodies, 259 transfused female patients (2.59%) were positive and identified as 46 Rh, 101 MNS, 23 P1PK, 22 Lewis, 1 Kidd, 50 multiple antibodies and 16 unidentified antibodies. The antibody screening results of pregnant women were positive for 285 cases (2.29%) and identified as 13 Rh, 53 MNS, 93 P1PK, 76 Lewis, 24 multiple antibodies and 26 unidentified antibodies. Our results revealed that the prevalence of unexpected antibodies, found in transfused patients and pregnant women, were not significantly difference ($P = 0.078$).

Thus, the prevalence of unexpected antibodies in each study may found to be different due to the group of study including laboratory testing techniques. However, the group of transfused patients and pregnant women may found to have unexpected antibodies higher than normal people. The result of this study may provoke the awareness and concern the presence of unexpected antibodies in transfused patients and pregnant women in order to provide better care or treatment at Taksin Hospital.

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THE PERSISTENCE AND EVANESCENCE OF RED CELL MIMICKING ANTIBODIES IN VIVO

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Background: Red cell mimicking antibodies (combine with autoantibodies specificity, or not combine) may become weak and undetectable over the course of time. They can cause a haemolytic reaction if the blood which contain corresponding antigen positive against mimicking antibody were transfused. So the research of persistence and evanescence of red cell mimicking antibodies is valuable in theory, and can offer valuable data for safe transfusion. Only a few studies have been published on this topic.
Aims: Through detailed research of some cases of mimicking antibodies (combine with autoantibodies specificity, or not combine). To analyze retrospectively and study the characteristics of mimicking antibodies and the rate of mimicking antibodies reduced in the patient's body. Interpret the clinical significance of the mimicking antibodies is important for patient transfusion management.
Methods: In our research, We detected and collected the cases of patients who produced mimicking antibodies from all of hospitals in Shanghai between 2005 and 2017. Several kinds of mimicking antibodies were tested in laboratory. If the mimicking antibodies were tested again after the first detection, we recorded these antibodies data included the strength of antibodies, the length of follow-up, and calculated the rate of evanescence.
Results: We retrieved the records of 12 antibodies (included mimicking anti-E/Ec,-Ce/C,-D,-S), which had been tested again after the initial detection, and found that 25.0% were non-persistent in average 29.9 days. The average rate of evanescence of mimicking antibodies is 0.377/per day (0.377 is agglutination reaction strength, see AABB Technical Manual 17th Editions Page: 873–874).
Summary/Conclusions: Mimicking antibodies did not see increased obviously after clinical blood transfusion treatment with red blood cells which have corresponding antigens. And after transfused negative red blood cells, mimicking antibody may became rapidly weakened. Blood transfusion treatment should be as far as possible to avoid mimicking antibody specificity, transfuse corresponding antigen negative of red blood cells will have obvious therapeutic effect on the patients.

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DETERMINATION OF IRREGULAR ANTIBODIES IN DONATED BLOOD AT A TERTIARY CARE CENTER IN RAJASTHAN, INDIA

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Background: Ever increasing efforts at improving blood safety have led to incorporation of regular screening protocols for detection of unexpected/irregular immune antibodies at various transfusion centers across the globe. Knowledge about the frequency of serum antibodies is important to minimize the risk of hemolytic transfusion reactions and to obtain timely and properly matched blood components for transfusion. This requires determination of irregular antibodies present in the serum as a part of immunohematological analysis in addition to ABO-Rh typing and extended phenotyping of red cell surface antigens.
Aims: 1. To generate blood bank donor data base for constitution of panel of blood donors to ensure the safety and effectiveness of clinical blood transfusion.
2. To determine the prevalence and specificity of unexpected/irregular anti-red blood cell antibodies among healthy blood donors, in order to find the rules.
3. To quantify the magnitude and specificity of the problem in the population and initiate measures to screen the population or a selected group in the population.
4. To obtain in-house antibodies from donated plasma for red cell typing to ensure compatible blood units for the patients having irregular antibodies in their serum.
Methods: This prospective study was conducted in the Department of Immuno-Haematology & Transfusion Medicine, Sardar Patel Medical College & Associated Group of Hospitals, Bikaner (Rajasthan), India among healthy blood donors during the period from 1st March, 2017 to 31st July, 2017. Donors' blood samples were tested for unexpected/irregular red blood cell antibodies (other than anti-A and anti-B antibodies) on automated immunohematology analyzer (Solid Phase Red Cell Adherence Technology) using commercially prepared pool cells and panel cells (3 and 14 cell panels), along with performing direct and indirect agglutination tests using gel cards (Column Agglutination Technology). SPSS version 16 and MS Excel were used for statistical data analysis.
Results: Out of the total 11,476 donor blood samples screened, presence of irregular antibodies was observed in total 6 samples (0.052%). On antibody identification, 2 samples were detected to have anti-K antibody, 1 was found to have anti-D antibody and 3 were detected to be positive for auto-antibodies. The positivity rate of irregular antibodies was found to be significantly higher among female blood donors (0.632%) compared with male blood donors (0.027%) (P = 0.000).
Summary/Conclusions: Determination of clinically significant irregular antibodies in donated blood is very useful for the patients receiving large volumes of plasma or whole blood as in massive blood transfusions and in pediatric patients. Irregular antibodies present in donors' plasma, upon transfusion, may lead to hemolytic transfusion reactions. These antibodies may cause incompatibility during cross matches causing unnecessary delay in the transfusion. That is why, 100% donated blood units must be tested for the presence of irregular antibodies in their serum as a part of pre-transfusion compatibility testing. The donated plasma, in which antibodies are detected, could be used as in-house antisera to reduce the hospital cost because commercial antibodies are very expensive.

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PRE-TRANSFUSION CONSULTATION IN PATIENTS WITH POSITIVE DIRECT COOMBS TEST

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Background: Understand the patient's immune history, especially those with direct Coombs test positive and with a recent history of blood transfusion. Through inquiry, we can quickly select effective test methods to detect and ensure the safety and timely treatment of blood transfusion.
Aims: Research methods and cross matching strategy for establishing blood samples of patients.
Methods: Serological methods: including saline medium, anti globulin, microcolumn gel method, condensation method and absorption and dispersion test. The antibodies in red blood cell antigen and serum (plasma) were screened and identified, if necessary, the absorption and dispersion test will be carried out for further identification It is good for the accuracy of methodology.
Results: In 2016, 147 patients with refractory blood transfusion were examined. There were 43 cases of specific antibodies, 119 cases of positive anti - positive (14 cases with recent blood transfusion history). Direct Coombs test positive is the main cause of difficulty in blood distribution, accounting for 80.95%. Recent transfusions resulted in Direct Coombs test positive patients, with a prevalence of clinically significant immune antibodies up to 92.86%.
Summary/Conclusions: Detailed interrogation of blood before cross transfusion is essentially, because of the patient's condition can be understood through interrogation, especially if there is a recent history of blood transfusion. If there is a recent transfusion and Direct Coombs test positive, then the possibility of immune antibodies produced by patients higher. In the test, select at least two or more methods for antibody screening. If necessary, further tests should be carried out to avoid the omission of weak antibodies, It can make transfusion safer and more effective.

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Abstract has been withdrawn

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ANALYSIS OF BLOOD IRREGULAR ANTIBODIES IN THE POPULATION OF CHINESE GUANGXI AREA

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Background: Blood transfusion is an important treatment to save patients' lives. With the development of transfusion medicine, increasingly importance has been attached to safe transfusion. Studies have shown that irregular antibody is an important problem affecting the clinical safety of blood transfusion.
Aims: To analyze the distribution characteristic of irregular antibodies in the patients of Chinese Guangxi Nanning area.

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Methods: 1206 patients with difficult blood cross matching were studied, all of them were from the hospitals of Guangxi Nanning area dated from January 2007 to December 2016, and showed positive by antibody screening test from hospitals. The irregular antibodies were detected and identify by salt water method, polybrene test and anti-globulin test.
Results: 516 patients were identified with irregular antibodies, the irregular antibody positive rate was 42.79% (516/1206). Among the antibody positive patients, 245 were male and 271 were female, 302 had autoantibodies and 214 had alloantibodies. Of 214 patients with alloantibodies, 178 had a history of blood transfusion and/or pregnancy, 25 neither had history of blood transfusion nor history of pregnancy, 11 were hemolytic disease of newborn. RBC alloantibodies specificity: 10 were anti-D, 2 were anti-DC, 12 were anti-Ce, 2 were anti-C, 5 were anti-e, 64 were anti-E, 24 were anti-Ec, 6 were anti-c, 12 were anti–Mur; 1 was combination of anti-c and anti-Mur, 1 was combination of anti-E and anti-Mur, 2 were combination of anti-Ce and anti-Mur, 1 was combination of anti-C and anti-Mur, 1 was combination of anti-e and anti-Mur, 1 was combination of anti-c anti-Mur and anti-Jk^b, 1 was combination of anti-E anti-Mur and anti-Jk^a, 4 were combination of anti-E and anti-Jk^a, 2 were combination of anti-E and anti-Jk^b, 2 were combination of anti-Ec and anti-P_1, 1 were combination of anti-Ec anti-Jk^a and anti-Fy^b, 1 was combination of anti-Ce and anti-M; 25 were anti-M, 1 was anti-N, 2 were anti-Jk^a, 2 were anti-Jk^b, 12 were anti-P_1, 9 were anti-Le^a, 2 were anti-Le^b, 2 were anti-Ce-like, 4 were anti-e-like, 1 was anti-Le^b–like, 1 was combination of anti-Wra and autoantibody.
Summary/Conclusions: Most of the patients with irregular antibodies were female. The main irregular antibodies in Guangxi Nanning region were in Rh system (58.41%, 125/214) and MNS system (17.76%, 38/214); There showed the highest compound irregular antibodies ratio of Rh system combination with MNSs system, especially Rh system combination with anti-Mur antibody .We should be combined with a variety of detection methods to screen and identify the irregular antibodies, which is conducive to detection and identification of different antibodies, and help to provide timely and safe blood transfusion for patients.

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Abstract has been withdrawn

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A CURIOUS CASE OF SUSPECTED ANTI PP1PK IN MAN WITH PROSTATE CARCINOMA

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Background: Anti $PP1P^k$ (formerly known as anti-Tj^a) has been documented to cause acute haemolytic transfusion reaction and spontaneous second-trimester abortion. This antibody is formed in the individuals with rare pp phenotype who lacks P, P1 and P^k antigen. The frequency of this phenotype is approximately 5–6 in one million populations. In this case report, we describe a case of 60 years old Malay gentleman with history Prostatic Carcinoma, who was electively planned for Robotic Prostatectomy. He needed 4 units of the crossmatched compatible blood prior to surgery. Due to inability to find the crossmatched compatible unit, the elective surgery needed to be postponed. The antibody which caused the incompatibility remains unknown due to technical limitations. Predeposit autologous blood donation was performed to collect blood for the patient prior to elective surgery.
Aims: To identify a complex case of irregular antibody to the high-incidence antigen.
Methods: Serological identification was performed on patient's sample using (i) antibody panel identification (ii) papain treatment (iii) panel cells using standard procedures. A serologic red cells phenotype for C, c, E, e, K, k, Fy^a, Fy^b, Jk^a, Jk^b, M, N, S, s, Le^a, Le^b, P^1 and pp antigen were performed according to the manufacturer's instructions. Known pp phenotype frozen packed cell and known Anti $PP1P^k$ serum was tested with the patient sample to confirm the findings.
Results: Patient blood group is Group O Rh (D) positive with negative P1 phenotype. Direct antihuman globulin test was negative. The antibody panel and papain treatment panel was pan reactive with negative auto-control result. Lysis was observed after immediate spin when tested with patient's serum in the tube method. Similarly, the reverse ABO grouping test gave comparable observations. Antigen typing analysis gave the following results: P1 Negative and Tj^a Positive. Extended phenotyping analysis gave the following results: CDe/CDe (R1R1), Jk(a+b+), Fy(a+b-), Le(a-b+), MN, ss, and kk. When patient's serum was crossmatched against with known pp phenotype frozen packed red cells, it was incompatible. Minor crossmatch between patient's packed cell and known serum containing anti-$PP1P^k$ was incompatible too. Laboratory testing failed to prove the antibody specificity due to the limitations. However, we cannot rule out the possibilities of having anti-P or anti-LKE antibody as well in this case.
Summary/Conclusions: Alloimmunizations within the P system (anti-$PP1P^k$ alloimmunization in patients belonging to the P group or anti-PP1 alloimmunization in patients belonging to the P^k group) remain extremely rare and potentially serious conditions. Our results indicate that serology is not sufficient to identify the irregular antibodies and require further testing. In this case we have to perform predeposit autologous blood donation to prepare for his surgery and sample was sent to International Blood Group Reference Laboratory, Bristol for further investigation and we are still awaiting for the results.

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DESCRIPTION OF ONE CASE OF DELAYED HAEMOLYTIC TRANSFUSION REACTION CAUSED BY THE "ENZYME-ONLY" ANTI-LITTLE E

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Background: The enzyme treated cells is not routinely used for the alloantibody screening and identification in the majority laboratories in China because the "enzyme-only" antibodies are considered as clinically irrelevance. The indirect antiglobulin test (IAT) is regarded as a sensitive and reliable method for detecting the significant alloantibodies, colloquially termed low-ionic strength saline- (LISS-) indirect antiglobulin test Liss/Coombs gel test. In this study, a patient with a delayed haemolytic transfusion reaction (DHTR) caused by anti-e undetected by the IAT, but detected by the IAT test using the enzyme-treated panel cells was found.
Aims: To report a case of DHTR cased by the "enzyme-only" anti-e.
Methods: A patient [female; 42 years of age with anaemia; blood group O, ccDEE] had been transfused a total of 16 units red blood cells in five times previously. After the last time of transfusion, the haemoglobin level increased. But 2 days later after the transfusion, a symptom of chills, high fever (up to 40 degrees), nausea, vomiting and clinical signs of haemoglobinuria and haemoglobinaemia were observed. And the haemoglobin level decreased, the haematocrit decreased and the indirect bilirubin level increased. Then, the antibody screening and identification records were retrospectively reviewed. The direct antiglobulin test (DAT) of the patient's red cells and all blood samples before and after transfusion were tested again. The antibody screening was performed using a commercial panel of enzyme untreated 3-cell panel and an enzyme-pretreated (papain) 3-cell panel in the LISS/Coombs gel (DiaMed-ID, Cressier sur Morat, Switzerland) test. Subsequently, the antibody identification was carried out with an enzyme-pretreated (papain) 11-cell panel in the LISS/Coombs gel test. Besides, a compatibility testing in the LISS/Coombs gel test with an enzyme-pretreated (papain) donors-cell designed to be transfused was carried out.
Results: A negative DAT result of the patient's red cells and a positive antibody screening result of the serum only with the enzyme-pretreated (papain) 3-cell panel in the LISS /Coombs gel test ("enzyme-only" specific RBC alloantibody) were obtained. Alloanti-e was identified in the antibody identification testing. Titre of anti-e was 16. All samples from the donors before transfusion have a e antigen positive blood group. Subsequently, 4 units of e-negative red cells were transfused to the patient. The haemoglobin level increased from 43 to 85 g/l, without an occurrence of haemolytic transfusion reaction. In the next 12 months, the patient did not need to receive a blood transfusion.
Summary/Conclusions: "Enzyme-only" alloantibody can caused an acute and delayed haemolytic transfusion reaction. The routine use of the enzyme-treated red blood cells in the antibody screening and identification testing is very helpful for the identification of the specific RBC alloantibodies to ensure the transfusion safety.

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IDENTIFICATION OF A CHINESE PATIENT WITH ANTI-JMH ALLOANTIBODIES

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Background: John Milton Hagen (JMH) blood group system (ISBT 026) includes five antigens expressed by the glycoprotein Semaphorin 7A encoded by the *SEMA7A* gene. Among them, JMH1 antigen is a high frequency antigen and JMH1-negative phenotype is extremely rare in all populations. The majority of JMH1-negative cases are of the acquired and transient phenotype often found in the elder persons and have a chance to produce a likely autoanti-JMH that is not clinically significant, while the inherited JMH1-negative phenotype was only observed in one family without alloantibodies. Besides, several probands of JMH-variant phenotype with JMH-like antibodies were reported resulting from the homozygous mutations of *SEMA7A* gene. In this study, a sample from one patient with a suspicious high frequency antibody was referred to the Reference laboratory for the antibody identification and blood crossmatching.
Aims: To clarify the specificity of the suspicious high frequency antibody in one Chinese patient.
Methods: Peripheral blood sample was collected from a 78 year-old gastric cancer patient. The routine antibody screening and identification were preformed. Then, a soluble recombinant Sema 7A protein was used to determine the specificity of the antibody by the inhibition test in the indirect antiglobulin test with the gel technique. The direct antiglobulin test (DAT) and the self control were also performed with gel technique. The cross-matching test was performed using the RBCs of one of his daughters and two JMH negative cells. Fourteen exons of *SEMA7A* gene were amplified and sequenced in the patient and his daughter with available RBCs.
Results: The serum of the patient had a weak (2+) agglutination with all panel cells except self-control. The DAT test was negative. A negative crosshatching was only obtained with the two JMH negative cells and his daughter's RBCs. The reaction of the serum with the JMH positive cells could be inhibited by the soluble recombinant Sema7A protein to indicate the presence of anti-JMH. The JMH antigen of the proband and his daughter was not determined for lacking the anti-JMH serum or monoclonal antibody. The sequence results of the *SEMA7A* gene revealed two novel heterozygous variants located in the promoter region (-194G>C) and the exon 14 (c.1989G>T, p.Leu663Phe) and one reported variant in the exon 5 (c.507C>T, p.Tyr169Tyr) in the patient and his daughter, while the same heterozygous variant (c.507C>T) was also identified in one control sample.
Summary/Conclusions: The specific alloanti-JMH was firstly identified in a Chinese patient. Two novel variants (-194G>C, c.1989G>T) were identified in the proband and his daughter. Whether the two novel variants are the molecular basis for the unusual JMH expression is still needed further investigation.

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A CASE REPORT OF RARE ANTI-TJA ANTIBODY

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Background: The P blood group system ranks the 4th after ABO, Rh and MNS. P blood group system P1PK contains P1 and PK antigens according to the nomenclature of the International Society of Blood Transfusion (ISBT). The rare p phenotype was first described by Levine and his co-workers in 1951, this antibody was called anti-Tja because it was found in the serum of a patient with gastric cancer and her sister. After reaction with other red blood cells, anti-Tja can rapidly cause hemolysis of the red blood cells. Therefore, only red blood cells with the same phenotype can be administered for these patients in the clinical blood transfusion. After the discovery of the anti-Tja, It was reported that the antibody was associated with the early abortion of pregnancy and can cause hemolytic transfusion reactions and hemolytic disease of the fetus and newborn.
Aims: The anti-Tja antibody serological characteristics of the patient was analyzed to provide a reference for the problem of blood transfusion.
Methods: The following techniques are used in blood group serology: ABO/RhD blood typing, phenotype identification, direct antiglobulin test (DAT), irregular antibody screening and antibody identification. Screened and identified cells used cover antigen systems, including Rh, MNS, Duffy, Kidd, Kell, Lewis, P1, Xg and Lutheran, as well as low-frequency antigens such as Mur and Di^a (Sanquin and Shanghai Hemo-Pharmaceutical &tBiological Co.), which have relatively high prevalence in Asia. Detection methods include conventional tube techniques, gel column test and polybrene method (Shanghai Hemo-Pharmaceutical &tBiological Co.).
Results: This case is female, 50 years old cervical adenocarcinoma patient, early habitual abortion history 13 times, no blood transfusion history. The ABO blood type of the patient is O and RhD positive. The result of DAT is negative. In irregular antibody screening and antibody identification, the serum agglutinated and hemolyzed all groups O red cells under 4 and 37°C, and all of 16 panel red cells representing known P1-positive and P1-negative groups were hemolyzed. After heating the patient serum to 56°C 30 min or the addition of ethylenediamine tetraacetic acid to serum, antigen antibody reaction is changed from hemolysis to agglutination, because of destruction of the complement fraction in the serum. The patient's blood type is considered to be a p phenotype, and the patient is cross-matched with the known p blood donor, which was previously confirmed by the Shanghai blood center. The hospital successfully completed the operation after collecting 400 ml of the patient's blood.
Summary/Conclusions: Through serological studies, the blood type of the patient was confirmed to be the rare p phenotype, and this antibody produced in her serum may led to her habitual abortions. With regard to the immunological factors, the anti-Tja antibody produced by the incompatible blood typing of the pregnant women and the fetal is an important cause of habitual abortion. For patients with such rare blood type, autologous blood transfusion is of great significance. Next, we will study the p phenotype of the patient and pedigree investigation through molecular biology.

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ANTI-DIA, A CLINICALLY SIGNIFICANT RED CELL ALLOANTIBODY DETECTED IN PATIENT: A CASE STUDY IN A TERTIARY HOSPITAL

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Background: A 57 year old Indian gentleman with end stage renal failure presented at the Emergency Department with complaints of chills and non-vertigo giddiness. He has had previous Anti-E red cell alloantibody detected in 2015 and was transfused with approximately 38 units of Red Blood Cells compatible using the Electronic Crossmatch method over a 9-year period prior to Anti-E being detected. Current Group and Crossmatch sample typed the patient as O Positive with a positive antibody screen results on Cell 2 (donor number 318730), a R2R2 cell, (screening cells lot 8SS292 from Ortho Clinical Diagnostics), yielding a strong 4 + reaction. Antibody exclusion panel yielded additional antibody(ies).
Aims: Presence of Anti-E and an additional clinically significant antibody(ies) was suspected. Specificity of the additional antibody(ies) needs to be determined in order to provide compatible blood products to the patient.
Methods: Based on our laboratory's protocol, the positive antibody screen result with an R2R2 cell suggests the presence of Anti-E. An exclusion panel was designed and performed with Panocell-16 lot number 45,028 (Immucor Inc) using the Column Agglutination Technique (CAT, Ortho Clinical Diagnostics) to exclude all other clinically significant red cell alloantibodies. The red cells were diluted to 0.8% before the addition of patient's plasma. The CAT cards were incubated for 15 min and centrifuged for 5 min prior to examination of reaction. Ten out of the 11 cells showed a negative reaction, which excludes all other clinically significant red cell alloantibodies. The one positive donor cell N4300 showed a strong positive reaction of 3+ despite being an rr phenotyped cell. Further investigation was initiated to determine the presence of an additional antibody against a low incidence antigen.
Results: Donor 4300 was typed as Di (a+) positive and this lead to investigating if the underlying newly produced red cell alloantibody was Anti-Di^a. Two other cells (Donor N565 and H1687) from Panocell-16 lot 29,844 and 05,540 respectively, were selected and gave a strong 3+ reaction. Both cells were typed as Di (a+). The investigations confirmed presence of Anti-Di^a in addition to the Anti-E.
Summary/Conclusions: Di^a antigen has a frequency of about 5% in Chinese (Reid, 2012) and 0.88% in Malaysian Indians (Cheong, Asian J Transfus Sci, 2013), which makes up about 70% and 10% respectively of the entire population of Singapore. This patient most likely has been exposed to the antigen during a transfusion episode. Antibody to Di^a is usually immunoglobulin G (IgG) but can also be immunoglobulin M (IgM) antibody. It is a clinically significant antibody and can cause hemolytic disease in the new born or hemolytic transfusion reaction. Obtaining compatible blood for this patient is not difficult and takes no challenge for subsequent transfusions in our Asian population since Di^a and E are both low incidence antigens. Unlike the E antigen, Di^a negative units cannot be antigen typed as

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commercial Anti-Di[a] reagent is not available in our institution. Hence, E negative, antiglobulin crossmatch compatible units are recommended for patient's subsequent transfusions. Inclusion of Di[a] antigen in the antibody screening cells is not mandatory currently by our local authorities and the need may have to be ascertained. Patient's retrospective records were reviewed and no signs of delayed transfusion reaction were observed. Patient was well and discharged from hospital uneventfully 2 days after Anti-Di[a] was detected.

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AUTOIMMUNE HEMOLYTIC ANEMIA – CLINICAL AND IMMUNOHEMATOLOGICAL PROFILE: EXPERIENCE FROM A TERTIARY CARE HOSPITAL IN NORTH INDIA

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Background: Autoimmune hemolytic anemia (AIHA) is characterized by increased red cell destruction and/or decreased red cell survival due to autoantibodies directed against self-antigens on red cells. AIHA is classified as warm AIHA, cold AIHA and mixed AIHA. Positive 'direct antiglobulin test' (DAT) remains the diagnostic hallmark in addition to laboratory evidence of hemolysis.
Aims: This study is an attempt to critically analyze the clinical and immunohematological profile and serological characterization of red cell bound autoantibodies.
Methods: The prospective study was conducted at Regional Blood Transfusion Center, Lady Hardinge Medical College (LHMC) & Associated Hospitals over the period of 4 years (2011-14) in 32 patients with clinical diagnosis of anemia under evaluation. DAT done using gel technology (Bio-Rad, Switzerland) was positive in all the patients Complete clinical details and laboratory parameters including CBC with peripheral smear and reticulocyte count, serum bilirubin, viral markers, ANA and serum LDH were analyzed. Further, extended blood grouping was done by tube method at three temperatures i.e. 4°C, 22°C & 37°C. Antibody screening and identification was done in saline phase and Coomb's phase. DAT was done with polyspecific Coomb's cards followed by DAT profile on monospecific cards. IgG class subtyping was performed wherever required. Adsorption, elution studies and DTT treatment were carried out if required. Antibody titers were done in the relevant cases.
Results: In our study, age ranged from 6 months to 55 years with female preponderance (M:F 0.62). Primary AIHA (66%) was more common than secondary AIHA (34%). The most common presenting symptom was weakness and fatigue. Hepatosplenomegaly was found in 21.2% of the patients. The hemoglobin values ranged from 1.8 to 5.3 g/dl with reticulocyte count varying from 1.2 to 12.8%. Majority of patients (95.7%) showed indirect hyperbilirubinemia. Auto-agglutination was present in 11 cases on naked eye examination of the EDTA sample at room temperature. DAT was positive in all the cases (2 + to 4 +).The most common antibodies in warm AIHA were 'IgG + C3d' (42.8%) followed by IgG (38%) alone. All the cases of cold AIHA showed presence of only complement C3d with only a single case showing DL antibody (PCH). Two cases of mixed AIHA had red cells coated with 'IgG + IgM + C3d'. IAT showed a pan-agglutinating reaction in 31 cases. Single case showed auto-antibody specificity (Auto anti 'e'). 3 cases had underlying allo-antibodies (Allo anti K, E, S). Based upon above findings, 78.1% of the cases were warm AIHA, 15.6% were cold AIHA while mixed AIHA was least common (6.25%).Out of 25 cases of warm AIHA, 4 cases had insignificant cold antibodies (16%).
Summary/Conclusions: The two important clues for the diagnosis of AIHA are DAT positivity and pan-agglutination on IAT. It is very important to look for underlying alloantibodies in all cases of AIHA. It is important to serologically characterize the autoantibody and determine the type of AIHA as the approach of clinician to the treatment differs in these patients.

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CLINICAL CHARACTERIZATION OF DIRECT ANTIGLOBULIN TEST OF AUTOIMMUNE HEMOLYTIC ANEMIA: A STUDY OF 44 CASES

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Background: The direct antiglobulin test (DAT) is a laboratory test that detects immunoglobulin and/or complement on the surface of red blood cells. There is no systematic and comprehensive clinical analysis of immunoglobulin and complement of autoimmune hemolytic anemia (AIHA) so far.
Aims: To investigated the clinical characteristics of immunoglobulin and complement on the surface of red blood cells of AIHA.
Methods: DAT test was performed on 44 patients with AIHA, and the positive samples were further identified the types of antibody and complement. The clinical parameters were analyzed among different antibody types, DAT (+) and DAT (−) AIHA.
Results: Twenty six cases were DAT positive, while 18 cases were DAT negative. The hemoglobin concentration (Hb), red blood cell (RBC) count and hematocrit in 26 DAT positive AIHA patients were significantly lower than those in 18 DAT negative patients ($P < 0.01$).While the mean corpuscular volume (MCV), the mean corpuscular hemoglobin (MCH) in DAT positive AIHA group were significantly higher than those in DAT negative group ($P < 0.05$), and the total bilirubin (TB), direct bilirubin (DB), indirect bilirubin (IB) were also higher compared to DAT negative group ($P < 0.01$). Regarding to the antibody intensities, our data showed that Hb, RBC and HCT in DAT 4(+) patients decreased significantly than those patients with DAT3(+) and DAT1(+) ($P < 0.05$). While the MCH in DAT 4(+) patients was higher compared to DAT 1(+) patients ($P < 0.05$), and the DB in DAT1 (+) was significantly lower than those with DAT3 (+) and DAT4 (+) ($P < 0.05$). Among the DAT positive patients, the type of IgG/C3d was the highest, accounting for 8 cases, followed by IgG and IgG/IgM/C3d with 6 cases and 5 cases respectively, then the IgG/C3d/IgA/IgM 3 cases, IgG/IgM/C3d/C3c 2 cases, IgG/C3d/C3c/IgA/IgM 2 cases. The Hb and HCT in multiple antibody with more than two antibodies and IgG/C3d were both significantly lower compared to IgG group ($P < 0.05$). While the MCH in Multiple antibody were obviously higher compared to IgG/C3d group ($P < 0.05$),the MCHC in Multiple antibody increased obviously than the IgG group ($P < 0.05$).
Summary/Conclusions: The DAT shows a variety of antibody types in AIHA. The severity of anemia and hemolysis of DAT positive AIHA patients is more severe than those of DAT negative patients. There is a certain relationship between the degree of hemolysis and the intensity of immunoglobulin.

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THE PREVALENCE OF AUTOIMMUNE COLD ANTIBODIES AMONG REGULARLY TRANSFUSED EGYPTIAN PATIENTS AND THE PREVALENCE OF ALLO-ANTIBODIES MASKED BY AUTOIMMUNE COLD ANTIBODIES

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Background: Autoimmune cold antibodies are clinically non-significant but they can interfere with the pre-transfusion tests like (antibodies screening, cross-matching, etc. . ...) by masking the presence of other antibodies that may be clinically significant.
Aims: The aim of the study is to find out:
The prevalence of autoimmune cold antibodies among regularly transfused Egyptian patients.
The prevalence of allo-antibodies masked by autoimmune cold antibodies.
Methods: The study conducted on transfusion dependent patients (1000 patients) with: Thalassemia, aplastic anemia, leukemia, renal failure and anemia of chronic diseases; presented to Egyptian NBTC for blood transfusion.
According to National Egyptian Testing Strategy, Patients were subjected to:
Screening for allo-antibodies against group O reagent RBCs (panel of 3 cells) using appropriate IAT at 37°C by CAT.
Antibody identification for positive screening samples against group O reagent RBCs (panel of 11 cells) using appropriate IAT at 37°C by CAT.
Screening at room temperature to detect the presence of cold antibodies.
Screening for allo-antibodies against group O reagent RBCs (panel of 3 cells) using appropriate IAT at strict 37°C with pre-warming of both serum & reagents(in case of positive results of Screening at room temperature).

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Antibody identification for positive screening samples to identify the specificity of allo-antibodies masked by autoimmune cold ones against group O reagent RBCs (panel of 11 cells) using appropriate IAT at strict 37°C with pre-warming of both serum & reagents.
Results: A total of 1000 patient were examined for the presence of autoimmune cold antibodies & allo antibodies that may be masked by them.
Out of a total of 1000 we found that 97 (9.7%) patients had autoimmune cold antibodies and the co-incident prevalence of masked allo-antibodies was as follows:
Autoimmune cold antibodies only: 67 out of 97 patients (69%).
Allo-anti D:2 out of 97 patients (2.1%).
Allo-anti C:4 out of 97 patients (4.1%).
Allo-anti E: 6 out of 97 patients (6.2%).
Allo-antic: 6 out of 97 patients (6.2%).
Allo-anti K: 4 out of 97 patients (4.1%).
Allo-anti Fy^a: 2 out of 97 patients (2.1%).
Allo-anti Jk^a: 4 out of 97 patients (4.1%).
Allo-anti S:2 out of 97 patients (2.1%).
Summary/Conclusions: Pre-warming techniques are helpful in differentiation between cold non-significant Abs and warm Abs, for proper pre-transfusion testing of the patients' samples.

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DISTRIBUTION OF I-NEGATIVE BLOOD DONORS IN NORTHEAST CHINA

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Background: The I antigen is the only antigen of the I blood group system, and the i antigen is attached to the Ii blood group. The i phenotype is seen in neonates, which gradually decreases with age. Most children change into I positive phenotype before they are 18 months, so I- negative adult (or i adult) is very rare. In Asia, i adults are associated with congenital cataracts.
Aims: To screen I-negative blood donors and investigate the distribution of I-negative blood donors in Northeast China.
Methods: I-negative blood donors screening was performed on 11,328 blood samples of blood donors in our blood center. The detection of I-negative blood sample is performed by the conventional tube method using IgM anti-I serum. Cord blood erythrocyte was used as negative controls.
Results: I-negative blood donors were screened in 3 (0.026%) cases out of 11,328 blood donors in our blood center. The ABO blood type of the three I-negative blood donors is A, B, O individually. Based on this, a rare blood type information bank of I-negative blood donors was established preliminarily in China.
Summary/Conclusions: The investigation of the distribution of the I-negative blood donors is the first report in China. The discovery of the I-negative blood donors and the establishment of information bank of I-negative blood donors are of great value to the emergency blood transfusion of patients with rare anti-I antibodies.

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JK (A-B-) SCREENING AMONG BLOOD DONORS IN NORTHEAST CHINA

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Background: A total of 3 Kidd blood group antigen are Jk^a, Jk^b, Jk3, they are the urea transporter proteins. There are four different phenotypes in the Kidd blood group system: Jk(a+b-), Jk(a+b+), Jk(a-b+), Jk(a-b-).The Jk(a-b-) phenotype erythrocyte is lack of the urea transporter proteins, it cannot concentrate urine as much as possible. The Jk(a-b-) phenotype is rare in most populations and there are significant differences in frequency in different regions.
Aims: To investigate the Jk (a-b-) phenotype in blood donors in Northeast China.
Methods: 2M urea and suspended red blood cells (2%>5%) were added to 3:1 by the method of test tube for screening of unhemolytic samples. To identify the serotypes used the monoclonal antibodies which against Jka and anti Jkb antibodies of IgM. Genotype was determined by PCR-SSP.
Results: Altogether, 59,865 blood samples from Harbin blood donors were tested and only one samples were resistant to the urea lysis. This samples shows hemolysis when the incubation period exceeded 20 min, and it shows completely hemolysis in 50 min. While 59,864 samples were completely lysed within 5 min. The Jk(a-b-) phenotype were screened at a frequency of 0.0017% in blood donors in Harbin. The Jk(a-b-) phenotype donors genotype is Jk*02/Jk*02.
Summary/Conclusions: The frequency of Jk(a-b-) phenotype in blood donors in Harbin was significantly lower than that in other areas reported in China. It is an effective measure which establish the Jk(a-b-) phenotype donors in this region,to solve the blood transfusion problem in patients with this type of blood group.

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ASSOCIATION BETWEEN CD4 CELL COUNT AND BLOOD GROUP ANTIGENS DISTRIBUTION IN HIV INFECTION

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Background: CD4 cell count has been a major predictor of disease progression and monitoring of the effectiveness of highly active antiretroviral therapy (HAART) in HIV infection along with the HIV-RNA level. Blood group antigens have been reported to be associated with many diseased conditions severally. Studies have suggested that ABO blood groups have an impact on infection status of the individuals possessing a particular blood group due to the significant associations observed when analyzed. However there is limited information on the relationship between these blood group antigens with CD4 cell count and haemoglobin genotype in HIV infection, hence the need for this study.
Aims: To determine the association between CD4 Cell Count, Blood Group Antigens and Haemoglobin Electrophoretic Pattern in HIV infection.
Methods: Exactly 360 subjects who are ≥16 years, consisting 240 newly enrolled seropositive patients and 120 apparently healthy blood donors were recruited for this study. Antibodies to HIV were determined using determine rapid HIV 1/HIV 2 test kit (Abbott), enzyme linked immunosorbent assay (ELISA) (GenScreen plus HIV Ag-Ab test kit, Paris) and Western blot (New-LAV Blot 1, Bio-Rad, France) for confirmatory test. ABO and Rhesus blood grouping was determined by standard tile and tube techniques using anti- A, Anti- B and anti- D reagents (Rapid Labs Ltd, UK). Haemoglobin genotype determined by alkaline cellulose acetate haemoglobin electrophoresis while CD4 cell count was estimated with Partec Cyflow analyser.
Results: The mean ± SD CD4 cell count obtained for both the patient and control group is 472.44 ± 14.48 (cells/mm^3) and 571.17 ± 314.26 (cells/mm^3) respectively. Of the 240 patients, 132(55%), 54(22.4%), 45(18.8%) and 9(3.8%) are ABO blood groups O, A, B and AB respectively while 204(85%) and 36(15%) are Rhesus Positive and Negative respectively. The patients group had 198(82.5%) with CD4 count ≥200-cells/mm^3; 42(17.5%) had CD4 count ≤ 200cells/mm^3 while all participants in the control group had CD4 count >200cells/mm^3. A significant association was observed between the CD4 cell count of the patients and their ABO blood group antigens (P = 0.04) and their haemoglobin genotype (P = 0.01) with blood group A followed by blood group AB and HbAA having the highest CD4 count.
Summary/Conclusions: This study reiterates the fact that blood group antigens are involved in immune protection against infectious disease. Blood group A which has been implicated to confer susceptibility in some diseased condition has also been observed to confer immunity through its association with CD4 cells in this study. Therefore further study is required to understand the mechanism by which CD4 cell count is increased in blood groups A and AB in HIV infected individuals.

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THE VARIATION OF BLOOD TYPE IN ACUTE MYELOID LEUKEMIA PATIENTS

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Background: Blood type is a hereditary trait in the human body which controlled by the blood group gene of the long arm of chromosome 9 and almost never change. But in some cases like leukemia (which is reported to be an acute myeloid leukemia, M2a), tumor, multiple myeloma and so on, the ABO blood group is likely to be weakened, and the pathological blood type may mutate.

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Aims: To explore the variation of blood type in M2a patients and discuss the blood transfusion methods and precautions in these patients with clinical blood type variation.
Methods: Statistical analysis the number of 893 patients with M2a who were hospitalized between January 2012 and June 2017, the ABO and RhD blood type, and the blood type and their changes before and after chemotherapy treatment. Finally, the genotype of blood group was identified in patients whose blood type changes. The antibody titer of serum blood group was determined after the blood type variation, and every chemotherapy cycle was tested one time.
Results: In statistical of acute myeloid leukemia patients, two cases have ABO blood serum type variation, respectively is: O to B, B to AB, female, aged 19 and 23, and blood type changes are in IA chemotherapy for the 6 week. Finally, the blood type were mutated blood type identified by the genotype.
Summary/Conclusions: The change of the antigen is the main reason of blood type variation. The antigen recovery after chemotherapy in the two M2a patients, and then blood type mutates, so blood type of such patients should be monitored closely during chemotherapy, and research on the reasons affected the antigens, which might be used on clinical application, and finally reduce blood type errors causing blood transfusion reactions.

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THE STUDY ON CORRELATION BETWEEN HIV INFECTION AND DUFFY BLOOD GROUP POLYMORPHISM IN NANJING AREA POPULATION

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Background: Duffy blood group was one of the cytokine receptors on red blood cell membrane, and as the HIV virus co-receptor involved in viral infection. The major polymorphisms of the Duffy blood group system: Fya and Fyb have only one amino acid difference (Asp42Gly). Whether Duffy polymorphism involved in HIV infection remains unclear.
Aims: To investigate the relationship between HIV infection and Duffy blood group polymorphism in Nanjing area population
Methods: Peripheral EDTA blood samples were collected from HIV positive individuals, and Duffy serological phenotype were performed with commercial anti-Fya and anti-Fyb reagents. The genomic DNA was extracted and Duffy genotyping were performed by PCR-SSP. The genotyping results were compared with the frequencies of Duffy gene in the normal population which have been reported before.
Results: 334 cases with HIV positive infection were collected. Among those, 301 were Fy(a+b-)(90.120%), 32 Fy(a+b+) (9.581%), 1 Fy(a-b+)(0.299%), and no Fy(a-b-) were found. Genotyping results were consistent with the serological results, and the gene frequency of FYA and FYB were 0.949 and 0.051, respectively. In the preliminary report, FYA and FYB gene frequencies were 0.9538 and 0.0462 in the normal population. Two groups of Duffy blood group typing showed no significant difference ($P = 0.573$, >0.05).
Summary/Conclusions: Duffy blood type has not significantly difference between HIV positive infection and normal population, and the association between Duffy polymorphism and HIV infection remains to be further studied.

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PREVALENCE OF RED CELL BLOOD GROUP ANTIGENS AMONG EGYPTIAN POPULATION AND OTHER POPULATIONS

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Background: Blood group antigens are macromolecules on the outer surface of the red blood cell, some of these antigens may be found on the surface of the tissues as they are not red cell specific. They may be either proteins or glycolipids but mostly glycoproteins .e.g. the ABO system specificity is determined by the oligosaccharide sequence, while the MN, kell, Duffy, Kidd and Diego system specificities are determined by the amino acid sequence.
Aims: The Knowledge of prevalence of different blood group antigens in any population helps to manage cases of allo-immunization especially multi–transfused patients .Those patients who suffer from thalassemia, sickle cell anemia, cancer,etc. are most likely to develop antibodies against minor blood group antigens,thus antigen negative blood is required for those patients who are already allo-immunized and sometimes with more than one antibody. Finding compatible units for such patients without having any knowledge of the prevalence of complicated antigens in any population is hard time consuming process.
Methods: A total number of 1200 regular blood donors in NBTC (national blood transfusion center)were subjected to RH system phenotyping (D, C, E, c, e) .The donation included both sexes with 4/1 male to female ratio (total number of males = 933,total number of females = 267) age ranging from 18 to 45 years. The classical method used for testing the blood group antigens was column agglutination technique using griffols neutral cassettes and mono specific anti-sera. This method is simple, specific,accurate and has a very high sensitivity. A 3 ml fresh non hemolysed EDTA blood sample was used for preparation of 5% RBCs suspension to detect RBC antigen. 10 microns of red cell suspension is added to 25 microns of mono specific anti-sera. Positive test results in case of red cell agglutination and negative test results in case of no agglutination of RBCs .
Results: Prevalence of phenotypes associated with blood group antigens over the last 5 years (2013 till 2017) was found to be:
Among Egyptians: R1r=33% (no. of patients=396) / R1R1=17% (no. of patients=204) / rr=14% (no. of patients=168) / Ror=12% (no. of patients=144) / Rzr=8% (no. of patients=96) / R2r=7% (no. of patients=84) / R2R2=1% (no. of patients=12).
Among Asians: R1r=8.5% / R1R1=52% / rr=0.1% / Ror=0.3% / Rzr=1.4% / R2r=2.5%. R2R2=4.4%.
Among Blacks: R1r=21% / R1R1=2% / rr=6.8% / Ror=46% / Rzr=0.2% / R2r=18.5% R2R2=0.4%.
Summary/Conclusions: The distribution of different blood group antigens in Egypt and other populations has a clinical significance in blood transfusion medicine and in hemolytic diseases. It is important in providing antigen negative blood to patients with medical conditions, who often require regular blood transfusion and thus saving a lot of lives through serving much more patients in less time.

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EXISTENCE OF GIANT PANDA ERYTHROCYTE BLOOD GROUP ANTIGEN

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Background: Allogeneic and xenogeneic blood transfusions with blood of Giant Panda and Asiatic black bear were proved to be very effective in previous cases, while the blood type which is the key point that determines the safety of blood transfusion has not been found.
Aims: In this study, allogeneic and xenogeneic cross matching tests were made to reveal the blood types of Giant Panda and Asiatic black bear.
Methods: Allogeneic agglutination phenomena among Giant Panda(16)/Asiatic black bear(40) blood samples and xenogeneic agglutination phenomena between Giant Panda (11) and Asiatic black bear(10) blood samples were observed in cross matching tests, then the reason of agglutination phenomena were analyzed by using human and mice blood.
Results: Three serological patterns($n = 1$; $n = 4$; $n = 35$) were found in Asiatic black bear tests, while weak agglutination was observed in Giant Panda tests. Agglutination patterns among the plasma of different Asiatic black bears and RBC of Giant Panda(11) were not regular. RBC of mice agglutinated with all of the plasma of Giant Panda and Asiatic black bear and the same phenomena were observed among the plasma of human with RBC of Giant Panda and among RBC of human with the plasma of Giant Panda. However, agglutination was not observed among the plasma of mice with RBC of Giant Panda.
Summary/Conclusions: First, there were at least three blood group antigens in Asiatic black bear species; second, agglutination phenomena in the allogeneic and xenogeneic cross matching tests of Giant Panda and Asiatic black bear were caused by blood group antibodies rather than interspecific antibodies. All above indicated that there were blood group antigens in Giant Panda.

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Abstract has been withdrawn

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THE ESTABLISHMENT OF A RARE BLOOD DONOR BASE

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Background: It is difficult for patients with rare blood type of blood supply, in order to ensure the security of clinical blood of patients with rare blood type, along with the advance of medical level, almost all in some developed countries set up a rare blood type blood donor database. In 2008, a number of blood centers in China jointly launched the China rare blood type database project, which carried out the screening and researching work of rare blood type. The establishment of blood protection system of rare blood type can provide scientific basis for clinical practice, timely obtain effective information, and win valuable time for emergency transfusion of patients.
Aims: to establish a rare blood type protection system in northeast of china, Minimize the risk of blood for rare blood patients.
Methods: 1. With RBC blood group serology and PCR method, the genetic testing for our province ABO, RhD, MNSs, Duffy, Kell, Dombrock, Diego, Kidd, Scianna, Colton, Lutheran, Yt 14 RBC blood group system such as 36 antigen for screening. 2. The blood samples of Harbin unpaid blood donors are screened for the rare blood group Jk (a - b -) of the Kidd blood group system. 3. The rare antibodies obtained in my laboratory were screened, and the serum of the clinical patients was screened for the li system rare antigen and Fya- antigen.
Results: 1. The rare blood type that will be screened, with computer and network platform as the carrier, combined with modern blood stations management, the establishment of a rare blood type in Heilongjiang province security system, including database and the physical library. 2. The screened rare blood types such as JK (a - b -) for frozen storage, -80 degrees has a 10 year period. To reply the urgent need of clinical blood, reduce the adverse reaction of blood transfusion and improve the curative effect.
Summary/Conclusions: The establishment of the north China rare blood type library, can be in clinical emergency blood use in patients with.

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INVESTIGATION ON THE DISTRIBUTION OF ABO, KELL AND P BLOOD GROUP ANTIGEN OF THE BLOOD DONORS IN LIUZHOU AREA

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Background: Landsteiner has first found the human red blood cell ABO blood group system since 1901, so far there has been found 36 red blood cell blood group systems confirmed by ISBT, including Kell and P blood group systems. As a genetic marker of human, red blood cell blood group has great significance in the fields of blood transfusion medicine, immunogenetics and forensic medicine. The distribution of ABO, Kell and P blood group systems exists difference among different race, nationality and region. Many scholars in China have studied the distribution of ABO, Kell and P blood group of random people in local region. The distribution of ABO, Kell and P blood group of the blood donors in Liuzhou area has not been reported. In order to investigate the distribution of ABO, Kell and P blood group antigen of the blood donors in Liuzhou area, we conducted this research.
Aims: To investigate the distribution of ABO, Kell and P blood group antigen of the blood donors in Liuzhou area.
Methods: Using research method of blood group population genetics, 1680 blood donors were enrolled into the investigation at random in Liuzhou area, and the distribution of ABO, Kell and P blood group antigen and their gene frequency were analyzed.
Results: The distribution of ABO blood group: O(0.429 2)>B(0.282 1)>A(0.231 6) >AB(0.057 1);the distribution of gene frequency: r(0.654 3)> q(0.188 1)> P (0.157 6). The distribution of Kell blood group: All were K-k+, K+k+ and K+k- could not be detected; the distribution of gene frequency: k was 1.000, K was 0.0000. The distribution of P blood group: P2(0.716 7)>P1(0.283 3); the distribution of gene frequency: P2(0.846 7)>P1(0.153 3).
Summary/Conclusions: The distribution of ABO, Kell and P blood group antigen of the blood donors in Liuzhou area accords with the distributive characteristic of blood group antigen of southern Chinese population.

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IMMUNODEFICIENCY MAY CAUSE ABO BLOOD GROUPING DISCREPANCY

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Background: ABO blood grouping discrepancy is frequently seen in clinical patients due to different causes. Immunodeficiency is one of the causes. A 24 years old young man with arthritis of the left shoulder who was going to have an operation, was found ABO blood grouping discrepancy, whose direct ABO blood group is O and reverse ABO blood group is AB. He had recurrent bacterial infections, such as pneumonia, arthritis, sinusitis since he was a little boy.
Aims: To study the cause of ABO blood grouping discrepancy of the young man who has immunodeficiency.
Methods: The amount of immunoglobulin isotypes (IgG, IgM, IgA, IgE) were detected and CD19+ B cells were grouped by flow cytometry in the laboratory.
Results: All of Ig isotypes are extremely low and the count of CD19+ B cells in the peripheral blood is significantly decreased. X-linked agammaglobulinemia(XLA) is suspected and confirmation test will be done by sequencing the *BTK(Bruton's tyrosine kinase)* to identify the exact mutation.
Summary/Conclusions: Immunodeficiency such as XLA may cause ABO blood grouping discrepancy due to the unexpected deficiency of the IgM anti-A and anti-B reciprocal antibodies.

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SEROLOGICAL AND GENE DETECTION OF B (A) BLOOD GROUP IN 3 CASES

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Background: B (A) blood type was first discovered in 1985, and its formation mechanism is that B allele occurs single base mutation on the basis of normal B gene sequence. The variant B gene has the ability to encode bifunctional active enzymes, which in turn exhibit a small A antigen specificity in addition to B antigen specificity.
Aims: Uses serological and molecular methods to identify B (A) blood type of 3 blood donors.
Methods: ABO serological-typing was tested using tube method and geno-typing was detected using DNA sequencing in 3 cases.
Results: 1 blood donors with positive AwB serology results, anti A1c with weak agglutination, the reaction of red blood cells and anti H antibodies were unsignificantly enhanced, no agglutination reaction with anti A1 antibody, it is found by sequencing of exon seventh B allele mutation in 640A>G, the ABO genotype was B (A) 04-001; 2 blood donors with positive AB serology results, anti A1c with weak agglutination, the reaction of red blood cells and anti H antibodies were significantly enhanced, no agglutination reaction with anti A1 antibody, it is found by sequencing of exon six B allele mutation in 700C>G, the ABO genotype was B (A) 07-001.
Summary/Conclusions: Identification of subtypes or rare blood type analysis in serological combined with molecular biological methods. Timely and accurate identification of ABO blood group and subtype, to ensure the safety of blood transfusion.

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RESOLVING ABO DISCREPANCIES: AN ANALYSIS OF 53 CASES

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Background: It is difficult to report the results of blood types when there are ABO discrepancies which can be affected by patient age, diagnosis, previous pregnancy, transfusion history, treatments, heredity etc.
Aims: The aim of this study is to resolve ABO discrepancies, analyze the reasons accordingly, and to provide proper blood products when needed.
Methods: Based on the three step analysis for ABO discrepancies, 53 cases of discrepant blood samples from January 2014 to December 2016, were identified by a series of serological tests. Retrospective analysis of patients was recorded and the results were categorized into four types of reactions respectively.
Results: Red blood cell autoantibodies accounted for 29 (54.7%) cases of ABO discrepancies, and precipitated extra reactions in both forward and reverse types or in the reverse type only; Unexpected alloantibodies and plasma protein abnormalities involved 11 (20.8%) cases of extra reactions in the reverse type, whereas weak antibodies led to 8 (15.1%) cases of weak reactions in the reverse type. In addition, weak ABO antigens, Para Bombay and ABO subgroups caused 5(9.4%)cases of weak or missing reactions in the forward type.
Summary/Conclusions: In this study, the major cause of ABO discrepancies was found to be red blood cell autoantibodies and the three step analysis helps to resolve ABO discrepancies. Blood type identification contributes to the selection of appropriate blood, which is essential to a safe and effective clinical blood transfusion process.

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ANALYSIS OF DISCREPANT ABO RESULTS IN BLOOD DONORS IN HEGANG

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Background: In the past nearly 2 years, hegang blood station has encountered positive and negative stereotypes in the actual work, conducted further analysis and identification, and summarized the results of the report
Aims: To analyze the situation of the positive and negative typing of blood donors in practical work, and exclude the unqualified blood. Then classify the persons who are not fit to give regular blood donation again.
Methods: Using serological methods to determine ABO blood type, in case of positive and negative types inconsistent, for positive definite form increase setting anti -AB, anti -A1, anti -H detection, for reverse typing use conventional stereotypes with O cells and its cells were detected, respectively, to observe the reaction at room temperature immediately, at room temperature for 15 min and 4 degrees and 37 degrees, the serum antibody screening with room temperature saline, Polybrene method; antibody screening positive cells and antibodies were identified in the spectrum at room temperature, saline Polybrene and indirect antiglobulin method; suspected subtype added experiment for absorption and elution test.
Results: In the 18,225 cases of unpaid blood donors, there are 20 patients with positive and negative stereotypes, including 2 cases of cold autoantibody, irregular antibodies are 8 cases, blood group antigen weakened is 1 cases, blood group antibody weakened are 2 cases, 1 case is abnormal plasma protein, 2 cases are ABO subtype, and rouleaux are 4 cases.
Summary/Conclusions: The conclusion of this investigation to identify ABO blood group with blood donors accounted for the proportion is 0.11%, 20 cases of positive and negative stereotypes do not fit in, cold autoantibody accounted for 10%, accounting for 40% of irregular antibodies, antigen antibody decreased 5%, decrease accounted for 10%, accounted for 5% of abnormal plasma protein, ABO subtype accounted for 10%. Rouleaux agglutination accounted for 20%. to avoid the waste of blood, the abnormal coagulation and false rouleaux condensation blood caused by the plasma protein was normally given out to use in the case of that the blood does not affect the normal distribution. With irregular antibody, high titer cold autoantibody blood can be prepared into washed red blood cells release clinical, finally will be marked in the donor's personal file, suggesting that when the blood donors ask for donating blood again, we should pay more attention, and we propose the blood donors should not donate routine blood again.

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Abstract has been withdrawn

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STUDY OF D-VARIANT AND RH PHENOTYPES IN RHD-NEGATIVE BLOOD DONORS IN JINAN

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Background: Knowledge of the distribution of Rh phenotypes and D-variants in a locality is vital for safe blood services. However, the distribution of Rh phenotypes and the relationship between phenotype and D-variant among blood donors in Jinan area is little explored.
Aims: Study the Rh phenotypes of RhD negative blood donors and analyze the relationship between phenotype and D-variant, To guide the clinical blood transfusion of RhD-negative individuals.
Methods: RhD(IgM) negative donors were screened out by using anti-D(IgM) reagents, The 588 RhD negative samples were confirmed by indirect anti-human globulin test (IAT). The Rh phenotypes were determined by serological tests in 588 blood donors.
Results: 16 D-variants were detected, among them 12 cases of ccEe, 3 cases of Ccee and 1 case of Ccee. Among 572 cases of RhD(IgG) negative donors, there were 333 (58.22%) cases of ccdee phenotype, 152 (26.57%) cases of Ccdee phenotype, 42 (7.34%) cases of ccdEe phenotype, 22 (3.85%) cases of Ccdee phenotype, 18 (3.15%) cases of CCdee phenotype, 4 (0.70%) cases of ccdEE phenotype, 1 (0.17%) case of CCdEe phenotype.
Summary/Conclusions: The RhD negative(IgM) individuals with E antigen were much more likely to be D-variants than others. The predominant Rh phenotypes of Rh-negative individuals were ccdee and Ccdee in Jinan, similar to the results of other areas in China but quite different from which of Caucasian and Japanese.

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Abstract has been withdrawn

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DEL PHENOTYPE DETECTION AND ANALYSIS FROM RHD NEGATIVE BLOOD DONORS IN NANCHANG AREA

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Background: DEL blood transfusion may cause anti-D antibodies to RhD negative recipients.
Aims: To study DEL phenotype distribution and RhD gene from RhD negative blood donors in Nanchang area.
Methods: Detected RhCE phenotypes of RhD negative blood donors confirmed by indirect anti-human globulin test. The donors with DEL phenotype were screened out with absorption-elution test and their RhCE phenotypes were also detected. 7 samples of DEL phenotype were chosen out for gene amplification with PCR-SSP.
Results: Among 84 RhD negative donors, 42 for ccee (50.0%), 29 for Ccee (34.5%), 11 for CCee (13.1%), 1 for CcEe (1.2%) and 1 for ccEe (1.2%). 22 DEL(26.2%)were found in 84 RhD negative blood donors. 15 for Ccee (68.2%) and 7 for CCee (31.8%). All 7 Del individuals carried RHD gene, with 5 heterozygote and 2 homozygote.
Summary/Conclusions: RHCE phenotypes of RhD negative donors in Nanchang area are polymorphic. The ratio of DEL phenotype is similar to the past related reports and DEL phenotype samples contain RhD gene.

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COMPARISON OF RHD BLOOD GROUP TEST AMONG THE TUBE INDIRECT ANTIGLOBULIN TEST, MICRO-COLUMN GEL METHOD AND THE GLASS BEAD METHOD

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Background: In the Rh blood group system, the antigenicity of D antigen was strongest. The expression of RhD antigen could be classified into D+, D– and D variant. Of D antigen, a decrease in number and variation in the structure caused D variant. D variant could stimulate RhD negative blood donors to produce antibodies, prone to immunological hemolytic transfusion reaction, as there was D antigen. According to the regulation of AABB, it was necessary to conduct a confirmatory test for the apparently RhD negative blood donors. At present, there were the classic

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tube indirect antiglobulin test, micro-column gel method and glass bead method in use and as conventional detection methods.
Aims: To investigate the differences and causes of results in RhD group tests with the three testing methods (mentioned above) in 3 blood donors samples, and to provide a reliable basis for improving accuracy of such type test results.
Methods: 265 RhD negative samples were collected during the period of January 2017 - June 2017. The RhD blood group of these samples was tested respectively by the tube indirect antiglobulin test and the micro-column gel method. And there were 3 samples with different consequences. The RhD blood group of these 3 samples were tested parallelly by the tube indirect antiglobulin test, micro column gel methods of 2 manufacturers, glass bead methods of 2 manufacturers, absorption-elution test and the genetic test.
Results: Of the 265 samples, 262 samples were consistent with the results of tube indirect antiglobulin test and micro-column gel method. For 3 cases with inconsistent results, tube indirect antiglobulin test combined with absorption-elution test, and genetic test showed the RhD blood group for Del, however, the results were RhD negative by micro-column gel method and glass bead method.
Summary/Conclusions: In the confirmation test of RhD blood group, there was missed detection by micro-column gel method and glass bead method. The result for the classic tube indirect antiglobulin test combined with absorption-elution test was more reliable, and genetic test was an effective complement to serological methods. In the meantime, combined use of various detected methods in order to secure the safety provided a reliable experimental basis in the clinical blood transfusion.

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DETERMINATION OF WEAK "D" ANTIGEN AMONG RHESUS NEGATIVE PAKISTANI BLOOD DONORS

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Background: Until the 19th century the blood transfusion procedure was unsafe, but this mystery was solved in 20th century with the discovery of ABO and Rh blood group antigens. The discovery of Rh system in 1939 was great to break through in transfusion medicine. It is the most significant blood group system after ABO. There is significant proportion of patients who's RBCs lack "D" antigen makes anti-D, if they have exposed to the "D" antigen by blood transfusion. Therefore, determination of Rhesus (Rh) phenotype is of critical importance. It causes hemolytic transfusion reactions and hemolytic disease of newborn.
Aims: Determination of weak "D" antigen, to highlight the importance of "Du Testing" among the Rhesus-negative blood donors.
Methods: This Cross-sectional Study was conducted at department of Hematology, Allama Iqbal Medical College, Lahore. Five different tertiary care and four secondary care hospitals of Lahore and six different institutes/universities including individuals visiting to blood banks for determination of their blood groups were included in study. Two ml of blood sample was collected in EDTA-containing vial and analyzed for determination of ABO/Rhesus (Rh) typing; all Rh-negative samples were further processed for the Du Testing according to standard protocol and commercially available monoclonal antibody.
Results: Out of total 55,874 participants only 8.0% (n = 4,454) were Rh-negative. Among these Rh-negative samples, 99.0% (n = 4,410) samples were negative even after "Du Testing". Only 0.98% (n = 44) were weak "D" positive, in Rh-negative samples.
Summary/Conclusions: Every Rh-negative individual should be processed for the detection of Weak "D" antigen by "Du Testing" as it may not be detected by immediate spin tube method.

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PRODUCTION OF MONOCLONAL ANTI-COMPLEMENT WITH HYBRIDOMA TECHNIQUE

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Background: Now anti-complement reagent (anti- C_3d) used for Anti-Human Globulin (AHG) reagent preparation of the National Blood Center, Thai Red Cross Society is produced by using polyclonal antibody from rabbit serum. This preparation process causes complicated and complex procedures, and wasted time. The antibodies are different for each production lot. Therefore, when monoclonal technique was developed, the National Blood Center, Thai Red Cross Society has adopted such techniques to do an experiment to be used instead of the conventional method of production, polyclonal antibodies.
Aims: To create cells secreting anti- C_3d and produce Anti-Human Globulin (AHG) reagents.
Methods: A mouse was injected with specific antigens. Then, the white blood cells from the mouse were fused with myeloma cells. The cell line secreting anti- C_3d was selected and cultured to increase the number of cell line. Serological properties were tested in potency and specificity with C_3b, C_3d, C_4b, C_4d coated cells; trypsinized cells and unsensitized cells including Immunoglobulin class, Then, they were mixed with Anti-IgG in appropriate ratio. When the AHG reagents was tested in patients with positive direct antiglobulin test (DAT) in comparison with the original reagents of the National Blood Center, Thai Red Cross Society (Polyclonal antibodies from rabbit serum) and from 3 foreign companies.
Results: The potency of anti- C_3d towards C_3b and C_3d coated cells was 1: 128 and 1: 256, respectively and negatively affected C_4b and C_4d coated cells, trypsinized cells, and unsensitized cells. When it was mixed with AHG reagents, it was tested with blood was equivalent to the reagent of 3 foreign companies. The original AHG was positive for 40 patients and negative for 41 patients.
Summary/Conclusions: The test results for type of Immunoglobulin class, anti- C_3d was found that it was IgG with a high potency of anti- C_3d at 1: 256. According to mixing anti-IgG to be prepared for AHG reagents, it was diluted about 100 times. The test results of blood of patients with positive DAT indicated that it was positive and negative for patients which was equivalent to the reagent of 3 foreign companies. The primitive AHG of the National Blood Centre, Thai Red Cross Society was positive for more than one sample which might be a false positive. This was due to the unwanted absorption of antibodies from anti C_3d prepared from rabbits was incomplete. In conclusion, monoclonal anti- C_3d had the properties to be prepared as AHG reagents instead of rabbit serum.

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STATISTICAL ANALYSIS OF DISTRIBUTION OF RH BLOOD GROUP ANTIGEN IN RHD NEGATIVE BLOOD DONORS IN KUNMING

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Background: To investigate the distribution of Rh blood group antigen in RhD negative blood donors in Kunming area
Aims: To provide reference basis for clinical science and safe blood transfusion.
Methods: A total of 689 blood samples which tested negative for RhD by saline screening method were subjected to serotype testing of RhC, c, E, and e antigens. Indirect antiglobulin test was performed for RhD confirmation.
Results: From 21st November 2015 to 18th, April 2016, there were 689 RhD negative samples indentified in 177117 blood donors. Among these donors, major distributions were dccee401 (58.2%), dCcee204 (29.6%), dccEe37 (5.4%), dCcEe9 (1.3%), dCCee37 (5.4%), dccEE1 (0.1%).
Summary/Conclusions: To understand the Rh blood group antigen distribution of RhD negative blood donation, a timely and effective reference basis is provided for construction of the rare blood group. It is important for guiding a scientific, reasonable and effective blood transfusion and supply.

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PRODUCTION OF MONOCLONAL ANTI-H REAGENT

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Background: For the development of monoclonal anti-H production by mouse hybridoma technique, clone 5A9 2E2 was selected because its potency (Titer) and specificity was good at every stage of the test. According to the study of the efficiency of reagents, it was found that the clone 5A9 2E2 was suitable to be prepared for the reagents.
Aims: To replace the complex method of producing anti-H produced from Ulex europaeus seeds including the inequal potency of Antibody in each production lot.
Methods: The supernatant clone 5A9 2E2 was tested and mixed with 4% BSA diluted to 1: 4 for monoclonal anti-H reagents. Potency (Titer) and stability of antibody were investigated in compared with the supernatant of Japan (murine monoclonal antibody) and anti-H of foreign companies as well as the supernatant produced from a lectin from Ulex europaeus of the National Blood Center, Thai Red Cross Society.
Results: It was found that the antibody potency of monoclonal anti-H reagent was 1:64. The net score was 67. The antibody potency of monoclonal anti-H reagent was greater than one produced from a lectin and a lectin of foreign companies. However, the potency was less than the potency of mouse monoclonal antibody of Japan. In the area of specificity, H Antigen of ABO and Parabombay can be verified. It was found that the weak or strong reaction depends on a amount of H substance in each blood type: O > A2 > A2B > B > A1 > A1B. The results were similar to the results of the National Blood Center, Thai Red Cross Society, mouse monoclonal antibody of Japan, foreign companies. In the area of stability, it was stable over a 12 month period.
Summary/Conclusions: According to the properties studied, it indicated that the specific potency of monoclonal anti-H reagent clone 5A9 2E2 was better or equivalent to anti-H from many sources including the reagents of the National Blood Center, Thai Red Cross Society. It can be diluted 4 times lower, leading to low production costs. Moreover, it was stable potency in accordance with the requirements of reagents. Therefore, it is appropriate to substitute the anti-H from a lectin.

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FREQUENCY OF RH PHENOTYPES AMONG THE BLOOD DONORS OF YI ETHNIC GROUP IN KUNMING

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Background: The Rh blood group system, which is composed of 54 antigens, is the most complex and polymorphic system. The five main Rh antigens, D, C, E, C, E were the most important, and group genetic studies showed that the frequency of Rh phenotypes in different populations varied. Kunming is the capital of Yunnan province in China, with ethnic minorities accounting for 15.44 percent of the city's permanent population. Yi is the most populous ethnic minority in Kunming, Yunnan province.
Aims: This study aims to demonstrate the antigen distribution of Yi nationality RhD, C, E, C and E in Kunming blood donation population, and provide basic data for the study of Yi nationality transfusion.
Methods: Date from total of 391 random blood donations of Yi ethnic group were analyzed, collected at the Yunnan Kunming Blood Center. Blood typing for the principal Rh antigens: D, C, E, and e, was performed, by agglutination using monoclonal antisera (Shanghai Hemo-Pharmaceutical &tBiological Co.). The Rh phenotypes were then determined according to modified Wiener terminology.
Results: From a total of (391) randomly phenotyped donors, data revealed that 214 (54.73%) were R1R1, 116 (29.68%) were R1R2, 27 (6.90%) were R1r, 20 (5.11%) were R2R2, 10 (2.55%) were R2r, 3 (0.77%) were R1Rz, 1 (0.26%) were R2Rz.
Summary/Conclusions: Although the study was conducted on a small scale, a distinctive frequency distribution was obtained. Concerning the Rh Phenotype Frequencies, the study showed that the most frequent phenotype in donors of Yi ethnic group in Kunming was R1R1 (54.73%) and the least frequent Rh phenotype was R2Rz (0.26%). Further investigation is required on a larger scale, to determine the phenotype frequency among the Yi ethnic group.

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PRODUCTION OF HUMAN MONOCLONAL ANTI-P1 BY USING HYBRIDOMA TECHNIQUE

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Background: The antibody now called anti-P1 was originally reported by Landsteiner and Levine in 1927. There are several other antigens in the system including P_1, P_2, p, P^k etc. An antibody to the P1 antigen is usually of the IgM class only reactive at low temperature and is rarely a cause of haemolytic transfusion reaction (HTR). The incidence of the P_1 antigen varies in different ethnic groups, in Caucasians 79% are positive, in the black population 94% are positive and in the population of Thailand 27% are positive.
Almost all P_1 negative are P_2. The p phenotype is less than 1 in a million. It is p individuals who can make the anti-P+P_1+P^k antibody that is called anti-Tj^a.
Aims: The production of human monoclonal anti-P1 by hybridoma technique is for P1 antigen testing on red blood cells.
Methods: Fusing B-lymphocytes which produced anti-P1 with human myeloma cells (JMS-3) enable the transformed cells to grow. The fusion cells that secreted antibody were selected for monoclone by limiting method. After selected the anti-P1 (1C9) cell line, we tested extensively for potency, specificity and stability to compare with monoclonal anti-P1 (HIRO-59) from Japanese Red Cross Society and polyclonal anti-P1 serum from donated blood.
Results: For potency testing, we found that monoclonal anti-P1 (1C9) has 100% specificity in titer 64 which is not only better than monoclonal anti-P1 (HIRO-59) which has a titer of 32 but also polyclonal anti-P1 from serum of donated blood which has a titer of 8. For stability testing, monoclonal anti-P1 (1C9) can be stored at 4°C and be kept for at least 12 months and at RT (22–24°C) can be kept for 8 months. In conclusion, human monoclonal anti-P1 (1C9) is capable for producing monoclonal anti-P1 (1C9) which is used for P1 antigen typing.
Summary/Conclusions: Human monoclonal anti-P1 (1C9) is capable for producing monoclonal anti-P1 (1C9) which is used for P1 antigen typing.

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APPLICATION OF AN ORTHOGONAL DESIGN FOR OPTIMIZATION OF A NEW AUTOMATED BLOOD TYPING SYSTEM

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Background: Automated systems for blood typing, as an important advance in the field of immunohematology, have been widely used in pre-transfusion laboratories. The final grouping accuracy of a system relies on combination of various parameters of different steps, including sampling, sample dilution, reagent dispensing, liquid mixing, incubation, centrifugation and so on. Orthogonal design is a popular experimental method that provides a convenient, fast and cost-saving resolution for optimization in many fields.
Aims: To identify an optimal combination of key parameters for ABO grouping before implementing the new system Microlab STAR BG into routine use.
Methods: Nine factors including mixing speed, mixing time, centrifugation speed, centrifugation time, shaking speed, shaking time, incubation time, incubation temperature and concentration of sample RBCs according to recommendations from instrument or reagent manufacturers were first optimized. Another three parameters including incubation speed, loading amount of diluted sample RBCs and monoclonal antibodies in forward typing, and the ratio of plasma and reagent RBCs in reverse typing were selected as factors to be optimized using an orthogonal array design via SPSS modeling. Incubation speed was set at 0 rpm, 150 rpm, 300 rpm, 450 rpm and 500 rpm based on the range of 0–500 rpm recommended by the instrument manufacturer. Loading amount of diluted RBCs and monoclonal antibodies in forward typing was set at 30 μl: 30, 40 μl: 40, 50 μl: 50 μl based on the 1:1 ratio recommended by the reagent manufacturer. The ratio of plasma and reagent RBCs was set at 30 μl:30 μl, 60 μl:30 μl and 90 μl:30 μl. SPSS 23.0 software was applied for the orthogonal modeling and data analysis, including direct-viewing analysis and variance analysis. Grouping accuracy was served to evaluate the performance of the optimized condition.
Results: The variance analysis showed that P values of incubation speed and loading amount of diluted RBCs and monoclonal antibodies in forward typing were 0.364 and 0.153, which indicated that these two factors did not significantly influence the ABO grouping. However, the P value of ratio of plasma and reagent RBCs

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was 0.000, indicating the ratio of plasma and reagent RBCs was a major influencing factor. Taking both results from direct-viewing analysis and variance analysis into consideration, 0 rpm was selected as the optimal incubation speed and the proportion 90:30 was selected as the optimal ratio for plasma and reagent RBCs. To minimize the cost of antibody reagents, 30 μl:30 μl was selected as loading amount for forward typing. Under optimal conditions obtained from orthogonal design, a set of 322 random donor samples were carried out to confirm the optimized combination, no erroneous results were found.
Summary/Conclusions: Optimization of a new system is a prerequisite to most analytical systems in immunohematology laboratories. The orthogonal design was successfully applied for determining the optimal parameter combination of the Microlab STAR BG analytical system and to guide the optimizing operation. It may also be applicable to other automated blood typing analyzers and other optimization problems in immunohematology laboratory.

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INVALID RESULTS ANALYSIS OF FULL-AUTOMATIC BLOOD TYPING INSTRUMENT

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Background: With the rapid development of blood stations, the automation and standardization of full-automatic blood type instrument can meet the needs of large numbers of samples. However, invalid results are often observed in the tests, so the cause analysis and solutions are very important for the safety of clinical blood transfusion.
Aims: To investigate the accuracy of full-automated blood typing instrument in the identification of blood types, and analyze the causes of invalid results in our tests and then formulate appropriate strategies.
Methods: From August 2016 to July 2017, 210055 blood samples were submitted to ABO and Rh (D) identification by GALILEO NEO full-automatic blood typing instrument. The samples of invalid results were reexamined by the tube method and the reasons were analyzed. Additionally, ABO blood typing discrepancies and Rh (D) negative samples were sent to blood typing laboratory for confirmation.
Results: Out of 210,055 blood samples, 208,785 (99.4%) were correctly identified, and 1270 (0.6%) were invalid. There were 78 samples with ABO blood typing discrepancies, and among them, cold antibody, specific antibody, weak antigen and weak antibody were 5 (0.4%), 19 (1.5%), 20 (1.6%) and 34 (2.7%), respectively. 223 (17.5%)cases were near the critical value. In the remaining 969 cases, there were 376 (29.6%) with non-added samples, 325 (25.6%) with negative control results, 90 (7.1%) lipemic samples, 154 cases with pollution (12.1%), and 24 (1.9%)with non-added reagents.
Summary/Conclusions: There were higher detection efficiency and accuracy of GALILEO NEO full-automatic blood typing instrument for large numbers of samples, and it was also convenient for manual check and trace. But we should pay attention to invalid results in our tests, and reexamined these results by the tube method. Manual check and technical training should be strengthened and blood donors should be consulted carefully before donation. Additionally, the daily maintenance and calibration of instrument can greatly reduce the invalid results.

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EVALUATION OF THE AUTOMATED BLOOD BANK SYSTEMS DAYMATE S AND IH-500 FOR ABO-RHD TYPING AND ANTIBODY SCREENING

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Background: Despite recent advances in laboratory automated instruments, the substantial part of workflow in the blood bank is still manually handled. Several automated immunohematological instruments which reduce minimize workload have been developed and are now available in the market.
Aims: This study was conducted in order to evaluate the performance of the DAYmate S (DAY medical SA, Switzerland) and IH-500 (Bio-Rad Laboratories, USA) for ABO-RhD Typing and Antibody Screening.
Methods: In June 2017, a total of 199 blood samples for ABO/Rh typing and 158 blood samples for unexpected antibody screening were collected. ABO-RhD blood type tests using the tube method and the two instruments were performed. Antibody screening tests using DiaMed-ID system (DiaMed GmbH, Switzerland) and the two instruments were performed.
Results: On ABO-RhD blood type results, a concordance rate of 99.5% was observed between the manual method and DAYmate S. A concordance rate of 96.0% was observed between the manual method and IH-500. All discordant samples showed cold antibody when manually tested at 4°C. On antibody screening tests, a concordance rate of 100% was observed between the manual method and DAYmate S. A concordance rate of 95.6% was observed between the manual method and IH-500. Six samples revealed positive results in IH-500 and negative results in manual method using the same gel card with IH-500.
Summary/Conclusions: Our study showed a different positive rate on the ABO-RhD blood typing and antibody screening test, depending on the instruments. These results could be originated from the difference of methods, reagent component or reagent storage temperature influencing the cold antibody detection rate. Thus, it is necessary to consider the pros and cons of each instrument and the adequate workflow for quality control when introducing an instrument.

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A COMPARISON OF CONVENTIONAL TUBE TECHNIQUE WITH ORTHO-VISION AUTOMATED TESTING PLATFORM FOR ABO ANTIBODY TITRATION- A PILOT STUDY

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Background: Significant variation in titer reporting between different laboratories has been reported previously. Performing ABO antibody titers is critical for determining both, the effectiveness of pre-transplant regimes and when titers are low enough to permit transplant.
Aims: To compare the ABO antibody titer results obtained by the automated testing (Ortho Vision) with conventional tube technique.
Methods: Isoagglutinin titers were tested in parallel by two different technologists in 91 patients (36-A, 54-B and 1-O blood group) using the conventional tube technique and the Ortho vision platform. Both IgM and IgG titers were reported. The concordance of the titers with both the technique was compared.
Results: The IgG and IgM antibody titers by CTT were in perfect agreement with IgG and IgM antibody titers reported by Ortho Vision in 20.8% and 32.9% of the cases respectively, IgG and IgM antibody titers by Ortho Vision were higher than those by CTT in 75.8% and 56% and lower then vision in 3.2% and 11.7% cases respectively. However, the difference between the results reported by the two techniques was not statistically significant ($P > 0.05$).
Summary/Conclusions: The automated platform Ortho Vision yielded comparable results with the gold standard method. We plan to study further the repeatability and variation of the titers with reference to patients blood groups and the type of transplantation, ie solid organ and hematopoietic stem cell transplantation.

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ASSESSMENT OF AUTOMATED BLOOD BANK ANALYZER IN PAKISTAN

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Background: The implementation of automated techniques improves the standardization of methods and the quality of achieved results. Human errors in manual processing are consequential causes of fatal transfusion complications. For that the automated systems were developed which are robotic instruments programmed to move all of the necessary blood samples for the given immunohematological assays in the correct sequence.
Aims: The aim of the blood bank of National institute of blood diseases and bone marrow transplantation, Karachi has been to evaluate the performance of Immucor Echo Galileo automated blood bank analyzer for the first time in Pakistan from January to June 2017. For assessment, we tested it in parallel with existing manual methods using (DiaMed ID-Micro Typing System).
Methods: We tested 840 healthy blood donors for ABO forward and reverse grouping. Five hundred and ninety five samples for antibody screening, 502 for cross match, 396 Rh Phenotype (D, C, c, E, e) and 50 samples for Direct Antiglobulin Test

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(DAT) using the Immucor analyzer and manual tube method. Furthermore, Echo performed the blood tests based on the principle of hemagglutination Capture^{R} solid phase technology in microplates.

Results: The automated Immucor Echo Galileo blood bank analyzer is a user friendly, time effective and data saving machine that shows high sensitivity for forward and reverse grouping, DAT, cross match, Rh phenotype and IgG type antibody. However, weak D testing and detection of IgM antibody showed very lower specificity with automated analyzer. Other disadvantages include small expiry of cells which delays the work and run around time is more as compared to manual methods.

Summary/Conclusions: In conclusion, the Immucor Galileo system allows a safe determination of routine blood tests with high reliability, specificity and a higher sensitivity compared to our previously used manual methods.

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EVALUATION OF ORTHO VISION MAX® ANALYSER® MAX ANALYSER– RESULTS OF FIELD APPLICATION TRIALS AT SIRIRAJ HOSPITAL, THAILAND

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Background: In 2006, Laboratory implemented fully automated immunohematology instruments, to improved laboratory operational efficiencies, safety checks and increased testing capabilities. Currently, AutoVue® Innova analysers are used for routine and specialized immunohematological testing in laboratory.

In 2015, a new ORTHO VISION MAX® ANALYSER is a recent addition to the list of fully automated immunohematology instruments available in Laboratory. Field Application Trials of the ORTHO VISION MAX® ANALYSER were undertaken by these laboratories as an opportunity to use the instruments in their respective settings and to provide feedback on the general ease of use and system efficiency of the analyser.

Aims: The ORTHO VISION MAX® is a new instrument designed to fully automate *in vitro* immunohematological testing of human blood using the ORTHO BioVue® System column agglutination test (CAT). Field Application Trials was conducted to evaluate the performance of the new instrument compared to the ORTHO AutoVue® Innova (AVI) System.

Methods: Field Application Trials of the ORTHO VISION MAX® ANALYSER was conducted in the Department of Transfusion Medicine, Faculty of Medicine, Siriraj Hospital, Thailand between 26 May, 2016 – 5 July, 2016. Two or more staff members completed on-line and extensive key operator training on the ORTHO VISION MAX® ANALYSER prior to commencement.

The Field Application Trial involved the completion of a series of structured test activities/cases using the ORTHO VISION MAX® ANALYSER and completion of the associated questionnaire for each case, per site. Test activities/cases were undertaken to evaluate:

- System Training.
- System Configuration.
- Quality Control (QC).
- Maintenance, Start-up and Shutdown.
- System Efficiency.
- Serial Dilution.

Results: Key findings from the Field Application Trials of the ORTHO VISION® analyser were:

- System training was easy and intuitive, leaving Key Operators confident to on-train other staff members.
- System configuration was easy to navigate and setup was largely intuitive.
- Quality Control menu is a vast improvement over the current system, but could be further expanded to provide additional functionality.
- Maintenance is easy to perform, and support documents provided are detailed enough to be useful.
- Efficiency of the ORTHO VISION® Max is greatly improved over the ORTHO AutoVue® Innova.
- Automating Serial Dilutions studies allows these time consuming and largely manual tests to be automated and standardized in an easy to use format.

Summary/Conclusions: Department of Transfusion Medicine, Faculty of Medicine, Siriraj Hospital, Thailand found the ORTHO VISION MAX® analysers to be efficient, easy to operate and provide significantly improved functionality and testing throughput compared to current ORTHO AutoVue® Innova analyser.

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EVALUATION OF THREE MICROCOLUMN AGGLUTINATION TECHNIQUE AUTOMATIONS FOR RED CELLS ALLOANTIBODIES DETECTION IN THAI BLOOD DONORS

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Background: Microcolumn agglutination technique automation was developed for pretransfusion tests to replace conventional tube test. This method is reported that it has the same sensitivity as conventional tube test for antibody screening test. Recently, microcolumn agglutination technique automation platforms have become widely used for red cell antibody detection in hospital blood bank in Thailand.

Aims: The aim of this study was to evaluate the performance of three microcolumn agglutination technique automated platforms in the detection of red cell alloantibodies in blood donors at National Blood Centre, Thai Red Cross Society.

Methods: In this study, we evaluated three microcolumn agglutination technique automated platforms (ORTHO VISION® Max; Ortho Clinical Diagnostics Pencoed,UK, Erytra; Grifols, Barcelona, Spain and IH-1000;Diamed GmbH, Cressier FR, Switzerland) for red cell alloantibodies detection compared with conventional tube test (Indirect Antiglobulin Test:IAT) using two cell screening (Thai Red Cross Society). One hundred and twenty known positive samples and 2,014 random samples were tested in parallel by tube test and three automated platforms. From 120 known positive samples, antibodies consisted of Anti-Mi^{a} (64), Anti-E (15), Anti-P$_1$ (10), Anti-Le^{a+b}(7), Anti-D (5), Anti-M (4), Anti-Mi^{a}+ Anti-E 4, Anti-S (3), Anti-Tj^{a} (2), Anti-Le^{a} (1), Anti-Di^{a} (1), Anti-Jk^{a} (1), Anti-Fy^{a} (1), Anti-P$_1$+Anti-Le^{b} (1), Anti-E+Anti-c+Anti-Mi^{a} (1)

Results: The performance of three automated platforms for detection of known positive samples was similar to random routine samples. In known positive samples, 119/120 (99.17%) sera were detected by ORTHO VISION® Max, 111/120 (92.50%) by Erytra and 97/120 (80.83%) by IH-1000. All 120 positive samples reacted in conventional tube test. From a total of 2,014 random samples, 60 samples reported positive results and 1,954 samples reported negative results by conventional tube test. The sixty positive red cell alloantibodies consisted of Anti-P$_1$(21), Anti-Le^{a+b} (15), Anti-M (7), Anti-Le^{a} (6), Anti-Le^{b} (5), Anti-Mi^{a} (5) and Anti-E (1) by conventional tube test. Fifty-five samples (91.67%) showed positive results by ORTHO VISION® Max, 20/60 (33.33%) by Erytra and 10/60 (16.67%) by IH-1000. From 1,954 negative sample, there are 3, 2 and 1 samples that showed false positive results by ORTHO VISION® Max, Erytra and IH-1000 respectively.

Summary/Conclusions: The sensitivity of all three microcolumn agglutination technique automated platforms for the detection of red cell alloantibodies in Thai blood donors was highest in ORTHO VISION® Max. However, the samples tested in this study were Thai blood donors using in-house screening cells from the National Blood Centre, Thai Red Cross Society. The microcolumn agglutination technique automation is appropriate for red cell alloantibody detection in a routine laboratory blood transfusion service.

P-400

EVALUATION OF THREE KINDS OF RED CELL PANELS FOR IRREGULAR ANTIBODY IDENTIFICATION

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Background: Irregular antibody screening is one of the important measures to ensure the safety of clinical blood transfusion. The various performance and efficacy characteristics of panel cells that can identify irregular antibodies must to be mastered.

Aims: To compare the differences and characteristics of the three commercially panel cells and to discuss the current status of irregular antibody identification.

Methods: According to the principle of irregular antibodies identification the characteristics of domestic and imported panel cells were evaluated.

Results: The appearance, hemolysis rate, antigen specificity, aseptic test and other aspects of the three panel cells were all in line with regulatory requirements. The domestic panel cells contained 10 groups of red cells and there were 25 blood type antigens in 10 blood type systems. One of the imported panel cells contained 11 groups of red cells and there were 26 blood type antigens in 9 blood type systems. Another imported panel cells contained 16 groups of red cells and there were 28 blood type antigens in 9 blood type systems. The domestic panel cells could detect most of the meaningful antibodies and some rare antibodies effectively, such as: Rh, MN, Lewis and other blood type system; Otherwise, the imported panel cells were better than those of domestic cells in the aspects of blood system, the number of

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blood type antigens and the level of antibody identification. The following antigens were the most obvious: C, e, K, k, s, Leb, Fya, Fyb, Jka. Imported panel cells were more scientific in terms of dose-effect arrangements than domestic cells and had an advantage in antibody identification.
Summary/Conclusions: The principle of irregular antibody identification is to use the smallest serum/panel cells to identify most of the meaningful common antibodies and some rare antibodies. Scientific and rational use of antibody identification panel cells can effectively prevent immune transfusion reactions and reduce the risk of neonatal hemolytic disease.

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Abstract has been withdrawn

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ANTIBODY SCREENING OF BLOOD DONORS

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Background: Since December 2016, Blood Transfusion Unit of Bekasi County has done antibody screening to all sample of blood donor. Antibody screening is an examination to find if there is any unexpected antibody in donor serum. We do antibody screening with magnetized method using Qwalys 3. The purpose of antibody screening is to select the blood donor sample which contain unexpected antibody, so the blood donors which will be transfused is safer and the transfusion reaction that caused by the reaction of antigen & antibody will be avoided.
Aims: To evaluate the result of antibody screening in 7 months.
Methods: Collect secondary data from database system from December 2016 to July 2017.
Results:

Months	Donor amount	Amount of positive antibody screening.
December 2016	4,814	3
January 2017	3,949	1
February 2017	4,840	2
March 2017	5,294	3
April 2017	4,954	6
May 2017	6,149	0
June 2017	2,271	1
Total	32,271	16

Summary/Conclusions: Based on the result of antibody screening in 7 months, from total 32.271 of blood donor sample, there were 16 positive results. The percentage of positive result of antibody screening is 0.049%. Since we found positive result for antibody screening, we suggest to do further examination to the samples which have positive result and antibody screening has to be done to all of blood donor samples.

P-403

STUDY ON THE STRATEGIES FOR DOUBTFUL ABO BLOOD GROUP

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Background: ABO blood typing in pre-transfusion testing is a major component of the high workload in blood banks that therefore requires automation. We often experienced discrepant results from an automated system, especially weak serum reactions. We evaluated the discrepant results by the methods of traditional test tube typing, absorption-elution and ABO genotyping to confirm ABO blood typing.
Aims: To analyze the results of doubtful ABO blood grouping, and investigate the strategies for doubtful ABO blood grouping for ensuring safety of clinical blood transfusion. traditional test tube typing, absorption-elution and ABO genotyping to confirm ABO blood typing.
Methods: 732 samples of blood donors from Liaoning Blood Centre were typed by automatic blood analyzer from the year 2010 to 2016. The methods of traditional test tube typing, absorption-elution and ABO genotyping were used to detect the samples for ABO groups.
Results: 717 (97.95%)cases of ABO group were confirmed by test tube typing, 457 cases (62.43%) with weak anti- B, 171 cases (23.36%) with weak anti - A. Irregular antibodies were found in 115 cases (15.71%), 18 (2.45%) cases with anti-M, 1 (0.14%)case with anti- E and 1 (0.14%) case with anti-Le (b) being identified. 15 cases of ABO subgroups were confirmed by ABO genotyping, 1 (0.14%)case with AB_w33, 1 (0.14%)case with A_x13B, 5(0.68%)cases with B_w, 2 (0.27%)cases with B_{el}, 2 (0.27%)cases with cisAB03, 1 (0.14%) case with cisAB06, 1 (0.14%)case with $B_{(A)}$, 1 (0.14%)case with para-Bombay A type and 1 (0.14%)case with antigen A-weakly expressed ABO*004 being confirmed.
Summary/Conclusions: The main reasons responsible for doubtful ABO blood groups by automatic blood analyzer were weak anti-B, anti-A and irregular antibodies. In Chinese in Shenyang region, subgroup B is more common than subgroup A. The samples of doubtful ABO blood groups by automatic blood analyzer should be analyzed by both traditional test tube typing and ABO genotyping.

P-404

ANALYSIS OF ABNORMALITIES IN ABO BLOOD TYPING OF CHANGSHA BLOOD DONORS

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Background: ABO blood antigen is the highest immunogenicity in human blood system antigens, and it is the key to ensure the safety of clinical transfusion. ABO subtypes, impaired blood antigens, irregular antibodies and human errors or technical reasons can lead to abnormalities of ABO blood typing. We did a retrospective statistical analysis of the abnormalities in ABO blood typing from laboratory medicine of Changsha Blood Center, and analyzed blood products returned by hospitals due to ABO blood typing abnormalities.
Aims: To analyze the abnormalities in ABO blood typing of Changsha blood donors, such as subtypes or missing detection of special antibody, ABO typing discrepancy, etc. We are aimed to explore the causes and laboratory treatment measures of the abnormalities, and to know about the blood scrap.
Methods: Microplate colorimetric method was used to screen the blood types of donors, and we use hand-holding tube method to examine the irregular blood types, such as irregular antibody identification, so as to judge whether the blood were available in clinical infusion.
Results: In Changsha blood center, there were 148 abnormalities in ABO blood typing of 563,293 blood donors from 2013.1.1 to 2017.1.31. There were 58 special antibodies in plasma, 26 with low antibody value, 24 subtypes, 5 false condensation and 35 with other causes. In total, 54 blood products were sent back by hospital because of blood type qualities. There were 34 special antibodies in plasma, 13 with cross-matching incompatibility, 4 subtypes, 3 wrong blood types.
Summary/Conclusions: The common reasons of abnormalities in ABO blood typing are nonspecific aggregation, subtypes and special antibodies. We should strengthen the training of laboratory to keep from abnormal test, and ensure blood safety.

Red Cell Immunology: Molecular

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Abstract has been withdrawn

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MOLECULAR BASIS OF THE B(A) PHENOTYPE AND ITS CLINICAL TRANSFUSION ANALYSIS

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Background: Numerous examples of A or B subgroup alleles have been described in many geographical and ethnic groups. Among them, the B(A) blood group is extremely rare. It is due to the inheritance of a single chromosome encoding an enzyme with both A and B transferase activity. Here, we describe a rare variant based on a nucleotide substitution c. 700G>C (p. P234A) in the B glycosyltransferase gene, identified in a patient with aberrant ABO phenotypes.
Aims: To investigate the serological characteristics and molecular characteristics of the B(A) phenotype, and to review safety issues in relation with clinical transfusion.
Methods: The individual was confirmed by standard serological techniques. The genotyping and sequencing were performed using polymerase chain reaction-sequence specific primer (PCR-SSP), direct sequencing and gene clones for Exon6 and exon7 of ABO locus respectively. A survey was carried out to investigate what is the actual position that the proband was used in the clinical transfusion.
Results: Both A and B antigens were detected on red blood cells of the proband and there was anti-A antibody in the serum. The result of PCR-SSP was *B/O02* phenotype. The direct DNA sequencing and gene clones revealed that the gene was B(A) 02/O02. The B(A)02 allele had one nucleotide change(C >G) at position 700 compared with that of *B101* allele, which resulted in an amino acid substitution (P234A).The results of cross match testing is in accordant between the proband and two donors with phenotype A_2B and there was no clinical adverse reaction after transfusion.
Summary/Conclusions: 700G>C of B allele can result in B(A) phenotype in the individual with the phenotype of A2B. The donors in the transfusion for the individuals with B(A) phenotype should include individuals with A_2B phenotype.

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CAUSE ANALYSIS AND IDENTIFICATION OF B (A) SUBTYPE MISSED DETECTION

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Background: PK7300 automatic are widely used in blood type testing. Limited to the methodology and the diversity of B(A) type serological characteristics, B(A) type are often undetected by PK7300 automatic. So we need to identified this blood type and analyze the missed reasons.
Aims: The missed B (A) blood group was analyzed and identified by serological and molecular biological characteristics, in order to find the missed reasons for PK7300 automatic.
Methods: The ABO blood group was detected by the method of blood group serology, the ABO blood group genotype was detected by PCR-SSP method, the coding region of exon 6 to exon 7 in ABO gene was amplified by polymerase chain reaction(PCR) and the PCR products were sequenced. The haplotypes were analyzed by cloning and sequencing.
Results: 1, Using the test tube method of the reagents of three different manufacturers, the results cannot be determined whether AsubB or B (A) type; 2, The results of the same factory reagent test with the automatic blood type instrument and board method are B type; 3, The result showed that the blood group is B(A)04/O02 by using PCR-SSP. 4, DNA sequencing showed 261G/del,526G/C,640G/A,646T/A,657T/C,681G/A,703A/G,771C/T,796A/C,803C/G,829G/A,930A/G,1096A/G heterozygotes in exon 6 to exon 7, it can be were B(A)04/O02, after cloning and sequencing, two alleles B(A)04 and O02 was showed.
Summary/Conclusions: Due to methodological limitations and serological characteristics of B (A) subtypes, ABO subtype are often undetected by PK7300 automatic method.

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MOLECULAR ANALYSIS OF FUT-1,2 IN CHINESE POPULATIONS FROM SOUTHEASTERN AND NORTHERN REGIONS OF CHINA WITH THE PARA-BOMBAY BLOOD TYPE

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Background: The incidence of the para-Bombay blood type in Chinese populations from the Southeast of China was estimated as 0.8%, and only one case has been reported out of a population of one million Chinese individuals in Northern China. Laboratory testing for para-Bombay generally perform anti-H and absorption-elution tests on peripheral venous blood and saliva samples. However, in some clinical conditions, such as a coma state or when it not clear whether the patient is conscious, it is very difficult to obtain a saliva sample.
Aims: The aim of this study was to evaluate the consistency between serotyping and molecular analysis in Chinese individuals with para-Bombay blood type. To observe differences in molecular features between Chinese individuals with para-Bombay blood from the Southeast and North of China, we conducted a molecular analysis of gene fragments from FUT-1, FUT-2, FUT-3, and ABO exon 6,7, on each para-Bombay blood sample.
Methods: Molecular analysis of gene fragments from FUT-1, FUT-2, FUT-3, and ABO exon 6,7 was used in conjunction with serotyping, including a saliva test and an absorption-elution test to examine A, B, and H substances and antigens, and further routine testing for ABO, H, and Lewis phenotypes.
Results: All eleven anti-H negative samples from Southeast China were identified as para-Bombay, by sequencing of FUT-1 and FUT-2. Of these, five samples were homozygous 547-548delAG, three samples were homozygous 880TT deletion, and three samples were heterozygous 547-548delAG combined with the heterozygous 880TT deletion. In contrast to para-Bombay samples from Northern China, three samples were heterozygous 586 C>T, heterozygous 328G>A and 649G>T, combined with the heterozygous 880TT deletion. The sequencing of FUT-2 confirmed 357C>T in fourteen samples, demonstrating that H, A, and B substances were secreted in the saliva, except in two samples which were reported as 385A>T (I129F) heterozygous, which is a weak secretor, and all samples from Northern China were 385A>T. The saliva test for A, B, and H substances, and the absorption-elution test examining the A, B, and H antigens on the surface of red blood cells completely matched the ABO exon 6, 7 sequencing results.
Summary/Conclusions: The incidence and mutation types of para-Bombay from various areas of China showed diversity, especially in para-Bombay samples from North of China, which comprised various mutation types. The sequencing of FUT-1, FUT-2, and ABO exon 6, 7 may become a useful tool to confirm the para-Bombay blood type.

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STUDY ON A-1,3-N-ACETYLGALACTOSEAMINOTRANSFERASE GENE 389 T > C VARIANT

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Background: The gene locus for the ABO blood group system encodes a glycosyltransferase. Weak expression of A/B histo-blood group antigens is often explained by single nucleotide substitutions at the ABO locus. We investigated the basis of weak A phenotype in a individual.
Aims: To study the effect of A antigen expression for 389 T > C variant of the α-1,3-N-acetylgalactosaminyltransferase of ABO gene by the ABO alleles' molecular genetic analysis in a individual was suspected as rare Ax subtype of ABO variant.
Methods: Serological assays were performed to identify ABO blood group on the samples of the proband. The extracted DNA was genotyped by sequence specific primer polymerase chain reaction (PCR-SSP) followed by directly sequencing for exon6 and exon7 of ABO locus to identify the ABO gene haploid of the proband. The bioinformatics analysis was carried out by biological analysis software to investigate the change of structure and function of enzymes influenced by the change amino acid.
Results: Weak A antigen was detected on red blood cells of the proband and there was anti-A and anti-B antibody in the serum. DNA sequencing showed 261delG and 389 T/C heterozygote from exon6 to exon7. By haplotype sequence analysis, two alleles *Ax22* and *O01* were obtained. The sequence of Ax22 allele had a nucleotides

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change (T to C) at position 389 compared with that of *A101* allele resulting in an amino acid substitution (L130P156). The 389 T>C mutation can disrupt the space conformation of its protein product, affecting the function of the protein.
Summary/Conclusions: T>C at nt389 of α-1,3-N-acetylgalactosaminyltransferase gene can result in the great reduction of A antigen activity. We can conclude that the amino acid 130 locates activity region of the transferase encoded by ABO gene.

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A NOVEL A ALLELE WITH C.538C>T MUTATION IDENTIFIED IN A3 PHENOTYPE

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Background: A3 is characterized by mixed-field reaction with a few large agglutinates among large amounts of free cells when typing with anti-A. The underlying mechanisms of A3 phenotype were diversified.
Aims: To perform serologic and genetic analysis to investigate the molecular mechanism of ABO serologic discrepancy detected in one A3 phenotype patient referred to our laboratory.
Methods: ABO serologic tests were done by the standard serologic protocol. Genomic DNA was isolated from the blood by a commercial kit according to the manufacturer's instructions. The *ABO* gene upstream CBF/NF-Y enhancer region, promoter, *ABO* +5.8-kb site, exons and adjacent introns were amplified and sequenced. PCR products of exon 6 to exon 7 and adjacent introns of the *ABO* gene were cloned and the DNA sequences were analyzed.
Results: The A antigen of red blood cells (RBCs) demonstrated mixed-field agglutination when tested with routine ABO blood group typing. The proband carried heterozygous missense mutations C>T at the position 467 and 538 in the *ABO* gene causing a substitution of Pro to Leu at position 156 (p.P156L) and Arg to Cys at position 180 (p.R180C), respectively which expressed A3 phenotype. No other mutations were found in the promoter, CBF/NF-Y enhancer region, other coding regions or the exon-intron boundaries of the *ABO* gene. Cloning sequencing confirmed that the novel c.538C>T single nucleotide polymorphism was associated with the *A102* allele. The Genbank nucleotide database accession numbers for the novel alleles was KX870187.
Summary/Conclusions: Our study suggested that the C>T substitution at position 538 of the *ABO* gene could decrease expression A antigen on RBCs causing the generation of the A3 phenotype. Correctly identify these subgroups to improve the accuracy of blood typing is important for compatible transfusion.

P-413

NOVEL MISSENSE MUTATION (662G>A) IN THE ABO GENE LEADING TO A BEL PHENOTYPE

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Background: ABO gene is organized in 7 exons, and exon 6 and 7 encode most of the sequence. ABO gene variant could alter the ABO transferase activity, which could cause the ABO discrepancies.
Aims: We interpreted here ABO discrepancies (forward O, reverse B) caused by missense mutation in the ABO gene.
Methods: ABO blood groups were determined by tube test. Forward typing was performed with anti-A, anti-B, anti-AB and anti-H. Reverse typing was determined by the tube method using A1 and B red blood cells. The presence of the weak B antigen on RBCs was determined by adsorption-elution test with human anti-B. The antibody was recovered by heat elution method (56°C, 10 min). Genomic DNA was extracted from blood sample using a commercial kit. Polymerase chain reaction and sequencing primer were designed, exon 6 and 7 were individually PCR amplified and analysis the result of direct sequence. dATP was added to 3' of PCR products which were subcloned within a plasmid vector, the reaction product was transformed into competent cells DH5a, cultured in a selective medium. Extracting plasmid form 10 clones respectively and sequencing the target fragment with T3 and T7 primers.
Results: The serological test result shows the ABO discrepancy: forward O and reverse B; The presence of anti-B in the equate indicated the presence of B antigen on test red cells. The variant allele was identified at the heterozygous form by direct sequencing. Cloning and sequencing were performed, and allele analysis shows a novel allele containing a single nucleotide substitution of G with A at position 662, which caused a amino acid change from Gly to Asp at position 221. The novel allele has been submitted to the GenBank by the accession number KU206323.
Summary/Conclusions: A single nucleotide substitution could be the reason of ABO discrepancies in serologic typing.

P-414

SEROLOGY AND MOLECULAR STUDY ON RARE AW AND BW BLOOD GROUP

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Background: Identifying genetic variants of the ABO gene can reveal new molecular mechanisms of the ABO blood group. Either A_w or B_w is a very rare ABO blood group phenotype and the genetic mechanism underlying it still remains largely unknown. Here we report the molecular genetic analysis of 83 unrelated ABO subgroup individuals from Qingdao and its surrounding areas.
Aims: To determine the genotypes of the blood samples whose serological results seem like A_w or B_w through DNA sequencing analysis.
Methods: Serologic investigations including serum transferase activity assay were performed with standard methods. The blood samples were preliminary genotyped by PCR-SSP. Complete Exon 6 and Exon 7 in the ABO genes were amplified by PCR and the PCR products were analyzed through direct DNA sequencing or sequencing after gene cloning to identify its genotype.
Results: The serological results of these blood samples were similar: red cells were weakly agglutinated by anti-A or anti-B. In the 83 rare ABO alleles, we identified 14 $A_{el}05$ (467C>T, 767T>C), two A_x13 (940A>G), 49 B_w03 (721C>T), eight B_w07 (1055G>A), eight B_x02 (905A>G), one B_w17 (784G>A), and a novel A_x26 (287G>T, 467C>T) subgroup allele. All of these were point-mutation alleles and the mutation sites were detected in Exon 6 and Exon 7 which resulted in the amino acid changes.
Summary/Conclusions: Through sequencing analysis, the haplotypes of the 83 samples typed as A_w or B_w in serological tests are identified. Amino acid substitutions resulted from point-mutations on ABO gene change highly conserved regions of the enzyme and may reduce the activity of the glycosyltransferases, leading to the A_w or B_w phenotype.

P-415

Abstract has been withdrawn

P-416

MOLECULAR MECHANISM RESEARCH OF A3 OR B3 BLOOD SUBTYPES IN SHANDONG PROVINCE

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Background: Molecular mechanism of A_3 or B_3 blood subtypes is not clear. And the research results of different regions are quite different.
Aims: Analysis the molecular mechanism of serological suspected A_3 or B_3 blood subtypes in Shandong Province from 2014 Jan. to 2017 Jun.
Methods: Serological tests were performed to characterize erythrocyte phenotype of all samples that we collected from Shandong Province. Polymerase chain reaction (PCR) direct sequence ABO gene enhancer (−3988 bp to −3645 bp), promoter (−149 bp to −2 bp), exons 1–7 and flanking sequence, and GATA binding site in I intron (~+5.8 kb) of suspected samples. For the samples with a mutation, haplotype sequencing is used to confirm again.
Results: By serological test, we suspected 38 samples of A_3 or B_3 subtype, which accounting for 10.1% of subtype samples in the same period. By DNA sequencing, we identified 7 samples of A_{307}, 1 samples of A_{304}, 2 samples of B_{301}, 2 samples of B_{305}, 1 sample of B_{310} and 1 sample of B_{311}. And 3 samples were confirmed as chimera by Short Tandem Repeats (STR). B_{303}, which more frequently in Taiwan area, and mutation in enhancer or promoter area reported in Japan were not detected in our samples.
Summary/Conclusions: Compared with other areas, molecular mechanism of A_3 blood subtypes in Shandong Province changed litter, while B_3 subtype varied greatly. The difference was obvious with Taiwan or Japan, while close to domestic cities such as Shanghai and Hangzhou.

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ANALYSIS ON SHANDONG PENINSULA POPULATION OF ABO GLYCOSYLTRANSFERASE WITH DUAL SPECIFICITY

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Background: To investigate the molecular genetic basis CisAB and B(A) subgroup of ABO blood group system in Shandong Peninsula population, China.
Aims: To investigate the molecular genetic basis CisAB and B(A) subgroup of ABO blood group system in Shandong Peninsula population, China.
Methods: Serological techniques were performed to characterize erythrocyte phenotype of discrepant samples that we collected from Shandong Peninsula population. Most samples were identified as $A_{weak}B$ or AB_{weak} phenotype by serological technology, PCR-sequence specific primer(PCR-SSP) method was used for genotyping these samples. Mutations of complete Exon 6 and 7 including flanking intron in the ABO genes were screened by PCR and directly sequencing. The haplotypes of exon 6 and 7 and intron 6 were also analyzed by cloning sequencing. We use Gentle software to analysis the sequences of these samples.
Results: Serologically, all these samples were inconformity between forward and reverse group methods. The forward group test showed that the samples were AB, while the reverse group test showed that the samples had the anti-B or anti-A1. Some of these samples were genotyped as O1/B by the PCR- SSP assay, and other samples were genotyped as A/B by PCR-SSP assay. All the samples were identified as CisAB or B(A) variants including genotypes by directly sequencing and haplotype sequencing. Eighteen heterozygous alleles were observed as CisAB: CisAB01var (1 samples), CisAB01 (17 samples). Forty-six heterozygous alleles were observed as B (A): B(A)04 (41 samples), B(A)02 (5 samples).
Summary/Conclusions: Haplotype B(A)04, CisAB01 and B(A)02 are most popular CisAB and B(A) alleles in Shandong Peninsula, China.

P-418

Abstract has been withdrawn

P-419

MOLECULAR GENETIC AND GENEALOGICAL ANALYSIS FOR A NOVEL AX ALLELE IN CHINESE

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Background: To elucidate a molecular genetic and genealogical analysis of members of an novel Ax family in Shandong Peninsula, China.
Aims: To elucidate a molecular genetic and genealogical analysis of members of an novel Ax family in Shandong Peninsula, China.
Methods: One Ax phenotypes were identified in a Shandong Peninsula Chinese W. family and two related family members, respectively. Serologic studies, sequencing of exon1-7, and molecular coining of exon6, exon 7 and intron 6 were carried out to determine their respective genotype characteristics. Genealogical analysis was conducted on this family.
Results: A novel Ax allele with nt287G>T mutation in the A102 background, which predicts one amino acid alteration of Trp96Leu, was identified in the Ax propositus. Mutation might give rise to weak expression of the A antigens on RBCs of this propositus. This novel Ax allele was demonstrated in other two family members.
Summary/Conclusions: There is different mutations of this rare Ax allele from the ABO subgroup alleles reported previously. A novel Ax allele with nt287G>T and mutation was demonstrated in a pedigree with the Ax phenotype. The mechanism leading to the formation of the Ax phenotype still awaits elucidation.

P-420

TWO NOVEL A2 ALLELE IDENTIFIED IN CHINESE BLOOD DONORS

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Background: The A2 is a relatively rare phenotype in the ABO blood group system in China, however, significant allelic heterogeneity among A2 and A2B donors has been discovered.
Aims: In this study, we determined the A2 alleles distribution and identified two novel A2 alleles and a rare allele A223 in Shandong Peninsula, China.
Methods: 6518 Chinese individuals of blood donors in Shandong Peninsula with group A and AB phenotypes were determined for A2 antigen based on the standard ABO serological test. The Exon6, 7 of the ABO genes were screened by PCR and directly sequencing. The haplotypes of exon 6,7 and intron 6 were also analyzed by cloning sequencing.
Results: 32 individuals were assigned as A2 and A2B phenotypes in 6518 individuals including 5 cases of A201, 24 cases of A205, 1 cases of A204, 1 cases of A223 and 2 cases of newly discovered subtype. One novel A2 allele has a 689G > T that results in the amino acid change G230P identical to A102 sequence. The new alleles registered number in Genbank is KX863521. Another novel A2 allele has a 692C>A that results in the amino acid change T231D identical to A102 sequence. The new allelecs registered number in Genbank is MF176160 The total frequency of the A2 phenotype among the group A and AB individuals was 0.49%, 0.29% of group A individuals were A2 phenotype, while 1.08% group AB individuals were A2B phenotype.
Summary/Conclusions: We report for the first time that a 689G > T and 692C>A mutations represents new molecular basis for the A2 blood type. The amino acid substitution may have effect on the expression of A antigen.

P-421

MOLECULAR GENETIC ANALYSIS OF A RARE A307 SUBTYPE OF ABO BLOOD GROUP SYSTEM

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Background: ABO blood group system is the most important in transfusion medicine and organ transplantation. Correct identification of ABO blood group system is important for compatible transfusion of red blood cells (RBCs). Serological analysis of ABO blood group type has been widely applied in clinical transfusion medicine, however, sometimes ABO typing discrepancy was observed between the forward and reverse type.
Aims: To explore the molecular mechanism of ABO typing discrepancy in a patient by ABO genotyping.
Methods: Routine serological test was carried out to analyze ABO blood group. Genomic DNA was extracted from the blood by a commercial kit according to the manufacturer's instructions. Exons 1–7 and adjacent introns of the *ABO* gene were amplified by polymerase chain reaction (PCR) and analyzed by directly sequencing. PCR products of exon 6 to exon 7 and adjacent introns of the *ABO* gene were cloned and analyzed.
Results: ABO forward typing showed the sample was weak A antigen and demonstrated mixed agglutination, while the reverse typing showed the sample contained anti-A1. The *ABO* gene sequencing showed a C>T substitution at the position 467 and 745 in exon 7 which caused a Pro to Leu exchange at codon 156 and Arg to Trp at codon 249, respectively. The cloning sequencing showed one allele was *A307*, and the other allele was *O01*. The ABO blood group genotype of the sample was *ABO* A307/O01*.
Summary/Conclusions: The 745 C>T substitution in the gene of *α-1,3-N-acetyl galactose* is the molecular mechanism leading to the decrease expression of A antigen of the A307 subtype. In order to ensure the safety of blood transfusion, the genotyping should be carried out when necessary.

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THREE CASES OF BW BLOOD GROUP SEROLOGY AND GENOTYPE ANALYSIS

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Background: Although ABO subtype is rare, it may seriously affect the safety and efficacy of blood transfusion. ABO subtype is one of the major causes of the positive and negative qualitative changes of ABO blood group. The correct identification of the ABO blood group is the first step in safe blood transfusion.
Aims: To identify 3 cases of Bw blood group with serology and genetic experiments.
Methods: Monoclonal antibody - A, B, D, H antibody and antireagent red blood cells were used to determine blood typing by serological method. Molecular biology pcr-ssp method was used to sequence the exons of ABO gene 6 and 7.
Results: In the three cases,the anti -B was weak,and the opposite A cell concentrated agglutination,and B cell was weakly concentrated; .The samples of the three samples were found to have B antigen on the surface of red blood cells by assimilate and diffuse experiments. 721C>T base mutation in exon seventh of B gene was found in a samples by sequencing, and 278C > T mutations in 2 cases.
Summary/Conclusions: Blood type and molecular biology experiment were combined to determine 1 case of Bw03/O, 2 blood type Bw12/O, Recommended infusion O type red blood cells and type AB plasma.

P-423

A CASE OF B SUBGROUP IDENTIFICATION AND FAMILY INVESTIGATION

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Background: ABO subgroup often lead to incorrect blood type identification Which may cause blood transfusion incident. In recent years, blood type detection is more accurate with the molecular biology techniques used. This make clinical blood transfusion safer and more effective.
Aims: Identify blood type of blood donors exactly to ensure the blood supplied to clinic is correct.
Methods: ABO blood-type was detected by serological test. Neutralization test was used to detect blood type substances. Genotyping was performed by PCR-SSP. Gene mutation was detected by gene sequencing. Family investigation was carried out for her parents and brother.
Results: Absorption-elution test detected B antigen. Neutralization test detected B, H substances. PCR-SSP confirmed genotyping O1B. Sequencing results showed nt695 site T/C missense mutation occurred compared with B_{101} sequence. This mutation was named $B_{w11.}$ Family survey showed her father also carried B_{w11} allele.
Summary/Conclusions: The blood type of blood donor was B_{w11} subtypes.

P-424

Abstract has been withdrawn

P-425

STABLE EXPRESSION OF HUMAN RHD PROTEIN ON THE MEMBRANE OF MOUSE 3T3 FIBROBLAST CELLS

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Background: Administration of polyclonal anti-D immunoglobulin is used to prevent RhD alloimmunization in pregnancies. Currently, the polyclonal anti-D is produced from the plasma of volunteer donors. The efforts to replace polyclonal anti-D with monoclonal anti-D have been severely hindered by the fact that the mechanism of action of anti-D has not been fully elucidated. The elucidation of the mechanism has in turn been hampered by lacking of suitable experimental models. The model most physiologically comparable to the human situation would be a transgenic male mouse with human RhD expression on mouse erythrocytes to mate with a wild type female mouse to produce a hemolytic disease of the fetus and newborn disease model. To generate a successful transgenic model, it is crucial to understand how surface RhD expression is achieved.
Aims: To express human RhD protein on the membrane of mouse cell lines using lentiviral transduction.
Methods: The coding sequences of human *RHD* (h*RHD*) and human *RHAG* (h*RHAG*) were cloned into pHeftig and pSINk lentiviral vector, respectively. In addition, a hybrid construct of *RHD* in which the human C-terminus was exchanged for the mouse sequence, was cloned. Mouse 3T3 fibroblasts, 1/11 erythroblasts and erythroleukemia MEL cell line were simultaneously transduced with h*RHD* and h*RHAG* constructs. By CRISPR/CAS9 approach the mouse *RHAG* (m*RHAG*) and mouse *RHc* (m*Rhc*) genes were disrupted in the 1/11 cells (either both or each gene separately), KO-cells were subcloned and lack of expression was verified by RT-PCR. After 72 h in culture, surface expression of hRhD and hRhAG was analyzed using human polyclonal anti-D serum, a panel of anti-D monoclonal antibodies (Moabs) including Bric69 and P3x241 and mouse Moabs anti-RhAG (clone 2D10 and LA18.18) respectively, by flow cytometry. 1/11 cells were also tested after induction of terminal erythroid differentiation (van den Akker, 2010).
Results: In all mouse cell-lines strong expression of hRhAG protein was accomplished. However, surface expression of hRhAG protein disappeared in 12 days after transduction in 1/11 erythroblast and erythroleukemia MEL cells, while it could be stably expressed on the membrane of 3T3 fibroblast cells. As expected, hRhD protein could not be expressed in the absence of hRhAG protein. hRhD protein could be stably expressed on the membrane of 3T3 fibroblast cells, but in contrast no surface expression of hRhD protein was detected on 1/11 erythroblasts and erythroleukemia MEL cells when the h*RHD* and h*RHAG* constructs were co-transduced in these cells, also not in the cells lacking endogenous expression of mRhc and/or mRhAG, neither in terminally differentiated 1/11 cells. Because for the expression of hRhD the interaction of its C-terminus with ankyrin is essential and mouse erythroid cells express different ankyrins, we tested whether the expression of hRhD could be obtained by replacing the human C-terminus with the mouse sequence. But again no RhD expression was seen in any of the 1/11 cells.
Summary/Conclusions: Whereas for the expression of human RhD in mouse fibroblast cells the cotransduction of human RhAG is sufficient, no hRhD expression in mouse erythroid cells could be obtained. We demonstrated that this is not due to the expression of mRhAG or mRhc, or lack of interaction with mouse erythroid ankyrin. We hypothesize that in the mouse erythroid cells either a co-factor needed for expression is missing or an inhibitor of expression is present. The present study offers us the tools to investigate this mechanism, and has direct implications for the construction of a transgenic *RHD* mouse.

P-426

MOLECULAR GENETIC ANALYSIS OF 779A>G AND 1227 G>A MUTATIONS OF RHD GENE RESPONSIBLE FOR A WEAK D

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Background: Rh blood group system is the most complex and immunogenetic blood group system. Prevalent RHD alleles varied in different populations. Here, a study was investigated with weak or discrepant D serologic typing by the molecular level in Chinese persons.
Aims: To analyze the molecular genetic basis of *RHD* alleles of an individual with rare weak D phenotype.
Methods: The regular serological methods and indirect antiglobulin test (IAT) were performed to characterize the RhD blood group. Mutations of the *RHD* gene were screened by polymerase chain reaction, reverse transcription polymerase chain reaction (RT-PCR) and DNA sequencing. The cDNA amplified product was TA cloned and haplotype analysis. The mutations were analyzed using Polyphen2, SIFT, HSF, DNAMAN 6.0 and Swiss-PdbViewer4.1.0 software.
Results: The RhD blood group of the proband was detected as weak D. The result of PCR amplification showed that all of 10 *RHD* gene exons were presence. 779 A/G and 1227A/G heterozygote were detected by gDNA and cDNA sequencing. After TA cloning and haplotypes sequencing analysis, two alleles weak D 779G and DEL were obtained. The sequence of weak D 779G had one nucleotide change (A to G) at position 779 in exon5 compared with that of *RHD* allele resulting in an amino acid substitution (H260R). The sequence of DEL had one nucleotide change (G to A) at position 1227 in exon9 compared with that of *RHD* allele resulting in no amino acid substitution. Both mutations can disrupt the space conformation of its protein product, affecting the function of the RhD protein.

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Summary/Conclusions: 779A>G mutation is responsible for the low expression of RhD antigen on red blood cell of the proband, and lead to weak D phenotype. Del (1227G>A) can also be designated as a particular weak D type.

P-427

IDENTIFICATION OF A NOVEL RHD*688C ALLELE ASSOCIATED WITH A PARTIAL D PHENOTYPE

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Background: The Rh blood group system is the most polymorphic system of the red blood cells (RBCs), consisting of 54 blood group antigens. Next to the ABO system, it is most clinically significant in transfusion medicine. Besides the common D+ or D− phenotype, a plethora of D variants also exist, which were commonly classified into weak D, partial D, and DEL phenotypes. In this study, the *RHD* genotype was analyzed in the Chinese donors with D variant phenotype and one novel *RHD* variant allele was identified.
Aims: To investigate the variant *RHD* alleles in the Chinese population.
Methods: The blood samples with D variant phenotype were collected from the Guangzhou Blood Center and Blood center of Shanxi province in China. The *RHD* and *RHCE* genotypes were analyzed by the developed Multiplex Ligation-dependent Probe Amplification (MLPA) genotyping assay. Further sequencing of ten exons of the *RHD* gene was conducted in the donor having a *RHD* variant allele that could not be defined by the MLPA analysis. The serological typing for D antigen was conducted by using the two monoclonal anti-D reagents (Clone Rum-1 and TH-28/MS-26) and the commercial panel anti-D reagents (D-Screen, Diagast).
Results: One donor collected from the Blood center of Shanxi province with a novel *RHD* variant allele defined by a missense mutation (c.688A>C, p.Ser230Arg) was identified by sequencing of *RHD* gene. *RHD*688C/RHD*01N.01* (*RHD* deletion) and *ccEe* genotypes were determined by the MLPA analysis combined with sequencing. RBCs of this donor negatively reacted with the monoclonal IgM anti-D (Clone Rum-1 for epD6.1), while weakly reacted with IgM/IgG anti-D((1+), Clone TH-28/MS-26) by the tube method. Besides, for the reaction with the panel anti-D (D-Screen), a positive agglutination with four monoclonal anti-D ((1 +) HM16 for epD6.4; (3 +) P3*21223B10 for epD9.1, P3*249 for epD2.1, P3*290 for epD3.1) and a negative agglutination with the rest five kinds of anti-D (Clone HM10 for epD6.6, P3*61 for epD6.1, P3*241 and P3*290 for epD5.4, P3*21211F1 for epD8.2) were detected by the tube method, while D+ control with a strong positive (4 +) reaction and D- control with a negative reaction.
Summary/Conclusions: The missense mutation defined the novel *RHD*688C* allele caused one amino acid change (p. Ser230Arg) located in the 4th extracellular loop of RhD protein, which indicated to result in a partial D phenotype.

P-428

Abstract has been withdrawn

P-429

POLYMORPHISM OF RHD GENE IN RHD NEGATIVE AND D VARIANT BLOOD DONORS IN QINGDAO AREA

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Background: Most studies of the molecular basis of D variants have been conducted in Chinese populations. There are no data on the molecular basis for RhD variants in Qingdao area populations.
Aims: To study the molecular mechanism and polymorphism of D gene of RhD negative and D variants in Qingdao area of Shandong province.
Methods: The *RHD* gene exon 1-10 were detected by PCR-SSP method in 357 D negative phenotype cases and 24 D variant cases of Qingdao voluntary blood donors confirmed by serological test. The samples which have all or partial *RHD* exons were then sequenced.
Results: There are 239 samples(69.08%) were whole *RHD* genes absence in 357 cases and 108 samples (31.12%) had some or all of 10 *RHD* exons. There are 10 alleles and genotypes respectively, *RHD*1227G>A* in 54 cases (15.56%), *RHD-CE- (2-9) -D* in 40 cases (11.53%), *RHD845G>A* 5 cases(1.44%), *RHD*711delC* in 2 cases (0.58%), *RHD-CE- (3-8)–D* in 1 case (0.29%), *RHD*3G>A* in 1 case (0.29%), *RHD845 A/G* in 1 case (0.29%), *RHD*1013T>C* in 1 case (0.29%), *RHD* 1227A/G* in 1 case (0.29%), in 1 case (0.29%), no mutations in all 10 *RHD* exons were found. There are 6 alleles and genotypes in 24 cases of D variants respectively, *RHD*845 G>A* in 15 cases (62.50%),*RHD*845G>A*; 1227 *G>A* in 1 case (4.17%); *RHD* 711delC* in 1 case (4.17%); *RHD* 711delC*; 1227 *G>A* in 1 case (4.17%),Dδa in 2 cases (8.33%), εβ in 2 cases (8.33%).In addition, we found 1 RhD negative and 2 D variant novel alleles.
Summary/Conclusions: There are rich genetic polymorphisms in RhD negative blood donors in Qingdao area, absence of the whole *RHD* genes is main.

P-430

POLYMORPHISM OF RHD MRNA SPLICEOSOME IN CHINESE POPULATION

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Background: RH of different genetic background between different races of human beings, so the races and the regions' expression of RH gene and the formation of antigen are different. RhD antigens are controlled by the RHD genes corresponding to the chromosome RHD gene in human (1p34-p36), including RHD and RHCE gene. The relative orientation of [RHCE gene (5 '- >3' - (3) – "<5 RHD]") that similarity of up to 96%, which are composed of 10 exons. The RHD and RHCE two genes of high similarity, so often because of genetic recombination between replication or lack of a lot of variation. The variation of the RhD antigens including the quality of different antigenicity of RhD. The RhD antigens can be divided into: D common antigen; weak D; incomplete D (partial D); enhanced based on D and so on. RhD antigens expression of diversity in order to investigate the RHD gene, according to the characteristics of RHD mRNA in NCBI Nucleotide genetic database to find the RhD mRNA reference sequence: JQ065733.1. We designed nine primers (Exon-1–2; Exon-1–3; Exon-1–4; Exon-1–5; Exon-1–6; Exon-1–7; Exon-3–8; Exon-3–10) that amplified by mRNA spliceosome. But the full-length RHD does not include the RHCE when amplifying the RHD gene. The study on simple research for the characteristics of RHD genes that based on China population's RhD antigen individual detection and research the RHD mRNA spliceosome different way. This finding revealed the polymorphism of RHD gene China population.
Aims: To investigate the RhD mRNA spliceosome in Chinese population and understand the characteristics of RhD gene.
Methods: We selected 150 cases of RhD positive blood samples in Chinese randomly and extracted mRNA in whole blood and reverse transcribed into cDNA. The cDNA as template that analysed the RhD mRNA spliceosome polymorphism between individuals.
Results: The 150 cases of RhD positive blood samples that the full-length RhD mRNA spliceosome for 100% (150/150), the RHD gene with deletion of exon-7 accounted for 85.3% (128/150), the deletion of exon-7,8,9 accounted for 96.0% (144/150), the deletion of exon-7,9 accounted for 0.7% (1/150), the deletion of exon-8,9 accounted for 8.7% (13/150), the deletion of exon-9 and the exon-10 accounted for 2.7% (4/150), the deletion of exon-6,7,8,9,10 accounted for 4.0% (6/150). But not found the deletion of exon-8 and the exon-7,8 spliceosomes in Chinese population.
Summary/Conclusions: The polymorphism of RhD mRNA spliceosome among different individuals in Chinese population. This results may provide a basis for the study of human RHD blood group genes and the expression of RhD antigens.

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MOLECULAR MECHANISM OF 960G>A MUTATION OF RHD GENE RESPONSIBLE FOR A WEAK RHD

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Background: RhD with high immunogenicity has important clinical significance. There are three variants of partial D, weak D and Del. Their identifications are not accurate by serological tests. We need to use molecular biology to identification and clarify molecular mechanisms.
Aims: To explore the molecular basis for an individual with a rare weak D phenotype by PK7300 automatic method.
Methods: The RhD blood group phenotype was detected by the method of regular serological methods. The exons of RHD gene were amplified with PCR and

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sequenced. The presence of Rhesus box was tested by PCR to determine the homozygosity of RHD gene.
Results: The RhD blood group was detected as weak D .Rh phenotype was CcDee, The 10 exons of the RHD gene and Rhesus box could be amplified by PCR ,and the genotype of RHD alleles was determined as RHD+/RHD−.The exons of the RHD gene were sequenced。and a 960G>A mutation in exon 7 was detected Compare with RHD * 00101 reference sequence, 1, 2, 3, 4, 5, 6, 8, 9, 10 exons without mutation. Therefore, the RhD blood group was confirmed as weak D 960A by both serological methods and DNA sequencing.
Summary/Conclusions: A weak D 960A has been detected. A 960G>A mutation in RHD gen e is responsible for the low expression of D antigen.

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A CASE STUDY OF RH BLOOD CHIMERISM

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Background: A 55-year-old male donor was found to have mixed field reaction in Rh typing. He had no prior history of blood transfusion or transplantation. The red cells of this healthy blood donor presented a typical mixed-field agglutination patterns with Anti-D, Anti-c and Anti-E. Based on this, further serological and genotyping analysis were applied to investigate the blood chimerism.
Aims: This study aimed to investigate the phenotype and molecular genetic basis of Rh blood chimerism.
Methods: In order to determine the Rh phenotype of two distinct cell populations, anti-E was used to separate mixed RBC populations. DAT testing was performed to confirm the cell population which is suitable for phenotyping and the antigen typing was performed using a standard tube method. The DNA extracted from buffy coat was genotyped using *RHD* zygosity test and PCR-SSP for *RHD*1227G>A* allele. The expression of the D antigen was analyzed by flow cytometry.
Results: After separation of RBC mixture, there were two cell populations typed as D-C+c-E-e+ and D+C+c+E+e+ by serological tests. Flow cytometry analysis revealed two main cell populations and the amount of D antigen expression were 5% D-positive cells (DCcEe) and 95% D-negative cells (dCCee). The result of PCR-SSP for *RHD*1227G>A* allele was positive, corresponding with the dCCee phenotype which was suspected to be D_{el}, and the *RHD* zygosity test was negative which indicated no *RHD* gene deletion.
Summary/Conclusions: Rh chimerism of this case is considered to be naturally occurred since the donor had no hematological diseases and twin sibling. The results support that there were two different Rh genotypes *DEL-Ce* and *D-cE* existing in the donor resulting in the expression of two populations of blood cells (DCcEe and dCCee).

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RHD GENOTYPING ANALYSIS IN THE CHINESE SOUTHERN HAN DONORS WITH D VARIANT PHENOTYPE

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Background: *RHD* genotyping analysis of D variants has systematically conducted in the Caucasian population, but not completely in the Chinese population. So far, less than 60 *RHD* weak or partial alleles have been identified in the Chinese population. *RHD*weak partial 15* and *RHD*DVI.3* are the most common *RHD* variant alleles, accounting for more than 65% of the reported Chinese D variants individuals.
Aims: To investigate the *RHD* genotype in the Chinese Southern Han donors with D variant phenotype.
Methods: Two kinds of anti-D reagents (Clone Rum-1 and TH-28/MS-26) were used to typing D antigen in routine testing. A sample with weak agglutination (≤2 +) compared with D+ controls (4 +) or discrepant serological typing using the two anti-D reagents was regarded as being of a D variant phenotype. The blood samples from 31 Chinese Han donors with the D variant phenotype were collected from the Guangzhou Blood Center. The *RHD* and *RHCE* genotypes were analyzed by the developed Multiplex Ligation-dependent Probe Amplification (MLPA) genotyping assay. Further sequencing of ten exons of the *RHD* gene was conducted in the donor with a *RHD* variant allele but not be detected by the MLPA analysis. The serological characteristic for the epitope of D antigen was further analyzed by using the commercial panel anti-D reagents (D-Screen, Diagast).
Results: A total of 13 kinds of genotypes with an allele described before, *RHD*DVI.3/01N.01* (*RHD* deletion) (n = 13), *RHD*DVI.3/DVI.3* (n = 1), *RHD*DVI.3/01N.03* (n = 1), *RHD*weak partial 15/01N.01* (n = 4), *RHD*960A/01N.01* (n = 3), *RHD*weak D type 23/01N.16* (n = 1), *RHD*weak D type 25/01N.01* (n = 1), *RHD*weak D type 71/01N.01* (n = 1), *RHD*weak D type 95/01N.01* (n = 1), *RHD*weak D type 111/01N.01* (n = 1), *RHD*weak D type 136/01N.01* (n = 1), *RHD*79_81delCTC/RHD*D-CE(2-10)* (n = 1) and *RHD*01EL.01/DFR2* (n = 1), were identified in the 30 donors by the MLPA analysis and *RHD* gene sequencing. Besides, one novel *RHD* variant allele defined by the missense mutation located in the exon 4 of the *RHD* gene (c.497A>G, p. His166Arg) was identified by sequencing in one donor with a partial D phenotype showed by the panel anti-D reagents (D-Screen). *RHD*497G/01N.01* and *Ccee* genotypes were determined by the MLPA analysis.
Summary/Conclusions: Except for the *RHD* variant alleles reported in the Chinese population before, *RHD*weak D type 23* and *RHD*weak D type 111* identified in the Japanese, and *RHD*weak D type 95* and *RHD*weak D type 136* detected in the French were firstly identified in the Chinese Han. The amino acid (p.166Arg) encoded by the novel missense mutation (c.497G) located in the 3rd extracellular loop of RhD protein, which also indicates to resulting in a partial D phenotype.

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ESTABLISHMENT OF A HIGH-RESOLUTION MELTING METHOD FOR RAPID GENOTYPING OF "ASIAN TYPE" DEL WITH THE RHD*1227A ALLELE

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Background: The DEL phenotype is often labeled as D negative phenotype, because it's detectable only by the adsorption-elution test using anti-D regents which is not routinely performed in the clinical laboratories and blood centers in China. Compared with the rare distribution in the Caucasian and Black populations, the DEL phenotype is very common in East Asia population with a frequency of 15%>32% in the apparent D negative individuals. More than 95% of Chinese DEL individuals carry the *RHD*1227A* (*RHD*01EL.01*) allele. An increasing number of studies proved that the epitopes of "Asian type" DEL are intact and could not immunized by D+ blood. Therefore, the individuals of "Asian type" DEL recipients should be treated as D+ individuals. Currently, PCR-SSP method and a real time PCR with a previous step of PCR-SSP and a last step of melting curve analysis has been developed for detection of *RHD*1227A*, but these methods are not suitable for routine genotyping for time-consuming or complicated operation steps needed.
Aims: To establish a high-resolution melting (HRM) method to genotyping "Asian type" DEL phenotype with the *RHD*1227A* allele.
Methods: One pair of *RHD* specific primers was designed to amplify the exon 9 of the *RHD* gene covering the missense mutation (c.1227G>A) with a length of 135 bp for PCR product. Real time PCR and high-resolution melting steps were performed in one closed-tube continuously on the Roche LightCycler® 480 qPCR instrument. Forty-five samples with a known genotype carrying the *RHD*1227A* allele, which obtained by using the *RH*-multiplex ligation-dependent probe amplification (*RH*-MLPA) assay and sequencing of *RHD* exon 9, were tested by the developed HRM method. Wild-type samples without *RHD*1227A* allele were used as the negative control.
Results: Among the 45 samples, three groups samples, including three samples having a homozygous *RHD*1227A*/*1227A* genotype, 1 sample having a *RHD*1227A*/*1227G* genotype and 41 samples having a *RHD*1227A*/*01N.01* (*RHD* deletion) genotype, and the wild-type control samples could be classified into four different groups having a different melting curves shape with each other by HRM. The genotypes for *RHD*1227A* of HRM were consistent with the results of MLPA. In summary, the total time of the genotyping process including the manual DNA extraction, the pipetting steps and the machine running time took less than 2 h.
Summary/Conclusions: HRM is a simple and rapid method for genotyping of "Asian type" DEL that is suitable to be used in the routine testing to prevent the unnecessary RhIg prophylaxis in the "Asian type" DEL pregnant women.

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P-435

A HYBRID RHCE A LLE LE, RHCE(1)-D(2)-CE(3-10), CAUSE AN UNUSUAL RHD– PHENOTYPE

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Background: D– lacks all RHCE antigens and is a very rare phenotype. The molecular background of D– is heterogeneous, including mutation or deletion of the RHCE gene and nonfunctional hybrid RHCE-D-CE alleles.
Aims: We describe a novel molecular background in a D - - proband of Chinese patient, with a potent anti-RH17.
Methods: Blood sample of a female patient was referred to our reference laboratory because of the presence of a antierythrocyte antibody. Rh typing of the patient and her family members were performed by standard hemagglutination techniques. Genomic DNA of the patient and her family members were extracted from peripheral blood cells using a fully automated device. RHD and RHCE alleles were identified by Polymerase chain reaction (PCR) assay and sequencing on the 10 exons of the RHD and RHCE genes.
Results: The patient, group A1, D+, had six deliveries and 4 children died soon after their birth. Serological studies showed an IgG 37°C-reactive alloantibody which reactivated with all cells in the panel. Propositus' red blood cells were positive for the anti-Ds tested and failed to react with anti-C, anti-c, anti-E and anti-e antisera. She was then diagnosed to be D– with a potent anti-RH17 in her serum. Adsorption-elution tests with polyclonal anti-C, anti-E, anti-c and anti-e were negative. The phenotype of Propositus' brother, daughter and niece were DCCee, DccEE and DCcEe respectively. Genomic DNA sequencing of the 10 RHCE exons revealed that a homozygous hybrid RHCE allele, RHCE(1)-D(2)-CE(3-10), was existed in the propositus. The exon 2 of RHCE was replaced by homologous RHD. Thus, this allele could not code normal RhCE protein and therefore resulted in RhD– phenotype. The same hybrid RHCE allele, RHCE(1)-D(2)-CE(3-10), was also found in one DNA chain of Propositus' daughter. RHCE genotyping of Propositus' brother, and niece did not show any mutation. And RHD genotyping of Propositus and all her family mentioned above did not show any mutation.
Summary/Conclusions: We report here a unique nonfunctional hybrid RHCE(1)-D(2)-CE(3-10) alleles was responsible for a silent RHCE gene encountered in D– people. To our knowledge, this is first description of this unusual phenotype in a Chinese patient.

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SEROLOGICAL SCREENING AND GENETIC ANALYSIS OF RHCE VARIANTS IN THE CHINESE DONORS

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Background: An increasing number of variant *RHCE* alleles have been identified, especially in the people of African origin. So far, more than hundreds of missense mutations or hybrid *RHD-CE* genes linked to *RHCE*ce, RHCE*cE, RHCE*Ce* and *RHCE*CE* alleles have been described to resulting in a weaken and/or partial expression of C, c, E, and e antigens, generation of low-prevalence antigens and/or absence of a high-prevalence antigen of Rh system. But for the Chinese population, the RhCE variants has not investigated systematically.
Aims: To screening the RhCE variants and conducting the molecular genetic analysis in the Chinese Southern Han donors.
Methods: Preliminarily, a monoclonal anti-E (MS80/MS258 diluted by 0.9% NaCl with 0.5% papain) was used to type the E antigen in the donors to obtain the E(+) samples with an automated analyzer (Microlab® STAR, Hamilton). Then, RhCE antigens were further typed by using two sets of monoclonal antibodies (anti-C, clone MS24 and P3 × 25513G8; anti-c, clone MS33 and 951; anti-e, clone MS16/MS21/MS63 and P3GD512/MS63; anti-E, clone 906) diluted by 0.9% NaCl with 0.5% papain with the automated analyzer. When a weaken expression or discrepant typing result of RhCE antigens obtained by two sets of antibodies was found in some samples, the RhCE antigens were tested again by using the standard serological method and further sequencing analysis of the ten exons of the *RHCE* gene was also performed.
Results: A total of 4,514 E(+) samples from the donors were obtained and five kinds of RhCE phenotype, CcEe (3592/4514, 78.2%), ccEE (542/4514, 12.0%), ccEe (357/4514, 7.9%), CCEe (43/4514, 0.95%) and CcEE (43/4514, 0.95%), were identified among them. The majority of E(+) samples has a CcEe phenotype, which is easier to find RhCE variants for the presence of two different *RHCE* haplotypes. Among the 4514 E(+) samples, two samples having a weaken expression of C antigen and CcEe phenotype were found but a normal *RHCE* sequencing result was obtained. While two samples having a weaken expression of e antigen and CcEe phenotype were identified and two variant *RHCE* alleles (*RHCE*Ce 667T* and *RHCE*Ce375G*) described before were found, respectively, by sequencing of *RHCE*.
Summary/Conclusions: The variant *RHCE*Ce 667T* and *RHCE*Ce375G* alleles were identified for the first time in the Chinese population. The further detailed phenotyping using more monoclonal anti-C and anti-e will be performed in these samples. For the two donors with a weaken C expression, The possible *DCE* haplotype might exist and account for the variant expression of C antigen.

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INHIBITION OF SUICIDAL ERYTHROCYTE DEATH BY INDIRUBIN -3′-MONOXIME

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Background: Qing Dai is a prized traditional Chinese medicine, whose major component, indirubin, and its derivative, indirubin-3′-monoxime (IDM), have inhibitory effects on the growth of many human tumor cells and especially pronounced antileukemic activities. However, the cytotoxicity influence of IDM in mature human erythrocytes is unclear.
Aims: This study aimed to evaluate the potential impact of indirubin-3′-monoxime, a potent inhibitor of cyclin-dependent kinases (CDKs), on erythrocytes and its underlying mechanisms.
Methods: Utilizing flow cytometry and confocal laser scanning microscopy, phosphatidylserine exposure at the cell surface was estimated by annexin V-FITC. The cell volume was expressed by the forward scatter of the flow cytometer. Fluo-3 fluorescence were used to bewrite changes of the cytosolic Ca^{2+} activity, reactive oxygen species (ROS) formation was assessed by DCFH-DA fluorescence, and ceramide profusion was evaluated by the conjugated-FITC specific antibodies.
Results: The 24-h exposure of human erythrocytes to IDM (12 μM) significantly decreased the percentage of annexin V-binding erythrocytes and the intracellular calcium concentration ($[Ca^{2+}]_i$). IDM (3–12 μM) did not significantly modify ceramide profusion and DCFH-DA fluorescence. Energy depletion significantly increased annexin V-binding and Fluo-3 fluorescence and diminished forward scatter, whereas effects significantly blunted by IDM (12 μM). Moreover, Ca^{2+} ionophore ionomycin similarly triggered eryptosis that was also significantly suppressed by IDM.
Summary/Conclusions: Indirubin-3′-monoxime is a novel inhibitor of suicidal erythrocyte death following treatment with Ca^{2+} ionophore ionomycin, energy depletion, accompanied by decreased $[Ca^{2+}]_i$. Thus, IDM may counteract anemia and impairment of microcirculation, at least in part, by inhibition of Ca^{2+} entry.

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PEPTIDE MIMOTOPES OF RHD ANTIGEN SELECTED VIA PHAGE DISPLAY PEPTIDES LIBRARY

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Background: Rh is the most complex and polymorphic of the blood group systems. At present, the reagents have many disadvantages such as low specificity and short shelf life when identify the irregular antibodies of the patients with AIHA and treat the patients with anti-D (such as RhD-HDFN).
Phage-Display technology is a powerful technique for the identification of peptides, proteins, or antibodies with affinity for a specific target. In several rounds of affinity selection or ''biopanning'', the phage library is gradually enriched for high affinity binders. The peptides enriched are identified by DNA sequencing. Depending on the sequence, the analysis of clones will result in the discovery of the most promising candidates.
In this study, bioinformatics analysis software and Internet analysis platform were used to analyze and predict the antigenic epitope of RhD protein, which provided the theoretical basis for the study of RhD structure and function. Highly specific

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phage clones which can specifically binding to this IgG monoclonal anti-D and its DNA sequences were obtained by Phage-Display technology.
Aims: This study was purposed to predict the antigenic epitopes of RhD protein and investigate the factors of screening peptide mimotopes of blood group D carbohydrate antigen via phage display peptide library, then to evaluate the best program for screening, in order to obtain the mimic antigen epitopes of RhD blood type antigens.
Methods: 1. To predict the antigenic epitopes of the RhD protein by bioinformatics technology. 2. By comparing the phage titers of eluent in each experiment with different elution time (4, 8, 12, 16 and 30 min). 3. By comparing the input/output ratios and P/N values in each round panning with different concentration of Tween20 in washing buffer and different blocking buffer types. 4. By analyzing the effect of screening to obtain the best program for screening. Then by using ELISA to analyze the binding forces of IgG monoclonal anti-D and eluted solution in each round panning. 5. Do the DNA sequencing to obtain the mimic epitopes of RhD blood type antigens, then do the antibody competition inhibition assay and agglutination inhibition test.
Results: 1. The predicted antigenic epitopes were found to locate at the protein sequence of amino acids 36-41, 100-104, 353-356 for RhD. 2. After comparing the influencing factors of screening, the best program for screening is "the elution time is 8 min, the washing buffer of 0.1%, 0.5%, 0.5% Tween20 and the blocking buffer of 0.5% BSA, 1% Gelatin, 5% skimmed milk in each biopanning round". 3. By using the best program, the binding forces of IgG monoclonal anti-D and eluted solution in each round panning are enhancing. 4. One major mimic peptides ("VHWDFRQWWQPS") with about 50% inhibition ratio were obtained. The phages with this peptide sequence are able to inhibit the agglutination of IgG monoclonal anti-D and D+ cells in a concentration dependent manner.
Summary/Conclusions: 1. The findings of the prediction would be helpful to predict the antigenic epitopes localized within human RhD. 2. The screening efficiency was affected by elution time, the concentration of washing buffer and the kind of blocking buffer. The screening procedure is optimized in this study. 3. The mimic peptides we obtained is able to inhibit the reaction of RhD blood type antigen and IgG monoclonal anti-D.

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Abstract has been withdrawn

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STUDY ON RHD GENE POLYMORPHISM AMONG RHD-NEGATIVE IN NINGXIA HUI POPULATION

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Background: The Rh blood group system is one of the most complex systems in the red blood groups. Its importance is second only to the ABO blood group system, which can cause severe hemolytic transfusion adverse reactions and hemolytic disease of the newborn. The RhD antigen is encoded by the RhD gene, which has obvious race and regional difference. The Chinese people, represented by ethnic groups in Mongolia, are made up of 56 nationalities. There are some differences in the structure of RhD gene in all ethnic groups. So, it is important to understand the structure and distribution of RhD gene in Rh negative individuals of Hui nationality in this region.
Aims: In order to know the RhD gene polymorphism among RhD-negative in Ning-Xia Hui Population.
Methods: Using saline and coombs method to Screen RhD- negative samples. To determining Del type by absorption elution method. At the same time, 10 exons of *RhD* gene were analyzed by PCR-SSP method.
Results: 91 cases (31.8%) were DeL type in the 286 RhD negative samples. There were 67 cases (73.6%) with Completely missing *RhD* gene and 24 cases (26.4%) with Partial deletion of *RhD* gene in DeL type. Among other 195 cases of RhD negative samples, there were 165 cases (84.6%) Completely missing *RhD* gene and 30 cases (15.4%) partial deletion of *RhD* gene. The *RhD* gene was detected in 50 Rh-positive samples of the control group. The gene frequencies of the DEL and RhD(-) loci were in Hardy-Weinberg equilibrium ($P > 0.05$) in Hui Population.
Summary/Conclusions: There were about one third of the DeL type in the RhD negative population of Ningxia Hui. It was complex polymorphisms in the RhD gene exon distribution. The genetic mechanism and RhD gene expression in RhD negative population of Han nationality need further study.

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Abstract has been withdrawn

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GENOTYPING ANALYSIS OF MNS BLOOD GROUP GP(B-A-B) HYBRID GLYCOPHORINS IN THE CHINESE SOUTHERN HAN USING A HIGH RESOLUTION MELTING ASSAY

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Background: MNS hybrid glycophorins GP(B-A-B) arise from gene conversions between the homologous genes of *GYPA* and *GYPB*. Some of them are common and clinical significant for Southeast Asians. But it is very hard to detect these hybrid glycophorins by serological testing because of lacking of commercial reagents.
Aims: In this study, we aim to develop a genotyping method for GP(B-A-B) hybrid glycophorins based on high resolution melting (HRM) analysis and apply it to genotype Chinese Southern Han population.
Methods: *GYPB* specific primers encompassed pseudoexon 3 were designed to develop a HRM method for *GYP(B-A-B)* genotyping. The DNA samples from 1000 Chinese Southern Han donors were collected. *GYP(B-A-B)* genotype was analyzed by HRM method. The multiplex ligation-dependent probe amplification (MLPA) assay was used to verify the genotyping results obtained by HRM in part of samples. Direct sequencing was conducted in the samples with a variant melting curve.
Results: Totally, 4 genotypes of *GYP(B-A-B)* hybrid gene were identified among 1000 samples including *GYP*Mur* heterozygote (n = 61), *GYP*Mur* homozygote (n = 2), *GYP*Bun* heterozygote (n = 1), and a novel *GYP(B-A-B)* hybrid allele (n = 1). The MLPA results showed high concordant with HRM results. In addition, five nucleotide polymorphisms (SNPs) including one reported SNPs and 4 novel SNPs were identified in 10 samples with a variant melting curve detected by HRM.
Summary/Conclusions: HRM analysis developed can successfully distinguish the different types of *GYP(B-A-B)* hybrid genes. It is a simple, rapid and sensitive method suitable for large scale screening of variations. The polymorphisms on the *GYPB* gene identified should be taken into consideration during the production of genotyping kits for Chinese population.

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GENOMIC CHARACTERIZATION OF THE JK GENE: DIFFERENT MOLECULAR OF THE JK (A+WB-) PHENOTYPE IN CHINESE POPULATION

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Background: The clinically important Kidd blood group locus encodes a urea transporter which is expressed on red cells and in the kidney. The Kidd gene is located on chromosome 18q12. The JK glycoprotein of 389 amino acid is encoded by exons 4 through 11 with the translation start point in exon4. JK^a and JK^b polymorphism was caused by a transition (G838A), resulting in an Asp280Asn amino acid substitution. We have reported that the molecular basis of the Kidd blood group polymorphism in Chinese population. A polymorphic nucleotide was noted at position T-68C in Intron7, beside exon9 G838A (280Asp/Asn). The G588A in exon 7, as well as the

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position -46 from the 3′ end of intron 9 were not found Jka-related polymorphisms in Chinese population.
Aims: To study the weakly Jk (a+^{W}b-) blood group phenotype associated with 130A at Exon 4 of the *JK* gene in Chinese population.
Methods: A lysis test was performed on 3800 samples by 2M urea. JK phenotypes of 100 controls were detected by standard serology and the urea hemolysis test. Genomic DNA was isolated from white blood cells with a DNA extraction Kit. The sequences covering 4–11 exons and partial introns of *JK* gene was amplified with genomic DNA by polymerase chain reaction, and fragments were directly sequenced.
Results: 3 samples Jk (a+^{W}b-) from 3800 Chinese blood donors whose RBC reacted weakly with anti-Jk^{a} and the urea hemolysis test was negative. Among the three Jk (a+^{W}b-) samples, two of which carried nt130G/G in exon 4 and *JKB/JKB*, the other one carried nt130G/A and *JKA/JKB*. The Kidd phenotypes of 100 controls were 24 Jk (a+b-), 43 Jk (a+b+) and 33 Jk (a-b+). Among 24 Jk(a+b-) samples 7 individuals carried 6 heterozygous nt130G/A, 17 homozygous nt130A/A and 1 homozygous nt130G/G. 7 of 43 Jk(a+b+) samples carried homozygous nt130G/G, the other 36 individuals were heterozygous nt130G/A. 33 Jk(a-b+) samples carried nt130G/G. Homozygous for intron8 84T/T was found in all 103 samples. In addition, a new mutation was found, the G was substituted for A at position -235 from the 3′ end of intron 4.
Summary/Conclusions: 130A in Exon 4 of the *JK* gene were not associated with Jk (a+^{w}b-) expression in this study. The allele occurrence with 130A is 37.5% in Chinese population. A novel allele with intron 4 -235A is 6.1%, Homozygous for intron8 84T/T in *JK* gene was common allele in Chinese population.

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LEWIS Y ANTIGEN REGULATES HEPATOCELLULAR CARCINOMA CELL MIGRATION VIA MODULATION OF STRESS FIBER FORMATION

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Background: Histo-blood group antigen Lewis Y (LeY) is an isomer of blood group antigen Lewis B, and their synthesis pathways are closely related to each other and share various fucosyl transferases (FUTs). Anchoring on various glycoproteins and glycolipids, LeY plays an important role in cell recognition, adhesion as well as migration. LeY is also demonstrated to be significantly overexpressed or re-expressed in several tumor tissues and cell lines and correlated with the clinical metastasis of tumors.
Aims: We analyze the effect of LeY on cell motility as well as the structure of cytoskeleton in hepatocellular carcinoma cells (HCC), to shed light on the molecular mechanism LeY utilizes to regulate cell migration, which can also be referenced in the studies of the clinical roles of other carbohydrate blood antigens. By confirmation of the role of LeY on cell migration, new transfusion strategies may be adapted with regard to possible roles of blood group antigens in tumor metastasis to improve transfusion safety.
Methods: The migration ability and expression of LeY of two HCC cell lines SMMC-7721 and HepG2 is assayed and compared. SMMC-7721 cells transfected with siRNA against FUT1 was analyzed for LeY expression and cell migration. LeY antibody blockage experiment is also conducted to verify the role of LeY in cell motility. Immunostaining analysis is carried out on SMMC-7721 cells with FUT1 knockdown for cytoskeleton change using Phalloidin which binds to F-actin. The results are visualized using Confocal Laser Scanning Microscopy.
Results: As shown by Western Blot results, LeY relative expression level in SMMC-7721 is significantly higher than in HepG2, which is positively correlated with their migration capacity detected by Transwell Assay. LeY blockage using anti-LeY antibody significantly inhibits cell motility as demonstrated by Transwell and Wound Scratch Assay. Furthermore, FUT1 knockdown in SMMC-7721 leads to significant decrease in LeY synthesis and cell migration, while stress fiber amount is markedly reduced compared to cells treated by negative control siRNA.
Summary/Conclusions: LeY antigen exerts positive effect on HCC cell migration, probably via regulating the formation of stress fiber structure. The underlying mechanism of signal transduction may involve the glycosylation of proteins carrying the LeY antigen. This provides us with novel ideas in studying other ABH blood group antigens. During transfusion practice, it may lower the risk of transfusion to take account of the significance of blood group antigens as tumor metastasis promoters.

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AMP DEAMINASE-3 VARIANTS IN AN AUSTRALIAN DONOR COHORT

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Background: The mechanism behind red blood cell (RBC) aging is largely unknown however, an enzyme that catalyzes purine nucleotide interconversion, AMP deaminase-3 (AMPD-3), has been reported to play a role in RBC longevity. Hortle and colleagues demonstrated that a gain of function mutation increased AMPD-3 enzyme, reduced RBC ATP, increased RBC GTP, decreased RBC lifespan and increased erythropoiesis in mice (Hortle et, al., Blood, 2016). We reviewed previously published mutations (R573C, Q433X, 600delT and 4311L), that were shown to decrease AMPD-3 enzyme activity. We hypothesize that this decrease in enzyme activity may impact on RBC lifespan.
Aims: To investigate the prevalence of AMPD-3 mutations in an Australian blood donor cohort.
Methods: Four AMPD-3 SNPs were chosen for investigation based on frequency and mutations that caused deletions/ frame shifts. Australian blood donors (n = 481) were genotyped using single nucleotide polymorphism (SNP) Taqman genotyping RT-PCR. R573C and Q433X were commercially available and 600delT and 4311L were custom designed (Invitrogen, Carlsbad, United States). Briefly, SNP assays containing fluorescent probes (VIC and FAM) were analysed performed according to manufacturer's instructions with positive (plasmid) and negative controls. All reactions were performed in a StepOnePlus™ instrument and analysed using StepOne Software v2.3 (Applied Biosystems, Foster City, United States).
Results: The prevalence of four known AMPD-3 SNPs were investigated in a cohort of Australian blood donors. Of the 4 SNPs investigated, R573C was reported to be the most prevalent (0.019%) in a Japanese cohort however we did not find this mutation in this Australian cohort. Both 600delT and Q433X were reported at a prevalence of 0.0007% in previous studies however these were not found. The V311L genotype was reported in a Polish cohort with a prevalence of 0.013%. In the Australian cohort studied we found nine individuals heterozygous for the V311L mutation indicating a prevalence of 0.019%.
Summary/Conclusions: Of the four AMPD-3 SNPs reported, only V311L was detected in an Australian blood donor cohort. Australia has a diverse multi-ethnic population, however, only the V311L mutation was found in Australian donors but not the "Japanese" R573C mutation. The V311L mutation is associated with decreased AMPD-3 enzyme function and therefore, further investigation of these nine blood donors into RBC AMPD-3 enzyme activity and impacts on nucleotide pool and RBC lifespan are warranted.

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THE KIDD BLOOD GROUP IS DISTRIBUTED AMONG THE BLOOD DONORS IN SOUTHERN ZHEJIANG

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Background: The Kidd blood group system is one of the most important blood group systems in blood transfusion. There are 2 common antigens on red blood cell: Jka antigen and Jkb antigen. Immunogenicity is weak, generally for foreign antigen immune ability of strong talent, anti -Jka and anti -Jkb. Anti -Jka and anti -Jkb can cause hemolytic disease of the newborn (HDN) and acute or delayed hemolytic transfusion reactions, and their antigen distribution shows polymorphism in all populations. At present, Jk (a-b-) rare blood type resource bank has not been established in the south of Zhejiang province.
Aims: The distribution pattern of Kidd blood group among donors in southern Zhejiang Province(Wenzhou, Taizhou and Lishui) was analyzed to provide a theoretical basis for establishing the rare blood bank of blood donors.
Methods: Of the 36235 donors, 5 Jk (a-b-) phenotypes were screened, with a frequency of 0.014%. In 672 blood donors, the distribution of the Kidd blood group phenotype (frequency) of Jk (a+b+) (0.4777) >Jk (a-b+) (0.3184) >Jk (a+b-) (0.2039), no Jk (a-b-), the gene frequencies of Jka and Jkb were 0.4428 and 0.5572, by 2 test, Kidd type of observation value and the expected value had no significant difference ($P > 0.05$), Hardy-Weinberg in good agreement.
Results: Of the 36235 donors, 5 Jk (a-b-) phenotypes were screened, with a frequency of 0.014%. In 672 blood donors, the distribution of the Kidd blood group phenotype (frequency) of Jk (a+b+) (0.4777) >Jk (a-b+) (0.3184) >Jk (a+b-) (0.2039), no Jk (a-b-), the gene frequencies of Jka and Jkb were 0.4428 and 0.5572, by 2 test,

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Kidd type of observation value and the expected value had no significant difference ($P > 0.05$), Hardy-Weinberg in good agreement.
Summary/Conclusions: The Kidd population distribution is polymorphic and conforms to the Hardy-Weinberg population genetic balance law. Jk (a-b-) phenotype is found in blood donors, especially for the establishment of Jk (a-b-) rare blood type resource pool for clinical use.

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MOLECULAR GENOTYPING OF RHCE, JK AND GYP.MUR BY POLYMERASE CHAIN REACTION – SEQUENCE SPECIFIC PRIMER

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Background: Accurate serological blood group phenotyping is limited in patients with recent red cell transfusion, administration of anti-CD38 or autoimmune haemolytic anemia. Typing of rare antigens is also hampered by the lack of commercial anti-sera. Molecular genotyping is thus strongly desired for those major blood groups with heterogeneous antigens among the local Chinese population, i.e., Rh(CE) and Kidd antigens, and also the relatively common GP.Mur phenotype which lacks commercial typing anti-sera. Commercial genotyping kits are not ideal for application in hospital blood banks as they are either too extensive or too scattered kits for individual blood groups.
Aims: This project was aimed to develop a fast and accurate Polymerase Chain Reaction-Sequence Specific Primer (PCR-SSP) protocol for one-step *RHCE, JK and GYP.Mur* genotyping in the local Chinese population.
Methods: Eleven sets of allele specific primers were designed for sequence-specific polymerase chain reaction (PCR-SSP) under same PCR condition in 8 tubes. Eighty samples which covered the most common Rh(CE) and Kidd phenotypes were tested. Seven transfused patients and 1 Jk(a-b-) patient were also included for the study. Nine Mi(a+) and 11 Mi(a-) samples were recruited for the *GYP.Mur* evaluation. As there lacks licensed anti-Mi^a, presence of *GYP.Mur* allele was confirmed by sanger sequencing. Primers were designed for detection of the 109 base pair (bp) insertion at intron 2 of *RHCE*C*, 203A for *RHCE*c*, 676C for *RHCE*E* and 676G for *RHCE*e* alleles in 4 tubes. It should be noted that *RHCE*C* typing by the 109 bp insertion at intron 2 is unsuitable for the Blacks as the false-negative rate can be up to 43.8%. Sets of combined primers, i.e., 838G for *JK*A* and *JK(C222)null*, 838A for *JK*B* and *JK(T871C)null, JK(IVS5-1 g>a)null* and *JK(G896A)null*, were run in 3 tubes. The forward primer for *GYP.Mur* binds to C at the 24th bp of pseudo exon 3 and the reverse primer binds to G at the 64th bp of the same exon. While the former anneals to *GYPB*, *GYPE*, *GYP.Mur*, and *GYP.Bun* but not *GYPA* and *GYP.HF*, the latter anneals to *GP.Mur*, *GYPA*, and *GYP.HF*, but not *GYPB*, *GYPE*, and *GYP.Bun*. Such combination of primers contributes a unique specificity for *GYP.Mur*. The amplifications were run with AmpliTaq Gold activation at 95°C for 5 min, 32 cycles for denaturation at 94°C for 30 s, annealing at 61°C for 60 s and extension at 72°C for 45 s, with a final extension at 72°C for 5 min. The targeted PCR products range from 88 to 263 bp.
Results: The *RHCE* genotyping showed no discrepancy with 80 samples of common Rh phenotypes which included R_1R_2, R_1R_1, R_2R_2, R_1r, R_2r, Ror, rr, r'r, r'r' and r"r. The *JK* genotyping agreed with 79 samples of common Jk(a+b-), Jk(a-b+) and Jk (a+b+) phenotypes. The *JK(IVS5-1 g>a)null* and *JK*B(838A)* were positive for a Jk (a-b-) sample. The *GYP.Mur* genotyping results agreed with 9 Mi(a+) and 11 Mi(a-) samples screened with in-house anti-Mi^a. Presence of *GYP.Mur* allele was further confirmed with sanger sequencing. The *RHCE* and *JK* genotyping results of the post-transfusion samples were concordant with the pre-transfusion phenotypes of 7 patients who received 2-17 units of red cells within 2 weeks.
Summary/Conclusions: The PCR-SSP for *RHCE*, *JK* and *GYP.Mur* demonstrated a reliable and fast supplement to serological typing methods in local hospital blood banks. It is also validated to be applicable for post-transfusion samples. Precautions should be observed if it is applied to non-Chinese patients.

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PARTICIPATION IN THE INTERNATIONAL EXTERNAL QUALITY ASSESSMENT IN MOLECULAR IMMUNOHEMATOLOGY FOR A CHINESE LAB: IMPLICATION FOR CLINICAL APPLICATION IN CHINA

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Background: Genotyping of blood groups can help solve many difficult serologically typing of red blood cell antigens. Introduced firstly since 1990, the molecular immunohematology method was rapidly being engaged in routine work in many developed countries and laboratory proficiency tests were established by external quality assessments (EQAs) since 2006. For the first time, a Chinese laboratory participated in the EQA in molecular immunohematology organized by INSTAND, a non-profit provider of proficiency tests, in October 2015.
Aims: The aim of this study is to improve the ability of genotyping in molecular immunohematology experiments and to facilitate clinical application of molecular methods in routine work in China.
Methods: PCR-Sequence Specific Primer (PCR-SSP) method was applied in testing 4 DNA samples distributed by INSTAND in Oct. 2015. The commercial kits purchased by our lab were able to detect the genotypes of RBC antigens from ABO, RHD, RHCE, Kell, FY, JK and MNSs systems. The genotyping results of the 4 samples were submitted by email to INSTAND within 2 weeks and then analyzed by specialists. The testing alleles and the criteria for passing the proficiency were supplied with samples by organizer.
Results: The genotyping results of the four DNA samples were as follows: no. 1, B/O1, KEL2/KEL2, JKA/JKB, FYA/FYA, MNs, Ccee, RHD Gen/gene (normal); no. 2, O1/O1, KEL2/KEL2, JKA/JKB, FYA/FYB, MNs, Ccee, RHD Gen/gene (normal); no. 3, O1/O1, KEL2/KEL2, JKA/JKB, FYA/FYA, NNs, CCee, RHD Gen/gene (normal); no. 4, O1/O1, KEL2/KEL2, JKA/JKB, FYB/FYB, MNs, CcEe, RHD Gen/gene (normal). The certificate was given to us demonstrating that we had fulfilled the requirements for the molecular diagnostic of gene from ABO, RHCE, Kell, FY, JK and MNSs systems. For RHD gene, we only fulfilled normal typing of RHD instead of further RHD zygosity, weak D, partial D analysis or typing, therefore probably failed to pass the proficiency.
Summary/Conclusions: External quality assessment can be helpful for transfusion medicine laboratories to do well in molecular immunohematology. Participating the EQA test can maintain reliable level for Chinese labs when introducing the genotyping technology, and further allow more clinical application in China.

P-450

SCREENING OF RARE BLOOD TYPES FROM ETHNIC MINORITY BLOOD DONORS IN XINJIANG, CHINA

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Background: Transfusion support for patients with rare blood types that lack high-frequency antigens and who have made alloantibodies against it is cumbersome, because compatible blood unit is always difficult to obtain within short term. It can be a special challenge for blood centers, particularly if a delay in transfusion is likely to be life-threatening. Hence, possibly donor of compatible unit should best be identified in advance.
Aims: This study aims to screen rare blood types Fy(a-),s-,Di(b-),Co(a-),Yt(a-),k- by molecular methods, and to investigate the distribution of Duffy, MNS, Diego, Colton, Yt, and Kell rare blood group antigens in ethnic minority blood donors from Xinjiang, China.
Methods: Four multiplex polymerase chain reactions (PCR) with DNA pooling was applied to screen for the rare *Fy^b* and *S*,*Di^a* and *Co^b*, *Yt^b*, *K* alleles in 1020 unrelated healthy donors of Xinjiang Minorities. The samples in the positive pools were further tested individually. Then the positive samples were tested via six sequence-specific primers polymerase chain reaction (PCR-SSP) assays for alleles *Fy^a*, *s*, *Di^b*, *Co^a*, *Yt^a*, *and k*.
Results: Among 1020 donor samples, 89 Fy(a-), 46 s-, 1 Di(b-),4 Yt(a-) donors were detected. The frequencies of *Fy^b*, *S*, *Di^a*, *Co^b*, *Yt^b*, and *K* alleles were 29.66%,21.32%, 2.30%、0.59%, 4.80%, and 1.18% respectively.
Summary/Conclusions: The rare blood types of Xinjiang Minorities obtained through screening can provide valuable information for compatible blood transfusion, and further enrich the National Rare Blood Bank of China.

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RARE AND NOVEL ALLELES IDENTIFIED IN AN INDIGENOUS AUSTRALIAN POPULATION: APPLICATION OF GENOTYPING TO IMPROVE TRANSFUSION SAFETY

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Background: Extended blood group sequence data for Indigenous Australians has never been reported. Recently published genomic datasets make it possible to review variants in blood group genes and provide variant analysis. Identification of novel variants and variants found only rarely in other populations provides an evidence base for improved transfusion management.
Aims: To analyse blood group variants from whole exome sequence data generated by Tang *et al.* (2016; https://doi.org/10.1038/sdata.2016.23) for 72 Western Desert Indigenous Australians of Martu ancestry.
Methods: Variants in genes for 36 blood group systems, and the transcription factors KLF1 and GATA1, were extracted from the dataset (.vcf), annotated and compared to those in public databases. Approval for data access and analysis was obtained from the *Telethon Kids Institute* (http://bioinformatics.childhealthresearch.org.au/AGHS). The possible impact of rare and novel missense variants were further investigated *in-silico* using the meta predictor, Meta-SNP.
Results: For the MNS system, a putative novel *GYPA* allele predicting 20Leu and 24Gly was identified in 5% (4/72) of individuals: a combination of residues associated with N (20Leu) and M (24Gly) antigens.
In the RH system, the 16Cys *RHCE* variant was identified at a homozygous level in 21% (15/72) of individuals with the E haplotype: in 3 individuals with EE and in 12 with eE. This frequency is similar to a historic report for the lack of antigenic component E^T in 30% of Western Desert individuals with the E haplotype. Also within the RH system, two novel *RHD* substitutions (predicting p.Ser67Leu and p.Ala202Thr) at the same sites associated with known weak D alleles, were identified separately and at a heterozygous level in 2 individuals.
In the DI (Diego) system a novel variant (p.Gly647Ser) that is predicted to have a functional effect, occurred at the same site associated with the Sw^a antigen and was observed at a heterozygous level in 5 individuals.
In the RHAG system, the RHAG:4 associated variant was found in 27 samples.
Predicted blood group phenotype profiles were similar to historic serology profiles. In addition we noted that for the DO (Dombrock) system, 67 individuals carried the Do^a variant (in Caucasians and Africans Do^b is higher frequency).
Summary/Conclusions: Blood group variants associated with known alleles, as well as rare and novel variants that have the potential to be immunogenic, were defined. The novel *GYPA* allele is defined by positions that carry the antigenic determinants for the immunogenic MN blood groups. The association between 16Cys/E haplotype with the historic E^T antigen requires investigation. The risk of developing anti-E^T in Indigenous Australians and associated transfusion reactions remains to be defined. Antibodies against RHAG:4 have been reported to cause severe HDFN. The variants analysed were for Indigenous Australians from Western Australia, the geographical distribution of these variants and more general applicability to transfusion practice remains to be defined.

P-452

APPLYING MOLECULAR IMMUNOHEMATOLOGY FOR DONORS AND PATIENTS IN NATIONAL BLOOD CENTER, MALAYSIA

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Background: The molecular basis of many blood group antigens is known, and it provides a means for predicting the red blood cell phenotype. Molecular typing is useful when serologic typing cannot be ruled out. We perform molecular typing for donors or patients samples which enable the examination of weak, unexpected, or unclear serologic findings.
Aims: To document our experience and observations when molecular techniques of RBC genotyping were first applied to a small scale for determining donors and patients RBC genotype. The observations can be helpful for improving the molecular technique and skills for future used.
Methods: A total of 91 samples with unresolved by serologic findings from 2013 to 2016 were included in this study. Red blood cell genotyping was performed with commercially available kits. Ready to use, BAGene polymerase chain reaction with sequence-specific priming (PCR-SSP) kits were used in this study.
Results: We analyzed 91 samples (71 patients and 20 donors) who unresolved results by serologic findings. All samples were performed with molecular testing in order to confirm either subgroup, Rh D status or red cell genotyping. From the findings, it was found that 22 (24%) out of 91 samples showed discordance results between serological and molecular testing. The discrepancies were found in ABO blood group (12), Rh blood group (4) and red cell genotypes (6) cases.
Summary/Conclusions: Molecular typing is a helpful supplementary techniques for resolving most of the common problems caused by discrepant or doubtful serologic results.

P-453

VALIDATION OF RED CELL GENOTYPING KIT RBC-FLUOGENE VERYFY (INNO-TRAIN)

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Background: Red cell antigen typing is an essential step in antibody identification and subsequent transfusion of antigen-negative blood.
Aims: Conventional serological typing methods are simple and cost-effective, but have limitations in patients that are Direct Antiglobulin Test (DAT) positive or recently transfused.
Methods: RBC-FluoGene vERYfy detection system is based on PCR SSP. This kit allows typing of Rh(D), Rh(C), Rh(c), Rh(E), Rh(e), $Rh(C^w)$, Kell, Kidd, Duffy, MNS, and Dombrock. 11 genotyping results were collated and analysed against known results using Chi Square Test, achieving P-value < 0.05 will be accepted as having been validated. 3 samples were sensitized with IgG (anti-D) to simulate strong positive DAT and compared with initial genotyping results. Genotyping kit is also evaluated using 4 recently transfused cases, with pre- and post-transfusion samples.
Results: Rh(C), Rh(c), Rh(E), Rh(e), Kidd, Duffy and MNS achieved the acceptable level of correlation. Rh(D) typing with one discrepant result, did not achieve the accepted level of correlation. $Rh(C^w)$, Kell and Dombrock typing required a wider sample size to achieve P -value < 0.05. Results for both DAT positive and recently transfused cases showed complete concordance.
Summary/Conclusions: RBC-FluoGene vERYfy genotyping kit can be implemented for laboratory use of Rh(C), Rh(c), Rh(E), Rh(e), Kidd, Duffy and MNS typing. Genotyping results of Rh(D), $Rh(C^w)$, Kell and Dombrock required further testing before the required P -value could be achieved.

Platelet Immunology

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PREDICTION OF LINEAR AND CONFORMATIONAL B CELL EPITOPES OF CD36 PROTEIN

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Background: CD36 (or Platelet glycoprotein χ, SCARB3) is a transmembrane glycoprotein, a multi-ligand receptor, possesses various biological functions, influences various diseases including angiogenesis, thrombosis, atherosclerosis, malaria, diabetes, steatosis, dementia and obesity. CD36 is an isoantigen that can immunize CD36-deficient individuals to produce anti-CD36 antibody that may mediate isoimmune thrombocytopenia. The use of neutralizing antibodies to block CD36 antigen causes inhibition of metastasis in human oral cancer. We studied the linear and conformational B cell epitopes of CD36 protein that may help to understand the pathogenicity and physiological mechanisms.
Aims: To predict the linear and conformational B cell epitopes of CD36 protein.
Methods: Based on the sequence of CD36 protein, transmembrane helices was predicted by the TMHMM and Tmpred web servers. CD36 B cell linear epitopes were first predicted by using ABCpred, COBEpro, BcePred, BepiPred web servers, and then located tertiary antigen in the 3D structure of CD36 that constructed by homology modeling of Swiss-Model Workspace. The conformational epitopes of CD36 were predicted by DiscoTope, CBTOPE, ElliPro and InterProSurf web servers.

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Results: The N-terminal 54–61,121 ~ 133,191 ~ 198,298 ~ 311,315 ~ 339,333 ~ 339 and 404 ~ 410 of CD36 protein were predicted to be the B-cell CD36 linear epitope region. The region of 35~49, 71~85, 101~109, 148~159, 187~197, 211~223, 241~248, 301~320, 356~366, 397~406, 425~432 of CD36 protein were predicted as the CD36 conformational antigens Epitope area.
Summary/Conclusions: These findings would be helpful for the identification of the location of the CD36 antigenic epitopes in CD36 protein.

P-455

FREQUENCY AND MOLECULAR BASIS OF CD36 DEFICIENCY IN SHE POPULATION OF CHINA

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Background: CD36 is a multifunctional membrane receptor and is expressed in several cell lines. Individuals who lack platelet (PLT) CD36 are at risk for immunization against this antigen, leading to several clinical syndromes. She is a minority in southern China.
Aims: This study aimed to investigate the frequency and molecular basis of CD36 deficiency in She population of China.
Methods: Whole blood samples were collected from She population by random, and the PLTs and monocytes were analyzed using flow cytometry to determine CD36 deficiency type. After genomic DNA was extracted, Exons 3 to 14 of CD36 gene including a part of relevant flanking introns were amplified. Polymerase chain reaction-sequence based typing were performed.
Results: 160 She samples were analyzed, 3 individuals failed to express CD36 on PLTs; no one expressed no CD36 on their monocytes. These results demonstrated that the frequencies of Type I (lacking CD36 expression on PLTs and monocytes) and Type II (lacking CD36 expression on PLTs only) CD36 deficiency among the study population were 0 and 1.9%, respectively. Nucleotide sequencing analysis revealed two different mutations were 380C/T and 1228-1239delATTGTGCCTATT.
Summary/Conclusions: The study findings have confirmed the fact that the frequency of CD36 deficiency in the she population is similar to Han population in South China. But significantly lower than the other minority population. Although the mutations are common types, there are unique characteristics of the frequency of CD36 deficiency in the She population.

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ANALYSIS OF CD36 MUTATIONS AMONG APHERESIS DONORS: A REGISTRY OF CD36 DEFICIENT INDIVIDUALS IN GUANGZHOU BLOOD CENTRE

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Background: Mutations of CD36 gene leading to the absence of CD36 expression on platelets and monocytes are known to be responsible for the production of anti-CD36 isoantibodies (also known as anti-Nak[a] antibodies) in different clinical settings of immune mediated thrombocytopenia such as fetal/neonatal alloimmune thrombocytopenia (FNAIT), platelet transfusion refractoriness (PTR) and post-transfusion purpura (PTP). Meanwhile, several cases of PTR or FNAIT caused by anti-CD36 antibodies have been reported in Chinese population. For the treatment of anti-CD36 mediated PTR and FNAIT, transfusion with CD36 negative blood cells are mandatory. Therefore, the development of a cohort of CD36 deficient blood donors should be established among Chinese Blood Centre.
Aims: In this study, CD36 deficient individuals should be identified among 2300 apheresis donors.
Methods: The surface expression of CD36 on platelets and monocytes should be identified using flow cytometry, and their molecular basis underlying the CD36 deficiency should be subsequently elucidated by nucleotide sequencing.
Results: In a total cohort of 2300 apheresis donors, 46 individuals with CD36 deficiency on platelets could be identified; 9 blood group A, 12 blood group B, 24 blood group O and one blood group AB. Among them, except the expression of CD36 on monocytes was not known in 16 donors, 9 individuals did show any CD36 expression, neither on platelets nor on monocytes (type I) and 21 donors failed to express CD36 on platelets but not on monocytes (Type II). Nucleotide sequencing analysis of CD36 gene for these deficient individuals revealed thirteen different mutations (1228-1239delATTGTGCCTATT, C1156T, 329-330del AC, A1163T, C268T, 198-205del GATCTTTG, 429 + 4insg, C220T, 1200-5inv49 bp, exon4delgCAAGTT, G1150C+1155delA, Dupl.1204-1246 and C1409T). Three mutations, 1228-1239del ATTGTGCCTATT, C1156T and 329-330delAC appear to be the most common mutations (56.25%) related to CD36 deficiency in Guangdong population.
Summary/Conclusions: Here, the first register of blood donors with CD36 deficiency could be established in the Guangzhou Blood centre. The availability of this cohort assures a good therapy options for the patients with PTR and fetus/neonates with FNAIT in our region. Further construction of such register with other blood centers should be discussed to improve the treatment of immune mediated thrombocytopenia in China. In addition, the registration of CD36 mutations underlying the CD36 deficiency may help us to define DNA methods for the screening of CD36 deficient in the future.

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THE ESTABLISHMENT OF CELL LINES DERIVED FROM PLATELETS CD36 DEFICIENT AND THEIR APPLICATIONS

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Background: CD36(GPIV) is a major glycoprotein of platelets, CD36 antigen deficiency individuals can immunized to CD36 through blood transfusion, pregnancy or transplantation and develop CD36 antibody (Ab). In Asians, CD36 Ab play important roles in immunity thrombocytopenia, such as CD36-related platelet transfusion refractoriness and neonatal alloimmune thrombocytopenia. CD36 antigen deficiency most cause by CD36 gene mutations, Currently, more than 30 kinds of CD36 gene mutations have be reported. Since the blood donors and patients with platelets CD36 deficiency cannot always maintain, the establishment of their immortalized cell lines will be able to permanently preserve the CD36 deficient genome DNA, to provide permanent experimental materials for the study of CD36 deficiency and related platelet immunology.
Aims: To establish the immortalized cell lines of platelets CD36 deficient, and to preserve these genetic resources permanently. It is of great significance for CD36 research and application of blood immunology, transfusion medicine and anthropology.
Methods: EB virus (Epstein-Barr virus) was used to transform the peripheral blood lymphocytes of blood donors and patients of platelets CD36 deficiency antigen into the immortalized cell lines, CD36 genotyping was by sequencing.
Results: We successfully established 131 platelets CD36 deficiency immortalized cell lines, CD36 genotypes include 329-330delAC, T538C, T200A, C1156T and C1409T et al. All the cell lines were revivified after being frozen in liquid nitrogen, mycoplasma PCR detection found no mycoplasma contamination, no gene-mutations were found between before and after the cell lines were established.
Summary/Conclusions: The immortalized cell lines from platelets CD36 deficiency blood donors and patients passage stably, the genes of these CD36 deficiency types are permanently preserved and can be used as permanent experimental materials for the study of CD36 deficiency and related platelet immunology.

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EXPRESSION OF CD36 AND CD36 DEFICIENCY IN THAI BLOOD DONORS

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Background: CD36 is a multifunctional membrane glycoprotein and expressed on the surface of several human cells, including platelets and monocytes. The expression of CD36 is variable among different individuals. There are two types of CD36 deficiency according to their surface expressions. In type I deficiency, CD36 is not expressed in both platelets and monocytes and in type II deficiency, CD36 is expressed on monocytes but not on platelets. Individuals who are CD36 type I deficiency may produce isoantibodies against CD36 which play a role on the mechanism

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of immune-mediated platelet disorders including fetal and neonatal alloimmune thrombocytopenia (FNAIT), post transfusion purpura (PTP) and platelet transfusion refractoriness (PTR). Therefore, the information on the frequency of CD36 deficient in a population is essential for evaluation of patients suffering from immune-mediated platelet disorders.

Aims: The objective of this study was to determine CD36 deficiency in 500 healthy blood donors at Blood Transfusion Center, Srinagarind Hospital, Khon Kaen, Thailand.

Methods: EDTA blood samples were collected from 132 platelet apheresis donors and 368 repeated whole blood donors. The expression of CD36 on platelets was determined by flow cytometry. For those individuals who did not show CD 36 expression on platelets, the expression of CD36 on monocytes was analyzed. The expression of CD36 on platelets and monocytes were assessed by median fluorescence intensity (MFI).

Results: Totally, 500 Thai blood donors were investigated. Positive CD36 expression was detected in 491 platelet individuals with the MFI ranged from 2.52 to 51.56. Mean and median of MFI ± SD in this group were 17.28 ± 9.59 and 15.46 ± 9.59, respectively. Nine donors lacked CD36 on platelets only (type II) and one of did not express CD36 on platelets and monocytes (type I). The results demonstrated that the frequencies of CD36 deficiency Type I and Type II in Thai blood donors were 0.2% and 1.6% respectively.

Summary/Conclusions: This study indicates that the incidence of CD36 deficiency in Thai blood donors is similar to Chinese population living in Shanghai (Type I: 0.2%, Type II: 2.0%), which is slightly lower than those of other Asian populations (Type I: 0.5–1.0%, Type II: 2.0–5.8%). This information may useful for the risk prediction of isoimmunization against CD36 antigen in our population. Interestingly, the expression of CD36 is highly variable among individuals. This study may provide baseline information to study the relevance of CD36 expression with immune thrombocytopenia and other diseases.

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THE SCREENING OF CD36 EXPRESSION BOTH ON PLATELETS AND MONOCYTES AMONG IRANIANS IN GUANGZHOU DISTRICT

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Background: CD36(also known as GPIV) deficiency is known to be responsible for the production of CD36 isoantibodies(anti-Nak^a) in different clinical settings such as fetal/neonatal alloimmune thrombocytopenia(FNAIT), platelet transfusion refractoriness(PTR) and post-transfusion purpura.

Aims: So far, there has no report about the frequency of CD36 deficiency in Iranians. Here, we screened CD36 expression both on platelets and monocytes among Iranians in Guangzhou district, and analyzed the genotype of the abnormal expression.

Methods: In total of 53 Iranian blood donors in 2016 were enrolled. Flow cytometry was used to detect the CD36 expression both on platelets and monocytes. Extracted DNA from the samples with abnormal CD36 expression, and then analyzed the genotypes using PCR-SBT.

Results: In 53 Iranians, we found 4 having abnormal CD36 expression, including 2 with low CD36 expression on platelets, 1 with CD36 deficiency expression both on platelets and monocytes (Type I deficiency), 1 with low CD36 expression on monocytes. So the frequency of CD36 deficiency was 1.89% (1/53). Nucleotide sequencing analysis of these 4 samples, 2 with low CD36 expression on platelets were discovered new gene mutations, exon 10 975T/G and exon11 del CTA respectively. The other 2 samples have normal CD36 genotypes.

Summary/Conclusions: In 53 Iranians, the frequency of CD36 deficiency was 1.89%, in 4 samples with abnormal CD36 expression, we found two new gene mutations, and their efficiencies on expression of CD36 antigen need to be continue confirmed.

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PREVALENCE STUDIES OF CD36 DEFICIENCY ON PLATELET AMONG MALAY, CHINESE, AND INDIAN BLOOD DONOR IN NATIONAL BLOOD CENTER, KUALA LUMPUR

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Background: CD36 is an isoantigen found on blood's platelet and monocyte. The gene is located on chromosome 7 q11.2. Frequency of CD36 negative individual is reported to be about 3% to 11% in Asian and <0.4% in Caucasian. Therefore anti-CD36 antibody could be developed in CD36 negative individual that responsible for platelet immune-mediated thrombocytopenia in Asian. However, up to date, there are no prevalence data regarding CD36 negative individual among Malay, Chinese and Indian population in Malaysia.

Aims: This study aims to determine the prevalence of CD36 negative platelet among Malay, Chinese and Indian of the Malaysia population.

Methods: Blood sample was collected in EDTA tube and platelet rich plasma was isolated using standard protocol. Platelet immunofluorescence test (PIFT) used in this study was adopted from ISBT Platelet working party and performed using standard procedure accordingly. Platelet preparation was carried out by washing with phosphate buffered saline ethylenediaminetetraacetic acid containing 0.2% bovine serum albumin (PBS/EDTA/02%BSA). Washed platelets were then incubated with CD36-FITC conjugated antibody clone FA6.152 (Beckman Dickinson, USA) for the detection of CD36 antigen. Flow cytometry technique was performed to measure CD36 antigen on the platelet membrane via FACS calibur TM analyzer (Becton Dickinson, USA) and the results were analyzed by BD CellQuestTM Pro (Becton Dickinson, USA) software.

Results: From this study, it has been observed that CD36 negative platelet does occurs among Malaysian, with the frequencies of 2.5% (5/200) among Malays (P -value<0.001), 3% (3/100) among Chinese (P -value<0.001) and 2% (2/100) among Indians (P -value<0.001). The overall frequencies of CD36 negative individuals among Malaysian are 2.5% (P -value<0.001).

Summary/Conclusions: This preliminary study reveals the prevalence of 2.5% of CD36 negative individual among Malaysian. Therefore, anti-CD36 could be one of the causes of immune-mediated thrombocytopenia among Malaysian. However, further investigation using large numbers of participant will improve the validity of this study in the near future.

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POLYMORPHISM OF HPA 1-17 ALLOANTIGEN SYSTEM I IN KOREAN POPULATION IN NORTHEAST CHINA

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Background: HPA gene polymorphism trigger platelet antigen changes and cause an abnormal immune, leading to clinical platelet transfusion immune response is increasing. Studying the difference of platelet genes in different populations is very helpful to improve the efficacy of platelet transfusion, reduce the incidence of immune thrombocytopenia.

Aims: The aim of this study is to investigate the genetic polymorphism of HPA 1–17 gene in the Korean population of northeastern China. To analyze the differential expression with other populations, then determine the antigen system which has significance in clinical.

Methods: In this study, 95 healthy and unrelated populations of Korean were selected as subjects. The human platelet antigens were investigated using the PCR with sequence specific primers (SSP). The genotype frequencies, gene frequencies were calculated respectively and statistical analysis.

Results: The HPA gene frequency was: 1a-1.00, 2a-0.910525, 3a-0.55263, 4a-0.994735, 5a-0.96316, 6a-0.994735, 15a-0.58947; 1b-0, 2b-0.089475, 3b-0.44737, 4b-0.005265, 5b-0.03684, 6b-0.005265, 15b-0.41053. The HPA-2、HPA-3 and HPA-5 differed from those observed in Han nationality of China. Compared with the UK, American blacks, American whites, the HPA-5, HPA-2, HPA-3 system had a significant difference ($P < 0.05$), respectively. The present data showed that the distribution of the HPA alleles among Han nationality is quite similar to that reported among Japanese and Argentineans.

Summary/Conclusions: The genotype of the HPA-1 antigen system in the Korean population was aa, and the corresponding allele HPA-b was not detected. The genotypes of HPA-2, 4, 5 and 6 antigen systems are mostly aa, the most frequent genotypes were aa for HPA-2, -4, -5 and -6 systems; HPA-3, 15 have a high degree of heterozygosity, prone to platelet mismatch and cause the homologous immunity. HPA-2, 3, 5 and 15 systems all have polymorphism, HPA-15 system is highly polymorphic and should be paid attention to in clinical infusion.

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ANALYSIS OF HPA1-17 AND HLA-A, B GENE POLYMORPHISMS AMONG BLOOD DONORS FROM THE BANK OF PLATELET GENBANK IN CHANGSHA BLOOD CENTER

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Background: Platelet refractoriness and lack of platelet increase after platelet transfusion are seen in patients receiving chronic platelet transfusion. Antibodies may develop against HLA class I antigens and/or human platelet antigens (HPA). HPA systems have more than seventeen bi-allelic antigen polymorphisms in which a base pair substitution leads to change in an amino acid of a glycoprotein expressed on the platelet. Cross-match-compatible platelets can improve corrected count increments (CCIs) in alloimmunized patients with platelet transfusion refractoriness(PTR). In addition, HPA and HLA typing in both patients and platelet donors are essential for the diagnosis and treatment of patients with PTR.
Aims: To study the polymorphisms of human platelet antigen (HPA) 1-17 and human leukocyte antigen (HLA)-A and -B loci among blood donors from The Bank of Platelet GenBank in Changsha Blood Center.
Methods: 535 blood donors from The Bank of Platelet GenBank in Changsha Blood Center were involved in our study. We used a PCR-based method to detect HPA-1 to HPA-17, and Gov platelet alloantigens. Polymorphisms of mentioned HPA systems were determined by polymerase chain reaction-sequence specific primers (PCR-SSP), and HLA systems were determined by sequence based typing (PCR-SBT).
Results: The frequencies of HPA-la, -1b, HPA-2a, -2b, HPA-3a, -3b, HPA-4a, -4b, HPA-5a, -5b, HPA-6a, -6b, HPA-Gova, -Govb were 1.0000, 0.0155, 0.9934, 0.0973, 0.7611, 0.7058, 1.0000, 0.0044, 1.0000, 0.0044, 1.0000, 0.0221, 0.6925 and 0.7699, respectively. The HPA-7-14 and HPA-16-17 showed no heterozygosity as the b allele was not detected in those loci. The most common HPA genotypic combination was HPA-(1-6,Gov) 1aa-2aa-3ab-4aa-5aa-6aa-Govab (0.2074). The frequencies of HLA-A and -B demonstrated that HLA-A2 (0.4683) and HLA-B40 (0.2342) are the highest frequencies at their respective loci.
Summary/Conclusions: The distribution of HPA and HLA polymorphisms show a significant ethnic and territorial difference. The alloimmunization due to HPA-2,-3,-6 and Gov antigens need to be emphasized in Changsha populations.

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DISTRIBUTION CHARACTERISTICS OF HPA GENE AND ESTABLISHMENT OF DATABASE IN BLOOD DONORS IN HEILONGJIANG OF CHINA

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Background: With the continuous popularization and application of component transfusion, platelet transfusion has become one of the important means of clinical transfusion treatment. Repeated transfusions of platelet incompatible blood groups may cause platelet isotype antibodies, resulting in ineffective platelet transfusion (PTR), post-transfusion purpura(PTP), and neonatal allo-immune thrombocytopenia (NAIT).Therefore, a platelet donor database is established to meet the needs of clinical blood type matched platelets, to avoid the generation of platelet isotype antibodies and to ensure the safety and efficacy of platelet transfusion.
Aims: To analyze the allele frequencies and its genetic polymorphisms of human platelet antigen (HPA) from 419 healthy unrelated voluntary donors in Heilongjiang population of China, to determine platelet antigen system with clinical significance by judging the incompatibility ratio of HPA, as well as to establish a database of donors HPA.
Methods: The human platelet antigens (HPA1-17) were investigated using the PCR with sequence specific primers (SSP). The HPA gene frequency and genotype frequency were calculated and compared with different populations at home and abroad.
Results: The results showed that the HPA-1a to -17a alleles were expressed in each of the samples. HPA-1b to -5b, 6b and 15b alleles were expressed; HPA-7b to -14b, -16b and -17b were not detected. In genotype, the expression of homozygous aa was mostly, and the high heterozygosity was expressed in HPA3 and HPA15. Chi square test showed that the frequency distribution of HPA gene in this region were all consistent with Hardy-Weinberg equilibrium.
Summary/Conclusions: Distribution of the allele polymorphism of HPA1-HPA17 in Heilongjiang population exhibits its own characteristics in comparison with different populations at home and abroad. We have established the local HPA genotype database of platelet donors in Heilongjiang area, which can provide the matching donors for clinical ineffective platelet transfusion.

P-464

Abstract has been withdrawn

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972 CASES OF HPA1-18 ANTIGEN GENE POLYMORPHISM INVESTIGATION IN HAN POPULATION OF NANJING CHINA

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Background: Since Newman PJ et al found that platelet PlA1 and PlA2 antigen was named HPA-1a/b, it has been discovered and named HPA1 ~ 29. Nanjing is located in the junction of the North and South of China, and in the East Asian region on the west coast of the Pacific Ocean. Through the investigation of HPA1-18 gene polymorphism of Nanjing Han population, we can not only carry out platelet matched transfusion,but also comprehend the differences of HPA polymorphism between Nanjing and the other parts of the world.
Aims: HPA alleles have different frequency distribution in different nations and regions. In order to understand the situation of HPA genetic polymorphisms among Nanjing population,which would facilitate clinical platelet matched transfusion.
Methods: In this study, we investigated the polymorphism of Human platelet antigen (HPA)1-18 in Chinese Nanjing Han population using the polymerase chain reaction with sequence-specific primers(PCR-SSP).Blood samples were collected from 972 unrelated healthy blood donors in Nanjing Han population, from age of 18–55.
Results: According to the analysis of 972 cases HPA-18 results among Chinese population in Nanjing region, the a and b loci in HPA-1~-6, -15 illustrated different polymorphism distribution. Among them, the frequencies of HPA-3a/-3b, HPA-15a/-15b present highly polymorphic, which are 0.567/0.434,0.507/0.493 respectively. While a allele frequency of HPA-1~-2, HPA-4~-6 is higher and b allele frequency is very low. The b locus of HPA-7~-14,-16~-18 were not be detected . According to the Hardy–Weinberg equilibrium, the P value of a, b locus of HPA-1,-2,-4,-5,15 did fit expectations ($P > 0.05$) while HPA-3,-5 did not($P < 0.05$). The results indicated that they were not from the same Mendel group. It may be due to the mobility and migration of the Chinese population in recent years.
The frequencies of HPA-1a/1b allele in Nanjing were 0.996399 and 0.003601 respectively, which are close to Japanese, Korean, Thailand, Vietnam, Taiwan and other region of east Asia. As the known information pointed, HPA-1a/-1b allele frequencies in East Asia are the highest, which were significantly higher than that of the Mediterranean coast, such as Greece, Lebanon, Morocco, Spain, Switzerland, Tunisian. It is also higher than some African and European countries. The frequency of polymorphism of HPA-3a/b and HPA-15a/b in Nanjing region is almost the same with that in all regions of the world ranging from 0.4 to 0.6.
Summary/Conclusions: The results show significant difference of HPA allele polymorphism in Nanjing area when compared with some countries.
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P-466

A MODIFIED MAIPA METHOD

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Background: The monoclonal antibody immobilization of platelet antigens assay (MAIPA) is the gold standard in detecting platelet-specific antibodies. However, detection of anti-HPA15 has proven difficult, primarily due to low and variable expression of the antigen.
Aims: A modified monoclonal antibody immobilization of platelet antigens assay has been reported, with higher sensitivity in detecting antibodies against human platelet antigens (HPA).The assay combined biotinylated monoclonal antibodies, streptavidin-coated beads with flow cytometry. This study is meant to prove the sensitivity and repeatability of the method, thus improve the ability in our laboratory.
Methods: 1) Platelets are incubated with plasma. 2) Incubation with biotinylated glycoprotein-specific antibody. 3) Platelet are lysed. 4) Incubation with streptavidin-coated beads. 5) Incubation with anti-human IgG-PE.5) The fluorescence is measured by flow cytometry, on 1000 gated beads.
Results: The beads-MAIPA present superior signal-to-noise resolution (<8-fold higher) to standard MAIPA. Moreover, it is more sensitive in quantification of anti-

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HPA-1a. The lower limit of quantification was 0.5 IU/ml, compared with 1 IU/ml for standard MAIPA.
Summary/Conclusions: The use of biotin and streptavidin as well as flow cytometry can improve the sensitivity of MAIPA. Thus, this modified method has the promise to be popularized.

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MICROARRAY PROFILING OF CIRCULAR RNAS IN HUMAN IMMUNE THROMBOCYTOPENIA PATIENTS

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Background: Dysregulated expression of circular RNAs (circRNAs) has been demonstrated as being implicated in a variety of human diseases. However, the expression profile of circRNAs in immune thrombocytopenia (ITP) has not been discussed.
Aims: Here, we aimed to determine circRNA profile in peripheral blood mononuclear cells (PBMCs) from ITP patients to improve our understanding of ITP pathogenesis.
Methods: Expression profile of circRNAs in PBMCs from 3 ITP patients and 3 healthy controls were analyzed by microarray assay. Several circRNAs were selected for validation using real time-quantitative PCR (qRT-PCR) in 30 ITP patients and 30 control subjects. MicroRNA (miRNA) target prediction software identified putative miRNA response elements (MREs), which were used to construct a network map of circRNA-miRNA interactions for the differential circRNAs.
Results: Among 184 differentially expressed circRNAs, 62 were upregulated and 122 were downregulated in ITP patients. The validation study demonstrated that hsa_circ_0008212, hsa_circ_0009181 and hsa_circ_0001439 levels were elevated in PBMCs of ITP patients whereas hsa_circ_0007011, hsa_circ_0008584 and hsa_circ_0000453 were downregulated. These circRNAs targeted complementary miRNA response elements. Hsa_circ_0008212 showed interactive potential with two immune-related miRNAs (miR-125b and miR-146a).
Summary/Conclusions: The present study performed circRNAs profiling of PBMCs from patients with ITP and the results may aid in the understanding of the regulatory mechanism of ITP.

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ESTABLISHMENT AND APPLICATION OF PLATELET ANTIGEN PANEL CELLS IN HEILONGJIANG

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Background: Human platelet antigen(HPA) has specific immunogenicity, which can produce corresponding antibodies through immune factors such as blood transfusion, pregnancy, and so on. Resulting in ineffective platelet transfusion, such as platelet immune abnormalities. To establish the specific platelet antigen panel cells in this region, and to lay the technical foundation for the identification of platelet isotype antibody, so as to provide the necessary technical guarantee for the diagnosis and treatment of platelet immune abnormality diseases.
Aims: To establish human platelet antigen panel cells and apply it to detect and identify platelet antibodies.
Methods: The genotype of platelet antigen HPA1-16 of 419 voluntary unpaid blood donors in Heilongjiang area was analyzed by PCR – SSP technique. According to the results the type O platelets were selected to establish platelet antigen panel cells.
Results: 8 platelet cells which phenotype are consistent with genotype and covered 13 antigens in HPA1-6 and 15 systems were selected, Platelet antigen panel cells has been established and applied to the detection and identification work.
Summary/Conclusions: The platelet specific antigen panel cells has been successfully established. The results provide an experimental basis for the diagnosis and research of platelet immune abnormalities.

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Abstract has been withdrawn

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ASSOCIATION OF HUMAN PLATELET ANTIGEN POLYMORPHISMS WITH PLATELET COUNT AND MEAN PLATELET VOLUME

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Background: Recent genome-wide association studies have identified a number of single nucleotide polymorphisms associated with platelet count and mean platelet volume (MPV). However, it remains unclear whether human platelet antigens (HPA) polymorphisms are associated with platelet count and MPV.
Aims: The aim of this study was to determine the association of the HPA-2, -3, -5 and -15 polymorphisms with platelet count and MPV.
Methods: The HPA were genotyped by 5′- nuclease assay in 139 healthy Chinese Han individuals, while platelet count and MPV from the same samples were measured using an hematology cell analyzer.
Results: The platelet count was significantly lower in the individuals with genotype HPA-2aa than in those with HPA-2ab ($P = 0.020$). While the platelet count was significantly higher in individuals with genotype HPA-5aa than in those with HPA-5ab ($P = 0.045$), and in individuals with genotype HPA-15aa than in those with HPA-15ab/bb ($P = 0.032$). However, genotype HPA-3aa and HPA-3ab/bb have no significant impact on platelet count ($P = 0.084$). In addition, the MPV was significantly lower in individuals with genotype HPA-5aa than in those with HPA-5ab ($P = 0.001$). The MPV did not differ among polymorphisms, such as HPA-2, -3 and -15. Furthermore, HPA-2, -5 and -15 polymorphisms are identified as independent factors for the platelet count, and HPA-5 polymorphism is shown as an independent factor for MPV.
Summary/Conclusions: This study demonstrated that HPA-2, -5 and -15 polymorphisms are associated with the platelet count, and HPA-5 polymorphism is associated with MPV. This finding will improve the understanding of the association of HPA polymorphisms with platelet-related diseases.

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DEVELOPMENT OF ANTI-FOULING DETECTION METHODS FOR NOVEL PLATELET LIGANDS AND PATHOGENIC PLATELET ANTIBODIES

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Background: Fibrinogen (Fg) binding to platelet integrin αIIbβ3 (GPIIbIIIa) for the past half century has been considered essential for platelet aggregation to occur. However, we have found that in the absence of Fg and VWF platelet aggregation persists but in β3−/− murine models platelet aggregation is arrested, indicating that this Fg-independent platelet aggregation is mediated by an unknown ligand(s) of αIIbβ, termed the X-ligand(s) (JCI 2000, JTH 2006, Blood 2009). Additionally, ligands of GPIbα, another major platelet surface receptor, have not been fully explored, especially under Fg −/− conditions. Therefore, it is completely unknown what these αIIbβ3 and GPIbα ligands are, how they regulate hemostasis/thrombosis, whether they affect platelet quality during storage, and whether they can be used for transfusion or if they cause adverse effects in patients following transfusion. Immune thrombocytopenia (ITP) is a common bleeding disorder caused primarily by autoantibodies against platelet αIIbβ3 and/or GPIbα. Currently intravenous immunoglobulin (IVIG) is a first line treatment for ITP. We first reported, and others have confirmed, that anti-GPIbα antibody-mediated ITP is often refractory to IVIG therapy (Blood. 2006). Approximately, 20–40% of ITP patients have autoantibodies against GPIbα, hence, identification would help guide physicians towards proper treatment and conserve IVIG. However, the "gold standard" assay for anti-platelet autoantibodies, MAIPA, is unreliable and is rarely clinically utilized, leaving clinicians to rely on empirical treatment. Furthermore, identification of these αIIbβ3 and GPIbα interactions remains challenging as many assays suffer from high background signal or false positives due to non-specific binding (NSB) interactions.
Aims: The aim of this study is to develop an anti-fouling (NSB resistant) detection strategy for the identification of novel ligands of αIIbβ3 and GPIbα and pathogenic anti-platelet antibodies.
Methods: We have developed novel organosilane and organothiol based self-assembling monolayer (SAM) coatings that feature either αIIbβ3 or GPIbα covalently and site-specifically immobilized. Organosilane monolayers can be applied to hydroxyl-terminated surfaces such as glass and silica, while, organothiol SAMs are used on metal surfaces such as gold and copper. The coatings were synthesized on glass

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slides, ferromagnetic silica beads, and gold SPR surfaces. Fluorescence spectroscopy/microscopy, flow-cytometry and SPR were employed for analysis.
Results: αIIbβ3 coated surfaces bound Fg as well as conformational and linear epitope specific anti-αIIbβ3 mAbs. GPIbα coated surfaces bound both conformational and linear epitope specific anti-GPIbα mAbs. Furthermore, αIIbβ3 coated beads were incorporated into murine wild-type and VWF/Fg −/+platelet aggregates, demonstrating the interaction with the yet unidentified x-ligand(s). Furthermore, using these coatings we developed a flow-cytometry assay for the detection of ITP autoantibodies from patient sera.
Summary/Conclusions: These data indicate αIIbβ3 and GPIbα adopt ligand binding conformations when immobilized on DETA. Moreover, DETA greatly resists non-specific interactions and maintains a very low background signal regardless of detection method, even without employing a blocking agent (e.g. BSA). Identification of novel αIIbβ3 and GPIbα ligands and SPR detection are ongoing. This work presents the first use of anti-fouling-monolayer attached platelet surface receptors, primarily integrins, and demonstrates the enormous potential that these synthetic coatings possess in research and diagnostics.

Granulocyte Immunology

P-472
ANALYSIS OF WHITE CELL ALLOANTIBODIES IN THE PATIENTS WITH TRANSFUSION REFRACTORINESS

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Background: Platelet transfusion is a necessary treatment of patients with cancer, hematological malignancies, bone marrow failure and hematopoietic stem cell transplantation. However, only a few studies mentioned that the specificities of HLA and HNA alloantibodies among these patients.
Aims: To analysis of white cell alloantibodies in the patients with transfusion refractoriness.
Methods: A total of 82 medical records of patients who were refractory to random single-donor apheresis platelets between January 2015 and December 2015 were enrolled. HNA and HLA antibodies in patients serum were screened by Luminex. The specificities of human leukocyte antigen (HLA) class I/II and HNA alloantibodies were determined by Luminex Single Antigen and by flow cytometry, respectively.
Results: Anti-HLA-I/II alloantibodies were found in 38 of 82 (46.34%) patients, including 10 of 82 (12.19%) with anti-HLA-I alloantibodies only, 2 of 82 (2.44%) with anti-HLA-II alloantibodies only and 26 of 82 (31.70%) with both anti- HLA-I and II alloantibodies. The highest expression of anti-HLA-I specificities was-A*9 (46.2%), -A*10(21%), -A*11(21%), -A*19(21%), -B*40(33.6%), -B*7(29.4%), -B*22 (25.2)%), -DRB1*04(29.4), - DRB1*09(18.7), respectively. Anti-HNA alloantibodies were found in 2 of 82 (2.44%) patients. The anti-HNA specificities were anti-HNA-2. One sample was anti-HNA antibodies with unknown allospecificities.
Summary/Conclusions: In this study, we tried to find the alloimmunization trends in patients after multiple transfusions and detect anti-HLA-I/II and HNA antibodies among these patients.

P-473
HEMOLYSIS POLARIZES MACROPHAGES INTO A M1 PHENOTYPE

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Background: Hemolytic anemia is caused by the increased destruction of red blood cells (RBCs) in vivo, which leads a great challenge to blood transfusion. RBC destruction happens mainly by a direct lysis. The hemolysis produces and accumulates debris. The immune response to RBC debris is not fully understood yet. Monocyte-macrophages are intravascular sweeper, and macrophages can develop into classically (M1) or activated cells (M2) under different stimulation. It is unknown that which subset of macrophages plays a role on the elimination of RBC debris.
Aims: Since there is a paucity of data about the effects of hemolysis on macrophage phenotype, we explored how RBC debris polarized macrophages.
Methods: RAW264.7 and THP1 cell lines were incubated with RBC lysate for 2 days. Then transmission electron microscopy (TEM) was employed to observe macrophage morphology. We also analyzed the morphological changes of with scanning electron microscopy (SEM).The IL-10, IL-12p70 and TNFα cytokines were measured by ELISA. The surface markers of M1 and M2 were analyzed by flow cytometry. Moreover, we established hemolysis model to affirm the macrophage phenotype in *vivo*.
Results: According to TEM, we found that the RBC lysate resulted in increased spikes on the surfaces of macrophages. The cytokines concentration of IL-12p70 and TNFα increased, while the IL-10 content decreased compared with that of control group. The $F4/80^+/CD206^+$cells number (M2) were fewer, while the $F4/80^+/CD206^-$ cell number (M1) increased than control group. Consistent with above observation, M1 cells increased and M2 cells decreased in bone marrow and spleen in hemolysis mice.
Summary/Conclusions: Collectively, the current investigation demonstrated that hemolysis polarized macrophages towards a pro-inflammatory M1 phenotype.

Fetal-Maternal Immunology

P-474
MOLECULAR ANALYSIS OF ANTI-D ALLOIMMUNIZATION IN D NEGATIVE PREGNANCIES

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Background: D- blood group is rare with a low frequency around 0.3% in Chinese population. For Chinese D- pregnant women, they only have less 3% chance to have a D- baby for the frequency of heterozygous of *D* genotype is only 6% in the Chinese D+ individuals. Thus, almost every pregnancy of D- Chinese women has a risk of anti-D immunization. Furthermore, lacking of treatment experience as well as expensive Rh immune globulin prophylaxis, make it difficult to treat and prevent HDFN caused by anti-D. However, anti-D production does not actually occur in D- Chinese pregnant women as frequently as expected. Anti-D alloimmunization seems to keep a low level according to recent studies in the Chinese D- pregnant women.
Aims: The aim of this study was to systemically analyze anti-D alloimmunization and the molecular genetic background in D- pregnant women from Shandong area of China.
Methods: 102 unrelated pregnant women from Shandong area were selected for this study who were serologically typed as D negative during May to July 2017. The D antigen (using anti-D reagents with clone numbers TH-28, MS-26 and Rum-1) as well as irregular antibody were further analyzed by IAT while RhCcEe phenotype was determined by saline method. For the samples with positive IAT test of D antigen, the adsorption-elution test was applied with two kinds of anti-D mixes and acid elution and detected the elution liquid in gel card to determine the DEL phenotype. DNAs of the DEL individuals were extracted. D variants and those which had produced anti-D antibodies, as well as additional 15 samples which had no anti-D antibody detected, were analyzed for *RHD* genotyping and/or zygosity using High-Resolution Melting (HRM), PCR-RFLP and Multiplex Ligation-dependent Probe Amplification (MLPA).
Results: For serological testing, 4 cases were identified as D variants by test of IAT and a panel anti-D reagents (D-Screen, Diagast) having ccEe phenotype. The RhCcEe phenotype of other 98 cases was as follow: ccee (n = 63), Ccee (n = 21), ccEe (n = 9), CCee (n = 4) and CcEe (n = 1). Ten cases were identified as a DEL phenotype and all possessed Ccee phenotype. Anti-D was detected in 10 cases with ccee (n = 7), Ccee (n = 1), CCee (n = 1) and CcEe (n = 1) phenotype, two of which simultaneously had anti-C antibody detected and the titer of anti-D ranged from 4 to 2048 fold respectively. The 10 cases which had anti-D antibodies were all not DEL or D variants. For molecular testing, 10 samples with DEL phenotype all showed *RHD*1227A* allele detected by HRM while the other samples showed negative result for this allele, including 1 sample of D variant, 4 having anti-D and 15 without anti-D samples. By PCR-RFLP, 12 samples including 10 DEL samples and 2 samples having anti-D showed hemizygous *RHD* gene, while the *RHD* deletion was showed in all other samples including 8 having anti-D and 16 without anti-D samples. Four samples were analyzed by MPLA and three of them with D variant phenotype were identified with a *RHD*weak partial 15/01N.01* (RHD deletion) genotype while one sample having anti-D was detected with a *RHD*01N.03/01N.01* genotype.
Summary/Conclusions: Anti-D alloimmunization did not occur in the pregnant women with DEL phenotype carried *RHD*1227A* allele, but only in the pregnant women with a true D- phenotype (10/98, 10.2%) in our study. The results provide more evidence to confirm "Asian type" DEL defined by *RHD*1227A* is a complete DEL that could not produce anti-D immunization. Therefore, "Asian type" DEL should be tested by the molecular method in routine D- confirmation testing to prevent unnecessary anti-D prophylaxis.

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THE GENOTYPING AND ALLOIMMUNIZATION STUDY OF RHD ANTIGEN AMONG RH NEGATIVE PERINATAL WOMEN

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Background: The Rhesus blood group system is the most polymorphic and immunogenic system of human blood group systems. It is also the most clinically significant blood group system except ABO system. The highly immunogenic RhD antigen is often involved in hemolytic disease of the fetal and newborn (HDFN) and other diseases. The HDFN caused by anti-D antibody will lead to anemia, edema, stillbirth of fetus and kernicterus, even death of newborn in RhD negative perinatal women. Timely testing of RhD antigen and antibody, monitoring of anti-D level will provide important information for the treatment of HDFN in RhD negative perinatal women.
Aims: The aim of the study is to explore the relationship between genotyping of RhD antigen and RhD isoimmunization in RhD negative perinatal women in Shandong area China, find the factors which affect the production of anti-D antibody among RhD negative pregnant women, seek for the regularity of RhD isoimmunization, adopt different prevention and treatment measures according to different situation of RhD isoimmunization and set up the correct prenatal testing, prevention and treatment strategy of RhD hemolytic disease of the fetus and newborn.
Methods: Detailed data of pregnancy history was collected of 1017 cases of unrelated pregnant women in 1 year. We identified the ABO and RhD types of pregnant women and their husbands with standard serological methods. With the negative result of RhD antigen for the first serological screen, we tested their D antigen with Indirect Antiglobulin Test (IAT), absorption-elution test and PCR-SSP methods to confirm the results of weak D type, partial D type and DEL types. The Rh CcEe antigens were also serologically typed to all D negative women and their husbands. The irregular antibodies were screened in all RhD negative women and the specificities and titer of the antibodies were identified if the result is positive.
Results: Among 1017 cases of maternal specimens, 266 cases were identified to be RhD negative by serological screening, of which 9 were recognized as weak D or partial D type, 56 were recognized as DEL type with further antiglobulin test, absorption-elution test and PCR-SSP methods. Among all of the pregnant women which were identified to be RhD negative by primary screening, 15 cases of them had produced the anti-D antibody. Proportion of RhD negative pregnant women with anti-D antibody in this study is 5.64%. Among the 15 cases of RhD negative pregnant women who had produced anti-D, those who carried compatible ABO antigens with their husbands(86.7%) are apparently more than those who carried incompatible ABO antigens with their husbands(13.3%, $P < 0.05$). Identified by serology and molecular biology methods, 15 cases of Rh-negative pregnant women who had produced anti-D were excluded from the DEL phenotype.
Summary/Conclusions: The RhD isoimmunization of D-negative pregnant women is affected by their pregnancy and delivery history, the couples' ABO blood group, being DEL type or not and so on. With compatible ABO types for the couples, more pregnancy and delivery history and not being DEL type apparently increase the possibility of RhD isoimmunization for the RhD negative pregnant women. The probability of producing anti-D is lower in DEL type pregnant women and therefore the antenatal testing of anti-D antibody and anti-D immunoglobulin prophylaxis is not necessary to them. Timely testing of RhD antigen and antibody will provide important information for the treatment of HDFN in RhD negative perinatal women.

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HAEMOLYTIC DISEASE OF NEWBORN CAUSED BY RH ANTIBODIES IN RH(D) POSITIVE MOTHERS

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Background: Despite the introduction of effective Rh immunization prophylaxis, anti-D is still the most common cause of Haemolytic Disease of the Newborn (HDN) deaths. Excluding RhD, most notably c, E, and K antibodies are most frequently implicated in severe cases of HDN, followed by antibodies belonging to the Duffy, Kidd, and MNS systems. In Yunnan Kunming, ABO and Rh(D) typing is routinely performed in pregnant females at hospital usually at 14–16 weeks of gestation, but antibody screening is not routinely performed in all gestation. Antenatal antibody screening is done only in Rh(D) negative mothers at hospital in Yunnan Kunming.
Aims: Analysis of antibody specificity and causes of haemolytic disease of newborn by Rh (D) positive mothers, for better prophylaxis and diagnosis of HDN.
Methods: From January 2015 to June 2017, a retrospective analysis was made on the detection of HDN among Rh (D) positive mothers in Kunming regional hospital.
Results: We report 8 cases of HDN of the Rh (D) positive mother, including 6 anti-E antibodies, 1 anti-Ec antibody and 1 anti-Ce antibody. Anti-E antibody accounts for the highest proportion, up to 75%. In addition, three of Rh (D) positive mothers were the same as their husband ABO. 2 cases to produce anti-E antibody cause HDN, including 1 case of mother for type B Rh (D) positive, CcDee, father B Rh (D) positive, ccDEE, while 1 case of mother for type B Rh (D) positive, CCDee, father B Rh (D) positive, ccDEE. 1 case produces anti-Ce antibody resulting in HDN, the mother is type A Rh (D) positive, ccDEE, father A type Rh (D) positive, CCDee. All three Rh (D) positive mothers had no history of blood transfusion, but had multiple miscarriages. Antigen-matched RBC units were used to exchange transfusions to avoid the alloantibody, and all 8 patients had a good effect of preventing of kernicterus in neonates with rapidly rising bilirubin.
Summary/Conclusions: These cases highlight the importance of routine antenatal antibody screening even in Rh(D) positive females. Rh (D) positive pregnant women can produce HDN, and the proportion of HDN caused by anti-E antibody is higher. For Rh (D) positive pregnant women, even if ABO blood type is the same as the husband, it may also produce HDN caused by other blood group antibody other than anti-D.

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TEST METHODS AND TREATMENT OF BLOCKED RHD ANTIGEN COMBINED WITH HDFN IN 3 NEONATES

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Background: ABO and RhD identification are especially important for transfusion and diagnosis of HDFN. Appropriate test methods and professional experience are crucial for detecting red cell antigens.
Aims: To discussion detecting methods and treatment of neonatal with blocked D antigen induced by high titer anti -D antibody.
Methods: Automated system, slide typing and tube typing combined with elution were used to detect RBC antigen. Unexpected antibodies and antibodies identification were detected by gel method.
Results: Two of the 3 neonates were twins. Two mothers had miscarriage history and transferred to our hospital at late pregnancy because of antibody screening positive. Anti-D was identified from maternal plasma and the titer was 1024. RhD antigens of 3 neonates were all negative identified by gel method (Grifols WADiana Compact, Spain), slide typing and tube typing, but anti-D and control column were all positive (4 +) tested by automated system (ORTHO AutoVue Innova, USA), which was a combination of findings that cannot be accepted. After 45°C elution, D antigen showed 2 + positivity by tube typing. RhD genotyping later corroborated the positive D antigen of the newborn. Hemolytic tests showed that direct anti-globulin test(DAT) of 3 neonates were 4 + positivity, and antibody screening positive. Then anti-D was identified from plasma and elution liquid of neonates. Hemolytic disease induced by ABO incompatible can be excluded according to ABO blood type of neonates and mothers. ABO blood group identical with neonates and RhD negative blood were given for exchange transfusion. 3 neonates were treated successfully then discharged.
Summary/Conclusions: It is necessary to detect irregular antibody regularly for RhD negative pregnant. If a high titer of anti-D antibody was detected, coupled with hemolysis test results and clinical manifestations of neonate, even though RhD antigen of neonate was negative when tested by various methods, blocked RhD antigen should be considered. ABO blood group identical with neonates and RhD blood type identical with mother should be given when transfusion or blood exchange.

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THE IGG ANTI-A/B TITER AND HLA ANTIBODY POST PARTURITION ASSOCIATED WITH HAEMOLYTIC DISEASE OF THE FETUS AND NEWBORN

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Background: Haemolytic disease of the fetus and newborn (HDFN) is characterized by the presence of IgG antibodies in the maternal circulation which can cause hemolysis in the fetus and newborn. The courses of HDFN may be ameliorated by maternal antibodies reacting with human leukocyte antigens (HLA) of the child in the maternal circulation.
Aims: To report the correlation between mother's IgG anti-A/B titer and HLA antibody post parturition and HDFN of maternal fetal ABO blood group incompatibility.
Methods: The samples were collected from clinical specimens of 91 cases of maternal fetal ABO blood group incompatibility of suspected HDFN from August 2014 to May 2016, of which 78 cases had histories of parturition (≥2), 13 cases of parturition for the first time. The titer of IgG anti-A/B antibody in mother serum was detected by micro-column gel agglutination assay. HLA antibodies in maternal serum were determined by Luminex. Three hemolytic tests were performed for the newborn (the test of sick newborn erythrocytic direct antiglobulin, detection of serum free antibody and the identification of sick newborn erythrocyte dispersion) to determine the possibility of HDFN.
Results: (1) The incidence of HDFN was 67% when the IgG anti-A/B titer was <64 in the multiple parturitions group, while there was no HDFN when the maternal titer was <64 in the first parturition group. (2) In the multiple parturitions group, the incidences of HDFN in the cases of following IgG anti-A/B titer groups <64, 64, 128, 256, 512, >512 were 67%, 75%, 93%, 92%, 96%, 100%, generally showed an upward trend. (3) In the multiple parturitions group, the positive rate of HLA antibody is 64%, and strong positive rate (cutoff≥3, and baseline ≥1000) was 32%. No HLA antibodies were detected in the first parturition group. (4) In the multiple parturitions group, the positive rates of HLA antibody in the cases of following IgG anti-A/B titer groups 64, 64, 128, 256, 512, >512 were 67%, 75%, 60%, 50%, 68%, 63%, showed no correlation trend.
Summary/Conclusions: (1) Even though the IgG anti-A/B titer was <64 for the women who have histories of parturitions and maternal fetal ABO blood group incompatibility, there is still the risk of HDFN. Accordingly, for the women who had histories of parturition and whose blood groupings were incompatible with their couples, the blood grouping and three hemolytic tests of the newborns should be performed in time after parturition, and the changes of bilirubin and hemoglobin in the fetus are closely observed. (2) The incidence of HDFN increased with the histories of parturition and the growth of IgG anti-A/B titer. The incidence of HDFN was extremely high when there were histories of parturition and IgG anti-A/B titer of >512. (3) The production of HLA antibodies was positively correlated with the histories of parturition, and there was no significant correlation with IgG anti-A/B titers, and there was insufficient evidence for the association with HDFN. (4) This study was limited by the number of samples, and should to be expanded to further study and draw more exact conclusions.

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Abstract has been withdrawn

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ANALYSIS OF RED CELL ALLOANTIBODY SPECIFICITIES DETECTED IN PREGNANT WOMEN IN SINGAPORE

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Background: Hemolytic Disease of the Fetus and Newborn (HDFN) is an alloimmune condition in which clinically significant maternal antibodies pass through the placenta and results in hemolysis of fetal or neonatal red cells. The haemolytic process in the fetus or neonate ranges from mild to severe anemia leading to hydrops fetalis and may even result in fetal death. According to reported literature, approximately 1% of pregnant women develop clinically significant red cell alloantibodies and there are more than 50 red cell antigens that can elicit production of red cell alloantibodies that have been reported to cause HDFN. There has been no previous data reported on the red cell alloantibody specificities detected among pregnant women in Singapore.
Aims: To determine the specificities of red cell alloantibodies found among pregnant women in Singapore.
Methods: Our study is a retrospective review of red cell alloantibodies identified from samples referred by Hospitals and clinics providing antenatal and maternity services in Singapore to the Health Sciences Authority-BSG Reference laboratory from January 1 to December 31, 2016. A total of 223 blood samples were received during the study period. Red cell antibody identification was performed using both tube and column agglutination technique (gel method).
Results: A total of 277 red cell alloantibodies were detected in the 223 samples received. Among 223 blood samples, 176 samples (79%) had single antibody, 42 samples (19%) had two antibodies and the remaining 5 samples (2%) had more than 3 antibodies identified. The most common red cell alloantibody identified is anti-Lea (19%) which was considered not to be clinically significant as the antibody does not cause HDFN. Clinically significant alloantibodies were identified in 129 samples (58%), whilst 94 samples (42%) had clinically not significant antibodies. Of the clinically significant alloantibodies identified, Anti-E antibody is the most common (16%), followed by warm acting Anti-M (13%) and Anti-D (12.6%). Due to the unusually high number of anti-D detected, records of samples which anti-D was identified were reviewed and it revealed that information on the use of Rh D Immune Globulin may not have been properly documented in some of the request forms and as such, we cannot determine if the anti-D identified during the study is due to alloimmunization or passive immunization from Rh D Immune Globulin.
Summary/Conclusions: Alloimmunization with anti-E and warm reacting anti-M were the most common clinically significant red cell antibodies identified in pregnant women in Singapore. Hence, pregnant women with clinically significant alloantibodies should be informed and advised on the implications, including the risk of possible Hemolytic Disease of the Fetus and Newborn (HDFN). The authors recommend that Hospitals and clinics providing antenatal and maternity services should properly fill up the reference laboratory request form and provide relevant information (e.g. use of Rh D Immune Globulin, IVIG, transfusion history, etc.) so that the reference laboratory can provide definite results and precise transfusion recommendations to patients with clinically significant antibodies.

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LABORATORY ANALYSIS OF NEONATAL HEMOLYTIC DISEASE CAUSED BY IRREGULAR ANTIBODIES

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Background: Blood group incompatible HDN(hemolytic disease of the newborn) refers to the alloimmune hemolysis disease caused by mother-child blood type mismatched. In all cases, ABO-HDN are the majority, and have milder symptoms without transfusion therapy. On the contrary, while Rh- and other HDN account for less percentage, their clinical symptoms are relatively severe. Hence, Rh- and other HDN should gain more attention from us than ABO-HDN.
Aims: To summarize and analyze the distribution characteristics of antibody in neonates caused by irregular antibodies in Wuhan.
Methods: We collected all HDN specimens in Wuhan Blood Center from 2006 to 2016 firstly, then, figured out those Rh- and other HDN cases, a retrospective analysis was conducted finally.
Results: Analysis of a total of 344 Rh- and other HDN cases showed that Rh-HDN was more frequently than other HDN (325 vs 19, 94.5% vs 5.5%). In Rh-HDN, anti-D and anti-E accounts for up to 64.3% (209 cases) and 22.8% (74 cases), the percentage of anti-C and anti-c were only 0.6%(2 cases) and 1.8%(6 cases). It is noteworthy that there existed rarely anti-Ec(28 cases) and anti-DC(6 cases). In other HDN, the antibodies responsible were complicated, including anti-M(12 cases), anti-Mur(1 case), anti-Jk^a(1 case), anti-Jk^b(1 case), anti-Jk3(1 case), anti-Fy^a(1 case), anti-Di^a(1 case) and anti-Le^a(1 case).
Summary/Conclusions: Anti-Rh immunoglobulin injection preventively of Rh-maternal should undertake in order to avoid developing antibodies during their next pregnancy. As to those women with multiple pregnancy and transfusion history, prenatal tests and diagnosis should enhanced in order to figure out potentially pathogenic irregular antibodies, especially those rarely ones. In addition, the titer of antibodies should strictly monitored every once in a while.

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THE IRREGULAR ANTIBODY ANALYSIS OF 93 CASES OF NON-ABO NEONATAL HEMOLYTIC DISEASE IN KUNIMNG, YUNNAN, CHINA

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Background: The most frequent cause of Haemolytic Disease of the Newborn (HDN) are due to IgG class anti-A and anti-B from ABO blood group system. Non-ABO blood type system can also lead to HDN. In addition to RH blood group systems, Kell, Kidd, Duffy and MNSs antibodies can cause non-ABO HDN. In these blood group systems, there are differences in the blood group distribution of different races and populations, and there may be differences in the antibodies that produce HDN. Kunming is the capital of Yunnan province in China, with the Han nationality of 84.56 percent of the population, with ethnic minorities accounting for 15.44 percent of the city's permanent population. The analysis and study of irregular antibodies of non-ABO hemolytic diseases in different areas are helpful to explore its clinical significance.

Aims: This paper provides a reference for prevention of HDN and blood exchange treatment in the region by analyzing the distribution of non- ABO HDN antibody in Kunming.

Methods: From June 2010 to June 2017, all cases of non-ABO HDN from 10 large tertiary hospitals in Kunming were analyzed retrospectively.

Results: The investigation found 93 cases of non-ABO HDN caused by irregular antibodies, including 89 irregular antibodies of Rh blood group system, 1 case as Lewis blood group System, 1 case for Kidd blood group system, 2 cases for M blood group system. In the irregular antibodies of Rh blood group system, 37 cases of anti-D, 41.58% (37/89); 25 cases for anti-E, accounting for 28.09% (25/89); 9 cases anti-Ec accounted for 10.11% (9/89); 5 cases anti-c, accounting for 5.62% (5/89); 5 cases anti-E and anti-D, accounting for 5.62% (5/89); 5 cases anti-C and anti-D, accounting for 5.62% (5/89); 2 cases anti-Ce, accounting for 2.24% (2/89); 1 cases anti-Rh17, accounting for 1.12% (1/89).

Summary/Conclusions: Rh blood group antibodies derived from mothers are the major cause of Non-ABO-HDN. The anti-D antibody in Kunming caused the largest proportion of HDN, and the proportion of anti-E was second. Further research is needed to investigate strategies to prevent primary maternal RBC alloimmunization to RhD and to other blood group antigens, as well as to develop approaches to mitigate the dangers of existing maternal RBC alloantibodies.

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SELECTING APTAMERS SPECIFICALLY BINDING TO ANTI-RHD ANTIBODIES

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Background: The maternal immune system can target the A, B, O, and Rh blood group antigens of its own blood circulating in both endogenous and exogenous, and then produce the corresponding antibodies, among which IgG antibody is the only one that can pass through the placental barrier and enters the fetal blood circulation, its molecular size is about 150 kDa to 160 kDa. If the maternal and fetal blood group incompatibility, then the IgG antibody will combined with the corresponding antigen on the surface of fetal red cells and activate the complement system rapidly, resulting in intravascular hemolysis, seriously endanger the fetus's life, is the main cause of hemolytic disease of the newborn. Although the HDN due to Rh blood group incompatibility are only accounted for about 14.6%, but it's more urgent and the mortality is higher than A, B, O blood group incompatibility if there is no timely treatment.

Aims: Therefore, we hope to find an aptamer that is highly compatible with human blood types, especially IgG antibodies that can be associated with the Rh(D) blood group, and then the blocking effect of the aptamer can be used to block the binding of IgG antibodies to the fetal red cell surface antigens. It may provide new effectively strategy for preventing or treating the HDN caused by Rh blood group incompatibility. So we aim to establish the screening methods of human Rh(D) blood IgG aptamers to make foundation for the subsequent synthesis of aptamers and prevention of hemolytic disease of the new born.

Methods: We used the Rh(D) blood group IgG antibody as target, by using the systematic evolution of ligands by exponential enrichment (SELEX) technology, single-stranded DNA library was constructed containing 52 random sequences. The nucleic acid molecules of high affinity with IgG class antibody were screened by using the nitrocellulose membrane. Meanwhile, the experimental conditions of amplifying the nucleic acid into double-stranded DNA library were explored and optimized, and the enrichment of nucleic acid molecules was examined by membrane binding assay. The affinity between Rh(D) blood type IgG class antibodies with its aptamer was preliminarily determined by gel shift assay.

Results: With the ongoing of screening, nucleic acid molecules enriched library got evolution toward the direction of enhanced affinity with the target molecule. After 11 loops, the nucleic acid molecule of strong affinity with Rh(D) blood group IgG antibody was found out, and it shown blocked belt after binding with IgG molecules in gel shift assay.

Summary/Conclusions: The SELEX technology is widely used for screening nucleic acid molecules as an aptamer. We selected the Rh(D) blood group IgG antibody, and the screening method on aptamer of human Rh(D) blood group IgG antibody based on SELEX technology was established initially, which provides the theoretical and experimental basis for the clinical prevention and treatment of HDN.

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Abstract has been withdrawn

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A CASE OF ANTI-G CONCOMITANT WITH ANTI-C, ANTI-E AND ANTI-JKA CAUSING HEMOLYTIC DISEASE OF NEWBORN FRATERNAL TWINS

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Background: Hemolytic disease of the newborn (HDN) is a common consideration in medicine of the newborn, especially among the jaundiced. Only very few reports on anti-G concomitant with anti-C, anti-e and anti-Jk^a in newborn exist, especially in fraternal twins. This report may add further information on the clinical significance of anti-Rh and anti-Kell antibody for the newborn.

A 39-year-old, no history of blood transfusion, gracida 2, para 3 woman with an abnormal antibody screen positive resulted in HDN of her fraternal twins. The brother of the fraternal twins was in urgent need of blood transfusion, his total bilirubin was 476 μmol/l by 6 days of age, hemoglobin (Hb) concentration of 51 g/l. The maternal was O group and DccEE, Jka-Jkb+ phenotype; her husband was O group and DCcee, Jka+Jkb+ phenotype; the brother of the fraternal twins was O group and DCcEe, Jka+Jkb+ phenotype; the sister of the fraternal twins was O group and DccEe, Jka+Jkb+ phenotype.

Aims: To study anti-G concomitant with anti-C, anti-e and anti-Jk^a causing hemolytic disease of newborn fraternal twins.

Methods: Serology detection for antibodies in sera of patients with red blood cell identification, direct antiglobulin test, serum antibody screening and antibody identification, absorption and elution, blood cross matching of blood group in blood transfusion before and after.

Results: In a case of severe HDN (hemoglobin concentration, 51 g/l) transfusions were required. The strongly positive direct antiglobulin test (DAT) of the newborn fraternal twins red blood cells led to the diagnosis of hemolytic disease. The routine antibody screen was incapable of being defined whether anti-C antibody exist, extended testing (IAT- papain treated) revealed the presence of anti-C, anti-e and anti-Jk^a hemolytic disease, maternal anti-e with a titer of 32, anti-C with a titer of 1, anti- Jk^a with a titer of 2.

Summary/Conclusions: This case report highlights a rare case of anti-G together with anti-C, anti-e and anti-Jk^a causing hemolytic disease of newborn fraternal twins. This report disseminates knowledge on improved identification method of Rh system antibody by antibody screen cells papain treated, transfusion compatible blood for newborn in a timely manner. Thus, in cases of HDN the IAT- papain treated screen cells should always be included in the diagnostic workup.

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HEMOLYTIC DISEASE OF A NEWBORN DUE TO RARE ALLOANTIBODY OF ANTI-DIB AND FAMILY PEDIGREE ANALYSIS

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Background: The Di^b antigen usually occurs with high incidence, except in certain Asian and South American Indian populations. In general, hemolysis caused by anti-Di^b is not severe and its clinical course is benign. The severity of hemolytic disease of the newborn (HDN) due to anti-Di^b ranges from no symptoms to severe jaundice that requires exchange transfusion.
Aims: To report a case of severe hemolytic disease of the newborn caused by anti-Di^b, and perform a pedigree analysis.
Methods: The baby was delivered by a 28-year-old Chinese woman who had a history of early miscarriage and a history of blood transfusion traced back to 14 years. The mother's irregular antibodies had not been screened during pregnancy. For the mother, an antibody screening and identification test was carried out, in addition to typing her primary red blood cell antigens. The newborn was required hospitalization because of jaundice with a total bilirubin (TB) level of 30.3 mg/dl and blood hemoglobin (Hb) level of 13.5 g/dl on the third day after birth. Direct anti-globulin test, elution test and antibody screening and identification test for the newborn were carried out. The genotype of the Diego blood group system (Di^a and Di^b) of the families were typed by the PCR-SBT method.
Results: The newborn and the mother's blood groups of the erythrocytes were O, CcDEe, and O, CcDEE, respectively. The neonate's erythrocytes reacted positively (3+) in the direct antiglobulin test (DAT) and the serum reacted positively (2+) in the indirect antiglobulin test (IAT). The mother's serum and the eluate of the neonate's erythrocytes were reactive at high titer (1:512) with the entire panel of erythrocytes. The results of the maternal family study were as follows: the mother and her sister was Di(a+b-), and her parents, a brother, and the baby were all Di(a+b+). Coincidentally, the baby's father is also Di(a+b+). This study diagnosed that the mother had a rare blood genotype of Di(a+b-) with allo-anti-Di^b, which was determined to cause the hemolytic disease of the newborn. The newborn was stabilized after receiving phototherapy and high dose iv immunoglobulin therapy, but not suitable for exchanging transfusion. When the baby was discharged on the 10th days after birth, the level of serum TB was 5 mg / dl.
Summary/Conclusions: The case of hemolytic disease was caused by anti-Di^b. But in China, most commercial antibody detection panels do not contain Di(b-) red cells, it is helpful to consider anti-Di^b in routine antibody screening and identification test to ensure the clinical antibody can be detected.

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THE IMMUNOLOGICAL ANALYSIS FOR ONE CASE OF FOETUS WITH ANAEMIA AND THROMBOCYTOPENIA CAUSED BY ANTI-CD36 ANTIBODIES

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Background: A Chinese female foetus was found with anaemia (haemoglobin 96 g/l) and thrombocytopenia (platelets 84×10^9/l) at 23 weeks of gestational age. Her mother, a CD36 deficient woman (Type I) with a history of foetal demise and/or hydrops several times, successfully gave birth to one hydropic foetus with severe anaemia and thrombocytopenia after intrauterine transfusions with CD36*null* RBC and platelet last year. This year the mother pregnant again, and the umbilical cord blood was obtained to analyse the impact of anti-CD36 antibodies on foetus.
Aims: To confirm the impact of anti-CD36 antibodies on this female foetus whose brother was found with severe anaemia and thrombocytopenia at 26 weeks of gestational age.
Methods: The antibodies binding on platelets and the surface expression of CD36 both on platelets and monocytes were measured using flow cytometry. A commercial Kit (PAKPLUS) was used to test anti-CD36 antibodies in umbilical cord or elution buffer of platelets. The molecular basis of CD36 was also analyzed by sequencing.
Results: Although anti-CD36 antibodies were detected in maternal serum at 23 weeks of gestational age by PAKPLUS, however, the intensity of anti-CD36 antibodies was a little bit lower than last year (OD, 1.484 vs 1.556). In contrast, anti-CD36 antibodies were not found in umbilical cord blood at 23 weeks and 27 weeks of gestation, respectively. Flow cytometry analysis showed normal CD36 expression on platelets as well on monocytes of the foetus. Interestingly, we found that the amount of CD36 expression on monocytes at 27 weeks of gestation decreased significantly in comparison to 23 weeks. Moreover, the platelets of foetus can directly react with goat anti-human IgG labelled with FITC using flow cytometry. By elution of foetus's platelets, the anti-CD36 antibodies were also detected by PAKPLUS (0.34 vs 0.111). Like her brother, nucleotide sequencing analysis of CD36 gene revealed heterozygous carrying both the mutant of 329–330*del* AC and the wild-type allele. To improve the anaemia serial intrauterine transfusions with RBC from CD36null donors were performed. Finally the baby (2600 g; Apgar scores 10) was delivered vaginally at 37 weeks of gestation with normal haemoglobin.
Summary/Conclusions: In this case, although the anti-CD36 antibodies could not be detectable in foetal serum at mid-gestation, the anaemia and thrombocytopenia can also happen because of the binding of CD36 antibodies on platelets. For suspected FNAIT cases, it's necessary to do a series of immunological analysis to confirm the impact of anti-CD36 antibodies on foetus.

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RARE B (A) SUBTYPE CAUSED HEMOLYTIC DISEASE OF THE NEWBORN AND ITS BLOOD TYPE IDENTIFICATION

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Background: So far, the significance of B (A) subtypes in clinical blood transfusion has been clarified, but hemolytic disease of the newborn caused by B (A) subtype has not been reported.
Aims: To analysis serological features of Hemolytic disease of the newborn caused by rare B (A) subtype and discuss the identification strategy of this blood group.
Methods: Serological methods were used to detect hemolytic disease of the newborn in 1 cases, and the characteristics of B(A) subtype were identified by serology,family surveys, molecular biology and so on.
Results: This is one case of hemolytic disease of the newborn caused by the rare B (A) subtype and only associated with relatively weak antigenicity of A antigen. We find 3 samples of B (A) subtype by family surveys. Genotyping and sequencing confirmed all of them were haplotype B(A) 07.
Summary/Conclusions: Rare B (A) subtype can cause hemolytic disease of the newborn, which has important clinical significance, and its identification requires a variety of methods.

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ANTI-JK3 INDUCED NEWBORN HEMOLYTIC DISEASE: INCLUDING PEDIGREE ANALYSIS

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Background: Natural or transfused immune anti-Jk3 present on Jk (a-b-) people were published, and anti-Jk3 has been responsible for severe immediate and delayed hemolytic transfusion reactions. Accidently, mother with anti-Jk3 never caused serve clinical problem on babies with direct comb test positive.
Aims: Pedigree analysis was conducted to reveal the genetic basis and serological features in a newborn with severe hemolytic disease caused by anti-Jk3 from the maternal plasma of an immunized Jk(a-b-) individual.
Methods: Kidd phenotyping and sequencing of whole gene *SCL14A1* were performed to evaluate the Kidd blood group. Antibody identification and direct agglutinin test were conducted to detect anti-Jk3 in maternal and newborn plasma. The red blood cells of the newborn were also eluated using chloroform.
Results: The serological types according to ABO, RhD, and Kidd systems were B+ with Jk(a-b+) in the newborn and the father, and O+ with Jk(a-b-) in the mother. The Kidd alleles were Jkb/Jkb(IVS5-1) in the newborn, Jkb(IVS5-1)/JKb(IVS5-1) in the mother, and Jkb/Jkb in the father. Anti-Jk3 was detected in maternal and newborn plasma. The direct agglutinin test showed a positive result for newborn red blood cells and the extracted solution also showed the presence of anti-Jk3. Thus, severe hemolytic disease of the newborn caused by anti-Jk3 was confirmed.
Summary/Conclusions: The Jk(a-b-) phenotype was possibly caused by a mutation in the intron 5 splicing site, contributing to exon 6 mRNA deficiency and anti-Jk3 production after immunization by pregnancy.

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ANALYSIS OF CLINICAL OUTCOMES OF COMPONENT TRANSFUSION IN VERY LOW BIRTH WEIGHT PRETERM INFANTS

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Background: Premature infants face double risk of bleeding tendency and thrombosis because of their special physiological characteristics. Component transfusion is considered to be an important treatment to maintain oxygen supply, improve blood coagulation function and increase survival rate in very low birth weight (VLBW) premature infants. The smaller the gestational age and the lighter the birth weight, the more blood transfusions are being used. Domestic scholars reported for the VLBW preterm infants which have anemia with clinical symptoms of hypoxia should be timely treatment of blood transfusion, and the other scholars believe that blood transfusion lead to negative impact on survival of VLBW premature infants, which should be careful in blood transfusion decisions.
Aims: A retrospective analysis was made on the clinical outcomes of very low birth weight (VLBW) preterm infants with different transfusion times within the first 30 days after birth. To investigate the effect of blood component transfusion on the survival rate of children within 7 days after birth.
Methods: Clinical data of 97 cases who diagnosed as very low birth weight premature infants and received component transfusions before discharge from January 2015 to December 2016 in our hospital neonatal center were collected by using information retrieval system. A total of 60 cases were retrospectively analyzed. Use the blood component transfusion times within 30 days as the threshold to divide the cases into experimental group (blood transfusion times more than 5, n = 20) and control group (blood transfusion times <5, n = 40).To compare the two groups' clinical observation indexes which included gender, birth weight, 1 min Apgar score, gestational age, the length of stay, blood transfusion times within 7 days after birth and complications, and use the Cox regression analysis to find out the factors influencing the prognosis of VLBW premature infants. Kaplan-Meier survival analysis was used to assess the classification variables of prognosis.
Results: (1) The difference of gender, 1 min Apgar score, the length of stay, bronchopulmonary dysplasia (BPD), and necrotizing enterocolitis (NEC) between the two groups were not statistically significant ($P > 0.05$); and the difference of gestational age, birth weight, postnatal transfusion times within first 7 days, and gastrointestinal bleeding were statistically significant ($P < 0.05$);(2) Single factor Cox regression analysis showed that gestational age, birth weight, postnatal transfusion times within first 7 days and gastrointestinal bleeding were the risk factors for VLBW premature death; Cox regression analysis showed that gestational age less than 30W and postnatal transfusion times within first 7 days more than 2 were risk factors for VLBW premature infant death; Kaplan-Meier survival analysis of two indexes showed that for the infants with gestational age less than 30W, the survival time was shorter and the survival rate was lower, infants with postnatal transfusion times within first 7 days more than 2 had shorter survival time and lower survival rate.
Summary/Conclusions: Gestational age less than 30W and postnatal transfusion times within first 7 days more than 2 are risk factors for the prognosis of VLBW premature infants.

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EPIDEMIOLOGY OF PEDIATRIC BURN AND TRANSFUSION STRATEGY FOR ESCHARECTOMY DURING THE STAGE OF SHOCK IN LARGE AREA BURN

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Background: Burn is an important cause of morbidity and mortality in pediatrics. As the development of various physiological organs in pediatrics is not perfect, the body's tolerance to body fluid loss is worse than that of adults. As a result, pediatrics are more likely to suffer from shock after large area burns. In shock stage, the application of the wound scab can significantly improve the prognosis and improve the cure rate of large area burns. But cutting the scab may mean a large amount of blood loss in the patient, so in addition to the infusion of plasma anti shock during the shock stage, the red blood cells need to be transfused in time. However, there are few reports about the transfusion strategy during the shock stage of large area burns in pediatrics.
Aims: To analyze the epidemiological characteristics of burn and scald in pediatric and to explore the transfusion strategy for escharectomy during the stage of shock with large area burn.
Methods: The clinical closed loop intelligent path management and information - evaluated system was used to retrieve 795 cases of burn and scald in pediatric under 14 years of age in our hospital from January 2014 to December 2016. The epidemiological characteristics of the gender, age, burn cause, burn site, burn season, burn place and resident location were retrospectively analyzed; 55 cases of extensive burn in shock stage were selected and compared with the control study, According to the different transfusion threshold divided into restrictive transfusion group (Hb≤70 g/l, n = 25) and liberal transfusion group (Hb≤100 g/l, n = 30), two groups of patients with clinical data, blood routine value, postoperative infection rate, complications, incidence of adverse transfusion events, length of stay hospital and 30 day mortality index were measured before and after blood transfusion.
Results: (1) 795 cases of burned pediatric in 1 ~<5 years of age group had the highest incidence, accounting for 83.6%; the main reason to burn was hot liquid scald (86.1%); see more complex to burn multiple sites (70.6%); burns below TBSA30% in mild and moderate burn area mainly (92.3%); burns most occurred at home (75%); the differences between groups were statistically significant ($P < 0.05$). (2) There was no significant difference in the general data of pediatric with extensive burn, shock stage and tangential excision ($P > 0.05$); (3) Compared with open blood transfusion group, the red blood cell count (RBC), hemoglobin concentration (Hb) and hematocrit (Hct) were significantly increased after 24 h transfusion in restrictive transfusion group, blood transfusion and the differences were statistically significant ($P < 0.05$);(4) the difference of postoperative infection rate, complications, incidence of adverse reactions, length of stay hospital and 30 days mortality rate between the two groups were not statistically significant ($P > 0.05$).
Summary/Conclusions: Supervision of pediatric (especially preschool) should be strengthened to reduce the harm of burns; Patients who were over >1 years old with severe burn and shock stage were treated with intravenous infusion of anti-shock fluid, Red blood cells were transfused according to the blood routine, condition and intraoperative blood loss of pediatric, Hb≤70 g/Las blood transfusion threshold is safe, effective and feasible.

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A CASE REPORT OF HDFN CAUSED BY RARE ANTI-E, ANTI-C, AND ANTI-A ANTIBODIES FROM TYPE B MOTHER

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Background: Hemolytic disease of the fetus and newborn (HDFN) is an alloimmune condition that these IgG antibodies produced by the mother pass through the placenta attack antigens on the red blood cells in the fetal circulation, breaking down and destroying the cells. The fetus can develop reticulocytosis and anemia.
Aims: To analyze the blood type and antibodies of a newborn and his mother, and to choose the appropriate blood for the newborn's treatment.
Methods: Microcolumn gel card was used to detect ABO, antibody screening, antibody identification, coombs test and antibody titer. Tube test was used to detect Rh type, antibody screening, antibody identification. Elution test was used to detect anti-A, anti-E, and anti-c.
Results: Blood type of mother is type B, DCCee. Neonatal blood type is type AB, DCcEe. Anti-E and anti-c antibody were detected in serum from mother and

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newborn by using antibody screening and antibody identification. The mother's antibody titer of anti-E and anti-c was 64 and 16 respectively before delivery. When type AB, DCCee leukocyte-poor red blood cells RBC was selected to do cross matching test with the newborn, the main side result was positive. By free and elution test, anti-A was found both in the newborn's serum and elution fluid and the titer was 4; the mother's antibody titer of IgG anti-A was 32. When type O or B, DCCee leukocyte-poor red blood cells RBC was selected to do cross matching test with the newborn, the result was consistent. We choose type O, DCCee washing RBC for the newborn's blood transfusion.

Summary/Conclusions: IgG anti-E, anti-c and anti-A were found both in the newborn's serum and elution fluid, and the neonatal hemolytic disease was caused by antibodies in and out of the ABO blood-group system. Generally, neonatal hemolytic disease happened in type O mother. The neonatal hemolytic disease of both in ABO and Rh blood-group system happened in type B mother was extremely rare.

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ANALYSIS OF BLOOD UTILIZATION IN CHILDREN WITH LEUKEMIA

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Background: Leukemia is a malignant clonal disease of hematopoietic stem cells. It is one of the most common malignant tumors in children under the age of 15. For over-proliferation and blocked differentiation and apoptosis, cloning of leukemia cells accumulate in bone marrow and other hematopoietic tissues, infiltrate other non-hematopoietic tissues and organs, and also inhibit normal hematopoietic function. This disease can be divided into acute and chronic leukemia. Acute leukemia of children and adolescents with more rapid onset of symptoms commonly includes fever, progressive anemia, significant bleeding tendency or joint pain. Therefore, varying degrees of anemia, hemorrhage, fever, hepatomegaly, splenomegaly, lymphadenectasis and bone pain can be seen.

For the decrease of normal blood cells, the leukemia patients always need frequently blood transfusions in their therapy. Blood transfusion plays an important role in reducing the complications associated with chemotherapy and protecting the vital organs. Therefore, it is very important to correctly grasp the principles and methods of blood transfusion for leukemia patients, especially in children.

Aims: Analyze the volume and types of blood used by children with leukemia from 2015 to 2016 in the Children's Hospital, in order to make the blood supply meet the requirement by monitoring the blood volume change.

Methods: Analyze the types and volume of blood that Blood Centre of Zhejiang Province provided for the Children's Hospital from 2015 to 2016. Analyze the types and volume of blood used by children with leukemia in the Children's Hospital from 2015 to 2016.

Results: The volume of blood supplied by blood centre is steady growing. In 2015 and 2016, the supply of red blood cell was 8,462.5 U and 9,239 U, while platelet was 17,500 U and 17,512 U, and plasma was 1,266,880 ml and 1,287,440 ml.

For leukemia children in Children's Hospital, in 2015 and 2016, the amount and its rate in blood supply as follows: red blood cell was 1,837.5 U (21.7%) and 2,352.9 U (25.5%), platelet was 13,066 U (74.7%) and 13,229 U (75.5%), and plasma was 497,450 ml (39.3%) and 525,330 ml (40.8%). The usage of platelet was the highest in blood supply in leukemia children.

Summary/Conclusions: Blood transfusion is important for treating serious injury, hemorrhage, and some blood system diseases. Thrombocytopenia is the most important cause of hemorrhage in patients with leukemia. Blood Center of Zhejiang Province has been widely publicizing the work of blood donation to lead the community care for leukemia children. More people are needed for donating platelets, and infusion platelets for children in hospital also need a reasonable plan before transfusion.

We should strengthen information construction and improve the statistical work. Meanwhile, we should make our blood supply more effective to response the changes of clinical blood demand, and then, ensure their requirements. Saving blood efficiently, avoiding insufficient blood supply, and reducing the risk of transfusion-transmitted diseases are our goals.

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TO ANALYZE THE CLINICAL EFFICIENCY AND RISK FACTORS OF RED BLOOD CELLS TRANSFUSION IN PEDIATRIC PATIENTS WITH ACUTE LYMPHOCYTIC LEUKEMIA

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Background: More and more children have get the hemolytic diseases nowadays. Pediatric transfusion is not effective as we think.

Aims: To investigate the clinical efficiency and risk factors of red blood cells transfusion in pediatric patients with acute lymphocytic leukemia in order to improve the effect of blood transfusion and serve as a guide for the clinical blood transfusion.

Methods: 108 cases of pediatric inpatients with red blood cells transfusion from January 2015 to January 2017 were analyzed by using univariate analysis and logistic regression analysis.

Results: The results of logistic regression analysis indicated that high clinical risk evaluation (OR 3.0, 95%CI 1.9~10.5),blood transfusion times over 6(OR 2.1, 95%CI 1.6~4.3), fever and infection (OR 1.6, 95%CI1.2~3.6) were risk factors associated with transfusion efficiency ($P < 0.05$).

Summary/Conclusions: The transfusion efficiency were not related to patients' sex and age, but were related to the clinical risk evaluation, transfusion times, fever and infection. With the high clinical risk evaluation, increasing times of transfusion, fever and infection, the efficiency of transfusion would decrease simultaneously.

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PROTECTION OF EPO ON THE NEURAL CELLS

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Background: Erythropoietin (EPO) is a kind of sialic acid containing specific glycoprotein hormones, fetal liver and adult kidney is a major organ synthesis and secretion of EPO, mainly through EPO and the specificity of the red cell membrane of erythropoietin receptor (erythropoietin receptor, EPOR) combined with and promote the blood circulation in the red blood cell proliferation, differentiation, mature and release, red blood cell precursors to restore hematopoietic ability, can promote erythropoiesis, red bone marrow progenitor cell growth and angiogenesis of red blood cell precursors inhibit apoptosis, engaging in process such as placenta angiogenesis. Studies have found that EPO besides promoting hematopoiesis also has the nerve to protect and repair function, and can improve the long-term prognosis of nerve.

Aims: EPO is critical to identify the activity of cell proliferation, apoptosis, invasion and metastasis of human U251 cells. Aim at finding the ways for further research on EPO and its role in nerve protection.

Methods: It was cultured the U251 cells *in vitro* and screen optimal concentration of EPO with MTT, then divided into negative control group and the experimental group (EPO optimum concentration for the experimental group), at last, observed the effect of EPO on the proliferation, migration, invasion and apoptosis capability of glial cells with cloning experiments, scratch test, transwell experiment and the cells flow cytometry.

Results: The result of MTT was that the optimal concentration of EPO was 300 IU/ml; cloning experiment finds the level of cell proliferation was increased; the result of the scratch test was that cell migration was increased; Transwell find the capacity of cell invasion was increased; cell flow cytometry find that the ability of apoptosis was reduced.

Summary/Conclusions: The experimentally derived EPO could promote the proliferation, migration and invasion of U251 cells, at the same time, it could inhibit it's apoptosis, so it could be seen the EPO had a protective effect on nerve cells, it provided a new way and theoretical basis for the intervention in the early of neonatal brain injury in clinical.

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EXCHANGE TRANSFUSION THERAPY IN TREATMENT OF SEVERE HEMOLYTIC DISEASE OF THE NEWBORN (HDN) AND RESULT ANALYSIS

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Background: *Hemolytic disease of* the *newborn* (*HDN*) happens when mother's blood group is incompatible with the newborn's. It's common seen in the newborns who have mismatched ABO and Rh blood group with their mothers.
Aims: To analyze the result of exchange transfusion therapy in treatment of neonatal hyperbilirubinemia.
Methods: 50 newborns with confirmed severe HDN (total bilirubin > 342 umol/l) between March, 2016 and March, 2017 are enrolled in the research. Blood tests are performed when HDN is diagnosed and after exchange transfusion therapy.
Results: Total bilirubin (from 412.58 ± 118.64 μmol/l to 192.37 ± 84.52 μmol/l) and indirect bilirubin (from 384.69 ± 114.29 μmol/l to 165.64 ± 70.58 μmol/l) significantly decreased after exchange transfusion therapy, $P < 0.05$. Erythrocytes count
(from 4.52 ± 0.78 to 6.21 ± 0.82) and hemoglobin (from 164.59 ± 31.42 to 189.26 ± 33.24)significantly increased after blood exchange therapy, $P < 0.05$.
Summary/Conclusions: Exchange transfusion therapy is an effective way to treat HDN.

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DESCRIPTION OF 10 CASES OF HDN CAUSED BY RH BLOOD GROUP ANTIBODY

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Background: Hemolytic diseases of the fetus and newborns (HDFN) are severe diseases, resulting from maternal red cell (RBC) alloantibodies directed against fetal RBCs. The incidence of hemolytic disease of newborns is very low in the Rh system. But the consequences are very serious and easy to lead to miscarriage and newborns are liable to neonatal jaundice, edema neonatorum, anemia, kernicterus and even death.
Aims: To retrospectively analyze the relationship among Rh antibody types, titers and Rh hemolytic diseases of newborns (Rh-HDN) in Taiyuan area.
Methods: The samples of hemolytic disease in Rh system of which puerpera had multiple pregnancies and no transfusion history and neonates had neonatal jaundice within 24 h were collected from January to June 2017 in Taiyuan area. A series of tests were performed, including ABO, Rh blood group tests, irregular antibody screening and identification and cross-matching tests.
Results: For the ten cases, the results were five cases of anti-D, three cases of anti-E, one case of anti-Ce, one case of anti-cE. There were three cases of postnatal or exchange transfusion (IgG antibody titer≥1024). Hemoglobin increased significantly after transfusion. None of the others received blood transfusions or the families disagreed with blood transfusions. Four cases had unexpressed hemolysis at birth (the titers ≤16). After blue light therapy, all of the total bilirubins were significantly lower. The duration of hospital stay was 3–24 days.
Summary/Conclusions: In the ten cases, HDN could be caused by the presence of antibodies of Rh blood group in pregnant women, including anti-D, anti-E, anti-Ec and anti-Ce. Rh hemolytic diseases of the newborns (Rh-HDN) were independent of different types of antibody and antibody titer.

Therapeutic Apheresis

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Abstract has been withdrawn

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CASCADE PLASMAPHERESIS FOR DESENSITIZATION IN ABO INCOMPATIBLE LIVER TRANSPLANT: A COMPARISON OF COBE SPECTRA AND SPECTRA OPTIA

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Background: The reduction of anti-A/anti-B antibodies titers to a safe level is essential pre-requisite for patients awaiting ABO incompatible (ABOi) liver transplantation (LT). In order to reduce the antibody titers, currently used therapeutic aphaeresis (TA) techniques are conventional plasmapheresis, cascade plasmapheresis (CP) and antigen-specific immunoadsorption. We used CP as the technique of choice. The Spectra Optia is a new continuous-flow centrifugal apheresis system developed by Terumo which is based on the COBE Spectra platform.
Aims: To compare the efficiency of Cobe Spectra vs Spectra Optia for CP for desensitization in ABOi LT recipients.
Methods: This retrospective observational study included CP procedures performed for desensitization in 4 ABOi LT recipients. CP procedures on the first two patients were done using the COBE Spectra machine and on the next two patients, were done using Spectra OPTIA. Desensitization protocol included Rituximab (dose-100 mg), given 21 days prior to stipulated day of the transplantation. CP/non-selective TA was performed daily/alternative day with daily estimation of antibody titers until the target titer of 8 or less was achieved.
Results: In all patients, antibody titers were lowered to a level below 1:8 by two to six CP sessions before LD (living donor) LT. Cobe Spectra had no secondary plasma device (SPD) program so manipulations had to be done, leading to an extensive circuit which increased the extracorporeal volume (ECV) and lead to hemodynamic instability. The Spectra Optia has a SPD program and also pressure monitoring for the same leading to a smaller circuit, less ECV and better control of the procedure. The mean time in minutes/plasma volume exchanged was 167.5 min for Cobe and 100.5 min for Optia. We also used a flow regulator (B Braun) to regulate waste fluid which improved control over the percentage waste fluid generated.
In initial procedures, when the antibody titers were high, CP led to decrease in antibody titers by a factor of 2 doubling dilutions. At lower titers (≤1:32) CP was less effective and there was resistance to further lowering of titers and conventional TPE was more effective in further reducing titers. Long-term titer levels remained stable in all transplanted patients.
Summary/Conclusions: In our limited experience doing CP procedures on Optia increased the efficiency and ease of doing the procedures with better hemodynamic stability. The use of cascade plasmapheresis is safe and effective for desensitization in ABOi LT recipients

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SUCCESSFUL LYMPHOPLASMAPHERESIS IN A CHILD WITH MACROPHAGE ACTIVATION SYNDROME: A CASE REPORT AND LITERATURE REVIEW

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Background: Macrophage activation syndrome (MAS) is a severe and life-threatening complication of rheumatic disorders, associated with high mortality rates. Traditionally, the treatment of MAS includes immunoglobulin, cyclosporine, cyclophosphamide, etoposide and plasma exchange. However, these managements are largely empiric and have provided conflicting results, there is great need for new and specific therapies for this condition. To our knowledge, no data are available concerning the use of lymphoplasmapheresis (LPE) in the treatment of MAS.
Aims: To investigate an effective method to save severe MAS patients' life and create a platform for medical therapy.
Methods: A 13-year-old boy was admitted to Xiangya hospital on account of high fever for eleven days. On the twelfth day after admission, the patient became life-threatening, with the rapid development of multi-organ failure, acute hemorrhagic and neurologic abnormality, which required the admission of the patient to the

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intensive care unit (ICU). Laboratory findings were: abnormal liver enzymes, increased myocardial enzyme and ferritin levels (>10000 μg/l), while the patient's heart rate was 45 beats/min. He was diagnosed of MAS and treated with methylprednisolone, cyclosporine, etoposide and plasma exchange. However, the child was still critical, as clinical symptoms and several laboratory findings were not improved. The COBE Spectra blood cell separator was used for blood components separation. Adjusting the parameter by the manual procedure, we combined the immune active cells removing with plasma exchange in LPE, therefore we could remove pathological plasma composition and activated macrophagocytes, at the same time we could selectively remove different immune active cells.
Results: We performed a LPE: 1,237 ml of the patient's plasma was removed and replaced 1,300 ml of frozen plasma, representing an exchange of 39.6% of the patient's total blood volume (approximately 3,127 ml). The LPE treatment was considered successful as no evidence of transfusion reaction or citrate toxicity was observed. Rapid decrease of lymphocyte and monocytes to normal levels were observed after the LPE. The patient showed a dramatic clinical improvement after the LPE and was discharged from the hospital with walking by himself.
Summary/Conclusions: In cases of MAS, prompt institution of LPE can be effective and life-saving. LPE is an innovative treatment that has been used to successfully treat patients of autoimmune diseases, it has also saved a number of extremely severe patients' lives. Future therapeutic protocols for the treatment of MAS can potentially include a combination of LPE.

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LYMPHOPLASMAPHERESIS IN THE TREATMENT OF THROMBOTIC THROMBOCYTOPENIC PURPURA

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Background: Thrombotic thrombocytopenic purpura (TTP) is caused by deficiency of ADAMTS13. Despite major progress in recent years in our understanding of the disease, many aspects around the pathophysiology of TTP are still unclear. Acquired thrombotic thrombocytopenic purpura (aTTP) characterized by low ADAMTS13 activity and the presence of anti-ADAMTS13 IgG antibodies. B lymphocytes (CD20) decreased during rituximab treatment has showed potential benefits for patients with aTTP. aTTP is mainly caused by an autoimmune mechanism. Plasma exchange (PE) plus corticosteroids as standard therapy has greatly reduced the mortality of TTP, However, abundant plasma is unavailable in most medical centers in China. Lymphoplasmapheresis (LPE) combined PE and lymphocyte apheresis, in addition to delete autoimmune antibody and B lymphocytes, most of the schistocytes and VWF–platelet aggregates are removed simultaneously.
Aims: To investigate the efficacy and safety of LPE in the treatment of aTTP.
Methods: Retrospective study of 12 aTTP patients received LPE plus corticosteroids. Platelet count, lactate dehydrogenases (LDH) level, hemoglobin, schistocytes, ADAMTS13 activity and its inhibitors were monitored. The number of LPE and total plasma volume, remission time and adverse effects were recorded.
Results: The median number of LPE was 4 (2–10) and median total plasma volume was 108.32 ml/kg (55.45–255.86 ml/kg). The median time of initial response of platelet was 3 days (2–6 days), and clinically kept increasing at a median time of 8 days (2–43 days). Severe adverse effects were not occurred during LPE therapy.
Summary/Conclusions: LPE may be more effective and safer than PE in the treatment of aTTP, especially for the severe hemolysis patients. However, randomized controlled trials are required in the further research.

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EFFECTS OF ADVERSE REACTIONS ON IMMEDIATE EFFECTS OF ARTIFICIAL LIVER THERAPY

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Background: The adverse effects of PE and MRAS treatment, and the immediate effects of adverse effects on the outcome of the treatment are unclear
Aims: To investigate the occurrence of adverse reactions in plasma exchange and molecular adsorbent recirculating system and their immediate effects on treatment.
Methods: 87 hospitalization patients with severe viral hepatitis received artificial liver treatment from July 2014 to November 2015 were divided into the plasma exchange group and MARS group, each group were divided into adverse-reaction group and no adverse-reaction group, according to whether adverse reactions occurred or not. Record the occurrence of adverse reactions during the treatment, and the test results before and after artificial liver.
Results: 1. Plasma exchange and MARS could effectively reduce the levels of ALT, TBIL, DBIL, Cr, Hb, PLT, and RBC; the ability of plasma exchange to reduce bilirubin concentration was higher than that of MARS, while the removal ability of creatinine was lower than that of MARS. 2. The rate of adverse reaction was 53% in the artificial liver, and plasma exchange had higher rate than MARS (61.43% vs 40.43%); the highest adverse reaction was hypocalcemia in the plasma exchange group and blood pressure response in the MARS group. 3. The incidence of adverse reactions was related to the type of treatment (plasma exchange or MARS), Ca^{2+} concentration and red blood cell count before treatment, and adverse reactions had effects on the changes of TBIL, DBIL, Cr, K^+ and WBC before and after treatment.
Summary/Conclusions: The artificial liver process is prone to adverse reactions, and adverse reactions have effects on the changes of TBIL, DBIL, Cr, K^+ and WBC values before and after treatment.

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EVALUATION OF ADVERSE TRANSFUSION EVENTS BY WHOLE BLOOD AND BY APHERESIS IN PEDIATRIC PATIENTS

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Background: In children the incidence of adverse transfusion reactions is higher than in adult patients. In this respect, blood collection and blood preparation have to be continuously improved to reduce the risk of transfusion related to adverse events especially in these patients.
Aims: The aim of this study was to assess the occurrence of adverse transfusion reactions of hemocomponents by whole blood and by apheresis in pediatric patients.
Methods: Between 2011 and 2015, n = 214 pediatric patients were transfused with hemocomponents obtained by whole blood and by apheresis collected from U.O.C. Division of Clinical Immunology, Immunohaematology and Transfusion Medicine of Università della Campania "Luigi Vanvitelli". Adverse event was described as minor allergic reaction, febrile episodes, vomiting, or presence of dyspnea and bronchospasm. Comparison between transfused components and presence/absence of adverse event were assessed using Pearson's chi-square test.
Results: Data from pediatric patients (n = 144 onco-hematologic and n = 70 thalassemic patients) were analyzed. Male gender was 56% (60% and 46% respectively in onco-hematologic and thalassemic patients), mean age was 12.0 ± 9.9 years (8.5 ± 5.3 and 19.4 ± 12.8, respectively in onco-hematologic and thalassemic patients). Median time of observation was 0.50 years (interquartile range (IQR) 0.11–0.83) in onco-hematologic patients, and 4.9 years (IQR 3.2–4.9) in thalassemic patients. In that period, 12,531 units of hemocomponents were transfused (2,662 in onco-hematologic and 9,869 in thalassemic patients). No difference in proportions of adverse events between whole blood and apheresis was observed (0.3% in whole blood and 0.3% in apheresis, P-value chisquared test = 0.98). Same results were found considering separately onco-hematologic (0.6% in whole blood and 0.3% in apheresis, P-value chisquared test = 0.49) and thalassemic patients (0.2% in whole blood and 0.3% in aphaeresis, P-value chisquared test = 0.57).
Summary/Conclusions: No difference in post-transfusion adverse events related to whole blood and apheresis both in onco-hematologic and thalassemic pediatric patients was observed. All blood components have shown the same transfusion safety in pediatric patients.

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IDENTIFICATION OF KEY GENES AND PATHWAYS IN PLATELETS OF GLIOMA PATIENT BY BIOINFORMATICS ANALYSIS

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Background: Platelet is an ideal screening object in the diagnosis of some disease, because it is easily isolated and anuclear which makes it more simple than nucleated cells in nuclear acids, besides, platelet contains additional nuclear acids by acquiring vesicles secreted by other cells such as tumor cell.

Aims: To find the differentionally expressed genes and related pathways in glioma patient' s platelets vs healthy controls.

Methods: We downloaded gene expression profiles of GSE31095 from GEO database. The GSE31095 dataset contains 20 platelet samples, including 12 healthy controls and 8 glioma patients. The differentionally expressed genes (DEGs) between the two groups was analysed with GEO2R online tool, then the gene ontology (GO) and Kyoto Encyclopedia of Genes and Genomes pathway (KEGG) enrichment analyses were performed with DAVID database and KOBAS3.0 software respectively, and protein–protein interaction (PPI) network of the DEGs was constructed by Cytoscape software.

Results: Totally, we identified 174 DEGs in glioma patient's platelets, including 153 down-regulated genes and 21 up-regulated genes. GO analysis results showed that DEGs were significantly enriched in biological processes (BP), including SRP-dependent cotranslational protein targeting to membrane, nuclear-transcribed mRNA catabolic process, nonsense-mediated decay and viral transcription; KEGG pathway analysis showed the DEGs were mainly enriched in ribosome, spliceosome and natural killer cell mediated cytotoxicity. We identified 15 hub genes,RPS5,RPL11, RPS15A,RPS7,RPL13A,RPL3,RPL5,RPL7A,RPS13,RPS15,RPS16,RPS21,RPS3,RPS3A, EEF1B2, from the PPI network, and further analysis revealed that 14 genes were involved in hsa03010:ribosome.

Summary/Conclusions: Taken above, we have identified DEGs candidate genes and pathways in glioma patients' platelets with integrated bioinformatics analysis, the results offer some indications for our understanding about the platelet change in glioma patients, and these candidate genes and pathways may be used in the diagnosis of glioma in the future.

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THE DIVERSITY OF KIR IN KOREAN ETHNIC GROUP OF CHINA

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Background: Killer cells immunoglobulin-like receptors (KIRs) are a family of inhibitory and activating receptors expressed mainly by natural killer (NK) cells and few subsets of T lymphocytes. KIRs are a major class of the receptors and recognize groups of HLA class I alleles (KIR ligands) .The KIR locus maps on chromosome 19q13.4 within the leukocyte receptor complex (LRC) and comprises a family of polymorphic and highly homologous genes that are tandemly arrayed over about 150 kb. To our best knowledge, no analogous study on the Northeast of China Korean ethnic group KIR Polymorphism has been performed so far.

Aims: This study describes the frequency distribution of 16 variable KIR genes in Korean ethnic group of China .

Methods: Genomic DNA from 202 unrelated blood donors from JiLin Province of China were included in this study. Genomic DNA was isolated from 5 ml of blood using the TIANamp Blood DNA kit (Tiangen).Polymerase chain reaction-sequence-specific primer (PCR-SSP) was performed to amplify the genomic DNA for presence or absence of 12 KIR genes (2DL1, 2DL2, 2DL3, 2DL5, 3DL1, 2DS1, 2DS2,2DS3, 2DS4, 2DS5, and 3DS1) and the pseudogene 2DP1, 3DP1. KIR2DS4 alleles were also typed as being either full length or having the 22-bp deletion that prevents cell surface expression. Two pairs of primers were used for each gene, selected to give relatively short amplicons of 100–800 bp, as previously described in full detail.

Results: In our study, twenty-nine different genotypes were observed and all 16 KIR genes 16 KIR genes were detected among the Korean ethnic individuals. Framework genes KIR3DL3, KIR2DL4, KIR3DL2 and the pseudogene KIR 3DP1 were present in all individuals. The most frequent non-framework KIR genes detected in the Korean ethnic group were: KIR2DL1(99%), KIR2DL3(97%),KIR3DL1 (94%), KIR2DS4 (94%) and the pseudogene 2DP1 (99%), whereas the frequencies of KIR2DS2, 2DS3, and 2DL2 were all less than 25%. The frequency of KIR2DS1, KIR2DL5 and KIR3DS1 was higher than Chinese Han populations (47% vs 33%, 42%vs 33%, 38%vs 29%). The most complex genotype no. 6 represented all 16 KIR genes. Among these genotypes, 46% of the individuals displayed genotype 1 (N = 87), but the other genotypes were only found in one to six individuals. The frequency of genotype 1 was less than other ethnic groups (42% vs 61%).

The group B haplotype occurred more frequently than the group A haplotype. B haplotypes outnumbered A haplotypes in frequency by approximately 1.3:1 (87 A haplotypes vs 115 B haplotypes),which was not similar to that observed in other Chinese Han populations.

Summary/Conclusions: The comparison of KIR frequencies between Korean ethnic group and a local Han population showed that the two populations showed similar frequencies for the most KIR genes. In contrast, the distribution of KIR haplotypes showed significant differences between them.

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MICA POLYMORPHISM AND LINKAGE DISEQUILIBRIUM ANALYSIS WITH HLA-B IN SHENZHEN HAN POPULATION

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Background: On one hand, a large number of studies and clinical trials have shown that, while taking into account the classic HLA matching, if the MICA gene also matched, the receptor's survival rate would be significantly improve after organ transplantation. On the other hand, the occurrence of post-transplant rejection has a clear correlation with serum MICA antibody in transplantation patients. Furthermore, in the population genetics, the distribution of MICA and HLA-B genes is related to the factors such as geographical and racial factors. So far, the data of MICA polymorphism of Han population in Shenzhen has not been reported.

Aims: To study the distribution characteristics of MICA in the blood donation population of Han nationality in Shenzhen and to analyze its linkage disequilibrium relationship with HLA-B gene.

Methods: The MICA and HLA-B genes were identified by PCR-SBT method in 143 randomly selected Han donors in Shenzhen, whilst the allele frequency, haplotype diversity and linkage disequilibrium parameters were analyzed by Pypop statistical software.

Results: We found 13 MICA and 35 HLA-B alleles in 143 blood donors, in which demonstrate that MICA * 008: 01(76/286) and HLA-B * 40: 01 (53/286) has the highest frequency, whilst MICA * 008: 01-HLA-B * 40: 01 (47/286) and MICA * 010-HLA-B * 46: 01 (46/286) are the most common haplotypes. We found 66 MICA-HLA-B haplotypes in total, among which there were 18 common haplotypes (MICA*008:01-HLA-B*40:01, MICA*010-HLA-B*46:01, MICA*002:01-HLA-B*58:01, MICA*045-HLA-B*13:01, MICA*019-HLA-B*15:02, MICA*009:01-HLA-B*51:01, MICA*010-HLA-B*15:01, MICA*002:01-HLA-B*38:02, MICA*008:01-HLA-B*13:02, MICA*012:01-HLA-B*54:01, MICA*019-HLA-B*27:04, MICA*009:01-HLA-B*51:02, MICA*010-HLA-B*15:11, MICA*019-HLA-B*40:01, MICA*008:01-HLA-B*46:01, MICA*008:01-HLA-B*38:02, MICA*008:01-HLA-B*37:01, MICA*010-HLA-B*48:01) and their frequency were all higher than 1%. After P value correction, it showed that these 18 MICA-HLA-B haplotypes had significant linkage disequilibrium in the population. No novel allele was found.

Summary/Conclusions: The presented allele frequencies, haplotype diversities and linkage disequilibrium parameters at high-resolution level in Shenzhen Han population can provide more insights into the studies and applications associated with MICA and HLA-B genes.

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DISTRIBUTION OF HLA-A, -B, AND -DRB1 ALLELE AND HAPLOTYPE AT HIGH RESOLUTION IN THE GUANGXI HAN POPULATION

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Background: The human leucocyte antigen (HLA) genes are the most polymorphic genes in the human genome. To date, May 2016, total of 15 020 HLA alleles have been identified according to the IMGT/HLA Database (release Version 3.23.0) (1). HLA has been found to be critical for infections resistance, autoimmune disease susceptibility, tumour resistance and immune responses to the allograft after solid organ or haematopoietic stem cell transplantation (HSCT) (2, 3). HSCT is a curative therapeutic method for blood malignancies and other blood disorders.

To 31th July 2016, China Marrow Donor Program (CMDP) has enlisted 2.2 million potential haematopoietic stem cell (HSC) donors and facilitated 5 845 HSC

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donations, including 240 patients beyond mainland China. No doubt, a better and precious characterization of HLA genotypes will improve CMDP and benefits transplant recipient patients.
Aims: This study designed to analyze the distribution feature of allelic and haplotypic polymorphisms of human leukocyte antigens at A, B and DRBl loci in Guangxi Han population.
Methods: A total of 1644 unrelated Han ethnic individual from Guangxi regions were genotyped by polymerase chain reaction-sequence based typing (PCR-SBT) for HLA-A, B and DRB1 loci. Allelic and haplotypic frequencies were estimated by maximum likelihood estimation method using Arlequin software 3.5.2.2, this software was also used to estimate linkage disequilibrium (LD). Phylogeny tree was constructed using the neighbour-joining (NJ) method with the standard genetic distances using the MEGA software 6.0. Departures from Hardy-Weinberg expectations are observed for all three loci in this population.
Results: A total of 37 HLA-A, 70 HLA-B and 37 HLA-DRB1 alleles were detected in 1644 samples. The most frequent alleles were A*11:01(28.86%), B*46:01(14.26%) and DRB1*15:01(13.32%). The most common three loci haplotype was A*33:03-B*58:01-DRB1*03:01(6.12%). A*02:07-B*46:01, A*11:01-DRB1*15:01 and B*58:01-DRB1*03:01 showed the strongest linkage disequilibrium. The phylogenetic tree analysis suggested that Guangxi Han population had a relative close genetic relationship with Guangxi Zhuang population and were genetically similar to southern Chinese populations.
Summary/Conclusions: We found that the HLA-A, B and DRB1 loci are highly polymorphic in Guangxi Han population, which will be useful for estimating the probability of HLA matching in transplantation as well as for the study of the forensic medicine and anthropology in Chinese population.

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ALLOIMMUNIZATION AMONG THALASSEMIA PATIENTS AT DR. CIPTOMANGUNKUSUMO HOSPITAL (RSCM)-JAKARTA, INDONESIA

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Background: Thalassemia is the most common blood genetic disorder worldwide. In Indonesia the fourth most populous countries in the world with 250 million people, approximately 0–9% β-Thalassemia carriers and 0–40% Hb AE among populations throughout the country potentially contribute around 50,000 β-Thalassemia and Thalassemia/HbE patients requiring blood transfusion per year. Despite the current technology and quality control practice, the availability and safety of blood transfusion is a matter of conjecture. Recent evidence that rare RBCs' antibodies were detected in multiple transfused thalassemia patients would become somewhat growing concern in clinical transfusion practice.
Aims: To study the occurrence of RBCs alloimmunization among multiple transfused thalassemia patients and identify blood group antibodies potentially developed and induce antigen-antibody reaction affecting patients.
Study was carried out in 88 multi-transfused patients recorded with β-thalassemia registered at Dr. Cipto Mangunkusumo Hospital-Jakarta. Patients' medical record were used for individual data information included the history of transfusion. Antibody screening and identification was tested using Capture R Ready Screen 3 cell and in case positivity using Capture R Ready ID with 14 cell panels (Immucor Inc. Norcross, GA) respectively.
Methods: Study was carried out in 88 multi-transfused patients recorded with β-thalassemia registered at Dr. Cipto Mangunkusumo Hospital-Jakarta. Patients' medical record were used for individual data information included the history of transfusion. Antibody screening and identification was tested using Capture R Ready Screen 3 cell and in case positivity using Capture R Ready ID with 14 cell panels (Immucor Inc. Norcross, GA) respectively.
Results: All 88 thalassemia patients were examined consisting of 43 (48.8%) males and 45 (51.2%) females. They were all RhD positive, but for the ABO blood group, there were 24 A, 20 B, 36 O and 8 AB respectively. They were previously transfused with blood matched only for ABO&D but seven of them developed allo-antibodies particularly those receiving first transfusion at their early ages. Age at first transfusion was significantly higher in alloimmunized than non-immunized patients ($P = 0.037$). Applying 10 year interval i.e. 1–10 years, 11–20 years, 21–30 years and 31–40 years, patients were divided into four groups with number of patients, i.e. Group I, 23 patients (26.14%)), Group II, 44 patients (50%), Group III, 18 patients (20.46%), and patients Group IV, 3 (3.4%) respectively. Among 23 patients of Group I, only one case of allo-immunization were detected. In Group II, five out of 44 patients were allo-antibodies positive, while in Group III, allo-antibodies were detected in one patients. However, no alloantibodies were detected in Group IV. Having considered gender grouping, despite small size number of patients, no significant differences was observed in rate of allo-immunization development between gender ($p > 0.05$).
Summary/Conclusions: This study re-emphasized the need for RBC antigen typing before first transfusion and provided antigen matched blood (at least for Rh antigens). Early institution of transfusion therapy after diagnosis is another means to prevent allo-immunization. Applying antibody screening is expected to reduce risk of allo-antibodies in thalassemia patients receiving recurrent blood transfusions.

Evidence Based Transfusion Medicine Practice

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LEVELS OF SERUM HEPCIDIN IN SICKLE CELL DISEASE SUBJECTS WITH MULTIPLE BLOOD TRANSFUSIONS

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Background: No effective physiological mechanism for excess iron excretion is known in humans. The predominant negative regulator of iron absorption in the small intestine and iron release from macrophages was recently discovered to be hepcidin which is a 25-amino acid peptide produced by hepatocytes. Conditions such as the sickle cell disease (SCD), where transfusions are frequently resorted to, can subsequently result in ready accumulation and circulation of exogenous iron as non-transferrin bound iron (NTBI) in tissues.
Aims: The present study is aimed at determining the alterations in serum hepcidin levels and red cell indices in sickle cell patients with multiple blood transfusions and any associations between hepcidin and degree of severity of sickle cell anemia.
Methods: The study was conducted on 150 subjects comprising of 50 sickle cell patients (SCD) in stable state, 50 SCD with vaso-occlusive crisis and 50 apparently healthy individuals with HbAA as control who were age, sex and socioeconomic standard matched. Complete blood count, hemoglobin electrophoresis and serum levels of hepcidin were performed on recruited subjects appropriately. The degree of severity (mild, moderate or severe) was determined using a scoring system incorporating annual number of blood transfusions, crisis episodes and presence of anemia, vaso-occlusive pain and organ complications.
Results: Mean serum hepcidin level was significantly higher in crisis and steady state sickle cell subjects than in controls ($P < 0.001$). Hepcidin level was found to be increasing with increasing severity of anemia in sickle cell disease subjects. The mean red cell indices showed negative correlation with serum hepcidin level in the sickle cell subjects. Irrespective of the gender, mean serum hepcidin level was higher in Hb-SS subjects with multiple transfusions than control groups ($P < 0.001$).
Summary/Conclusions: In this study, we observed that serum hepcidin levels were higher in Hb-SS subjects with multiple blood transfusions than individuals with Hb-AA. Also, significant association was observed between hepcidin levels and severity of anemia in sickle cell disease subjects.

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15 YEARS EXPERIENCE OF A COLLABORATIVE PROGRAM TO IMPROVE BLOOD MANAGEMENT PRACTICES IN AUSTRALIA

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Background: Blood collected in Australia comes from voluntary, unpaid blood donors; and, is provided to the patient at no charge. Collaboration of national and

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local state governments stakeholders is essential to enhance best practice in patient blood management (PBM), and the stewardship of a precious resource.
Aims: To describe the collaborative approaches used by the Blood Matters Program since 2002, to improve health service outcomes, including meeting national standards and guidelines in Victoria, Australia. Blood Matters is a collaboration between the Victorian Department of Health and Human Services, and the Australian Red Cross Blood Service.
Methods:
- Developing strategies that optimize appropriate product use, and person-centered care, using governance frameworks and standards
- Fostering and strengthening partnerships and stakeholder relationships promoting best practice
- Analyzing data and disseminating findings across the sector to raise awareness, influence practice, promote efficiencies, and provide value
- Providing and promoting expert knowledge sharing and collaboration across the sector, including specific initiatives, tools and specialized advice
Results: With over 15 years in the local blood sector, Blood Matters is recognized as a leader in improving blood management:
- Establishment of Transfusion Nurses, Trainers & Safety Quality Officers roles across 43 health services, who now systematically improve transfusion practice and sustain change in their respective health services
- Over 150 successful graduates of the Graduate Certificate in Transfusion Practice, creating a wealth of highly skilled blood experts across the health system
- More than 1,700 incidents reported and analysed in the voluntary haemovigilance system (Serious Transfusion Incident Reporting), allowing strategies and recommendations to be made to improve the overall safety of the transfusion process
- Reduction of Victorian red blood cell wastage and meeting national targets; with Victorian red cell wastage rate sitting below national average for the first time in April 2016
- Increased communication between the Blood Service, health services, jurisdictions and government agencies
- Support three additional jurisdictions to improve PBM through haemovigilance reporting, supporting transfusion nurses and practice audits
Summary/Conclusions: With a small financial investment and a focus on collaborative work across the blood and health sector, and embedding blood champions, much has been achieved. Ongoing government and Blood Service support, and collaboration, are essential to continue to enhance best practice in PBM and improve stewardship of blood.

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DEVELOPMENT AND CLINICAL APPLICATION OF PRECISION BLOOD TRANSFUSION MANAGEMENT SOFTWARE

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Background: The rationality and accuracy of blood transfusion is a difficult problem in clinical practice. Inappropriate transfusions waste valuable blood resources and disadvantageous situation of the clinical transfusion management is expected to be reversed.
Aims: To develop management software that fulfills to precise transfusion and applies it to clinical practice so as to promote rational use of blood.
Methods: According to the guideline of blood transfusion and logical function formula of transfusion theory, the blood transfusion management software was developed to give specific functionality including the audit for pre-transfusion indications. Meanwhile, the software can run automatic calculation by given some parameters and generates a recommended transfusion value of red blood cell, hemoglobin, plasma, cryoprecipitate and platelet for different individuals. In addition, it can conduct the evaluation of curative effect on erythrocyte and platelet transfusion.
Results: Blood Precision Transfusion Management Software was successfully developed. Preliminary application results show that, the appropriateness of blood transfusion including blood transfusion indications and transfusion amount has achieved significant improvement compared with the data before use this software (2/80 vs 15/80, $P < 0.05$). In addition, there was a significant decrease in the average blood transfusion amount per inpatient and the average blood transfusion amount per operation ($P < 0.05$).
Summary/Conclusions: The Blood Precision Transfusion Management Software can significantly improve the rationality of clinical blood transfusion, and perform more accurate and effective blood transfusion and save blood resources. It is worthy of further popularization in clinical practice in future.

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CLINICAL APPLICATION OF PLATELET MATCHING IN PATIENTS UNDERGOING SURGICAL TREATMENT

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Background: Platelet matching is an effective strategy in the treatment of ineffective platelet transfusion. It has been shown that platelet matching can improve the platelet transfusion efficiency significantly in patients with hematologic diseases or cancer. Our researches focus on the patients who undergoing surgery which intraoperative massive blood transfusion including liver transplantation or valve replacement and who platelet transfusion is needed postoperative, to investigate the positive rate of platelet antibody and the effect of platelet matching in these patients, and to provide clinical evidence for finding effective ways to improve effective rate of platelet transfusion in these patients.
Aims: To explore the effect of clinical application of platelets match in patients who need platelet transfusion after surgery.
Methods: Randomly select 45 patients who need platelet transfusion after surgery in our hospital in 2017 as experimental group, and adopt Solid-phase agglutination test to undertake platelet antibody screening as well as platelet matching test, and infuse the matched platelets. Meanwhile select 45 patients who need platelet transfusion after the same surgery as control group and infuse random platelets. The effect of platelet infusion of two groups is evaluated by 24 h corrected count increment (CCI), and compare the efficiency of platelet transfusion between the two groups to evaluate the clinical effect of platelet matching in surgical patients.
Results: The positive rate of 45 patients' platelet antibody screening in experimental group is 53.3%(24/45), and 24 h CCI in experimental group is 12.7 ± 1.9; while in control group 24 h CCI is 7.0 ± 1.5. The difference between the two group is statistical significance ($P < 0.05$). In experimental group, the effective rate of matched platelet transfusion is 71.1%(32/45), while in control group the effective rate of random platelet transfusion is 44.4%(20/45). The difference between the two group is statistical significance ($\chi 2 = 6.6$, $P < 0.05$).
Summary/Conclusions: Surgical patients are likely to produce platelet antibodies due to intraoperative infusion of blood products, and platelets matching could improve the effective rate of platelet transfusion.

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PATTERNS IN BLOOD PRODUCT UTILISATION IN ORTHOTOPIC LIVER TRANSPLANTATION IN A TERTIARY HOSPITAL

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Background: Transfusion requirements in orthotopic liver transplantation (OLT) have reduced dramatically over the last two decades. Minimising bleeding and reducing unnecessary transfusions are key goals in the perioperative period. Advances in surgical and anaesthetic techniques including point of care testing have all contributed to a decline in allogeneic blood product use. There are limited studies that have shown a change in use of blood and blood products in OLT patients.
Aims: The aim of the study was to examine the changing trends in blood product use in orthotopic liver transplantation (OLT) patients over the years
Methods: A retrospective study of patients undergoing OLT at Flinders Medical Centre between 2003 and 2014 was conducted. Over the 12 year period there was no change in number of male patients or patient's age or MELD score. OLT patient details were extracted from the liver transplant database, patient demographics and morbidity from the hospital database and blood product use from the transfusion laboratory information system
Results: While there was no change in RBC transfusion rates, a high proportion (80%) of OLT patients received FFP at the start of the study period compared to only 28% in 2014 ($P < 0.001$) with a significant fall in the number of FFP units transfused from a median of 4 (2–5) in 2003 to 0 (0–1.5) in 2014 ($P < 0.001$). There was negligible use of cryoprecipitate between 2003 and 2007. However, the median dose of cryoprecipitate gradually increased to reach a peak in 2012, median of 4 doses (IQR 1.5–7) and subsequently dropped to a median of 2 doses in 2014 ($P < 0.001$). The proportion of patients receiving platelets initially started to increase from 2006 (69.2% of patients) and reached a peak in 2010 (85%) and remained stable until 2013 followed by a drop in 2014 (56%) ($P < 0.008$). The median volume of intraoperatively scavenged RBC reinfused in 2014 was 477 ml (240–736 ml) compared to a median of 1330 ml (778–2855 ml) in 2003 ($P < 0.001$).

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Summary/Conclusions: The single centre analysis shows a large reduction in FFP and an increase in use of platelets and cryoprecipitate over a 12 year period. This reduction has been associated with reduced intraoperative crystalloid use, but more importantly with the routine use of intra-operative viscoelastic assays to manage coagulopathy and guide appropriate transfusion practice during PLT.

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SEROLOGY APPROACHES TO RESOLVE ABO DISCREPANCY IN A THAI FAMILY

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Background: ABO blood group system is the most clinically important in transfusion medicine. Not only in acute hemolytic transfusion reactions and hemolytic disease of the fetus and newborn, but in paternity testing as well. The absence of H antigen on red cell membrane and in secretions together with potent anti-H in plasma result in Bombay phenotype which is very rare. In our population, another phenotype called para-Bombay is more common. They have low level of ABH, weak anti-H in plasma and can secrete ABH in secretions. Thus, result in ABO discrepancy which may cause wrong interpretation in paternity testing.
Aims: To demonstrate the serological solving in a Thai family with ABO discrepant results between the baby and the parents.
Methods: A 2-day-old newborn was found to be AB, Rh positive but the father and the mother were B and O, respectively. Due to the discrepant results, blood samples (clotted and EDTA) and saliva of the parents were sent to the Reference Laboratory, National Blood Centre, Thai Red Cross Society in Bangkok for further investigation.
Results: ABO cell grouping of the parents and the baby were performed by conventional tube test (CTT) with anti-A, anti-A1, anti-B, anti-A, B and anti-H. The results confirmed group B of the father and group AB of the baby. Interestingly, the mother's red cells gave negative reaction with all ABO antisera tested. Her plasma gave 4+ reaction with B cells and weak reaction with all panel cells tested at IAT, autocontrol was negative. To solve ABO discrepancy, saliva test was performed which A and H substances were detected. We concluded that the mother was A, para-Bombay with weak anti-H. Therefore, the baby inherited A antigen from the mother and B antigen from the father.
Summary/Conclusions: In order to solve ABO discrepancy, anti-H and anti-A1 should be included in ABO cell grouping using CTT. In addition, saliva test results could help confirm the correct ABO group.

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ASSESSMENT OF THE EFFECTIVENESS OF POINT OF CARE TESTING (ROTEM) IN BLOOD AND BLOOD COMPONENT THERAPY OF PATIENTS IN NATIONAL HOSPITAL OF SRI LANKA

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Background: Thromboelastometry (TEM) performed on the Rotational thromboelastometry (ROTEM) system, is an enhancement of the traditional thromboelastography (TEG) method. The rapid assessment of coagulation is the main advantage of ROTEM and it allows the effective monitoring of haemostatic treatment in real time. It is important to assess the effectiveness of this point of care testing in blood and blood component therapy, for timely treatment of patients with acute need of blood.
Aims: Assessment of the effectiveness of point of care testing (ROTEM) in blood and blood Component therapy
Methods: A descriptive retrospective study was carried out in blood bank of National Hospital of Sri Lanka (NHSL) during six months (July 2015 to January 2016) period.
Results: During the period of this study Blood bank of NHSL received 667 ROTEM samples for analysis. Samples were sent from Neuro trauma/surgery, Cardiothoracic, intensive care units, operation theatres, medical and surgical wards. Most of the samples (80.6%) were sent from Neurotrauma/ surgical, cardiothoracic and surgical ICUs and they were 59.2%, 11.5% and 9.9% respectively. ROTEM investigations are usually carried out for the management of patients with brain hemorrhage, post operative bleeding and active bleeding after trauma. 81% of the samples received had normal findings. Among the abnormal samples, abnormalities were corrected with the blood and blood components prescribed according to ROTEM results, in 97% of cases. Samples were sent to assess the coagulation problems of patients. But most of patients had no indication for blood component therapy according to ROTEM results. In 2015 total FFP and CPP usage have decreased but platelet, Cryo and RCC usage have increased compared to the usage in 2014
In both Neuro trauma/ surgery units and Cardiac units FFP usage is reduced and there is increased usage of platelet, cryoprecipitate and RCC. There is a significant percentage increase in FFP usage in cardiac surgery units ($P < 0.001$), but Cryoprecipitate usage has significantly reduced ($P < 0.01$) .In Neuro trauma/ surgery units there is a statistically significant percentage increase seen in platelet usage and changes in other components are not significant.
Summary/Conclusions: Samples for analysis were sent from bleeding patients or patients with bleeding tendency but most had normal finding and blood and blood components were not prescribed, showing proper patient blood management.
According to literature, most of the studies have shown that there is a reduction in plasma, cryo and platelet usage after introduction of ROTEM. Even though it has been shown that patient management using ROTEM is cost effective, component usage has increased with introduction of ROTEM testing in our setting. Patient management using ROTEM based algorithms have been implemented in most of the settings and there is reduced component usage after implementation of this new technology. In our setting, it was found that prescription of blood and blood components were done without using proper documented protocols or algorithms.
Component usage during the study period showed a change to what observed in previous year. It is recommended to introduce algorithm based management and it should be followed up with an assessment of the cost effectiveness after a reasonable period of time.

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BLOOD COMPONENT USAGE OF MITRAL VALVE SURGERY AT 22 GENERAL HOSPITALS IN CHINA

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Background: Since 2010, mitral valve surgery always top the cardiac valve operations in China. To better understand the baseline information of blood component use in mitral valve surgery, a survey was conducted at 22 tertiary general hospitals in China.
Aims: To better understand the baseline information of blood components transfusion at mitral valve operation in China.
Methods: A retrospective study of blood component transfusion of mitral valve surgery in 2015 was conducted in 22 Grade 3A general hospitals from 22 selected provinces in China. Paper-based questionnaire including the essential information, blood component use, perioperative bleeding and perioperative medication of the patient who had mitral valve surgery was distributed to every selected hospital by organizing center (Chinese Institute of blood transfusion). The organizing center collected the survey after all the hospitals were completed and then analyzed the data.
Results: In total, 1052 questionnaires were completed in 22 general hospitals and 86% of these patients were diagnosed with valvular heart disease in which 47.29% were caused by rheumatic heart disease. Transfusion rates of Red blood cells, plasma, platelet, cryoprecipitate were 74.05%, 78.23%, 12.26%, 20.34%, respectively and female was significantly higher than that of male ($P < 0.05$). Blood components (red blood cells, plasma, platelet, cryoprecipitate) transfused per mitral valve surgery were 3.68 unites, 676.63 milliliters, 0.80 unites and 2.12 unites, separately. When only considering the blood transfusion surgery, the usage of blood components per surgery were 4.96 unites, 855.33 milliliters, 5.33 unites and 9.43 unites, respectively.
Summary/Conclusions: General collection and analysis of such data not only provide basic information of blood component usage in mitral valve surgery but also can guide and standardize clinical blood transfusion. Further prospective or retrospective study was also needed to observe the trends of blood component transfusion in mitral valve operation.

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PLATELET TRANSFUSION PRACTICES IN ADULT SURGICAL, HAEMATOLOGY AND ONCOLOGY PATIENTS: A CLINICAL AUDIT

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Background: Platelet transfusions are indicated for the prevention and treatment of hemorrhage in patients with thrombocytopenia or platelet function defects. Platelet transfusion should be judicious and must be given only when there is clear clinical justification in order to reduce risks associated with transfusion, costs from production and possible shortages.
Aims: The aim of the study was to evaluate current practice of platelet transfusions in adult surgical, hematology and oncology patients against standards drawn from the BCSH (British Committee for Standardization in Hematology) guidelines for platelet transfusions. This would help to identify platelet ordering practices of physicians and the areas that need improvement.
Methods: A retrospective analysis of platelet transfusion therapy given to adult surgical, hematology and oncology patients at Aga Khan University Hospital was done in the month of June 2015. Medical charts of all patients who received platelet transfusion during study period were reviewed. The record was evaluated to review primary diagnosis, platelet count and indications for platelet transfusion. Current practice of platelet transfusion were compared against standards drawn from the BCSH.
Results: A total of 141 episodes of platelet transfusion occurred in the service of surgery, hematology and oncology patients. Out of which 72% (n = 101/141) of transfusion occurred in hematology patients, 10% (n = 14/141) in oncology patients and 18% (n = 26/141) in surgical patients. Among hematology and oncology patients the transfusion were mostly done with the prophylactic intent comprising 90% and 71% respectively. Whereas in surgery 73% of transfusion was done for therapeutic purpose. Eighty-nine percent (n = 90/101) of transfusions done in hematology was justified as per the BCSH guidelines. In oncology department 71% of transfusion done were rational. Whereas in surgical department only 31% of platelet transfusion were adherent to guidelines. Platelets were transfused in these patients having normal platelet count; but with active bleeding due to other surgical correctable causes.
Summary/Conclusions: The audit demonstrated a high rate of adherence of platelet therapy to BCSH standard in hematology and oncology patients. However the policy was significantly breached in surgical patients indicating an area for intervention in the form of physician's education to rationalize the use of platelet product.

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TRANSFUSION THERAPY IN HEMATOPOIETIC STEM CELL TRANSPLANTATION

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Background: Hematopoietic stem cell transplantation (HSCT) is one of the most effective methods to treat malignancies and hematological and inherited disorders. One of the main problems after HSCT is long-lasting pancytopenia, which usually leads to massive transfusion therapy. The presence of AB0-incompatibility between the donors of hematopoietic stem cells and recipients in a significant number of patients requires an individual approach in the implementation of transfusion therapy.
Aims: Analyze the results and optimize the algorithm of transfusion therapy in AB0-incompatible HSCT.
Methods: Over 2000–2015, 915 patients with HSCT (allogeneic related – 186, allogeneic unrelated – 469, haploidentical – 138 and autologous – 122) were included in the study. In 842 cases of HSCT 8,864 doses of red blood cells were transfused. In 795 HSCT cases (86.8%) – 10,636 doses of platelets (aphaeresis 2.0 x 10^{11} – 7,046 doses, dose from whole blood 0.6 x 10^{11} – 3,439 doses, pooled – 151 doses). 1,321 doses of fresh frozen plasma (FFP) – in 228 (24.9%) cases of HSCT. Transfusions were carried out according to the rules of Russian Association of transfusiologists and the instructions of Ministry of Health of the Russian Federation. Before HSCT, recipients underwent immunohematological screening: typing of the AB0 system, antigens of Rh system and Kell, and a direct antiglobulin test. Blood transfusions were performed after individual compatibility tests with indirect antiglobulin tests and in the post-transplant period, according to the engraftment status.
Results: 18.5 doses of blood components per patient were transfused in allogeneic HSCT cases, in autologous cases – 14.7 doses. Leukofiltrated and irradiated red blood cells, aphaeresis pathogen-inactivated FFP, and platelet concentrates prepared with aphaeresis technology and with leukocyte depletion and additive solutions are more clinically effective and safe for the patient. Side effects of blood transfusion included: 1 case of non-immune TRALI after blood transfusion, 7 cases of delayed hemolysis after blood transfusion in AB0- and Rh-incompatible (C, c, E, e) HSCT, and several cases of mild transfusion reactions. A selection scheme for transfusion of blood components was proposed, based on the results of patients' immunohematological tests.
Summary/Conclusions: The results of the current study and worldwide experience have allowed for the development and optimization of a transfusion support algorithm for autologous and allogeneic HSCT. It is recommended to use AB0 compatible, leukofiltrated, irradiated and pathogen-inactivated blood components.

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SONOCLOT ANALYSIS IN PATIENTS WITH CARDIAC SURGERY AND ITS CORRELATION WITH CONVENTIONAL COAGULATION TESTS

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Background: Sonoclot Analysis in Patients with Cardiac Surgery and Its Correlation It is important to monitor the coagulation status in patients undergoing cardiac surgery. The conventional coagulation tests including prothrombin time (PT), activated partial thromboplastin time (APTT), fibrinogen concentration (FIB), and platelet count (PLT) are mostly applied in clinical laboratories. The Sonoclot analyzer traces the transition of whole blood, from fluid to viscous clot, with a vibrating probe. Both clotting and late fibrinolytic state can be assessed with variables including activated clotting time (ACT), clot rate (CR), and platelet function (PF).
Aims: To study the correlation between the results of Sonoclot Analysis and the results of conventional coagulation tests in patients with cardiac surgery.
Methods: The test results of 281 patients with cardiac surgery were reviewed retrospectively. Pearson correlation was performed between Sonoclot variables (ACT, CR and PF) and conventional variables (PT, APTT, FIB, and PLT), then linear regression model was used to evaluate the association between conventional variables and Sonoclot variables.
Results: There was significant positive correlation between ACT and APTT or TT, the correlation coefficients were 0.622 ($P < 0.01$) and 0.528 ($P < 0.01$) respectively, and the regression equations were $Y = 1.524X+118.184$ and $y = 2.076x+141.336$. There was no significant correlation between ACT and PT, or FIB, or PLT. There was significant negative correlation between CR and APTT or TT, the correlation coefficients were -0.433 ($P < 0.01$) and -0.415 ($P < 0.01$) respectively, and the regression equations were $Y = -0.193X+30.344$ and $Y = -0.296X+28.164$. There was significant positive correlation between CR and FIB or PLT, the correlation coefficients were 0.449 ($P < 0.01$) and 0.206 ($P < 0.01$) respectively, and the regression equations were $Y = 8.241X+5.757$ and $Y = 0.041X+16.376$. There was significant negative correlation between PF and PT or APTT, the correlation coefficients were -0.2093 ($P < 0.01$) and -0.213 ($P < 0.01$) respectively, and the regression equations were $Y = -0.098X+3.385$ and $Y = -0.010X+2.558$. There was significant positive correlation between PF and FIB or PLT, the correlation coefficients were 0.389 ($P < 0.01$) and 0.615 ($P < 0.01$) respectively, and the regression equations were $Y = 0.771X+0.608$ and $Y = 0.013X+0.392$.
Summary/Conclusions: There is significant correlation between Sonoclot parameters and conventional coagulation parameters in patients with cardiac surgery.

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A PLATELET TRANSFUSION REFRACTORINESS CAUSED BY GBA GENE MUTATION IN GAUCHER DISEASE

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Background: Gaucher disease is a glucosylceramide lipidosis caused by deficient acid β-glucosidase enzymatic activity mainly due to GBA gene mutations. In this case, we describe a patient with type 1 Gaucher disease who experienced platelet transfusion refractoriness and Sequencing of GBA genomic identified one novel homozygous mutation, c.484A> G (p.Met162Val).

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Aims: To analysis the mutation characteristics of GBA gene in Gaucher disease led to one case of platelet transfusion refractoriness.
Methods: A 38-year-old female patient was diagnosed with anemia and thrombocytopenia. After she showed platelet transfusion refractoriness, bone marrow smear, β-glucocerebrosidase activity in leukocytes, B-ultrasonography, Sequencing technology and pedigree investigation were performed on the proband and family. β-glucocerebrosidase activity in leukocytes was detected by Dried Blood Spot. DNA of the proband and family was extracted from the peripheral blood and subjected to sequencing using the Sequencing technique.
Results: Pedigree investigation showed that the father, mother, son, daughter and sister of this proband were all heterozygous mutation for the GBA gene. Microscopic findings demonstrated Gaucher cells (accounted for 6.0%) in bone marrow and β-glucocerebrosidase activity in leukocytes was 3.78 nmoL/(h·mgPro)(lower than normal). Meanwhile B-ultrasonography showed spleen size 6.2 × 25 cm (slightly splenomegaly); Sequencing of GBA genomic and cDNA identified one novel homozygous mutation, c.484A> G (p.Met162Val).
Summary/Conclusions: The patients of Gaucher disease may appear platelet transfusion refractoriness due to the hypersplenism. In addition, the GBA gene mutation had its local characteristics and was the main pathogenic missense mutation, which contributes to Gaucher disease of the family.

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VERIFICATION TEST REPORT OF TWO NONINVASIVE DEVICES MEASURING HB

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Background: The measurement of hemoglobin (Hb) concentration in the blood plays an important role in the detection, evaluation, and management of chronic and acute anemia. Hb is usually measured as a part of the routine Complete Blood Count (CBC) test from a blood sample. Current methods for Hgb measurements generally do not provide an instant result and are not suitable for home use. We focus on the verification of two noninvasive cFDA approved devices (TensorTip MTX, Israel and Masimo Radical-7) as part of the exploration of selecting a noninvasive device to be used in the blood transfusion department.
Aims: The aim of the test case was to verify the accuracy of hemoglobin (Hb) measurements of each device independently with the hospital reference (Sysmex device model XN9000).
Methods: The Radical-7 Pulse CO-Oximeter ((Masimo Corporation, Irvine, CA) is an optical device that uses the principle of spectrophotometry to calculate Hb concentration. The TensorTip MTX is an optical device based on optical technology of real time color image sensor and uses four monochromatic light sources. The color image sensor in the TensorTip MTX provides temporal spatial color information for the calculation of the Hgb levels and other parameters. The subjects were selected from healthy volunteers. The Sysmex device model XN9000 was considered to be the gold-standard measure. Investigators recorded Hb levels by the Radical-7 or the TensorTip MTX according to the instructions for use. Blood samples were extracted simultaneously for Hb measurements by the hospital lab reference device. We compare to the accuracy of Hb between the two new devices and the reference method.
Results: We enrolled 97 subjects to the tests, including 71 subjects detected with TensorTip MTX vs reference, 26 subjects with Masimo Radical-7 vs reference. The Mean absolute difference between the TensorTip MTX device and the reference is 0.994 g/dL. The outcome of the mean positive difference is 0.507 g/dL and for the mean negative difference is -0.487 g/dL.
The TensorTip MTX measurements are not significantly different from the reference (P = 0.934).
Masimo Radical-7: The mean absolute difference result is 1.442 g/dL. The mean positive and negative differences are 0.096 g/dL and -1.346 g/dL, respectively, indicating unbalanced measurements around the reference.
The Masimo outcome is significantly different from the Lab reference (P = 4.772E-05).
Summary/Conclusions: The current verification test clearly shows that the Tensortip MTX device provides results within the accepted industrial criteria and a better accuracy than the Masimo Radical-7 when compared to the hospital reference device. However, the effect of less data with the Masimo on the results cannot be eliminated. Noninvasive Hb detection equipment can be used for human health management and patients of requiring blood transfusion.

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A MULTICENTER STUDY OF PLASMA AND PLATELET USE IN CHINA BASED ON A SAMPLING OF 24 HOSPITALS

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Background: There are a little information regarding plasma and platelet use in the China, it is needed to survey for practice improvement of plasma and platelet therapy.
Aims: The purpose of this study was to survey detailed medical information of plasma and platelet using from Grade 3A hospitals in China and operation to identify patterns of blood use that may be used for blood program planning and transfusion audits.
Methods: A blood utilization survey was performed by collecting detailed medical information from 24 Grade 3A hospitals (tertiary hospital) for all patients transfused with plasma and platelet, one each from the 24 selected provinces in China, from January 1, 2014 to December 31, 2014. A standard data collection form was distributed to the selected hospitals by the coordinating center (Chinese Institute of Blood Transfusion).
Results: The distribution of major clinical departments of plasma clinical application is 42% of surgery, 27% of medicine, 15% of ICU, 10% of gynaecology and obstetrics, 3% of pediatrics and 3% of other departments; The distribution of major clinical departments of platelet clinical application is 17% of surgery, 48% of medicine, 6% of ICU, 19% of gynaecology and obstetrics, 6% of pediatrics and 4% of other departments.
Summary/Conclusions: This is the first blood utilization survey performed in China in large scale. This data provided the baseline status of blood use in China; it can be used to guide future epidemiology study as well as clinical trials.

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ANALYSIS OF CURRENT STATUS OF CHRONIC RED BLOOD CELL TRANSFUSION

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Background: Chronic red blood cell (RBC) transfusion is widely practiced as part of supportive care in patients with chronic anemia. However, there were few reports on the actual situation of that.
Aims: We aimed to study the current status of chronic RBC transfusion.
Methods: Among patients who received RBC transfusions at a regional hospital from January 2011 to December 2015, patients who were consistently transfused with more than 15 units of RBC for more than one year were included. We retrospectively analyzed the status of RBC transfusion, including the laboratory findings related to transfusion using medical records.

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Results: A total of 58 patients were included in the study. The mean age of patients was 66.4 years, and there were 24 hematologic malignancies (41.4%), 20 malignant tumors (34.5%), and 14 chronic diseases (24.1%). The mean hemoglobin level at the beginning of transfusion was 7.0 g/dL, and eight patients (13.8%) showed RBC alloimmunization as the transfusion proceeded. The mean serum ferritin level at the initial phase of transfusion was 475.3 ng/ml and increased to 1,462 ng/ml during the late phase of transfusion.
Summary/Conclusions: Since the number of elderly patients treated with chronic diseases including tumors is expected to increase significantly, patients with chronic transfusions are also expected to increase. It is necessary to research and prepare measures such as establishment of guidelines.

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TO DETERMINE THE EFFECTIVENESS OF ALBUMIN ADMINISTRATION IN BURN INJURIES

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Background: Albumin, a natural colloid, is commonly used for fluid resuscitation and treatment on burn and critically patients due to less anaphylactic reaction and less infection rate compared with fresh frozen plasma (FFP). It decreases " fluid creep", a phenomenon of using more resuscitation fluid than predicted. Fluid resuscitation is mainly to maintain organ functions and to decrease edema through minimal volume intake. Albumin is an ideal fluid to reach the goal by oncotic pressure and improve hypoalbuminemia and supply nutrients, so it is reported to reduce mortality, compartment syndrome and other complications.
On the contrary, some studies indicated that the group using albumin as resuscitation fluid in burn patients, and patients with hypoalbuminemia has higher mortality rate. Color Dust explosion at Formosa Fun Coast waterpark happened in Taiwan on June 27, 2015, caused 15 deaths and 484 injuries. Large amount of albumin was used for treatment in this devastating event.
Aims: To determine whether albumin use is safe and effective.
Methods: The single variable and multiple Cox proportional hazard models to estimate hazard ratios (HRs) and corresponding 95% confidence intervals (CIs) were performed. The study was used SPSS statistical package.
Results: In univariate analysis, only transfusion of albumin increased length of stay (LOS) ($P < .05$). After adjusting cofounding factors such as TBSA%, inhalation injury and blood product transfusion, albumin administration are independent risk factors to increase LOS. The use of albumin decreased the probability of discharge in patients with mild status, including the absence of inhalation injury and acute kidney failure within 5 days admission, meeting <3 items of ABA sepsis criteria at admission and concurrent involvement of the above 3 conditions. (All $P < .01$).
Summary/Conclusions: Severe burn patients with larger TBSA were reported to have poor prognosis and hypoalbuminemia which led to organs dysfunction, so using albumin could increase the probability of discharge in patients with TBSA50%. Increasing bronchial circulation induced airway mucosal edema and pulmonary transvascular flux of fluid, neutrophils and inflammatory mediators, and then increased reactive oxygen species (ROS) and cell damage following inhalation injury. Albumin can reduce extravascular volume by its oncotic pressure and the level of ROS by its antioxidant properties. Those clarify the therapeutic effects of using albumin significantly in presence of inhalation injury.
Acute renal failure (ARF) mainly is caused by hypovolemia state postburn and can be prevented by adequate fluid resuscitation, so the albumin use in the patients without ARF had lower hazard ratio of discharge rate.
In addition, the patients meeting count of ABA sepsis criteria < 3 items had lower hazard ratio of discharge rate which is consistent with the result of albumin use in sepsis patients in ALBIOS study and EARSS study and subsequent findings which indicate the use of albumin didn't increase a survival rate, although hemodynamic improvements are obvious.
Sum up, our study not only provides us another aspect that albumin use in burn patients through demonstrating that albumin use in patients with mild status might be detrimental, but also opens up new horizons which in turn lead us to a deeper thinking the indications of albumin in different clinical conditions.

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EFFECTIVITY AND SAFETY OF SMALL DOSE RED BLOOD CELLS TRANSFUSION IN SEVERE HEMOLYTIC ANEMIA PATIENTS

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Background: Severe hemolytic anemia is a disease threaten to human life and health. Transfusion is one of the main therapy. However, large doses of xenogenous red blood cells infused may aggravate hemolysis and kidney damage. Without transfusion, patients may arise serious hypoxia, even multiple organs failure. In our study, we discovered that small dose red blood cells transfusion could improve hemoglobin of the severe hemolytic anemia patients.
Aims: To observe the therapeutic result and safety of severe hemolytic patients infused small dose of homotype red blood cells.
Methods: In the study, we screened the small dose red blood cells (0.5U to 1U) with the same antigen of DEeCc and MNS for these severe hemolytic anemia patients. After blood transfusion, we observed with the variation of hemoglobin, bilirubin and creatinine value in the first day and the third day.
Results: 17 patients were included, which treated with small dose red blood cells in Xiangya Hospital from June, 2014 to June, 2017. The average age was 42.7 years (from 16 years to 69 years). The average hemoglobin, total bilirubin and creatinine of these patients were 34 g/l, 59.1μmmol/l and 111.1μmmol/l before transfusion. After transfusion, the hemoglobin value was increased and the bilirubin values were decreased. The variation had statistical significance ($P<0.05$). The alter of creatinine value had no statistical significance ($P>0.05$).
Summary/Conclusions: Conclusions: Treated severe hemolytic anemia patients with small dose of red blood cells could improve hemoglobin, and it wasn't aggravate hemolysis and renal damage. So we deemed it was a safety and effective therapeutic method.

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Abstract has been withdrawn

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CLINICAL EFFECTS OF WASHED RED BLOOD CELLS SUSPENDED IN MAP AND NORMAL SALINE

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Background: Normal saline washed red cells are a special component prepared only on demand for patients with antibodies to plasma protein and those who have severe allergic reactions when transfused with blood products. The normal saline washing is done thrice times by manual or automated methods, the final product should be suspended in normal saline or MAP red blood cells storing solution. But it is unclear that clinical effects of washed red blood cells suspended in MAP and normal saline
Aims: To explore clinical effects of washed red blood cells suspended in MAP red blood cells storing solution and normal saline and to provide theoretical basis for clinical accuracy of blood transfusion;
Methods: Retrospective analysis was performed on 300 patients who were treated with washed red blood cells from 2016.6 to 2016.12. These patients were divided into two groups, one group was infused with washed red blood cells suspended in normal saline (saline group), the other group was infused with washed red blood cells suspended in MAP (MAP group). All patients were infused with 2U corresponding washed red blood cell products, then Blood Routine Tests were measured before transfusion and 24 h after transfusion. The changes of hemoglobin (Hb) levels were

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observed before and after treatment in two groups, At the same time, the adverse reaction of blood transfusion was observed.
Results: After blood transfusion, Hb levels of patients in two groups were improved, and compared with before and after transfusion were significantly difference ($P < 0.05$); but there was no significantly difference between the two groups ($P > 0.05$); There were two patients who had adverse reaction of blood transfusion in MAP group, the other group had no adverse reaction of blood transfusion, but the incidence of adverse reactions of blood transfusion was no significantly difference between the two groups ($P > 0.05$);
Summary/Conclusions: Two kinds of washed red blood cells were safe and effective, but for patients with renal dysfunction, the elderly patients with recurrent transfusion of hemolytic anemia, it is suggested that as far as possible to infuse with washed red blood cells suspended in normal saline in 24 h.

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ANALYSIS OF INFLUENTIAL FACTORS OF CLINICAL PROGNOSIS IN TRAUMATIC PATIENTS

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Background: The effect of blood transfusion on the prognosis and prognosis of trauma patients is not clear and needs further study.
Aims: To investigate the prognostic factors of trauma patients by retrospective case-control study on clinical data of trauma patients.
Methods: Retrospective analysis of all the trauma patients' clinical data in the first affiliated hospital of Nanchang university from January 1, 2013 to December 31, 2016, a total of 419 cases were included. According to the units of RBC transfusion within 24 h after admission, the patients were divided into transfusion group and non-transfusion group, and including 265 cases of transfusion group, 154 cases of non-transfusion group. Analyzed and compared two-group patients' clinical data and clinical prognosis indicators, and used Logistic regression analysis to analyze the risk factors related to in-hospital mortality, to explore the influence factors of clinical prognostic of trauma patients.
Results: There was no significant difference in sex, age and cause of injury between the patients with transfusion and non-transfusion ($P > 0.05$). The difference between two groups with injury severity score (ISS), glasgow coma scale (GCS), heart rate (HR), systolic blood pressure (SBP), hemoglobin (Hb) and blood coagulation indexes were statistically significant ($P < 0.05$). The 60 days mortality rate, total mortality rate, the rate of nosocomial infection, time of mechanical ventilation, ICU stay and total length of stay in patients with transfusion group were significant higher than those in non-transfusion group ($P < 0.05$), no adverse reactions of blood transfusion occurred in transfusion group. Univariate logistic regression analysis showed that trauma type, ISS, HR, SBP, HB, Fbg, INR, RBC and plasma transfusion volume, nosocomial infection, time of mechanical ventilation, ICU stay and total length of stay was the risk factor of in-hospital death in trauma patients ($P < 0.05$). Multivariate logistic regression analysis showed that trauma type, RBC transfusion volume and ICU stay were the independent influence factors of in-hospital death in trauma patients ($P < 0.05$).
Summary/Conclusions: Increased RBC transfusion, open trauma, and prolonged ICU hospital stay were independent risk factors for trauma patients' in-hospital mortality. Traumatic patients with early positive hemostasis, a reasonable blood transfusion can reduce the total blood use, reduce the incidence of complications of patients, improve patient clinical prognosis and reduce patient mortality.

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EXTENSIVE TRANSFUSION OF RHD-POSITIVE BLOOD IN AN "ASIA TYPE" DEL PATIENT

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Background: Among D variants known as DEL, the most common is *RHD* (1227G>A), which is also referred to as "Asia type DEL". Transfusions of red blood cells from Asia type DEL donors were reported to induce anti-D alloimmunization. Patients who carry the DEL variant (*RHD* 1227G>A) have been reported to receive transfusions from RhD-positive donors without the risk of alloimmunization. Recently, the Korean national transfusion guideline (2016) allowed the transfusion of RhD-positive blood products to Asia type DEL patients when shortage of RhD-negative blood is eminent and in cases of emergency
Aims: We report our experience of extensive use of RhD-positive blood products in a patient with Asian type DEL blood type
Methods: RhD typing was done with monoclonal–polyclonal blend anti-D (Bioclone, clone MAD2, Ortho Clinical Diagnostics, Raritan, NJ, USA) and human IgG/IgM Monoclonal anti-D (Bioscot, Millipore, Livingston, UK). Extended phenotyping of Rh was done with monoclonal anti-C (Bioclone, Ortho Clinical Diagnostics), -c, -E and –e (Bioclone, Ortho Clinical Diagnostics). Genotyping was done by PCR on 4 loci of the *RHD* gene (promoter, intron 4, exon 7 and exon 10) and direct sequencing on exon 9.
Results: A 29-year-old man was admitted to our hospital with cardiac arrest due to ventricular tachycardia. He was diagnosed with severe heart failure due to left isomerism with complex congenital heart disease. ABO and RhD blood grouping results were group A and RhD-negative. Extended Rh phenotyping was CcdEe, and antibody screening test was negative. *RHD* genotyping confirmed the patient as Asia type DEL (*RHD* 1227 G>A) variant. On 11 day of admission, transfusion orders was issued due to persistent thrombocytopenia and hemorrhage at the removal site of intravenous central line. Inventoried RhD-negative platelets was short, and one unit of RhD-positive SDP was transfused along with injection of 1,500 IU of Rhogam. Following antibody identification tests showed the presence of anti-D, which converted to negative after 70 days. The patient received heart transplantation and received a total of 37 blood transfusions for 17 days after. Inventory of RhD-negative red blood cells was short, and 4 units of RhD-positive blood were transfused to account for emergencies. Antibody screening and direct coombs tests were persistently negative during our follow up for three months after RhD-positive blood transfusion.
Summary/Conclusions: We have demonstrated that RhD-positive blood can safely be transfused to Asia type DEL patients (1227G>A) without risk of anti-D alloimmunization.

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ELEVATED IMMUNOSUPPRESSIVE ACIDIC PROTEIN CORRELATES WITH HEPATIC ENZYME LEVELS IN PATIENTS RECEIVING ALLOGENIC BLOOD TRANSFUSION

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Background: Allogenic blood transfusions produce generalized immunosuppression in the recipient. Previous reports suggest that transfusion of blood in patients with no known liver diseases produce an increase in transaminases. This may be result from increased vascular permeability due to possible leukocyte antibodies or secondary causes.
Aims: The relationship between the immunosuppressive effect of blood transfusion and liver function enzymes was observed in this study.
Methods: Institutional ethical approval and informed consent of participants was obtained in this cross sectional study. Participants were trauma patients receiving care in a tertiary hospital facility. 5 ml of pre and post transfusion (48 h) blood samples was obtained from consenting subjects into K3 EDTA and Plain sample bottles after the completion of a structured questionnaire. Blood samples were also obtained from units transfused to the subjects. Platelet counts, aspartate aminotransferase (AST), alanine transferase (ALT), alkaline phosphate (ALP), plateletcrit (PCT), platelet distribution width (PDW), mean platelet volume (MPV), Total Protein estimation and Albumin estimation was determined on the samples using standard protocols. Thrombopoietin and Immunosuppressive acidic protein (IAP) was determined by ELISA.
Results: There was a significant rise ($P < 0.05$) in the mean value of IAP observed after transfusion (712.10 ± 512 μg/l) compare to pre-transfusion values (662 ± 190 μg/l). It was observed that Platelet counts, MPV, Total protein and Albumin levels were significantly lower ($P \leq 0.05$) in patients who received transfusion of banked blood, while AST, ALT, ALP, and thrombopoietin levels were significantly

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increased ($P \leq 0.05$) in patients receiving allogeneic blood transfusion. There was no correlation between the number of units received and the levels of the hepatic biomarkers but a strong correlation was observed with the levels of IAP.
Summary/Conclusions: The transfusion of allogeneic blood is associated with immunosuppression, which is evidenced by elevated levels of immunosuppressive acidic protein. The rise in levels of post transfusion IAP correlates with hepatic enzyme levels in patients receiving allogeneic blood. This has implications for patients with graft vs host disease or immunodeficiency requiring blood transfusion.

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HOW DO WE TREAT REFRACTORY CASES OF THROMBOTIC THROMBOCYTOPENIC PURPURA

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Background: Thrombotic Thrombocytopenic Purpura (TTP) is a rare but serious disease condition that can have a high fatality rate if not treated promptly. The characteristic feature associated with TTP is deficiency of von Willebrand factor (vWF) cleaving protein ADAMTS13 (a disintegrin and metalloproteinase, with a thrombospondin type 1 motif, member 13) which in turn activates the thrombotic system causing microvascular thrombi, macroangiopathic haemolytic anemia (MAHA) and organ ischemia.
The current protocol for treatment of TTP basically includes intensive plasma exchange with Plasma as a replacement fluid along with immunosuppressant such as steroids, however in few cases, the patients might not show clinical improvement or deteriorate after prolonged plasma exchanges. In such cases, second line of treatment in the form of rituximab can be considered.
We present a case who was diagnosed with TTP and was being treated for same with plasma exchanges and steroids, the patient responded well initially and his platelet count rose to 150,000/ul on day 8 however, as soon as the plasma exchange was tapered the platelet count decreased to 40,000/ul on day 10.
Aims: To study the treatment protocols to be followed in cases who are refractory to conventional Plasma exchanges.
To study the efficiency of Rituximab as a second line of treatment in cases who are refractory to conventional plasma exchanges.
Methods: A 34-year-old male patient presenting with chief complains of pain in abdomen, fever, purpuric rashes and two episodes of loss of consciousness within 24 h was referred to our hospital. On further investigations, Laboratory reports showed anemia and thrombocytopenia and peripheral smear showed schistocytes, suggestive of microangiopathic hemolytic anemia, and raised serum lactate dehydrogenase (S-LDH); the features were suggestive of classic pentad of TTP, no other specific investigations were done, the patient was diagnosed to be case of Thrombotic Thrombocytopenic purpura (TTP) and was urgently treated with plasma exchanges and steroids.
Results: During treatment, the patient responded well to plasma exchanges and his platelet count rose from 15000/ul baseline to 150000/ul on day 8, but on tapering the plasma exchanges the platelet count decreased to 40000/ul on day 10, indicating it to be case of refractory TTP.
The patient was then started with rituximab 375 mg/m2 per week for 4 weeks, which increased the clinical condition of patient significantly.
Summary/Conclusions: The case emphasizes the role of rituximab in cases with refractory TTP and concludes that rituximab should be considered in TTP patients who fail to respond after 7–14 days of standard treatment with daily plasma exchanges and glucocorticoids.

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Abstract has been withdrawn

Haemorrhage and Massive Transfusion

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THE EFFECT OF STORED PLATELET TRANSFUSION ON THE INFLAMMATORY PROCESS OF HEPATIC TISSUE IN A SERUM AMYLOID A-LUCIFERASE TRANSGENIC MICE

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Background: Serum amyloid A (SAA), a major acute phase protein predominantly secreted by hepatocytes, increased up to 1000-fold after infection or trauma. Platelets are essential for primary hemostasis and are also involved in inflammation and immune regulation. The storage temperature (such as 22°C or −80°C) and storage time (0 – 5 days at 22°C) of platelets is associated with the intensity of inflammatory response after platelet transfusion.
Aims: to determine the effect of stored platelet on inflammatory process of hepatic tissue by using SAA transgenic mice.
Methods: Male SAA-luciferase transgenic mice were subjected to a volume-controlled hemorrhage by withdrawal of 30% of blood volume (21 ml/kg body weight) over 1 h, and then were transfused with fresh platelets (FPs), liquid-stored platelets (LPPs), Cryopreserved platelets (CPPs), or fresh frozen plasma (FFP) respectively at a volume equal to the blood loss. A control group received no transfusion. The induction of liver luciferase expression was detected at 1 h, 3 h, 6 h, 9 h, 24 h, 48 h and 72 h after transfusion by bioluminescent imaging system.
Results: The changes of SAA luciferase activity with time were similar among groups with a trend of increasing first and then decreasing. However, the peak time was different with 6 h after FFP or FPs transfusion, 9 h after LPPs transfusion and 24 h after CPPs transfusion. The peak SAA luciferase activity was highest in the LPPs group, and then in the CPPs group, followed by the FPs group and the FFP group. Compared with the control group, the SAA luciferase activity was lower in the FFP group at each time point, but higher in other groups.
Summary/Conclusions: Transfusion with LPPs or CPPs contributes to SAA expression in hepatic tissue. *In vivo* monitoring the SAA-luciferase activity could be use to investigate the effects of platelet transfusion on the inflammatory response of the liver.

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EARLY CORRECTION OF COAGULOPATHY CAN REDUCE THE 28 DAY MORTALITY IN ADULT PATIENTS WITH MAJOR BURN

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Background: Early stage correction of coagulopathy has not been found to reduce the 28 day mortality rate in adults with major burn.
Aims: Investigate the relationship between early correction of coagulopathy and the 28 day mortality in patients with major burn.
Methods: 503 cases of patients with burn injury from January 2015 to May 2017 were found in Jiangxi province burn center, 73 cases of major burn transfusion patients with coagulopathy on admission were included in the study, according to early coagulopathy (on admission 48 h) was corrected or not, the patients were divided into the experimental group (early coagulopathy correction group, n = 40) and control group (early coagulopathy without correction group, n = 33), compared with two groups of basic clinical data and clinical prognosis indicators, logistic regression analysis and survival curve were also used.
Results: (1)The gender, age, weight, burn causes, inhalation injury, INR, APTT, 48 h urine volume and fluid volume after admission, the amount of frozen plasma transfusion and INR in 48 h after admission were not significantly different ($P > 0.05$);

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the two groups in TBSA, III degree burn area, 24 h urine volume and fluid volume, the amount of fresh frozen plasma and APTT in 48 h were significantly different ($P < 0.05$); (2) Comparison showed related prognostic index: two groups' burn ICU hospitalization time and hospitalization time were not statistically significant different ($P > 0.05$);mechanical ventilation time of experimental group were longer than the control group, the 28 day mortality rate was lower than that of the control group, the differences were statistically significant ($P < 0.05$); (3) The results of Logistic regression univariate analysis showed TBSA, III degree burn area, inhalation injury, length of stay, mechanical ventilation time, the amount of frozen plasma and the amount of fresh frozen plasma transfusion and correction of coagulopathy in 48 h after admission had effects on 28 days mortality of major burn transfusion patients with coagulopathy ($P < 0.05$); the two groups in gender, age, weight, burn causes, burn ICU hospitalization time, INR, APTT on admission, the amount of fresh frozen plasma transfusion after 48 h, INR and APTT after 48 h had no effects on 28 day mortality ($P > 0.05$); (4) The results of multivariate Logistic regression analysis showed that mechanical ventilation time, length of stay, inhalation injury were not independent influence factors ($P > 0.05$);TBSA, III degree burn area,and the amount of frozen plasma transfusion and correction of coagulopathy in 48 h after admission were independent influence factors ($P < 0.05$). (5) Survival curve analysis of major burn transfusion patients with coagulopathy showed that with different TBSA, different III degree burn area, different amount of frozen plasma transfusion and whether coagulopathy corrected or not in 48 h, the difference in 28 day survival rate were statistically significant (χ^2=24.128, 41.688, 26.055, 18.452, $P < 0.05$)

Summary/Conclusions: TBSA, III degree burn area, the amount of frozen plasma transfusion and correction of coagulopathy in 48 h after admission are independent influence factors on 28 day mortality. Increasing the amount of frozen plasma transfusion, especially the fresh frozen plasma transfusion, can achieve the dual purpose of anti shock and correction of coagulopathy, improve the prognosis and reduce the mortality of 28 days.

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ANALYSIS OF INFLUENCE OF CLINICAL OUTCOMES OF CRYOPRECIPITATE INFUSION IN PATIENTS WITH EARLY POSTPARTUM HEMORRHAGIC SHOCK

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Background: The effect of cold precipitation on the prognosis and prognosis of postpartum hemorrhage and hemorrhagic shock is uncertain.

Aims: To investigate the influence of cryoprecipitate on clinical outcome of patients with early postpartum hemorrhagic shock.

Methods: Retrospectively analyze 63 patients with early postpartum hemorrhagic shock in our hospital. According to the presence of cryoprecipitate infusion divided into research group (transfusion of red blood cells, plasma, platelet and cryoprecipitate, $n = 34$) and the control group (infusion of red blood cells, plasma, platelet, $n = 29$), and analyzed two groups of patients' clinical data characteristics, blood transfusion volume, liquid input volume, the curative effect after transfusion, and clinical outcome prognosis related indicators.

Results: 1) The age, delivery mode, pathogenesis, hemostatic and surgical procedures of two groups had no statistical significant difference ($P > 0.05$); 2) Compare before and after transfusion: the level of fibrinogen on research group had statistical differences 0.83 ± 0.22 vs 1.46 ± 0.32 ($P < 0.05$), while the difference had no statistically significance on control group, PT and APTT of two groups had improved after transfusion ($P < 0.05$). Compare research group and control group: bleeding amount (ml) were 1476 ± 481.45 vs 1972 ± 633.78 ($P < 0.05$), the average infusion dosage of red blood cells (U) respectively were 7.53 ± 1.48 vs 9.72 ± 2.19 ($P < 0.05$), the infusion dosage of plasma (ml) respectively were 826 ± 104.43 vs 1041 ± 221.89 ($P < 0.05$), the platelet infusion volume (U) respectively were 24 ± 3.0 vs 26 ± 5.3 ($P > 0.05$), liquid input volume (ml) respectively were1305 ± 256 vs 2143 ± 309 ($P < 0.05$), the improvement rate were 88.2% vs 62.0% ($P < 0.05$), the incidence of last-stage shock were 2.9% vs 13.8% ($P < 0.05$), the incidence of pulmonary edema were 8.8% vs 24.1% ($P < 0.05$), the average hospitalization days (d) were 10.73 ± 2.46 *vs* 17.88 ± 3.15 ($P < 0.05$).

Summary/Conclusions: Early infusion of cryoprecipitate can correct hypofibrinemia and control bleeding timely, can reduce the volume of red blood cells and plasma in patients with early postpartum hemorrhagic shock, then reduce the incidence of complications, and have a better clinical outcome of patients.

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ESTABLISHMENT AND DIAGNOSTIC PERFORMANCE EVALUATION OF MASSIVE BLOOD TRANSFUSION PREDICTION SCORING SCHEME FOR TRAUMA PATIENTS

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Background: Early start massive transfusion protocol to prevent and correct blood coagulation dysfunction, reduce blood loss of trauma patients would improve patient survival rate. But accurate prediction of early trauma patients with massive blood transfusion requirements is still a huge challenge for clinicians.

Aims: Through retrospective analyze the clinical data of trauma patients with blood transfusion, to explore the influence factors of red blood cell (RBC) transfusion volume and establish a prediction scoring scheme of massive blood transfusion and evaluate its clinical prediction accuracy.

Methods: According to the RBC transfusion volume in 24 h after admitted to the hospital, we divided the 265 trauma patients which were admitted in our hospital from January 1, 2013 to December 31, 2016 into massive transfusion group (RBC transfusion≥18u) and non-massive transfusion group (RBC transfusion<18u). Analyzed the clinical data of two groups and made regression analysis of related factors of RBC transfusion, to establish and evaluate massive blood transfusion prediction scoring scheme.

Results: There was no significant difference in age, sex and cause of injury between the two groups ($P < 0.05$), but there were significant differences in other clinical indexes between the two groups ($P < 0.05$). Multivariate regression analysis shows that the type of trauma, injury severity score, heart rate, hemoglobin, prothrombin time, fibrinogen, and base excess were the independent predictor of RBC transfusion in trauma patients with blood transfusion ($P < 0.05$). Massive blood transfusion prediction scoring scheme was established and the prediction score of massive blood transfusion 0–8. Receiver operating curve analysis results show the area under the curve was 0.914, when the score was 4 points, the sensitivity and specificity were 88.9% and 79.9%.

Summary/Conclusions: Massive blood transfusion prediction scoring scheme has a better sensitivity and specificity, start massive blood transfusion protocol and give transfusion therapy when the trauma prediction score more than 4 points can decrease blood loss and blood transfusion, improve the clinical prognosis of trauma patients and the survival rate of patients.

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INVESTIGATION OF THE RELATIONSHIP BETWEEN THE AMOUNT OF FROZEN PLASMA INFUSION AND THE PROGNOSIS OF PATIENTS WITH MAJOR BURN

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Background: Whether the amount of frozen plasma transfusion during shock stage affect 30 day mortality in major burn are remain unclear.

Aims: Investigate the relationship between the amount of frozen plasma infusion at shock stage and the prognosis of patients with major burn

Methods: By using the whole closed-loop intelligent path management and information-evaluated retrieval system (self-developed),463 cases of patients with large area burn from January 2015 to January 2017 were found in Jiangxi province burn center, 86 cases were included in the study. Using the median amount of infusion of fresh frozen plasma as the packet threshold divided the patients into experimental group (frozen plasma transfusion <2000 ml and $n = 41$) and control group (frozen plasma transfusion≥2000 ml and $n = 45$).Compare two groups' basic clinical data and clinical prognosis indicators. Use the logistic regression analysis simultaneously.

Results: (1) the differences between the two groups in age, weight, TBSA, III degree burn area, urine volume, fluid volume, burn causes and inhalation injury number were not significant ($P > 0.05$); men were more than women in the two groups ($P < 0.05$); (2) comparison showed related prognostic index: two group burn ICU hospitalization time and hospitalization time were not significantly different ($P > 0.05$);mechanical ventilation time of experimental group were longer than the control group, the 30 day mortality rate was higher than that of the control group, the differences were statistically significant ($P < 0.05$); (3) the results of Logistic regression univariate analysis showed TBSA, III degree burn area, mechanical ventilation time, inhalation injury and the amount of frozen plasma transfusion during shock stage had effects on 30 days mortality ($P < 0.05$); (4) the results of

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multivariate Logistic regression analysis showed that TBSA and the amount of frozen plasma transfusion during shock stage were independent influence factors ($P < 0.05$).
Summary/Conclusions: TBSA and the amount of frozen plasma transfusion during shock stage were independent influence factors of 30 days mortality in patients with major burn. In the early stage of large area burn patients, frozen plasma should be used as the anti shock therapy as soon as possible to improve the prognosis and reduce the mortality in 30d.

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THE EFFECTS OF AUTOLOGOUS BLOOD TRANSFUSION ON COAGULATION FUNCTION IN PATIENTS WITH CARDIAC SURGERY

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Background: The effect of autologous blood transfusion on coagulation function is not clear.
Aims: Analyze the effect of blood transfusion on the function of coagulation in patients with cardiac surgery.
Methods: 216 patients with cardiac surgery using autologous blood transfusion technique were divided into 3 groups according to the intraoperative autologous blood transfusion. The blood transfusion volume of group A was < 200 ml, the blood transfusion amount of group B was 200 ml - 500 ml, and the blood transfusion amount of the group C was > 500 ml. Test the level of hemoglobin (Hb), red blood cells deposited (Hct), platelet count (Plt), prothrombin time (PT), partial prothrombin time (APTT) and fibrinogen (FIB) before the operation, and the levels before and after autologous transfusion .
Results: The Hb and Hct of group A were improved before autologous transfusion ($P < 0.05$), and the Plt and coagulation function indexes were not significantly different ($P > 0.05$). In group B, Hb and Hct were improved before autologous transfusion, PT and APTT extended, FIB decreased ($P < 0.05$). In group C, the Plt and coagulation function decreased more obviously after the autologous transfusion ($P < 0.05$), and the Hb and Hct were significantly improved after the autologous transfusion ($P < 0.05$).
Summary/Conclusions: Intraoperative autologous blood transfusion can effectively reduce the dosage of allogeneic blood transfusion in cardiac surgery patients, but for surgery patients with severe blood loss, blood coagulation function of patients should be monitored and fresh plasma, platelets or other blood coagulation factors should be supplied promptly.

P-550
MEASURING THE EFFECT OF RIVAROXABAN WITH THROMBOELASTOGRAPHY

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Background: The direct factor Xa inhibitor rivaroxaban does not need routine drug monitoring, but a rapidly available test might be beneficial in situations such as thrombotic events or bleeding complications, the need for urgent surgery or drug overdose.
Aims: The aim of this study was to assess the value of thromboelastography (TEG) in determining rivaroxaban administration.
Methods: The whole blood samples obtained from 20 healthy volunteers were spiked with rivaroxaban to yield samples with different drug concentrations ranging from 0 ng/ml to 600 ng/ml. Each sample was performed TEG kaolin test. Whole blood samples obtained from 20 patients before rivaroxaban therapy and 3 h after therapy were also perform TEG kaolin test.
Results: There was a strong correlation between rivaroxaban plasma concentrations and the TEG parameters R time (spearman correlation coefficient 0.79). R time was significantly prolonged in samples obtained 3 h after rivaroxaban therapy compared to those obtained before therapy ($P < 0.05$).
Summary/Conclusions: TEG could be a valuable test for rapidly determination of the effect of rivaroxaban at the point of care when needed.

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TO INVESTIGATE THE RISK FACTORS OF OBSTETRIC TRANSFUSION

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Background: Obstetric hemorrhage is the major leading cause for the death of obstetrical patients.
Aims: To investigate the risk factors of obstetric transfusion.
Methods: 9000 cases of obstetrical inpatients from December 2015 to December 2016 were analyzed by using univariate analysis and logistic regression analysis.
Results: The results of logistic regression analysis indicated that cesarean delivery (OR 3.6, 95%CI 1.2~5.6),abnormal placentation (OR 25.4, 95%CI 17.8~31.0), preeclampsia (OR 11.0, 95%CI 6.7~19.2), leiomyomas (OR 16.5, 95%CI 9.3~21.1) and prenatal hemoglobin less than 70 g/l (OR 35.1, 95%CI 29.6~43.2) were risk factors associated with transfusion ($P < 0.01$). The abnormal placentation and prenatal hemoglobin less than 70 g/l had the strongest correlation with transfusion.
Summary/Conclusions: The risk factors mentioned above, especially abnormal placentation and prenatal hemoglobin less than 70 g/l, were significant risk factors associated with obstetric transfusion. Clinicians should pay more attentions to the prenatal checks and care , and assure the reasonable and safe transfusion for the obstetric patients.

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ANALYSIS OF BLOOD USAGE OF 465 CASES BURNED PATIENTS IN CHINA

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Background: Burns often cause high temperature damage to the skin tissue. Large area of deep burn can cause blood volume reduction, lack of energy supply, coagulation abnormalities and red blood cell loss and other systemic changes. Therefore, burn treatment often requires blood transfusion.
Aims: To analyze the actual blood usage in burned patients, providing useful reference for rational clinical use of blood.
Methods: Retrospective analysis of 465 cases of burned patients with types and amount of transfusion, in the local hospital from January 2014 to December 2015.
Results: There are 306 males (65.81%) and 159 female (34.19%) in 465 cases. The age range is from 2 months to 90 years old (Table 1). The amount of red cell transfusion was 1547.50 U, the amount of plasma transfusion was 5235.20 U, and the amount of platelet transfusion was 14 U. In this research, under 3 years of age accounted for the largest proportion of patients (34.40%). The maximum amount of blood transfusion was 40–49 age group (1446.5U, 20.81% of the total). The maximum per capita output was 50–59 age group (30.51U). The red blood cell and plasma volume were the most in patients with extremely severe burn patients (5.18U and 18.41U, separately).
Summary/Conclusions: Actually there is not a standardized blood transfusion program for burn patients in China. The clinician carries transfusions through experiences and blood stock. As a result, the use of blood is abused and non-standard. According to the patients themselves, a scientific and reasonable component transfusion scheme is needed.

		Sexes		Degrees			
Ages	Numbers	Male	Female	Mild	Moderate	Massive	Severe
<3	160	106	54	0	30	81	49
3~12	63	30	33	0	14	20	29
13~17	3	2	1	0	1	1	1
18~29	32	21	11	0	8	7	17
30~39	52	39	13	2	12	21	17
40~49	64	48	16	2	15	25	22
50~59	47	36	11	3	9	17	18
≥60	44	24	20	7	15	7	15

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Degrees	Numbers	Red Blood Cells (U)	Plasmas (U)	Platelets (U)
Mild	14	2.64 ± 4.5	9.61 ± 17.43	0 ± 0
Moderate	104	2.53 ± 3.59	4.69 ± 11.53	0 ± 0
Massive	179	2.11 ± 3.78	8.54 ± 13.42	0 ± 0
Severe burns	168	5.18 ± 9.16	18.41 ± 28.28	0.08 ± 0.58

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MULTI-ANTIBIOTIC-INDUCED AUTOIMMUNE HAEMOLYTIC ANAEMIA IN A PATIENT WITH RARE RH-NULL SYNDROME

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Background: Drug induced immune haemolytic anaemia (DIHA) and rare blood groups both cause a huge challenge in clinical transfusion. The frequency of DIHA was 10% in drug induced haematological status and fatality rate of DIHA was 6.6% in China. Patients with the Rh_{null} phenotype are negative for RhD and RhCE antigens, resulting in major challenges with transfusion because anti-Rh 29 reacts with D, C, c, E, and e antigens, which are typically present in these patients. In a study in Sweden, the estimated incidence of Rh_{null} was 0.0001–0.0002 for the amorph haplotype. Because the RhD-negative phenotype is present in less than 1% of the population in Central Asia, we expect that the incidence of this phenotype may be lower in Chinese individuals than in Caucasians.
Aims: To present a case of sulbactam combine with cefoperazone and Moxifloxacin induced severe autoimmune haemolytic anaemia (AIHA) in a patient with Rhnull syndrome.
Methods: The haematological clinical condition was examined using Coombs test, and measurements of reticulocyte count, bilirubin level, serum haptoglobin. To confirm the drug type that led to the haemolytic disease, drug-sensitized O Rh-null cells reacting to the patient's plasma, using the micro-agglutinin test, were identified. Phenotyping and genotyping of RhD, RhCE, and sequencing exons of RhD, RhCE and RhAG, were conducted to identify the Rh-null blood type.
Results: Case summary: A 53-year-old man was diagnosed with severe AIHA, along with a haemoglobin (Hb) level that had dropped from 106 g/l at Day 0, due to antibiotic therapy, to 44 g/l at Day 7 and a reticulocyte count that was 3.75 g/l at Day 8. Erythropoietin, folic acid, and iron were continuously supplied because Rh-null syndrome presents with stomatocytes which, in turn, lead to a shortened RBC survival period, and to avoid the risk of immunization through blood transfusion. DIHA was suspected by the medical transfusion laboratory staff and corticosteroid treatment was started immediately at Day 7, apart from antibiotic therapy. The Hb level increased uneventfully from 53 g/l at Day 9 to 81 g/l at Day 15.
Laboratory examination for transfusion: The patient's plasma reacted to cefoperazone-sensitized O Rh-null cells, and moxifloxacin-sensitized O Rh-null cells were positive 4 + . AB plasma from a donor reacted to cefoperazone-sensitized O Rh-null cells, and moxifloxacin-sensitized O Rh-null cells were trace positive. The patient's plasma combination with cefoperazone and moxifloxacin reacted to the O Rh-null cells, giving a positive 4 + result. The AB plasma combination with cefoperazone and moxifloxacin reacted to O Rh-null cells, giving a trace positive result. A direct Coomb's test and auto-control for the patient were positive 1 + , and the patient's plasma reacted with O Rh-null cells to give a positive 2 + result. Antibody identification reported anti-f, a shift to anti-c, followed by anti-Rh29 along with anti-E, before the Hb level declined, and phenotyping of CCDee RBC products (the same as the patient's genotyping) were used on transfused blood from the previous month.
Summary/Conclusions: Moxifloxacin and cefoperazone can induce autoimmunity and mimic primary haematological diseases. In general, autoimmune haemolytic anaemia leading to severe haemolysis is rare, but antibiotic and NSAID drug therapies should be used with caution. In patients with a rare blood group, drug treatments should be administered with care and caution.

Adverse Events, Including TRALI

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IN VIVO IMAGING THE ACTIVATION OF ALVEOLAR MACROPHAGES DURING THE PROGRESSION OF TRANSFUSION-RELATED ACUTE LUNG INJURY

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Background: Transfusion-related acute lung injury (TRALI), a syndrome of respiratory distress caused by blood transfusion, is the leading cause of transfusion-related mortality. Alveolar macrophages (AMs), as one of the resident macrophage populations, are critical to the initiation and resolution of lung inflammation. However, the role of AMs in the pathogenesis of TRALI-associated lung failure is incompletely understood.
Aims: To track in real time the activation of AMs in the progression of TRALI and uncovered the mechanism by which AMs contribute to the pathogenesis of 34-1-2s-induced lung injury.
Methods: A novel lentiviral vector (Lenti-IL6P-luc-P2A-AcGFP) encoding the firefly luciferase (Fluc) together with the enhanced green fluorescent protein (eGFP) reporter gene under control of IL-6 promoter regulatory elements was developed. Lenti-IL6P-luc-P2A-AcGFP was administered to recipient BALB/c mice by intratracheal (IT) instillation. The localization of transgene expression was investigated by bioluminescence imaging and flow cytometry. LPS was used to induce the activation of IL-6 promoter. To induce TRALI, BALB/c mice were first challenged with i.p. lipopolysaccharide (LPS), and then transfused with MHC I mAb (34-1-2s) via the tail vein. Bioluminescence imaging was executed using an IVIS Spectrum system.
Results: The combination of reporter genes allows for independent tracking of transduced (GFP+) cells by flow cytometry in addition to continuous, real-time assessment of IL-6 activation levels by luciferase expression. We confirmed that the Lenti-IL6P-luc-P2A-AcGFP treatment result in transduced AMs in mouse lung and the kinetics of IL-6 activation correlated nicely with imaging-detected luciferase activity. We next used this reporter mouse model to investigate the impact of TRALI on AMs. Our results indicated that the TRALI progress is accompanied with IL-6 activation in AMs.
Summary/Conclusions: The major findings of this study are that (a) AMs was activated in the TRALI progress, which will help to understand the pathogenesis of TRALI. (b) a stable reporter expression mouse model for *in vivo* imaging of IL-6 activation in AMs is successfully established, which provide the basis for study of the pathogenesis of a variety of inflammatory lung disease.

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THE DYNAMIC CHANGE IN POTASSIUM LEVEL AFTER RED BLOOD CELL TRANSFUSION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Background: There is no unanimous consensus on whether the potassium concentration [K+] in stored red blood cell (RBC) supernatant affects the post-transfusion potassium level of the recipient.
Aims: We conducted this meta-analysis to assess the dynamic change in potassium levels following transfusion.
Methods: We searched the PubMed, Embase, Web of Science and China National Knowledge Infrastructure (CNKI) databases from January 1971 to June 2016. Forest plots were constructed for the included studies. The overall effects were evaluated by using the Z-score. Subgroup and sensitivity analyses were used to discover the source of heterogeneity.
Results: The meta-analysis included 2,057 participants from 38 studies. The pooled result showed that the blood potassium levels of patients were significant different between pre- and post-transfusion [MD = -0.17, 95%CI (-0.29 to -0.04)] and that the whole blood potassium levels of patients during transfusion were not different to those pretransfusion [MD = -0.06, 95% CI (-0.32 to 0.21)]. Although the [K+] was

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significantly increased in patients at 1–4 h [MD = -0.43, 95% CI (-0.48 to -0.37)] and 12 h post-transfusion [MD = -0.31, 95% CI (-0.47 to -0.15)] compared to that in patients pretransfusion, there was no difference in the [K+] in patients pretransfusion and 24-h post-transfusion [MD = 0.11, 95% CI (-0.17 to 0.38)]. The blood [K+] in the post-transfusion samples from the massive transfusion (MT) group was significantly higher[MD = -0.09, (95% CI, -0.13 to -0.06)] than that in those from the non-massive transfusion group.

Summary/Conclusions: We demonstrated that there was a significant risk of hyperkalemia after blood transfusion, particularly in the MT group. The incidence of hyperkalemia was significantly increased at 1–4 h and 12 h post-transfusion. Our research could help clinicians design better strategies to avoid and treat transfusion-associated hyperkalemia.

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REPORTING AND EVALUATING TRANSFUSION-RELATED ACUTE LUNG INJURY (TRALI) CASES AT A TERTIARY CARE CENTRE OVER LAST FIVE YEARS

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Background: Transfusion-related acute lung injury (TRALI) is a syndrome characterized by the development of new onset acute respiratory distress with hypoxemia during or up to 6 h after completion of a blood transfusion. Transfusion-related acute lung injury (TRALI) is a rare but potentially fatal complication of blood product transfusion.

Aims: The diagnosis and reporting of TRALI will allow better understanding of the incidence, clinical course and associated mortality of this reaction. Furthermore, by identifying cases of TRALI, steps can be taken to prevent further cases of TRALI by investigating donors involved in these cases and deferring them from further donations if they are found to be implicated.

Methods: This is a retrospective review of blood transfusion reactions during past five years [January 2010 to September 2014]. The record of the total number of transfusions was obtained from electronic blood bank information system. All TRALI cases were identified from manual review of reported transfusion reaction forms. For detailed information of all TRALI cases, medical record charts of patients were reviewed. The record of donors implicated in TRALI cases was derived from Blood bank system.

Results: Total number of transfusions during study period was 291,041. Six cases of TRALI were reported during this period. Rate of TRALI per 1000 units transfused was calculated to be 0.02%. The mortality associated with TRALI was 33.3%. TRALI occurred following transfusion of fresh frozen plasma in 4 patients, cryoprecipitate in 1 patient and packed red blood cells in 2 patients. In all cases, the donors were male.

Summary/Conclusions: The rate of TRALI reported to our blood bank was found to be 0.02% which is very low as compared to international data. This shows that TRALI is under reported in our setting. Education and reinforcement is required to create awareness about importance of diagnosing and reporting TRALI cases.

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THE INCIDENCE AND TYPE OF TRANSFUSION REACTIONS IN SICHUAN PROVINCE IN CHINA BASED ON THE US HEMOVIGILANCE REPORTING SYSTEM

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Background: The incidence and type of adverse transfusion reactions in China have not been well studied. We investigated the reported transfusion reactions of 66 hospitals in Sichuan province in 2015.

Aims: To better understand the actual condition of transfusion reactions in China and provide data support for establishment of Chinese future hemovigilance system.

Methods: The reported transfusion reactions in 66 hospitals in 2015 of Sichuan province were studied retrospectively based on the US hemovigilance reporting definitions. These hospitals distributed in 12 different cities in Sichuan and we selected them randomly. We reviewed the patients' case history and tried to evaluate and classify the transfusion reactions again. We not depended on the hospital's reported results and limited to them entirely, any signs or symptoms related to transfusion reactions were analyzed all over again.

Results: A total of 632 transfusion reactions were reported between 1Jan 2015 and 31Dec 2015 and 596 (94.30%) reactions were identified as appropriate for study inclusion according to definitions by the US hemovigilance reporting system. Of these 596 transfusion reactions, 253 (42.45%) were febrile non-hemolytic transfusion reactions, 305 (51.17%) were allergic transfusion reactions, 11 (1.85%) were transfusion associated dyspnea reactions, 1 (0.17%) were hypotensive transfusion reaction, and 26 (4.36%) were other/unknown reactions. No transfusion-associated circulatory overload,transfusion-related acute lung injury,hemolytic transfusion reaction and transfusion-transmitted infection were observed.

Summary/Conclusions: Febrile non-hemolytic transfusion reaction and allergic transfusion reaction were the most common types of transfusion reactions in Sichuan. Some patients' transfusion related information and follow-up check results in case history were not complete and affected the judgment of transfusion reactions in some sense unavoidably. Further study was also needed to better observe the actual condition of transfusion reactions in China and the national hemovigilance system should be established as soon as possible.

P-558

FACTORS AFFECTING RE-TRANSFUSION RATE AFTER INTERRUPTING A TRANSFUSION DUE TO AN ADVERSE TRANSFUSION REACTION

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Background: Because a blood transfusion always has potential risks, then, when a transfusion is being considered, the benefits for the patient should be more important than the risk of that transfusion. Therefore, if a patient had clinically required a transfusion, the appropriate blood component should be transfused. However, adverse transfusion reactions (ATR) may cause an insufficient transfusion benefit due to the interrupted blood infusion.

Aims: To review clinician's attitude regarding their re-transfusion decision for patients following an ATR and interruption of the original transfusion, we analyzed re-transfusion rates (RTR) for similar blood components and the associated factors affecting re-transfusion.

Methods: All patients with a suspected ATR at Ajou University Hospital are reported to the hospital Blood Bank. Patients with interrupted blood component transfusions due to an ATR or suspected ATR between January 1998 and December 2016 were enrolled in the study. ATR details and patient's clinical data were collected retrospectively from ATR report forms and the laboratory information system. Hemoglobin, platelet count and prothrombin time results before blood transfusion and within 24 h following the interrupted blood transfusion were included for analysis. Comparisons of RTRs by gender, blood component, ATR classifications and the improvement of laboratory data were done using a chi-square test with a P value <0.05 considered statistically significant.

Results: Total RTR during the admitted period was 45.8% (66/144). The ratio of male:female with RTR was 83:61 and mean age (standard deviation) was 48.9 (18.9) years (P = 0.99). The RTR by blood components was 49.5% (48/97) for red blood cell products, 55.6% (15/27) for platelet products and 15.0% (3/20) for fresh frozen plasma (P = 0.01). However, RTRs within 24 h and 48 h were only 30.6% (44/144) and 37.5% (54/144) respectively. Total RTR according to the ATR classifications were 42.1% (24/57) for allergic reactions; 44.7% (21/47) for febrile transfusion reactions; 33.3% (4/12) for respiratory distress; 55.6% (5/9) for hematuria; 83.3% (5/6) for chest discomfort; 50.0% (2/4) for hypotension; and 55.6% (5/9) for others (P = 0.51). There was no significant difference between total RTRs in cases without improvement of laboratory data after transfusion interruption (53.3% [16/30]) and those with improvement (54.5% [42/77], P = 0.91).

Summary/Conclusions: Although a transfusion effect was not observed due to an interrupted blood transfusion following an ATR, many clinicians did not consider re-transfusion for similar blood components, especially for FFP. And no specific factors were associated with the decision for re-transfusion after an ATR interrupted blood transfusion. Therefore, clinicians should be educated not to overestimate the need compared to the risk of blood transfusion when deciding on a transfusion of blood products.

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RETROSPECTIVE STUDY OF 276 CASES OF ACUTE TRANSFUSION REACTION IN A CHINESE WOMEN AND CHILDREN'S HOSPITAL

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Background: Acute transfusion reactions present as adverse signs or symptoms during or within 24 h of a blood transfusion. Some need resolve promptly without specific treatment or complications, while some are rare, but can be life threatening. Prompt recognition of an immune-mediated transfusion reaction is fundamental to improving patient outcome.
Aims: Through retrospective analysis of all acute transfusion reaction cases in a Women's and Children's Hospital over recent 2 years, to understand the characteristics of acute transfusion reaction in an Chinese women and children's Hospital, helping provide effective strategy for blood transfusion for women and children.
Methods: Collating the blood transfusion data and the transfusion reaction information from the Blood Bank laboratory of West China Second University Hospital from January 2015 to December 2016. Parameters reported in this study include: transfused blood type, transfusion volume, transfusion aim, transfusion efficiency and disease diagnosis, incidence of transfusion reaction, etc. A total of 276 cases of acute transfusion reaction were reported by the clinician over past 2 years were enrolled in this study.
Results: For these 276 cases of acute transfusion reaction, there were 34403 person-times transfused during the past 2 years, the incidence of transfusion reaction is 0.8% (276/34403). Classify the transfusion reaction according to the severity, the mild transfusion reaction is accounting for 63.77% (176/276), the moderate is 36.23% (100/276), and there was no serious blood transfusion reaction yet. For different blood components, the platelet was the most common transfusion reaction, was 2.60% (149/5727), and followed by plasma (0.76%), red blood cells (0.32%) and cryoprecipitate (0.16%). In different populations, the most susceptible to blood transfusion reaction were the pediatric patients, and the incidence was 0.98%; of which the main patients were those with blood system disease, accounting for 67.76%. The rate of patients with the history of blood transfusion or pregnancy of all was 93.84%.
Summary/Conclusions: Women and children patients with transfusion reactions is given priority to mild transfusion reaction, platelets are the most common blood components leading to transfusion reactions. Pediatric patients are the most susceptible to blood transfusion reaction in women and children. The proportion of patients with blood system disease, blood transfusion history, or pregnancy history was in the majority.

P-560

SEPTIC PLATELET TRANSFUSION REACTIONS

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Background: Septic platelet transfusion reactions (SPTRs) are the most common, serious risk of transfusion today in Egypt. Although public concern has largely focused upon the declining risks of the transmission of viral agents causing hepatitis or AIDS, the persistent risk that bacterial contamination of platelets may cause SPTRs is generally not appreciated. National blood transfusion service previously reported that the risk of an SPTR approaches 1 in 4,200 platelet transfusions, while the risk of hepatitis has declined to less than 1 in 100,000 and that for AIDS approaches 1 in 1,000,000.
Although transfusion medicine experts are well aware of this persistent problem, preventing these reactions remains a difficult goal. A consensus conference documented some potential solutions, but most of the recommended interventions in donor screening or component manipulation are not yet available. In addition, many of these events occur sporadically, evade detection because the patient is already receiving antibiotics, or are not reported.
Aims: To Highlight the Importance of Single donor platelets over pooled or random platelets to reduce the reactions during transfusion.
Methods: Cases of SPTR were obtained from transfusion reaction records maintained in the Transfusion Medicine Division of the therapeutic unit in the national blood transfusion center in Egypt.
Cases of SPTR must meet the following criteria: a transfusion reaction with positive bacterial culture from the residual contents of the platelet bag and confirmation of the positive platelet culture by
1) isolation of the same bacterial species with the same antibiotic sensitivities from the patient's blood or from one or more of a platelet pool's constituent units or
2) positive Gram's stain of the residual contents of the PC bag.
Platelet components: One equivalent unit of platelets is defined as 5.5 × platelets. A PC was assumed to contain one equivalent unit of platelets; SDPs were counted and equivalent units in each SDP were calculated
Platelet transfusions Blood bank records were reviewed to determine the number of Platelet components and the equivalent units of SDPs transfused.
Poisson distribution was used to determine whether recipients of Platelet components transfusions were more likely to have a septic transfusion reaction than recipients of SDP transfusions
Results: In the last 12 years, the use of SDPs increased from 51.7 percent to 99.4 percent of all platelet transfusions. SPTRs fell from three events in 1 year to the current rate of one event per year. The incidence of SPTRs decreased from 1 in 4,818 transfusions to 1 in 15,098 transfusions. The rate of SPTRs due to PCs was 5.39 times higher than that of SPTRs due to SDPs.
Summary/Conclusions: The use of SDPs is a simple means of reducing SPTRs. Other measures such as sterilization will be required to eliminate all SPTRs.

Haemovigilance and Transfusion Safety

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TRANSFUSION OF RHD POSITIVE UNITS TO A DEL PATIENT: A CASE REPORT IN A LOCAL TERTIARY HOSPITAL

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Background: The RHD antigen is highly immunogenic and can cause severe immune haemolytic anemia in transfusion. The frequency of RhD(−) phenotype in Asians is <0.5%. This severely limits the supply of RhD(−) blood in Asia. Approximately 30% of RhD(−) Asian individuals carry a RHD variant, termed "Asia type" DEL. It is postulated that individuals with DEL variants express normal RHD phenotype and thus pose virtually no risk of forming anti-D.
Aims: Literature search has not revealed reports pertaining to DEL variant patients receiving large volume of RhD(+) red blood cells (RBC) and how this exposure may contribute to Anti-D formation. A DEL variant patient in our facility received RhD (+) blood products and development of Anti-D was retrospectively evaluated.
Methods: A 41 year old Chinese male with liver cirrhosis secondary to chronic Hepatitis B infection was admitted in December 2012 for a cadaveric liver transplant. In 2011, he was typed as RhD(−) with a negative antibody screen on 4 antibody screening tests. He was transfused with 4 units RhD(+) Platelets (PLT). Anti-D prophylaxis was not administered. He did not develop Anti-D when he was typed on 4 subsequent occasions. In the immediate pre-transplant, 10 units of RhD(−) RBC was requested for the surgery. Due to rarity of RhD(−) donors in Asia, it was decided to test the patient for DEL. Weak and Partial RhD phenotype was performed with 2 different types of Anti-D reagent (Diagast TOTEM and Bioscot) and Ortho ABORH/Rev Cassette. RhD adsorption and elution was performed. A saliva sample for DNA extraction was used for RHD genotyping by BLOODchip method (Progenika) for confirmation. Phenotyping for the other RH antigens using serological technique was also performed.
Results: Weak and partial RhD was not detected with serological methods. Adsorption and elution showed the presence of RhD antigen on his red cells. Genotyping showed Del *RhD1227A* mutation. Phenotyping revealed cCee antigen expression. The "Asia type" DEL was determined on this patient. Due to the difficulty in obtaining RhD(−) RBC and despite the presence of DEL, he went on to receive RhD(+) blood products (12 units of RBC, 4 random PLT and 2 pooled PLT) for his liver transplant. Anti-D prophylaxis was not administered during or after his surgery. For prevention of graft rejection, he received a combination of immunosuppressants (mycophenolic acid, tacrolimus and prednisolone) post operatively. Antibody screen performed in the immediate 2 weeks post operatively and 6 months later did not reveal Anti-D. There was also no evidence of hemolysis detected post transfusion.
Summary/Conclusions: This case supports the biochemical observations that the DEL variants express normal RHD and may have low risk of Rh alloimmunization. A lower rate of alloimmunization was also seen in heavily immunosuppressed RhD(−) patients (Komal, Transfusion, 2017). This may be one of the contributory factors in the lack of alloimmunization in this patient. In an Asian population, where RhD(−)

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blood is in short supply, the use of RhD(+) blood may be relatively safe in a "Asia type" DEL patients. Adopting such a policy can lessen the need for RhD(−) blood in "Asia type" DEL patients. However, additional evidence needs to be gathered to implement such a practice. Other DEL types may still be at risk of Rh alloimmunization. The ''Asia type'' DEL needs to be specifically detected in a RhD(−) Asian patient to clearly define this group of patients where the practice applies.

P-562

Abstract has been withdrawn

P-563

RATE OF RED CELL ALLOIMMUNIZATION IN TRANSFUSED PATIENTS IN A SINGLE LOCAL INSTITUTION

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Background: Alloantibodies production in patients can occur after transfusion or pregnancy. Alloimmunization rate has been reported to range from 0.3% to 38% (Walker, AABB, 2011) depending on different study groups and sensitivity of testing methods. Prior to transfusion, the compatibility testing or crossmatch has been an important procedure to determine the safety of red cells transfusion.
Aims: A retrospective analysis of patients who developed red cell antibodies and their transfusion history prior to red cell antibody formation was collected. Our aim is to determine the rate of alloimmunization in patients who received prior transfusions; the most common type of antibodies and the number of red cell transfusions before antibody(ies) formation.
Methods: Blood group and antibody screening was performed using column agglutination technique (CAT) using the AutoVue System from OrthoDiagnostics. Electronic crossmatch (EC) was performed using HCLL Transfusion from Mediware which was validated on site for EC procedure. The Immediate Spin (IS) crossmatch was performed by the traditional tube technique. Antibody identification was performed with commercial panels cells (Ortho or Immucor) using CAT. The following data for each recipient who developed antibody(ies) were captured from July 2015 to June 2016 - number of red cell transfusion before recipient develop antibody(ies) and specificity of antibody(ies).
Results: A total of 3858 patients received red cell transfusions in our institution. 344 patients developed antibody(ies) during the one year period. 84 had documented prior transfusions in our institution, crossmatch being performed either by IS or EC. 260 had no known transfusion in our institution (these patients may have received transfusion in another hospital); hence they will be excluded from this analysis. Total rate of alloimmunization in our institution was 9% (344/3858). Of which 2.2% (84/3858) had prior known transfusions - a total of 66 significant alloantibodies were detected, the two most common being Anti-MIA (30/66, 45%) and Anti-E (24/66, 36%). Majority of the patients 55% developed antibodies between 0 to 5 transfusions (36/66), 21% (14/66) developed antibodies between 6 to 10 transfusions and 23% (15/66) developed antibodies after more than 10 transfusions. Between 1 to 5 transfusions, 19 developed Anti-MIA and 12 developed Anti-E.
Summary/Conclusions: Our data show that rate of alloimmunization in patients who received prior transfusion to be 2.2%. This is similar to the reported range (2% to 9%) (Tormey, Transfusion, 2008) for multiply transfused patients. The usage of EC and IS XM for transfusion did not show an increase in alloimmunization rate. The most common antibodies detected, Anti-MIA is not unexpected as the MIA antigens has been shown to appear at higher frequency in Chinese and Southeast Asians, 15% (Reid, 2012). Anti-MIA has been reported to cause haemolytic transfusion reaction and haemolytic disease of the new born; hence the usage of MIA cells has been included in the screening cells in our institution. This may have accounted for the high frequency detected in our patients. The Rh blood group system has been shown to be most immunogenic; hence Anti-E was the second most detected antibody. Due to high prevalence of D antigen (99% in Asians), there was no alloanti-D detected in this analysis. Most patients (36) developed antibodies within 0 to 5 transfusions in our institution, correlation between number of units transfused and alloimmunization rate has not been clearly associated.

P-564

Abstract has been withdrawn

P-565

PRELIMINARY ESTABLISHMENT AND APPLICATION OF PCR-SSP METHOD FOR IDENTIFICATION OF LANGEREIS BLOOD GROUP

R Lv

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Background: By establishing sequence-specific primer PCR (PCR-SSP) method of Langereis blood group to identify Lan blood group system, and to screen the Lan blood group of blood donors in Hefei.
Aims: By establishing sequence-specific primer PCR (PCR-SSP) method of Langereis blood group to identify Lan blood group system, and to screen the Lan blood group of blood donors in Hefei. Then support some suggestion to clinical transfusion and tumor therapy by the comparison of the different frequency between home and broad.
Methods: The ABCB6 full genomic sequence was obtained from Genbank of NCBI (National Center for Biotechnology Information) web site. And to get the information of ABCB6 gene mutations responsible for Lan(−) blood group phenotype from the current literature. After finding the position of mutation sites in the full sequence, intercept the a series of sequence that before and after the mutation sites as the destination sequence, then use the primer 5.0 software to design two pairs of primers which based on mutational type and wild type respectively. Choose six mutation sites to design 18 specific primers in total, and their specificity tested by blast of NCBI. Several main factors such as Mg^{2+} concentration, annealing temperature, dosage of Taq DNA polymerase, times of cycles and dosage of template DNA that influence the PCR results were investigated and optimized. At last, use this method and DNA of hematopoietic stem cell donors to screen the Lan blood group in Hefei.
Results: Based on the study mentioned above, five mutation sites of ABCB6 gene mutation that responsible for Lan(−) blood group were amplified as 434 bp, 356 bp, 221 bp, 323 bp, 365 bp positive band respectively, under the Mg^{2+} concentration of 2 mM, Taq DNA polymerase of 0.5U, DNA template of 1ul, annealing temperature of 56°C in the total volume of 25ul PCR reaction. And the sixth one was amplified as 262 bp under the DNA template of 1.5 ul, annealing temperature of 64 in the total volume of 25ul PCR reaction.
Summary/Conclusions: The PCR-SSP method of Lan blood group was preliminary established successfully, the positive band of six detected mutation sites was clear. It is exact that identifying the Lan blood group is based on its molecular characteristics. This method is feasible and dominant for Lan blood group whose serological method is not widespread at present.

P-566

CASE-CONTROL STUDY ON HIV SCREENING POSITIVE RATE AND ITS CORRELATES AMONG VOLUNTARY BLOOD DONORS IN A CERTAIN CITY IN LIAONING PROVINCE

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Background: Blood transmission was one main transmission route of HIV main transmission, was an important threat of blood safety. In recent years, although blood transmission of HIV has got better control, clinical transfusion of HIV infection occasionally happened. In this research, we analyzed HIV infection from 2002 to 2016 among voluntary blood donors in a certain city in Liaoning province and its factors, supplied the blood system with status of the epidemic and the data references such as interventions in formulating and improving for the screening of HIV-positive blood donors, improving safety blood donation and self-protection awareness of the masses and blood safety.
Aims: We intended to understand characteristics of HIV screening positive rate among voluntary blood donors in order to provide evidence for evaluating transfusion safety and further improving blood safety.
Methods: We retrospectively analyzed HIV screening positive rate and its changing trend from 2002 to 2016 among voluntary blood donors in a certain city in Liaoning province. 1:5 paired case-control study and conditional logistic regression analysis

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were also used to explore influencing factors associated with HIV infection among this subpopulation.
Results: From 2002 to 2016, there were altogether 1, 302, 161 voluntary blood donors in a certain blood bank in Liaoning province, among which 215 were detected as HIV positive. The average HIV screening positive rate was 16.51/100, 000, and it significantly increased from 3.92/100, 000 between 2002 and 2004 to 27.29/100, 000 between 2014 and 2016(P for trend<0.001). 86 HIV-positive blood donors between 2014 and 2016 were chosen as the case group, while 430 eligible HIV-negative blood donors were chosen as the control group paired with the donation year. Compared with HIV-negative blood donors, the proportions of the younger, male, being unmarried, non-local residents, and first-time donors of HIV-positive donors were significantly higher (all $P < 0.05$). According to conditional multivariate logistic regression analysis, aged between 18 and 25 years old, aged between 26 and 35 years old, male, being unmarried, and first-time donors were more likely to get infected with HIV infection among voluntary blood donors (all $P < 0.05$).
Summary/Conclusions: HIV screening positive rate of voluntary blood donors in the study site was relatively high and showed a significantly increasing trend, which implies that the proportion of blood donation from HIV-related high-risk population is growing as well. Though the window period of nucleic acid testing is quite short, there still exists a certain degree of hidden risk regarding blood safety. Relevant health departments should list the younger, male, unmarried, and first time donors among voluntary blood donors as the key screening and intervention subpopulations aims to further improve blood safety.

P-567

EVALUATION OF TRANSFUSION REACTION INCIDENCE FOR THE LAST 3 YEARS IN THE SANGLAH GENERAL HOSPITAL DENPASAR BALI INDONESIA

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Background: Transfusion reaction is one of the indicators of hospital service quality that must be monitored and reported. Data on the adverse effects of transfusion delivery nationally have not been reported. One of the factors that become obstacles in the implementation of hemovigilance program in Indonesia is the variation in the ability of blood service between regions. Monitoring and reporting of transfusion reaction incidence is very importance not only for the patients' safety but also very useful in improving the production quality of blood products.
Aims: The purpose of the study was to evaluate the incidence of transfusion reactions that are beneficial in improving the blood service system including improving the production quality of blood products. This study is also expected to be the pioneer and driving force in the implementation of the national hemovigilance program in Indonesia.
Methods: In each unit of transfused blood is always included a transfusion monitoring form which contains data for transfusion reaction reporting. If the patient undergoes a transfusion reaction, a completed monitoring form is filed on the patient's record and one form is sent to the blood bank along with the patient's blood sample and the remaining transfused blood unit. Blood bank officers will respond and inputting data in the system. The transfusion reaction data is reported every month, and once every 3 months a meeting is held to discuss the progress of achieving the quality indicators as well as making the planning for improvement.
Results: Over the last 3 years, from July 2014 until June 2017 there were 313 cases of transfusion reactions of 88,385 units of blood transfused. Type of blood products transfused consists of 110 units of WB, 62,214 of PRC, 1906 of WRC, 4424 of FFP, 19,197 of TC, 1032 of TC apheresis, and 970 of Cryoprecipitate. The incidence of transfusion reactions per unit transfused WB is 1: 110, PRC 1: 322, WRC 1: 477, FFP 1: 492, TC 1: 193, TC apheresis 1: 206 and cryoprecipitate was 1: 323. Seen from the type of blood group, 1: 285 transfusion reactions occur in transfusion of type A, 1: 275 on group B, 1: 306 on type O, and 1: 184 group of AB. The incidence of severe transfusion reactions occurred in 0.64% of cases, moderate were 7.99% and 91.37% were mild cases. Types of transfusion reactions include fever as much as 35.46%a, urticaria and other allergic reactions of 48.24%, 11.50% tightness, fever with chills of 2.88% and hypotension of 1.9%.
Summary/Conclusions: Safe and beneficial blood transfusions require standardization at all stages of service both from the donor and patient aspects. From the aspect of the patient, standardization must be performed starting from the determination of indication to the transfusion monitoring. Mistakes in determining indications, identification and pre-transfusion tests can trigger the emergence of severe transfusion reactions. Other components such as blood preparation procedures, screening, storage, distribution, transfusion delivery and monitoring techniques also play a major role in the incidence of transfusion reactions. Good coordination among physicians, nurses, blood banks and all related units is helpful in establishing a safe and effective transfusion chain. At the national level, the implementation of hemovigilance programs is needed to formulate various policies related to blood services so that blood transfusion can provide the greatest benefit with the least risk.

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CLINICAL TRANSFUSION PRACTICE IN SICHUAN, CHINA: A CROSS-SECTIONAL SURVEY

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Background: Blood transfusion is a vital healthcare activity of modern healthcare and blood is a scarce and costly resource. However, it is also associated with significant clinical risks, such as infectious diseases and transfusion related immunologic reactions. Safe, effective and appropriate transfusion practices are important to patients who need transfusion treatment.
Aims: To get full knowledge of clinical transfusion practice in different level hospitals in Sichuan, China from 2011 to 2015.
Methods: This survey was performed by means of a questionnaire which consisted of three parts: basic conditions of hospitals, utilization of different blood products and application of autologous blood transfusion (ABT). Data was collected from hospitals with scale, geographic, and even religious diversity in Sichuan, China. Data analysis was conducted in 3 groups according to the official classification of hospital (level 3, level 2 and below level 2).
Results: The questionnaires were send to 500 hospitals and 384 (76.8%) hospitals answered the questions completely. The mean number of bed, outpatient, inpatient and operation per hospital per year of level 3 hospitals were significant highly than that of level 2 hospitals and hospitals below level 2 ($P<0.01$ for all comparison) and the mean units of blood used per hospital per year in level 3 hospitals was significant highly than other two level hospitals (both $P<0.01$) in the past five years. The usage of whole blood showed significant decreasing trend ($P = 0.047$) and cryoprecipitate showed significant increasing trend ($P = 0.045$) from 2011 to 2015. The blood products used most in all hospitals were Red Blood Cells (RBC) and plasma and in level 2 and level 3 hospitals, the inpatient number and operation number were extremely associated with the used units of plasma and RBC (all $r \geq 0.442$; $P<0.01$). The plasma used per operation per year by level 3 hospitals and RBC used per inpatient per year by level 2 hospitals both showed a decreasing trend from 2011 to 2015 ($P = 0.047$ and $P<0.001$). In the past five years, the ratio of plasma to RBC units transfused (plasma:RBC) of level 3 hospitals was not lower than 1:1.8 and the autologous blood transfusions (ABT) rate was not higher than 42.16% in all the three level hospitals.
Summary/Conclusions: The transfusion service level of hospitals in Sichuan, China was closely related to the grade of hospital and it has improved a lot in the past 5 years, especially in level 2 and level 3 hospitals. As the products which used most in all hospitals, RBC and plasma used by level 2 and level 3 hospitals were definitely associated with the number of inpatient and operation and plasma may be overused in level 3 hospitals. Otherwise, whole blood still being used in some hospitals and the autologous blood transfusions rate in Sichuan province still very low. For improving the level of clinical blood transfusion, it's necessary for the managers of hospitals in Sichuan province to pay more attention to directing clinical blood usage, evaluating clinical transfusion appropriateness and promoting ABT.

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ABD PAD®: EVALUATION OF A NEW MANUAL ABO / RH BLOOD GROUPING METHOD

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Background: As part of the CE marking of the ABD PAD® device, which allows the ABO / Rh group to be confirmed using a new proprietary technology, DIAGAST, associated to the Etablissement Français du Sang de Nantes (French Blood Service) organized a performance evaluation of the product.

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Aims: The purpose of this evaluation was to perform the ABO and Rhesus group control on 1124 blood donations by comparing the ABD PAD® method with the routine reference method. The main aspects studied were the visual interpretation, time for getting the results, traceability, safety and ergonomics.
Methods: The ABD PAD® was compared with the EUROBIO Euclone® method associated with the BIO-RAD Héma-6® plates.
Results: The DMDIV ABD PAD® and the PAD Buffer meet the acceptance criteria: 100% matching samples.
The ABD PAD® method is faster, safer and more ergonomic compared to the routine reference method.
Summary/Conclusions: The ABD PAD® by DIAGAST is a new, innovative and reliable method for blood grouping.

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THE RETROSPECTIVE EXPLORATION OF POLYMORPHISM DISTRIBUTION OF BLOOD DONORS' MILTENBERGER BLOOD TYPE AND GP.MUR BLOOD CLINICAL INFUSION

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Background: Sub system of MNS blood group system is the Miltenberger system, gp.mur is a version of the system, the crowd in Europe is rare, the clinical significance of small, but in some Eastern populations in the distribution of high frequency, it has been shown that the anti-mur antibody can cause hemolytic transfusion reaction (HTR) and hemolytic disease of the newborn (HDN). At present, China has not yet been a routine screening of clinical blood transfusion before Mur antigen, clinical effect and so on gp.mur polymorphism distribution of the Miltenberger blood group system to improve the safety of clinical blood transfusion in China has important significance.
Aims: To establish a simple and rapid detection method of Miltenberger blood group system, and investigate the polymorphism distribution of Miltenberger blood group system to donate blood. For gp.mur positive blood patients were retrospectively reviewed. The clinical significance of gp.mur positive blood.
Methods: According to the molecular basis of Miltenberger blood group system, specific primers were designed using polymerase chain reaction sequence specific primer (PCR-SSP) screening and sequencing, molecular biology detection method of pcr-ssp-sequencing to establish the Miltenberger blood group system. Serum and verify accuracy of the method using anti -mur standard .A retrospective survey of gp.mur positive blood transfusion in blood donors from 2011–2013, Miltenberger blood group identification and anti-Mur antibody testing and clinical transfusion effect analysis.
Results: This study successfully established the molecular biological detection method of pcr-ssp- sequencing of Miltenberger blood group system. Serological methods to verify fully comply with. Methods 24 cases of Miltenberger blood group positive individuals were screened by PCR-SSP from the voluntary blood donors in Henan nationality, accounting for 0.7%.12 patients with gp.mur positive blood transfusion were observed and followed up, of which there were male and female in 4 cases, and the Miltenberger blood group was negative for gp.mur .The sera of patients with anti -mur antibody were 2 and 4, respectively .
Summary/Conclusions: Group Miltenberger blood group system detection blood donors of Han nationality in Henan Province, e gp.mur phenotype, the frequency of 0.9%. Infusion of gp.mur positive red blood cells of the patients developed anti -mur antibody, can cause transfusion reactions, recommend routine screening before blood transfusion of gp.mur blood group.

P-571

HOW TO REGULATE BLOOD TYPE AND SEROLOGY DETECTION TECHNOLOGY IN THE HOSPITALS WITH ON-SITE ASSESSMENT AS THE STARTING POINT

J Pu

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Background: Through analysis of the situation of on-site operation assessment of blood type and serology test projects in clinical blood hospitals in Qinghai province, to explore the role of on-site assessment in regulating blood type and serology detection technology in clinical hospitals.
Aims: Through analysis of the situation of on-site operation assessment of blood type and serology test projects in clinical blood hospitals in Qinghai province, to explore the role of on-site assessment in regulating blood type and serology detection technology in clinical hospitals.
Methods: The samples taken at the assessment in 2006 were high-titer IgG antibodies. Thereafter, to improve the detection level of each hospital, the titer of IgG antibody was reduced year by year. Retrospective comparative analysis of the situation of the on-site assessment in 23 clinical hospitals in Xining area in 2006 and 2016 was made.
Results: Compared to 2016, there was only one hospital performing anti-stereotyped tests; all hospitals didn't carry out Rh(D) stereotypes, antibody screening test and non-brine medium matching test. In the on-site single-blind trial assessment, the anti-stereotyped samples with weakened antibodies in 23 hospitals were failed to be detected; in polybrene medium matching trial, 7 with high-titer antibody samples were detected, and 16 were failed to be detected. While in 2016, all the 23 hospitals with low-titer antibody sample matching were detected.
Summary/Conclusions: Practice has proved that on-site assessment promoted the standardization process of blood type and serological testing in clinical blood hospitals and played a positive role in development of local clinical blood transfusion management, ensuring the safety of clinical blood transfusion.

P-572

15 YEARS EXPERIENCE OF POTASSIUM ADSORPTION FILTER FOR RED BLOOD CELLS

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Background: For potassium overdose, one of the risks of red blood cells (RBCs) transfusion, the potassium adsorption filter was developed in 1998 and was granted manufacturing approval and marketing permission in Japan as a medical device in 2002.
Aims: To confirm safety of the filter: Because of a new product, all cases were asked for post-marketing surveillance mandated from Japanese government..
Methods: We have sent a questioners mail all hospitals to use this filter.
Results: Post-marketing survey was implemented from May 20, 2002 to January 17, 2005 (2 years and 8 months). In 526 cases 463 filters were used. Average age of patients were 27.6 years (0 - 101 years old; Male: 296 & Female 230). In only one case, hypotension with systolic pressure of 75 mmHg was reported, but not severe.
Summary/Conclusions: Medical device approval: Safety was confirmed in this survey, insurance was applied in 2012, and reimbursement price was obtained as medical equipment in Japan.
Indications of this filter at present:
1. Transfusion to premature infants / neonates
2. Exchange transfusion against fetus or premature neonates
3. Children undergoing extracorporeal circulation
4. Patients who needs emergency rapid massive transfusion on life
5. To remove excessive potassium from transfused RBCs which were raised potassium level due to irradiation or long-term preservation (>5 weeks)

Current use situation: The number of filters used in 2016 was about 65,000, reaching 2% of the number of RBCs used in Japan. For the future expansion, addition of renal failure (dialysis patients) into insurance coverage is considered. Also, CE marking has been obtained and FDA approval obtainment is considered for overseas expansion.

P-573

REPORT OF INVESTIGATION ON THE 50 GENERAL HOSPITALS' INFORMATION SYSTEMS USAGE IN CHANGSHA

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Background: In the daily work of the hospital, the Internet has also been widely used. The hospital information management system (HIS), laboratory information management system (LIS), and the hospital transfusion management system (BIS) to achieve seamless docking, the three systems and blood transfusion management information system will become the trend of development.
Aims: To investigate the use of information systems in medical institutions at all levels in Changsha and to understand for information networking between hospitals

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and blood stations in order to provide a basis for improving the clinical transfusion management.
Methods: Through on-site visits and filling out questionnaires, the use of information systems, software functions, software upgrade and networking requirements of 50 medical institutions at all levels were investigated. There are 20 three -level hospitals,20 two-level hospitals and 10 first-level hospitals.
Results: He utilization rate of hospital management information system (HIS) and laboratory management system (LIS) of three- level,two-level and the first-lever hospitals were 100%, 70% and 30% respectively. The rate of information docking for HIS and LIS was 100%, 35% and 20% respectively. The utilization rate of blood transfusion management system (BIS) and the rate of docking with HIS and LIS were 65% in the tertiary hospitals, and the utilization rate of BIS system in the secondary level and first-level hospital were zero percent. 13 three-level hospitals which using the BIS system have three new functional requirements, such as blood utilization evaluation, blood transfusion adverse reactions and blood transfusion monitoring. 37 clinical hospitals at all levels which didn't using the BIS system have the use of demand, and hope that the function is comprehensive. In the 37 hospitals which using the HIS and LIS systems, the establishment of Hierarchical Authorization accounted for 81.1%;Login password replacement frequency was set to 3 months, six months, 1 years, accounting for 5.4%, 40.6%, 2.7% respectively, and no requirements and never replaced were 29.7% and 21.6%. Among the 50 hospitals surveyed, the demand for blood bank information system networking was 98%;Hospitals hope to solve the problems through networking, mainly including understanding of the blood product inventory, the application of blood transfusion, the investigation of adverse reactions in blood transfusion and the tracking of the quality of blood products. The willingness of hospitals to open the blood bank, including blood transfusion products inventory, blood transfusion and difficult samples of related testing information, adverse reactions retroactive blood transfusion; Only 48% of hospitals would open blood transfusion records to blood bank. The main obstacles of information sharing are the cost, technology and information security aspects.
Summary/Conclusions: The construction of information system in all levels of hospitals in our city needs to be strengthened, especially the extensive application and function upgrading of blood transfusion management information system (BIS). The security of information system cannot be ignored. Strengthen the construction of networking between medical institutions and blood stations information systems, break through bottlenecks and achieve the goal of sharing resources, mutual benefit and win-win results.

P-574

THE STUDY OF IRON DEFICIENCY IN A BLOOD DONOR POPULATION IN TLEMCEN

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Background: Blood transfusion is an essential substitutive therapy for many patients. To reduce the risk of infectious transfusion, WHO recommends the use of regular blood donors. However, these donors constitute a population at risk of anemia due to iron deficiency. This latter is not covered by almost of haemovigilance donor systems.
Aims: The aim of the present study is a determination of the prevalence of the deficiency among donors of whole blood from Tlemcen, Algeria.
Methods: This analytical cross-sectional study concerns the 91 donors of whole blood. These were selected to perform a blood count and a dosage of ferritinemia. Statistic analyzes using SPSS software, Excel by applying the Pearson correlation and the chi2 test.
Results: The prevalence of iron deficiency was 35.2%, anemia was 22%. Ferritinemia was strongly related to the different characteristics of the donation, mainly in women.
Summary/Conclusions: Haemovigilance systems focus their efforts on protection against iron deficiency. Unlike hemoglobin, serum ferritin remains the most sensitive and early marker for donor protection against this deficiency.

P-575

ANALYSIS OF TRANSFUSION'S CURATIVE EFFECT OF AIHA PATIENTS WITH RBC AUTOANTIBODIES

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Background: For autoimmune hemolytic anemia patients with positive autoantibodies, the efficacy of blood transfusion is poor in patients with primary and secondary cross incompatibility.
Aims: To investigate the curative effect of RBC transfusion for AIHA patients with RBC autoantibodies
Methods: 32 cases of hematologic patients with positive DAT and positive RBC autoantibodies were collected and set as study group. 32 cases of routine transfusion patients with negative DAT results were set as control group. Parameters of Hb, RBC count and Hct of 24 h before and after transfusion in the two groups were detected and compared.
Results: The values of Hb, RBC count and Hct of the patients in study group showed no significant differences before and after transfusion (P>0.05).While the results of these parameters showed significant difference before and after transfusion in control group (P<0.05).Comparing the two groups, the indices of hemogram and transfusion effective rate were significant different (P<0.05).
Summary/Conclusions: RBC transfusion for patients with RBC autoantibodies is ineffective, and transfusion should only be applied when there is an indication. On the other hand, only in an emergency situation, patients with RBC autoantibodies may have RBC transfusion, which can maintain their Hb levels. They should not be refused transfusion because of the cross matching incompatibility.

P-576

ACUTE BLOOD TRANSFUSION REACTIONS IN A TERTIARY CARE HOSPITAL IN PAKISTAN- AN INITIATIVE TOWARDS HEMOVIGILANCE

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Background: Hemovigilance programs have been implemented in many countries and adverse events associated with blood transfusion are published in their annual reports. Pakistan has no established program at present.
Aims: In this study we report acute transfusion reactions at our hospital.
Methods: A cross sectional study was done and all adverse reactions that were reported to the blood bank was included from January 2014 to March 2016. An adverse response in the patient, related to administration of blood (within 24 h) was considered as immediate transfusion reaction.
Results: During the study period 20,956 blood and its components were issued. Total of 32(0.15%) adverse reactions were documented. The reactions reported were mainly allergic reactions, febrile non-hemolytic reactions, bacterial contamination, and acute hemolytic reactions. Allergic reactions were the most common adverse event and were observed in 15(46.8%) of the cases. Febrile non-hemolytic transfusion reaction was the second most common seen in 9(28%).
Summary/Conclusions: The low incidence indicates underreporting and the need for a formal hemovigilance system. International benchmarking between different medical systems is helpful to identify areas in the transfusion process, which have to be changed to improve transfusion safety.

P-577

ADVERSE REACTIONS AND LEUCODEPLETION IN THALASSAEMIA MAJOR PATIENTS

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Background: Blood transfusion is a life-saving intervention but can also cause adverse reactions in the recipients. The transfusion reaction surveillance systems as part of the national haemovigilance systems are not well established in Pakistan. The multi-transfused thalassaemia patients are particularly prone to develop these

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reactions. Thalassaemia is a major health care challenge in Pakistan and the transfusion dependant thalassaemia patients are the biggest consumer of its transfusion services. Leucocyte-free blood transfusion is especially recommended in these chronic recipients to reduce transfusion reactions. However, the leucodepletion practice is not common in Pakistan resulting in higher incidences of transfusion reactions including febrile non-haemolytic transfusion reaction, urticaria, etc. In the absence of leucofilters some centres use washed red cells to reduce the incidence of reactions at the risk of causing infections. The centres that do use leucofilters are often of poor quality.

Aims: To estimate the prevalence and rate of transfusion reactions in thalassaemia patients and to assess the efficacy of leucocyte removal from packed red blood cells by Leucolite™ filters at the bedside.

Methods: This was a prospective multi-centre study carried out from Mar–Jul 2017, at three thalassaemia centres, i) Thalassaemia Centre, Shaheed Zulfiqar Ali Bhutto (SZAB) Medical University, Islamabad; ii) Sundas Foundation Thalassaemia Centre, Lahore; and iii) Fatimid Foundation, Lahore. A standardized patient haemovigiliance reporting form developed by SBTP was used to collect the adverse reactions data. The selected patients were 70 thalassaemia patients who had history of frequent mild to moderate immediate transfusion reactions [febrile non-haemolytic reaction (30%) and allergic rashes (21%)]. Pre- and post-transfusion blood samples were collected for white blood cell count. Ethical approval of the study was granted by the Ethical Review Board of the SZAB Medical University. The clinical records and the reaction workup was done to rule out the haemolytic reactions. Statistical analyses were done through Microsoft Excel 2013.

Results: 70 Leucolite™ filters (by GLT Medical; Co. Ltd. China) were used during transfusion to selected 70 patients. These patients were routinely given pre-medication [Solu-Cortef (100 Mg) and Avil Injections] to avoid transfusion reactions, but with the filter use no pre-transfusion medications was given to observe the effectiveness of the leucofilters. No haemolytic and delayed transfusion reaction was reported during the study. The age of transfused red cell concentrates ranged from 5–27 days (mean 13.39 days). The pre- and post-filtration white blood cell counts showed an overall mean percentage of leucodepletion as 97.7%. None of the patients using filter developed the transfusion reaction.

Summary/Conclusions: Leucodepletion by using bedside filter is very effective for eliminating non-febrile transfusion reactions. In resource limited settings, leucoreduction using the buffy-coat method is also effective in reducing the transfusion reactions. Leucoreduction of blood components in thalassaemic patients is helpful in preventing transfusion reactions. Use of sub-standard filters needs to be regulated. This is an on-going study, leucofilters will be used consistently on the thalassaemia and other chronic recipients (oncology and renal patients) to assess the effectiveness of leucodepletion and co-relate it with financial impact including prolonging duration of interval of transfusion.

P-578

EVALUATION OF QUALITY CONTROL OF BLOOD COMPONENTS AT OUR BLOOD BANK

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Background: Quality control is the backbone of quality assurance program. As part of it, quality control of blood components ensure availability of a sufficient supply of blood, its components of high quality with maximum efficacy and minimum risk to both donors and patients, hence, ensuring overall optimum safety of blood.

Aims: The objective of the study was to evaluate quality control of blood components preparation at our blood bank.

Methods: An observational cross sectional study was done at the blood bank of National Institute of Blood Diseases and Bone Marrow Transplantation Karachi to analyze the quality control of blood components. Total 100 units of each blood components were randomly selected during this study period. Packed red cell units were evaluated for hematocrit and culture. Random platelets and mega units were evaluated for pH by paper method and yield. Fresh frozen plasma (FFP) and cryoprecipitate (CP) was evaluated for unit volume, factor VIII, fibrinogen concentration, while cryosupernatant (CS) was evaluated for unit volume and factor XI concentration.

Results: Among the blood products, mean packed red cells hematocrit was 74.8 with SD± 2.84, mean platelet count was 889.5 ± 127, pH was > 6.2 in 98% of platelets units and platelet yield was found to be 7.3 ± 1.20. Mean FVIII and fibrinogen level was found to be 89.6 ± 5.2 and 2.8 ± 0.49 for FFP respectively. For cryoprecipitate, mean Factor VIII and Fibrinogen levels was found to be 108 ± 11.98 and 3.2 ± 0.34 respectively. The mean Factor IX in cryosupernatant was found to be 86.5 ± 5.76.

Summary/Conclusions: The quality of blood components preparation at our blood bank is well within the standards. All those involved in blood transfusion related activity must be aware of the importance of quality management for its successful implementation.

P-579

ANALYSIS OF 1256 BAGS OF BLOOD PRODUCTS RETURNED IN 2016

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Background: In Guangzhou City, the growth of clinical blood demand is much more than the growth of blood supply, blood supply shortages have become an ordinary state. We found that the blood returned amount in 2016 reached to 1256 bags, accounting for 0.21% of the total blood supply, so we analyzed the reasons for the returned 1256 bags of blood products, designed to reduce blood returned rate and reduce waste of resources.

Aims: To analyze the causes for the returns of blood products from hospitals supplied by the Guangzhou Blood Center, aiming at reducing the complexity of the circulation process of qualified blood products and avoiding the release of defective blood.

Methods: In total, 1256 bags of suspended blood cells, single platelets, frozen plasma, and cold precipitated blood samples were collected from the blood samples of Guangzhou Blood Center. For each type of blood samples, the scrapping rate and the re-issued rate were collected.

Results: First, the scrapping rates of suspended red blood cells, single platelets, frozen plasma and cold precipitates were 45.0%, 2.9%, 91.4% and 71.4% respectively, and the re-issuing rates were 55.0%, 97.1%, 8.6% and 28.6%; Secondly, for the suspended red blood cells returned, 50.6% were due to deployment, 26.4% were due to blood clot, and 15.2% were due to irregular antibodies; For the single platelet, 97.1% were returned for deployment, while 2.9% were for leakage; For the frozen plasma and the cold precipitate returned, blood bag leakage accounted for 53.7% and 45.2%, fiber precipitation accounted for 37.2% and 26.2%, and deployment accounted for 8.6% and 28.6% respectively.

Summary/Conclusions: The establishment of direct distribution channels between hospitals could reduce the complexity of the circulation process of blood products and ensure the effective use of blood. The frozen plasma and cold precipitates are highly brittle and are easy to break. To avoid the waste of resources, the staffs of the blood center are required to strictly follow the operating procedures and the staffs of the hospitals should take the receiving responsibilities, which need to be defined.

P-580

CORRELATION BETWEEN TH1/TH2 CYTOKINES PRODUCTION AND TRANSFUSION REQUIREMENTS IN PATIENTS UNDERGOING ORTHOPEDIC SURGERY

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Background: Blood transfusion in the perioperative period has frequently been significantly associated with immunosuppression in patients after surgery. Moreover, mediating mechanisms underlying it are unknown. So confirm and minimize the possibility of blood transfusion may be important to improve prognosis for surgery patients.

Aims: This study aims to explore the relationship of transfusion requirement and the balance of Th1/Th2 cytokine in orthopedic surgery patients, which may benefit for future improvement of treatment.

Methods: We collected data for orthopedic patients received surgery. Plasma concentrations of several Th1 and Th2 cytokines (IL-2, -6, -10, TNF-α) were detected by enzyme-linked immunosorbent assay (ELISA), and then evaluated their clinical relevance to transfusion requirements.

Results: A total of 168 orthopedic surgery patients were recruited in this study. Overall, 36 patients (21.43 percent) received blood transfusion. Immunological analysis indicated that transfusion patients had significantly lower plasma concentrations of TNF-α (0.76 vs 1.67 pg/ml, P = 0.025), no significant concentration difference of IL-2, -6 and -10 were noted between two patient subgroups. Lower TNF-α

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expression induced a shift of Th1/Th2 balance toward Th2 dominance suggesting an immunosuppression result in blood transfusion requirement in perioperative period.
Summary/Conclusions: Our findings indicated that transfusion requirements is associated with immunosuppression which contributes to poor prognosis in orthopedic patients.

P-581
TWO CASES OF AEL GENOTYPE IN OBSTETRICAL TRANSFUSION PREPARATION
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Background: There were 2 pregnant women in obstetrics specimens, who have discrepancy in the results of ABO blood typing.
Aims: To explore the significance of ABO subtype identification.
Methods: Blood samples were identified by gel microcolumn assay, conventional tube test, adsorption and elution tests and gene direct sequencing.
Results: In routine serological blood typing, the red cell typing of both pregnant women were O, whereas serum typing were A. In adsorption and elution tests, presence of A antigen was found on Ael red cell. Finally, the two pregnant women have been proved to be Ael04/ O02 and Ael01/O02 by direct sequencing analysis. In obstetrics transfusion, O washed RBC and A plasma were the best chose when there wasn't homo-blood donor.
Summary/Conclusions: Ael is a rare subtype of ABO blood type, characterized by the least amount of A antigen on the red cell and anti-A1 in the plasma. In routine serological blood typing, the blood cell always showed weak agglutination or no agglutination. In a result, the ABO typing was wrongly identified to be O, which bring the risk to patients in transfusion, especially in obstetrical and neonates department. When there were individuals with discrepancies in the results of ABO phenotyping, ABO genotyping is needed for an accurate evaluation of their blood type, in order to provide scientific basis for transfusion safety.

Alternatives to Blood Transfusion

P-582
COMPARISON OF THREE COMMERCIAL RED CELL GENOTYPING KITS
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Background: Genotyping of red cell antigens has potential benefits when compared to serological method. Genotyping is useful for rare blood group antigens, chronic transfused patients, patients with positive direct antiglobulin test, solving ABO discrepancy and evaluation risk of hemolytic disease of the fetus and newborn. However, some limitations still remain due to unknown or newly defined polymorphism or many allelic variants of blood group antigen which result in mistypings or discordant results with serological method.
Aims: To evaluate the efficiency in red cell genotyping of three commercial kits.
Methods: In this study during 2015 to 2017 a total of 40 samples from regular blood donors at National Blood Centre, Thai Red Cross Society were genotyped for red cell antigens using three commercial kits. Each kit used different methods and could genotype different number of red cell antigens. The commercial systems evaluated were Fluidic Microarray with Luminex (kit "A", 37 antigens genotyped), BeadChip BioArray with BioArray Solution machine (kit "B", 35 antigens genotyped) and Mass Spectrometry with Mass Spectrometer machine (kit "C", 47 antigens genotyped), respectively. From the genotype results, the predicted red cell phenotypes analyzed from software were then compared with red cell phenotypes using serological method.
Results: The predicted red cell phenotypes of Rh, Kell and Diego systems were all concordant when compared to serological method. In Kidd and MNS systems, only the results from kit "A" were concordant with serological method. But in the Duffy system, the results shown sample with a novel variant. For kit "B", the results of MNS system were concordant with serological method but 5 DNA samples gave discordant results in Kidd system. Also the result of Duffy system due to limitation of the primers. For kit "C", discordant results were due to S antigen and 5 samples for Kidd system. Interestingly, of the three kits tested, only kit "A" could detect Mi^a antigen which is important for our population.
Summary/Conclusions: The three commercial red cell genotyping kits had different limitation of primers, resulting in these discordant results. The selection of appropriate kit for red cell genotyped should be considered with local population in mind.

P-583
ANALYSIS ON GENETIC POLYMORPHISM OF HUMAN PLATELET ANTIGEN HPA1-6,15 SYSTEMS TO PLATELET TRANSFUSIONS
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Background: To study the genetic polymorphism of human platelet antigens in Zhejiang.
Aims: Understand the transfusion effect on platelet gene matching group in platelet transfusion.
Methods: Platelet antigen genotyping was performed by sequence-specific primer polymerase chain reaction (SSP-PCR) for the HPA1-6,15 systems in 400 platelet donors and 54 patients of Zhejiang. The effectiveness of different platelet transfusion for 54 patients of platelet antibody screening positive.
Results: Among HPA1-6,15 Systems, HPA-3 has the highest b gene frequency,its frequency is 0.51, HPA-15 has the highest heterozygosity, HPA-15ab 0.33, others systems is mainly aa homozygous genotype, more than 90%. Platelet gene matching group has higher CCI and PPR than the group that receives random platelet group.
Summary/Conclusions: The effectiveness of patients transfusion gene matching platelet prevent from occurrence of platelet transfusion refractoriness and improve the effective rate of transfusion.

P-584
TO COMPARE THE CLINICAL EFFECT OF RESTRICTIVE BLOOD TRANSFUSION WITH THAT OF OPEN BLOOD TRANSFUSION DURING OPERATION
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Background: We study the clinical effect of restrictive blood transfusion and that of open blood transfusion during operation as to use accurately limited quantities of blood.
Aims: Observing and Comparing the clinical effect of blood transfusion between restrictive and open intra-operative.
Methods: We have selected 124 blood transfusion patients during operation in Qiqihar First People's Hospital from December 2015 to December 2016, 124 cases were randomly divided into two groups respectively named as observation group and control group .62 cases
Of control group were adopted the open transfusion strategy, as well 62 cases of The observation group were applied restrictive transfusion way, and then we compared these items such as the actual blood transfusion rate, total blood transfusion per patient, and Adverse reactions caused by blood transfusion in the two groups.
Results: These aspects of observation group like total quantity of blood transfusion, total blood transfusion each patient, the incidence of adverse reactions were significantly inferior to that of the control group ($P < 0.05$).
Summary/Conclusions: Compared with open transfusion strategy, it is obvious that intraoperative restrictive transfusion way can reduce blood transfusion rate, decrease the per capita amount of blood transfusion and the incidence of adverse reactions led by blood transfusion, so the intraoperative restrictive blood transfusion way is worthy of popularization and application in clinical.

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Cellular Therapies Stem Cell and Tissue Banking, Incl. Cord Blood

P-586

THE COMPARATIVE STUDY OF THE INTERVENTION OF THE MOUSE BRAIN INFARCTION MODEL IN TWO STATES OF THE UMBILICAL CORD BLOOD

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Background: Human umbilical cord blood mesenchymal stem cells can repair the injury of nerve cells caused by ischemia and hypoxia,but which state of cells has a better treatment effect: primary isolation or induced differentiation is not yet known.

Aims: Two kinds of cells were treated respectively through the animal model, and the curative effect was evaluated by behavior, histology and molecular biology, and find the best state of cell therapy

Methods: The separation of quadruple bags by means of density gradient centrifugation *in vitro* culture and differentiation of human umbilical cord blood mesenchymal stem cells; construct the rat model of cerebral infarction; experimental animal groups for cell therapy, balance beam test in rats balance and motor coordination; the ability of learning and memory in the water maze test in rats with spatial localization; TTC staining cell calculation 14d after treatment of cerebral infarction volume; rats survival cerebral infarction area week neurons Nissl staining; GFAP immunohistochemical staining of reactive astrocytes were observed after treatment; CD34 immunohistochemical staining was used to observe the angiogenesis after treatment; Ki-67 immunohistochemical staining was used to observe the effect of cell therapy on cell proliferation. ELISA method was used to determine the secretion of inflammatory factors IL-1 β, IL-6 and TNF- β14 days and 60 days after cell treatment

Results: The isolated cells of MRI detection model the balance beam test and balance beam walking test results showed that cerebral infarction in rats induced by cell therapy group, balance ability and walking ability improved the water maze test results showed that the rats spatial orientation ability and the ability of learning and memory, learning ability is better than the induction group the primary observation group, 14d infarct volume. TTC staining: induction group and primary group were 42.39 ± 2.91% and 45.66 ± 5.23% and Nissl staining neurons survival induced by cerebral infarction group center than the primary group significantly reduced tissue edema process Reduce the number of cells decreased; the hippocampus is not obvious, the arrangement of relatively neat immunohistochemical detection of CD34, GFAP in induced group high expression, low expression of GFAP and ELISA in the brain tissue of inflammatory factor IL-1 beta, IL-6, the expression of TNF- in 60 days when IL-1 β, TNF- β, IL-6 in treatment group induced low expression.

Summary/Conclusions: Ectomesenchymal cells umbilical cord blood can relieve cerebral infarction infraction area in rats model of inflammation, promote the regeneration of the infarction area microcirculation blood vessels and cells, cells induced by culture, the intervention of the effect is better than that of the original generation group.

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P-589

THE CREATION OF AN UNRELATED DONORS BONE MARROW REGISTRY IN PAVLOV FIRST SAINT-PETERSBURG STATE MEDICAL UNIVERSITY

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Background: Allogeneic stem cell transplantation (alloHSCT) is an effective method for the treatment of hematological malignancies, hematopoiesis insufficiency, inherited diseases and some solid tumors. Demand for alloHSCT in Russian Federation in children and adults is more than 4 000 cases per year. Due to low fertility levels in Russia, for only 10–15% of Russians can an identical HLA related donor be found. The subsequent search for an unrelated donor can be carried out using the online platform BMDW Search & Match Services. However, for only 80–85% of Russians can an HLA-compatible unrelated donor be found in the international database, largely in connection with the polymorphism of the population of Russian Federation. In this light, there are clear prerequisites for the creation and development of a unified Union of independent States (CIS) register of bone marrow donors, which would increase the availability of alloHSCT for citizens of Russia and the CIS.

Aims: To increase the efficiency of successful searches for and activation of unrelated bone marrow donors in Russian registers by creating an online search platform – "Bone Marrow Donor Search" (BMDS, www.dmds.info).

Methods: Since 2012, the BMDS online search platform has been in development at the Raisa Gorbacheva Memorial Institute for Children's Oncology, Hematology and Transplantation, which, together with other specialized public health institutions, has been incorporating its implementation into clinical practice. Currently, the BMDS includes anonymized data from approximately 60 000 HLA-phenotypes of potential bone marrow donors from 13 Russian and Kazakh bone marrow donor registries.

Results: From November 2012 to April 2017 in various Russian and Kazakh transplant centers 144 cases of alloHSCT were performed from an unrelated donor found in the Russian registers. 88.2% cases were fully compatible – a 10/10 match for the 5 HLA genes loci between donor-recipient; in 11.8% of cases, the match was partially compatible – 9/10. Full compatibility of system AB0 erythrocyte antigens was noted in 20.6% of the cases. The content of $CD34^+$ in the graft was $1.2–12.0 \times 10^6$, median 5.0×10^6. Engraftment was observed in 79.4% of the cases.

Summary/Conclusions: The use of the BMDS online platform increased the efficiency of searches for unrelated HLA-compatible donors in Russian registers for patients in the Russian Federation and the CIS. Further development of the donor base and the inclusion of newly created bone marrow donor registers will further improve the effectiveness of searches for an unrelated donor in Russia, and reduce the time and associated financial costs.

Collection, Processing, Storage and Release

P-590

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REVIEW OF TECHNICAL ASPECTS OF PERIPHERAL BLOOD STEM CELL COLLECTION IN PEDIATRIC DONORS: OUR EXPERIENCE

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Background: Pediatric Peripheral Stem Cell collection has purported to be a modality of treatment in many malignant and non-malignant conditions.
Aims: The aim of this study was to review the technical aspects related to Pediatric PBSC collection including vascular access, flow rate, anti-coagulation, calcium replacement and blood volume processed.
Methods: This was a retrospective observational review of 34 PBSC procedures done in 31 children less than 16 years for various indications. Informed consent was taken from respective guardians. Mobilization was attempted with G-CSF and Plerixafor was added if there was a failed mobilization with G-CSF. Pre-procedure CD34 count was carried out & apheresis was commenced only if peripheral blood CD34 count was ≥20/μl. Femoral venous access was selected with HD triple lumen central venous catheter using mild sedation with Ketamine +/- Midazolam according to hospital guidelines by Pediatric consultant. All procedures were done in ICU under continuous bedside monitoring. COM.TEC (Fresenius- Kabi) cell separator equipment with P1YA kit and continuous Calcium Gluconate infusion was used in all cases. RBC priming was done when required
Results: Of the 31 Pediatric donors (F:M; 20:11) the mean age was 99.5 months with youngest being 13 months old. 16 (51.6%) donors were ≤25 kg. The average pre-procedure TNCC was 48.52 × 10^3/ul, (23.1 to 95.7 x 10^3 /ul). The average pre-procedure CD 34 count was 87.15/ul, slightly lower for female donors (86.02/ul) but the difference was not significant. Highest mobilization of CD 34 cells was seen in children aged less than 5 year (123.7 /ul The average volume processed was 5900 ml. Average TNCC &tCD 34 of product was 237.58 x 10^3/ul & 1019.32 x 10^3/ul respectively. Average hematocrit & Platelet counts of the collected product were 15.7%. & 18 x 10^5/ul. In 28 (90.32%) single procedure was done and only in 3 donors it was done twice. The average dose administered was 5.79 × 10^6/kg with average engraftment time for neutrophils was 14.88 days (11–7 days) while for platelets it was 14.58 days (9–30 days).
Summary/Conclusions: For pediatric PBSC extractions, we recommend a triple lumen HD catheter preferably Femoral as it is less prone to complications. As ECV is an important factor, in children with low TBV, red cell priming should be additionally done with cross matched, irradiated, leucodepleted fresh RBC (< 5 days collection). ACD-A is a preferable anticoagulant as it lacks systemic side effects of heparin. Moreover, the only known commonest complication of citrate toxicity can be well taken care of by continuous calcium infusion based on pre-procedural ionized calcium levels. Volume per cycle is an important determinant, if low it leads to more number of cycles being processed for the same volume of whole blood thereby increasing the product volume, RBC and platelet contamination & increased DMSO requirement for cryopreservation. Blood Flow rates should be kept optimum around 1 ml / kg to avoid complications. Recruitment of progenitor cells may be age-specific and hence more predominant in younger children than in their older counterparts.

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PRELIMINARY COMPARISON OF THE IMPACT OF PBMC COLLECTION ON BLOOD DONORS AND COLLECTION EFFICIENCY BETWEEN 5 DIFFERENT BLOOD CELL SEPARATORS

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Background: In anti-tumor immunotherapy, DC is the most powerful antigen presenting cell. DC is widely distributed *in vivo*, but less in number (only about 1% of peripheral blood mononuclear cells, PBMC). DC can be induced from CD34 + cells or monocytes and proliferate. The adherent mononuclear cells in peripheral blood were used as precursor cells of DC, and higher purity DC could be obtained by cytokine induction.
Ficoll liquid separation is the traditional method which is mainly used for small amount of blood, while small amount of PBMC is difficult to meet the needs of clinical treatment. Large amount of blood has to be divided into many centrifuge tubes, repeated open operation increase the chance of contamination.
Nowadays, the technology of blood cell separator is very advanced. Collection of blood components has been automated and applied to various fields of clinical practice. Blood separator can also be used to collect PBMC from large amount of peripheral blood. In this study, PBMC cells from healthy donors were collected by different blood cell separators, and then the DC cells were cultured *in vitro*.
Aims: This study aims to compare the effects of the collection of peripheral blood mononuclear cells (PBMCs) using five types of blood cell separators, and investigate changes in donors' blood-related indicators.
Methods: The PBMCs of 60 healthy donors were collected using five types of blood cell separators, including MCS+, Amicus, COBE, COM.TEC and OPTIA. Blood routine test was conducted on donors before and after the collection, and was also conducted on the final collection of products. The collection effects of these separators were analyzed by comparing collection efficiency, product content, the amount of collected circulating blood, and product contamination. The collection effects of these separators and their influences on the donors were discussed.
Results: The amount of collected PBMCs ($\times 10^9$/l) in the five groups was 4.11 ± 1.58, 6.61 ± 2.22, 6.00 ± 1.79, 5.77 ± 2.07, and 5.67 ± 1.99, respectively. The decrease value of P1t ($\times 10^9$/l) was 95.58 ± 40.90, 16.83 ± 19.05, 66.83 ± 43.47, 54.67 ± 18.32, and 39.92 ± 17.94, respectively. Furthermore, collection efficiency was 0.50 ± 0.13, 0.43 ± 0.11, 0.46 ± 0.12, 0.50 ± 0.09, and 0.54 ± 0.13, respectively.
Summary/Conclusions: Understanding the impact of different blood cell separators on the donors, as well as its collection efficiency and product contamination rate, and assessing these before collection, would contribute to improving the collection quality of PBMCs. Enough PBMCs can be collected using all five blood cell separators, which meets culturing requirements and clinical treatment requirements. In addition, the collection process is safe and effective for donors.

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COMPARISON OF IN VITRO CHARACTERISTICS OF PLATELETS STORED IN REFRIGERATED WITH A COMMERCIAL PLATELET ADDITIVE SOLUTION CONTAINING N-ACETYLCYSTEINE OR VITAMIN C

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Background: Because of the potential risk of bacterial contamination, platelets have a very short shelf life of 5 days stored at 20–24°C. So, there have been many trials storing platelets in refrigerated or frozen, but cold-stored or frozen-thawed platelets after transfusion have known cleared rapidly by hepatic macrophages or hepatocytes. We need platelets circulating in vivo for long days, having a long shelf life, not contaminated by bacteria.
Aims: We developed and evaluated platelets stored in refrigerated with platelet additive solutions (PAS) containing antioxidants such as N-acetylcysteine (NAC), vitamin C.
Methods: In the study, 50 mM NAC or 100 mM vitamin C mixed in PAS (80 mM sodium chloride, 30 mM sodium acetate, 10 mM sodium citrate, 5 mM potassium chloride, 3 mM magnesium chloride, 26 mM sodium phosphate) added to refrigerated platelets. Prepared platelet-rich plasma (PRP) was stored at either room temperature

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(RT) or cold temperature (CT) mixed with or without PAS containing NAC or vitamin C for 9 days. We measured parameters showing changes in metabolism (pH, glucose, lactate), platelet activation (CD62P, annexin V), oxidative stress (production of reactive oxygen species, intracellular calcium, mitochondrial membrane potential), and sialidase activity in PRP of day 0–9.

Results: Metabolic parameters such as pH, glucose, lactate were significantly better in cold-stored platelets (CT-PLT) than RT-stored platelets (RT-PLT). The pH was changed in CT-PLT from 7.3 to 7.0, but in RT-PLT from 7.3 to 6.6 during storage. Lactate production was increased from 1 mmol/l to 6 mmol/l in RT-PLT, but from 1 mmol/l to 4 mmol/l in CT-PLT. Glucose was decreased from 150 mg/dl to 10 mg/dl in RT-PLT, but from 150 mg/dl to 110 mg/dl in CT-PLT. CD62P expression on CT-PLT with PAS+vitamin C at Day 7 was the lowest (32.22–1.85%), in control 43.24–0.15%. Annexin V binding was also the lowest in PAS+vitamin C treated CT-PLT (52.90–7.07%), which was similar to controls (55.96–5.63%). In rewarming of cold-stored platelets, PAS+vitamin C treated CT-PLT showed the lowest CD62P (38.99–1.05%), but Annexin V result after rewarming was the lowest in PAS+NAC+vitamin C treated CT-PLT (53.92–1.99%). Oxidative stress indices were the lowest in PAS+vitamin C CT-PLT (ROS production, DCFH-DA, MFI, control vs vit-C 2.0 vs. 1.1; intracellular Ca, control vs vit-C 1.07 vs. 1.03; MMP FL2/FL control vs vit-C, 0.3 vs. 0.15). Sialidase activity in rewarming was increased to 110 RFU in control but, in NAC+PAS CT-PLT to 90 RFU.

Summary/conclusions: It revealed that NAC or vitamin C treated CT-PLT prevented platelet activation and reduced the sialidase activity upon rewarming and on prolonged storage. We believe that NAC, vitamin C, or those combination can be a good candidate for an additive solution to retain platelet characteristics during cold storage and may allow for extension of storage shelf life.

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Abstract has been withdrawn

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FIRST EXPLORATION OF THE KEY FACTOR IN THE PBMC COLLECTION QUANTITY USING DIFFERENT BLOOD CELL SEPARATOR

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Background: In anti-tumor immunotherapy, DC is the most powerful antigen presenting cell. DC is widely distributed *in vivo*, but less in number (only about 1% of peripheral blood mononuclear cells, PBMC). DC can be induced from CD34 + cells or monocytes and proliferate. The adherent mononuclear cells in peripheral blood were used as precursor cells of DC, and higher purity DC could be obtained by cytokine induction.

Ficoll liquid separation is the traditional method which is mainly used for small amount of blood, while small amount of PBMC is difficult to meet the needs of clinical treatment. Large amount of blood has to be divided into many centrifuge tubes, repeated open operation increase the chance of contamination.

Nowadays, the technology of blood cell separator is very advanced. Collection of blood components has been automated and applied to various fields of clinical practice. Blood separator can also be used to collect PBMC from large amount of peripheral blood. In this study, PBMC cells from healthy donors were collected by different blood cell separators, and then the DC cells were cultured *in vitro*.

Aims: This study aims to investigate the effects of various factors on the collection of peripheral blood mononuclear cells (PBMCs) from blood donors using five types of blood cell separators, and discuss the collection effects of these five models.

Methods: The influence of 13 factors on the collection of PBMCs were investigated, including blood donor body weight, collector volume, amount of blood collected, the number of collection cycles, collection time, the amount of pre-collection white blood cells, the amount of pre-collection red blood cells, the hematocrit of pre-collection red blood cells, pre-collection hemoglobin content, amount of pre-collection platelets, absolute amount of pre-collection lymphocytes, absolute amount of pre-collection monocytes, and absolute amount of pre-collection mononuclear cells.

Results: For the Amicus blood cell separator, the absolute amount of pre-collection mononuclear cells of blood donors should be given attention. For the MCS + blood cell separator, the collector volume and absolute amount of pre-collection mononuclear cells of donors should be given attention. For the COBE blood cell separator, the absolute amount of pre-collection mononuclear cells of donors should be given attention. For the COM.TEC blood cell separator, the absolute amount of pre-collection lymphocytes of donors should be given attention. For the Optia blood cell separator, the blood collection amount and absolute amount of pre-collection lymphocytes of donors should be given attention.

Summary/Conclusions: Understanding the influencing factors of different blood cell separators and evaluating these before collection would contribute in improving the collection quality of PBMCs.

P-600

THE TENTATIVE EXPLORATION OF COLLECTING PBMC FROM HEALTHY DONORS AND CULTURING PBMC-DERIVED DCS *IN VITRO*

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Background: In anti-tumor immunotherapy, DC is the most powerful antigen presenting cell. DC is widely distributed *in vivo*, but less in number (only about 1% of peripheral blood mononuclear cells, PBMC). DC can be induced from CD34 + cells or monocytes and proliferate. The adherent mononuclear cells in peripheral blood were used as precursor cells of DC, and higher purity DC could be obtained by cytokine induction.

Ficoll liquid separation is the traditional method which is mainly used for small amount of blood, while small amount of PBMC is difficult to meet the needs of clinical treatment. Large amount of blood has to be divided into many centrifuge tubes, repeated open operation increase the chance of contamination.

Nowadays, the technology of blood cell separator is very advanced. Collection of blood components has been automated and applied to various fields of clinical practice. Blood separator can also be used to collect PBMC from large amount of peripheral blood. In this study, PBMC cells from healthy donors were collected by different blood cell separators, and then the DC cells were cultured *in vitro*.

Aims: To study the cell harvest and viability of the PBMC collected from healthy donors' by different blood cell separators, and then, study the impact of cell harvest and viability on DCs culture *in vitro* .

Methods: To collect healthy donors' PBMC with 5 different blood cell separators including Amicus, COM.TEC, Cobe Spectra, MCS+, and Spectra Optia. 12 donors were treated as a group and for each group a type of blood cell separator was used. The donors are all regular unpaid blood donors who donate platelets every 15 days. Circulating blood volume was between 5 and 9 liters. Collecting time was between 2 h and 3 h. Mobilizers were not used. The final PBMC product volume was between 100 ml and 120 ml. The number of PBMC cell harvest was between 1.4 10^9 and 5.8 10^9.

Before being cultured, PBMC were preserved under 22°C for 18 h or 28 h. Apoptosis ratio of pre-culturing PBMC were observed by microscope. After 6 days of cultivation, viability of DC cells was observed by flow cytometry. Anti-$CD11c^+$ antibodies were used to mark DC. Viability of DC was detected by DAPI staining. At the sixth day, regarded as qualified for culture, the number of active DC must be no less than 1 10^8.

Results: Of 60 samples, there were 12 donors whose PBMC harvest was less than 2 10^9, meanwhile, there were also 12 donors whose PBMC viability were lower than 95%, while these 12 donors were not the same as above. There were 13 donors whose DCs' viability didn't meet standard. Through χ^2 test, PBMC harvest and viability played significant role in cultivation of DCs. The viability of DCs was significantly higher in the group when PBMC harvest was more than 2 10^9($\chi^2 = 11.13$, $P < 0.01$), or PBMC viability was higher than 95%($\chi^2 = 14.74$, $P < 0.01$).

Summary/Conclusions: PBMC can be collected effectively and safely by blood cell separator, and the viability of DCs *in vitro* cultivation was acceptable. For DC cultivation, PBMC harvest and viability are key factors.

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THE RECOVERY OF DONORS' BLOOD ROUTINE INDEX AFTER PBMC APHERESIS WITH BLOOD SEPARATORS

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Background: In anti-tumor immunotherapy, DC is the most powerful antigen presenting cell. DC is widely distributed *in vivo*, but less in number (only about 1% of peripheral blood mononuclear cells, PBMC). DC can be induced from CD34 + cells or monocytes and proliferate. The adherent mononuclear cells in peripheral blood were used as precursor cells of DC, and higher purity DC could be obtained by cytokine induction.

Ficoll liquid separation is the traditional method which is mainly used for small amount of blood, while small amount of PBMC is difficult to meet the needs of clinical treatment. Large amount of blood has to be divided into many centrifuge tubes, repeated open operation increase the chance of contamination.

Nowadays, the technology of blood cell separator is very advanced. Collection of blood components has been automated and applied to various fields of clinical practice. Blood separator can also be used to collect PBMC from large amount of peripheral blood. In this study, PBMC cells from healthy donors were collected by different blood cell separators, and then the DC cells were cultured *in vitro*.

Aims: To compare the changes of routine blood test results in healthy volunteers after the collection of peripheral blood mononuclear cell with five different blood cell separators.

Methods: 12 healthy volunteers were collected the peripheral blood mononuclear cell respectively. Using five kinds of blood cell separators Amicus, Com.tec, Mcs plus, COBE and Optia the blood to do the routine blood test before the collection, after the collection 2 weeks and after the collection 4 weeks and observe the changes after the collection.

Results: There is no obvious variation of the blood cells count before and after collection in 2 weeks except the WBC. And there is no obvious variation of the blood cells count before and after collection in 4 weeks.

Summary/Conclusions: The test of white blood cells from volunteers shows that the figures of the two-weeks interval group are higher than themselves before the collection. While the figures are still fluctuate within the normal range, and the 4-weeks interval has less effect on the healthy volunteers.

P-602

IMPACT OF DIFFERENT PRESERVATION CONDITIONS ON THE APOPTOSIS RATE OF PBMC COLLECTED BY BLOOD SEPARATOR

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Background: In anti-tumor immunotherapy, DC is the most powerful antigen presenting cell. DC is widely distributed *in vivo*, but very few in number (only about 1% of peripheral blood mononuclear cells, PBMC).Recently, several clinical trials have reported some promising benefits of using the CAR-T cell therapy in the treatment of a variety of cancers. PBMC cells can serve as the source of T cells for CAR-T and DC. Ficoll liquid separation is the traditional method which is mainly used for small amount of blood, while small amount of PBMC is difficult to meet the needs of clinical treatment. Large amount of blood has to be divided into many centrifuge tubes, repeated open operation increase the chance of contamination.

Nowadays, the technology of blood cell separator is very advanced. Collection of blood components has been automated and applied to various fields of clinical practice. Blood separator can also be used to collect PBMC from large amount of peripheral blood. In this study, PBMC cells from healthy donors were collected by blood cell separator .

Aims: To study the optimum condition to preserve PBMC collected by blood cell separator. Different storage conditions refer to the combination of different temperature and different storage time. The purpose of the study is to find the best preservation time and temperature.

Methods: To collect 12 healthy donors' PBMC with Spectra Optia cell separator using MNC program. The donors range from 27 to 56 years old, including 8 men and 4 women. They are all regular unpaid blood donors who donate platelets every 15 days. Circulating blood volume was between 5 and 8 liters. Collecting time was between 2 h and 3 h. Mobilizers were not used. The final PBMC product volume was between 100 ml and 115 ml. The number of PBMC cells was between 1.8 10^9 and 4.9 10^9. We took 6 ml from the product and divided it into 6 equal parts. 3 samples were preserved at 4°C and the other three at 22°C. After 5 h, two samples were took out, each of 4°C and 22°C. Apoptosis ratio of PBMC were analyzed by flow cytometry (FasAria II, BD). Then at 18 h, and finally at 28 h. Anti-$CD45^+$ antibodies were used to mark total PBMC, anti-$CD3^+$ antibodies labeled T lymphocytes, and anti-$CD14^+$ antibodies labeled monocytes. The apoptosis ratio of different kinds of cells was detected by 7aad. All reagents were purchased from BD company.

Results: Taking monocytes as an example, the mean apoptotic ratio was 4.9 ± 1.3% (4 5 h), 3.3 ± 0.5% (4 18 h), 4.1 ± 1.6% (4 28 h) and 4.0 ± 1.2%(22 5 h),9.8 ± 1.5% (22 18 h), 20.2 ± 10.4% (22 28 h). As for T lymphocytes, the ratio was 1.9 ± 0.5% (4 5 h), 4.0 ± 1.0% (4 18 h), 5.3 ± 1.2% (4 28 h) and 1.7 ± 0.4% (22 5 h),4.3 ± 0.6% (22 18 h), 9.5 ± 1.8% (22 28 h).Within 5 h, there were no significant difference between 4°C and 22°C preservation for all cells. But it's worth noting that at 18 h the apoptosis rate of 22°C were higher than 4°C significantly. And after 28 h this feature was more obvious. The apoptosis ratio of 4°C preservation grew slowly while at 22°C grew rapidly. Under the same condition, the apoptosis ratio of monocytes was always higher than T lymphocytes.

Summary/Conclusions: Compared with 22°C,4°C was better for PBMC preservation. Under 4°C, the average apoptosis ratio of PBMC remained within 5% at 28 h.

P-603

ANALYSIS OF BLOOD NUCLEIC ACID DETECTION IN CHANGSHA AREA

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Background: Blood related infectious diseases prevention and control has been the focus of world attention, select the appropriate detection technology can effectively improve blood safety. This paper reviews the nucleic acid testing (NAT) in the application of blood screening.

Aims: By analyzing the nucleic acid results of blood donors in Changsha, the significance of the nucleic acid testing (NAT) in blood screening was discussed.

Methods: 1) First of all, the samples of blood donors were tested two times of HBsAg, anti -HCV and anti -HIV by ELISA method; 2) Using Roche nucleic acid or KELONG blood screening system, mixed samples were tested for HBV-DNA, HCV-RNA, and HIV-RNA; 3) Mixed inspection reactive samples were split test.

Results: The results showed that in January 2017-June 2017, 73009 samples were tested in Changsha blood center, 84 pools positive and 41 specimens split positive (all of which were positive for HBV-DNA), and the resolution rate was 48.81% (41/84).

Summary/Conclusions: Therefore, the testing of blood nucleic acid can effectively reduce the risk of blood transfusion, and ensure the safety of the blood.

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THE CHANGE OF BLOOD DONORS' ELECTROLYTE CONCENTRATION AFTER PBMC APHERESIS WITH DIFFERENT BLOOD SEPARATORS

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Background: In anti-tumor immunotherapy, DC is the most powerful antigen presenting cell. DC is widely distributed *in vivo*, but less in number (only about 1% of peripheral blood mononuclear cells, PBMC). DC can be induced from CD34 + cells or monocytes and proliferate. The adherent mononuclear cells in peripheral blood were used as precursor cells of DC, and higher purity DC could be obtained by cytokine induction.

Ficoll liquid separation is the traditional method which is mainly used for small amount of blood, while small amount of PBMC is difficult to meet the needs of clinical treatment. Large amount of blood has to be divided into many centrifuge tubes, repeated open operation increase the chance of contamination.

Nowadays, the technology of blood cell separator is very advanced. Collection of blood components has been automated and applied to various fields of clinical practice. Blood separator can also be used to collect PBMC from large amount of peripheral blood. In this study, PBMC cells from healthy donors were collected by different blood cell separators, and then the DC cells were cultured *in vitro*.

Aims: Make comparisons of blood donors' electrolyte concentration between three time points: before PBMC apheresis, 15 min after PBMC apheresis, and 2 weeks after PBMC apheresis.

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Methods: To collect healthy donors' PBMC with 5 different blood cell separators including Amicus, COM.TEC, Cobe Spectra, MCS+, and Spectra Optia. 12 donors were treated as a group and for each group a type of blood cell separator was used. The donors are all regular unpaid blood donors who donate platelets every 15 days. Circulating blood volume was between 5 and 9 liters. Collecting time was between 2 h and 3 h. Mobilizers were not used.
Electrolyte concentration of blood donors was detected at three time points: before apheresis, 15 min after apheresis and two weeks after apheresis. And then the data was compared.
Results: The data of 15 min after apheresis indicated that, concentration of Blood $K^+$and blood Cl^- decreased significantly after the collection ($P < 0.05$), while there was no obvious change of blood Na^+ concentration ($P > 0.05$).AS for the data of two weeks after apheresis, there was no obvious difference compared with the data detected before apheresis. This indicated that the electrolyte concentration of the blood donors had recovered after two weeks.
Summary/Conclusions: After PBMC apheresis, there was a transient reduction of the donors' blood K^+ and blood Cl^- electrolyte concentration. Appropriate intervention prior to apheresis or corresponding treatment after apheresis was necessary.
Two weeks after PBMC apheresis, the electrolyte concentration of the blood donors had recovered. This proved that collecting PBMC with blood separator was safe.

Clinical Applications

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CD8 + FOXP3 + REGULATORY T CELLS DERIVED EXOSOMES AND THEIR POTENTIAL ROLES IN THE IMMUNE MODULATION

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Background: Exosomes are defined as one type of membrane vesicles secreted into extracellular space by most types of cells and are reported to involve in varies kinds of biological process. Human periphery blood $CD8^+FOXP3^+$ Tregs cells are reported as more stable regulatory cells with greater inhibition effects. However, $CD8^+FOXP3^+$Tregs derived exosomes and their functions involved in $CD8^+$Tregs mediated immune-modulation were seldom reported.
Aims: To isolate the $CD8^+CD103^+$Tregs derived exosomes and investigate their functions in $CD8^+CD103^+$Tregs mediated immune-modulation
Methods: $CD8^+$ T cells were freshly purified from PBMCs, cultured with anti-CD3/CD28 antibody packaged beads and IL-2, and then cultured in the Tregs polarizing condition with TGF-β and rapamycin. The harvest cells were co-cultured with dendritic cells stimulating CD4 + CD25- effector cells in the transwell plate. The supernatant derived from CD8 + Tregs was collected and ultrafiltrated by centrifugation and the remaining solution was precipitated with PEG. The harvest precipitation was resuspended in PBS and exosomes were analyzed by SEM and NTA. Exosome surface marker CD63, CD81, TSG101 and other proteins expression were evaluated by flow cytometry and western blot. microRNA was isolated and miR-155, let-7b, let-7d were measured by qPCR. The precipitated exosomes were further purified by CD63 immunoaffinity capture and co-cultured with effector cells to investigate their function in immune modulation.
Results: As compared with direct contact co-culture, separated $CD8^+$Treg cells could suppress the proliferation of effector cells with a small decline ($P > 0.05$), which means some non-contact factors involved in the $CD8^+$ Treg mediated immune modulation. A total number of $4.57 \pm 0.52 \times 10^8/10^6$ cells exosomes were harvest. Electron microscope analysis demonstrated a kind of round-shaped membrane vesicle 50–150 nm in diameter (145.1 ± 6.7 nm by NTA). Western blot experiments confirmed the expression of those exosome surface markers CD63, TSG101. Besides, TGF-beta and NOX2 were also expressed on $CD8^+FOXP3^+$Treg derived exosomes. Compared with their IL-2 induced effecter $CD8^+$ counterparts, $CD8^+$Treg derived exosomes contained higher levels of miR-155, let-7b and let-7d ($P < 0.05$). And the addition of $CD8^+$ Treg cells derived exosomes demonstrated a dose-depend inhibition on the proliferation of $CD4^+$ effector cells.
Summary/Conclusions: Extracellular vesicles ranged in 50–150 nm diameter were secreted by $CD8^+CD103^+$Treg cells and those exsomes could involve in $CD8^+$Treg mediate immune modulation.

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Abstract has been withdrawn

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RNA-SEQ-BASED GENE EXPRESSION PROFILING OF MEGAKARYOCYTES FROM HUMAN CORD BLOOD CD34 + CELLS *EX VIVO* EXPANDED IDENTIFIES NEW TRANSCRIPTION REGULATORS OF MEGAKARYOPOIESIS

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Background: Megakaryocytes (MKs) are the sole progenitors for platelets. The processes of differentiation and platelet formation from megakaryocytes still remain to be elucidated. While some key megakaryocytic transcription factors (TFs) have been identified, the complete network of megakaryocytic transcriptional control is poorly understood.
Aims: To understand cellular mechanism in the megakaryopoiesis, using RNA-Seq technology we examined the mRNA level of transcriptional regulators in megakaryocytes derived from human cord blood CD34 + hematopoietic cells.
Methods: CD34 + cells were isolated using density gradient centrifugation and magnetic activated cell sorting (MACS). Cultures were stimulated with only recombinant human TPO (100 ng/ml). After 7, 14 and 18 days, the MK fraction was selected by immunomagnetic sorting from the non-MK fraction using an anti-CD41a monoclonal antibody. RNA-Seq-derived gene expression data was performed on uncultured samples (day 0), cultured but unselected samples (days 7), and cultured, selected samples (days 14, days 18) by using the next-generation sequencing (NGS) platform. RT-PCR and immunofluorescence were used to verify the expression of transcription factors.
Results: The transcriptional kinetics of most known megakaryocytic transcription factors have been identified including GATA1, FLI1, and MAFG. 9 genes were upregulated specifically in Mk cells compared with CD34 + cells and unselected early cells. Most of these upregulated TFs either have not previously been associated with megakaryopoiesis or their role in the Mk lineage remains ambiguous .These include MXD1, MEF2C, BACH1, which are strongly upregulated in Mk cells. Protein expression and nuclear localization were confirmed in megakaryocytic cells for three of the novel candidate megakaryocytic transcription factors.
Summary/Conclusions: This study reveals a global gene expression profile of *in vitro* human CB -derived megakaryopoiesis. Some of these genes may play regulatory roles during the development of CB-derived megakaryopoiesis. Our observations may help to expand our understanding of transcriptional regulation in megakaryocytic differentiation of stem and progenitor cells.

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ALPHA-DYSTROGLYCAN PLAYS IMPORTANT ROLES IN PLATELET AGGREGATION AND HEMOSTATIC PLUG FORMATION

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Background: Fibrinogen and von Willebrand factor (VWF) have been considered essential for platelet adhesion, aggregation and hemostatic plug/thrombus formation. However, it has been shown that platelets lacking fibrinogen, and even lacking both fibrinogen and von Willebrand factor (VWF), can still aggregate *in vitro* and *in vivo*. Suggesting other unidentified proteins can mediate platelet aggregation and platelet plug formation independent of fibrinogen and/or VWF. Through screening published platelet proteomics data, we identified a potential candidate, alpha-dystroglycan (α-DG). α-DG is a glycoprotein found in the dystroglycan complex, which bridges the intracellular actin cytoskeleton with extracellular matrix proteins like laminin. Despite vast reporting on the role of α-DG and the dystroglycan complex in myocytes, its role in platelet aggregation and hemostasis remains to be explored. Hence, we hypothesize that additional proteins, such as alpha-dystroglycan (α-DG), are involved in platelet aggregation and hemostatic plug formation.

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Aims: To identify whether α-DG contributes to platelet adhesion, aggregation, and hemostatic plug formation.
Methods: Flow cytometry and western blotting were used to assess the expression of α-DG on platelets. Light transmission aggregometry was used to investigate the effect of α-DG blocking antibodies on platelet aggregation in both platelet-rich plasma and washed platelets with a variety of agonists. A murine laser-injury intravital microscopy was used to evaluate the effect of anti- α-DG antibodies within flowing blood *in vivo*. Collagen-coated perfusion chambers were used to study anti-α-DG antibodies in human blood under flowing conditions. The effects of anti- α-DG antibodies on the viscoelastic properties and kinetics of coagulation were assessed using thromboelastography.
Results: We found α-DG was expressed on the surface of human and murine platelets (resting and activated). Using light transmission aggregometry, we observed that a polyclonal antibody against full length α-DG markedly inhibited platelet aggregation in both platelet-rich plasma and gel-filtered platelets. To localize the specific functional domains on α-DG that contribute to platelet aggregation, an α-DG N-terminus polyclonal antibody was employed and a similar decrease in platelet aggregation was observed. Furthermore, a monoclonal antibody specific to the N-terminus laminin binding site in α-DG was able to inhibit platelet aggregation, suggesting an important role of the laminin binding site in mediating hemostasis. We performed intravital microscopy with vessels of the mouse cremaster muscle using a laser injury model and found that injection of the polyclonal antibody against α-DG into mice decreased the growth of thrombi, which were consistently washed away after platelet plug formation (embolization) at the site of injury. Preliminary data showed that the α-DG polyclonal antibody decreased clot strength and inhibited human platelet adhesion and aggregation in collagen coated perfusion chambers.
Summary/Conclusions: α-DG is a novel protein that plays a functional role in platelet aggregation and hemostatic plug formation. The interactions between the N-terminus of α-DG with platelet receptor(s) and/or sub-endothelial matrix proteins are currently under investigation.

P-609

PRELIMINARY RESULTS OF HLA-TYPING IN DONORS AND RECIPIENTS

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Background: Organ transplantation and hematopoietic stem cell transplantation (HSCT) often are the main methods of treatment for patients with end-stage renal, hepatic and cardiac failure, malignancy, and hematological and hereditary diseases. In the Republic of Tajikistan, more than 500 patients are in need of kidney transplantations per year, HSCT – about 100 patients, liver transplants – more than 200 patients. However, during any given year only a few dozen of these patients are able to receive kidney and liver transplants. HSCT is performed in foreign clinics for a limited number of patients, and there the main limiting factor is the absence of HLA-compatible donors due to ethnic and genetic features. In organ and HSCT, the fact of graft engraftment and rejection is determined by the degree of biological compatibility between donor and recipient. The median time before transplant rejection in full HLA-compatible allogeneic relative transplantation is 22.4 years and 4.6 years in allogeneic organ transplantation.
Aims: To determine HLA typing in donors and recipients amongst a sample drawn from the Republic of Tajikistan.
Methods: HLA class I (A and B) phenotype identification was performed in lymphocyte suspension with the complement-dependent lymphocytotoxic test in the Terasaki microplates (Gisans, St. Petersburg). For molecular genetic typing of HLA class I and II antigens in blood samples PCR-SSP method was used, which is based on polymerase chain reaction using sequence-specific primers with PROTRANS kits (Germany). Also, HLA system antibodies (sensitization index) identification and cross-match tests for tissue antigens between donors and recipients were performed.
Results: Obtained results showed that genes A2, A11, A1, A3 dominate in the HLA-A locus. Genes A19, A23, A24, A26, A32, A33, A36 and others are less often detected. Genes B35, B13, B51, B49, B17, B14, B38, B12 were found predominantly in HLA-B locus. Genes B7, B8, B11, B15, B16, B18, B22, B40, B41, B60, B61 were detected much less often. Most European peoples have a dominance of the genes A1, A2, A3, B7 and B35. Comparative analysis shows a significant difference in genes between the citizens of the Republic of Tajikistan in comparison with Europeans, especially at the locus B. DRB1*03, DRB1*13, DRB1*04, DRB1*15 genes predominate among locus DRB1*DR. The remaining genes (DRB1*05, DRB1*11, DRB1*09, DRB1*14, DRB1*08 and others) are less frequently detected and only in single cases. Citizens of European countries mainly have genes DRB1*04, DRB1*07, DRB1*08, DRB1*10, DRB1*11, DRB1*12, DRB1*14, DRB1*16, DRB1*18. The results of anti-HLA antibodies search were positive in 18.2% of recipients, and the results of cross-match with donor lymphocyte samples were positive.
Summary/Conclusions: There are a number of differences in the distribution of HLA antigens among the population of the Republic of Tajikistan in comparison with Russian Federation, the Republic of Kazakhstan and European countries. The level of HLA-incompatibility underlines the need for the creation of a national register of typed donors.

P-610

Abstract has been withdrawn

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IDENTIFICATION OF MITOXANTRONE DIHYDROCHLORIDE AS AN ANTI-HSV COMPOUND

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Background: HSV is one of the most prevalent human pathogen that causes of oral-labial infection, urogenital lesions and life-threatening encephalitis. There are some antiviral medicines with activity against HSV-1 and HSV-2, however, more and more evidences show that the incidence of drug resistance is increasing.
Aims: The aim of our study was to identify new anti-HSV compound with potential use for further drug development.
Methods: We screened a chemical library of around 1500 compounds to identify inhibitors of HSV-1 induced toxicity. Cell was pretreated with compounds and their viability was determined by measuring ATP levels to reflect the inhibition effect against HSV-1 induced necrosis. Viral titers were determined by plaque assay. Further experiments including western blot analysis, Q-PCR analysis and Luciferase assay were performed to explore the antiviral mechanism of the compounds.
Results: Here, we identified a small molecular, mitoxantrone dihydrochloride (henceforth MD) with potency against HSV-1-induced death. Viral titers and the GFP-labeled HSV-1 were potently reduced by pretreatment of MD in both human and mouse cell lines. Moreover, MD efficiently blocked the expression levels of viral proteins in various cells. Notably, MD inhibited the expression of immediate-early gene *ICP0*, *ICP22*, *ICP27* and *ICP47* which are essential for the expression of early and late viral gene products. Notably, transcriptional expression of genes regulating HSV replication such as *UL5*, *UL9*, *UL29*, *UL30*, *UL42* and *UL52* were also significant reduced in the presence of MD. We also found MD had no influence on the NF-κB activity in host cells and *ICP0* promoter activity of HSV-1.
Summary/Conclusions: Taken together, our study revealed that MD has potent efficacy against HSV induced toxicity through inhibition of the essential immediate-early gene. Therefore, MD could be used as a candidate in the development of anti-HSV drugs.

P-612

FACILE SYNTHESIS OF SILVER NANOPARTICLES AS ANTI-LEUKEMIA AGENT VIA PRODUCTION OF REACTIVE OXYGEN SPECIES

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Background: Silver nanoparticles have been reported owning anti-cancer effect, mainly through elevating reactive oxygen species (ROS), and these particles could also display a synergistic effect against cancer cells with chemotherapeutic drugs. However, whether and how these particles could inhibit the growth of acute myeloid leukemia (AML) cells is unclear. Moreover although several investigators have reported that silver nanoparticles have potential cyto-toxic effect against leukemic cell lines, however, there has been scarce investigation on the effect of silver

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nanoparticles exposure on clinical isolates of cancer or leukemia. Meanwhile the mechanism underlying silver nanoparticles-induced biological effects has not been clearly elucidated.
Aims: To achieve a rapid large scale synthesis of silver nanoparticles by microwave irradiation and investigate whether and how silver nanoparticle could inhibit the growth of AML cells.
Methods: Silver nanoparticles were synthesized via microwave irradiation at 150 °C using soluble starch as capping agent. L-lysine or L-arginine served as reducing agent. A series of characterization of silver nanoparticles including surface charge, UV-vis spectrum and so on were studied. Three leukemic cell lines and the bone marrow aspirations of AML patients had been chosen for research, treated with silver nanoparticles or control. Cell viability was measured using CCK-8 assay. The intracellular generation of reactive oxygen species (ROS) and cell apoptosis were measured by flow cytometry. The mitochondrial membrane potential (MMP) was investigated using the fluorescent lipophilic cationic dye JC-1. DNA damage was assayed by confocal laser scanning microscopy.
Results: Silver nanoparticles were synthesized with uniform size. Viability of normal cell and leukemic cell were studied to analyze the cytotoxic effect of silver nanoparticles. AML cells were collected from AML patients. Apoptosis, losses of mitochondrial membrane potential (MMP), the generation of ROS and DNA damage could be enhanced obviously treated by silver nanoparticles with chemotherapeutic drugs, as well as the results treated with silver ions alone. As a chelator, N-acetyl-L-cysteine (NAC) could bind the silver ions to reverse the production of ROS and DNA damage, which validated the important role of silver nanoparticles served as anti-leukemia agent.
Summary/Conclusions: The cytotoxic effect of silver nanoparticles on AML cells and their underlying mechanism and might have significant impact on AML treatment.

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HUMAN UMBILICAL CORD DERIVED MESENCHYMAL STEM CELLS CO-CULTURE WITH CD4 + T CELLS OF HEALTHY DONORS AND SYSTEMIC LUPUS ERYTHEMATOSUS PROMOTE IL-17 EXPRESSION

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Background: Inflammation initiated by T helper 17 (Th17) cells orchestrate the pathogenesis of autoimmune diseases. As a promising candidate for therapeutic treatment of autoimmune disease, human umbilical cord-derived mesenchymal stem cells (hUC-MSCs) appear to directly suppress T cell activation, but the effect of hUC-MSCs on the differentiation of human CD4 + T cells is not clearly known.
Aims: The purpose of this study is to evaluate the effect of hUC-MSCs on the human CD4 + T cells of healthy donors and systemic lupus erythematosus.
Methods: Cytokines and mRNA levels for retinoic acid receptor-related orphan receptor C (RORC) were quantified by ELISA and RT-PCR respectively, and cell sorting was performed by flow cytometry. Twelve SLE patients (ten females, two males) were recruited at the Beijing Hospital. Diagnosis of SLE was established according to the 1982 revised American Rheumatism Association criteria (ARA).
Results: IL-17 production was increased when non-fractionated human peripheral blood mononuclear cells (hPBMCs) isolated from 10 healthy donors that were activated by PHA and then co-cultured with hUC-MSCs (2.18 ± 0.77 vs 17.47 ± 3.81, hUC-MSCs only: 2.43 ± 0.88; $P < 0.05$). RORC mRNA levels were increased significantly by the addition of hUC-MSCs at day 5 (7.77 ± 0.79 vs 19.07 ± 1.91; $P < 0.01$). Flow cytometry analyzed the percentages of Th17 cells when CD4 + T cells were co-cultured with or without hUC-MSCs ($13.67 \pm 3.79\%$ vs $5.82 \pm 1.53\%$, $P < 0.01$). In co-cultures of hUC-MSCs and stimulated CD4 + T cells, PGE2 was increased dramatically, compared with CD4 + T cells cultured alone ($P < 0.05$). No significant difference ($P > 0.05$) was found in IL-17 production between activated CD4 + T cells/hUC-MSCs in the presence of IL-23 (1429 ± 166.6) or not (1374 ± 208.3) in healthy donors and SLE patients. The addition of hUC-MSCs to cultures of CD4 + T cells resulted in decreased IFN-γ (46.63 ± 4.27 vs 9.23 ± 1.70; $P < 0.01$) and enhanced TGF-β production (333.3 ± 43.72 vs 938.3 ± 187.5; $P < 0.05$). Furthermore, we detected the percentages of Th1 and regulatory T cells (Treg) separately using flow cytometry. These data were consistent with the data from ELISA analysis (Th1: 17.40 ± 2.07 vs $5.71 \pm 2.92\%$, $P < 0.01$; Treg: 11.35 ± 2.05 vs $24.49 \pm 4.17\%$, $P < 0.05$).
Summary/Conclusions: HUC-MSCs promote the expression of IL-17 but not IL-23 production from hPBMCs/CD4 + T cells of healthy donors, which might decrease the production of Th1 and increase the production of Treg. PGE2 is also critical for IL-17 production by hUC-MSCs. This finding provides some controversial effects of BM-MSCs reported recently, and gives a new insight into the potential immunomodulative function of MSC-based clinical studies.

P-614

POST-OPERATIVE ATYPICAL HEMOLYTIC UREMIC SYNDROME: A CASE REPORT

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Background: aHUS (atypical Hemolytic Uremic Syndrome) accounts for 5–10% of HUS cases. Many etiologies have been known to trigger the disease but post-operative HUS is a relatively newer and uncommon entity. In the postoperative period, it is crucial to differentiate this entity from other causes of anemia, thrombocytopenia, and renal failure. It carries a poor prognosis, with mortality as high as 25% and progression to end-stage renal disease in about 50% cases. We report a case where post-operative period was complicated by aHUS and emphasize the role of Therapeutic plasma Exchange (TPE) in improving clinical outcome. To the best of our knowledge only 4 cases have been reported in the literature and we report this as the fifth case.
Aims: To evaluate the role of Therapeutic plasma Exchange in achieving hematological remission in post-operative aHUS.
Methods: Therapeutic Plasma Exchange on Com-Tec (Fresenius) Apheresis System using PL1 kit.
Results: A 39 year old female with previous bad obstetrics history underwent hysterectomy for Grade IV Endometriosis. She developed sudden severe hypotension in peri-operative period compounded by renal dysfunction on first post-operative day. This was accompanied by micro-angiopathic hemolytic anemia and thrombocytopenia. DCT and ICT were negative. Coagulation profile was normal. Patient was afebrile, had no neurological abnormality and normal ADAMTS 13 levels precluding the diagnosis of Thrombotic Thrombocytopenic Purpura (TTP). Complement levels were within normal range. We performed 9 (7 consecutive days and 2 alternate day) Therapeutic Plasma Exchange which gradually improved the hematological profile with normalization of Platelet counts (90 on post-operative day (POD) 1 to 160 ($\times 10^3$/ul) on POD 13) and Serum LDH levels (3103 IU/l on POD 3 to 195 IU/l on POD 17). However despite early intervention with hemodialysis patient did not recover from the renal insult and has persistently raised serum creatinine levels requiring maintenance hemodialysis.
Summary/Conclusions: Post-operative aHUS is an uncommon disorder with only 4 cases reported in literature. High clinical suspicion is pivotal in early recognition to prevent significant mortality associated with this condition. Prompt medical intervention with Therapeutic Plasma Exchange (TPE) is life saving in these patients.

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THE POTENTIAL OF HYPOXIA PRECONDITIONING ON THE NEURAL GENES EXPRESSION PROFILING IN HUMAN UMBILICAL CORD BLOOD MESENCHYMAL STEM CELLS

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Background: In recent years, human umbilical cord blood-derived mesenchymal stem cell (hUB-MSCs) has been regarded as an alternative source for stem cell therapy.
Aims: In this study, we evaluated the effect of hypoxia preconditioning (HPC) on the expression of Nt-3, GFAP, Nestin, Oct-4 and Nanog genes and proliferative capacity of hUB-MSCs in comparison with normoxic conditions.
Methods: HPC + Hypoxia protocol includes cultured hub-MSCs for 15 min at 2.5% O2 and after that reoxygenation for 30 min at 21% O2 (HPC), and then hypoxia preconditioned hUB-MSCs subjected to 2.5% O2 for 72 h (Hypoxia).

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Results: Conclusively, the results showed that hypoxic preconditioning is an effective strategy for enhancing proliferation capacity of hUB-MSCs; and also can trigger expression of some of the neural genes.
Summary/Conclusions: In addition, the concept of involvement of oxygen tension in the expression of some of the neural genes of hUB-MSCs would be a good sign of enhanced neural differentiation potential *in vitro*.

P-616

ENHANCED ANTITUMOR ACTIVITY OF RGD-MODIFIED ADENOVIRUS MEDIATED LIF AND IL-24 CO-EXPRESSION IN HUMAN MYELOID LEUKEMIA CELLS

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Background: Myeloid leukemia is a common type of leukemia in adults, which is associated with a high relapse rate and poor prognosis. Molecular mechanisms for cancer development and progression are due to multiple defects in a cell that include gene alterations, gene silencing, and alterations in the cell signaling pathways. Gene therapy represents a promising therapeutic modality for cancers. Multigene-based combination therapy represents a valid and innovative approach especially for hematological malignancies treatment. Interleukin-24 (IL-24), a member of the IL-10 cytokine gene family, causes growth suppression and apoptosis in various tumor cells. Leukemia inhibitory factor (LIF) is a multifunctional cytokine, it has been shown to have growth inhibition effect on some kinds of leukemia cells. Based on the antitumor features of LIF and IL-24, we developed a LIF/IL-24 bicistronic recombinant adenovirus Ad.RGD-LIF-IL-24 and assessed its effect on myeloid leukemia cells.
Aims: To construct a RGD-modified recombinant adenoviral vector co-expressing LIF and IL-24 and explore its enhanced antitumor effect on human leukemia cells and its mechanism.
Methods: The RGD-modified adenovirus Ad.RGD-LIF, Ad.RGD-IL-24 and Ad.RGD-LIF-IL-24 were constructed by using the method of homologous recombination. LIF or IL-24 gene was transduced into human myeloid leukemia cells via recombinant adenoviral vector. The effect of ectopic expression of IL-24 or LIF on the growth of the myeloid leukemia cells was determined by CCK-8 assay. Cell-cycle distribution and apoptosis of the leukemia cells were determined by FCM. Quantitative Real-Time PCR analysis was used to detect the expression levels of p21 and E2F1 gene. Western blot method was used to detect the expression of p53, Bax, Caspase-3 and Bcl-2 gene expression after adenovirus infected.
Results: The RGD-modified adenovirus Ad.RGD-LIF, Ad.RGD-IL-24 and Ad.RGD-LIF-IL-24 were constructed successfully. They mediate gene transfer into MEG01 and K562 leukemia cells with high efficiency. Treatment with recombinant adenovirus could potentially induce leukemia cells cell-cycle arrest. LIF and IL-24 gene expression could inhibit the growth of MEG01 cells and induce cell apoptosis through the regulation of Bax, Bcl-2, p53, E2F1 and caspase-3 gene expression. Besides, the double gene expression has a synergistic effect.
Summary/Conclusions: RGD-modified recombinant adenovirus could mediate gene transfer into myeloid leukemia cells with high efficiency. Ad.RGD-LIF-IL-24 had growth inhibition and cell cycle arrest effects on myelogenous leukemia cells. LIF and IL-24 gene co-expressing had significant synergetic effect on apoptosis induction of the leukemia cells. Furthermore, they showed enhanced effect on up-regulating the expression of Bax, caspase-3 and p53, down-regulating the expression of Bcl-2 and E2F1, which may be responsible for its synergetic antitumor effect on myeloid leukemia cells.

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THE INFLUENCE OF MDSCS DEPLETION ON CD8 + T-CELLS IN LUNG CARCINOMA MOUSE MODEL

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Background: Recently, there has been a growing concern about bone marrow suppression induced by tumor initiation and growth. Myeloid-derived suppressor cells (MDSCs), as a significant element in marrow suppression, are precursor of dendritic cells, macrophages and granulocytes. MDSCs facilitate tumor development and tumor immune escape by inhibiting cellular immune response, especially depress number and function of $CD8^+$T-cells.
Aims: The aim is to explore the effect of MDSCs on number and function of $CD8^+$T-cells, tumor growth *in vivo* and survival in Lewis lung cancer mouse model.
Methods: LLC tumor cells were inoculated into the flank of C57 mice. MDSC were depleted by intravenous administration of Gr-1 blocking antibody. Tail blood was collected on specific time and then $CD8^+$T-cells and MDSCs percentages were detected by flow cytometry. Tumor growth and survival of tumor baring mice were also recorded.
Results: On day 7 after tumor inoculation, the percentages of MDSCs from tumor baring mice were significantly higher than control mice ($23.87 \pm 1.35\%$ VS $7.600 \pm 0.677\%$, $P < 0.001$) and $CD8^+$T-cells were declined than control ($7.906 \pm 0.428\%$ VS $11.22 \pm 0.606\%$, $P < 0.001$). Gr-1 blocking antibody was administrated to deplete MDSCs. On day 3 after administration, the percentages of MDSCs were rapidly decreased ($1.578 \pm 0.299\%$ VS $38.34 \pm 3.214\%$, $P < 0.001$) and $CD8^+$T-cells were significantly higher than control mice (9.464 ± 0.820 VS 4.024 ± 0.488, $P < 0.001$). On day 10 after administration, MDSCs percentages were higher than previous, but lower than control mice ($31.98 \pm 3.595\%$ VS $63.66 \pm 5.763\%$, $P = 0.0016$) and $CD8^+$T-cells were deceased than previous but higher than control mice ($5.806 \pm 0.554\%$ VS $2.894 \pm 0.330\%$, $P = 0.0019$). Tumor growth in mice with MDSCs depletion was markedly inhibited compared with control group ($P < 0.05$). The survival of mice with MDSCs depletion was notably longer than control mice ($P = 0.036$, median survival 46 days VS 53 days).
Summary/Conclusions: MDSCs depletion in LLC tumor baring mice eliminated the inhibition of $CD8^+$T-cells by MDSCs. Tumor growth was delayed and survival was prolonged than control mice.

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THE EFFECTS OF TH9, TH17, CD4 + CD25 + FOXP3 + REGULATORY T CELLS IN PERIPHERAL BLOOD OF PATIENTS WITH CHRONIC LYMPHOCYTIC LEUKEMIA

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Background: Chronic lymphocytic leukemia (CLL) is the most common form of leukemia, which is characterized by clonal expansion of $CD5^+$ B cells in periphery and lymphoid tissues. CLL progression is mainly affected by the immune status of the CLL patients. Several immune cells and mediators contribute to immunopathogenesis of CLL, such as Th1 and Th2 cells .Increased absolute number of total T lymphocytes has been shown in CLL patients which was associated with expansion of $CD8^+$ T cells, result in a low CD4:CD8 ratio. In some autoimmune diseases, $CD4^+CD25^+FoxP3^+$ regulatory T (Treg) cells maintain self-tolerance and inhibit antitumor responses, T helper (Th)9, (Th)17 cells may enhance inflammatory and antitumor responses.
Aims: The present study aims to investigate the presence and significance of Th9, Thl7, Treg cells and the related cytokines (IL-9, IL-17, TGF-β) in peripheral blood of patients with adult Chronic Lymphocytic Leukemia.
Methods: Peripheral blood samples were obtained from 48 CLL patients from the Second Hospital of Dalian Medical University (Dalian, China), including 14 females, 34 males; mean age, 66 ± 12 years, excluding those with a current infection. Control samples were collected from 40 healthy volunteers (13 females, 27 males; mean age, 61 ± 7 years), none of which suffered from any immune diseases. The percentages of Th9, Th17 and Treg cells in peripheral blood were detected with Flow cytometry. The serum levels of IL-9, IL-17 and TGF-β were examined by ELISA. Data analysis was carried out using SPSS 18.0 statistical software. Descriptive data was summarized and analyzed.
Results: Compared with healthy subjects, the percentages of Th9, Thl7 and concentrations of IL-9, IL-17 were all significantly increased in patients with CLL [$(1.29 \pm 0.33)\%$ vs $(0.70 \pm 0.27)\%$, $P < 0.05$; $(2.05 \pm 0.41)\%$ vs $(0.99 \pm 0.34)\%$, $P < 0.05$; $(26.79 \pm 7.21)\%$ vs $(16.31 \pm 5.07)\%$, $P < 0.05$; $(10.64 \pm 3.85)\%$ vs $(7.19 \pm 2.66\%)$, $P < 0.05$];the percentages of $CD4^+CD25^+FoxP3^+$ cells and concentrations of TGF-β were lower in patients [$(4.73 \pm 0.88)\%$ vs $(7.19 \pm 1.84)\%$, $P < 0.05$; $(3.92 \pm 1.33)\%$ vs $(6.61 \pm 1.91)\%$, $P < 0.05$]. Moreover, the percentages of Th9, Thl7 and concentrations of IL-9, IL-17 were all negatively correlated lymphocytic accounts ($\gamma s = -0.352$, $P = 0.041$; $\gamma s = -0.388$, $P = 0.029$; $\gamma s = -0.439$, $P = 0.013$; $\gamma s = -0.404$, $P = 0.026$) of patients with CLL; the percentages of $CD4^+CD25^+FoxP3^+$ cells and concentrations of TGF-β were positively correlated with lymphocytic accounts ($\gamma s = 0.413$, $P = 0.023$; $\gamma s = 0.408$, $P = 0.027$) of patients with CLL.
Summary/Conclusions: The abnormality Th9, Thl7 and Treg cells and their related cytokines (IL-9, IL-17 and TGF-β) changes in patients with CLL,which may be involved in immunological pathogenesis of CLL.

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Clinical Immunogenetics HLA in Transfusion Medicine

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THE SEROLOGICAL CHARACTERISTICS AND FAMILY GENETICS OF A NOVEL HLA ALLELE, HLA-A*26:82

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Background: To date, there are 16,429 HLA alleles documented according to the IMGT / HLA Sequence Database in January 2017, and more than 80% of them were submitted in the last 10 years. Besides identification of new sequences,many of the novel HLA alleles' profiles have not included their serological reactivities. HLA-A*26:82 was first detected in our laboratory during our HLA typing for a potential donor from China Bone Marrow Donor Program (CMDP). For further study, the serological characteristics and inheritance investigation were performed.
Aims: To investigate the sequence, family genetics and the antigen serological characteristics of a novel HLA allele, A *26:82.
Methods: The routine HLA tying for the potential donors from CMDP were performed by bi-allelic Sequence-Based Typing method, using a commercial kit (ROSE Europe GmbH, Frankfurt, Germany). In the cases, there was no fully matched HLA typing result obtained, allele group specific HLAssure-SE SBT Typing Kit (Texas Biogene Inc., Taipei, Taiwan) was employed to identify the nucleotide sequences of the novel allele. Fresh blood samples were collected of the proband and his family members with their consent, in order to analysis the serological reactivities and the possible haplotype associations bearing the novel allele. The HLA serological specificity was indicated by One Lambda (ASN72D)HLA kit.
Results: No fully matched result was obtained at HLA-A locus in HLA typing for a donor, which suggested the possible existence of a novel allele. The latter analysis indicated that the proband have a novel nucleotide sequence at HLA-A locus, the new sequence was mostly close to those of HLA-A *26:01:01:01, but 1 nucleotide substitution in exon 4, by nt 746 C-A (codon225 ACC-AAC), that resulted in one amino acid substitution,Thr- Asn. The novel HLA-A allele was officially named as HLA-A*26:82 by the WHO Nomenclature Committee for Factors of the HLA System. The HLA typing results of the propand and his daughter were assigned as HLA-A*26:82,33:02; B*48:01, 15:01; DRB1*11:01, 12:02 and A*26:82, 23:01; B*48:01, 50:01; DRB1*11:01,13:02, respectively. The haplotype associations to A*26:82 could be A*26:82-B*48:01-DRB1*11:01. According to the serological reactivity patterns, the antigen encoded by the novel HLA-A allele showed the characteristics of A26.
Summary/Conclusions: A novel HLA-A allele, HLA-A*26:82, was identified and associated to the haplotype, A*26:82-B*48:01-DRB1*11:01, and the encoded antigen showed the serological specificity of A26.

P-620

IDENTIFICATION OF A NOVEL HLA ALLELE, HLA-A*24:198 BY SEQUENCE-BASED TYPING IN A CHINESE HEMATOPOIETIC STEM CELL DONOR

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Background: HLA, as a consequence of human evolutionary mechanisms that diversify the peptide-presenting function of the major histocompatibility complex, is the most polymorphic in the genome. To date, more than 3900 HLA-A alleles have been identified according to IPD-IMGT/HLA Database.
Aims: To identify and confirm a novel HLA allele in a hematopoietic stem cell donor of Liaoning, China.
Methods: The routine HLA genotyping was performed by sequence-based typing SBTexcellerator®, Genome Diagnostics B.V., Arnhem, Holland). HLA-A locus specific amplification was performed from exon 1 through exon 8 by thermal cycling the amplification primer mix and template DNA using the Long Rang PCR kit (Qiagen, Hilden, Germany).
The exons 2, 3, and 4 were sequenced in both directions using SBTexcellerator® sequence primer, and Big Dye® Terminator v3.1 Reaction Kit (Applied Biosystems, Torrance, CA). The final reactions were purified using Ethanol/EDTA precipitation to remove unincorporated sequence primers and residual nucleotides and run on ABI PRISM® 3730xl Genetic Analyzer (Applied Biosystems, Foster City, CA). Sequencing data was analyzed using SBTengine® software.
Results: The obtained heterozygous sequence showed the presence of a novel allele, as the nucleotide sequence did not match with any known allelic combination. And only one nucleotide was heterozygous in the homozygote, so we deduced that one of two homozygous alleles was a novel allele. The result was that one allele was *HLA-A*24:02:01*, and the other one was a new *HLA-A*24* allele.
Summary/Conclusions: The new sequence differs from *HLA-A*24:02:01* by a single nucleotide substitution in exon 2 at position 157 (G>A); This mutation results in one codon change: at codon 29(GAC>ACC) where an Aspartic Acid (D) is substituted by an Asparagine (N). The nucleotide sequence was submitted to GenBank and given the accession number JQ392563. The name *A*24:198* has been officially assigned by the WHO Nomenclature Committee for Factors of the HLA system.

P-621

SEQUENCE ANALYSIS OF A NOVEL HLA ALLELE HLA-DRB1*13:161

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Background: With the rapid development of genetic immunology and molecular biology, the understanding and elucidation of the structure and function of HLA system has been gradually improved. At the same time, new alleles are constantly being discovered. These new findings not only rich HLA family, but also find a breakthrough point for the study of genetic superiority or disappearance of national gene.
Aims: To identify the novel HLA allele HLA-DRB1*13:161 and to analysis the nucleotide sequence of the abnormal reaction pattern.
Methods: The sample from volunteer of Chinese Marrow Donor Program were detected using PCR-SBT method. The ambiguous novel HLA allele was confirmed with single stranded SBT method, then DNA sequencing was performed to identify the difference between the novel allele and HLA- DRB1*13:12:01 allele. size: 12.0pt; line-height: 150%'>breakthrough point for the study of genetic superiority or disappearance of national gene.
Results: The sequence of the novel allele was different from all alleles in the HLA databases. After analysis, the novel allele has 1 nucleotide change from the DRB1*13:12:01 at nucleotide 230 where G→T (codon 53 CGG→CTG) resulting in a coding change, 53 Arg is changed to Leu.
Summary/Conclusions: The result suggested that the allele was a novel allele and it has been officially named HLA-DRB1*13:161 by the World Health Organization (WHO) HLA Nomenclature Committee.

P-622

Abstract has been withdrawn

P-623

ASSOCIATION STUDY OF HLA ALLELES (A, B, DRB1) AND HIV-1 INFECTION PROGRESSION IN THE HAN POPULATION OF HUBEI PROVINCE, CHINA

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Background: HIV-1 infected individuals show different disease process. As many as 95 percentage of HIV-1 infected people progressed to representative patients 2 to 10 years later, the natural disease course of other 5 percentage tends to over 10 years, these infected persons show no characteristic symptom of AIDS, maintain the amount of CD4 + T cells in normal without antiviral treatment[1]. The influencing factors of HIV-1/AIDS disease progression mainly have hereditary factors including HLA specificity, auxiliary receptor polymorphism of HIV-1 and viral factors. Recently research abroad has been demonstrated that multiple HLA genotype have associations with AIDS disease progression[2-5], while domestic research differ sharply, this is mainly due to regional and ethnic differences in the distribution of HLA alleles[6-9]. The correlation of HLA alleles polymorphism analysis and AIDS disease progression remains elusive in Han nationality from Hubei province of China gives

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a hand for the research of HIV-1 infected people who didn't progressed for a long time.
Aims: Firstly, we obtained gene frequencies of HLA alleles (A, B, DRB1) of HIV-1 infected group and AIDS patients group in Hubei province, China, then, comparing this two groups to seek out any difference, at last, we made a preliminary research of association of HLA Alleles (A, B, DRB1) and HIV-1 infection progression in HIV-1 infected group.
Methods: An amount of 424 HIV-1 seropositive individuals were chosen as study subjects. We divided this individuals into two groups – HIV-1 infected or AIDS patients according to clinical diagnosis standard. HLA-A, B, and DRB1 allele typing was performed using polymerase chain reaction-sequence-specific oligonucleotide (PCR-SSOP) and polymerase chain reaction-sequencing based typing (PCR-SBT) techniques. Arlequin ver3.0 was used to acquire the allele and haplotype frequencies of HLA-A, B, and DRB 1, whereas Epi Info 7 and SPSS18.0 was used to analyze the differences of HLA alleles between HIV-1 infected and AIDS patients.
Results: Logistic regression analysis revealed that B*57:01 maybe show a protective effect on HIV-1 infection, whereas DRB1*04:05 maybe show a positive effect on HIV-1 infection. All of these effects were independent of age, gender, and infection way of the host. Furthermore, Some of the common HLA haploid individual may have impact on HIV-1 infection. Homozygous HLA Alleles (A, B, DRB1) as well as HLA_Bw4-Bw6 showed unconspicuous effect on HIV-1 infection.
Summary/Conclusions: These results confirmed that HLA gene polymorphism influence AIDS disease progression. The correlation study of HLA alleles polymorphism analysis and AIDS disease progression gives a hand for the research of HIV-1 infected people who didn't progressed for a long time. Furthermore, this study will open up avenues for AIDS prevention and precise cure.

P-624

THREE HLA NOVEL ALLELES WERE IDENTIFIED BY SEQUENCE-BASED TYPING IN CHINESE INDIVIDUALS

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Background: More and more novel alleles have been confirmed with the development of HLA gene typing and the increasing capacity of the china bone marrow bank. Up to January 20, 2017, 1830HLA-A alleles and 4647 B alleles have been reported in the HLA gene database EBI (http://www.ebi.ac.uk/imgt/hla/) .
Aims: To identify three novel alleles of HLA by using group specific primer sequencing (GSSP).
Methods: Using the method of group specific primer sequencing to confirm the suspicious samples which performed by PCR-SBT sequencing but no exact result. The measured sequences were submitted to gene library of GenBank (www.ncbi.nlm.nih.gov/GenBank) and compared to the known allele sequences in the highest homology, then submitted the sequences to EBI (http://www.ebi.au.uk/imgt/hla/) to obtain submission numbers and official gene nomenclature.
Results: 3 new alleles were different from all known HLA alleles, of which two were the base mutation at 266 and 164 of exon2, which resulted in the change of amino acids. The third novel allele is caused by the base deletion at 287 in exon 2, resulting in the change of the reading frame of the gene and the change of amino acid and the termination of the translation in advance.
Summary/Conclusions: 3 cases of suspected new alleles were identified, and were assigned officially by the HLA Nomenclature Committee, assigned A*11:152, B*55:77 and B*44:237N.

P-625

STUDY ON THE ANALYSIS OF HIGH-RESOLUTION HLA-B ALLELES FROM 4253 HEMATOPOIETIC STEM CELL DONORS OF HAN NATIONALITY FROM HUBEI POPULATION

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Background: HLA is a group of closely linked, dominant genetic systems that have been found in the human genome to date. HLA is an important immune recognition and response system. It is of great significance in transplantation immunity, tumor immunity, infection immunity and so on.
Aims: analyze the genetic polymorphism, common and Well-documented (CWD) and rare alleles of high-resolution HLA-B alleles by analysis from hematopoietic stem cell (HSC)donors in Chinese.
Methods: PCR-sequence-based typing method was applied for HLA-B high-resolution genotyping of 4253 unrelated Han nationality from Hubei healthy donors of hematopoietic stem cells in Hubei branch of Chinese National Marrow Donor Program registry.
Results: 90 alleles of HLA-B locus were found. The frequencies of the most common alleles were B46:01(0.1526), B58:01(0.0824), B40:01(0.0756), B51:01 (0.0656). 82alleles of HLA-B were CWD, which account for 99. 8% of total number of samples. and a few rare alleles not reported in Chinese population were found
Summary/Conclusions: The results of high-resolution, CWD and rare alleles reflect the HLA distribution in Han nationality from Hubei population, and provide reference data for the study on population genetics of populations in China.

P-626

DETECTION OF FALSE POSITIVE OF HPA GENOTYPE BY PCR-SSP METHOD

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Background: PCR-SSP is one of the most common technologies for HPA genotyping. However, nonspecific amplification often appeared for this method due to the complex primer design and competition between amplified templates. This amplification leaded to false positive, resulting in error genotyping.
Aims: By analyzing the difference between specific amplified bands and nonspecific amplified bands, the false positive recognition capability was improved to ensure the correct genotype result.
Methods: PCR-SBT was used to confirm whether the PCR-SSP result was true positive, and the internal reference band was regarded as true positive band. Genetool software was used to analyze the difference between specific amplified bands and nonspecific amplified bands in the picture, and transformed the image data to the digital data. SPSS18.0 software was used to analyze the digital data.
Results: Consistency of specific amplification was better and the bands were clear.DNA length obtained by Genetool software was consistent with the reagent specification ($P > 0.5$),and the intensity was stronger than that of the nonspecific amplification ($P < 0.01$).Nonspecific amplification was blurred, smeared, multiple strip coexistence and DNA length was not consistent with reagent specification. C
Summary/Conclusions: The above method can effectively distinguish the specific and nonspecific amplification, reduce the typing error and get the correct classification results.

P-627

IDENTIFICATION OF TWO NOVEL GENOMIC FULL-LENGTH SEQUENCES OF HLA-E IN CHINESE INDIVIDUALS

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Background: HLA-E is one of the non-classical HLA class I genes. The whole gene of HLA-E include 8 exons, 7 introns and 5′promoter and 3′-UTR. At present, the study of polymorphism analysis mainly aims at the variation in exon 3 of HLA-E, which determines HLA-E*01:01 or HLA–E*01:03. However, studies about identification of the polymorphisms on genomic full-length HLA-E and its novel alleles are rarely reported.
Aims: To establish a method for identifying HLA-E genomic full-length sequence, and detect novel full-length alleles in healthy blood donors in Shenzhen, China.
Methods: Peripheral blood DNA were extracted from the whole blood of all research subjects. The amplification primers (EU:5′-GCCCAGCCAGGACTAATTTCT-3′;EL:5′-CAGGGAAGGAACACAGGTCAG-3′) and walking sequencing primers were designed in conserved regions according to the sequences of HLA-E released on the IMGT/HLA database. We use a high-fidelity and long-fragment PCR reaction system to amplify a 3.6 kb fragment covering the whole gene of HLA-E, and PCR product were sequenced by walking sequencing primers on Sanger sequencer ABI3730. Sequencing fragments were assembled and checked by Seqman Software, and the typing results were obtained by comparing the sequences with known allele sequences released on the IMGT/HLA database.

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Results: During this research, we successfully established the method for amplifying genomic full-length sequence of HLA-E and sequence-based typing assay. Two novel HLA-E alleles with the length 3.6 kb had been nominated by WHO HLA Nomenclature committee as HLA-E*01:01:01:06 and HLA-E*01:01:01:07. Compared with the most related allele HLA-E*01:01:01:01, HLA-E*01:01:01:06 has one nucleotide change at nt-26 (G->T) in 5′-promoter, and HLA-E*01:01:01:07 has one nucleotide change at nt3345 (T->C) in 3′-UTR.
Summary/Conclusions: The polymorphism data of genomic full-length HLA – E in Chinese individuals needs to be explored and extended, and the method we developed here supply the key technique for population genetics studies of HLA-E.

P-628

Abstract has been withdrawn

P-629

Abstract has been withdrawn

P-630

HLA-C HIGH RESOLUTION GENOTYPE OF HAN POPULATION IN HUBEI REGION

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Background: The human leukocyte antigen (HLA) complex contains the most polymorphic genes in the human genome. The classical HLA class I (HLA-A, HLA-B, HLA-C) and class II genes (HLA-DRB1, HLA-DQA1, HLA-DQB1) define the specificity of adaptive immune responses. Genetic variation at the HLA genes is associated with susceptibility to autoimmune and infectious diseases and plays a major role in transplantation medicine and immunology.
Aims: To research the distribution features of HLA-C high resolution alleles in the Han population from Hubei region.
Methods: Next-generation Sequencing (NGS)technologies was applied for HLA-C high resolution genotype of 3391 unrelated and healthy donors in the Han population from Hubei region, and gene frequencies were calculated by counting and maximum likelihood method.
Results: A Total of 41 HLA-C alleles were observed and the most frequent alleles (>0.0500)were C*01:02, C*07:02, C*03:04, C*08:01, C*06:02, C*03:03, C*03:02 and C*04:01.
Summary/Conclusions: Get the distribution features of HLA-C high resolution allele in the Han population from Hubei region. These data will provide an outline of the HLA characteristic of healthy individuals in our region and help bone marrow transplantation patients find suitable HLA-matched donors.

P-631

COMPARISON OF THE THREE GENERATIONS OF SEQUENCING TECHNOLOGY USED IN HLA GENOTYPING AND ITS PROSPECT

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Background: Sequencing technology gives genotype directly, thus it is used in HLA genotyping widely. Significant development of sequencing technology improved the efficiency and definition of HLA genotyping in a great extent.
Aims: The three generations of sequencing technology, Sanger sequencing, next-generation sequencing (NGS) and next-next-generation sequencing, used in HLA genotyping were compared.
Methods: Here we summarized the application of PCR-SBT and next-generation sequencing (NGS)in HLA genotyping, also with their advantages and disadvantages. On the other hand, the broad applying prospect of next-next-generation sequencing in this field was also discussed.
Results: Sanger sequencing, as a golden standard, is the most popular method used in HLA genotyping now, but the equivocal result is an important disadvantage. Next-generation sequencing (NGS)is rising fast due to its high-throughput and data accuracy. Next-next-generation sequencing is a future strategy for HLA genotyping in expectation of further simple operation, easy to achieve and more accurate result.
Summary/Conclusions: Both of sanger sequencing and next-generation sequencing are commonly used in HLA genotyping, their merit and demerit are complementary.

P-632

DISTRIBUTION OF HLA ANTIGEN IN PLATELET DONORS IN CHANGZHOU AREA

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Background: With the increasing use of platelets in the clinical, the infusion problems been more and more attention, but the platelet transfusion refractor (PTR) phenomenon is more and more common, severe input of allogeneic platelets in the body was Rapid damage or even endanger the lives of patients, so the study and solve the problem of PTR has become an important issue of clinical blood transfusion. HLA antibodies are the most common immune factors that cause PTR, accounting for 80% of all immune factors and 11.7% of all etiologies. There are HLA-A and B antigens on platelet membrane. HLA antigen can be produced by repeated transfusion due to strong antigenicity. According to statistics, more than 10 times the blood transfusion, antibody positive rate of 30% -85%. If patients with HLA antibody infusion of random platelets, the antibody can react with the input platelet, resulting in thrombocytopenia, resulting in transfusion of platelets ineffective, and produce adverse blood transfusion reactions. At present, the effective solution is to select HLA-compatible or compatible platelets. Therefore, you can choose with the patient ABO, Rh blood type isotype, and HLA antigen matched platelet donors, infusion of white light irradiation alone can achieve good results.
Aims: HLA-antibody detection was performed on fixed-platelet donors and HLA-antibody detection was available for patients with platelet transfusion. A single platelet was collected from a platelet donor using HLA-free antigen for HLA antibody to achieve better efficacy
Methods: 125 cases of platelets were collected and HLA genotype was detected by PCR-SSP. The positive cases were analyzed statistically
Results: HLA-B locus was found in 11 sites, mainly A02, A11, A24-based; HLA-B site detected 23 sites, mainly B40, B15, B46, B13-based; HLA-DRB1 Point detection of 13 sites, mainly DRB1 09, DRB1 12, DRB1 15-based. The HLA-A, B, DRB1 loci have polymorphism in the existing database of free-platelet donors in Chagzhou.
Summary/Conclusions: Due to the high genetic polymorphism of HLA, it is very difficult to find a perfectly matched donor. We can provide a number of non-HLA-specific antigen-free single platelets for patients who produce HLA antibodies with multiple platelets in order to achieve a better therapeutic effect. The establishment of a platelet donor HLA library will have far-reaching implications for addressing the problem of platelet ineffective infusion.

P-633

IDENTIFICATION A NOVEL HLA ALLELE HLA-B*15:275

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Background: HLA (human leukocyte antigen) system is the most abundant genetic system known to humans so far. The HLA system is composed of a series of closely linked loci consisting of highly polymorphic genetic complex. HLA gene is the necessary condition for the immune system to recognize itself and its own and maintain the stability of the environment *in vivo*. HLA is distinctly racially specific, so the identification of HLA in the region of new alleles can be improved with the opportunity to find donors.
Aims: To confirm the novel allele HLA-B*15:275 and analyzed the nucleotide sequence of the abnormal reaction pattern. tunity to find donors.
Methods: The HLA typing of sample DNA was performed by PCR-SBT. The ambiguous novel HLA allele was confirmed with single stranded SBT method, then DNA sequencing was performed to identify the difference between the novel allele and HLA-B*15:18:01 allele. Finally, it was modeled by Swiss-Model to three-dimensional structure of HLA Molecule.

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Results: The novel allele is not the same with all known HLA-B allele sequence. After analysis, there were two nucleotide differed from the HLA-B*15:18:01. One is at position 486 where G→C (codon 138 ACG->ACC) resulting in a coding change, The other is at position 527 where A->T (codon 152 GAG-> GTG) resulting in a coding change, 152Glu is changed to Val.
Summary/Conclusions: The result suggested that the allele is a novel allele that it has now been officially named B*15:275 by the World Health Organization (WHO) HLA Nomenclature Committee.

P-634

NUCLEOTIDE SEQUENCE ANALYSIS OF A NOVEL HLA-A*35:230 ALLELE

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Background: HLA is the most complex human genetic polymorphism system, it is a series of closely linked genes composed of highly polymorphic complex. The earliest research on HLA typing was based on serology and cytology, the most accurate is also the most thorough PCR-SBT technology as the gold standard for genotyping currently.
Aims: To confirm the novel allele HLA-B*35:230 and analyzed the nucleotide sequence of the abnormal reaction pattern.
Methods: The HLA typing of sample DNA was performed by PCR-SBT. The ambiguous novel HLA allele was confirmed with single stranded SBT method, then DNA sequencing was performed to identify the difference between the novel allele and HLA allele.
Results: The novel allele is not the same with all known HLA-B allele sequence. After analysis, there was 3 nucleotides differed from the HLA-B*35:88 gene sequence occurred in the third exon, at position 379 where C→G resulting in a coding change, 103 Leu is changed to Val; at position 499 where A→T resulting in a coding change, 143 Thr is changed to Ser; at position 512 where G→T resulting in a coding change, 147 Trp is changed to Leu.
Summary/Conclusions: The result suggested that the allele is a novel allele that it has now been officially named HLA-B*35:230 by the World Health Organization (WHO) HLA Nomenclature Committee.

P-635

CORRELATION BETWEEN KILLER CELL IMMUNOGLOBULIN-LIKE RECEPTOR AND SEVERE FEVER WITH THROMBOCYTOPENIA SYNDROME

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Background: Sever fever with thrombocytopenia syndrome (SFTS) is an emerging infectious disease in East Asia. SFTS is caused by a novel bunya virus, SFTS virus (SFTSV), which could be transmitted through tick bite. Shandong Province is the second highest incidence area of SFTS in China. There were 761 SFTS cases reported in Shandong Province from 2011 to 2014 with an overall mortality rate of 12%. SFTS is a hemorrhagic fever disease with fever and thrombocytopenia as the main clinical manifestations. The body temperature of most SFTS cases usually exceeds 38°C. Over 70% of the patients have fever over 39°C. Patients often had headaches, muscle aches, gastrointestinal symptoms such as loss of appetite, nausea, vomiting, abdominal pain, diarrhea and hematochezia, leukopenia, liver and kidney dysfunction. The majority of patients has a good prognosis and recovered. On the contrary, some patients have a poor prognosis because of basic diseases, older ages, neuropsychiatric symptoms, bleeding tendency and hyponatremia. Those patients who had severe bleeding tendency and in critical condition might die of multiple organ failure. In multivariate analysis, the odds for SFTS were 2.4–4.5fold higher in patients who reported tick bites or presence of tick in the living area. The killer cell immunoglobulin-like receptors (KIR) are interacting with human leukocyte antigens (HLA), regulating the activations of natural killer (NK) cells and certain T-cell subsets in response to microbe infection. There are accumulated literatures reporting on the association of KIR with different infectious diseases. but the correlation between KIR and SFTS had not yet been investigated.
Aims: To investigate killer cell immunoglobulin-like receptor (KIR) for a correlation with Severe Fever with Thrombocytopenia Syndrome (SFTS).
Methods: A total of 85 confirmed SFTS patients and 286 unrelated healthy individuals from Shandong province were genotyped by PCR-sequence specific primer (PCR-SSP) for KIR. Allelic frequency were estimated by direct counting and compared using χ^2 test.
Results: The frequency of KIR2DL2 and KIR2DS2 in SFTS patients are significantly lower than that in control group ($P < 0.05$). And the odds ratio of KIR2DL2 and KIR2DS2 are 0.1215 and 0.3181 respectively.
Summary/Conclusions: KIR2DL2 and KIR2DS2 might have correlated with the occurrence of SFTS.

Histocompatibility in Stem Cell Transplantation

P-636

GENETIC BACKGROUND AND SEROLOGICAL CHARACTERISTIC OF A NOVEL HLA ALLELE, HLA-A *01:71

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Background: Over the past ten years, more and more novel HLA alleles have been found, and the total number of HLA alleles has featured an accelerated growth, from 2683 in January 2007 to 16429 in January 2017. However, many new alleles now are lacking corresponding serological specificity. HLA-A*01:71 was a novel allele that our laboratory detected out in HLA typing for China Bone Marrow Donor Program (CMDP), followed by genetic pedigree and serological specificity investigations.
Aims: To identify the serological characteristic and investigate the genetic background of the novel allele HLA-A *01:71.
Methods: The routine HLA typings for the potential donors from CMDP were performed by bi-allelic Sequence-Based Typing method, using a commercial kit (ROSE Europe GmbH, Frankfurt, Germany). In the cases of no fully matched HLA typing results, group specific HLAssure-SE SBT Typing Kit (Texas Biogene Inc., Taipei, Taiwan) was employed to identify the nucleotide sequences of the novel allele. Fresh blood samples were collected from the proband and his family members with consent, in order to analyze the serological reactivities and the possible haplotype associations to the novel allele. The HLA serological specificity was indicated by One Lambda (ASN72D) HLA kit.
Results: No exact result was obtained at HLA-A locus in HLA typing for a donor. The latter analysis indicated that the proband had a novel nucleotide sequence at HLA-A locus, which was most close to the sequence of HLA-A *01:01:01:01, but for one nucleotide substitution in exon 2, by nt 261 G-C (codon63 GAG->GAC), resulting in one aminoacid substitution,Glu- Asp. The novel HLA-A allele was officially named as HLA-A*01:71 by the WHO Nomenclature Committee for Factors of the HLA System. The HLA typing results of the proband and his son were assigned as HLA-A*01:71,11:01;B*13:02,38:02;DRB1*07:01,09:01 and A*11:01, 30:01; B*13:02, 38:02; DRB1*07:01, 09:01, respectively. The HLA typing results of the proband's wife were A*11:01, 30:01; B*13:02, 39:01; DRB1*07:01, 09:01. Because all members of the HLA-DRB1′s genotyping results were same, it was unable to determine the genetic haplotype. However, according to the local HLA gene polymorphism research, haplotype A*30:01-B*13:02-DRB1*07:01 has the highest distribution frequency. Presumably, the related haplotype to A*01:71 could be A*01:71-B*13:02-DRB1*07:01. According to the serological reactivity patterns, the novel HLA-A allele showed the characteristic of A1.
Summary/Conclusions: A novel HLA-A allele, HLA-A*01:71, was identified, and was most likely associated with the haplotype, A*01:71-B*13:02-DRB1*07:01. Its serological specificity was A1.

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STUDY ON THE POLYMORPHISM OF HLA-A, B, DRB1 IN HUNAN HAN POPULATION AND ANALYSIS OF THE GENETIC DISTANCE

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Background: The HLA (human leukocyte antigen) system is the major histocompatibility complex (MHC) of humans. It consists of several loci of genes located on the short arm of chromosome 6 at 6p21.3. It is polymorphic and different among populations. To study the polymorphism of HLA-A, B, DRB1 loci and to calculate the genetic distance can provide the invaluable tools for the study of anthropology and history.
Aims: To study the polymorphism of HLA-A, B, DRB1 loci in 14 Hunan Han populations and to calculate the genetic distance among the 14 populations.
Methods: The HLA-A, B, DRB1 low genotyping (n = 12995) was done by the polymerase chain reaction with sequence specific primer and sequence specific oligonucleotide probes technique. The genetic distance between among the 14 populations were calculated by the Nei formula. The cluster analysis and phylogenetic tree were figured out by weighted pair-group method using arithmetic averages (WPGMA) method.
Results: 66 alleles were detected in total. The most frequent HLA-A, B, DRB1 alleles in 14 Hunan Han populations are A*11, A*02, A*24, A*33, B*40, B*46, B*15, B*13, DRB*09, DRB1*15, DRB1*12, DRB1*04 and DRB1*08 respectively. There are no distinct differences among the 14 Hunan Han populations. The genetic distance between Changsha and Yiyang is the nearest (1879), while Zhangjiajie and Xiangxizizhizhou is the farthest (44398). 4 clusters were achieved in the analysis.
Summary/Conclusions: The polymorphism of HLA-A, B, DRB1 are very multiplicity. The result of cluster analysis is similar to the anthropological and historic study of Hunan population. It provides the invaluable tools for the study of Hunan population.

P-638

VALIDATION OF THE CELL CONCENTRATION IN THE CRYOPRESERVATION HEMATOPOIETIC STEM CELL

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Background: Hematopoietic stem cells – apheresis are cryopreserved in transplantation setting, the techniques are usually set limits of the concentration of nucleated cells (NC) below 160 $\times$ 10^6 NC per ml and Dimethyl sulfoxide (DMSO) concentration of 10 percent in total medium. However, the standard NC concentrations have large volume to be reinfused with high costs of processing. Moreover, the large volume of reinfusing may be associated with significant side effects, mostly related to DMSO content. To simply reduce this unfavorable side effect, may be possible by increasing the final NC concentration to the certain limit and yet still be able to cryopreserve NC.
Aims: To evaluate the NC cell viability when final NC concentration for cryopreservation procedure was increased to be more than 160 $\times$ 10^6 NC/ml.
Methods: Sixty-seven bags of Hematopoietic stem cells from 16 patients, 7 males and 9 females with a median age of 49 years (range, 17- 64 years), were collected by regular apheresis system (COM.TEC Blood cell Separator FRESENIUS KABI) and cryopreserved in cryopreservation bags (Cryocyte, 500 ml, Origen Biomedical). The bags were separated into group 1 and 2, forty-three bags of standard NC concentrations were group 1 and twenty-four bags of high NC concentrations were group 2. The observed results of group 2 was calculated to obtain the final nucleated cell (NC) concentration in the range between 161 $\times$ 10^6 and 200 $\times$ 10^6 NC per ml. The cells were then frozen in the liquid phase of nitrogen for one month. Before reinfused, the number of viable NC was manually counted by Trypan blue staining (Sigma Chemical Co.)
Results: The mean (SD) of NC concentration before freezing was 129 (30) x 10^6 NC per ml (median, 132 $\times$ 10^6/ml; range, 49 $\times$ 10^6/ml - 160 $\times$ 10^6/ml) for group 1 and 178 $\pm$ 11 $\times$ 10^6 NC per ml (median, 178 $\times$ 10^6/ml; range, 162–196 $\times$ 10^6/ml) for group 2. The mean (SD) of viable cell before reinfused was 97 (1.5) percent (median, 97 percent; range, 91–99 percent) and 96 (1.8) percent (median, 96 percent; range, 92–99 percent) for group 1 and group 2, respectively. Both NC concentrations were no significant difference in the viability (P = 0.124).
Summary/Conclusions: This study showed that the viability is not different between standard NC concentrations and high NC concentrations. Therefore, the cryopreservation at high cell concentration would be an alternative ways for cryopreserved technique. This is also useful to minimize the total product volume to be infused which in turn decrease the risk of DMSO-related complications.

P-639

RARE HLA ALLELES AMONG HAN POPULATION IN SHAANXI REGION IDENTIFIED BY ION TORRENT-NGS

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Background: The Human Leukocyte Antigen (HLA) is the most polymorphic region of the human genome. Compared with Sanger-sequencing Based Typing (SBT) methods, Next Generation Sequencing (NGS) has significant higher throughput and depth sequencing characteristics, becoming increasingly attractive for HLA typing.
Aims: The aim of this study was to evaluate the presence of rare HLA alleles detected by SBT in Shaanxi region using NGS platform, and estimate haplotype pairs containing the rare alleles using their phenotypes and reference database.
Methods: 1040 healthy potential stem cell donors from Chinese Marrow Donor Program (CMDP, Shaanxi Province) in 2016 were typed at high resolution using SeCore SBT kits (ThermoFisher Life Technologies). Exon 2–4 (HLA-A, B, C), 2 and 3 (DQB1) and 2 (DRB1) were sequenced, and data were analyzed by uTYPE 7.2. Four DNA specimens from peripheral blood carrying rare alleles were performed and identified by the NXType™ workflow (One Lamda, Inc.) and the accompanying TypeStream™ software V1.1 on the Ion Torrent NGS platform. The amplification of 5'UTR–3'UTR full length of HLA-A, B, C loci and partial length of HLA-DRB1, DQB1 loci were sequenced using Ion Torrent S5 530 chip. We used the phenotype and "Be The Match Registry Haplotype Frequencies" table (https://bethematch.org/) as the reference database to predict haplotype pair containing the rare allele.
Results: According to IMGT/HLA Database 3.26, the complete matched SBT pattern of the four cases was B*40:113/*48:01 (case 1), B*40:01/*58:19 (case 2), DRB1*13:02/*13:10 (case 3) and DQB1*04:01/*05:102 (case 4), respectively. Ion Torrent S5 NGS confirmed the B*40:113, B*58:19 and DRB1*13:10 alleles, which were identified for the first time among Han population in Shaanxi region. However, case 4 with DQB1*04:01/*05:102 result implied by SBT was actually verified as DQB1*04 novel/*05:02 with a nucleotide substitution in exon 3 leading to an amino acid change. The very rare allele B*40:113, existing in Caucasian with a 0.0002% frequency, presented a strong linkage with C*03:04. According to the phenotype, B*40:113-C*03:04 was estimated as the highest-ranking haplotype pair for case 1's HLA-B–C haplotype blocks. Haplotype A*33:03-B*58:19-C*03:02, mainly distributed in Malay (0.0526%), was the highest-estimated relative frequency haplotype pair for case 2 with Shaanxi Han ethnic based on its phenotype. Case 3 had a phenotype characteristic that HLA-DRB1*13:10 and DQB1*06:03 were in very strong linkage disequilibrium.
Summary/Conclusions: The results demonstrate the feasibility and significant benefit of HLA typing by NGS as providing phase-defined and complete sequencing information. A large number of novel and rare alleles have been identified, which enriched the HLA database greatly and provided more information for donor selection.

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HLA HAPLOTYPES RECOMBINATION IDENTIFIED IN POTENTIAL HEMATOPOIETIC STEM CELL TRANSPLANTATION PATIENT FAMILIES FROM CHINESE HAN POPULATION: UTILITY OF NEXT-GENERATION SEQUENCING METHOD

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Background: Allogeneic hematopoietic stem cell transplantation (HSCT) from an human leukocyte antigen (HLA)-identical donor can be curative for eligible patients with non-malignant and malignant hematological disorders. In recent years, HLA-mismatched haploidentical relative donor, is increasingly considered as a viable therapeutic option for patients lacking HLA-matched donors. Recombination in the human MHC is thought to have played a role in generation of novel HLA haplotypes. We present here the identification of HLA haplotypes recombination event occurring in potential hematopoietic stem cell transplantation patient families from Chinese han population.
Aims: To analyze the human leukocyte antigen complex class I (-A, -C, and -B) and class II (-DRB1, and -DQB1) linked haplotypes of Chinese Han population, and to study the recombination events of the five classical loci in the inheritance of HLA haplotypes.
Methods: A total of 1409 peripheral blood samples were collected from 297 families in Chinese Han population who came to Shenzhen Blood Center for HLA typing. HLA-A/B/ C/ DRB1 / DQB1 alleles were typed with microbeads SSO assay based on Luminex platform using a commercially available LABType® SSO HLA Typing Kits

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(One Lambda, Inc., Los Angeles, CA), then the recombination sites were analyzed by family study. The samples of 78 individuals from the families with exchange recombination were selected and sequenced with Next-generation sequencing (NGS) HLA typing method, to judge whether the recombination was interallelic exchange or interlocus exchange. In brief, the whole gene of HLA-A, -B, -C, and the essential genomic regions of HLA-DRB1, and DQB1, were PCR-amplified in a single PCR for each locus. Amplicons of each sample were pooled and processed using NGSgo® DNA library preparation workflow for Illumina MiSeq. During library preparation the samples were fragmented, barcoded and multiplexed. Finally paired-end sequencing was performed on the Illumina MiSeq platform using a standard flow cell and data was analysed using HLA typing software NGSengine®.

Results: Among 815 offspring individuals of 297 families in Chinese Han population, 14 individuals with HLA-A-C-B-DRB1-DQB1 linked haplotypes had a recombination rate of 1.718%. Among 14 HLA haplotypes recombined family, 5 of them are found to have a crossover between HLA- A and -C loci and 9 of them have a crossover between HLA- B and -DRB1 loci. 6 of these recombination events occurred in the most common haplotypes A*3303-Cw*0302-B*5801-DRB1*0301-DQB1*0201 of Chinese Han population。Among 14 cases of recombination, 8 of them were formed by a crossover between maternal chromosomes, 6 cases a crossover between paternal chromosomes. 5 individuals with an exchange between A/C loci are all females. Among 9 cases with an exchange between B/DRB1 loci, 8 of them were males and 1 case was female.

Summary/Conclusions: This research enlarged the HLA haplotypes genetic map of Chinese Han population. It preliminarily locates the position and discipline of the occurrences of recombination, proving the valuable genetics data to the further research of the mechanism of HLA recombination. Meantime, NGS presents a significant technical advantage for accurate high-resolution HLA typing, and it is useful for resolving common ambiguities observed using current SBT method.

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THE SEROLOGICAL CHARACTERISTICS AND FAMILY GENETICS OF A NOVEL HLA ALLELE, HLA-B*13:26

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Background: Identification of the HLA new alleles has been an ongoing activity of HLA science from the beginning. With the advent of high throughput and high resolution typing methods the rate of new allele discovery has accelerated. As to July 2017, there have been 4,828 HLA-B alleles documented according to the IMGT/HLA Database, and most of them were submitted in the recent decades. Besides the identifications of their new sequences, many of the novel HLA alleles' profiles have not included their serological characteristics. The HLA-B*13:26 was first detected in our laboratory during our routine HLA typing of a potential HSCT donor of Han ethnic from Shandong province of northeast China. For further study its serological characters and inheritance background, the serological reactivity and family investigation were performed.

Aims: To investigate the novel sequence and its family genetics and the antigen serological characteristics of the novel HLA allele, B*13:26.

Methods: The new allele was initially detected from a registered donor of China Marrow Donor Program during a routine HLA SBT typing using ROSE-HLA-SBT commercial kit (ROSE Europe GmbH, Germany), analyzed by Conexio Genomics software (Conexio Genomics Pty Ltd, Fremantle, Australia). As to clarify the unusual HLA-B SBT typing results, the donor and the family were typed again using new DNA isolates by HLAssure™ SE SBT HLA B Typing Kit (Texas BioGene Inc. Texas, U.S.A), sequence analyzed by AccuTypeTM SBT software (Nova BioSoft LLC, Oklahoma, U.S.A.). The serological specificity was typed by Terasaki HLA Class I Typing Tray (One Lambda, Inc. Canoga Park, CA, U.S.A.).

Results: The initial SBT results showed that there was no fully matched assignment was obtained at HLA-B locus for the subject, which suggested the possible existence of a novel allele. The repeated single allele-specific sequencing results were consistent with the initial SBT results and showed that the subject has a novel nucleotide sequence at HLA-B locus. The new sequence was mostly close to that of HLA-B *13:01:01, but has 4 nucleotides substitutions in exon 3 at: nt 387 G→C (codon105 CCG→CCC), 412 A→G (codon114 AAC→GAC), 419,420 TA→CC (codon116 TTA→TCC); which result in the amino acid exchanges of Asn114→Asp, Leu116→Ser. A nucleotide BLAST search in the IMGT/HLA Database confirmed the existence of this new HLA-B*13 allele. The name HLA-B*13:26 has been officially assigned by the WHO Nomenclature Committee. This follows the agreed policy that subject to the conditions stated in the most recent nomenclature report (Marsh et al. 2010), names will be assigned to new sequences as they are identified. The complete HLA typings of the subject donor and his daughter were A*26:01, B*13:26, DRB1*07:01; A*02:01, B*15:11, DRB1*09:01 and A*02:01, B*15:11, DRB1*09:01; A*02:01, B*48:01, DRB1*15:01 respectively. So the novel allele-bearing haplotype should be A*26:01, B*13:26, DRB1*07:01. According to the serological reactivity patterns, the antigen encoded by the novel B*13:26 allele showed most likely the characteristics of B13.

Summary/Conclusions: A novel HLA-B allele, HLA-B*13:26, was identified. The haplotype expressing the new allele is A*26:01, B*13:26, DRB1*07:01. The serological specificity of the novel allele showed most likely the characteristics of B13.

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IDENTIFICATION OF THE NOVEL HLA-B*27:103 ALLELE IN A CHINESE BONE MARROW DONOR

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Background: The human leukocyte antigen (HLA) genes are the most polymorphic in the human genome. As of April 2016 a total of 10730 HLA classIalleles and 3743 HLA classIIalleles have been assigned. There are more than 4200 alleles HLA-B alleles assigned (1).In this study, we describe a novel allele HLA-B*27:103 found in a Chinese bone marrow donor.

The human leukocyte antigen (HLA) genes are the most polymorphic in the human genome. As of April 2016 a total of 10730 HLA classIalleles and 3743 HLA classIIalleles have been assigned. There are more than 4200 alleles HLA-B alleles assigned. In this study, we describe a novel allele HLA-B*27:103 found in a Chinese bone marrow donor.

Aims: we describe a novel allele *HLA-B*27:103* found in a Chinese bone marrow donor in this study.

Methods: The donor of China Marrow Donor Program (CMDP) was originally genotyped for the HLA-A,-B and DRB1 loci at high resolution level using commercial polymerase chain reaction sequence-based typing (PCR-SBT) kits according to the manufacturer's instruction.

Results: The HLA-B*27:103 has one nucleotide change from B*27:04:01 at position 121 (C>T).

Summary/Conclusions:: The name B*27:103 has been officially assigned by the WHO Nomenclature Committee in March 2013.

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PRELIMINARY ANALYSIS ON ANTIGEN POLYMORPHISM STUDY OF PLATELET DONOR REGISTRY WITH KNOWN HLA TYPING IN ZHUANG POPULATIONS IN GUANGXI AREA

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Background: Our lab established a voluntary apheresis platelet donor registry in the Guangxi Zhuang population who live in Guangxi, then investigated the allele and haplotype frequencies of HLA-A, and -B loci.

Aims: To resolve the platelet transfusion refractoriness (PTR): Methods: The blood sample of 172 Zhuang regular apheresis platelet donors in Guangxi region were genotyped by PCR-SSOP, allele frequencies were obtained by direct counting method and was compared with results from various regions. The Hardy-Weinberg equilibrium (HWE) test at each locus was performed by Arlequin software package, the P-values are considered significant at the 0.05 level.

Results: The P values of the HLA-A and HLA-B loci showed that the HLA allelic distribution in Guangxi Zhuang population was in accordance with Hardy-Weinberg

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equilibrium ($P > 0.05$), a total of 11 HLA-A alleles, 24 HLA-B alleles and 58 A-B haplotypes were identified.
Summary/Conclusions: The Guangxi Zhuang population of platelet donor registry was genetically close to southern Chinese populations, but it still keeps its unique genetic characteristics, which can rapidly find compatible donors for patients with PTR.

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DETERMINATION OF DONOR-SPECIFIC ANTIBODIES (DSA) STRENGTH TO USE FOR VIRTUAL CROSSMATCH

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Background: Human leukocyte antigen (HLA) antibodies can be the barrier to organ transplantation. Donor-specific antibodies (DSA) can cause antibody-mediated rejection. The strength of DSA correlates with the risk of antibody mediated rejection both pre- and post transplantation. Theoretically, the sample with HLA-DSA as identified by separately HLA class I and II should be positive by other crossmatch techniques. Many centers have attempted to define the strength of DSA as mean fluorescent intensity (MFI) values that can predict crossmatch results. However, the strength of DSA detected was different among laboratories.
Aims: The aim of this study was to determine the strength of HLA-DSA associated with cytotoxicity crossmatch (CDCXM) and flow cytometric crossmatch (FCXM). Then, to define a cutoff value for DSA strength that can predict a positive virtual crossmatch results.
Methods: We retrospectively analyzed the results performed at Histocompatibility and Immunogenetics Laboratory, National Blood Centre Thai Red Cross Society, during 2014 to 2016. Totally 83 samples included 18 samples of APHIA Quality Assurance Program (QAP) and 65 samples of patients received living donor related kidney transplants. All samples had been crossmatched by CDCXM and 19 samples by FCXM. We detected HLA-DSA using Luminex single bead-antigen (Labscreen LABScreen®, Onelamda, Canoga Park, CA) assay. To analyze the strength of HLA-DSA, if the sample contained more than one antibody, the highest MFI value of such sample with positive FCXM will be considered. To determine the cutoff of HLA-DSA, the lowest MFI value detected as positive by FCXM was selected.
Results: A total of 83 samples were composed of 44 samples of absent HLA-DSA and 39 samples of present HLA-DSA. Among those 16 of 39 samples were positive for CDCXM and 19 of 39 samples were positive for FCXM. For a positive CDCXM, the lowest MFI of HLA class I – DSA was 17066 and HLA class II – DSA was 8428. Regarding a positive FCXM, the lowest MFI of HLA class I - DSA was 1056 and HLA class II – DSA was 824. Additionally, the MFI of HLA class I- DSA was 1056 and HLA class II – DSA was 824 that gave 100% consensus with FCXM QAP samples. It was formed that 26 samples of tested serum had HLA class I – DSA and 16 samples of HLA class II – DSA, those showed the positive predictive values (PPV) and negative predictive values (NPV) for HLA class I – DSA were 100% and 92%, respectively. Interestingly, the PPV and NPV of HLA class II - DSA was 100% and 100%, respectively.
Summary/Conclusions: The minimum MFI value of a positive FCXM tested samples for HLA class I and II were 1056 and 824, respectively and considered as cutoff value for the positive HLA antibodies. The MFI strength of HLA-DSA that gave positive FCXM may be used for virtual crossmatch. However, these strength values may be varied which depend on various factors such as sample size of the study, kit reagents and the methodology.

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